

NONPARAMETRIC STATISTICAL PROCESS CONTROL

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MARIEN ALET GRAHAM

with website

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Nonparametric Statistical Process Control

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WILEY

This edition first published 2019
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9600 Garsington Road, Oxford, OX4 2DQ, UK

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Library of Congress Cataloging-in-Publication Data

Names: Chakraborti, Subhabrata, author. | Graham, Marien Alet, author.
Title: Nonparametric statistical process control / Subhabrata Chakraborti, Marien Alet Graham.
Description: Hoboken, NJ : John Wiley & Sons, 2019. | Includes bibliographical references and index. |
Identifiers: LCCN 2018023685 (print) | LCCN 2018032639 (ebook) | ISBN 9781118890677 (Adobe PDF) | ISBN 9781118890578 (ePub) | ISBN 9781118456033 (hardcover)
Subjects: LCSH: Nonparametric statistics. | Process control—Statistical methods.
Classification: LCC QA278.8 (ebook) | LCC QA278.8 .C445 2018 (print) | DDC 519.5–dc23
LC record available at https://lccn.loc.gov/2018023685

Cover Design: Wiley
Cover Image: © Rachelle Burnside/Shutterstock

Set in 11/13pt CMR10 by SPi Global, Chennai, India

10 9 8 7 6 5 4 3 2 1

To the memory of my parents,

Himangshu and Pratima

To my wife, Anuradha, and our son, Siddhartha Neil

SC

To the memory of my grandmother, Marie Lochner

MAG

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Preface

Statistical process control (SPC) methods have a long and successful history. Starting around the late 1930s, led by Walter Shewhart and Edward Deming, these methods have stood the test of time and have revolutionized many facets of industrial production, not just in the United States but also in countries around the world. Over the last several years, research in SPC, particularly on control charts, has seen a phenomenal growth. Control charts are no longer confined to manufacturing and are now applied for process control and monitoring in a wide array of applications, from education to disease mapping and crime prevention. During his visit in the Department of Statistics at the University of Pretoria in South Africa in 2004 on a Fulbright fellowship, the first author had an opportunity to develop and teach a postgraduate course on SPC. That visit and the course turned out to be the beginning of a very successful decade-long relationship, including the establishment of new undergraduate and postgraduate courses and the completion of several honors essays, master's theses, and two doctoral dissertations under his supervision. In this long period of teaching and collaborative research, it was realized that, although there were good textbooks on SPC, most of their focus was on the applied side, and much of the more recent progress on the technical developments, particularly those that have taken place over the last decade or so, were not fully embraced or covered. Much of the relevant details were scattered all over the literature. This often created an inconvenient gap for users and researchers. Hence, the idea for a new book on SPC was born. The goal was to write a book that would bring some of the more recent developments, including the use of computers and simulations, within the reach of both the researchers and the practitioners of SPC.

During this period of time, another interesting development took place. The vast majority of the control charts found in the literature are developed under the assumption that the process is normality distributed. While these are useful, within the quarters where SPC is applied, there has been some uneasiness among both users and researchers, since the process distribution was often unknown (or the normality assumption was untenable), and hence the efficacy of the parametric control charts was either unknown or in question. Thus, as has happened in the fields of classical statistical estimation and hypothesis testing over the last many years, the subject of nonparametric SPC (NSPC) came into the forefront of research some fifteen or so

years ago. Now, the field is grown, so much so that researchers from many parts of the world are developing new methodologies that are challenging the conventional wisdom in SPC. The first author, his students, and collaborators have played, and continue to play, a key leadership role in pushing this exciting methodological front forward. Yet, there remains much work to be done.

In this book, written with Marien Graham, the second author and a former doctoral student from the University of Pretoria, an attempt is made to cover some of the key ideas in the field of univariate control charts. Although the focus is on the newer nonparametric control charts, the reader is first introduced to the main classes of parametric (normal theory) control charts and the associated theory, so that a proper foundational background can be laid.

The book is aimed at the advanced undergraduate (honors) and postgraduate (masters' and doctoral) students and researchers working in SPC. The reader is expected to have at least a basic background in statistical methods as well as in probability and statistical inference (mathematical statistics). A related background in mathematics (knowledge of calculus) is assumed. The goal is to write a book for the student, the interested practitioner, the self-learner and the researcher, so a careful balance is struck between theory and practice. It is intended to be a vital source of reference for the researcher so that relevant theories are presented along with many references, including some of the most recent advances in the literature. Another important distinguishing feature of this book is its use of computers and software, including simulations and graphics to illustrate concepts and research ideas in SPC.

The book is partly based on the research done by the authors with their collaborators and the lecture notes developed over several years of teaching at the Department of Information Systems, Statistics and Management Science, University of Alabama, at the Department of Statistics and the Department of Science, Mathematics and Technology Education at the University of Pretoria. At the Department of Statistics, these courses have been taught successfully over the last several years and now comprise an integral part of the statistics honors and masters' programs.

In a nutshell, in this book, quality control methodology, mostly control charts, are presented from a statistician's viewpoint, striking a careful balance between theory and practice, explanations and implementations. The first chapter gives a review of and the background to mathematical statistics covering well-known topics such as basic probabilities, random variables and their distributions, expected values, moments, variance, skewness, the central limit theorem, order statistics, and so on. The second chapter is an introductory chapter to the basic concepts and theory of statistical quality control. Here we discuss some basic terminology and concepts, such as the types of variability encountered, the definition of a control chart, the

different types of control charts, the design of control charts, the sample size, sampling frequency, the choice of control limits and so on. In the third chapter parametric control charts are considered where one has knowledge about the underlying process distribution, which is typically the normal distribution. In the fourth chapter the distribution-free (nonparametric) charts are discussed for the case where the underlying process distribution is unknown. Finally, in Chapter 5, some miscellaneous nonparametric control charts are discussed. Throughout the book guidance is given on the choice of control chart, the choice of the charting constants (i.e. control limits), and other quality-related matters. Required formulas and tables are provided.

The book has been several years in the making. A work of this magnitude couldn't be completed without the support and cooperation from many, who have contributed with their time, work and knowledge, directly or indirectly, enriching our knowledge and understanding of the subject matter. They all deserve our gratitude. This group of people includes our current and former students, colleagues, research collaborators and friends. It would be hard to list them all individually, we name a special few. The authors are especially thankful to Professor Nico Crowther and Professor Chris Smit from the Department of Statistics, University of Pretoria, for their interest, enthusiastic support and cooperation, over the long haul, without which much of the impetus for this project would simply not exist. The support from the South African Research Chairs Initiative (SARChI) is also gratefully acknowledged. Finally, the chain reaction that started all of these activities would most likely not have happened without the first author's visit to the University of Pretoria on a Fulbright Fellowship awarded by the Council of International Exchange of Scholars, USA. That visit was also made possible with the strong support from the University of Alabama, Tuscaloosa, and in particular from Professor Edward Mansfield, the Head of the Information Systems, Statistics and Management Science department at the time, where the first author is a professor. We owe a lot to these people and organizations.

We would like to acknowledge the editorial and production team at Wiley, including Blesy Regulas, for their support. We are grateful to Ms. Debbie Jupe, a former editor at Wiley, who played a key role in initiating this project. We are also thankful to the authors and publishers who have given us permission to reproduce their work, such as tables, figures, etc., in our book.

Last and not the least, we thank our family members for their sacrifices, encouragement and support during the years this book was written. Without their full cooperation, this project couldn't have been completed.

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About the companion website

The companion website for this book is at

www.wiley.com/go/chakraborti/Nonparametric_Statistical_Process_Control



The website includes:

- SAS programs
- Microsoft Excel data files

CHAPTER 1

Background/Review of Statistical Concepts

Chapter Overview

This chapter gives an overview of some key statistical concepts as they relate to statistical process control (SPC). This will aid in familiarizing the reader with concepts and terminology that will be helpful in reading the following chapters.

1.1 BASIC PROBABILITY

The term **probability** indicates how likely an event is or what the chance is that the event will happen. Most events can't be predicted with total certainty; the best we can do is say how likely they are to happen, and quantify that likelihood or chance using the concept of probability. A probability is a real number between (and including) zero and one. When an event is certain to happen, its probability equals one, whereas when it is impossible for the event to happen, its probability equals zero. Otherwise, the event is likely to happen or occur with a certain probability, expressed as a fraction between zero and one. For example, when a coin is tossed, there are two possible outcomes, namely, that a head (H) or a tail (T) can be observed. Note that an **outcome** is the result of a single trial of an experiment and the **sample space** (S) constitutes all possible outcomes of an experiment (the sample space is exhaustive). In the coin tossing example, the sample space is given by S = {H,T}. If the coin is unbiased (or fair), the probability (P)

Nonparametric Statistical Process Control, First Edition.
Subhabrata Chakraborti and Marien Alet Graham.
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Companion website: www.wiley.com/go/chakraborti/Nonparametric_Statistical_Process_Control

of observing a head is the same as the probability of observing a tail, each of which equals $\frac{1}{2}$. The probability of the set of all possible experimental outcomes in the sample space must equal one. In this example, this is evident since $P(\mathrm{H}) + P(\mathrm{T}) = 0.5 + 0.5 = 1$. When all experimental outcomes in the sample space are equally likely, this is referred to as the **classical method** of assigning probabilities, which is illustrated in the coin example. Another example of the classical method of assigning probabilities is when a dice is thrown. In this case, the sample space is given by $\mathrm{S} = \{1,2,3,4,5,6\}$ and if the dice is unbiased (or fair) the probability of observing a one on the dice is the same as observing any other value on the dice that equals $\frac{1}{6}$. Mathematically, we can write

$$\mathrm{P}(\mathrm{E}_1) = \frac{1}{6} \quad \mathrm{P}(\mathrm{E}_2) = \frac{1}{6} \quad \mathrm{P}(\mathrm{E}_3) = \frac{1}{6} \quad \mathrm{P}(\mathrm{E}_4) = \frac{1}{6} \quad \mathrm{P}(\mathrm{E}_5) = \frac{1}{6} \quad \mathrm{P}(\mathrm{E}_6) = \frac{1}{6}$$

where E_i defines the ith experimental outcome, i.e.

$E_1 = 1$ Observed value on the dice is a one
$E_2 = 2$ Observed value on the dice is a two
$E_3 = 3$ Observed value on the dice is a three
$E_4 = 4$ Observed value on the dice is a four
$E_5 = 5$ Observed value on the dice is a five
$E_6 = 6$ Observed value on the dice is a six

Again, note that the probability of the set of possible experimental outcomes equals one since

$$\begin{aligned}&\mathrm{P}(\mathrm{E}_1) + \mathrm{P}(\mathrm{E}_2) + \mathrm{P}(\mathrm{E}_3) + \mathrm{P}(\mathrm{E}_4) + \mathrm{P}(\mathrm{E}_5) + \mathrm{P}(\mathrm{E}_6)\\ &\quad = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{6}{6} = 1.\end{aligned}$$

In the two examples given above, the experimental outcomes are equally likely. Let's consider an experiment where the experimental outcomes are not equally likely. Suppose that a glass jar contains four red, eight green, three blue, and five yellow marbles. If a single marble is chosen at random from the jar, what is the probability of choosing a specific color, say, a red marble? In general, the probability of an event occurring is calculated by dividing the number of ways an event can occur by the total number of possible experimental outcomes.

$$P(\text{Red}) = (\text{Number of red marbles})/(\text{Total number of marbles}) = \frac{4}{20}$$

$$P(\text{Green}) = (\text{Number of green marbles})/(\text{Total number of marbles}) = \frac{8}{20}$$

$$P(\text{Blue}) = (\text{Number of blue marbles})/(\text{Total number of marbles}) = \frac{3}{20}$$

$$P(\text{Yellow}) = (\text{Number of yellow marbles})/(\text{Total number of marbles}) = \frac{5}{20}$$

Again, note that the probability of the set of possible experimental outcomes equals one since

$$P(\text{Red}) + P(\text{Green}) + P(\text{Blue}) + P(\text{Yellow})$$
$$= \frac{4}{20} + \frac{8}{20} + \frac{3}{20} + \frac{5}{20} = \frac{20}{20} = 1.$$

When all the experimental outcomes are not equally likely, this is referred to as the **relative frequency method** of assigning probabilities, which is illustrated in the marble example.

Next, we consider random variables and their distributions that play the most important roles in statistics and probability.

1.2 RANDOM VARIABLES AND THEIR DISTRIBUTIONS

A **random variable**, denoted as X, can take on a value, or, an interval of values, with an associated probability. The random variable can be univariate (one) or bivariate (two) or even multivariate (more than two). There are two major types of random variables, namely, discrete and continuous. Although there are situations where there can be a mixed random variable, which is partly discrete and partly continuous, we focus on the discrete and continuous variables here. To illustrate a **discrete random variable**, let's consider the coin example where either a head or a tail can be observed in a trial (a coin toss). Suppose that a coin is tossed five times and the random variable X denotes the number of heads that are observed. Then X can only take on integer values S = {0,1,2,3,4,5} and, accordingly, X is a discrete random variable. Another example of a discrete random variable would be an X that denotes the number of members in a household. Alternatively, a **continuous random variable** can take on values within some range. The probability of a continuous variable taking on any specific value is zero. If X denotes the height of a tree, then it is possible for a tree to have a height of 2.176 m or even 2.1765482895 m; the number of decimal places depends on the accuracy of the measuring instrument. Thus X can take on values other than only integer values, within some range of values and, accordingly, X is a continuous random variable. Another example of a continuous random variable would be if X denotes the lifetime of a light bulb.

A random variable has an associated **probability mass function** (pmf) if discrete or a **probability density function** (pdf) if continuous. First, we define the **cumulative distribution function** (cdf) before defining the pmf and pdf for discrete and continuous random variables, respectively.

Every random variable has a cumulative distribution function (cdf) that defines its distribution. The cdf is a function that gives the probability that a random variable X is less than or equal to some real value x, that is, $F(x) = P(X \leq x)$. In the case of a discrete random variable, the cdf is calculated by adding the probabilities up to and including x, whereas for a continuous random variable, the cdf is calculated by finding the area (integrating) under its pdf up to x. The cdf is a monotone non-decreasing right-continuous function, which is a step function for a discrete random variable (see Figure 1.1) and is a continuous function for a continuous random variable (see Figure 1.2). For Figure 1.1, it should be noted that $x_1, x_2, \ldots, x_k, \ldots$ with $x_1 < x_2 < \ldots x_k < \ldots$ are the discrete values that the random variable X can take on. For more details on the properties of a cdf see any mathematical statistics book.

The pmf of a discrete random variable X is a function that gives the probability $f(x)$ that the random variable X takes on the value of x, that is, $f(x) = P(X = x)$. More formally, let $f(x_i) = P(X = x_i) \forall\, i$, satisfying the following two conditions

$$0 \leq f(x_i) \leq 1 \,\forall\, i \text{ and } \sum_i f(x_i) = 1.$$

The pdf of a continuous random variable X is the first derivative of the cdf $F(x)$. That is, $f(x) = \dfrac{d}{dx}F(x)$. More formally, the pdf must satisfy the following two conditions

$$0 \leq f(x) \leq 1 \,\forall\, i \text{ and } \int f(x)dx = 1.$$

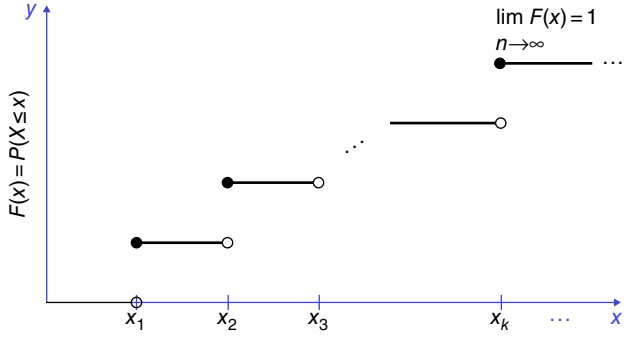


FIGURE 1.1 *The cdf for a discrete random variable.*

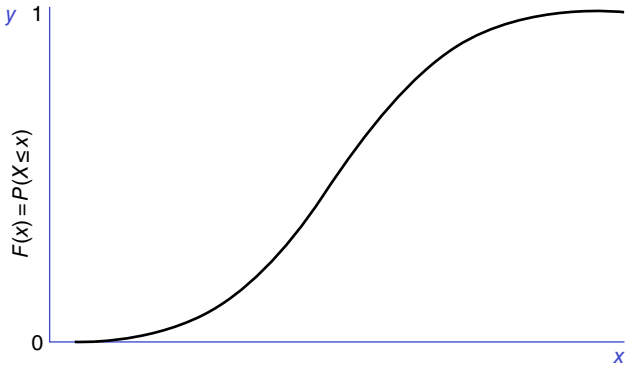


FIGURE 1.2 *The cdf for a continuous random variable.*

The cdf (or equivalently the pmf and the pdf) describes the distribution of a random variable over its values or its range or domain of values, that is, how the total probability (which equals one) is distributed or spread out over the values or the range of values of the random variable(s). Probabilities may be either marginal, joint, or conditional. A **marginal probability** is the probability of the occurrence of a single event. It may be thought of as an unconditional probability since it is not conditioned (or dependent) on another event. An example of a marginal probability is the probability that a red card is drawn from a deck of cards, which is given by $P(\text{Red}) = \dfrac{26}{52} = \dfrac{1}{2}$, since 26 out of 52 cards, that is, half the cards in a deck of cards, are red. A **joint probability** is the probability of the joint occurrence (or the intersection) of at least two events. The probability of the intersection of two events, A and B, may be written as $P(A \cap B)$, for example, the probability of drawing a red ace from a deck of cards is given by $P(\text{Red} \cap \text{Ace}) = \dfrac{2}{52}$, since there are two red aces in a deck of 52 cards, namely, the ace of hearts and the ace of diamonds. A **conditional probability** is the probability of event A occurring, given that event B occurs, and is denoted by $P(A|B)$, for example, the probability of drawing an ace, given that the card is red, is given by $P(\text{Ace} \,|\, \text{Red}) = \dfrac{2}{26}$, since there are two aces in the total of 26 red cards, namely, the ace of hearts and the ace of diamonds. The definition of a conditional probability $P(A|B)$ is given by

$$P(A|B) = \frac{P(A \cap B)}{P(B)}. \tag{1.1}$$

This formula shows the relationship between the marginal, the joint, and the conditional probability. Returning to the example of the deck of cards, for example, $P(\text{Ace} \,|\, \text{Red}) = \dfrac{P(\text{Ace} \cap \text{Red})}{P(\text{Red})} = \dfrac{2/52}{1/2} = \dfrac{2}{26}$, which is the same

answer as found previously. Typically, a marginal probability relates to an event associated with a single (scalar) random variable, whereas both joint and conditional probabilities relate to events associated with two or more random variables, that is, in a bivariate or a multivariate distribution.

Next, consider the distribution of two random variables, X and Y. We may like to know how the two random variables relate to each other, say, how one might affect or influence the other. We thus use the term **joint probability distribution** or the **joint distribution** to describe the probability distribution of the two random variables, as they vary jointly or together, over the mass points or the ranges of the random variables under consideration. The joint distribution can then be calculated from the joint probabilities that two events, related to the two random variables, occur simultaneously. For example, for two discrete random variables, X and Y, the **joint** pmf can be written as

$$\begin{aligned} P(X = x, Y = y) &= P(X = x|Y = y)P(Y = y) \\ &= P(Y = y|X = x)P(X = x) \end{aligned}$$

where $P(X = x|Y = y)$ and $P(Y = y|X = x)$ are the conditional probabilities of X given $Y = y$, and of Y given $X = x$, respectively, and $P(X = x)$ and $P(Y = y)$ are the marginal probabilities of X and Y, respectively. On the other hand, for two continuous random variables the **joint** pdf can be written as

$$f_{X,Y}(x, y) = f_{X|Y}(x|y)f_Y(y) = f_{Y|X}(y|x)f_X(x)$$

where $f_{X|Y}(x|y)$ and $f_{Y|X}(y|x)$ are the conditional pdfs of X given $Y = y$, and of Y given $X = x$, respectively, and $f_X(x)$ and $f_Y(y)$ are the marginal pdfs of X and Y, respectively. The latter, that is, a **marginal probability distribution**, is defined next. Often, when confronted with the joint probability distributions of two random variables, we wish to restrict our attention to the distribution of the individual variables. These probability distributions are called the marginal probability distributions of the respective individual random variables and each of them can be obtained by summing (or integrating) the joint distribution over the range of the other variables. Two events are said to be **independent** if the occurrence of the one event does not affect the probability of the other event. In this case, the joint distribution for discrete independent random variables is given by $P(X = x, Y = y) = P(X = x)P(Y = y)$, and for continuous random variables we have $f_{X,Y}(x, y) = f_X(x)f_Y(y)$. Finally, note that the joint cdf of two or more random variables (discrete or continuous) can be obtained by summing (or integrating) the relevant values (or the range of values) of the joint pmf (or the pdf) of the corresponding random variables. In the continuous case, the joint pdf is obtained by calculating the partial derivative

of the joint cdf. Any standard probability book can be consulted for more details.

The **expected value** of a random variable is a (one) measure of its location or central tendency. For a random variable X, the expected value or the mean of its distribution is given by $E(X) = \sum_i x_i P(X = x_i)$ when X is discrete and $E(X) = \int x f(x) dx$ when X is continuous. Some well-known properties of expected value are as follows. If X and Y are random variables and a is any constant, then (i) $E(a) = a$, (ii) $E(X + a) = E(X) + a$, (iii) $E(aX) = aE(X)$, and (iv) $E(X + Y) = E(X) + E(Y)$. The reader can consult any standard probability and statistics book for these details.

The expectation of a discrete random variable has important applications in SPC as a chart performance measure. This is because the random variable called the run-length, to be defined later, of any control chart is a discrete random variable.

Next, we consider the **conditional expectation** in a bivariate distribution with two random variables, X and Y. The conditional expectation of X given Y is given by $E(X|Y = y) = \sum_x x.P(X = x|Y = y)$ when X and Y are two discrete random variables. When the random variables are continuous, $E(X|Y = y) = \int x f X|Y(x|y) dx$.

An important result relates the unconditional or the marginal expectation, say, of X, that is, $E(X)$, to the conditional expectation of X given Y, that is, $E(X|Y)$. It can be shown that $E(X) = E_Y(E_X(X|Y))$. This result shows that for any two random variables the unconditional expectation of one variable can be obtained by taking the expectation of the conditional expectation of that variable given the other variable. This result plays an important role in studying the performance of control charts when parameters are estimated, as will be seen in later chapters. Note that the conditional expectation, $E_X(X|Y)$, is a function of Y and $E_Y(E_X(X|Y))$ and is obtained simply by taking the expectation of this function over the distribution of Y.

We show proof of this result when X and Y are discrete variables with a joint distribution. The continuous case can be treated in a similar way.

Proof:

$$
\begin{aligned}
E(E(X|Y)) &= \sum_y E(X|Y = y) P(Y = y) \\
&= \sum_y \left(\sum_x x P(X = x|Y = y) \right) P(Y = y) \\
&= \sum_y \sum_x x P(X = x|Y = y) P(Y = y) \\
&= \sum_y \sum_x x P(Y = y|X = x) P(X = x)
\end{aligned}
$$

$$= \left(\sum_x xP(X = x) \right) \left(\sum_y P(Y = y|X = x) \right)$$
$$= \sum_x xP(X = x)$$
$$= E(X).$$

Along with the expectation, the variance of the conditional distribution is also important. The conditional variance of Y given $X = x$ is given by $var(Y|x) = E((Y - E(Y|x))^2|x)$, which can equivalently be written as $var(Y|x) = E(Y^2|x) - (E(Y|x))^2$. A useful result is that the variance of the unconditional distribution of, say, Y, can be obtained from the expectation and the variance of the conditional distribution of Y given X

$$var(Y) = E(Y - E(Y))^2) = E_X\left[var(Y|X)\right] + var_X(E(Y|X)).$$

The verification of these results is left to the reader.

Some common discrete and continuous distributions are given in Tables 1.1 and 1.2, respectively, along with general formulae to calculate their means, medians, and variances. Column (a) of Tables 1.1 and 1.2 displays the name and parameters of each distribution. Column (b) displays the pmf and the pdf for the discrete and continuous probability

Table 1.1 Some common discrete probability distributions.

(a) Distribution	(b) Probability mass function (pmf)	(c) Mean	(d) Variance
Binomial $X \sim BIN(n, p)$ $x \in \{0, 1, 2, \ldots, n\}$ $n \in \aleph$ $0 \leq p \leq 1$	$\binom{n}{x} p^x (1-p)^{n-x}$	np	$np(1-p)$
Poisson $X \sim POI(c)$ $c > 0$ $x \in \{0, 1, 2, \ldots\}$	$\frac{e^{-c}c^x}{x!}$	c	c
Geometric $X \sim GEO(p)$ $0 < p < 1$ $x \in \{1, 2, 3, \ldots\}$	$(1-p)^{k-1}p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$

Table 1.2 Some common continuous probability distributions.

(a) Distribution	(b) Probability density function (pdf)	(c) Mean	(d) Median	(e) Variance
Standard Normal $X \sim N(0,1)$ $x \in (-\infty, \infty)$	$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$	0	0	1
Student's t $X \sim t(v)$ $x \in (-\infty, \infty)$ $v > 0$ denotes the degrees of freedom	$\frac{\Gamma\left(\frac{v+1}{2}\right)}{\sqrt{v\pi}\,\Gamma\left(\frac{v}{2}\right)} \left(1 + \frac{x^2}{v}\right)^{-\left(\frac{v+1}{2}\right)}$	0	0	$\frac{v}{v-2}$ for $v > 2$
Gamma $X \sim GAM(\alpha, \beta)$ $x \in [0, \infty)$ $\alpha > 0$ and $\beta > 0$ denote the shape and scale parameters	$x^{\alpha-1} \frac{e^{-x/\beta}}{\Gamma(\alpha)\beta^{\alpha}}$	$\alpha\beta$	No simple closed form	$\alpha\beta^2$

(*continued overleaf*)

Table 1.2 (*continued*)

(a) Distribution	(b) Probability density function (pdf)	(c) Mean	(d) Median	(e) Variance
The gamma function (denoted $\Gamma(\cdot)$) is defined for all complex numbers (except for negative integers) as $\Gamma(t) = \int_0^\infty x^{t-1}e^{-x}dx$. Note that $\Gamma(t) = (t-1)\Gamma(t-1)$ with $\Gamma(1/2) = \sqrt{\pi}$ and $\Gamma(t) = (t-1)!$ when $t \geq 1$ is a positive integer, with the convention that 0!=1.				
Beta $X \sim Beta(a, b)$ $x \in (0, 1)$ $a > 0, b > 0$	$\frac{x^{a-1}(1-x)^{b-1}}{\mathrm{B}(a,b)}$	$\frac{a}{a+b}$	$I^{[-1]}_{\frac{1}{2}}(a,b)$ $\approx \frac{a-\frac{1}{3}}{a+b-\frac{2}{3}}$ for $a, b > 1$	$\frac{ab}{(a+b)^2(a+b+1)}$
Where the function $\mathrm{B}(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$, $a,\ b > 0$ is called the beta function.				
Logistic $X \sim Logistic(\alpha, \beta)$ $x \in (-\infty, \infty)$ $-\infty < \alpha < \infty$ and $\beta > 0$ denote the location and scale parameters	$\frac{e^{-(x-\alpha)/\beta}}{\beta(1+e^{-(x-\alpha)/\beta})^2}$	α	α	$\frac{\pi^2}{3}\beta^2$

Log-Logistic $X \sim$*Log-Logistic*(α, β) $x \in [0, \infty)$ $\alpha > 0$ and $\beta > 0$ denote the scale and shape parameters	$\frac{(\beta/\alpha)\,(x/\alpha)^{\beta-1}}{\left(1+(x/\alpha)^{\beta}\right)^2}$	$\frac{\alpha\tau}{sin(\tau)}$ $\tau = \frac{\pi}{\beta}$ $\beta > 1$ else undefined	α	$\alpha^2\left(\frac{2\tau}{sin(2\tau)} - \frac{(\tau)^2}{(sin(\tau))^2}\right)$ with $\tau = \frac{\pi}{\beta}$ for $\beta > 2$
Laplace or Double Exponential $X \sim DE(\alpha, \beta)$ $x \in (-\infty, \infty)$ $-\infty < \alpha < \infty$ and $\beta > 0$ denote the location and scale parameters	$\frac{1}{2\beta}e^{-\frac{\lvert x-\alpha\rvert}{\beta}}$	α	α	$2\beta^2$
Uniform $X \sim U(a, b)$ $x \in [a, b]$ $-\infty < a < b < \infty$	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$

distributions, respectively. For the discrete distributions, Columns (c) and (d) display the mean and the variance of each distribution, respectively. For the continuous distributions, in Table 1.2, Columns (c), (d) and (e) display the mean, median, and variance, respectively. Note that, for discrete distributions, the percentiles, which includes the median, need to be defined so that they are unique. This is why they are omitted in Table 1.1, but we consider them in later chapters.

Some notes on Table 1.2 follow. The gamma distribution is positively skewed and the skewness of the gamma distribution increases as the shape parameter α decreases. Also note that the $GAM(1,1)$ distribution is the exponential distribution with mean 1, $EXP(1)$. The skewness of a distribution is typically defined in terms of the moments and is given later.

The contaminated normal (CN) distribution is a mixture of two normal distributions, which has interesting applications in SPC; however, it is not included in Table 1.2 since the formulae are too complex to fit into the table. The definition and some properties of the CN distribution are given below. The pdf of the CN distribution is given by

$$f(x) = (1-\varepsilon)\phi(\mu_1, \sigma_1) + \varepsilon\phi(\mu_2, \sigma_2)$$

where $\phi(\mu, \sigma)$ is the pdf of a normal distribution with mean μ and variance σ^2 and $0 < \varepsilon < 1$ denotes the level of contamination. It can be shown that the expected value and the variance of the CN distribution are given by

$$(1-\varepsilon)\mu_1 + \varepsilon\mu_2$$

and

$$(1-\varepsilon)(\mu_1^2 + \sigma_1^2) + \varepsilon(\mu_2^2 + \sigma_2^2) - ((1-\varepsilon)\mu_1 + \varepsilon\mu_2)^2$$

respectively.

The verification of these results is left to the reader. For more information on the contaminated normal distribution, the reader can consult a standard mathematical statistics book. ■

1.3 RANDOM SAMPLE

Much of statistical inference is based on the key idea of a random sample. A **random sample** is a sample of observations taken from a population so that each observation from the population has an equal chance of being selected in the sample. Stated differently, the selection of each element should be governed purely by chance with no predictability. Put in a more mathematical way, a random sample is a set or a collection of random variables that are independent and identically distributed, denoted as **i.i.d**.

Another important concept is the **moment** of a random variable, which may be used as a descriptor of the location, the variance and the shape of its probability distribution. The moments can be used to identify a distribution; the reader can consult a standard mathematical statistics book for more details. The rth raw moment is given by $E(X^r)$ for $r = 1, 2, \ldots$, so that the first raw moment, for $r = 1$, is $E(X)$, the **mean** of the distribution, that is, the expected value, which is a measure of location or the central tendency of a distribution. The mean is typically denoted as μ. Depending on the range of the distribution, the mean μ can take on any value over the real line, which is illustrated in Figure 1.3 for a continuous symmetric distribution. Indeed, it can be shown that for any symmetric distribution, continuous or discrete, the point of symmetry equals the mean of the distribution.

The central moments are defined in terms of the deviations from the mean of the distribution. The rth central moment is given by $E(X - \mu)^r$. While the first central moment is zero, the second central moment $E(X - E(X))^2$ is the **variance**, denoted as σ^2, which is a measure of the spread of a distribution around its mean. Typically, one reports the mean (which is a measure of location) and the standard deviation of a distribution, where the standard deviation, denoted as σ, is the square root of the variance, that is, $\sigma = \sqrt{\sigma^2}$. Figure 1.4 illustrates the effect on the distribution as the spread changes. Note that a larger variability leads to a flatter curve. One can show that $\sigma^2 = E(X - E(X))^2 = E(X^2) - (E(X))^2$, the latter expression may be preferred in some situations. We leave the verification of this to the reader.

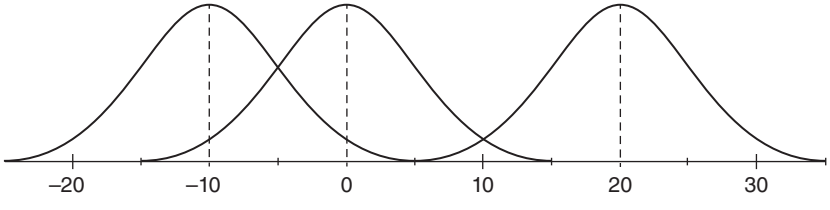


FIGURE 1.3 *Illustration of some different numerical values that the mean can take on.*

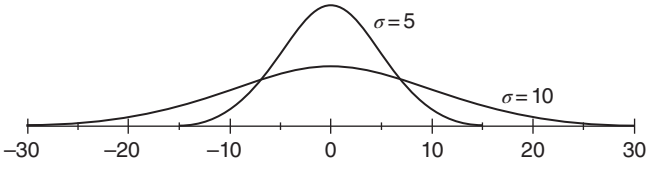


FIGURE 1.4 *Illustration of the effect on the distribution as the spread changes.*

Other moments for higher values of r are used in the description of the shape or the tail behaviour of a distribution. For example, the normalized third central moment $E(X-\mu)^3/\sigma^3$ is taken as a measure of the **skewness** of the distribution. A positive (negative) value of the skewness measure indicates a positively (negatively) skewed distribution. However, when the distribution is symmetric, the third central moment in the numerator of the skewness measure is zero and hence the skewness measure is zero, as it should be. Skewness can be best illustrated with the use of figures. Panel (a) of Figure 1.5 shows a continuous pdf distribution that is skewed to the left, or negatively skewed. In this case, the left tail is longer, more of the mass of the distribution is concentrated on the right, and it has relatively few low values. Alternatively, Panel (b) of Figure 1.5 shows a continuous pdf that is skewed to the right, or positively skewed. In this case, the right tail is longer, the mass of the distribution is concentrated on the left, and it has relatively few high values. Panel (c) of Figure 1.5 shows a symmetric distribution where the left and right tails are equal in length.

Next, we consider **linear combinations of independent random variables**, their distributions, and moments. These results are important in finding the mean and the standard deviation of the sample mean, which is a very popular and useful statistic in SPC. Suppose $X_1, X_2, \ldots, X_n$ are n independent random variables with corresponding means $\mu_1, \mu_2, \ldots, \mu_n$ and variances $\sigma_1^2, \sigma_2^2, \ldots, \sigma_n^2$., respectively. Consider the linear combination $Y = \sum_{i=1}^{n} a_i X_i$ where $a_1, a_2, \ldots, a_n$ are real constants. Then the mean and the variance of the linear combination Y are given by

$$\mu_Y = \sum_{i=1}^{n} a_i \mu_i$$

and

$$\sigma_Y^2 = \sum_{i=1}^{n} a_i^2 \sigma_i^2$$

respectively. While the first and the second moments of Y are easily found this way, in general, finding the distribution of a linear combination is more involved. However, it is well known that when the $X's$ are independently normally distributed, Y is also normally distributed with $E(Y) = \mu_Y$ and a variance σ_Y^2. This is a very useful result when making statistical inference using a linear combination of random variables, which is generally not true for many other distributions in practice. Note that the independence condition can be relaxed to include correlations among the X's and in that case the variance formula for the linear function Y of correlated random variables can be obtained by suitably modifying the above formula. This is an interesting result and the reader can consult any standard mathematical statistical book for more details.

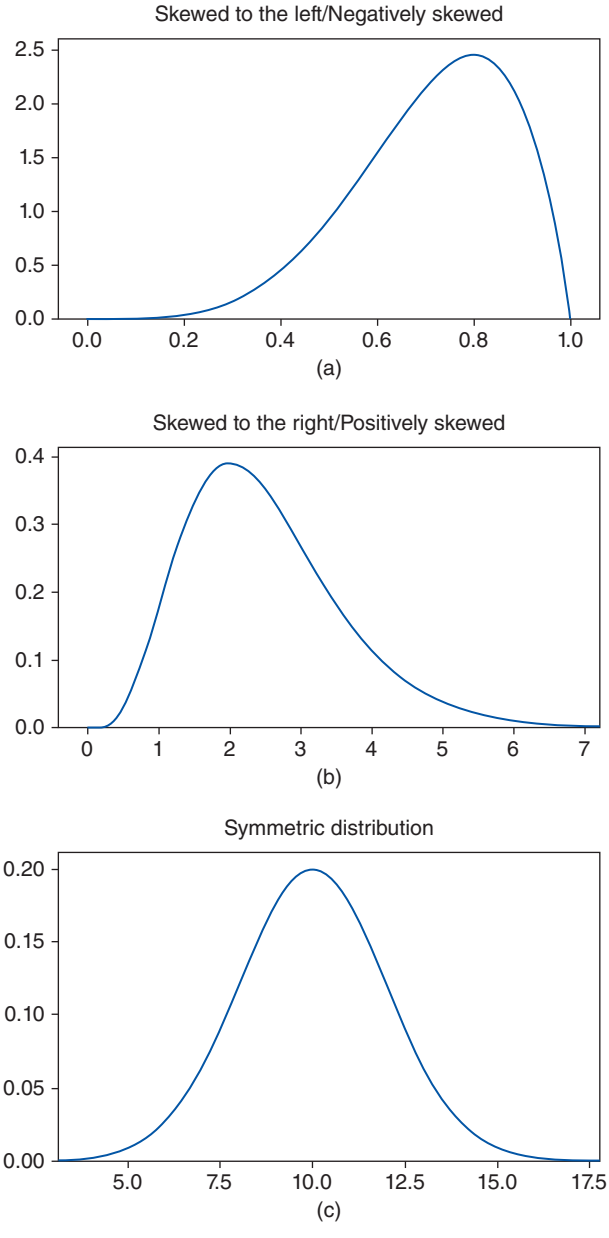


FIGURE 1.5 *Asymmetric pdfs (Panels (a) and (b)) and a symmetric pdf (Panel (c)).*

One important direct application of this result gives the distribution of the mean $\overline{X}$ of a random sample from a normal distribution with mean μ and standard deviation σ, which is a linear function of iid random variables $X_1, X_2, \ldots, X_n$, with all weights $a_i = \dfrac{1}{n}$, $\mu_i = \mu$ and $\sigma_i = \sigma$. Thus, for a

random sample of size n from a normal distribution with mean μ and standard deviation σ, the sample mean $\overline{X}$ follows a normal distribution with mean μ and variance σ^2/n (standard deviation $\sigma/\sqrt{n}$).

The **moment generating function** is a useful method for finding the moments of random variables, including linear functions of random variables. Moment generating functions allow us to obtain statistical results by directly working with the probability density functions or the cumulative distribution functions.

Next, we consider one of the most important results in statistical inference, called the **central limit theorem**, which states that the average of n independent random variables approximately follows a normal distribution, for large values of n, regardless of the distribution of the individual variables, for example, if $X_1, X_2, \ldots, X_n$ are n independent random variables each with mean μ and variance σ^2, we write $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ for sufficiently large values of n (typically for $n \geq 30$). Note that this result allows us to make statistical inference based on the average of a random sample of a sufficiently large sample size n using the normal distribution tables, even if that sample was obtained from a population that did not follow the normal distribution. The topic of statistical inference is discussed next.

1.4 STATISTICAL INFERENCE

Statistical inference means drawing conclusions about a population based on data from a random sample that is a subset of the population. Typically, population parameters are unknown and, consequently, sample statistics (also referred to as point estimators) are used to obtain estimates of these unknown population parameters. Table 1.3 lists some frequently encountered population parameters in Column (a) along with the corresponding point estimators in Column (b). The absolute difference (or the ratio) between the population parameter and the point estimator is referred to as the sampling error, and this is given in Column (c) of Table 1.3. Typically, for a location parameter, the sampling error involves the difference, whereas for a scale parameter, the sampling error is expressed as a ratio.

It may be noted that one important parameter of interest, not shown in Table 1.3, is the population median, which is also the 50th percentile of the distribution. Thus, 50% of the cumulative probability must be at or below the median. While the median can be defined uniquely for most continuous distributions, that is not the case for the discrete distributions, which has jumps in its cdf (see Figure 1.1), and one needs to define the median (a percentile) so that it is unique. We mentioned this point before Table 1.1, and we discuss this later in Chapter 2. The population median

Table 1.3 Population parameters, the corresponding point estimators, and sampling errors.

	(a) Population parameter	(b) Sample statistic (Point estimator)	(c) Sampling error
Mean	μ	$\overline{X} = \frac{\sum_{i=1}^{n} X_i}{n}$	$\lvert\overline{X} - \mu\rvert = \lvert\mu - \overline{X}\rvert$
Variance	σ^2	$S^2 = \frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{n-1}$	S^2/σ^2
Standard deviation	$\sigma = \sqrt{\sigma^2}$	$S = \sqrt{S^2}$	S/σ
Proportion	p	$\overline{p}$	$\lvert p - \overline{p}\rvert = \lvert\overline{p} - p\rvert$

is estimated by the sample median and the sample median is defined as the observation in the middle (of the sample ordered from the lowest to the highest) when the sample size is odd and as the average of the two middle observations when the sample size is even. The sample median is generally the preferred estimator of the center (location) of a skewed distribution form a robustness point of view. The median plays an important role in SPC and will be discussed further later in the book.

While the point estimators are useful starting points for unknown parameters, a more informative approach is to use interval estimation and the corresponding interval estimators. Let us consider the population mean and the population proportion for illustration purposes. The sample mean $\overline{X}$ is typically used as a point estimator of μ, which provides no indication of the variability inherent in $\overline{X}$. Thus, one might prefer an interval around $\overline{X}$ to estimate μ. Such an interval is called an **interval estimator** and the process of estimating μ with it is called **interval estimation**. The interval estimator is usually found through **confidence intervals**, which is a topic of great interest in statistical inference. The confidence interval is constructed from two quantities, obtained from adding and subtracting a **margin of error** to and from the point estimator. The reader is referred to a book of statistics for more details on confidence intervals. Note that, whereas estimators are random variables, their numerical values calculated for a given set of data are called estimates. Similarly, since the sample proportion $\overline{p}$ may be used as the point estimator for p, one can use $\overline{p} \pm$ some margin of error for the purposes of interval estimation of p. Subtracting the margin of error will provide the lower limit of the interval, and adding the margin of error will provide an upper limit for the interval. For the population mean μ, there are two different scenarios, one for when the population standard deviation σ is

known and one for when the population standard deviation σ is unknown. For the σ known case we use $\overline{x} \pm z_{\alpha/2}{}^{\sigma}/\sqrt{n}$ where $z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of the standard normal distribution. For the σ unknown case we use $\overline{x} \pm t_{\alpha/2,\,n-1}\,{}^{s}/\sqrt{n}$ where $t_{\alpha/2,\,n-1}$ is the upper $\alpha/2$ percentage point of the t-distribution with degrees of freedom $df = n - 1$. For the population proportion, there is only one interval, which is given by $\overline{p} \pm z_{\alpha/2}\sqrt{\dfrac{\overline{p}(1-\overline{p})}{n}}$. The interval estimator provides a range (interval) of values for the unknown parameter based on the data. From the interval, one can get a single value (a point estimate, such as the midpoint of the interval) as well as an idea of the uncertainty and the variation in the estimate from the margin of error. The charting statistic for a control chart is often based on the point estimator of the parameter being monitored. The control limits may be viewed as being constructed from the confidence intervals with some suitable adjustments.

We now discuss some important statistical properties of point estimators. The first property of an estimator is the **bias** of an estimator, which is the difference between the estimator's expected value and the true value of the parameter being estimated. Mathematically, $\text{Bias}_\theta(\hat{\theta}) = E_\theta(\hat{\theta} - \theta)$, where $\hat{\theta}$ denotes an estimator, θ denotes the parameter being estimated and E_θ denotes the expected value over the distribution of $\hat{\theta}$. An estimator is said to be **unbiased** if its bias is equal to zero for all values of θ. Let's consider the mean, proportion, and standard deviation for illustration purposes. Since it can be shown that $E(\overline{X}) = \mu$, the sample mean $\overline{X}$ is an unbiased estimator of the population mean μ. When considering the proportion, again, since one can show that $E(\overline{p}) = p$, the sample proportion $\overline{p}$ is an unbiased estimator of p. Alternatively, the sample standard deviation S is a biased estimator of σ, since it can be verified that $E(S) \neq \sigma$, in general. However, assuming that the data come from a normal distribution, many unbiased estimators of σ can be found, such as $\hat{\sigma} = \dfrac{S}{c_4}$ or $\hat{\sigma} = \dfrac{R}{d_2}$ where c_4 and d_2 are some constants, S is the sample standard deviation given by $\sqrt{\dfrac{\sum (X_i - \overline{X})^2}{n-1}}$ and R is the sample range, namely, the difference between the largest and the smallest sample values. Thus, $R = X_{max} - X_{min}$, where X_{max} and X_{min} denote the largest and the smallest sample values, respectively.

The constants c_4 and d_2 can be found for a specified parametric distribution, and are called unbiasing constants so that $E(\hat{\sigma}) = \sigma$. For the normal distribution, expressions for the unbiasing constants have been obtained and their values have been tabulated in various books, such as in Montgomery (2009). We provide these tables in the Appendix.

Next, we consider the property of **sufficiency**. Let $X_1, X_2, \ldots, X_n$ be a random sample from a probability distribution (a pdf or a pmf) $f(x;\theta)$ with an unknown parameter θ. The statistic $T = T(X_1, X_2, \ldots, X_n)$ is said

to be **sufficient** for θ if the conditional distribution of $X_1, X_2, \ldots, X_n$, given the statistic T, does not depend on θ. Mathematically, given that $X_1, X_2, \ldots, X_n \sim f(x_1, x_2, \ldots, x_n; \theta)$, the statistic T is sufficient for θ if $f(x_1, x_2, \ldots, x_n|t; \theta) = f(x_1, x_2, \ldots, x_n|t)$. From an intuitive point of view, we prefer an estimator based on a sufficient statistic, which uses all the information about a parameter available in the random sample and should be the starting point when looking for an estimator. Sufficiency is best explained using an example.

Example 1.1 Suppose $X_1, X_2, \ldots, X_n \sim POI(\theta)$ and let $T = T(X_1, X_2, \ldots, X_n) = \sum_{i=1}^{n} X_i$. Then

$$\begin{aligned} & f_{X_1, X_2, \ldots, X_n|T}(x_1, x_2, \ldots, x_n|t) \\ & = P((X_1, X_2, \ldots, X_n) = (x_1, x_2, \ldots, x_n)|T = t) \\ & = \frac{P((X_1, X_2, \ldots, X_n) = (x_1, x_2, \ldots, x_n) \cap T = t)}{P(T = t)} \end{aligned}$$

from Equation 1.1. However,

$$P((X_1, X_2, \ldots, X_n) = (x_1, x_2, \ldots, x_n) \cap T = t)$$

$$= \begin{cases} 0 & T(x_1, x_2, \ldots, x_n) \neq t \\ P(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n) & T(x_1, x_2, \ldots, x_n) = t. \end{cases}$$

Therefore, $P((X_1, X_2, \ldots, X_n) = (x_1, x_2, \ldots, x_n)) = \prod_{i=1}^{n} \frac{e^{-\theta}\theta^{x_i}}{x_i!} = \frac{e^{-n\theta}\theta^{\sum x_i}}{\prod(x_i!)} = \frac{e^{-n\theta}\theta^t}{\prod(x_i!)}$.

Note that $T(X_1, X_2, \ldots, X_n) = \sum_{i=1}^{n} X_i$ so that $P(T = t) = \frac{e^{-n\theta}(n\theta)^t}{t!}$, since $T \sim POI(n\theta)$.

Thus $\frac{P((X_1, X_2, \ldots, X_n) = (x_1, x_2, \ldots, x_n)}{P(T = t)} = \frac{t!}{(\prod(x_i!))\, n^t}$, which does not depend on the parameter θ, indicating that $T = T(X_1, X_2, \ldots, X_n) = \sum_{i=1}^{n} X_i$ is a sufficient statistic for the parameter θ.

Another important property of an estimator is **consistency**. An estimator $\hat{\theta}$ of a parameter θ is said to be **consistent** if it converges in probability to the parameter. We do not discuss the various concepts of the convergence of a sequence of random variables; therefore, the reader is referred to work such as that of Hogg, McKean, and Craig (2005) for more details. Intuitively, a consistent estimator is the one that gets "closer" to the parameter as the sample size gets larger, and is preferable since one expects better, more precise estimates with more data.

The **mean squared error** (MSE) is another important concept relating to an estimator, which may or may not be unbiased. The MSE is defined as $E(\hat{\theta}-\theta)^2$, where $\hat{\theta}$ denotes the estimator and θ denotes the parameter being estimated. Intuitively, we want the error $\hat{\theta}-\theta$ to be small, which means that we want the squared error $(\hat{\theta}-\theta)^2$ to be small, which implies that a smaller MSE is more desirable. It can be verified that $MSE(\hat{\theta}) = E(\hat{\theta}-\theta)^2 = var(\hat{\theta}) + Bias^2(\hat{\theta})$, so that when an estimator is unbiased, its MSE and variance are the same.

To summarize, in order to find a good charting statistic to construct the control limits, we often look for the best estimator of the parameter being monitored in the sense of being sufficient, unbiased, and having the smallest variance. Accordingly, we want the estimator to be a **minimum variance unbiased estimator** (MVUE), which is often based on sufficient statistics.

Next, we briefly discuss what are called **order statistics** and their statistical properties. Order statistics play an important role in descriptive and inferential statistics; several books are available in the literature on this. For example, popular descriptive statistics such as the median, the first and the third quartiles (also known as the 50th, the 25th and the 75th percentile, respectively) in a sample of data are order statistics that help us better understand the data distribution in terms of its shape, location and spread. The popular statistical graphic, the boxplot, is a visual display of five order statistics, the minimum, the second quartile, the median, the third quartile and the maximum. As introduced earlier, the range, as an estimator of standard deviation, is a function of two order statistics, the largest and the smallest. Other than using the range and some of its variants, the use of order statistics, in general, is fairly recent in SPC and is more common in the area of nonparametric SPC, as will be seen later. In general, the ordered values of a **random sample** $X_1, X_2, \ldots, X_n$ usually arranged from the lowest to the highest, denoted by $X_{(1:n)} < X_{(2:n)} < \ldots < X_{(n:n)}$, are known as the order statistics. Thus, as noted earlier while defining the range of a random sample, $X_{(1:n)}$ denotes the smallest or the minimum and $X_{(n:n)}$ denotes the biggest or the maximum of the sample observations. Other ordered values such as the median occur in the middle. It is typically assumed that the population is continuous, so theoretically there are no ties among the ordered values. Our goal is to study the distribution of order statistics. Suppose that $X_1, X_2, \ldots, X_n$ denotes a random sample of size n from a continuous pdf $f(x)$. The pdf of the kth order statistic $X_{(k:n)}$ is given by

$$g_k(x) = \frac{n!}{(k-1)!(n-k)!}(F(x))^{k-1}(1-F(x))^{n-k}f(x). \tag{1.2}$$

The range of the pdf, in general, is $-\infty < x < \infty$.

As noted earlier, important special cases of the order statistics are the minimum and the maximum values of a sample, as well as the sample

median. For the sample minimum, we set $k = 1$ into Equation 1.2 to obtain

$$g_1(x) = n(1 - F(x))^{n-1} f(x),\ -\infty < x < \infty. \tag{1.3}$$

For the sample maximum, we set $k = n$ into Equation 1.2 to obtain

$$g_n(x) = n(F(x))^{n-1} f(x),\ -\infty < x < \infty. \tag{1.4}$$

It is emphasized that these formulas work for any continuous random variable with a pdf $f(x)$. When considering the sample median, it is easier to consider an odd sample size so that the order statistic is unique. Accordingly, for simplicity, we let n be odd and set $k = \dfrac{n+1}{2}$ into Equation 1.2 to obtain

$$g_k(x) = \frac{n!}{\left(\frac{n-1}{2}\right)!\left(\frac{n-1}{2}\right)!}(F(x))^{(n-1)/2}\,(1 - F(x))^{(n-1)/2},\ -\infty < x < \infty. \tag{1.5}$$

When the sample size is even, the typical convention is to define the median as the average of the middle two order statistics. Its pdf, for example, can be found from the joint pdf of these two order statistics.

In general, the joint pdf of two order statistics $X_{(k:n)}$ and $X_{(l:n)}$ with $k < l$ is given by

$$g_{kl}(x, y) = \frac{n!}{(k-1)!(l-k-1)!(n-l)!}(F(x))^{k-1}(F(y) - F(x))^{l-k-1} \times (1 - F(y))^{n-l} f(x) f(y),$$

where $-\infty < x < y < \infty$. For derivations of these formulas and for more explanations and examples, see Gibbons and Chakraborti (2010).

Another useful result in the area of order statistics is the **probability integral transformation** (PIT). The PIT states that if $U = F_X(X)$, where F_X denotes the cdf of a continuous random variable X, then

1. U has a uniform (0,1) distribution, and

2. $U_{(r)} = F_X(X_{(r)})$ is the rth order statistic of a random sample from the uniform (0,1) distribution function and has the $Beta(r,\ n - r + 1)$ distribution with the pdf

$$f_{U_{(r)}}(u) = \frac{1}{B(k, n-k+1)} u^{r-1}(1-u)^{n-r}, 0 \le u \le 1.$$

Order statistics for discrete distributions can be defined and their distributions can be obtained. Interested readers can consult a book on order statistics.

1.5 ROLE OF THE COMPUTER

The computer plays a role in much of everything we do today. In statistics, with the availability of modern computers and software, a lot of the mathematical results/derivations that had to be worked out previously by hand using mathematical statistical techniques and algebraic manipulations, which were often complicated, can now be attempted/obtained empirically and thus implemented with relative ease. This is not to suggest that the mathematical derivations are not important; it is that the computer-based approaches can provide valuable and interesting insights into the problem, complementing theoretical derivations and making solutions perhaps more achievable, particularly when it comes to working in higher than two dimensions. This "modern" approach has revolutionized all areas of statistics, and SPC is no exception. Much of this "new" work is based on what is called simulation or Monte Carlo simulation, which can often provide significant and interesting insights that may or may not always be (easily) available under classical mathematical derivations. Many books have been written on the subject and several software packages are now available to implement the simulation algorithms. Basically, the simulations can provide insights and lead to solutions to a variety of problems by allowing one to perform statistical sampling "experiments" on the computer and repeat them many times within a very short period of time. Results from such experiments are then analyzed to find answers to the questions being asked. In Chapter 2 Monte Carlo simulations are discussed in detail as it is applied in SPC together with some other, more theoretical approaches used by researchers.

CHAPTER 2

Basics of Statistical Process Control

Chapter Overview

In this chapter, we focus on the basic concepts, terminology, and some of the underlying statistical theory of univariate control charts. This includes topics such as control limits, placement of the control limits, sample size, sampling frequency and how to form rational subgroups. We start with the classical Shewhart control charts. Some of the more advanced, yet practical, topics such as the performance and the statistical design of a control chart are also introduced.

2.1 BASIC CONCEPTS

To describe and understand the underlying theory, we begin with a discussion of some important terminology.

2.1.1 Types of Variability

Within any process, a certain amount of natural variability, no matter how well the process was designed, developed, or maintained, will always be present. For example, not all car doors painted by the same machine under the same settings will be identical. Not all five-pound bags of potatoes packed by the same plant will contain exactly five pounds of potatoes, no matter how carefully the bags are packed. Such variability is inherent to

Nonparametric Statistical Process Control, First Edition.
Subhabrata Chakraborti and Marien Alet Graham.
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Companion website: www.wiley.com/go/chakraborti/Nonparametric_Statistical_Process_Control

a process and cannot be (or need not be) totally eliminated, possibly not without incurring an enormous cost for a minimal return. Conversely, there can also be other kinds of variability present within a process that is disruptive, which can result in a deterioration of quality. The goal of statistical process control (SPC) is to identify the source(s) of this variability, examine the root cause(s), and eliminate or minimize them so that the quality of the product is improved and maintained at a satisfactory level.

Dr. Walter A. Shewhart acknowledged the difference between these two types of variability and was the first to introduce the terminology of a common cause and a special cause (see Shewhart, 1926, 1931).

The term **common causes** (or **chance causes**) refers to those sources of variability that are inherent or naturally present in a process. The variation, due to such causes, is uncontrollable or ever present and is also called the "background noise" within a process, which might be due to the cumulative effect of many small and undetectable, but unavoidable, causes.

Special causes (or **assignable causes**) are those sources of variability that are not part of the common causes or the natural variability of a process and can directly affect the quality of a process. In a manufacturing process, for example, special causes can typically arise from defective raw materials, improperly adjusted or controlled machines, or operator errors.

The **total variation** present in a process is the sum of the variation due to these two sources, that is, the variation due to common causes and the variation due to special causes.

A process is considered in a state of statistical control or **in-control** (IC) if it is operating according to what is targeted or expected in the presence of common causes. Thus, when a process is IC, only random deviations or fluctuations occur around some target level. We write IC to denote the IC state (status) of a process. When a process is not IC, it is said to be **out-of-control** (OOC). The OOC state of a process is denoted by OOC. This is when there are reasons to believe that the presence of some special causes influences the process and thus needs to be identified and, if possible, eliminated so that the process is restored to the IC state.

To emphasize, the goal of SPC is to minimize and/or eliminate the assignable causes of variability so that the process operates only under common causes and while meeting or exceeding any assigned specifications. Three things are generally necessary to achieve this goal: (i) **identification**: detecting the presence of any special causes of variation from a "signal" from the chart; (ii) **trace**: finding the reason and being able to trace the source of a signal; and (iii) **action**: the ability to take any corrective action that is necessary to bring the process back IC. A control chart, which is based on sound statistical principles, is a graphical tool with a visually appealing interface and it provides an objective way of understanding the process

variability, and thus maintain and improve quality. Thus, a control chart plays a major role in SPC.

2.1.2 The Control Chart

A control chart is a graphical display consisting of the plot of successive values of a suitable summary statistic. This statistic is called a charting statistic or a plotting statistic, which is an estimator of the quality parameter of interest, and is calculated from each of some successive samples of measurements (data) randomly taken on the quality characteristic on the vertical axis versus the sample number (or the time) on the horizontal axis, along with three lines drawn on the display that are called control limits. Typically, the control limits consist of a **center line** (CL) and two horizontal lines, one on either side of the CL. The line above the CL is called the **upper control limit** (UCL), whereas the line below the CL is called the **lower control limit** (LCL). These three lines aid the user to decide when a process remains IC or goes OOC.

The values of the **charting statistic** are plotted on the control chart to monitor the process. In order to find the appropriate charting statistic, one has to understand the process control problem, define the process parameter of interest to be monitored, and choose a suitable statistic, which is often (based on) an estimator of that parameter that has desirable statistical properties such as unbiasedness, minimum variance, etc., as discussed in Chapter 1. Typically, when a charting statistic plots or falls on or outside either of the control limits, it signals the possibility of the presence of some assignable causes, and at that point the process may be declared OOC. This event is called a **signaling event** or an alarm. A **false alarm** is when the process is declared OOC when in fact it is IC. In SPC applications, the probability of a false alarm is referred to as the **false alarm rate** and is abbreviated as *FAR*. Once a signal is observed, an investigation is generally started to find the source of the signal, and corrective action is taken if necessary. To the contrary, when the charting statistic plots between the upper and the lower limits of the process and shows no systematic pattern over time, the process is said to be IC. In other words, no signal is observed on the control chart. The corresponding event is called a **non-signaling event**. A typical control chart is shown in Figure 2.1.

The CL is drawn at either the expected value of the charting statistic when the process is IC, or at its estimator for the parameter of interest. The UCL and LCL are placed at some distance above and below the CL, such that if the process is truly IC, the probability of a point falling on or outside the control limits is very small. The degree of smallness depends on the context at hand and the seriousness of the consequences. This means

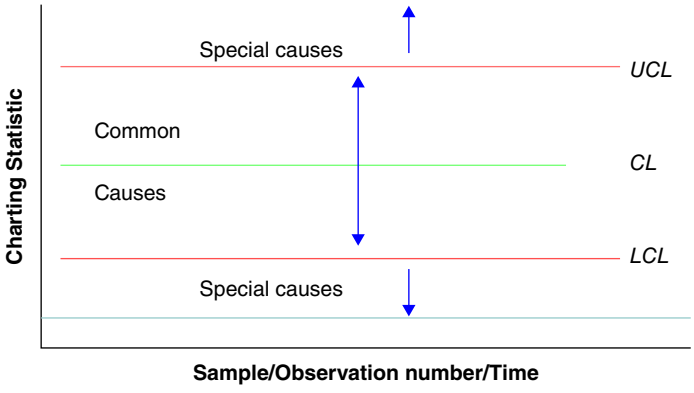


FIGURE 2.1 *A typical control chart.*

that, by design, nearly all but a very small number of the plotted points, that is, the values of the charting statistic from the various samples taken from the process, are expected to fall between the control limits when the process is truly IC. When all of the points plot between the control limits, it typically indicates the absence of any special causes of variation and no corrective action is generally necessary. This is when the process is declared to be in statistical control or simply IC. However, note that, even when all the points plot between the control limits, there might still be problems, for example, when all the points plot between the control limits and yet the plot shows some non-random pattern, or when the points plot either below or above a target value (what may be expected), there may be concern that special (assignable) causes might be present and the process might be OOC.

Figure 2.2, for example, displays a Shewhart control chart in which all the points fall between the UCL and LCL, and the pattern of the plot appears to be random. Thus, no signaling events are observed here and there is no evidence of any anomalous patterns among the points. Hence the process can be assumed to be IC with respect to the parameter being monitored and monitoring of the process continues uninterrupted. However, in Figure 2.3, even though all of the points plot between the UCL and LCL, there is a clear cyclical pattern. Consequently, there is real concern that some assignable causes might be present (i.e., the process might be OOC), thus a search for such assignable causes may be necessary. Cyclical patterns may result from environmental changes such as changes in temperature. It can also be a result of operator fatigue, or fluctuation in voltage or pressure, or some other variable in the machinery. The chemical properties of raw material can also play a role. Figure 2.4 shows another example where all of the points plot between the UCL and LCL; however, the points seem to cluster around the CL. This is referred to as stratification and there is a concern

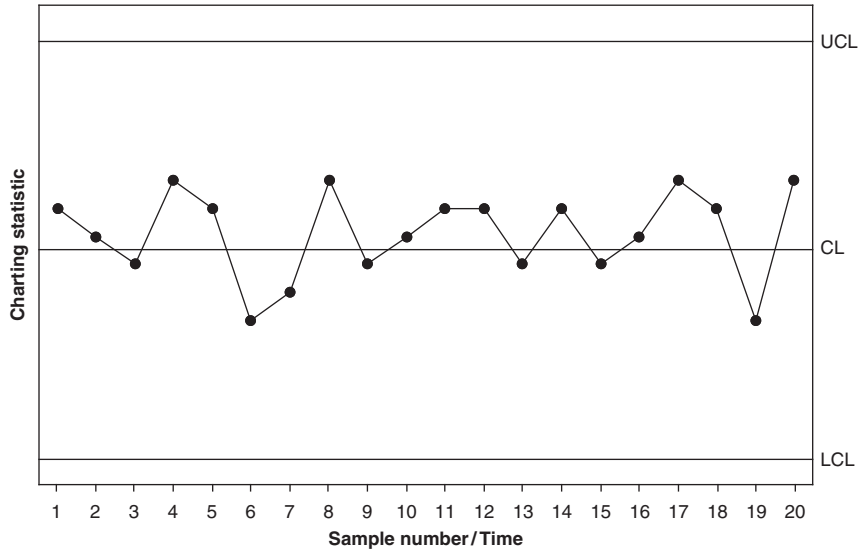


FIGURE 2.2 *A Shewhart control chart with a random pattern.*

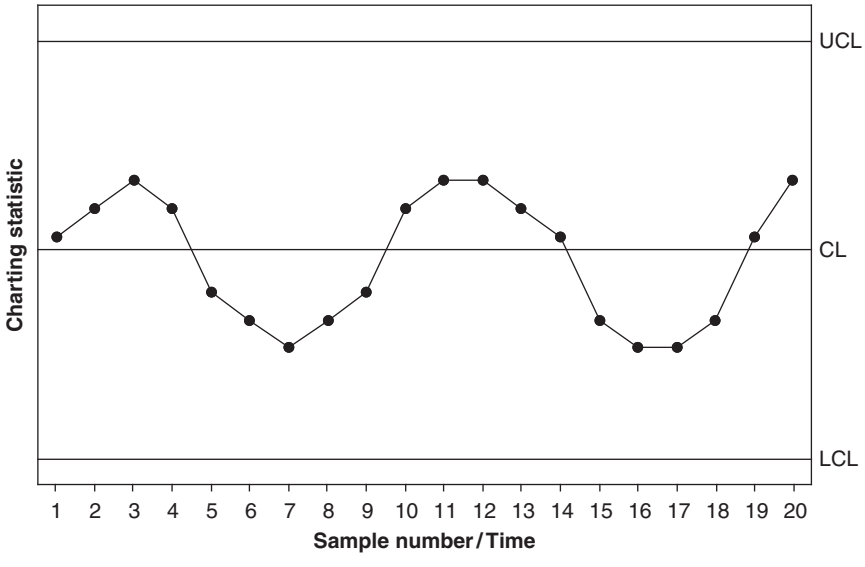


FIGURE 2.3 *A Shewhart control chart with a cyclical pattern.*

that the process might be OOC because there is a marked lack of variability. Stratification may result from the incorrect calculation of control limits or from incorrect subgrouping; rational subgrouping is discussed later in the book. It may also be a result of not recalculating the control limits after process improvement. Figures 2.3 and 2.4 are two examples where all of the points plot between the two control limits; however, it is clear that there

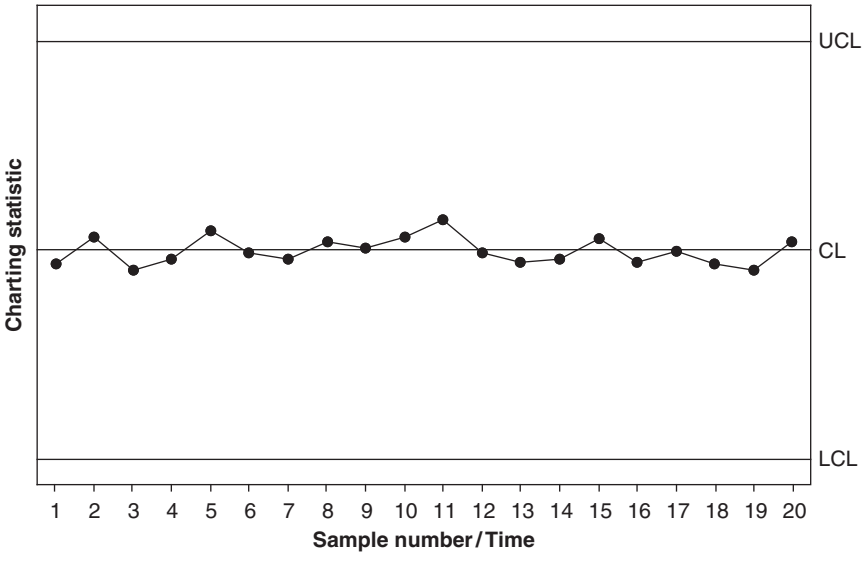


FIGURE 2.4 *A Shewhart control chart with a stratified pattern.*

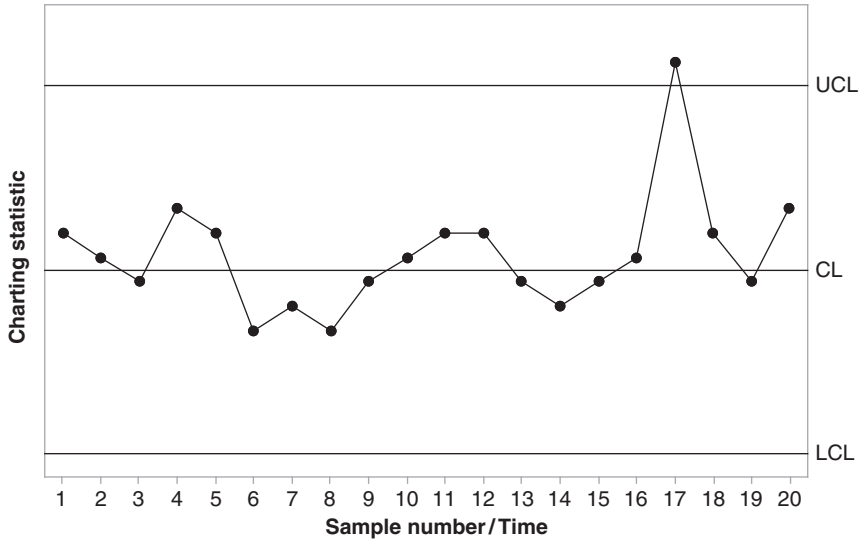


FIGURE 2.5 *A Shewhart control chart with one point plotting outside the upper control limit.*

is a problem. We will discuss the issue of patterns in more detail in a later section. In Figures 2.3 and 2.4, two control charts were illustrated where all of the points plot between the control limits; however, in Figure 2.5 a control chart is illustrated where one point plots outside the *UCL* or *LCL* (so there is a signaling event), indicating that the process may be OOC.

Note that it is customary to join the points on a control chart with straight-line segments for easier visualization.

Control charts play a crucial role in various fields and industries worldwide, particularly since the Second World War. Montgomery (2009) lists five key reasons for their popularity:

- Control charts are a proven technique for improving productivity.
- Control charts are effective in defect prevention.
- Control charts prevent unnecessary process adjustment.
- Control charts provide diagnostic information.
- Control charts provide information about process capability.

He goes on to say, "Control charts are among the most important management control tools; they are as important as cost controls and material controls. Modern computer technology has made it easy to implement control charts in any type of process, as data collection and analysis can be performed on a microcomputer or a local area network terminal in real-time, on-line at the work center."

We begin with how to construct a univariate Shewhart control chart. Let W be a sample statistic that measures some quality characteristic of interest and suppose that the IC mean and variance of W are μ_w and σ_w^2, respectively. Then the UCL, CL, and LCL of a Shewhart chart for μ_w when μ_w and σ_w are specified are

$$\begin{aligned} UCL &= \mu_w + k\sigma_w \\ CL &= \mu_w \\ LCL &= \mu_w - k\sigma_w \end{aligned} \tag{2.1}$$

where k is the "distance" of the upper (lower) control limit from the CL, expressed in terms of the standard deviation of W. The constant k is called the charting constant. These control limits are called k-sigma limits and the control charts developed according to these principles are called k-sigma Shewhart control charts. A popular choice for the constant k is 3, and this yields the 3-sigma Shewhart control chart.

2.1.3 Construction of Control Charts

The charting statistic has been mentioned previously, and statistical considerations such as unbiasedness, minimum variance, robustness, and efficiency are generally employed in the choice of this statistic. In order to monitor the

central tendency or the location of a process, for example, the sample mean ($\overline{X}$) or the sample median ($\tilde{X}$) is often used. Alternatively, the sample range (R), the sample standard deviation (S), and the sample variance (S^2) are used to monitor the process spread, standard deviation, or variance, respectively. Once a charting statistic is chosen, we use its probability distribution, along with some desired chart performance criteria, to be discussed soon, to set up the control limits. Many of the classical control charts have been developed under the assumption that the process distribution is normal. Sometimes, the charting statistic is an average or a statistic for which the central limit theorem can be used to invoke approximate normality without an explicit normality assumption about the process distribution. When normality or some other distributional assumption (such as exponentiality) is made about the process, the control chart is called **parametric**. When such a distributional assumption cannot be made or justified, alternative control charts are available, which are called **nonparametric** or **distribution-free**. Such charts are discussed in detail in Chapter 4.

2.1.4 Variables and Attributes Control Charts

Variables are characteristics that we are interested in monitoring. In order for a variable to be monitored, it must be measurable. There are mainly two types of variables: quantitative and qualitative (categorical). Quantitative variables can be continuous or discrete. In SPC, the terminology **variables data** refers to measurements on quantitative continuous variables. Examples include data on variables such as length, width, temperature, weight, volume, etc., each of which is a continuous variable. Depending on the parameter, the charting statistic can be the sample mean ($\overline{X}$), the sample median ($\tilde{X}$), the sample range (R), the sample standard deviation (S), or the sample variance (S^2). For a variables control chart, the charting statistic follows a continuous distribution.

Conversely, the terminology **attributes data** refers to recordings or measurements on quantitative discrete variables. Examples include the number of errors or mistakes made in completing a loan application, or the number of medical errors made in a hospital. For an attributes control chart, the charting statistic is discrete and follows a discrete distribution, for example, the charting statistic can be the fraction nonconforming (p), the number nonconforming (np), the number of nonconformities (c), or the average number of nonconformities per unit (c).

Attributes control charts have the advantage of allowing for quick summaries of various aspects of the quality of a product, that is, the engineer may simply classify products as acceptable or unacceptable based on various quality criteria. Thus, attributes charts sometimes bypass the need

for expensive, precise devices and time-consuming measurement procedures. Also, this type of chart tends to be more easily understood by managers who are unfamiliar with quality control procedures; therefore, it may provide more persuasive evidence (to management) of quality problems.

In general, however, when all other considerations are equal, variables data and variables control charts are generally preferred and recommended as variables data and the corresponding charts provide more information about the process. A flow chart is given in Figure 2.6 to assist the practitioner in selecting the appropriate Shewhart control chart.

2.1.5 Sample Size or Subgroup Size

Data must be collected before a control chart can be implemented. In practice, the data are usually collected in batches or samples, each typically with n observations, taken at random at equally spaced time points. The batches are also called subgroups and such data are referred to as **subgrouped** data. The sample size n generally is a small number, around five, although there are applications in the modern environment where every individual data point can be and is inspected. Such data are commonly referred to as **individual data**. In this case $n = 1$. Some cases where individual values are monitored include, for example, where:

- Production may be extremely slow and collecting more than one observation is not practical.
- There is no rational way for subgrouping the data.
- It may be that a sample of more than one is not justified economically.

We will treat the individual observations case later under control charts for individual observations.

2.1.6 Rational Subgrouping

A **rational subgroup** is a sample or a subgroup taken from the process so that the variation among the measurements, or observations within the sample, is only due to the inherent or natural variability of the process, and no additional variation due to any special causes is present within the sample measurements. Rational subgroups are those subgroups that contain or enclose only the common/chance causes. The variation among the readings should be due to the natural process variation. There should be no opportunity for assignable/special causes to add to this variation.

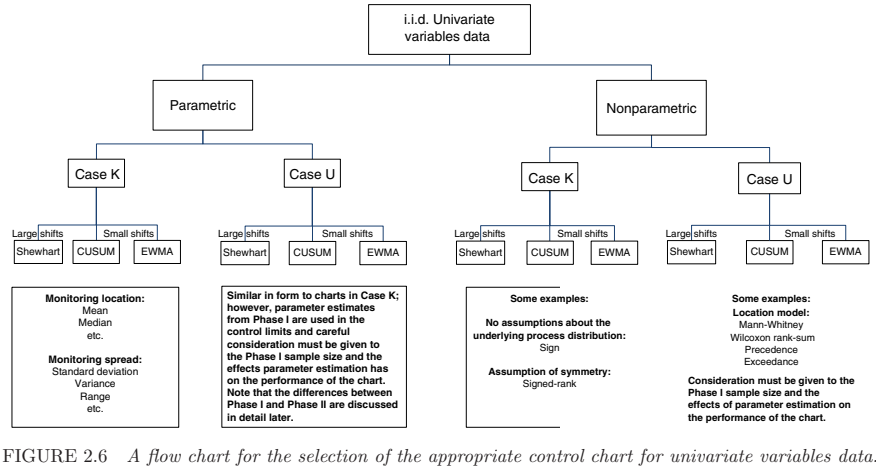


FIGURE 2.6 *A flow chart for the selection of the appropriate control chart for univariate variables data.*

Example 2.1 Rational Subgrouping

Suppose we have two operators operating the same machine at different times and we want to pick up on operator-to-operator differences using samples of size 3.

• Operator 1

∗ Operator 2

Then the correct way of rational subgrouping would be, for example:

●●●●● ∗ ∗∗∗ ∗ ●●●●● ∗∗ ∗∗∗

In this way, the subgroups are chosen to isolate each operator's performance.

An incorrect way of rational subgrouping would be, for example:

●●●●● ∗∗∗∗∗ ●●●●●

In doing it this way, any differences between the operators will be effectively masked.

Figure 2.7 represents a process that consists of five machines that pool their output together into one "bucket." Suppose that a random sample is drawn from the "bucket" to form a subgroup of size 5. We could have, for example, ○▲□○● or □□●■▲. Such subgroups will not be able to detect machine-to-machine differences. This is referred to as a masking effect, since the machine-to-machine differences will effectively be masked. One solution is to take a subgroup of size 5 from machine 1, then a subgroup of size 5 from machine 2, and so forth. Taking subgroups in this manner ensures that the control chart is capable of detecting machine-to-machine differences.

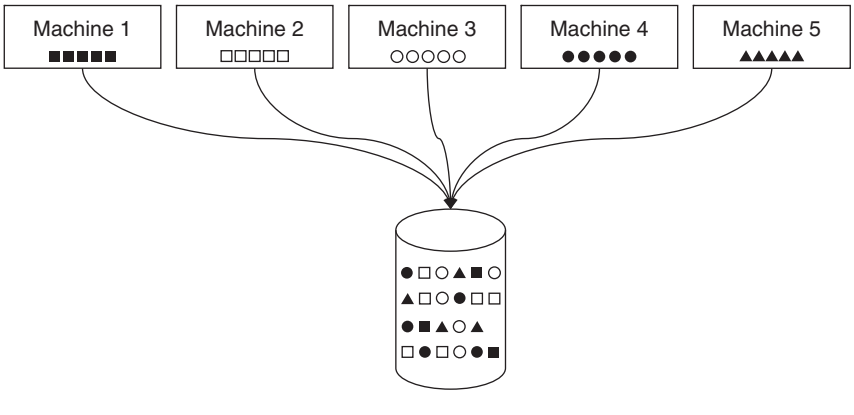


FIGURE 2.7 *An illustration where the differences will be effectively masked.*

2.1.7 Nonparametric or Distribution-free

From a historical point of view, much of the early work in statistics was based on the assumption of normality of the underlying distribution. The work in SPC, by and large, is no exception. While most traditional variables control charts have been parametric, that is, they are based on some parametric model assumption about the process distribution, such as normality, there are many situations in practice where such an assumption is not viable or justifiable and can, in fact, be counterproductive. In such situations, it may be preferable to use an alternative set of control charts called **nonparametric** or **distribution-free**. As in Gibbons and Chakraborti (2010), the term nonparametric is not intended to imply that there are no parameters involved, in fact, quite the contrary. While the term distribution-free seems to be a better description of what we expect from these charts, that is, they remain "valid" for all continuous distributions, nonparametric is perhaps the term more often used. By valid, we mean that the IC performances of these charts remain unchanged for all continuous distributions, regardless of their shapes. Thus, nonparametric control charts have IC robustness guaranteed, which is a major selling point for them over parametric charts. Moreover, it has now been demonstrated that nonparametric charts often have competitive OOC performance relative to their parametric counterparts, and thus it seems like a win–win proposition to use them in practice.

In the statistics literature, there is now a rather vast collection of nonparametric hypothesis tests and confidence intervals, and these methods have been shown to perform well compared to their Normal Theory counterparts. Remarkably, even when the underlying distribution is normal, the asymptotic relative efficiency of some nonparametric methods relative to the corresponding (optimal) Normal Theory methods can be as high as 0.955 (see, for example, Gibbons and Chakraborti, 2010, p. 508). For some other heavy-tailed and skewed distributions, the efficiency can be 1.0 or even higher. It may be argued that nonparametric methods will be "'less efficient" than their parametric counterparts when one has a complete knowledge of the process distribution for which that parametric method was specifically designed. However, the reality is that such information is seldom, if ever, available in practice.

Parametric control charts are the most commonly used as they have been developed since the beginning of SPC, starting around the 1940s. They are available in many software packages. However, from a practical point of view, it seems natural to develop and use nonparametric methods, including control charts in SPC applications. Quality practitioners would be well advised to have these techniques in their toolkits.

The advantages of a nonparametric control chart can be summarized as follows

1. It is easy to implement, i.e., simplicity.

2. No need to assume a particular parametric distribution for the underlying process.

3. The IC run-length distribution is the same for all continuous process distributions.

4. It is more robust and outlier resistant.

5. It may be more efficient in detecting changes when the true distribution is markedly non-normal, particularly with heavier tails.

6. Estimating the variance to set up charts for the location parameter may not be always necessary.

From a technical point of view, most nonparametric procedures, including control charts, require the population to be continuous in order to be distribution-free. Thus, in an SPC context, we consider the so-called nonparametric variables control charts. While the nonparametric charts offer many advantages, they are not without their critics. Some of the perceived shortcomings of the nonparametric control charts, along with our responses, parenthetically, are as follows

1. They will be "less efficient" than their parametric counterparts when one has a complete knowledge of the process distribution for which that parametric method was specifically designed (but such complete knowledge is seldom, if ever, available).

2. One usually requires special tables when the sample sizes are small (computer software can be written and made available to overcome this).

3. Nonparametric methods are not as well known among researchers and practitioners in quality (this may be rectified with more training and visibility).

Although the nonparametric control charts form the main focus of this book, in order to set the stage, we first discuss some of the details of the

traditional parametric charts in Chapter 3. Following this, in Chapter 4, we discuss univariate nonparametric control charts, which are designed to track the location of a continuous process; very few charts are available for scale and more work is needed for this case. The field of multivariate control charts is interesting and the body of literature on nonparametric multivariate control charts is growing. However, we don't focus on these in this book; the reader is referred to a recent book by Qiu (2014) that has a good discussion of some of the parametric and nonparametric control charts.

The basic point is this: if the underlying model can be assumed, and that turns out to be the correct model, a parametric statistical procedure, such as a parametric control chart, would be more efficient than a nonparametric control chart, which does not make a model assumption. However, the reality is that, in practice, such a model assumption may often not be available or justifiable, and, as such, the nonparametric control chart may often be the only viable alternative to use.

2.1.8 Monitoring Process Location and/or Process Scale

Two important classical problems in the usual SPC are monitoring the process mean and/or the process standard deviation. In the nonparametric setting, we consider, more generally, monitoring the center or the location parameter, and/or a scale parameter of a process. The location parameter represents a typical value and could be the mean or some percentile, such as the median of the distribution. The latter is especially attractive when the underlying distribution is expected to be skewed. Let $F(x)$ denote the unknown cumulative distribution function (cdf) of the monitored continuous variable X. In analogy with the parametric, mainly the normal distribution case, it is assumed that F follows either:

1. a location model, with a cdf $F(x-\theta)$, where $x \in (-\infty, \infty)$ and $\theta \in (-\infty, \infty)$ is the location parameter; or

2. a scale model, with a cdf $F\left(\frac{x}{\tau}\right)$, where $x \in (-\infty, \infty)$ and $\tau > 0$ is the scale parameter; or

3. a location-scale model with cdf $F\left(\frac{x-\theta}{\tau}\right)$, where $x \in (-\infty, \infty)$ and $\theta \in (-\infty, \infty)$ and $\tau > 0$ are the location and the scale parameter, respectively.

Thus, in the nonparametric setting, the interest is to monitor θ or τ or both under these model assumptions based on random samples or subgroups

of size $n \geq 1$ of data, usually taken at equally spaced time points, or with $n = 1$, corresponding to the individual measurements case. To highlight the analogy in the usual parametric control charting problems, F is assumed to be the cdf (Φ) of the standard normal distribution, whereas in the nonparametric setting, for variables data, F is some unknown continuous cdf. Although the location-scale model seems to be a natural model to consider paralleling the Normal Theory case with mean and variance both unknown, most of what is currently available in the nonparametric statistical process control (NSPC) literature deals mainly with the location model. As noted earlier, this will be the main focus in this book.

2.1.9 Case K and Case U

Note that for the parametric control charts, one makes a distributional assumption for the underlying process, such as normality, but the parameters of the distribution may be known/specified or unknown. When the underlying parameters are known or specified, this is referred to as the "standard(s) known case" and is denoted by **Case K** for easy reference. By contrast, if the parameters are unknown and need to be estimated, this situation is referred to as the "standard(s) unknown case" and is denoted by **Case U**. Typically, the parameter estimation is done in Phase I with reference data. For parametric charts, both cases are studied in Chapter 3. For nonparametric charts, although we do not make specific model assumptions such as normality, we still have Case K and Case U, as is discussed in Chapter 4. We will first consider Case K before considering the case more typically found in industry, which is Case U. We start with a discussion on parametric variables control charts in Chapter 3.

2.1.10 Control Charts and Hypothesis Testing

There are similarities between control charts and traditional inference tools such as hypothesis testing and confidence intervals. The reader should recall that when using a Shewhart chart a process is declared IC when the charting statistic falls between the control limits, whereas a process is declared OOC when the charting statistic falls on or outside either control limit. Thus, as with testing hypothesis, there is a yes/no decision based on a value of a charting statistic and decision regions.

Suppose that the charting statistic T, with standard deviation σ_T, is an unbiased point estimator for a process parameter θ. That is, $E(T) = \theta$ and $VAR(T) = \sigma_T^2$. Then, there is a connection between using a Shewhart control chart based on a charting statistic T and a hypothesis test of the form

$$H_0\colon \theta = \theta_0 \text{ versus } H_1\colon \theta \neq \theta_0.$$

For example, if T is the sample mean (i.e., $\overline{X}$) and we wish to test the hypothesis that the process mean μ is equal to some specified value μ_0, the hypotheses testing problem is

$$H_0\colon \mu = \mu_0 \text{ versus } H_1\colon \mu \neq \mu_0$$

where H_0 is the null hypothesis and H_1 is the alternative hypothesis.

Typically, H_0 is rejected if, and only if $\dfrac{\sqrt{n}\left|\overline{X} - \mu_0\right|}{\sigma_0} \geq k$ or, alternatively, if and only if

$$\overline{X} \geq \mu_0 + k\frac{\sigma_0}{\sqrt{n}} \tag{2.2}$$

or

$$\overline{X} \leq \mu_0 - k\frac{\sigma_0}{\sqrt{n}}. \tag{2.3}$$

Note that the right hand sides of Equations 2.2 and 2.3 are in fact the k-sigma control limits of Equation 2.1. Thus, if $\overline{X}$ plots between the two control limits, we conclude that the process mean is in statistical control, which is the same as saying that we fail to reject the null hypothesis $H_0\colon \mu = \mu_0$. Conversely, if $\overline{X}$ plots on or outside either control limit we conclude that the process mean is not in statistical control, which in turn implies the same null hypothesis $H_0\colon \mu = \mu_0$ is rejected in favor of the alternative hypothesis $H_1\colon \mu \neq \mu_0$. Thus, the two OOC regions of a Shewhart control chart can be viewed as a graphical display of the rejection region or the critical region for a hypothesis test of the form $H_0\colon \theta = \theta_0$ versus $H_1\colon \theta \neq \theta_0$. Therefore, a control chart can be viewed as testing the null hypothesis $H_0\colon \theta = \theta_0$ repeatedly at a number of points in time. There are various perspectives on and controversies regarding these issues in the literature, see, for example, an article by Woodall (2000), along with the contributions from various discussants.

The hypothesis-testing framework can be useful in studying the performance of a control chart, for example, as in the context of hypothesis testing, here too we can think of errors and probabilities of errors. The **Type I error** is the event that the process is declared OOC when, in fact, the process is IC. Similarly, the **Type II error** is the event that the process is declared to be IC when, in fact, the process is not. This is represented in Table 2.1.

The corresponding probabilities of these two errors (or events) are

$$\alpha = P(\text{Type I error}) = P(\text{Declare process is OOC} \mid \text{process is actually IC})$$

and

$$\beta = P(\text{Type II error}) = P(\text{Declare process is IC} \mid \text{process is actually OOC})$$

respectively.

Table 2.1 Representation of Type I and Type II errors.

	Declare process is	
Status of process	In-control	Out-of-control
In-control	Correct decision	Type I error
Out-of-control	Type II error	Correct decision

In SPC applications, the Type I error is commonly called a **false alarm** and the associated probability is referred to as the false alarm rate (*FAR*). The *FAR* is discussed in detail in Section 2.1.12. As the Type I error probability is more serious in a hypothesis test, the *FAR* is deemed more serious in SPC and one usually tries to control the *FAR* at a rather small value, for example, a *FAR* value of 0.0027 is often used in the industry, which is similar to using an α value of 0.05 in traditional statistical analysis.

The probability of a Type II error is referred to as the β – *risk*. The two probabilities of error are inversely related to one another. If the *FAR* increases, the β – *risk* decreases and vice versa. The question arises: Why are α and β important when it comes to the choice of control limits?

If we widen the control limits (select a larger value of k) it will lower *FAR* which will decrease the risk of a Type I error which will decrease the risk of a point falling on or outside the control limits when no assignable causes are present. Similarly, if we widen the control limits (select a larger value of k) it will increase the risk of a Type II error which will increase the risk of a point falling between the control limits when the process is really OOC.

2.1.11 General Steps in Designing a Control Chart

In the previous sections, many issues regarding control charting have been discussed, such as whether location and/or scale is to be monitored, whether attributes or variables control charts should be used, the choice of sample size, rational subgrouping, etc. Here, in Figure 2.8, we summarize the general steps that need to be followed when designing a (any) control chart before introducing the different control charts.

Let $X_{i1}, X_{i2}, \ldots, X_{in}$ denote a random sample (measurements on some quantitative quality characteristic) of size $n \geq 1$, taken from a process at time $i = 1, 2, 3, \ldots$

2.1.12 Measures of Control Chart Performance

Since a control chart is used to monitor a process, and the quality of the process output is vital, the performance of any control chart is an important

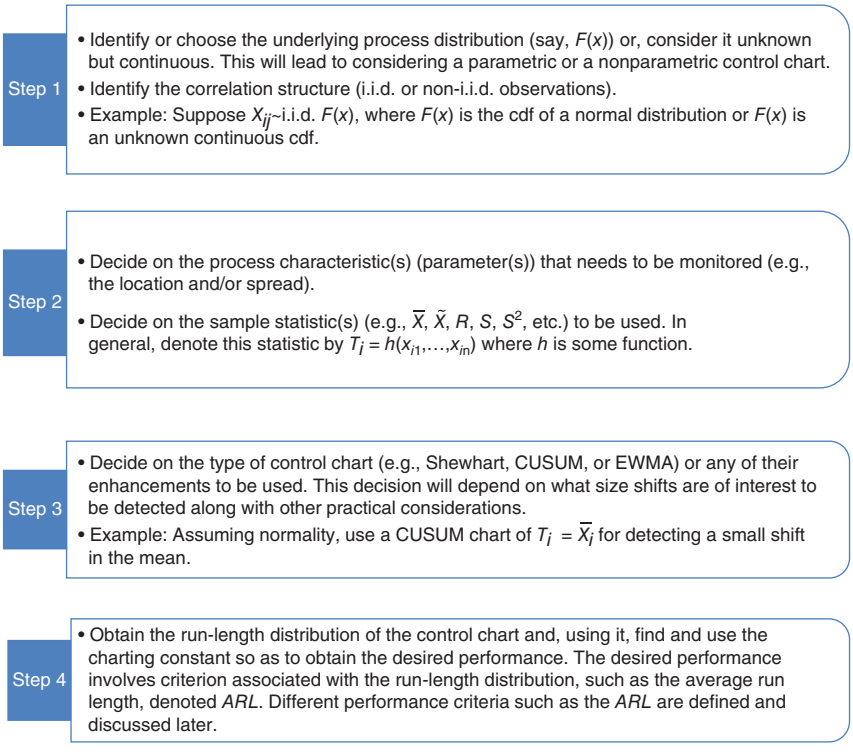


FIGURE 2.8 *General steps in designing a control chart.*

consideration. Various characteristics of the run-length distribution are generally used to assess chart performance. The **run-length** is a random variable that denotes the number of points (charting statistics) that must be plotted (or the number of subgroups that must be collected, each of which yields, say, one charting statistic) before the first OOC signal is given by the chart (or the first signaling event is observed). The run-length is denoted by N. Since the run-length is a counting random variable, it is discrete, has a probability mass function (pmf), and we examine various attributes of this pmf to evaluate chart performance. For example, we calculate its mean or the expected value, the so-called average run-length (denoted ARL), to understand average performance, its standard deviation, which is the standard deviation of the run-length distribution (denoted $SDRL$) to understand the variability in performance. It can be seen that the run-length pmf is skewed to the right and thus we also consider its percentiles such as the median, the so-called median run-length (denoted MRL), for a more comprehensive understanding of chart performance. The ARL is perhaps the most well-known chart performance measure in practice, but the MRL has been gaining ground more recently, while other proposals have been made. We discuss some of these issues in more detail later.

For the discussions to follow, it is important to first highlight the fact that SPC usually consists of two parts (or phases or stages), namely, Phase I (or Stage I) and Phase II (or Stage II), in which monitoring objectives are different so that the control charts are constructed under different performance criteria. From this point forward, we will refer to phases separately to make matters clearer. The first phase, known as Phase I or the retrospective phase, is mainly exploratory and is typically used to establish control or stability of a process based on an analysis of historical or retrospective data. Typically, these data are carefully analyzed using descriptive and inferential statistical methods, including control charts. Then unknown parameters are estimated, distributional assumptions are checked, and control limits are calculated. The Phase I control charts are calculated using a slightly different criterion, more will be said on that later. Then the charting statistic from each available sample is calculated and plotted on the control chart. If all the charting statistics are within the control limits and no systematic patterns are evident, the process is declared to be IC. If, however, one or more of the control statistics fall on or outside the control limits, one typically investigates those samples for assignable causes. If any are present, those samples can be discarded and the control limits are recalculated using the remaining samples. The charting statistics are then plotted on the revised control chart. This process is repeated until we are satisfied that the process is IC, after which the data at hand are called reference data; these form the basis on which any unknown parameters are estimated and the control limits are calculated using these estimates. The next part of SPC, namely, the future monitoring of the process, is referred to as the prospective phase or Phase II. Thus, Phase I analysis typically involves issues related to the design, estimation (if necessary), and establishment of control limits in an iterative manner, whereas Phase II involves future process monitoring using parameter estimates and setting up of control limits with the information and other assumptions developed in the course of the Phase I analysis.

Understanding the statistical performance of a control chart is important in both phases. In Phase I, typically one examines the false alarm probability (FAP) or the probability of *at least one* false alarm (a signal when the process is IC). In Phase II, however, some attributes of the run-length distribution, such as the ARL, the $SDRL$, or the MRL, are used to measure and describe chart performance. First, we discuss the measures of control chart performance for Phase I (i.e., the FAP), after which we discuss the measures of control chart performance for Phase II.

2.1.12.1 False Alarm Probability (FAP)

Without any loss of generality, we assume that m denotes the "final" number of reference samples at the end of a Phase I analysis when the process is deemed to be IC. Thus, the reference data set is assumed to have $N = mn$

observations. An OOC situation is indicated when a charting statistic falls on or outside either of the control limits. As noted earlier, this important event is often called a signal or a signaling event. It is convenient to consider the complementary event, that is when a subgroup does not signal, and is called the non-signaling event. Thus, for the ith subgroup, $E_i = \{L\hat{C}L < C_i < U\hat{C}L\}$, $i = 1,2,\ldots,m$, denotes the non-signaling event. Note that $L\hat{C}L$ and $U\hat{C}L$ denote the estimated Phase I control limits calculated using the reference data, and that C_i denotes the charting statistic from the ith reference sample. As discussed earlier, the *FAP* is usually the recommended chart design criterion adopted in Phase I (see Chakraborti, Human, and Graham 2009). The *FAP* can be expressed as follows

$$\begin{aligned} FAP &= P(\text{At least one false alarm from the } m \text{ subgroups}) \\ &= 1 - P(\text{No signal among the } m \text{ subgroups}|\text{IC}) \\ &= 1 - P(E_1, E_2, \ldots, E_m|\text{IC}) \\ &= 1 - P\left(\bigcap_{i=1}^{m}\{L\hat{C}L < C_i < U\hat{C}L\}|\text{IC}\right). \end{aligned} \tag{2.4}$$

The last step of Equation 2.4 equals

$$1 - \int_{L\widehat{C}L}^{U\widehat{C}L}\int_{L\widehat{C}L}^{U\widehat{C}L}\cdots\int_{L\widehat{C}L}^{U\widehat{C}L} f_{C_1,C_2,\ldots,C_m}(c_1, c_2, \ldots, c_m)dc_1 \ldots dc_m$$

if the joint distribution of the C_i's is continuous and where $f_{C_1,C_2,\ldots,C_m}(c_1, c_2, \ldots, c_m)$ denotes the joint probability density function of the charting statistics $C_1, C_2, \ldots, C_m$ when the process is IC. Note that the joint distribution is used to calculate the *FAP*, since the m signaling events in this case are dependent, as each charting statistic is compared to the same limits, which are constructed from the same set of m reference samples.

Similarly, if the C_i's are discrete, then $P(\cap_{i=1}^{m}\{L\hat{C}L < C_i < U\hat{C}L\}|\text{IC})$ can be re-written as $P(\cap_{i=1}^{m}\{L\hat{C}L + 1 \le C_i \le U\hat{C}L - 1\}|\text{IC})$ and, consequently, Equation 2.2 equals

$$= 1 - \sum_{c_1=L\widehat{C}L+1}^{U\widehat{C}L-1}\sum_{c_2=L\widehat{C}L+1}^{U\widehat{C}L-1}\cdots\sum_{c_m=L\widehat{C}L+1}^{U\widehat{C}L-1} f_{C_1,C_2,\ldots,C_m}(c_1, c_2, \ldots, c_m)dc_1 \ldots dc_m$$

where $f_{C_1,C_2,\ldots,C_m}(c_1, c_2, \ldots, c_m)$ denotes the joint pmf of the charting statistics $C_1, C_2, \ldots, C_m$ when the process is IC.

Thus, the calculation of the *FAP* requires knowledge of the joint distribution of the charting statistics, when the process is IC. Derivation of this distribution and determination of the control limits (and the associated charting constants; which differs, depending on the type of control chart

under consideration) pose some important practical challenges, particularly in an SPC context where the number of comparisons (m) can be as few as 25 or as many as 300 or more.

The *FAP* and similarly the signaling probability, that is, the probability of at least one signal from the m subgroups, are the only performance measures typically considered for Phase I. Next, we discuss the performance measures for Phase II.

2.1.12.2 False Alarm Rate (*FAR*)

The **Type I error** is the event that the process is declared OOC when, in fact, the process is IC. This is the concept used primarily in hypothesis tests. In SPC applications, the Type I error is commonly called a **false alarm** and the associated probability is referred to as the **false alarm rate**, which is abbreviated as ***FAR***. The value $FAR = 0.0027$ is a typical industry standard, but other values such as 0.01 or 0.001 are also used depending on the particular application.

2.1.12.3 The Average Run-length (*ARL*)

As noted already, the performance of a control chart is studied via its run-length distribution, and the most popular performance measure is the expected value of the run-length distribution, the so-called **average run-length** (ARL). The ARL of a control chart is the expected number of charting statistics that must be plotted (subgroups that must be collected) before the control chart signals for the first time. When the process is IC, the expected number of subgroups that must be collected before the control chart signals for the first time is called the IC average run-length and is denoted by ARL_{IC}. Conversely, the OOC average run-length, denoted by ARL_{OOC}, is the expected number of samples to be collected before a control chart signals for the first time, when the process has gone OOC. Intuitively, for an efficient control chart, the ARL_{IC} should be large and the ARL_{OOC} should be small. Hence, when comparing the performance of two or more charts, the ARL_{IC} of the charts is fixed at an acceptably high level and the control chart with the smallest ARL_{OOC} for some specified shift (δ) in the process parameter performs the best. Note that this concept is similar to fixing the probability of a Type I error of some hypothesis tests at a nominal value, such as 0.05, and choosing among them the test with the highest power at some value of the parameter under the alternative hypothesis. In Figure 2.9, we plot two ARL curves against varying sized shifts, corresponding to two different control charts.

From Figure 2.9, it can be seen that the ARL_{IC} values of the two charts are fixed at a nominal $ARL_{\text{IC}} = 370$, which indicates that, when the process is IC, the first signal would be observed, on average, at every 370th plotted point. The nominal value 370 is the industry standard. With this being

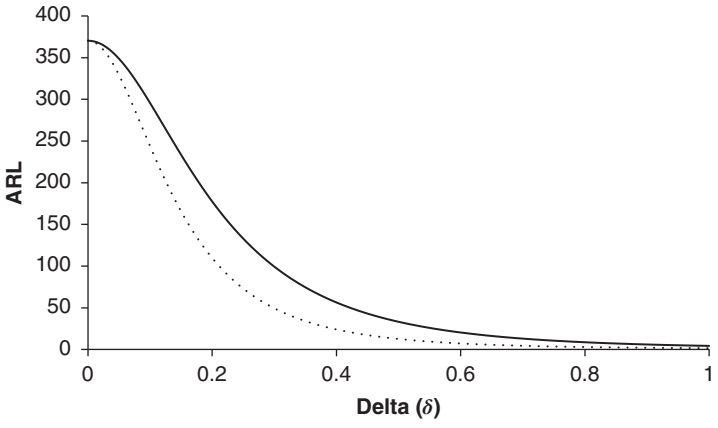


FIGURE 2.9 *Two* ARL *curves corresponding to two different control charts against varying sized shifts.*

said, the point of fixing the ARL_{IC} values is so that we can find the more powerful of the two charts by examining the ARL_{OOC} values. The most powerful chart is the one that gives the smallest ARL_{OOC} for any given shift. From Figure 2.9, it can be seen that the control chart indicated by the dotted curve is the better choice as it performs the best uniformly for all values of the shift.

2.1.12.4 Standard Deviation of Run-length (*SDRL*)

Another measure of chart performance is the standard deviation of the run-length distribution, denoted as ***SDRL***. Clearly, for a desirable control chart, we want the *SDRL* to be small and the *ARL* to be equal or close to the nominal ARL_{IC} value.

2.1.12.5 Percentiles of Run-length

Although the *ARL* is the most commonly used chart performance indicator, noting that the run-length distribution is significantly right-skewed, many researchers have recommended examining a number of percentiles, including the 5th, 25th, 50th, 75th, and the 95th percentiles, to better characterize the run-length distribution; this is discussed next.

The evaluation of the percentiles of the run-length distribution is important for a better understanding of chart performance, since the run-length distribution is significantly right-skewed. Recall the discussion about the median, which is the 50th percentile, in Chapter 1. We noted that, in general, the percentiles need to be defined so that they are unique, which may be an issue for discrete distributions, since the cdf has jumps at the mass points. To this end, since the run-length random variable N is discrete, we start by

defining the 100*qth percentile ($0 < q < 1$) as the smallest positive integer t, such that $P(N \leq t) \geq q$. In other words, the 100*qth percentile of the distribution is the smallest positive integer t, so that the cdf of N at t is at least equal to q. The most popular percentiles to be investigated are the 5th, 25th, 50th, 75th, and the 95th percentiles, which span the distribution and include the first, the second, and the third quartiles. The second quartile or the 50th percentile is the median, and the median run-length is denoted by *MRL*. The idea of looking at percentiles in SPC goes back to Barnard (1959). Many authors, including Chakraborti (2007) and Khoo *et al.* (2011), have advocated the use of percentiles, such as the median, for the assessment of chart performance. Suppose that the *MRL* is used as a measure of typical chart performance. The *MRL* may be the preferred measure from a robustness point of view, since, as we noted earlier, the run-length distribution is significantly right-skewed. Then, an efficient control chart should have a large MRL_{IC} and a small MRL_{OOC}. This is similar in spirit to the approach when using the *ARL* as the performance measure. When comparing the performance of charts, among the charts with an acceptably high nominal MRL_{IC}, the control chart with the smallest MRL_{OOC} at some specified shift (δ) in the process parameter is preferred. In Figure 2.10, we plotted two *MRL* curves corresponding to two different control charts against varying sized shifts.

From Figure 2.10, it can be seen that the MRL_{IC} values of the charts are fixed at a nominal $MRL_{\text{IC}} = 350$, which indicates that the first false alarm will be observed within the first 350 plotted points at least 50% of the time. This is an interpretation that the ARL_{IC} does not have. Fixing the MRL_{IC} at 350 is the industry standard. With this being said, since both charts have the same MRL_{IC} value, we can find the more efficient of

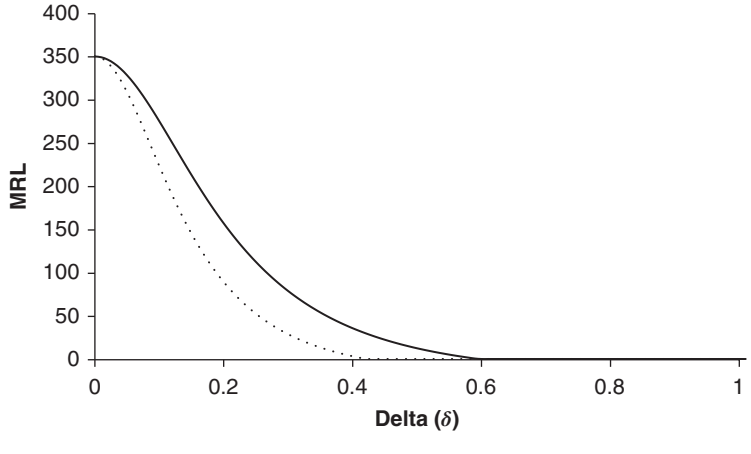


FIGURE 2.10 MRL *curves corresponding to two control charts for varying shift sizes.*

the two charts by examining the MRL_{OOC} values. The more efficient of the two charts will be the chart that gives the smallest MRL_{OOC} for any given shift. From Figure 2.10, it can be seen that the control chart corresponding to the dotted curve performs uniformly better.

While the *ARL* and the *MRL* indicate typical chart performance, and the run-length distribution is markedly skewed, for a better understanding of performance of the chart, the entire run-length distribution needs to be examined and assessed. To this end, boxplot-like graphs are sometimes employed (see Radson and Boyd, 2005); examples are shown in Figure 2.11. Each boxplot shows the mean (*ARL*) of the distribution as a black line, the standard deviation (*SDRL*) as a circle, and the median (*MRL*) as a gray line inside the box. The "whiskers" are extended to the 5th and the 95th percentiles (marked X's) instead of the usual minimum and maximum.

From Figure 2.11, for example, it can be seen that Charts 1 and 2 have a similar run-length distribution, which is different from that of Charts 3 and 4 in both location and spread. Note also that, in each case, the *SDRL* values are closer to the *ARL* values.

The disadvantages of using the *ARL* are summarized below. The use of the *MRL* will ensure a guaranteed control over the *FAR* in the sense that no more than 50% of the false alarms will be realized before the MRL_{IC}. Moreover, there is practically no disadvantage to using the *MRL* to describe typical chart performance. However, note that, while using the *MRL* to characterize the typical chart performance or the center of the run-length distribution, the spread is measured by the interquartile range (IQR), which is the difference between the third and the first quartiles.

When comparing the performance of charts based on the *MRL*, the nominal MRL_{IC} is fixed at an acceptably "high" level and the chart with the smaller MRL_{OOC} is preferred. To this end, note that the run-length

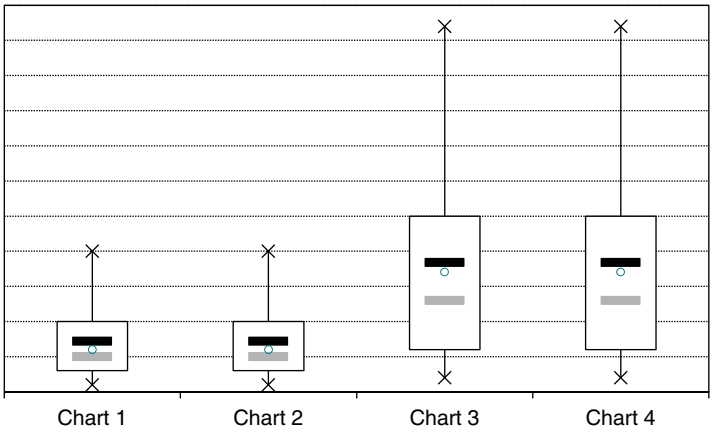


FIGURE 2.11 *Boxplot-like graph summarizing the run-length distribution of control charts.*

distribution of a Shewhart chart, for Case K, follows a geometric distribution with success probability $p = \frac{1}{ARL_{\text{IC}}}$. For industry standard values for a nominal ARL_{IC} such as 370, $p = \frac{1}{370} = 0.0027$. Consequently, distribution with success probability p, the cdf is $P(N \leq t) = 1 - (1-p)^t$, it follows that the MRL_{IC} equals $\frac{\ln(0.5)}{ln(1-p)}$. For example, for $p = 0.0027$, the $MRL_{\text{IC}} = \frac{\ln(0.5)}{ln(1-0.0027)}$, which equals 256.37. As another example, for $p = 0.002$, which corresponds to an ARL_{IC} of 500, the MRL_{IC} equals $\frac{\ln(0.5)}{ln(1-0.002)} = 346.2$. Therefore, keeping parity with the traditional Shewhart chart and for simplicity, we recommend using 250 or 350 as the target nominal MRL_{IC}. Other nominal MRL_{IC} values corresponding to nominal ARL_{IC} values, such as 370, can be found similarly.

Next, we provide some comments on the *ARL* and the *MRL*.

Note 2.1 Some Comments About the ARL and the MRL

1. The run-length distribution of a chart is highly right-skewed and thus conclusions based on the *ARL* about typical chart performance can be misleading.

2. Difficulty of interpretation. A control chart having, say, an ARL_{IC} of 500 will have about 63% of all the run-lengths lower than 500 (this can be seen by calculating the 63rd percentile), but 50% of all the run-lengths will be lower than 348 since the corresponding MRL_{IC} is 348.

3. There may be problems where the *ARL* (mean of a distribution) does not exist mathematically; see, for example, Chakraborti, Van der Laan, and Van de Wiel (2004) and Graham *et al.* (2012).

4. High standard deviation. Again, since the run-length distribution is highly right-skewed, in many cases there is a high *SDRL*, which is undesirable. Moreover, from a practical point of view, this means there can be some extremely large run-length values, and therefore some run-length characteristics can't be computed within a reasonable amount of time, i.e., using the *ARL* as performance measure can be very time-consuming. This is particularly problematic in the IC case.

5. Lack of robustness. As is well known, the *ARL* being an average isn't a robust measure and, consequently, using a winsorized or trimmed mean may be necessary in some applications. The interested reader is referred to the literature for more details.

In conclusion, we advocate the use of percentiles, such as the median, in the assessment of chart performance. For a more comprehensive look at the run-length distribution, one should consider examining multiple performance measures, including the *ARL*, the *SDRL*, and the percentiles. Perhaps a boxplot-like graphic, as mentioned earlier, can be useful in this regard for a quick visualization.

2.1.12.6 Average Number of Samples to Signal (*ANSS*)

The **average number of samples to signal** (*ANSS*) is defined (see Reynolds and Stoumbos, 2000) as the expected number of samples of n observations taken from a specified time point to the time that the chart signals. Usually for *ANSS* (as well as for *ANOS* and *ATS*; discussed below), the measurements are taken from the start of the process, that is, at time zero. Similar to the *ARL* and *MRL*, for an efficient control chart the $ANSS_{\text{IC}}$ should be large and the $ANSS_{\text{OOC}}$ should be small. To conserve space, graphs are omitted; however, when comparing two or more charts, the $ANSS_{\text{IC}}$ is fixed at an acceptably high level, and the control chart with the smallest $ANSS_{\text{OOC}}$ for some specified shift (δ) in the process parameter performs best.

2.1.12.7 Average Number of Observations to Signal (*ANOS*)

The **average number of observations to signal** (*ANOS*) is defined (see Reynolds and Stoumbos, 2000) as the expected number of individual observations taken from a specified time point to the time that the chart signals. The following important relationship exists between the *ANOS* and the *ANSS*, namely, $ANOS = (n)(ANSS)$. Similar to the *ANSS*, for an efficient control chart the $ANOS_{\text{IC}}$ should be large and the $ANOS_{\text{OOC}}$ should be small. Again, to conserve space, graphs are omitted; however, when comparing two or more charts, the $ANOS_{\text{IC}}$ is fixed at an acceptably high level, and the control chart with the smallest $ANOS_{\text{OOC}}$ for some specified shift (δ) in the process parameter performs best.

2.1.12.8 Average Time to Signal (*ATS*)

The **average time to signal** (*ATS*) is the number of time periods that occur until a signal is generated on the control chart (see Khoo, 2004). The *ATS* is necessary, since the *ANSS* and the *ANOS* may be insufficient if there exist some time periods during which we have no inspection. For samples

that are taken at equally spaced intervals with a length of time l,

$$ATS = ARL \times l.$$

If we conduct a 100% inspection, $n = l$ and therefore $ATS = ARL \times n$ (see, for example, Wu, Luo, and Zhang, 2006). It is important to note that l is defined as the length of the period that we are dealing with. This period can refer to a day, a week, an hour, etc. These results are based on fixed sampling intervals. Formally, we can define ATS as the expected length of time until the chart signals, measured from the start of the monitoring process. So, if we were working in days, the ATS is the average number of days to the first signal detection.

We need to take into account whether the process is IC or OOC when we obtain our ATS. For an IC process with all the parameters either known or given to us, we have

$$ATS_{\text{IC}} = ARL_{\text{IC}} \times l$$

So that

$$ATS_{\text{IC}} = \frac{1}{FAR} \times l.$$

Similarly, we have for an OOC process

$$ATS_{\text{OOC}} = \frac{1}{P(\text{Signal in a period})} \times l.$$

De Vries and Conlin (2003) define the target ATS to be the desired IC performance that we want. As an example, a target ATS of 365 days and a period length of 7 days implies a target Type I error rate of $\frac{l}{\text{Target } ATS} = \frac{7}{365} = 0.0192$. Note that this is higher than the nominal $FAR = 0.0027$.

Similar to the ARL, for an efficient control chart the ATS_{IC} should be large and the ATS_{OOC} should be small. Again, to conserve space, graphs are omitted; however, when comparing two or more charts, the ATS_{IC} is fixed at an acceptably high level, and the control chart with the smallest ATS_{OOC} for some specified shift (δ) in the process parameter performs best.

2.1.12.9 Number of Individual Items Inspected (I)

This is just another way of presenting the results that we got for the ARL. Here we are expressing the ARL in terms of the number of individual items that have been inspected. If we suppose that we have samples of size n,

we obtain that the number of individuals inspected is just the product of the average number of samples taken and the sample size. So, we have $I = (ARL)(n)$.

2.1.13 Operating Characteristic Curves (*OC*-curves)

The operating characteristic (OC) is the probability that a control chart will fail to detect a shift in the process. Failure to detect a shift is seen as the event when a chart gives no signal on the first subsequent observation following the shift. The operating characteristic is a probabilistic measure of how quickly a change in the process can be detected by a control chart.

We now define the operating characteristic as $P(\text{No signal}|\text{Shift of size } \delta) = \beta$. The operating characteristic is plotted against δ, which gives us the operating characteristic curve (OC curve). The OC curve is useful to examine the influence of the size of the shift on the performance of the chart. Obviously, OC curves can also be used with great success to compare different charts. For a satisfactory chart, we would like to see that a larger shift corresponds to a smaller probability such that the charting statistic will plot between the LCL and the UCL, that is, will give no signal. Conversely, the signaling probability $(1 - \beta)$ can also be plotted.

In Figure 2.12, we plotted two OC curves corresponding to two different control charts against varying sized shifts. For both charts note that we start with $\beta = 1$ for $\delta = 0$. This was expected, since we definitely wanted the process not to give a signal if no shift had occurred (i.e., if the process was still IC). As δ increases, the values of β decreases for both charts. So, we see that, as the shift in the process parameter increases, the probability

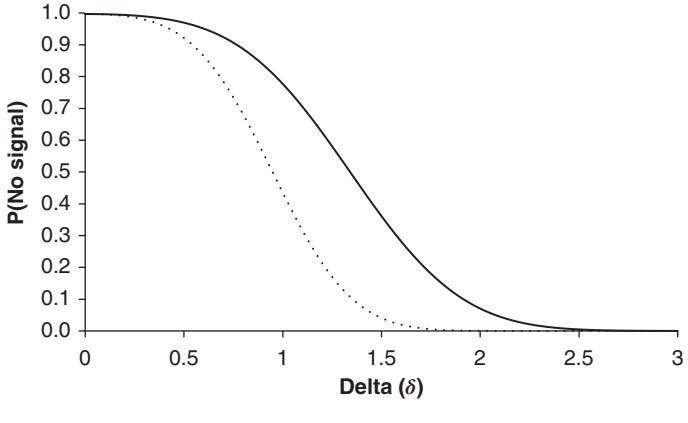


FIGURE 2.12 *Two* OC *curves corresponding to two different control charts against varying sized shifts.*

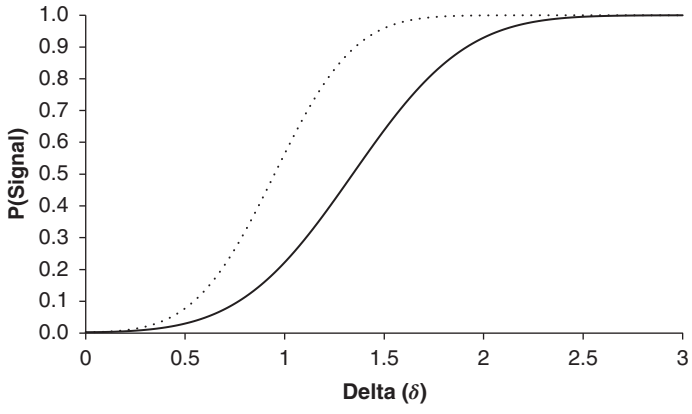


FIGURE 2.13 *Two probabilities of a signal, corresponding to two different control charts, against varying sized shifts.*

of not getting a signal decreases quickly; this is desirable. When comparing the two charts, on the basis of their OC curves, we find that since the dotted curve always lies below the solid curve, the former chart is uniformly better.

To make the same point in another way, for the same two charts, the two signaling probabilities are plotted and shown in Figure 2.13. For both charts it is seen that the probability of observing a signal increases as the size of the shift in the process parameter increases, which is desirable. Now, since the dotted curve dominates the solid curve, which is consistent with what we said for Figure 2.12, the corresponding chart is uniformly better.

Now that different performance measures have been discussed, we discuss two practical cases or situations where control charts are applied and studied, namely, when the process parameters of interest are known or specified while the process is IC, and when they are unknown and need to be estimated before the process monitoring can start.

2.1.14 Design of Control Charts

Once a parameter to be monitored is identified and a charting statistic is selected, we need to select the sample size n, the frequency of sampling, and the form of the control limits. This is referred to as the design of a control chart. We now discuss the various components of chart design.

2.1.14.1 Sample Size, Sampling Frequency, and Variable Sample Sizes

The sample size and the sampling frequency (the length of time between two consecutive samples) must be specified before a chart can be implemented. These depend on considerations such as time and cost, and the size of the

shift to be detected. Both can impact the statistical performance of the chart. Larger samples can detect smaller shifts but are more expensive and time consuming, whereas smaller samples can be sufficient if the expected shift is large.

Frequency of sampling, that is, how often to collect data, is impacted by conditions on the ground, so to speak, that is, again, time and cost and expected shift. Obviously, it is desirable to take large samples as frequently as possible, but this is probably not going to be feasible. We can generally take smaller samples at shorter time intervals (more frequency) or larger samples at longer time intervals (less frequently). Current industry practice favors the former approach. Automated data collection facilities have made this process much easier, so much so that it is not uncommon to sample and inspect every item that is produced.

The appropriate sample size can be determined by looking at the underlying process variability and the size of the shift. Generally, a process with a relatively large inherent variation requires reasonably larger samples to detect an OOC condition as opposed to a process with a relatively small inherent variation. In general, it is also easier to detect a small shift using larger samples as opposed to when the process shift is relatively large, for which smaller samples would be sufficient.

However, the sampling frequency is also important. Taking large samples very frequently is possibly the most desirable course of action to detect any process change. However, this is not economically feasible. Therefore, the problem of deciding on the sample size and the frequency of sampling becomes a problem of allocating sampling effort, that is, we either take small samples more often (over shorter time intervals) or take larger samples less frequently (over longer time intervals).

In practice, samples of size 1 to 25 are often used for variables charts, with $n = 5$ being a very popular choice. In spite of this, sometimes a particular process may impose its own constraints on the sample size and force $n = 1$.

Occasionally it can be that some of the samples have different number of observations, for example, even though $n = 5$, it could be that a sample had to be taken with $n = 4$ at some time points due to some practical constraints. If this is the case, we have variable sample sizes and there are typically three options.

Average sample size can be used to calculate the control limits, which results in a set of approximate, but constant, control limits. It is assumed that future sample sizes do not differ greatly from the previous sample sizes when using these control limits. After all, these control limits must apply to all subgroups. This approach works best with large samples and when the sample sizes do not vary more than 25% from the average sample size. However, since variables control charts are usually based on small samples, changes of 25% or higher are therefore not unlikely. A subgroup of size

$n = 3$, for example, represents a 25% decrease in the sample size if the average sample size is $n = 4$.

Problems arise when a point plots close to any of the approximate control limits. A point that plots outside one of the approximate control limits might well be inside if we calculated the exact control limits. The opposite is also possible, that is, a point that plots inside the approximate control limits might, in fact, plot outside if we calculate the exact control limits.

Example 2.2 Shewhart $\overline{X}$ Control Chart for Variable Sample Sizes

Suppose that Column (a) of Table 2.2 presents the individual measurements of 15 samples that were simulated from a normally distributed process with a *known* IC mean of $\mu_0 = 5$ and a *known* IC standard deviation of $\sigma_0 = 2$. However, note that the sample sizes (n_i) for these samples, as displayed in Column (b), are different and should be taken into account when calculating the control limits. If we are to use the average sample size method, we need to find the average sample size, that is, $\overline{n} = \frac{1}{15}\sum_{i=1}^{15} n_i = 4.33 \approx 4$. Consequently, a set of **approximate** (but constant) 3-sigma control limits is given by

$$UCL = \mu_0 + 3\frac{\sigma_0}{\sqrt{\overline{n}}} = 5 + 3\frac{2}{\sqrt{4}} = 8$$

Table 2.2 Measurements from a process with variable sample sizes.

Sample	(a)					(b)	(c)
i	X_1	X_2	X_3	X_4	X_5	n_i	$\overline{X}_i$
1	0.46	2.80	7.87	—	—	3	3.710
2	9.68	5.60	2.10	9.14	2.02	5	5.708
3	6.18	4.93	5.27	7.54	—	4	5.980
4	3.94	5.53	5.32	4.33	4.63	5	4.750
5	6.43	7.93	3.92	8.64	—	4	6.730
6	5.81	0.89	5.50	2.85	3.98	5	3.806
7	1.96	6.00	5.65	—	—	3	4.537
8	1.37	1.28	7.17	3.74	6.93	5	4.098
9	7.94	4.01	4.69	7.65	—	4	6.073
10	8.33	5.75	6.01	4.57	—	4	6.165
11	3.59	1.41	2.34	9.71	4.57	5	4.324
12	4.11	7.43	3.50	7.68	0.66	5	4.676
13	8.77	5.53	6.37	9.50	—	4	7.543
14	4.97	6.00	8.62	2.73	—	4	5.580
15	4.81	3.40	8.29	5.20	5.37	5	5.414

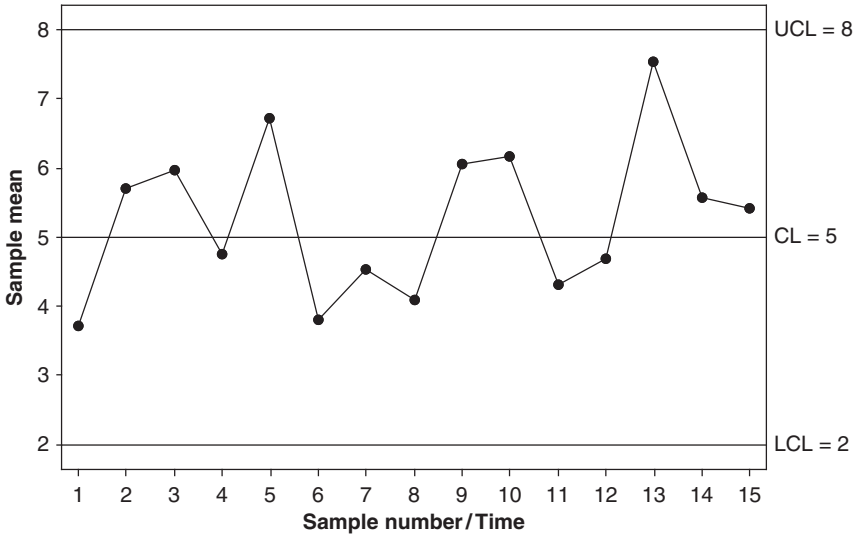


FIGURE 2.14 *A Shewhart $\overline{X}$ control with approximate control limits in the case of variable sample sizes.*

and

$$LCL = \mu_0 - 3\frac{\sigma_0}{\sqrt{\overline{n}}} = 5 - 3\frac{2}{\sqrt{4}} = 2$$

with a center line $CL = \mu_0 = 5$.

Figure 2.14 displays the sample means, $\overline{X}_i$, of Column (c), plotted on a Shewhart $\overline{X}$ control chart together with these approximate control limits.

In Figure 2.14, all of the points plot between the control limits, and there are no anomalous patterns in the data. Hence, we conclude that the process is functioning in statistical control.

2.1.14.2 Variable Control Limits

Although the former approach to variable sample sizes makes the calculations and interpretation of the control charts easier, with the availability of current computer technology it is natural to try to calculate the exact control limits based on the actual size n_i of each sample. Here, the control limits are calculated separately for each subgroup based on the subgroup size n_i. This implies that the spread between the *UCL* and the *LCL* will vary (increase or decrease) as the sample size varies. However, the varying width is merely an indication of the varying amount of information available from each sample, as portrayed by the different sample sizes. Consequently, the visual representation of these control limits is not as attractive as those based on the average sample size with constant control limits. On the one hand, having fixed rather than varying control limits is definitely more pleasing to the eye. On the other hand, the interpretation of these

variable control limits is still the same. If a point plots on or above the UCL or on or below the LCL, it is regarded as a signal of a possible OOC process, whereas a point that plots between the two control limits is regarded as no signal and is associated with an IC process.

Example 2.2 Continued

For the Shewhart $\overline{X}$ control chart with exact control limits, in the case of variable sample sizes, the different sample sizes (n_i), as displayed in Column (b) of Table 2.3, are taken into account when calculating the control limits. For these data, the 3-sigma variable control limits can be found from $LCL_i \,/\, UCL_i = \mu_0 \pm 3\frac{\sigma_0}{\sqrt{n_i}}$ for $i = 1,2,3,\ldots,15$ and are presented in Columns (d) and (e) of Table 2.3.

To illustrate the calculations, consider sample number 1 with $n_1 = 3$. The 3-sigma variable control limits are calculated as follows

$$LCL_1 = 5 - 3\frac{2}{\sqrt{3}} = 1.536 \text{ and } UCL_1 = 5 + 3\frac{2}{\sqrt{3}} = 8.464.$$

Whereas, for sample number 2 with $n_2 = 5$, the 3-sigma variable control limits are calculated as follows

$$LCL_2 = 5 - 3\frac{2}{\sqrt{5}} = 2.317 \text{ and } UCL_2 = 5 + 3\frac{2}{\sqrt{5}} = 7.683.$$

Table 2.3 Measurements from a process with variable sample sizes.

Sample	(a)					(b)	(c)	(d)	(e)
i	X_1	X_2	X_3	X_4	X_5	n_i	$\overline{X}_i$	LCL_i	UCL_i
1	0.46	2.80	7.87	—	—	3	3.710	1.536	8.464
2	9.68	5.60	2.10	9.14	2.02	5	5.708	2.317	7.683
3	6.18	4.93	5.27	7.54	—	4	5.980	2.000	8.000
4	3.94	5.53	5.32	4.33	4.63	5	4.750	2.317	7.683
5	6.43	7.93	3.92	8.64	—	4	6.730	2.000	8.000
6	5.81	0.89	5.50	2.85	3.98	5	3.806	2.317	7.683
7	1.96	6.00	5.65	—	—	3	4.537	1.536	8.464
8	1.37	1.28	7.17	3.74	6.93	5	4.098	2.317	7.683
9	7.94	4.01	4.69	7.65	—	4	6.073	2.000	8.000
10	8.33	5.75	6.01	4.57	—	4	6.165	2.000	8.000
11	3.59	1.41	2.34	9.71	4.57	5	4.324	2.317	7.683
12	4.11	7.43	3.50	7.68	0.66	5	4.676	2.317	7.683
13	8.77	5.53	6.37	9.50	—	4	7.543	2.000	8.000
14	4.97	6.00	8.62	2.73	—	4	5.580	2.000	8.000
15	4.81	3.40	8.29	5.20	5.37	5	5.414	2.317	7.683

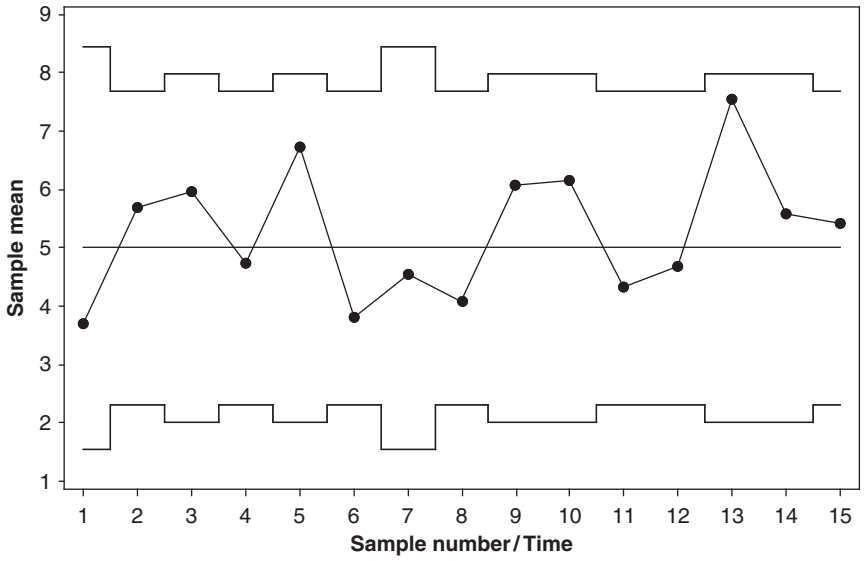


FIGURE 2.15 *A Shewhart $\overline{X}$ control chart with exact control limits in the case of variable sample sizes.*

Figure 2.15 displays the sample means, $\overline{X}_i$, of Column (c), plotted on a Shewhart $\overline{X}$ control chart together with these variable control limits. As we can see from Figure 2.15, the distance between the control limits widens and narrows according to the particular sample or subgroup size.

In Figure 2.15 all of the points plot between the control limits and there are no anomalous patterns in the data. Hence, we conclude that the process is functioning in statistical control.

2.1.14.3 Standardized Control Limits

A third approach to solving the problem of varying sample sizes is to use standardized charting statistics, which results in standardized control limits. Let T_i be the charting statistic for the ith subgroup with IC mean $\mu_{T,\text{IC}}$ and IC standard deviation $\sigma_{T,\text{IC}}$. The standardized charting statistic for subgroup i is found by calculating $Z_i = (T_i - \mu_{T,\text{IC}})/\sigma_{T,\text{IC}}$, $i = 1, 2, 3, \ldots$ These standardized charting statistics are assumed to approximately follow a standard normal distribution, that is, $Z_i \sim N(0, 1)$, and hence the LCL and the UCL can be taken to be simply –3 and +3 respectively, if we use the 3-sigma approach to calculate the control limits.

The additional effort in calculating the standardized charts is offset by the major advantage that we now have constant control limits, no matter what the sample size for the subgroups is. However, a possible difficulty with the interpretation of the standardized charting statistics, or the standardized control charts, is that they are no longer in the original scale of measurement.

Table 2.4 Measurements from a process with variable sample sizes.

Sample	(a)					(b)	(c)	(d)
i	X_1	X_2	X_3	X_4	X_5	n_i	$\overline{X}_i$	Z_i
1	0.46	2.80	7.87	—	—	3	3.710	−1.117
2	9.68	5.60	2.10	9.14	2.02	5	5.708	0.792
3	6.18	4.93	5.27	7.54	—	4	5.980	0.980
4	3.94	5.53	5.32	4.33	4.63	5	4.750	−0.280
5	6.43	7.93	3.92	8.64	—	4	6.730	1.730
6	5.81	0.89	5.50	2.85	3.98	5	3.806	−1.335
7	1.96	6.00	5.65	—	—	3	4.537	−0.401
8	1.37	1.28	7.17	3.74	6.93	5	4.098	−1.008
9	7.94	4.01	4.69	7.65	—	4	6.073	1.073
10	8.33	5.75	6.01	4.57	—	4	6.165	1.165
11	3.59	1.41	2.34	9.71	4.57	5	4.324	−0.756
12	4.11	7.43	3.50	7.68	0.66	5	4.676	−0.362
13	8.77	5.53	6.37	9.50	—	4	7.543	2.543
14	4.97	6.00	8.62	2.73	—	4	5.580	0.580
15	4.81	3.40	8.29	5.20	5.37	5	5.414	0.463

Example 2.2 Continued

If we use the standardized charting statistics in Column (d) of Table 2.4, we again obtain a set of **constant** control limits. To illustrate the calculations, consider sample number 1 with $n_1 = 3$. The first charting statistic, which is the standardized sample mean, is calculated as follows

$$Z_1 = \frac{\overline{X}_1 - \mu_0}{\sigma_0/\sqrt{n_1}} = \frac{3.710 - 5}{2/\sqrt{3}} = -1.117.$$

Figure 2.16 displays the standardized sample means, Z_i, of Column (d), plotted on a Shewhart $\overline{X}$ control chart together with these standardized control limits.

In Figure 2.15, all of the points plot between the control limits and there are no anomalous patterns in the data. Hence, we conclude that the process is functioning in statistical control.

2.1.15 Size of a Shift

The size of a shift refers to the absolute amount (or distance) by which the process parameter being monitored has changed, at the same time taking into account the process variability. For example, suppose we are monitoring the process mean of which the IC process mean is μ_0 and the IC process

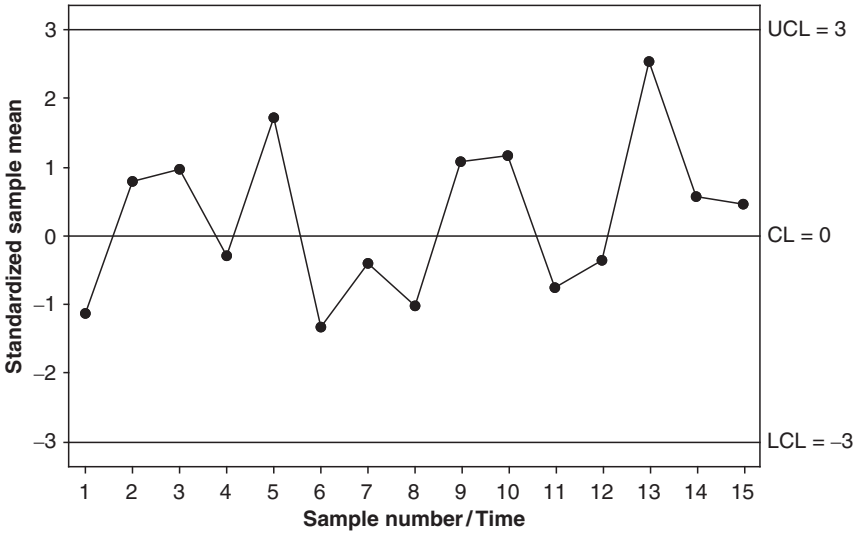


FIGURE 2.16 *A Shewhart $\overline{X}$ control chart with standardized control limits in the case of variable sample sizes.*

standard deviation is σ_0, and we are interested in detecting whether the process mean undergoes a sustained shift to a new value μ_1. Then, it is customary to express the "new"' process mean (μ_1) in terms of the "old" IC process mean (μ_0) as $\mu_1 = \mu_0 + \delta\sigma_0$. Thus, the **absolute size** of the shift in the process mean is $|\mu_1 - \mu_0| = |\delta\sigma_0|$, whereas the **relative size** of the shift in the process mean is $\delta = |\mu_1 - \mu_0|/\sigma_0$, expressed in standard deviation units.

Instead of working with the absolute shift in the process mean, it is more convenient to work with the relative shift in the process mean. For example, suppose we are monitoring two independent processes (A and B) with the same IC process mean, $\mu_A = \mu_B = 10$, say, but process A and process B have different standard deviations, $\sigma_A = 1$ and $\sigma_B = 2$, say. If we are trying to detect whether the mean of process A has changed to 11 and the mean of process B has changed to 12, the absolute size of the shifts in the process means are different, that is, $|11 - 10| = 1$ for process A and $|12 - 10| = 2$ for process B. However, the relative size of the shifts in the process means are the same, that is, $\delta = |11 - 10|/1 = |12 - 10|/2 = 1$.

Thus, we had two different processes (A and B) and we were trying to detect two seemingly different shift sizes, but expressing these shifts in terms of standard deviation units led us to believe that we were really trying to detect the same size in shift – the former being the absolute size of a shift and the latter being the relative size of a shift. Therefore, in future, we will refer to the relative size of a shift simply as the size of the shift and specifically mention when we refer to the absolute size of a shift.

When δ is less than 2.0, it is customary to refer to the shift in the process parameter as a small shift, whereas for larger values of δ the shift is considered large.

2.1.16 Choice of Control Limits

An important consideration in implementing the control chart is the choice of the charting constants. For the Shewhart chart, the charting constant is k and the choice of k directly affects the statistical properties of the chart. Increasing k, for example, or moving the control limits further away from the CL, decreases the probability of declaring a process OOC (when the process is IC), that is, decreasing the FAR, but at the same time, widening the limits increases the probability of not declaring a process OOC (when the process is really OOC), that is, increasing the β – risk. Conversely, moving the control limits closer to the CL will have the opposite effect. That is, in this case, the FAR increases while the β – risk decreases.

2.1.16.1 *k*-sigma Limits

We presented the Shewhart k-sigma limits earlier in the chapter. With these limits, we can calculate the FAR (α) and the β – risk (β) for any choice of k. However, since the control limits on a typical Shewhart control chart are usually set at three standard deviations from the CL (the mean), as an example, we focus our attention on this specific choice of control limits. Recall that the "3-sigma" control limits are given by

$$\begin{aligned} UCL &= \mu_T + 3\sigma_T \\ CL &= \mu_T \\ LCL &= \mu_T - 3\sigma_T. \end{aligned}$$

Assuming that the sampling distribution of the charting statistic T is normal or approximately normal, we can calculate the FAR (i.e., α). However, it is more convenient to calculate the complementary probability $1 - \alpha$, which is the probability of a no-signal when the process is IC. Note that

$$\begin{aligned} 1 - \alpha &= P(LCL < T < UCL) = P(\mu_T - 3\sigma_T < T < \mu_T + 3\sigma_T) \\ &= P(-3 < Z < 3) = 0.9973. \end{aligned}$$

So, the FAR or the probability of a Type I error (α), which is the probability that the charting statistic T falls outside the 3-sigma control limits, is 1 – 0.9973 = 0.0027 when the process is actually IC. This implies that we will observe, on average, an incorrect OOC signal or a false alarm in 27 out of 10 000 or roughly 1 in every 370 samples from the process.

2.1.16.2 Probability Limits

As an alternative to using a fixed or a specified constant k (such as 3) in the Shewhart control limits, which leads to some *FAR* (such as 0.0027) under an assumed distribution (normality), one might be interested in specifying a *FAR* and determining the corresponding k (and hence the control limits). Such control limits are called **probability limits** and are particularly valuable when the sampling distribution of the charting statistic used to monitor the process parameter of interest is skewed.

If we want a *FAR* of 0.0027, for example, the Shewhart control limits are to be symmetrically placed at a distance of k-sigma units from the *CL*, where k is found from

$$P(T \geq \mu_T + k\sigma_T) = P(Z \geq k) = 0.00135$$

or equivalently from

$$P(T \leq \mu_T - k\sigma_T) = P(Z \leq -k) = 0.00135$$

with $Z \sim N(0,1)$. Thus, the required constant k is simply the 99.865th percentile of the IC distribution of the standardized statistic Z, which gives us the 99.73% probability limits. Either of these equations yields $k = 3.0$ and the resulting chart is the familiar 3-sigma Shewhart control chart. Hence, the 99.73% *FAR*-probability limits are the same as the 3-sigma limits. Similarly, for a *FAR* of 0.01, we find $k = 2.58$ so that the 99% *FAR*-probability limits are the same as the 2.58-sigma control limits. Hence, the probability limits in this case uniquely correspond to certain k-sigma limits and vice versa, and this is because the IC distribution of T is assumed to be normal, which is symmetric. This is not the case, however, when the IC distribution of T is asymmetric. For example, suppose that the charting statistic T has a chi-square distribution with 5 degrees of freedom when the process is IC. In this case, one can use the 95th and the 5th percentiles of the IC distribution to find the 90% *FAR*-probability limits; the upper-tail probabilities for the chi-square distribution are given in Table F of Appendix A. However, one can also use the 97th and the 7th percentiles of the chi-square distribution with 5 degrees of freedom. The two sets of control limits, with the same *FAR* of 90%, are different.

To summarize, in the fixed limits approach, the distance of the control limits from the *CL* is specified by the user, which leads to a certain *FAR* (α), whereas with the probability limits the user specifies a desired nominal *FAR* value and that subsequently determines the distance (k) of the control limits from the *CL*.

It may be noted that the above calculations assume an **equal-tailed** (or **symmetric**) approach, in the sense that the probability that the charting statistic plots above the *UCL* is equal to the probability that it plots below

the LCL. Another way to view this is by recalling the fact that the UCLs and the LCLs are both at a distance of k standard deviations from the CL. This approach seems reasonable when the distribution of the charting statistic is symmetric. In general, however, one can take

$$\alpha_L + \alpha_U = \alpha$$

where

$$\alpha_L = P\,(\text{Charting statistic plots below}\, LCL \mid \text{Process is actually IC})$$

and

$$\alpha_U = P(\text{Charting statistic plots above}\, UCL \mid \text{Process is actually IC}).$$

These two probabilities are referred to as α-lower (or $\boldsymbol{FAR_L}$) and α-upper (or $\boldsymbol{FAR_U}$), respectively. The specification of these is particularly meaningful when using a charting statistic that has a skewed distribution.

In cases where only an upward or a downward shift is of interest so that one is looking for a one-sided control chart, or when an upward (or downward) shift is less (or more) of interest than a downward (upward) shift, an unequal assignment of the FAR might be useful. In the former case of a one-sided chart, either $\alpha_U = 0$ or $\alpha_L = 0$ and in the latter case $\alpha_L + \alpha_U = \alpha$ where, for example, $0 < \alpha_U < \alpha_L < 1$. Under this **unequal–tailed** (or **asymmetric**) approach, and assuming a normal distribution for the charting statistic, we need to find the solutions to the equations $P(Z \leq -z_{\alpha_L}) = \alpha_L$ and $P(Z \geq z_{\alpha_U}) = \alpha_U$ in order to find the LCL and the UCL, respectively. The required constants $-z_{\alpha_L}$ and z_{α_U} can be easily found as the $100\alpha_L$ and the $100\alpha_U$ percentiles of the Z distribution, respectively. Another alternative in a situation like this might be to consider a one-sided (upper or lower) control chart with only upper (or lower) control limits, where either a fixed or a probability limits-based approach can be used to determine the required constants.

Example 2.3 Unequal-tailed (or Asymmetric) Control Limits

Suppose that the charting statistic of interest has a normal distribution and assume that α_L is specified to be 0.0015, whereas α_U is specified as 0.0005. Using $\alpha_L + \alpha_U = \alpha$, we find the FAR to be 0.002. Solving for $z_{0.0015}$ and $z_{0.0005}$ from $P(Z \leq -z_{0.0015}) = 0.0015$ and from $P(Z \geq z_{0.0005}) = 0.0005$, we obtain $z_{\alpha_L} = z_{0.0015}$=2.97 and $z_{\alpha_U} = z_{0.0005} = 3.29$, respectively. Thus, the control limits can be determined from Equation 2.1 as

$$UCL = \mu_T + 3.29\sigma_T$$

and

$$LCL = \mu_T - 2.97\sigma_T.$$

In this case, the *LCL* is closer to the *CL* than the *UCL*. This approach might be used when an upward shift in a particular process parameter is of more concern than a downward shift. Next, consider a one-sided upper control chart.

Example 2.4 One-sided Upper Control Limit

Suppose that we are only interested in detecting an upward shift in the mean of a process that follows a normal distribution. Then, if the nominal *FAR* is specified as 0.0025, let $\alpha_L = 0$ and $\alpha_U = 0.0025$. Subsequently, we need to find $z_{0.0025}$ from $P(Z \geq z_{0.0025}) = 0.0025$, which yields $z_{\alpha_U} = z_{0.0025} = 2.81$, the 99.75th percentile of the Z distribution. Thus, the *UCL* is set at $UCL = \mu_T + 2.81\sigma_T$ with no *LCL* on the control chart.

In many situations, the true underlying distribution of the quality characteristic is unknown, and consequently the sampling distribution of the charting statistic is also unknown. Therefore, calculating the exact probability limits is problematic. Note that, in such a case, one can use k-sigma limits for some given value of k, but in finding the *FAR* of that control chart, this could result in facing the same problem. However, if the charting statistic is an average, or can be expressed as an average, approximately, the **central limit theorem** (which implies that the average of n i.i.d. random variables approximately follows a normal distribution regardless of the underlying distribution of the individual variables) can be used. For example, if $X_1, \ldots, X_n$ are i.i.d. with mean μ and variance σ^2, then, $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ approximately, for sufficiently large values of n, regardless of the population distribution. Thus, for example, the 3-sigma control limits for the mean can be used which are assumed to yield a *FAR* of approximately 0.0027. However, if the true underlying process distribution is markedly non-normal, the properties of a control chart (such as the *FAR*) based on the approximate limits can be highly affected, particularly since the sample size n is generally not large in a typical process control application. In such situations, the user has other options available. Some of these are discussed later in this book.

CHAPTER 3

Parametric Univariate Variables Control Charts

Chapter Overview

In this chapter, we study parametric variables control charts. These are charts that are based on an assumption about the underlying process distribution (such as normality) or charts that are based on some approximation (such as normality, via the central limit theorem) about the distribution of the charting statistic. There are three main classes of parametric variables control charts: the Shewhart chart, the cumulative sum (CUSUM) chart and the exponentially weighted moving average (EWMA) chart, each of which is generally used for a specific type of shift detection purpose in mind. We describe some of the charts in detail and the relative advantages and disadvantages of these charts, that are well documented in the literature and are touched on later. This background is useful since analogs of many of these parametric variables charts are considered in Chapter 4 for the nonparametric setting.

Variables charts are based on charting statistics that have a continuous distribution and we focus on these charts here. In addition to the variables charts, there are also attributes charts based on charting statistics with a discrete distribution. These include, for example, the fraction nonconforming chart (called the p chart) and the control chart for nonconformities (called the c chart), as described in Montgomery (2009, pp. 290, 308). These control charts, although quite useful in practice, are not considered in this book. This chapter also considers the various ways to sensitize a control chart (such as runs-rules or sensitivity rules, warning limits, interpreting patterns

Nonparametric Statistical Process Control, First Edition.
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on a control chart, etc.) and ends off with a discussion on the robustness of parametric control charts.

3.1 INTRODUCTION

Let $X_{i1}, X_{i2}, \ldots, X_{in}$ denote a random sample (measurements on some quantitative continuous quality characteristic or variable, such as temperature, diameter, etc.) of size $n \geq 1$, taken from a process at time $i = 1, 2, 3, \ldots$ For parametric variables control charts, the (form of the) underlying process distribution is assumed to be known and the most common assumption in practice is that the distribution is normal, although other distributions, such as the exponential, Weibull, etc., are possible and have been studied. However, since most control charts in the literature are for the normal distribution, we only consider them here and refer to these charts as parametric charts or normal theory charts from this point on. Suppose that a process follows a normal distribution with mean μ and variance σ^2, which may be known (specified) or unknown. If the in-control (IC) process parameters are known or specified, this situation is referred to as Case K and is typically indicated by adding a subscript zero to the parameter symbols (e.g., μ_0 and σ_0^2) to the mean and the variance, respectively. If the IC process parameters are unknown, they are to be estimated before monitoring can begin. This is typically done in a retrospective analysis of the available data, called a Phase I study, where a process is stabilized or brought under control and thus a set of reference data is generated before prospective process monitoring can start in Phase II (see Section 2.1.12 for more details on Phase I and Phase II). This situation where process parameters are unknown, is referred to as Case U, and the process mean and the process variance, estimated from the Phase I reference data, are typically denoted by $\widehat{\mu}$ and $\widehat{\sigma}^2$, respectively.

First, we discuss the three basic parametric variables control charts for the known parameter case. The unknown parameter case is discussed later, in Section 3.8.

3.2 PARAMETRIC VARIABLES CONTROL CHARTS IN CASE K

Recall that there are three main classes of parametric variables control charts: the Shewhart chart, the CUSUM chart, and the EWMA chart, each of which is generally used for a specific type of shift detection purpose in mind. We describe these charts in more detail in each of the three sections that follow.

3.2.1 Shewhart Control Charts

Among the many control charts used in practice, the Shewhart charts are the most popular because of their simplicity, ease of application, and the fact that these versatile charts are quite efficient in detecting moderate to large shifts. These charts were originally proposed by Walter Shewhart in 1926. To describe the Shewhart chart in general, suppose that a process location parameter θ, such as the mean, is to be monitored using a charting statistic T, which is a good point estimator of θ, statistically speaking. Further suppose that the expected value and the standard deviation of T are μ_T and σ_T, respectively. Statistical considerations often lead us to take T to be an unbiased estimator of θ so that $\mu_T = \theta$. Then, a general formula for the center line (CL) and the control limits of a Shewhart control chart are

$$\begin{aligned} UCL &= \theta + k\sigma_T \\ CL &= \theta \\ LCL &= \theta - k\sigma_T \end{aligned} \tag{3.1}$$

where $k > 0$ is the charting constant, which is a chart design parameter that determines the "distance" of the control limits from the CL, expressed in terms of the standard deviation. Hence, these control limits are often called ***k*-sigma** limits. A Shewhart k-sigma control chart is the graphic that displays these three limits as straight lines along with the realized (calculated) values of the charting statistic T for a number of samples or over time. Note that, in a Shewhart chart, the upper and the lower control limits are symmetrically placed around the CL. Such control limits are more meaningful when the distribution of T is symmetric or approximately so, which goes well with the assumption in a Shewhart chart that either the process distribution is normal or that T has a distribution that is approximately normal with mean θ. Suppose, for example, that θ is the process mean μ to be monitored and the IC value of μ is μ_0. In this case, T is taken to be the mean $\overline{X}$ of the sample and then the k-sigma limits are given by $\mu_0 \pm k\,\sigma_0/\sqrt{n}$, where σ_0 is the known process standard deviation, since $\sigma_{\overline{X}} = \sigma_0/\sqrt{n}$. The rationale behind the k-sigma limits is that $\overline{X}$ is exactly (or approximately) normally distributed when the process distribution is normal (or by virtue of the central limit theorem). When the charting statistic plots on or outside of either the upper or the lower control limits, we say that a signal has been observed or that a signaling event has taken place and the process is declared to be out-of-control (OOC).

According to the objective of detecting increases and/or decreases in the process parameter μ_0, control charts are often designed in a one-sided or two-sided form. A one-sided upper (or lower) chart is used to detect upward (or downward) changes, while a two-sided control chart is used to detect some change in the process, which could be either upward or downward.

Thus, the use of a one-sided control implies that one is interested in the shift or change in a known direction, that is, an increase or a decrease. A lower one-sided Shewhart chart signals that a downward shift occurred if $T_i \leq LCL$ (the chart only has an LCL), whereas an upper one-sided Shewhart chart signals that an upward shift occurred if $T_i \geq UCL$ (the chart only has a UCL). From this point forward, we consider two-sided charts since they are more general (as the practitioner often may not know the direction of the shift) and since these have received the most attention in the literature. Moreover, two-sided charts can easily be adapted to yield one-sided charts.

Various refinements and enhancements of the Shewhart chart have been considered in the literature; we discuss some of these in the later sections. An illustrative example of the Shewhart control chart is provided later as well.

Note 3.1 Charting Constant: The 3-sigma Principle
Shewhart (1926, 1931) recommended that the charting constant k be taken to be equal to 3, which yields the well-known 3-sigma Shewhart $\overline{X}$ control chart. It is evident that such a choice assumes, either implicitly or explicitly, an underlying normal distribution. Hence, the 3-sigma Shewhart $\overline{X}$ control chart is a parametric control chart. The choice for k usually depends on some criterion related to the performance of the chart. In his discussion, Shewhart motivated the choice of $k = 3$ using a combination of both empirical and theoretical evidence. Assuming that the underlying process is normally distributed, using properties of the normal distribution, we have

$$P\left(\mu_0 - 3\frac{\sigma_0}{\sqrt{n}} < \overline{X} < \mu_0 + 3\frac{\sigma_0}{\sqrt{n}}\right) = P\left(-3 < \frac{\overline{X} - \mu_0}{\frac{\sigma_0}{\sqrt{n}}} < 3\right)$$
$$= \Phi(3) - \Phi(-3) = 0.9973$$

where Φ denotes the cumulative distribution function (cdf) of the standard normal distribution. This means that 99.73% of the values of the sample means will fall within a range of $\pm 3\sigma_{\overline{X}}$ from the IC mean; the 3-sigma principle is illustrated in Figure 3.1. In other words, the chance of a false alarm, that is, the chance of the charting statistic $\overline{X}$ falling on or outside the control limits, when the process is actually IC, equals $1 - 0.9973 = 0.0027$ exactly,

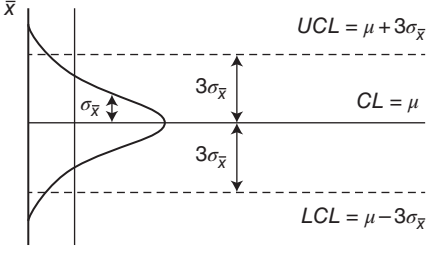


FIGURE 3.1 *The 3-sigma principle.*

when the process is normal, or approximately, when the sample size is sufficiently large. The chance of a false alarm is called a false alarm rate (FAR). The value 0.0027 is an industry standard, and can be found from Tables D1 and D2 in Appendix A. The FAR value 0.0027 implies that a false alarm is expected in every 27 out of 10 000 or 3 out of 1000 samples, but other values are possible and may be desirable depending on the situation at hand.

Some authors, see, for example, Nelson (2003), have asked the question "When should the limits on a Shewhart control chart be other than a CL $\pm$3-sigma?" Nelson (2003) goes on to answer this question as, "Whenever one has a good reason." Here, we elaborate on this. When the charting constant k is less than 3, the control chart has narrower control limits, which leads to a higher FAR. Conversely, when the charting constant k is greater than 3, the control chart has wider control limits, which leads to a lower FAR. There are other arguments in the literature regarding when $k \neq 3$ should not be used.

The Shewhart charts are known to be more effective in detecting sudden or abrupt large shifts (see, for example, Montgomery, 2009, p. 400).

3.2.2 CUSUM Control Charts

While the Shewhart charts are widely known and often used in practice because of their simplicity and effectiveness in detecting moderate to large shifts, other charts, such as CUSUM charts, may be more useful in certain situations for detecting smaller, persistent kind of shifts. These charts, sometimes labeled time-weighted charts, are more naturally appropriate in the process control environment in view of the sequential nature of data collection. The CUSUM control charts were first introduced by Page (1954) (although not in its present form) and have been studied by many authors over the last 60 years. See, for example, Barnard (1959), Ewan and Kemp (1960), Johnson (1961), Goldsmith and Whitfield (1961), Page (1961), Ewan (1963), Hawkins (1992, 1993), and Hawkins and Olwell (1998). These are some examples, and since the introduction of CUSUM charts in 1954 by Page, there has been an incredible amount of work on CUSUM charts (see the overview in the *Encyclopaedia of Statistics in Quality and Reliability* by Ruggeri, Kenett, and Faltin (2007a) and the citations therein, for example). The CUSUM charts are typically based on the CUSUMs of a statistic or of differences of a statistic from its IC expected value, and are calculated progressively as the data accumulate over time. For example, the CUSUM chart for the mean is typically based on the CUSUM of the deviations of the individual observations (or the subgroup means) from the specified value of the IC target mean.

To describe the CUSUM chart in more detail, assume as before that $X_{i1}, X_{i2}, X_{i3}, \ldots, X_{in}$ denote the ith sample or a sample (subgroup) of size $n \geq 1$ at sampling instance (time) i, $i = 1,2,\ldots,$ from a process with a

known or specified IC process mean μ_0 and a known or specified IC process standard deviation σ_0. Let

$$T_i = \psi(x_{i1}, x_{i2}, \ldots, x_{in}) \tag{3.2}$$

be a statistic constructed using the data in the ith sample, $i = 1,2,\ldots$ where ψ is some function. The statistic in Equation 3.2 is referred to as the basic (pivot) statistic, which would be an estimator of the process mean; for example, T_i could be the sample mean. As we noted earlier, one can form either a CUSUM of the sample means (for subgrouped data) or of the individual observations ($n = 1$) or of the differences of the sample means or the individual observations from μ_0. The CUSUM chart for individual observations is described below. Here $n = 1$ and $T_i = X_i$. The adaptation to sample means is straightforward and will be commented on later. The CUSUM chart is formed by plotting C_i where

$$C_i = \sum_{j=1}^{i} (x_j - \mu_0)$$

which can be written as

$$C_i = (x_i - \mu_0) + \sum_{j=1}^{i-1} (x_j - \mu_0)$$

$$C_i = (x_i - \mu_0) + C_{i-1}.$$

There are two ways to represent CUSUMs, namely, (i) the **tabular** (or **algorithmic**) CUSUM and (ii) the **V-mask** form of the CUSUM. We only consider the **tabular** CUSUM, since the **V-mask** form of the CUSUM is no longer used in practice. Montgomery (2009, p. 403) states, "Of the two representations, the tabular CUSUM is preferable." Since we only consider the tabular CUSUM, the word "tabular" will be dropped from this point forward and we will only refer to the CUSUM chart.

The *upper one-sided* CUSUM chart is used to detect increases in the mean and it works by accumulating differences between X_i and μ_0. Hence, for the upper one-sided CUSUM chart, we use the charting statistic

$$C_i^+ = \max[0,\ X_i - \mu_0 - K + C_{i-1}^+] \ \text{ for } \ i = 1, 2, 3, \ldots \tag{3.3}$$

to detect positive shifts (increases) in the mean from μ_0, with starting value $C_0^+ = 0$ and where $K \geq 0$ is called a reference or a slack value. It is common to express K in terms of the process standard deviation, so that $K = k\sigma_0$. The control chart issues a signal for the first i such that $C_i^+ \geq H$, where the constant $H > 0$ is called the decision interval.

Similarly, the *lower one-sided* CUSUM is of interest for detecting decreases in the mean and it works by accumulating differences from μ_0

that are below target

$$C_i^- = \min[0,\ X_i - \mu_0 + K + C_{i-1}^-] \text{ for } i = 1, 2, 3, \ldots \tag{3.4}$$

or

$$C_i^{-^*} = \max[0,\ \mu_0 - K - X_i + C_{i-1}^{-^*}] \text{ for } i = 1, 2, 3, \ldots \tag{3.5}$$

and is used to detect negative shifts or deviations from μ_0 with starting value $C_0^- = C_0^{-^*} = 0$. Here, a signaling event occurs for the first i such that $C_i^- \leq -H$ (if Equation 3.4 is used) or $C_i^{-^*} \geq H$ (if Equation 3.5 is used). For a more visually appealing chart, Equation 3.4 will be used in this book to construct the lower one-sided CUSUM from this point forward. The *two-sided* CUSUM chart is of interest when both increases and decreases in the mean are of interest. This chart signals for the first i at which either one of the two inequalities, $C_i^+ \geq H$ or $C_i^- \leq -H$ is satisfied. Note that for the CUSUM chart, there are some counters, N^+ and N^-, which indicate the number of consecutive periods that the CUSUM's C_i^+ and C_i^- have been non-zero before signaling, which helps in identifying at what point in time the shift may have taken place. If a signal is given by the upper CUSUM (i.e., C_i^+) at sample number 29 and, say, the corresponding counter is equal to 7 (i.e., $N^+ = 7$), then we would conclude that the process was last IC at sample number 29 – 7 = 22, so the shift likely occurred between sample numbers 22 and 23. The CUSUM charts have the added advantage that one can obtain an estimate of the new process mean, $\widehat{\mu}$, following a shift. The quantity $\dfrac{C_i^+}{N^+}$ is an estimate of the amount by which the current mean, after a shift, is above $\mu_0 + K$ when a signal occurs with C_i^+. The quantity $\dfrac{C_i^-}{N^-}$ is an estimate of the amount by which the current mean, after a shift, is below $\mu_0 - K$ when a signal occurs with C_i^-. Thus, if the upper CUSUM signaled, we would estimate the new process average as

$$\widehat{\mu} = \mu_0 + K + \frac{C_i^+}{N^+} \tag{3.6}$$

whereas, if the lower CUSUM signaled, we would estimate the new process average as

$$\widehat{\mu} = \mu_0 - K - \frac{C_i^-}{N^-}\ . \tag{3.7}$$

The constants K and H are needed in order to implement the CUSUM chart; this is discussed next.

The constants K and H are referred to as the design parameters of the CUSUM chart and are typically chosen so that the chart has a specified nominal ARL_{IC} (such as 370) and is capable of detecting a shift in the mean, especially a small shift, as soon as possible. The first step in this direction is to choose K, which is called the reference value. Since $K = k\sigma_0$,

and σ_0 is assumed to be known, by specifying k we are, in fact, specifying K. Accordingly, from this point forward, we only consider the choice of k. Note that the latter comment also holds for the design parameter H because H is also typically defined as $H = h\sigma_0$ with $h = 4$ or 5 providing good ARL properties (see Montgomery, 2009, p. 408); so, by specifying h we are, in fact, specifying H. For the design parameter k, let us consider the parametric CUSUM chart for monitoring the known mean of a normal distribution ($\mu_0 = 0$, without loss of generality) with standard deviation $\sigma_0 = 1$ (again, without loss of generality), on the basis of individual data ($n = 1$). In this case, the IC distribution of X_i follows a normal distribution with mean $\mu_0 = 0$ and standard deviation $\sigma_0 = 1$. To examine the impact of the choice of k, we examine a graph of some ARL_{OOC} values calculated for the normal distribution, calculated by simulation, in Figure 3.2, setting the nominal $ARL_{\text{IC}} = 500$, for $\mu_1 = 0.1$, 0.25, 0.5, and 1.0. Note that μ_1 represents the increased values of μ to be detected "quickly" from $\mu_0 = 0$; hence, μ_1 represents the true shift in the mean, where $\mu_1 = \mu_0 + \delta\sigma_0$ with $\delta = 0.1$, 0.25, 0.5, and 1.0, respectively.

From Figure 3.2, several interesting observations can be made. On the one hand, when the shift is small (see Panels (a) and (b)) and a larger value

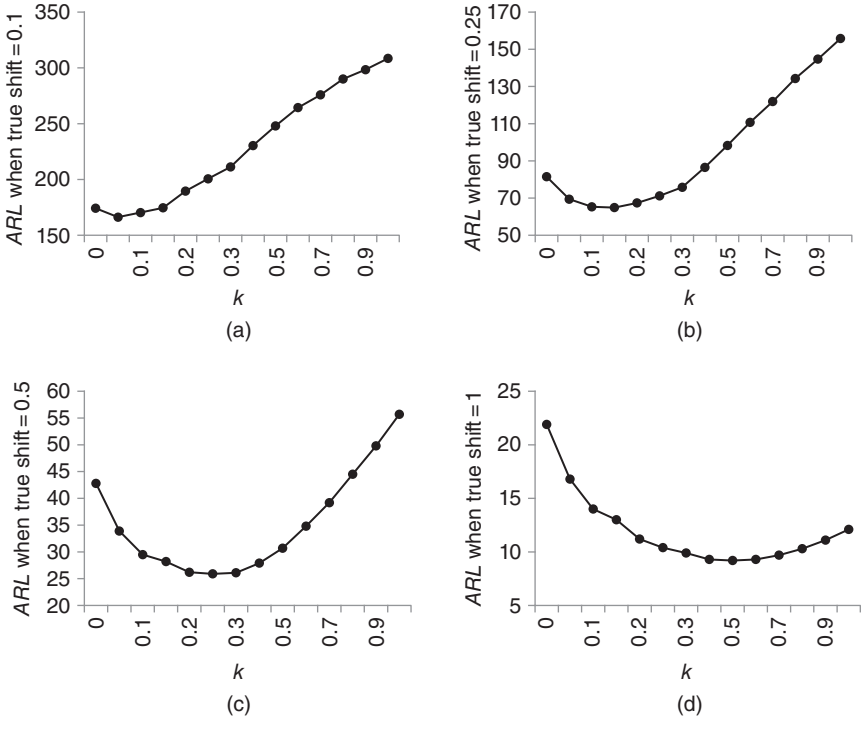


FIGURE 3.2 ARL_{OOC} *values of the traditional CUSUM chart with the nominal* $\text{ARL}_{IC} = 500$ *for different values of* k *and* $\mu_1 = 0.1$, *0.25, 0.5, and 1.0.*

of k is chosen, the ARL_{OOC} values become unacceptably high. On the other hand, if the shift is large (see Panels (c) and (d)) and a smaller value of k is chosen, the ARL_{OOC} values are also high, but not as high as in the latter case. This suggests that when there is little or no a priori information regarding the size of the shift, a small value of k is a safer choice (to protect against any unnecessary delay in detection).

The choice of k for the parametric CUSUM chart for the normal mean has been discussed by many authors; see, for example, Lucas (1985), Hawkins and Olwell (1998), Kim *et al.* (2007), and Montgomery (2009). Lucas (1985) states, "The CUSUM parameter k is determined by the acceptable mean level (μ_a) and by the unacceptable mean (μ_d) level which the CUSUM scheme is to detect quickly. For normally distributed variables the k value is chosen half way between the acceptable mean level and the unacceptable mean level." In the more recent literature, see, for example, Montgomery (2009), it is agreed that, in the normal theory setting, k is typically chosen relative to the size of the shift that we want to detect, that is, $k = \frac{1}{2}\delta$, where δ is the size of the shift in the mean expressed in standard deviation units.

After choosing a value of k, the next step is to find the decision interval h in conjunction with the chosen k so that a specified nominal ARL_{IC} is attained. Note, however, that for a discrete random variable, the chances are that h cannot always be found such that the specified nominal ARL_{IC} is attained exactly and hence, using a conservative approach, h is found so that the attained ARL_{IC} is less than or equal to the specified nominal ARL_{IC}. The decision interval, h, can be found using a grid search algorithm using, say, 100 000 Monte Carlo simulations using some statistical software such as SAS®[1] or R®.[2]

At this point, it should be noted that the detection capability of the CUSUM chart depends on the proper design (tuning) of the chart. The proper design of a CUSUM chart involves obtaining the optimal combination of the CUSUM chart parameters (k and h) by minimizing the OOC average run-length (denoted by ARL_{OOC}) for a specified value of the shift size δ and a nominal IC average run-length (denoted by ARL_{IC}). In other words, one may obtain many k and h combinations that attain a specified nominal ARL_{IC}, but the optimal pair is that for which the ARL_{OOC} is the lowest.

Note that this general discussion regarding the CUSUM chart is for Case K, that is, when the process parameters are known. In Case U, the process parameters are unknown and need to be estimated. Naturally, this will impact the choice of k and h. The reader is referred to Jones, Champ, and Rigdon (2004), who evaluated the performance of the CUSUM chart with estimated parameters. The authors also provide a method for approximating the run-length distribution and its moments.

[1] *SAS Version 9.2.* SAS Institute Inc., Cary, NC.
[2] https://www.r-project.org/.

Table 3.1 CUSUM chart parameters: (h, k) values for a nominal ARL_{IC} value of 370 and 500, respectively.

370	k	0.25	0.50	0.75	1.00	1.25	1.50
	h	8.01	4.77	3.34	2.52	1.99	1.61
500	k	0.25	0.50	0.75	1.00	1.25	1.50
	h	7.27	4.39	3.08	2.32	1.83	1.47

In this section, we have discussed the CUSUM chart for individual measurements. However, if rational subgroups of size $n > 1$ are taken, then simply replace X_i with $\overline{X}_i$ and σ_0 with $\sigma_{\overline{X}} = \dfrac{\sigma_0}{\sqrt{n}}$ in the equations above.

The reader is referred to Hawkins and Olwell (1998) for a detailed overview on parametric CUSUM charts. They also provide software and some very useful tables with ARL_{IC} values for various values of h and k; see their Tables 3.1 and 3.2 on pages 48 and 49, respectively. Hawkins (1993) further provides a table of (h, k) values for a nominal ARL_{IC} value of 370. Using these tables, we provide the (h, k) values for a two-sided CUSUM chart in Table 3.1 for a nominal ARL_{IC} value of 370 and 500, respectively. For example, for a shift of about $1\sigma_0$ in the process mean, taking $k = 0.5$ and $h = 4.77$ gives an $ARL_{\text{IC}} = 370$.

It should be noted that as with the Shewhart chart, a lot of work has been done on enhancing the performance of the CUSUM chart. One of these is referred to as the fast initial response (*FIR*) feature. The *FIR* enhancement feature was originally proposed in Lucas and Crosier (1982) for the parametric CUSUM chart. The *FIR*, or head-start (HS), feature is used when one wants to improve the sensitivity of a CUSUM at process start-up. This is done by setting the starting values C_0^+ and C_0^- equal to some non-zero value, typically we set $C_0^+ = \dfrac{H}{2}$ and $C_0^- = -\dfrac{H}{2}$. This is called a 50% HS.

With so much work done with the parametric control charts, it is natural to consider analogs using nonparametric charting statistics. This approach has led to nonparametric CUSUM charts (denoted by NPCUSUM), which are discussed in the next chapter.

3.2.3 EWMA Control Charts

Another popular class of control charts is the EWMA charts. The EWMA charts also take advantage of the sequentially (time-ordered) accumulating nature of the data, arising in a typical SPC environment, and are known to be efficient in detecting smaller shifts, but may be easier to implement than the CUSUM charts (see, for example, Montgomery, 2009, p. 419). The

classical EWMA charts for the mean were introduced by Roberts (1959) and they contain the Shewhart charts as a special case. The literature on EWMA charts is enormous and continues to grow at a substantial pace (see, for example, the overview in the *Encyclopaedia of Statistics in Quality and Reliability* by Ruggeri, Kenett, and Faltin (2007a) and the references therein). Some more recent references include Capizzi and Masarotto (2012) and Ross *et al.* (2012).

To describe the EWMA chart in more detail, assume, as before, that $X_{i1}, X_{i2}, X_{i3}, \ldots, X_{in}$ denote a random sample (subgroup) of size $n \geq 1$ on the process output at each sampling instance $i = 1,2,\ldots$, from a process with a known IC process mean μ_0 and a known IC process standard deviation σ_0. As in the case of the CUSUM chart, we consider the individual observations case so that $n = 1$ and we monitor $X_1, X_2, X_3, \ldots$ with $T_i = X_i$ for $i = 1,2,\ldots$ The adaptation to sample means is straightforward and is commented on later.

The charting statistic for the EWMA control chart is defined as

$$Z_i = \lambda X_i + (1-\lambda)Z_{i-1} \text{ for } i = 1, 2, 3, \ldots \tag{3.8}$$

where $0 < \lambda \leq 1$ is a constant called the smoothing parameter. The starting value Z_0 is typically taken to be the IC value of the process mean, that is, $Z_0 = \mu_0$.

To illustrate that the charting statistic Z_i is a weighted average of all the previous statistics, Z_{i-1} may be substituted by $Z_{i-1} = \lambda X_{i-1} + (1-\lambda)Z_{i-2}$ into Equation 3.8 to obtain

$$\begin{aligned} Z_i &= \lambda X_i + (1-\lambda)(\lambda X_{i-1} + (1-\lambda)Z_{i-2}) \\ &= \lambda X_i + \lambda(1-\lambda)X_{i-1} + (1-\lambda)^2 Z_{i-2} \\ &= \lambda X_i + \lambda(1-\lambda)X_{i-1} + (1-\lambda)^2(\lambda X_{i-2} + (1-\lambda)Z_{i-3}) \\ &= \lambda X_i + \lambda(1-\lambda)X_{i-1} + \lambda(1-\lambda)^2 X_{i-2} + (1-\lambda)^3 Z_{i-3}. \end{aligned}$$

This method of substitution is called recursive substitution. By continuing the process of recursive substitution for Z_{i-j}, $j = 2,3,\ldots$, we obtain

$$Z_i = \lambda \sum_{j=0}^{i-1} (1-\lambda)^j X_{i-j} + (1-\lambda)^i Z_0.$$

The expected value and the variance of the charting statistic Z_i are given by

$$E(Z_i) = \mu_0 \tag{3.9}$$

and

$$var(Z_i) = \sigma_0^2 \left(\frac{\lambda}{2-\lambda}\right)(1-(1-\lambda)^{2i}) \text{ for } i = 1, 2, \ldots \tag{3.10}$$

respectively (see Appendix 3.1 for the derivations). Hence, the exact control limits and the CL of the EWMA control chart are given by

$$\begin{aligned} UCL &= E(Z_i) + L \times SD(Z_i) = \mu_0 + L\sigma_0\sqrt{\left(\frac{\lambda}{2-\lambda}\right)(1-(1-\lambda)^{2i})} \\ CL &= E(Z_i) = \mu_0 \\ LCL &= E(Z_i) - L \times SD(Z_i) = \mu_0 - L\sigma_0\sqrt{\left(\frac{\lambda}{2-\lambda}\right)(1-(1-\lambda)^{2i})} \end{aligned} \tag{3.11}$$

where $L > 0$ is a charting constant. Note that the EWMA control limits are set at L standard deviations away from the IC mean (the IC expected value of the charting statistic). When $\lambda = 1$, the control limits reduce to the Shewhart chart limits $\mu_0 \pm L\sigma_0$ and the EWMA chart reduces to the Shewhart chart.

Calculating and implementing the exact EWMA control limits may be somewhat cumbersome. Alternatively, one can use the so-called steady-state control limits (which are typically used when the EWMA chart has been running for several time periods so that the term $(1-(1-\lambda)^{2i})$ in Equation 3.11 approaches unity) are given by

$$\begin{aligned} UCL &= \mu_0 + L\sigma_0\sqrt{\left(\frac{\lambda}{2-\lambda}\right)} \\ &\text{and} \\ LCL &= \mu_0 - L\sigma_0\sqrt{\left(\frac{\lambda}{2-\lambda}\right)}. \end{aligned} \tag{3.12}$$

The two-sided EWMA chart is constructed by plotting Z_i against the sample number i (or time). If the charting statistic Z_i falls between the two control limits, that is, $LCL < Z_i < UCL$, the process is considered IC. If the charting statistic Z_i falls on or outside one of the control limits, that is, $Z_i \leq LCL$ or $Z_i \geq UCL$, the process is considered OOC and a search for assignable causes is necessary.

The two-sided EWMA chart can be easily modified to construct a one-sided EWMA chart in much the same way as was done for a CUSUM chart. For example, an upper one-sided EWMA charting statistic is given by $Z_i^+ = \max[\mu_0, \lambda X_i + (1-\lambda)Z_{i-1}]$ for $i = 1, 2, 3, \ldots$ with starting value $Z_0^+ = \mu_0$. If the Z_i plots on or above the UCL, the process is declared OOC and a search for assignable causes is necessary.

Also, like the CUSUM chart, the design parameters of the EWMA chart, L and λ, are chosen so that the chart has a specified nominal ARL_{IC} and is capable of detecting a specified amount of shift, specially a small shift,

as soon as possible, in terms of the shortest ARL_{OOC}. Montgomery (2009, p. 422) states that, "The optimal design procedure would consist of specifying the desired IC and OOC average run-lengths and the magnitude of the process shift that is anticipated, and then to select the combination of λ and L that provide the desired ARL performance." The constant λ ($0 < \lambda \leq 1$) is the smoothing parameter and is selected depending on the magnitude of the shift to be detected. The constant $L > 0$ is the distance of the control limits from the CL (the larger the value of L, the wider the control limits and vice versa) and is selected in combination with the value of the smoothing parameter λ. With regard to the implementation of the EWMA chart, the first step is to choose λ. The recommendation is to choose a small λ, say, equal to 0.05, when small shifts are of interest. If moderate shifts are of greater concern, choose $\lambda = 0.10$, whereas $\lambda = 0.20$ should be chosen if larger shifts are of interest (see, for example, Montgomery, 2009, p. 423). After λ is chosen, the second step involves choosing L so that a desired ARL_{IC} is attained. This can be done using a grid search algorithm using, say, 100 000 Monte Carlo simulations using some statistical software such as SAS® or R®.

Table 3.2 provides some (λ, L) combinations that give ARL_{IC} values close to the desired values of 370 and 500, respectively, for individual data. These were found using Monte Carlo simulations, as described above.

The shift detection capability of the EWMA chart, like that of the CUSUM chart, depends on proper designing (tuning) of the chart. A proper design of an EWMA chart involves obtaining the (optimal) combination of the EWMA chart parameters (λ and L), by minimizing the ARL_{OOC} for a specified value of the shift size δ and for a given nominal ARL_{IC}. In other words, one may obtain many λ and L combinations that yield a specified nominal ARL_{IC}. The optimal pair, out of these pairs, is that one for which the ARL_{OOC} is the lowest.

Note that this general discussion regarding the EWMA chart is for Case K, that is, when the IC values of the process parameters are known. In Case U, the process parameters are unknown and need to be estimated. The process of estimation impacts chart performance and the chart design

Table 3.2 (λ,L) combinations that give ARL_{IC} values close to the desired nominal values of 370 and 500.

Shift size	Nominal $ARL_{IC} = 370$	Nominal $ARL_{IC} = 500$
Small	(0.05, 2.492)	(0.05, 2.615)
Moderate	(0.10, 2.703)	(0.10, 2.814)
Large	(0.20, 2.860)	(0.20, 2.962)
	Not provided for $\lambda = 0.25$	(0.25, 2.998)
	Not provided for $\lambda = 0.40$	(0.40, 3.054)

parameters, and therefore the optimal choice needs to be re-evaluated in this case. The reader is referred to Jones, Champ, and Rigdon (2001), who evaluated the performance of the EWMA chart with estimated parameters. We take up the issue of parameter estimation later in this chapter.

In this section, we have discussed the EWMA chart for individual measurements. However, if rational subgroups of size $n > 1$ are taken, then simply replace X_i with $\overline{X}_i$ and σ_0 with $\sigma_{\overline{X}} = \dfrac{\sigma_0}{\sqrt{n}}$ in the equations above.

A lot of research has been done to improve the performance of the EWMA chart by adding enhancements. One of these is the *FIR* feature. The *FIR* enhancement feature was originally proposed in Lucas and Crosier (1982) for the parametric CUSUM chart. Then, Lucas and Saccucci (1990) proposed a similar feature for the parametric EWMA chart. An *FIR* feature is used as an antidote to start-up problems and those processes that lack corrective action after the previous OOC signal. Lucas and Saccucci (1990) stated that, for the EWMA chart, the *FIR* feature is most useful when λ is small. In Table 3.3, we give the different types of *FIR* features for the EWMA chart found in the literature so far.

With so much work available on parametric control charts, it is clearly natural to consider analogs of these charts using nonparametric charting

Table 3.3 Different types of *FIR* enhancement for EWMA charts and the corresponding articles.

Type of *FIR* enhancement	Description	Article
(i) Fixed control limits with head-start (HS)	$0\% < \text{HS} \leq 100\%$	Lucas and Saccucci (1990)
(ii) Time-varying (TV) control limits with HS	$0\% < \text{HS} \leq 100\%$	Rhoads, Montgomery, and Mastrangelo (1996)
(iii) TV control limits with exponential-type *FIR*	$FIR_{\text{adj}} = 1 - (1-f)^{1+a(t-1)}$	Steiner (1999)
(iv) TV control limits with modified exponential-type *FIR*	$MFIR_{\text{adj}} = \left(1-(1-f)^{1+a(t-1)}\right)^{1+\frac{1}{t}}$	Haq, Brown, and Moltchanova (2014)

statistics. This line of thinking has led to several nonparametric EWMA (NPEWMA) charts, which are discussed in the next chapter.

Before going into more detail and giving illustrative examples on the parametric CUSUM and EWMA charts, we first focus on the parametric Shewhart control charts.

3.3 TYPES OF PARAMETRIC VARIABLES CHARTS IN CASE K: ILLUSTRATIVE EXAMPLES

3.3.1 Shewhart Control Charts

3.3.1.1 Shewhart Control Charts for Monitoring Process Mean

For monitoring the mean of a process, we typically use the sample mean ($\overline{X}$). An example follows.

Example 3.1 A Parametric Shewhart $\overline{X}$ Control Chart

Shewhart charts are typically applied with subgroup data. Column (a) of Table 3.4 presents some simulated data from a normal distribution, which represent measurements taken from 25 independent samples, each of size 5 ($n = 5$) on a type of wafer. Suppose that, from engineering considerations, the IC mean dimension of the wafers μ_0 is specified to be 1.5 cm and the IC process standard deviation σ_0 is known to be equal to 0.1 cm. The mean of each sample is shown in Column (b) of Table 3.4. To illustrate the calculations, consider sample number 1. The first charting statistic, the mean of sample 1, is calculated as follows

$$\overline{X}_1 = \frac{1.3235 + 1.4128 + 1.6744 + 1.4573 + 1.6914}{5} = \frac{7.5594}{5} = 1.5119.$$

To apply the formulas for the control limits in Equation 3.1, note that the expected value of $\overline{X}$ (i.e., $\mu_{\overline{X}}$) is simply the specified process mean μ_0, whereas the standard deviation of $\overline{X}$, namely, $\sigma_{\overline{X}}$, is equal to $\frac{\sigma_0}{\sqrt{n}}$. Thus, the CL and the k-sigma control limits for a Shewhart $\overline{X}$ control chart are given by

$$\begin{aligned} UCL &= \mu_0 + k\frac{\sigma_0}{\sqrt{n}} \\ CL &= \mu_0 \\ LCL &= \mu_0 - k\frac{\sigma_0}{\sqrt{n}}\,. \end{aligned} \tag{3.13}$$

Table 3.4 Measurements used to construct the Shewhart $\overline{X}$ control chart and the subgroup means.

Sample	(a)					(b)
i	X_1	X_2	X_3	X_4	X_5	$\overline{X}_i$
1	1.3235	1.4128	1.6744	1.4573	1.6914	1.5119
2	1.4314	1.3592	1.6075	1.4666	1.6109	1.4951
3	1.4284	1.4871	1.4932	1.4324	1.5674	1.4817
4	1.5028	1.6352	1.3841	1.2831	1.5507	1.4712
5	1.5604	1.2735	1.5265	1.4363	1.6441	1.4882
6	1.5955	1.5451	1.3574	1.3281	1.4198	1.4492
7	1.6274	1.5064	1.8366	1.4177	1.5144	1.5805
8	1.4190	1.4303	1.6637	1.6067	1.5519	1.5343
9	1.3884	1.7277	1.5355	1.5176	1.3688	1.5076
10	1.4039	1.6697	1.5089	1.4627	1.5220	1.5134
11	1.4158	1.7667	1.4278	1.5928	1.4181	1.5242
12	1.5821	1.3355	1.5777	1.3908	1.7559	1.5284
13	1.2856	1.4106	1.4447	1.6398	1.1928	1.3947
14	1.4951	1.4036	1.5893	1.6458	1.4969	1.5261
15	1.3589	1.2863	1.5996	1.2497	1.5471	1.4083
16	1.5747	1.5301	1.5171	1.1839	1.8662	1.5344
17	1.3680	1.7269	1.3957	1.5014	1.4449	1.4874
18	1.4163	1.3864	1.3057	1.6210	1.5573	1.4573
19	1.5796	1.4185	1.6541	1.5116	1.7247	1.5777
20	1.7106	1.4412	1.2361	1.3820	1.7601	1.5060
21	1.4371	1.5051	1.3485	1.5670	1.4880	1.4691
22	1.4738	1.5936	1.6583	1.4973	1.4720	1.5390
23	1.5917	1.4333	1.5551	1.5295	1.6866	1.5592
24	1.6399	1.5243	1.5705	1.5563	1.5530	1.5688
25	1.5797	1.3663	1.6240	1.3732	1.6887	1.5264

Substituting $\mu_0 = 0.5$, $\sigma_0 = 0.1$, $n = 5$, and $k = 3$ into Equation 3.13, the CL and the control limits for the Shewhart $\overline{X}$ chart are given by

$$UCL = 1.5 + 3\frac{0.1}{\sqrt{5}} = 1.634$$

$$CL = 1.5$$

$$LCL = 1.5 - 3\frac{0.1}{\sqrt{5}} = 1.366.$$

Figure 3.3 displays the sample means, $\overline{X}_i$ from Column (b) of Table 3.4, which are the charting statistics for the Shewhart $\overline{X}$ control chart, plotted

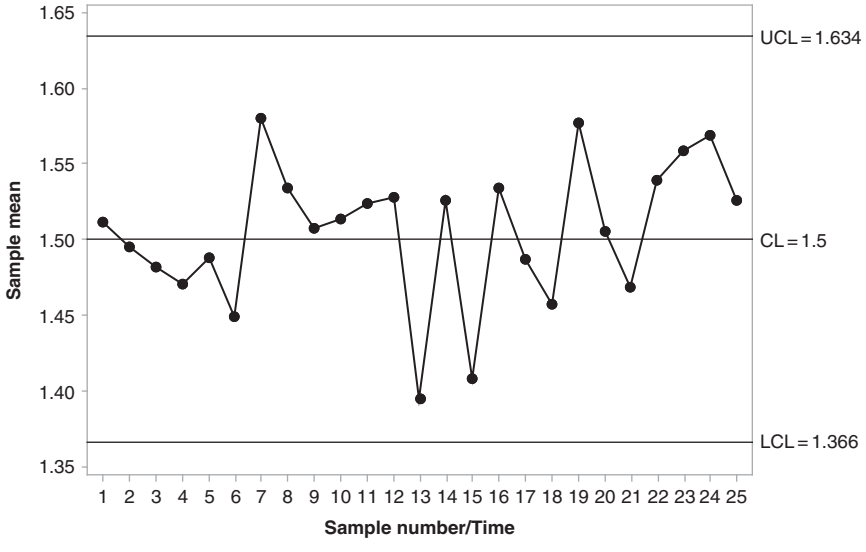


FIGURE 3.3 *A Shewhart $\overline{X}$ control chart for monitoring process location.*

on the chart together with the control limits. This display, which was created with the statistical software package Minitab®, version 17, is typically referred to as a Shewhart $\overline{X}$ control chart or an "X-bar" chart.

Once the control chart is constructed, a decision is made about the status of the process. In Figure 3.3, none of the sample means plots outside the control limits and the points do not seem to exhibit any non-random pattern. Hence, we conclude that the process is functioning in a state of statistical control (IC) with respect to the target mean.

3.3.1.2 Shewhart Control Charts for Monitoring Process Variation

Variation is an important aspect of any analysis and thus it is necessary to monitor the process variation or spread and ensure that it is IC. Moreover, as we see in Equation 3.1, the Shewhart control limits for the process mean depend on the process standard deviation. Thus, unless the standard deviation remains IC, the control chart for the mean will not be very informative. So, we need to monitor the variance or the standard deviation using a control chart.

There are several possible statistics that can be used to monitor variation. The most popular choices are the sample range (R), the sample standard deviation (S), and the sample variance (S^2).

Typically, we use a control chart to monitor the process mean together with a control chart to monitor the process variation. If the variation is IC, we go ahead and examine the control chart for the mean. For example,

a Shewhart $\overline{X}$ chart for the mean is often used together with a Shewhart R chart for the spread. Note that, for illustration, we consider the Shewhart R chart even though recent literature recommends using a different spread chart, such as the Shewhart S chart; see, for instance, Mahmoud *et al.* (2010). We do this because the Shewhart R chart is simple and continues to be used in the industry.

In Case K, the values of μ and σ are known or are specified so that they can be used to construct the respective control charts. We illustrate the Shewhart R and S charts for the known standard deviation σ_0.

Example 3.2 A Parametric Shewhart R Control Chart

The same data used to illustrate the Shewhart $\overline{X}$ chart are used for the illustration of the Shewhart R chart. Column (a) of Table 3.5 presents the measurements taken from 25 independent samples on wafers each of size ($n = 5$) from a normal distribution. The range of each sample is shown in Column (b) of Table 3.5. The standard deviation of each sample is also given in Column (c) of Table 3.5. However, this will only be used in Example 3.3 where the Shewhart S chart is illustrated. For the Shewhart R chart, in order to illustrate the calculations, consider sample number 1. The first charting statistic, the range of sample 1, is calculated as follows:

$$R_1 = \max_1 - \min_1 = 1.6914 - 1.3235 = 0.3679.$$

For the Shewhart R control chart, the CL and the control limits are given by

$$\begin{aligned} UCL &= D_2\sigma_0 \\ CL &= d_2\sigma_0 \\ LCL &= D_1\sigma_0 \end{aligned} \tag{3.14}$$

where D_1, d_2, and D_2 are constants that depend on the sample size n (see Table C in Appendix A for these constants). For $n = 5$, we get $D_1 = 0$, $d_2 = 2.326$, and $D_2 = 4.918$. Supposing the specified value of $\sigma_0 = 0.30$ into Equation 3.14, the CL and the control limits for the data in Table 3.5 are given by

$$\begin{aligned} UCL &= D_2\sigma_0 = (4.918)(0.3) = 1.4754 \\ CL &= d_2\sigma_0 = (2.326)(0.3) = 0.6978 \\ LCL &= D_1\sigma = (0)(0.3) = 0. \end{aligned}$$

Figure 3.4 displays the sample ranges, R_i, of Table 3.5, Column (b), which are the charting statistics for the Shewhart R control chart, plotted on the chart together with the control limits.

In Figure 3.4, all the points plot between the control limits but since the points all fall at or below the CL, it raises some strong suspicion that process variation is not IC.

Table 3.5 Measurements used to construct the Shewhart R and S control charts and the charting statistics.

Sample	(a)					(b)	(c)
i	X_1	X_2	X_3	X_4	X_5	R_i	S_i
1	1.3235	1.4128	1.6744	1.4573	1.6914	0.3679	0.1635
2	1.4314	1.3592	1.6075	1.4666	1.6109	0.2517	0.1111
3	1.4284	1.4871	1.4932	1.4324	1.5674	0.1390	0.0565
4	1.5028	1.6352	1.3841	1.2831	1.5507	0.3521	0.1389
5	1.5604	1.2735	1.5265	1.4363	1.6441	0.3706	0.1412
6	1.5955	1.5451	1.3574	1.3281	1.4198	0.2674	0.1168
7	1.6274	1.5064	1.8366	1.4177	1.5144	0.4189	0.1614
8	1.4190	1.4303	1.6637	1.6067	1.5519	0.2447	0.1077
9	1.3884	1.7277	1.5355	1.5176	1.3688	0.3589	0.1439
10	1.4039	1.6697	1.5089	1.4627	1.5220	0.2658	0.0988
11	1.4158	1.7667	1.4278	1.5928	1.4181	0.3509	0.1548
12	1.5821	1.3355	1.5777	1.3908	1.7559	0.4204	0.1682
13	1.2856	1.4106	1.4447	1.6398	1.1928	0.4470	0.1699
14	1.4951	1.4036	1.5893	1.6458	1.4969	0.2422	0.0937
15	1.3589	1.2863	1.5996	1.2497	1.5471	0.3499	0.1568
16	1.5747	1.5301	1.5171	1.1839	1.8662	0.6823	0.2423
17	1.3680	1.7269	1.3957	1.5014	1.4449	0.3589	0.1432
18	1.4163	1.3864	1.3057	1.6210	1.5573	0.3153	0.1289
19	1.5796	1.4185	1.6541	1.5116	1.7247	0.3062	0.1195
20	1.7106	1.4412	1.2361	1.3820	1.7601	0.5240	0.2230
21	1.4371	1.5051	1.3485	1.5670	1.4880	0.2185	0.0819
22	1.4738	1.5936	1.6583	1.4973	1.4720	0.1863	0.0832
23	1.5917	1.4333	1.5551	1.5295	1.6866	0.2533	0.0922
24	1.6399	1.5243	1.5705	1.5563	1.5530	0.1156	0.0431
25	1.5797	1.3663	1.6240	1.3732	1.6887	0.3224	0.1482

Example 3.3 A Parametric Shewhart S Control Chart

The same data are used to illustrate that the Shewhart $\overline{X}$ chart is used for the illustration of the Shewhart S chart. Recall that Column (a) of Table 3.5 presents the measurements taken from 25 independent samples on wafers of each of size ($n = 5$) from a normal distribution. The standard deviation of each sample is shown in Column (c) of Table 3.5. To illustrate the calculations, consider sample number 1. The first charting statistic, the standard deviation of sample 1, is equal to 0.1635. For the Shewhart S control chart, the CL and the control limits are given by Montgomery (2009, p. 253).

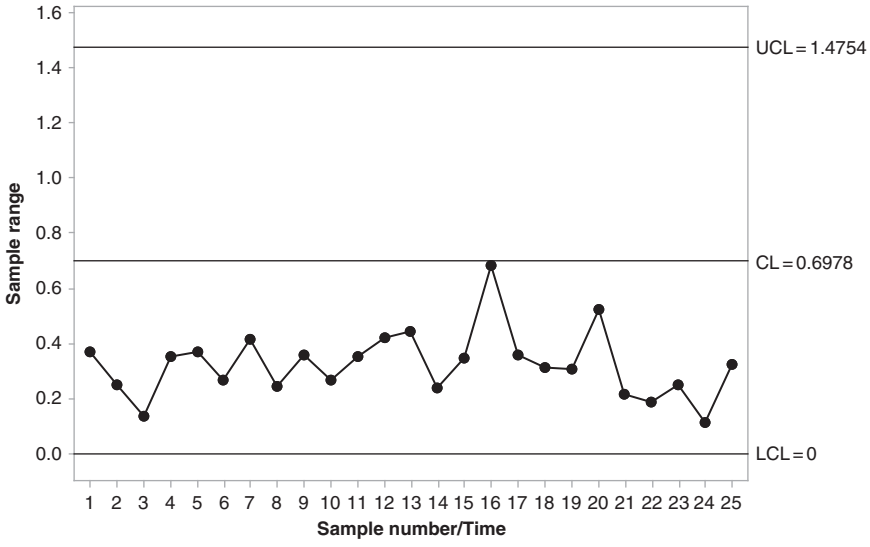


FIGURE 3.4 *A Shewhart* R *control chart for monitoring the process variation.*

$$\begin{aligned} UCL &= B_6\sigma_0 \\ CL &= c_4\sigma_0 \\ LCL &= B_5\sigma_0 \end{aligned} \tag{3.15}$$

where $B_5 = c_4 - 3\sqrt{1-c_4^2}$ and $B_6 = c_4 + 3\sqrt{1-c_4^2}$ and c_4 is a constant that depends on the sample size n (see Table C of Appendix A for these constants).

For $n = 5$, we get $B_6 = 1.964$, $c_4 = 0.94$, and $B_5 = 0$ from Table C (Appendix A), so, again, supposing the specified value of $\sigma_0 = 0.30$ and using Equation 3.15, the CL and the control limits for the S chart for the data in Table 3.5 are given by

$$\begin{aligned} UCL &= (1.964)(0.3) = 0.5892 \\ CL &= (0.94)(0.3) = 0.282 \\ LCL &= (0)(0.3) = 0. \end{aligned}$$

Figure 3.5 displays the sample standard deviations, S_i, of Column (c) from Table 3.5, which are the charting statistics for the Shewhart S control chart that are plotted on the chart together with the control limits.

By examining Figure 3.5, we reach the same conclusion as with the R chart, that is the process variation is not IC, even though the points all plot between the control limits and there are no anomalous patterns in the data.

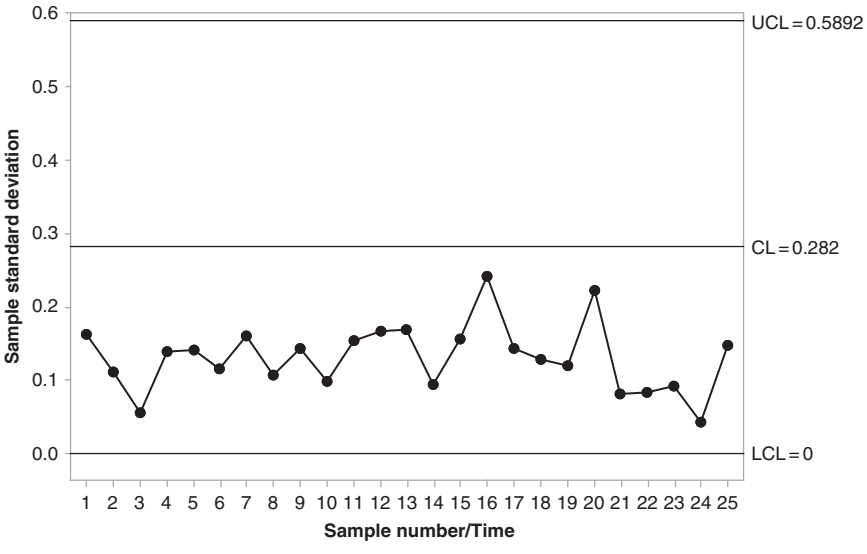


FIGURE 3.5 *A Shewhart* S *control chart for monitoring the process variation.*

It is often the case in practice that the Shewhart control limits are expressed in terms of multiples of the standard deviation (or standard error) of the charting statistic. The multiple 3 is, of course, the most ubiquitous one. The Shewhart R and S charts presented above use the multiple 3 and are based on the fact that the distribution of the charting statistic (R or S) can be approximated by the normal distribution via the central limit theorem. While this may be a reasonable approach in some situations, and has been used for many years, for typically recommended sample sizes, such as 5, the lower control limit turns out to be negative and is therefore set to be equal to 0 since the standard deviation σ can't be negative. This practical "fix" creates a slight problem, though, since the control chart with a lower control limit of zero is not able to detect any decreases in the process variation. Both the R and the S 3-sigma charts presented here share this deficiency for sample sizes less than or equal to 6. One solution to this problem is to use probability limits where the charting constants are found for a given FAR. Thus, for a nominal FAR of, say, 0.002, we need to find the 0.001 and the 0.999 percentage points of the IC distribution of R/σ (or S/σ) and construct the control limits from there. Assuming a normal distribution, such percentiles have been calculated and given in Grant and Leavenworth (1980) for some nominal FAR values. Montgomery (2009, p. 242) states that it is most common to use k-sigma control limits in the US, whereas probability limits are more common in the UK and parts of Western Europe. In our upcoming section on the dispersion control charts for the parameter(s) unknown case, we use the probability limits.

An important issue with regard to monitoring the process mean is the fact that, since the standard deviation appears in the control limits for the mean, we examine the standard deviation first, and when that is found to be IC, we examine the chart for the mean. Thus, one can make the argument that in fact both the mean and the standard deviation are jointly (or simultaneously) monitored while monitoring the mean. Hence, as can be seen from examples above, two charts ($(\overline{X}, R)$ or $(\overline{X}, S)$) are used at the same time, basically as one charting scheme, to make a decision about the status (IC or OOC) of the process. The process is considered IC whenever both charts plot within their respective control limits and display random patterns. Conversely, the process is declared OOC when at least one of the charts shows an OOC situation. If the standard deviation is OOC, there is no point considering the mean chart; if the standard deviation is IC, the mean chart is examined to see if the process mean is IC or not. According to the current practice, the Shewhart $\overline{X}$ chart for the mean and the Shewhart R chart for the standard deviation are constructed independently, each using 3-sigma limits. This is a common practice and is recommended in most textbooks. However, this may be problematic, since the *FAR* of the scheme would be considerably higher (almost doubled) than that of each of the two charts, which is problematic. The issue here is the effect of unaccounted multiple testing. As two charts are used simultaneously in the decision-making process, unless proper adjustments are made to the control limits, the charting scheme does not deliver the specified nominal ARL_{IC}. In fact, some recent work has shown that the attained values can be far lower than the nominal value, which means that there will be many more false alarms than what is nominally expected. This can significantly reduce the effectiveness of the control charts. The reader is referred to Diko, Chakraborti, and Graham (2016) for more details and some corrected (adjusted) charting constants.

3.3.2 CUSUM Control Charts

An example for the parametric CUSUM control chart for individual data follows.

Example 3.4 A Parametric CUSUM Control Chart

Column (a) of Table 3.6 presents 30 measurements generated from a process that is normally distributed. The first 20 of these observations were drawn at random from a normal distribution with mean $\mu_0 = 10$ and standard deviation $\sigma_0 = 1$. The last 10 observations were drawn from a normal distribution with mean $\mu_1 = 11$ and standard deviation $\sigma_0 = 1$. Thus, $\mu_1 = \mu_0 + \delta\sigma_0 = 10 + (1)(1)$ so that $\delta = 1$. To illustrate the CUSUM chart, suppose that we are interested in detecting an increase (a shift) in the mean

Table 3.6 Data and the charting statistics used to construct the parametric CUSUM control chart.

Observation number, i	(a) x_i	(b) C_i^+	(c) N^+	(d) C_i^-	(e) N^-
		0		0	
1	9.45	0	0	−0.05	1
2	7.99	0	0	−1.56	2
3	9.29	0	0	−1.77	3
4	11.66	1.16	1	0	0
5	12.16	2.82	2	0	0
6	10.18	2.50	3	0	0
7	8.04	0.04	4	−1.46	1
8	11.46	1.00	5	0	0
9	9.20	0	0	−0.30	1
10	10.34	0	0	0	0
11	9.03	0	0	−0.47	1
12	11.47	0.97	1	0	0
13	10.51	0.98	2	0	0
14	9.40	0	0	−0.10	1
15	10.08	0	0	0	0
16	9.37	0	0	−0.13	1
17	10.62	0.12	1	0	0
18	10.31	0	0	0	0
19	8.52	0	0	−0.98	1
20	10.84	0.34	1	0	0
21	10.90	0.74	2	0	0
22	9.33	0	0	−0.17	1
23	12.29	1.79	1	0	0
24	11.50	2.79	2	0	0
25	10.60	2.89	3	0	0
26	11.08	3.47	4	0	0
27	10.38	3.35	5	0	0
28	11.62	4.47	6	0	0
29	11.31	5.28	7	0	0
30	10.52	5.30	8	0	0

of size $1\sigma_0$, which is a medium size shift. Note that, here, we can use K and k interchangeably, since $\sigma_0 = 1$. The same argument holds for H and h. Hawkins (1993; Table 1, p. 465) recommends that, for a shift of about $1\sigma_0$ in the process mean, taking $k = 0.5$ and $h = 4.77$ gives an $ARL_{\text{IC}} = 370$. Accordingly, we apply the parametric CUSUM using $k = 0.5$ and $h = 4.77$.

The CUSUM charting statistics, C_i^+ and C_i^-, are given in Columns (b) and (d) of Table 3.6, respectively, whereas the corresponding quantities, N^+ and N^- are given in Columns (c) and (e), respectively. To illustrate the calculations, consider period 1. The charting statistics for the first period are calculated as follows

$$\begin{aligned} C_1^+ &= \max[0,\ X_1 - (\mu_0 + K) + C_0^+] = \max[0,\ 9.45 - (10 + 0.5) + 0] \\ &= \max[0, -1.05] = 0 \end{aligned}$$

and

$$\begin{aligned} C_1^- &= \min[0,\ X_1 - \mu_0 + K + C_0^-] = \max[0, 9.45 - 10 + 0.5 + 0] \\ &= \max[0, -0.05] = 0. \end{aligned}$$

Figure 3.6 displays the CUSUM statistics, C_i^+ and C_i^-, of Columns (a) and (c) respectively, which are plotted on a parametric CUSUM control chart together with the decision interval ($H = 4.77$).

From Figure 3.6, it can be seen that the CUSUM chart signals at sample number/period 29. The CUSUM chart also indicates when the shift probably

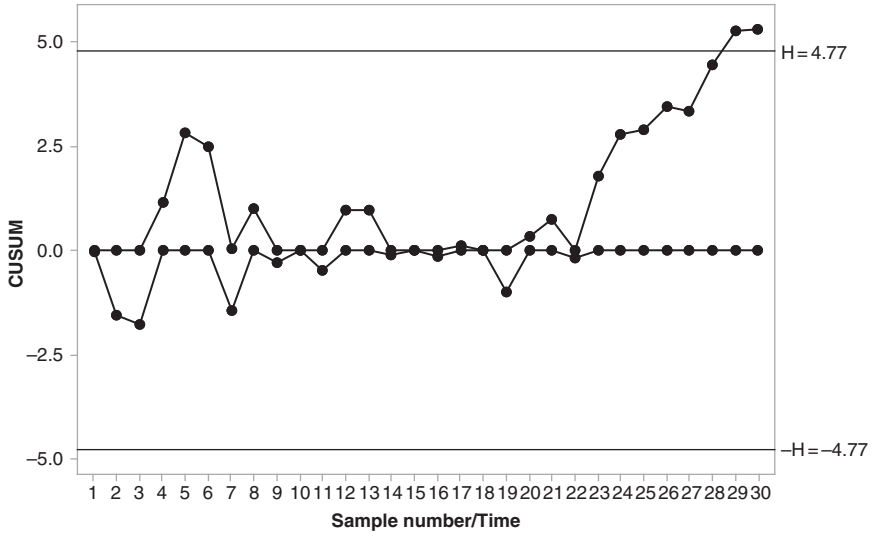


FIGURE 3.6 *A parametric CUSUM control chart.*

occurred. Since $N^+ = 7$ at period 29, we would conclude that the process was last IC at period 29 – 7 = 22, so the shift likely occurred between periods 22 and 23.

In this example, we've illustrated the parametric CUSUM for individual measurements. However, if rational subgroups of size $n > 1$ are taken, then we would simply replace X_i with $\overline{X}_i$ and σ_0 with $\sigma_{\overline{X}} = \dfrac{\sigma_0}{\sqrt{n}}$ in the previous equations. In this case, the optimal k and h pair may differ from the pair used in this example for individual data, for example, for $n = 1$ Hawkins (1993) recommends that, for a shift of about $1\sigma_0$ in the process mean, taking $k = 0.5$ and $h = 4.77$ gives an $ARL_{\text{IC}} = 370$. Using a search algorithm, we find that for $n = 5$, using the same k and h values, that is, $k = 0.5$ and $h = 4.77$, gives an $ARL_{\text{IC}} = 295$. Using a search algorithm, we find that for a sample size of 5 the pair $(k, h) = (0.5, 4.8)$ should be used in order to obtain $ARL_{\text{IC}} = 370$. Note however, that this CUSUM chart uses simulated data from a normal distribution, which may be hard to justify in practice, particularly for individual data.

3.3.3 EWMA Control Charts

An example for the parametric EWMA control chart for individual data follows.

Example 3.5 A Parametric EWMA Control Chart

The same data that were used to illustrate the parametric CUSUM chart are used to illustrate the parametric EWMA chart. Column (a) of Table 3.7 presents 30 measurements taken for a process that is normally distributed. The first 20 of these observations were drawn at random from a normal distribution with mean $\mu_0 = 10$ and standard deviation $\sigma_0 = 1$. As before, the last 10 observations were drawn from a normal distribution with mean $\mu_1 = \mu_0 + \delta\sigma_0 = 10 + (1)(1) = 11$ and standard deviation $\sigma_0 = 1$. Say we are interested in detecting a shift of size $1\sigma_0$, that is, a medium shift. The EWMA charting statistic is shown in Column (b), whereas the time-varying lower and upper control limits are shown in Columns (c) and (d), respectively. We will apply the parametric EWMA control chart with $\lambda = 0.1$, $L = 2.703$ (see Table 3.2; this gives an ARL_{IC} value = 370) and starting value $Z_0 = 10$ to the data in Table 3.7. To illustrate the calculations, consider observation number 1. The first charting statistic is calculated as follows

$$Z_1 = \lambda X_1 + (1 - \lambda)Z_0 = (0.1)(9.45) + (1 - 0.1)(10) = 9.945.$$

Substituting $\mu_0 = 10$, $\sigma_0 = 1$, $\lambda = 0.1$, $L = 2.703$, and $i = 1$ into Equation 3.11, the CL and the exact control limits for the data (for $i = 1$)

Table 3.7 Constructing the parametric EWMA control chart: the data, the charting statistics and the control limits.

Observation number i	(a) X_i	(b) Z_i	(c) LCL_i	(d) UCL_i
		10.00		
1	9.45	9.945	9.730	10.270
2	7.99	9.750	9.637	10.363
3	9.29	9.704	9.576	10.424
4	11.66	9.899	9.533	10.467
5	12.16	10.125	9.500	10.500
6	10.18	10.131	9.475	10.525
7	8.04	9.922	9.456	10.544
8	11.46	10.076	9.441	10.559
9	9.20	9.988	9.429	10.571
10	10.34	10.023	9.419	10.581
11	9.03	9.924	9.412	10.588
12	11.47	10.078	9.406	10.594
13	10.51	10.122	9.401	10.599
14	9.40	10.049	9.397	10.603
15	10.08	10.053	9.394	10.606
16	9.37	9.984	9.391	10.609
17	10.62	10.048	9.389	10.611
18	10.31	10.074	9.388	10.612
19	8.52	9.919	9.386	10.614
20	10.84	10.011	9.385	10.615
21	10.90	10.100	9.384	10.616
22	9.33	10.023	9.384	10.616
23	12.29	10.249	9.383	10.617
24	11.50	10.375	9.383	10.617
25	10.60	10.397	9.382	10.618
26	11.08	10.465	9.382	10.618
27	10.38	10.457	9.382	10.618
28	11.62	10.573	9.381	10.619
29	11.31	10.647	9.381	10.619
30	10.52	10.634	9.381	10.619

in Table 3.7 are given by

$$UCL = 10 + 2.703(1)\sqrt{\left(\frac{0.1}{2-0.1}\right)(1-(1-0.1)^{2(1)})} = 10.270$$

$$CL = 10$$

$$LCL = 10 - 2.703(1)\sqrt{\left(\frac{0.1}{2-0.1}\right)(1-(1-0.1)^{2(1)})} = 9.730.$$

The steady-state control limits can be calculated using Equation 3.12 and are given by $LCL = 9.380$ and $UCL = 10.620$, respectively.

Figure 3.7 displays the EWMA statistics, Z_i given in Column (b) of Table 3.7, which are plotted on a parametric EWMA control chart together with the time-varying control limits. It can be seen that the EWMA chart signals at period 29, which is similar to the CUSUM chart considered in Example 2.4.

In this example, we've illustrated the parametric EWMA chart for individual measurements. However, if rational subgroups of size $n > 1$ are taken, then simply replace X_i with $\overline{X}_i$ and σ_0 with $\sigma_{\overline{X}} = \dfrac{\sigma_0}{\sqrt{n}}$ in the previous equations.

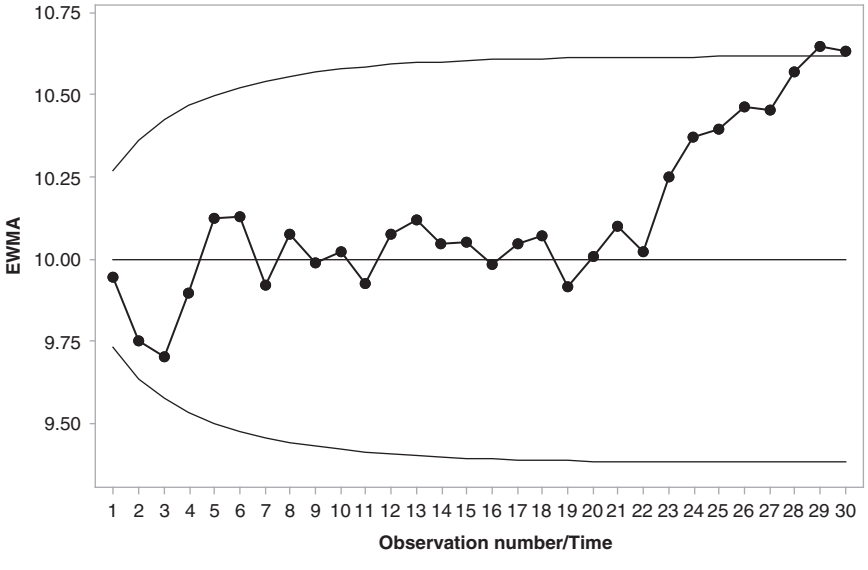


FIGURE 3.7 *A parametric EWMA control chart.*

3.4 SHEWHART, EWMA, AND CUSUM CHARTS: WHICH TO USE WHEN

As we have noted, the Shewhart charts are the most popular in practice because of their simplicity, ease of application, and the fact that these versatile charts are quite efficient in detecting moderate-to-large shifts. While the Shewhart charts are the most widely known and used control charts in practice because of their simplicity and global performance, other classes of charts, such as the CUSUM and EWMA charts, are useful and sometimes more naturally appropriate in the process control environment in view of the sequential nature of data collection. These charts, typically based on the cumulative totals of a charting statistic, obtained as data accumulate, are known to be more efficient for detecting small to moderate magnitudes of shifts in the process. Thus, the CUSUM and the EWMA control charts have been developed as alternatives to the Shewhart chart to detect small, persistent shifts in mean. A natural question to ask is, "Between the CUSUM and the EWMA chart, which control chart is the most effective and should be used when?" Most of the available literature suggests that the performance of the CUSUM and the EWMA charts are very similar. However, each has its advantages and disadvantages.

The EWMA chart has what is known as the **inertia** problem which may be a practical issue. Before the concept of inertia is explained, we first clarify that both the EWMA and the CUSUM charts have the problem of inertia. The CUSUM chart doesn't have such a significant inertia problem since it uses resets (see Yashchin, 1987, 1993). A more detailed explanation of this follows after the term "inertia" has been defined. The term "inertia" refers to a measure of the resistance of a chart to signaling a particular process shift, for example, if the EWMA charting statistic happens to be closer to the LCL at the time when an upward shift occurs, the time required to reach the UCL will be longer than if the EWMA statistic was closer to the CL. It has been shown that EWMA charts have more of an inertia problem than the CUSUM charts (see, for example, Woodall and Mahmoud, 2005). This is particularly the case when we are interested in both upward and downward shifts (i.e., two-sided control charts) as the EWMA is implemented by means of a single charting statistic, as opposed to a CUSUM procedure, which uses two separate (upper and lower) charting statistics. Therefore, although the EWMA chart is easier to implement in practice, its "worst-case" OOC performance is worse than that of a CUSUM chart. Although there have been some recommendations in the literature on how to overcome the problem of inertia, such as using a one-sided EWMA procedure with resetting (see Spliid, 2010), using EWMA charts in conjunction with Shewhart limits (see Woodall and Mahmoud, 2005), or using an adaptive EWMA (AEWMA) approach (see Capizzi and Masarotto, 2003), these

refinements are not typically used in practice. Woodall and Mahmoud (2005) proposed a measure of inertia, referred to as the **signal resistance**, to be the largest standard deviation from target not leading to an immediate OOC signal. It is highly recommended that the inertia properties of control charts be considered important in control chart selection and that the signal resistance be calculated for that reason. By calculating the signal resistance for EWMA and CUSUM charts, Woodall and Mahmoud (2005) concluded that the EWMA chart has worse inertial properties than the CUSUM chart, in the sense that the signal resistance values can be considerably higher. Another advantage of the CUSUM chart is that the time of the shift can be pinpointed by making use of the quantities N^+ and N^-, which indicate the number of consecutive periods that the CUSUM's C_i^+ and C_i^- have been nonzero. For example, if a signaling event occurred on sample number 29 with the corresponding quantity being $N^+ = 7$ at sample number 29, we would conclude that the process was last IC at sample number $29 - 7 = 22$, so the shift likely occurred between sample numbers 22 and 23. Another advantage to using the CUSUM chart is that one can obtain an estimate of the new process mean following the shift, for example, continuing with the previous example, if the IC process mean equals 10 ($\mu_0 = 10$), the reference value is taken to be 0.5 ($K = 0.5$) and the value of $C_{29}^+ = 5.28$, then the new process mean is estimated using $\hat{\mu} = \mu_0 + K + \dfrac{C_{29}^+}{N^+} = 10 + 0.5 + \dfrac{5.28}{7} = 11.25$. Thus, the IC process mean of 10 has shifted upward to the value of 11.25 sometime between sample numbers 22 and 23. With the CUSUM chart having all of these advantages, why would one use the EWMA chart? Hawkins and Wu (2014) perhaps stated it best and we quote their findings here, "One thing is clear – if the shift occurs at or near the beginning of the process (the 'initial-state') then the EWMA is a better choice than the CUSUM chart. No matter what size of shift is monitoring process is designed for or that actually happened, the EWMA always responds faster than the CUSUM."

3.5 CONTROL CHART ENHANCEMENTS

3.5.1 Sensitivity Rules

While the Shewhart control chart is effective in detecting large process shifts, it has been shown that it lacks sensitivity in detecting small process shifts. The papers by Koutras, Bersimis, and Maravelakis, (2007) and Park and Seo (2012) present literature reviews on Shewhart charts with supplementary sensitizing rules based on runs and scans to improve the effectiveness of Shewhart control charts for detecting small shifts. Sensitizing rules are signaling rules designed to detect some improbable and/or

non-random pattern of the charting statistics on a control chart. We start by discussing the well-known Western Electric rules and, following this, we discuss other supplementary sensitizing rules based on runs and scans.

The Western Electric rules make use of warning limits, typically set at a distance of 1-sigma or 2-sigma from the *CL* to determine whether the process is IC or OOC. The Western Electric rules (Western Electric Company, 1956) are listed below. The chart will signal if any one of these conditions is met.

1. One or more points on or outside the control limits.

2. Two of three consecutive points plot outside the 2-sigma warning limits but still inside the control limits.

3. Four of five consecutive points above the 1-sigma limits.

4. A run of eight consecutive points on one side of the *CL*.

The Western Electric rules are illustrated in Figures 3.8–3.11. The question may arise about how these rules are set up. For example, why do we consider a run of eight consecutive points on one side of the *CL* in Western Electric Rule number 4 instead of nine consecutive points on one side of the *CL?* All of these rules are designed so that they will approximately have the same probability of a false alarm, which is also close to the nominal value

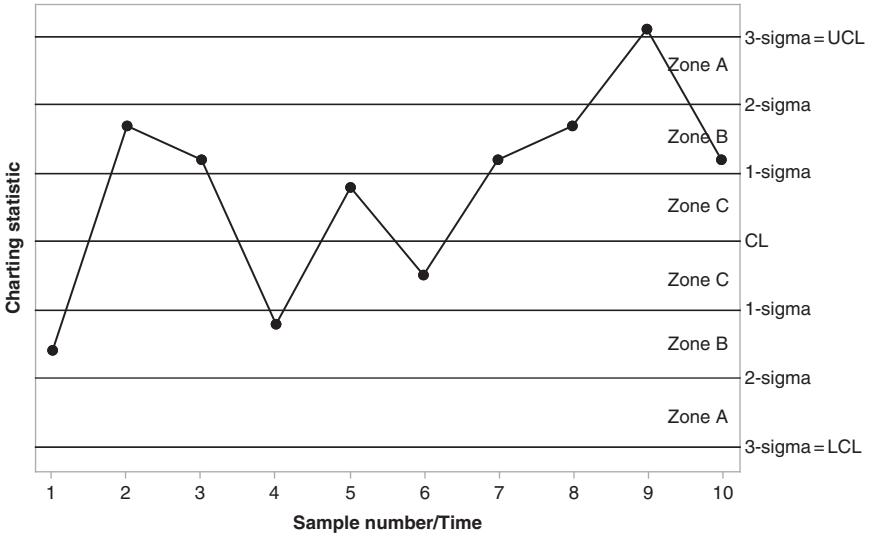


FIGURE 3.8 *Western Electric rule number 1.*

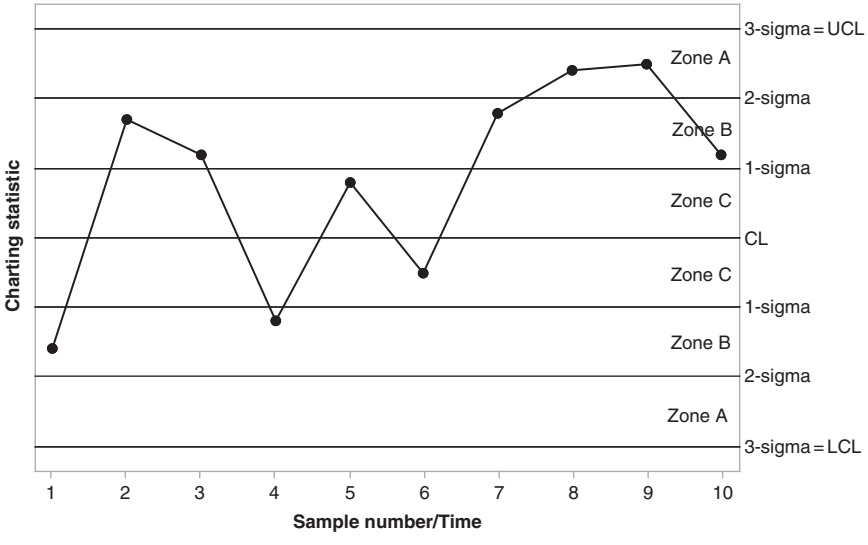


FIGURE 3.9 *Western Electric rule number 2.*

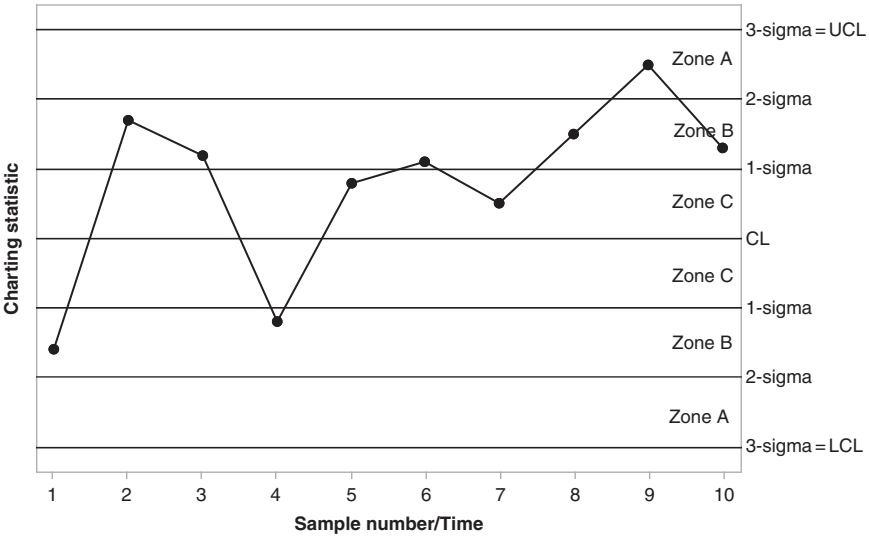


FIGURE 3.10 *Western Electric rule number 3.*

that is typically taken to be 0.0027. For Western Electric Rule number 4, for example, if we make the assumption that the *CL* of a control chart is equal to the IC median of the distribution, then the probability of a charting statistic to plot above and below the *CL* are equal, thus P (Charting

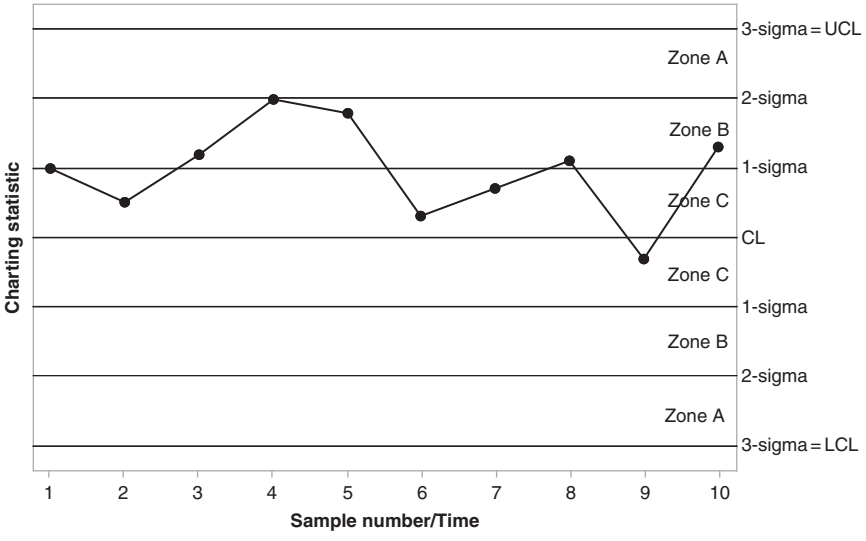


FIGURE 3.11 *Western Electric rule number 4.*

statistic plots above the CL) $= P$ (Charting statistic plots below the CL) $= 0.5$. Thus,

$$
\begin{aligned}
&P(8 \text{ points plot above the } CL) \\
&= P(1\text{st point is above})\, P(2\text{nd point is above}) \cdots P(8\text{th point is above}) \\
&= (0.5)(0.5)\cdots(0.5) \\
&= (0.5)^8 \\
&= 0.0039
\end{aligned}
$$

which is close to the typically desired nominal value of 0.0027.

Western Electric rule number 1 is illustrated in Figure 3.8. This rule gives a signal if one or more points plot on or outside the LCL or UCL. The control chart in Figure 3.8 signals at sample number 9. Although this rule is illustrated for a possible upward shift in the process, it also gives a signal if the charting statistics had a perfect mirror-image copy around the CL, indicating a possible downward shift in the process.

Western Electric rule number 2 is illustrated in Figure 3.9. This rule gives a signal if two of three consecutive points plot outside the 2-sigma warning limits but still inside the control limits. The control chart in Figure 3.9 signals at sample number 9. Although this rule is illustrated for a possible upward shift in the process, it also gives a signal if the charting statistics had a perfect mirror-image copy around the CL, indicating a possible downward shift in the process.

Western Electric rule number 3 is illustrated in Figure 3.10. This rule gives a signal if four of five consecutive points plot outside the 1-sigma warning limits but still inside the control limits. The control chart in Figure 3.10 signals at sample number 10. Although this rule is illustrated for a possible upward shift in the process, it also gives a signal if the charting statistics had a perfect mirror-image copy around the *CL*, indicating a possible downward shift in the process.

Western Electric rule number 4 is illustrated in Figure 3.11. This rule gives a signal if a run of eight consecutive points on one side of the *CL* occurs. The control chart in Figure 3.11 signals at sample number 8. Although this rule is illustrated for a possible upward shift in the process, it also gives a signal if the charting statistics had a perfect mirror-image copy around the *CL*, indicating a possible downward shift in the process.

Some other sensitizing rules have been suggested in the literature and these include:

- Six points in a row steadily increasing or decreasing.
- Fifteen points in a row in Zone C (both above and below the *CL*); see Figures 3.8–3.11 for an illustration of the different zones.
- Fourteen points in a row alternatively up and down.
- Eight points in a row on both sides of the *CL* with none in Zone C; see Figures 3.8–3.11 for an illustration of the different zones.
- An unusual or non-random pattern in the data.
- One or more points near a warning or control limit.

In Table 3.8, we list some possible causes of control chart signals.

3.5.2 Runs-type Signaling Rules

Runs-type signaling rules are used to improve the Shewhart chart's sensitivity for detecting smaller process shifts, along with preserving the simplicity of the Shewhart charts. These are also called supplementary rules as they are used to supplement an original chart to make it more sensitive to detecting process changes. The original Shewhart chart uses the charting statistic from the current sample to check if there is a signal or not in order to determine the state of the process; thus, this chart is also known as the *1-of-1* chart (based on the *1-of-1* rule). While this is reasonable, it might be worthwhile to consider the charting statistics from a few of the previous samples to see if there is a pattern in the signals. For example, if we have signals from two back to back samples (a run of two), a stronger evidence of

Table 3.8 Possible causes of control chart signals.

Signal	Possible Cause
One point outside the control limits followed by no anomalous patterns	Measurement of recording error Isolated/temporary event such as substitute staff
Fifteen points in a row in Zone C (both above and below the *CL*); see Figures 3.8–3.11 for an illustration of the different zones	Data have been manipulated, i.e., extreme values are not being recorded or are being recorded incorrectly
Trends, e.g., six points in a row steadily increasing or decreasing	The gradual drifting of data can indicate wear and tear on measuring equipment or on machinery. It may also result from machine warm-up and cool-down or inadequate maintenance
Cyclical pattern; see Figure 2.3	This may result from environmental changes such as changes in temperature It can also be a result of operator fatigue, or fluctuation in voltage or pressure or some other variable in the machinery Chemical properties of raw material can also play a role
Stratification; see Figure 2.4	Stratification may result from the incorrect calculation of control limits or from incorrect subgrouping It may also be a result of not recalculating the control limits after process improvement
A sudden jump	This could be a result of damaged equipment, new staff, etc.

change would emerge. Charts using such signaling rules are called runs-rules enhanced charts. Two of the popular signaling rules are the *2-of-2* and *2-of-3* runs-type signaling rules and these runs-rules enhanced charts are labeled the *2-of-2* and *2-of-3* charts.

Note that the *k-of-k* ($k \geq 2$) runs-type signaling rule is the general case of the *2-of-2* runs-type signaling rule. The *k-of-k* runs-type signaling rule signals where k consecutive charting statistics all plot on or outside the control limit(s). A generalization of the *k-of-k* runs-type signaling rule is the *k-of-w* ($1 \leq k \leq w$) runs-type signaling rule, which signals where k of the last w charting statistics plots on or outside the control limit(s).

While the runs-rules enhanced charts can make the Shewhart chart more sensitive to a change, there may be some potential costs. The *FAR* may be increased and the resulting chart might lack the ability to immediately detect a large process shift. The *2-of-3* signaling rules, for example, require at least the last two or three charting statistics to signal, which may be potentially costly to the practitioner since the chart's ability to detect a large shift in the process is delayed until at least three samples are collected. Thus, the cost and the benefit need to be carefully weighed before one decides to use a supplementary runs-rules enhanced chart.

Many parametric control charts with runs-type signaling rules have been proposed in the literature. The interested reader can take a look at, for example, Roberts (1958), Champ and Woodall (1987), Champ (1992), Klein (2000), Shmueli and Cohen (2003), Acosta-Mejia (2007), Khoo and Ariffin (2006), Lim and Cho (2009), Cheng and Chen (2011), Santiago and Smith (2013) and others. However, the focus of our discussion is on nonparametric control charts and, accordingly, we focus on nonparametric control charts to monitor location and/or scale with runs-type signaling rules. Overall, to learn about the advantages and disadvantages of Shewhart-type control charts supplemented by runs-type signaling rules, refer to Nelson (1985). Koutras *et al.* (2007) show that the overall probability of a false alarm when h decision rules are used is

$$\alpha = \prod_{i=1}^{h}(1 - \alpha_i), \tag{3.16}$$

provided that all h decision rules are independent, with rule i having probability α_i such that the charting statistic plots on or outside the control limits when the process is IC. Note that Equation 3.16 is not valid in the case of the typical runs-rules; see Nelson (1985) and Montgomery (2009).

3.5.2.1 Signaling Indicators

Let the charting statistic for the ith subgroup be denoted by Q_i for $i = 1, 2, 3, \ldots$ Let the indicator functions for the one-sided control charts be denoted by

$$\xi_i^+ = I(Q_i \geq UCL) = \begin{cases} 1 \text{ if } Q_i \geq UCL \\ 0 \text{ if } Q_i < UCL \end{cases} \tag{3.17}$$

and

$$\xi_i^- = I(Q_i \leq LCL) = \begin{cases} 1 \text{ if } Q_i \leq LCL \\ 0 \text{ if } Q_i > LCL \end{cases} \tag{3.18}$$

respectively. Thus, ξ_i^+ equals 1 and indicates a signal for an upper one-sided chart when Q_i plots on or above the *UCL*. Similarly, ξ_i^- takes a value 1 and indicates a signal for a lower one-sided chart when Q_i plots on or below the *LCL*. Thus, these binary variables are called signaling indicators. For a two-sided control chart, the signaling indicator may be denoted by

$$\xi_i = \begin{cases} 1 & \text{if } Q_i \geq UCL \\ 0 & \text{if } LCL < Q_i < UCL \\ 2 & \text{if } Q_i \leq LCL \end{cases} \tag{3.19}$$

so that ξ_i indicates whether the charting statistic of Q_i plots on or above the *UCL* ($\xi_i = 1$), between the two limits ($\xi_i = 0$), or on or below the *LCL* ($\xi_i = 2$). The values 1 and 2 indicate that there is a signal, on the high or the low side, respectively, whereas the value 0 indicates there is not a signal and that the process is IC.

3.5.2.1.1 The 1-of-1 Runs-type Signaling Rule The most commonly used and least complex charts are the usual Shewhart charts, called *1-of-1* charts, which are not supplemented by any runs-rules. However, they are known to be insensitive in detecting small process shifts. These charts signal when the event A_1 or A_2, or $(A_1 \cup A_2)$ occurs, respectively, where

1. The *1-of-1* runs-type signaling rule for an upper one-sided chart: A_1:$\{Q_i \geq UCL\} \leftrightarrow \{\xi_i^+ = 1\}$.

2. The *1-of-1* runs-type signaling rule for a lower one-sided chart: A_2:$\{Q_i \leq LCL\} \leftrightarrow \{\xi_i^- = 1\}$.

3. The *1-of-1* runs-type signaling rule for a two-sided chart: A:$\{Q_i \geq UCL \cup Q_i \leq LCL\} \leftrightarrow \{A_1 \cup A_2\} \leftrightarrow \{\xi_i^+ = 1 \cup \xi_i^- = 1\} \leftrightarrow \{\xi_i = 1 \cup \xi_i = 2\}$.

Figure 3.12 illustrates points (i) and (ii) of the *1-of-1* runs-type signaling rule for an upper one-sided and a lower one-sided chart, respectively. Panel (a) detects an upward shift at time $i = 7$ when Q_7 plots above the *UCL*, whereas Panel (b) detects a downward shift at time $i = 7$ when Q_7 plots below the *LCL*. For each chart the process is declared OOC and a search for assignable causes can be started.

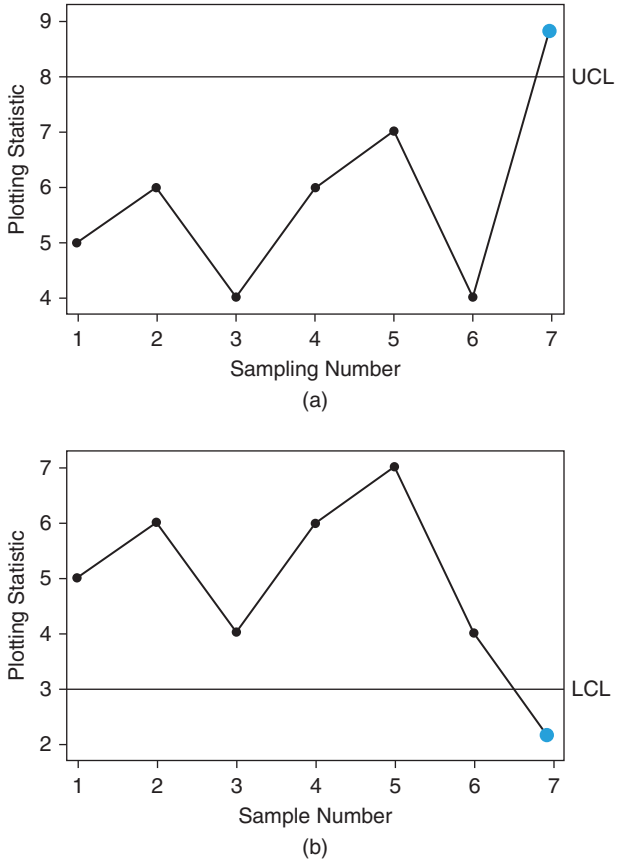


FIGURE 3.12 *The* 1-of-1 *runs-type signaling rule for an upper one-sided and a lower one-sided chart.*

Figure 3.13 illustrates point (3) of the *1-of-1* runs-type signaling rule for a two-sided chart. Both charts signal at time $i = 7$; at first an upward shift is detected when Q_7 plots above the *UCL*, and then a downward shift is detected when Q_7 plots below the *LCL*.

3.5.2.1.2 The k-of-k and k-of-w Runs-type Signaling Rules Several runs-type signaling rules, such as the *2-of-2* and *2-of-3* runs-rules, have been investigated by many authors and proved that their "runs-rules enhanced" charts outperform the *1-of-1* chart.

One-sided k-of-k and k-of-w Runs-type Signaling Rules To illustrate, we start with the *2-of-2* runs-type signaling rule. This rule uses the last two charting statistics, Q_{i-1} and Q_i, to determine whether the process is IC

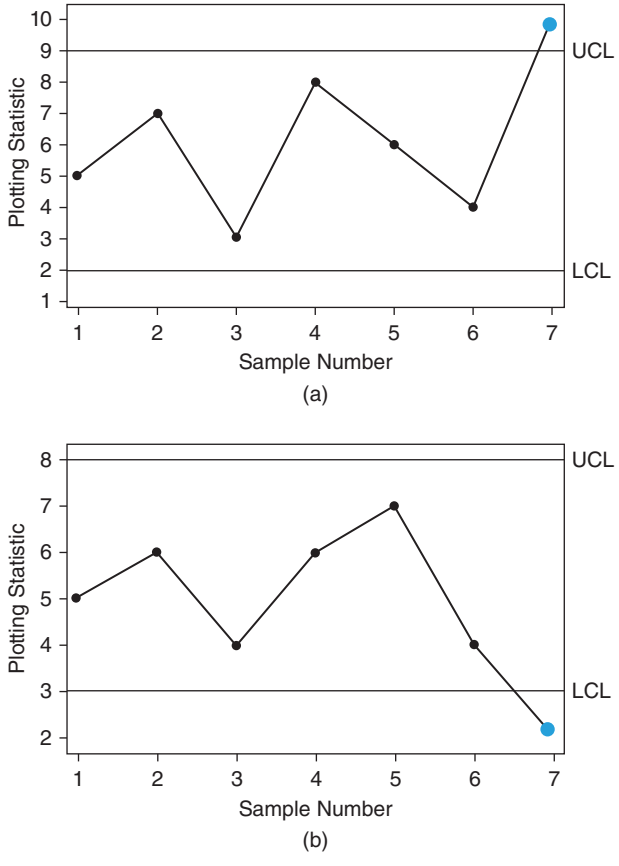


FIGURE 3.13 *The* 1-of-1 *runs-type signaling rule for a two-sided chart.*

or OOC. For the *2-of-2* runs-type signaling rule, an upper one-sided chart signals when event B_1 occurs and a lower one-sided chart signals when event B_2 occurs where

1. an upper one-sided chart supplemented by a *2-of-2* runs-type signaling rule: B_1:$\{\xi_{i-1}^{+} = \xi_i^{+} = 1\}$

2. a lower one-sided chart supplemented by a *2-of-2* runs-type signaling rule: B_2:$\{\xi_{i-1}^{-} = \xi_i^{-} = 1\}$.

Panels (a) and (b) in Figure 3.14 illustrate these signaling events B_1 and B_2, and both signal at time $i = 7$ where two consecutive points plots above

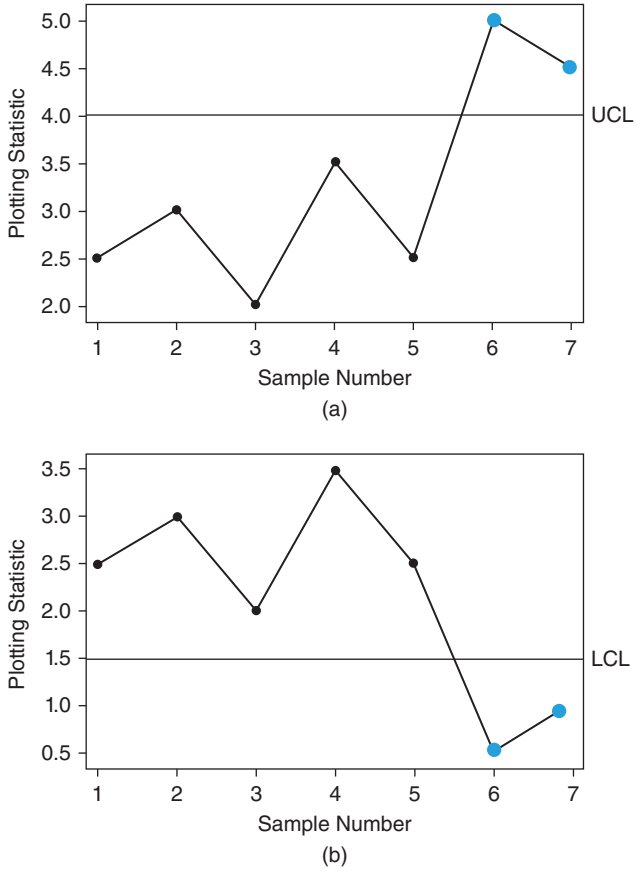


FIGURE 3.14 *The* 2-of-2 *runs-type signaling rule for an upper one-sided and a lower one-sided chart.*

(below) the *UCL* (*LCL*). For each chart, the process is declared OOC and a search for assignable causes can be started.

Next, we illustrate the *2-of-3* runs-type signaling rule for an upper one-sided chart and a lower one-sided chart, respectively. These charts use two of the last three charting statistics Q_{i-2}, Q_{i-1}, and Q_i to determine whether the process is IC or OOC. Although there are three ways that two of the three charting statistics can result in a signal, only on two of these possibilities does the last charting statistic plot on or above (below) the upper (lower) control limits. The one-sided charts supplemented by the *2-of-3* runs-type signaling rule signal when event $C(+) = \{C_1 \cup C_2\}$ or $C(-) = \{C_3 \cup C_4\}$ occurs where:

An upper one-sided chart supplemented by the *2-of-3* runs-type signaling rule
(1) C_1: $\{Q_{i-2} < UCL$ and $Q_{i-1} \geq UCL$ and $Q_i \geq UCL\} \leftrightarrow \{\xi_{i-2}^{+} = 0, \xi_{i-1}^{-} = 1, \xi_i^{+} = 1\}$
(2) C_2: $\{Q_{i-2} \geq UCL$ and $Q_{i-1} < UCL$ and $Q_i \geq UCL\} \leftrightarrow \{\xi_{i-2}^{+} = 1, \xi_{i-1}^{-} = 0, \xi_i^{+} = 1\}$
A lower one-sided chart supplemented by the *2-of-3* runs-type signaling rule
(3) C_3: $\{Q_{i-2} > LCL$ and $Q_{i-1} \leq LCL$ and $Q_i \leq LCL\} \leftrightarrow \{\xi_{i-2}^{+} = 0, \xi_{i-1}^{-} = 1, \xi_i^{+} = 1\}$
(4) C_4: $\{Q_{i-2} \leq LCL$ and $Q_{i-1} > LCL$ and $Q_i \leq LCL\} \leftrightarrow \{\xi_{i-2}^{+} = 1, \xi_{i-1}^{-} = 0, \xi_i^{+} = 1\}$

Figure 3.15 illustrates these signaling events C_1, C_2, C_3, and C_4 of an upper one-sided chart and a lower one-sided chart supplemented by the *2-of-3* runs-type signaling rule, respectively. The upper one-sided charts in Panel (a) signal at time $i = 7$ where two of the last three points plot above the *UCL*. For each chart, the process is declared OOC and a search for assignable causes can be started. Likewise, the lower one-sided charts in Panel (b) signal at time $i = 7$ where two of the last three points plot below the *LCL*. For each chart, the process is declared OOC and a search for assignable causes can be started.

Some of the patterns of the *2-of-3* runs-type signaling rules are excluded. If the first and the second charting statistics plot outside the control limits, but the third (or last) charting statistic plots between the control limits, the process cannot be declared OOC, because the process has shifted back to being IC and, accordingly, these patterns are excluded from the *2-of-3* runs-type signaling rules. Refer to Figure 3.16, where the upper one-sided chart supplemented by the *2-of-3* runs-type signaling rule and the lower one-sided chart supplemented by the *2-of-3* runs-type signaling rule are excluded as signaling events.

Two-sided Charts Supplemented by k-of-k and k-of-w Runs-type Signaling Rules A two-sided chart has both an upper and a lower control limit, thus it can detect either an upward or a downward shift. To illustrate, we start with the *2-of-2* runs-type signaling rule. A two-sided chart supplemented by a *2-of-2* runs-type signaling rule is much the same as that of the one-sided charts in the sense that it uses the last two charting statistics, Q_{i-1} and Q_i, to determine whether the process is IC or OOC, but it is able to detect both an upward or a downward shift. A two-sided chart

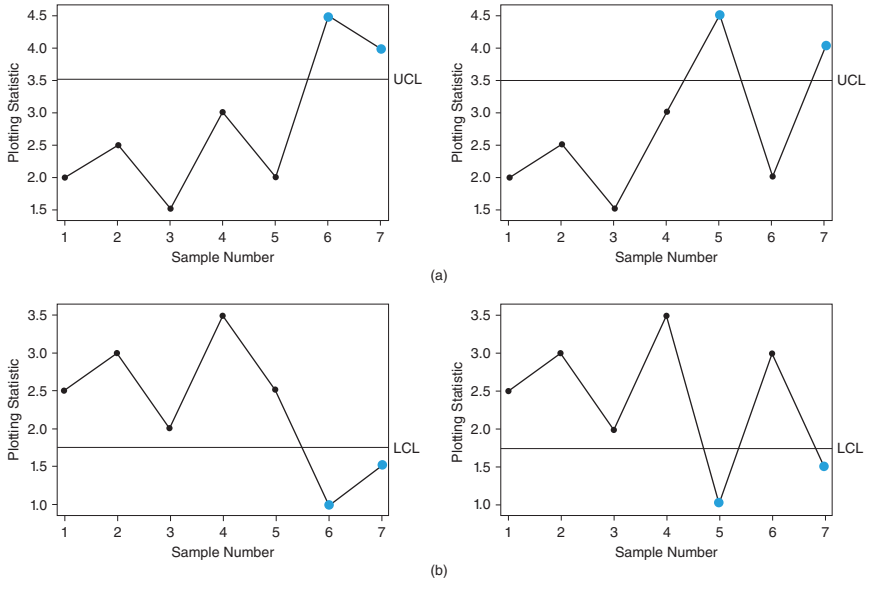


FIGURE 3.15 *The* 2-of-3 *runs-type signaling rule for an upper one-sided chart and a lower one-sided chart.*

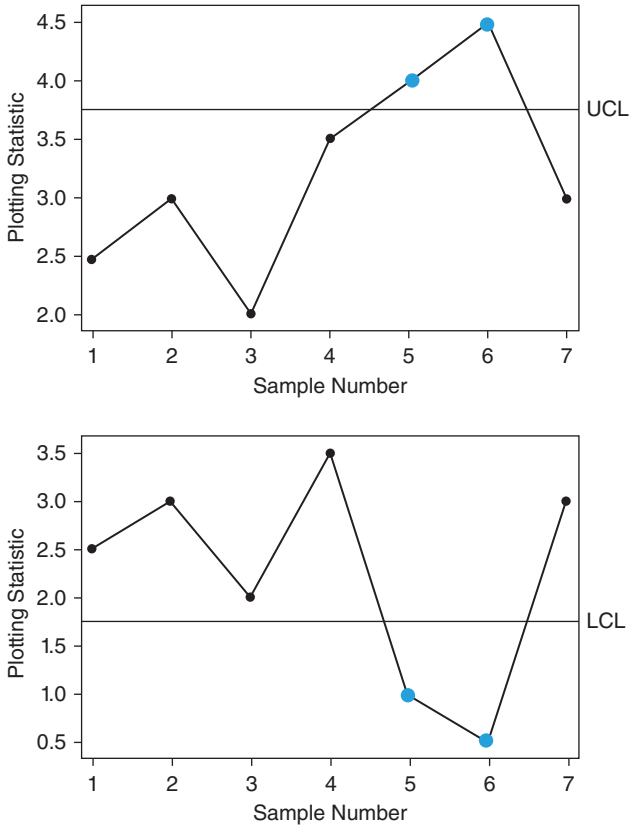


FIGURE 3.16 *The* 2-of-3 *runs-type signaling rules that are excluded for an upper one-sided and a lower one-sided chart.*

supplemented by the *2-of-2* runs-type signaling rule signals when event $D_i,\ i = 1, 2, 3, 4,\ \{D_1,\ D_2, D_3$ or $D_4\}$ occurs where

1. both charting statistics plot on or above the UCL: D_1: $\{Q_{i-1} > UCL$ and $Q_i > UCL\} \leftrightarrow \{\xi_{i-1} = 1, \xi_i = 1\}$

2. both charting statistics plot on or below the LCL: D_3: $\{Q_{i-1} \leq LCL$ and $Q_i \leq LCL\} \leftrightarrow \{\xi_{i-1} = 2, \xi_i = 2\}$

3. the first charting statistic plots on or above the UCL and the second plots on or below the LCL: D_3: $\{Q_{i-1} \geq UCL$ and $Q_i \leq LCL\} \leftrightarrow \{\xi_{i-1} = 1, \xi_i = 2\}$

4. the first charting statistic plots on or below the LCL and the second plots on or above the UCL: D_4: $\{Q_{i-1} \leq LCL$ and $Q_i \geq UCL\} \leftrightarrow \{\xi_{i-1} = 2, \xi_i = 1\}$.

Figure 3.17 illustrates these signaling events D_1, D_2, D_3, and D_4 for the two-sided chart supplemented by *2-of-2* runs-type signaling rules, respectively. Panels (a) and (b) illustrate events D_1 and D_2; both charts signal at time $i = 7$, where two consecutive points plots above (below) the UCL (LCL). Panels (c) and (d) illustrate events D_3 and D_4 when an upward shift is immediately followed by a downward shift, or vice versa.

Next, we illustrate a two-sided chart with *2-of-3* runs-type signaling rules. Similar to one-sided chart, a two-sided chart uses two of the last three charting statistics, Q_{i-2}, Q_{i-1}, and Q_i, to determine whether the process is IC or OOC. There are up to 12 scenarios where exactly two of the three charting statistics can plot outside the control limits; however, we will only focus on four signaling events E_i, $i = 1, 2, 3, 4$, $\{E_1,\ E_2,\ E_3$ or $E_4\}$ that may occur where:

Two of the three charting statistics plot on or above the *UCL*

(1) E_1: $\{LCL < Q_{i-2} < UCL$ and $Q_{i-1} \geq UCL$ and $Q_i \geq UCL\} \leftrightarrow$
$\{\xi_{i-2} = 0, \xi_{i-1} = 1, \xi_i = 1\}$
(2) E_2: $\{Q_{i-2} \geq UCL$ and $LCL < Q_{i-1} < UCL$ and $Q_i \geq UCL\} \leftrightarrow$
$\{\xi_{i-2} = 1, \xi_{i-1} = 0, \xi_i = 1\}$

Two of the three charting statistics plot on or below the *LCL*

(3) E_3: $\{LCL < Q_{i-2} < UCL$ and $Q_{i-1} \leq LCL$ and $Q_i \leq LCL\} \leftrightarrow$
$\{\xi_{i-2} = 0, \xi_{i-1} = 2, \xi_i = 2\}$
(4) E_4: $\{Q_{i-2} \leq LCL$ and $LCL < Q_{i-1} < UCL$ and $Q_i \leq LCL\} \leftrightarrow$
$\{\xi_{i-2} = 2, \xi_{i-1} = 0, \xi_i = 2\}$

Figure 3.18 illustrates these signaling events E_1, E_2, E_3, and E_4 for a two-sided chart supplemented by *2-of-3* runs-type signaling rules. Panel (a) illustrates events E_1 and E_2, where exactly two out of the three charting statistics plot on or above the UCL, whereas Panel (b) illustrates events E_3 and E_4, where exactly two out of the three charting statistics plot on or below the LCL.

Figure 3.19 illustrates eight of the 12 scenarios that are excluded as signaling events. Panel (a) shows a possible swing detected, whereas Panel (b) illustrates a downward or upward trend in the process. As mentioned in Figure 3.16, the four events illustrated in Panel (c) will be excluded as signaling events due to the fact that the last point plots between the control limits; therefore, the process cannot be declared OOC, because the process has shifted back to being IC.

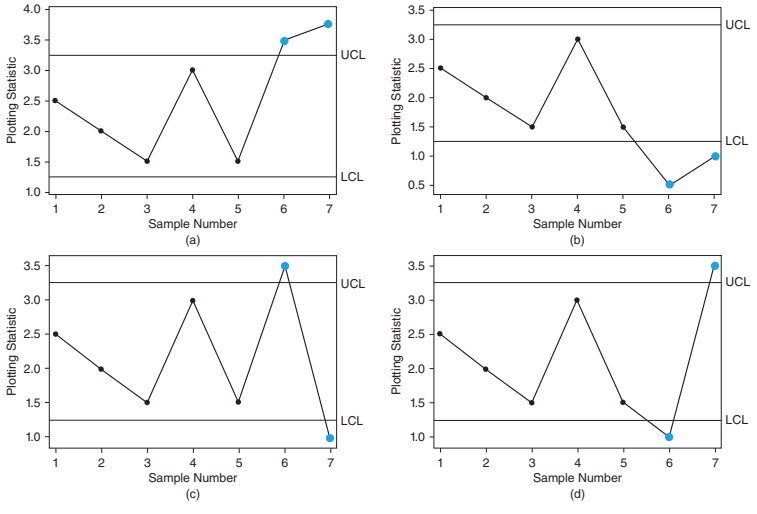


FIGURE 3.17 *The* 2-of-2 *runs-type signaling rule for a two-sided chart.*

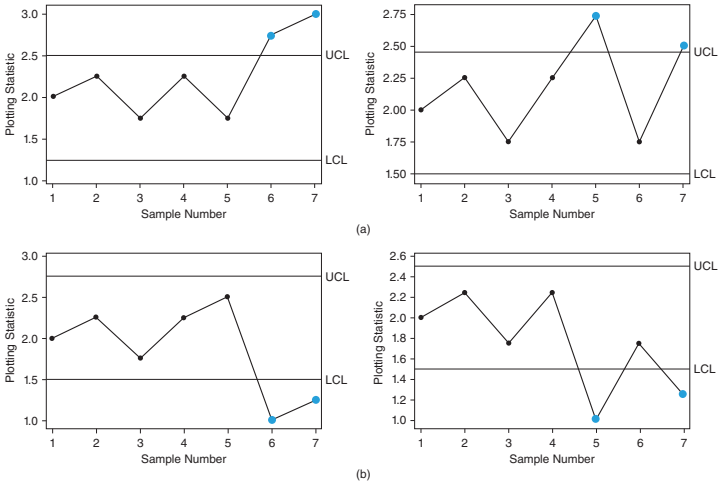


FIGURE 3.18 *The* 2-of-3 *runs-type signaling rule for a two-sided chart.*

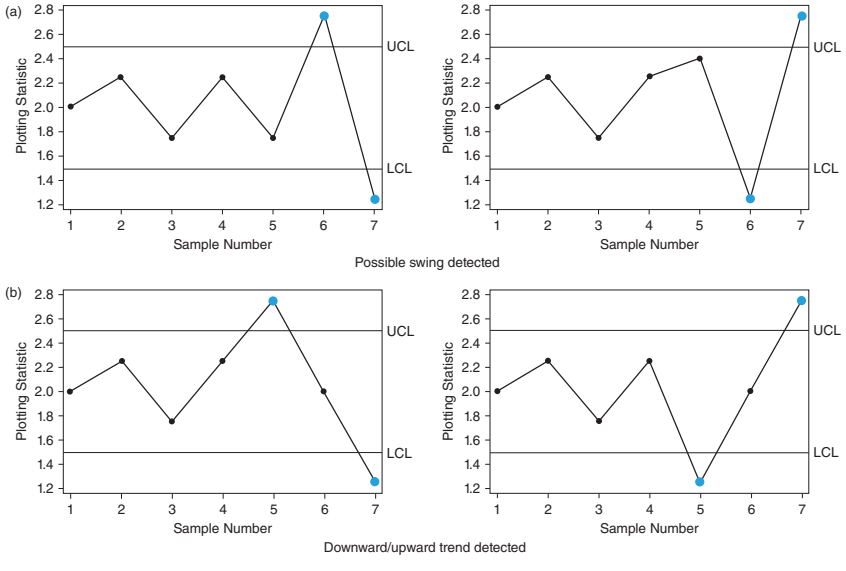


FIGURE 3.19 *The* 2-of-3 *runs-type signaling rules that are excluded for a two-sided chart.*

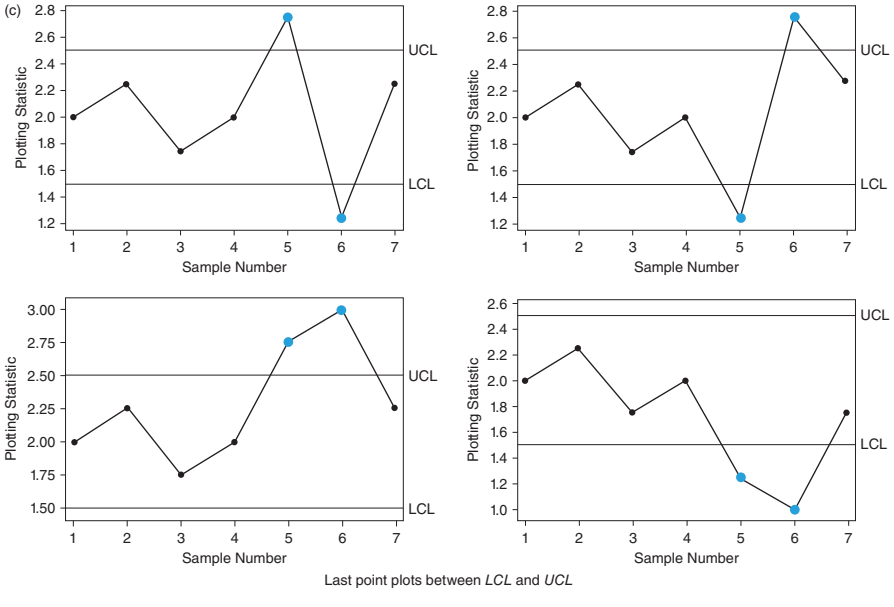


FIGURE 3.19 *(Continued)*

3.6 RUN-LENGTH DISTRIBUTION IN THE SPECIFIED PARAMETER CASE (CASE K)

The performance of a control chart is analyzed via the run-length distribution and associated characteristics such as the various moments and percentiles. We discuss the methods of calculating the run-length distribution of control charts in this section. The discussion is general and applies to all types of control charts, including parametric and nonparametric control charts and the parameter known (Case K) and unknown (Case U) cases. We deal with Case K first.

3.6.1 Methods of Calculating the Run-length Distribution

There are four methods to calculate (or at least approximate) the run-length distribution of a control chart. Specifically, in Case K, these are

1. the exact approach (for Shewhart and some Shewhart-type charts);
2. the Markov chain (MC) approach;
3. the integral equation approach; and
4. the computer simulations (the Monte Carlo) approach.

A discussion on each method follows. We consider first the case of Shewhart charts.

3.6.1.1 The Exact Approach (for Shewhart and some Shewhart-type Charts)

For the Shewhart and some Shewhart-type control charts, it is easy to calculate the characteristics of the run-length distribution exactly in Case K. For the two-sided 3-sigma Shewhart $\overline{X}$ chart, the run-length, which is the number of samples until the first signal, follows a **geometric distribution** with probability $P\left(\frac{\sqrt{n}\,|X-\mu|}{\sigma} > z_\alpha\right)$, which equals the *FAR*, denoted α, when the process is IC. We provide more details on this later. Thus, in Case K, all IC properties of the Shewhart $\overline{X}$ control chart can be characterized in terms of the *FAR*, α, and obtained from the properties of the geometric distribution. Thus, for example, the IC *ARL* equals $ARL_{\text{IC}} = \frac{1}{\alpha}$, the IC *SDRL* equals $SDRL_{\text{IC}} = \frac{1-\alpha}{\alpha^2}$, and so on.

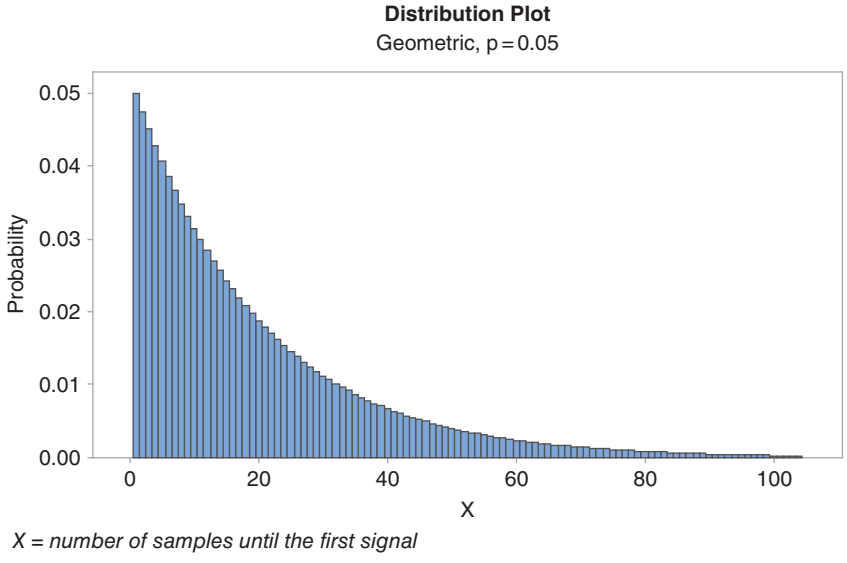


FIGURE 3.20 *Run-length distribution of a Shewhart control chart for the mean in Case K with* $\alpha = 0.05$.

Figure 3.20 has a display of the run-length distribution for $\alpha = 0.05$. The right-skewed nature of the distribution is obvious. For smaller values of α, such as 0.0027, which is fairly typical in the industry, the run-length distribution is even more right-skewed with an even longer right tail.

For some well-known control charts like the CUSUM, the EWMA, and the Shewhart charts with runs-rules, a more common approach to determining the run-length distribution has been the clever application of the idea of a **Markov chain** (MC). The MC theory is more commonly used in the area of stochastic processes and the control charting literature has borrowed some important results from this literature. We describe some of the concepts and the key results below.

3.6.1.2 The Markov Chain Approach

Some theory and concepts in the area of what is called a finite homogeneous MC have been applied to derive the run-length distribution and various characteristics of the run-length distribution of some time-weighted charts, such as the CUSUM and the EWMA. The reader can look up, for example, Fu and Lou (2003) and Balakrishnan and Koutras (2002) for a more detailed treatment of the topics; however, most existing SPC books don't seem to treat this important topic. In this section, some of the basic terminologies, definitions, results, and theorems are provided that give the necessary background for calculating the run-length distribution of some control charts via the MC approach. These theorems and results are critical to the following chapters; illustrations are provided.

The MC approach entails that the charting statistic is viewed as following a MC (as a stochastic process), with a **state space** S and a **transition probability matrix** M. The state space consists of two types of states:

1. one **absorbing** state (i.e., this state is entered when the chart signals, i.e., when the charting statistic is greater than or equal to the UCL, or less than or equal to the LCL for a two-sided chart); and

2. ν **transient** or non-absorbing states, so that there are $\nu + 1$ states in total.

The $(\nu + 1) \times (\nu + 1)$ transition probability matrix, M, is often written in a partitioned form

$$M_{(\nu+1)\times(\nu+1)} = \begin{pmatrix} Q_{\nu\times\nu} & | & \underline{p}_{\nu\times 1} \\ - & - & - \\ \underline{0}'_{1\times\nu} & | & 1_{1\times 1} \end{pmatrix} \tag{3.20}$$

where the sub-matrix $Q_{\nu\times\nu}$ contains all the probabilities of going from one transient state to another and is called the essential transition probability matrix. The column vector $\underline{p}_{\nu\times 1}$ contains all the probabilities of going from each transient state to the absorbing state; $\underline{0}'_{1\times\nu}$ is a row vector of zeros, consisting of the probabilities of going from the absorbing state to each transient state (which are all zero), and the scalar value 1 is the probability of staying in the absorbing state once it has been entered. Note that the key component in using the MC approach is to obtain the essential transition probability matrix $Q_{\nu\times\nu}$.

In the context of MCs, the run-length random variable N of a control chart is viewed as the waiting time for the MC to enter the absorbing state for the first time. Using this analogy, and assuming that the process starts IC when the chart is implemented, the probability mass function (pmf), the expected value (ARL), the standard deviation ($SDRL$), and the cdf of N are given by (see Fu and Lou, 2003, Chapter 5)

$$P(N = t) = \underline{\xi} Q^{t-1}(I - Q)\underline{1} \text{ for } t = 1, 2, 3, \ldots \tag{3.21}$$

$$E(N) = \underline{\xi}(\mathrm{I} - Q)^{-1}\underline{1} \tag{3.22}$$

$$SD(N) = \sqrt{\underline{\xi}(I + Q)(I - Q)^{-2}\underline{1} - (E(N))^2} \tag{3.23}$$

$$P(N \leq t) = 1 - \underline{\xi} Q^t \underline{1} \text{ for } t = 1, 2, 3, \ldots \tag{3.24}$$

respectively, where $I = I_{\nu\times\nu}$ is the identity matrix, $Q = Q_{\nu\times\upsilon}$ is the **essential transition probability matrix**, $\underline{1} = \underline{1}_{\nu\times 1}$ is a column vector with all elements equal to one, and $\underline{\xi} = \underline{\xi}_{1\times\nu}$ is a row vector called the initial

probability vector, which contains the probabilities that the MC starts in a given state. The vector $\underline{\xi} = (\xi_{-s}, \ldots, \xi_s)$ is typically chosen such that $\sum_{i=-s}^{s} \xi_i = 1$.

The zero-state and steady-state modes of analysis are used to characterize the short- and long-term run-length properties of a control chart, respectively. That is, the zero-state run-length is the number of charting statistics plotted before a signal is given when the process begins in some initial state; however, the steady-state run-length is the number of charting statistics plotted before a signal is given when the process begins and stays IC for a very long time. Then, at some random time, an OOC signal is observed; see, for example, Champ (1992) and Machado and Costa (2014a, b). Let $q_{1\times\nu}$ denote the row vector of initial probabilities associated with the zero-state mode, that is

$$\underline{\xi} = \underline{q}_{1\times\nu} = (0\,0\ \ldots\ 1\ \ldots\ 0\,0) \tag{3.25}$$

so that the initial state element on the transition probability matrix corresponds to the value of 1 in Equation 3.25. It should be noted that the 1 is in the middle of the row vector so that the process starts IC. This is done by forcing the initial state to be as close to the *CL* as possible, that is, the selection of the middle value corresponding to the middle discretized subinterval which, in turn, is the closest one can start to the *CL*. In the steady-state, the vector $\xi = q_{1\times\nu}$ is replaced by a vector $s_{1\times\nu}$, that is, the steady-state initial probability vector. This is given by

$$\underline{\xi} = \underline{s}_{1\times\nu} = (\mathrm{s}_1\,\mathrm{s}_2\,\ldots\,\mathrm{s}_M) \tag{3.26}$$

where the elements in Equation 3.26 sum to unity. For example, for $v = 5$, the $\underline{\xi}$ may be equal to $\underline{s}_{1\times5} = (0.15\ \ 0.2\ \ 0.3\ \ 0.2\ \ 0.15)$. This would indicate that one would start closer to the *CL* with the highest probability of 0.3, or one would start further away from the *CL* with smaller probability 0.2, and, finally, one could start close to the control limits with the smallest probability of 0.15. The smallest probability is assigned to starting close to the *UCL* and close to the *LCL*, respectively, because this is not ideal. Another example could be, for $v = 7$ we have $\underline{s}_{1\times7} = (0\ \ 0\ \ 0.1\ \ 0.8\ \ 0.1\ \ 0\ \ 0)$. In this example, we have some elements in ξ that equal zero and other elements that are non-zero. Here, it indicates that one would start closer to the *CL* with the highest probability of 0.8, or one would start further away from the *CL* with smaller probability 0.1, and, finally, one could start close to the control limits with zero probability.

From Equation 3.24, it is clear that in order to apply the MC approach to find the run-length distribution for a given control chart one needs to specify the essential transition probability matrix Q. Once the Q matrix is found, the run-length distribution and associated characteristics can be obtained easily using the formulas given above for a given initial probability vector.

Now, in order to calculate the elements of Q, the state space first needs to be identified. For an attributes control chart with a discrete charting statistic this is easy, since the number of states will be fixed. However, for a variables control chart, which uses a charting statistic with a continuous distribution, the states of the process thus need to be defined carefully. The steps of how this is typically done are provided below.

It should be noted that, with runs-type signaling rules, the number of states is defined by the type of rule used, for example, whether one uses a *2-of-2* or a *2-of-3* runs-rules. There is no need to discretize the interval between the LCL and the UCL into $\nu = 2s + 1$ subintervals (as explained in Step 1 below), since, as stated before, for runs-type signaling rules the number of states depends on the type of rule. Also, it should be noted that, with CUSUM charts, the number of states is a fixed number (see Example 3.6). There is no need to discretize the interval between the LCL and the UCL into $\nu = 2s + 1$ subintervals (as explained in Step 1 below), since, as stated before, for CUSUM charts the number of states will be a fixed number. The steps below apply to the EWMA charts.

Step 1: Divide the interval between the LCL and the UCL into $\nu = 2s + 1$ subintervals of width $2\tau = (UCL - LCL)/\nu$ (see Figure 3.21), where each subinterval corresponds to a transient (or an accessible) state of the MC. State j is said to be accessible from state i if, and only if, starting in state

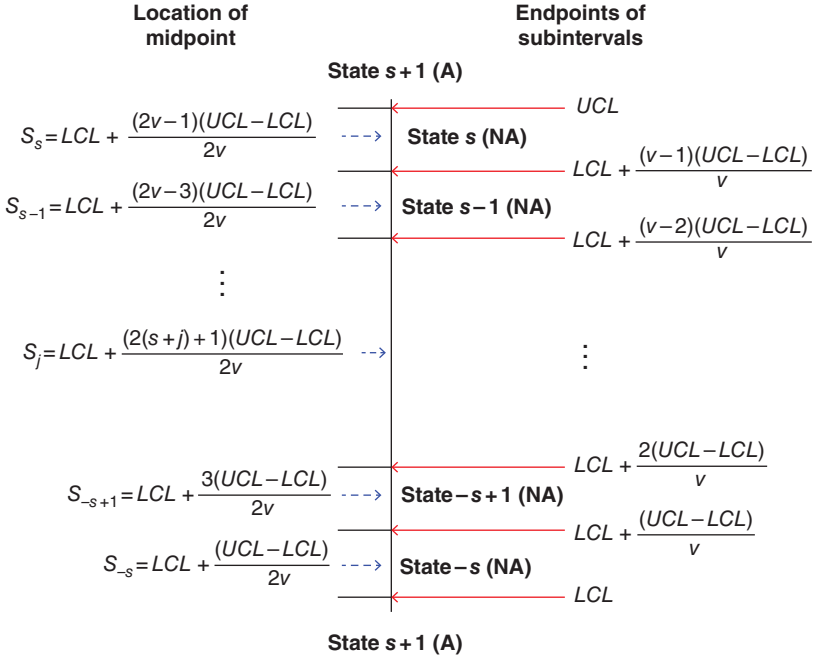


FIGURE 3.21 *Partitioning of the interval between the* LCL *and the* UCL *into* $\nu = 2s + 1$ *subintervals.*

i, it is possible that the process will at some stage enter state j. Note that $2\tau = (UCL - LCL)/\nu$ will equal $2\tau = 2UCL/\nu$ so that $\tau = UCL/\nu$ if the control limits are symmetrically placed around zero (i.e., $-LCL = UCL$). It should be noted that taking larger values of ν would result in more accurate answers. A detailed discussion on the choice of ν is given in Note 3.2.

Step 2: Choose the number of discretized subintervals ν to be an odd positive integer so that there is a unique middle entry.

Step 3: Declare the charting statistic, Z_i, to be in the transient state j at time i if $S_j - \tau < Z_i \leq S_j + \tau$ for $j = -s, -s+1, \ldots, s-1$ and $S_j - \tau < Z_i < S_j + \tau$ for $j = s$, where S_j denotes the midpoint of the jth interval.

Step 4: Calculate the one-step transition probabilities (p_{ij}'s) where p_{ij} denotes the probability of moving from state i to state j in one step at any point in time.

Step 5: Construct the transition probability matrix, consisting of the one-step transition probabilities, to find the run-length distribution.

Note that the midpoints (defined in Step 3) can be found using the following general calculation formula

$$S_j = LCL + (2(s+j)+1)\tau \text{ and } S_0 = 0$$

since we assume (for simplicity) that $-LCL = UCL$. The charting statistic is said to be in the absorbing or OOC state (i.e., $j = s+1$) if Z_i falls on or outside the control limits. This region is considered absorbing since the process is stopped when a signal is given by the chart. Hence, the process is declared to be OOC whenever Z_i is in the absorbing state, whereas the process is considered IC whenever Z_i is in a transient state (also referred to as an IC state). Therefore, the IC region consists of v non-absorbing states, whereas the OOC region is treated as a single absorbing state.

As explained before, in order to apply the MC approach, we need expressions for the signaling probabilities so that we can set up the transition probability matrix. The elements of the essential transition probability matrix Q are called the one-step transition probabilities; $Q = [p_{ij}]$ for $i, j = -s, -s+1, \ldots, s-1, s$. In order to calculate these probabilities once the states have been defined or identified, we assume that the charting statistic is equal to the midpoint S_i whenever it is in state i. This is an approximation that becomes more accurate when the numbers of subintervals increase, and is commented on later. For each control chart under consideration, the one-step transition probabilities are calculated differently. Next, we give another example of using the MC approach to calculate the run-length distribution of a CUSUM chart.

Example 3.6 Run-length Distribution of a CUSUM Chart for the Normal Mean with Individual Data

Let us consider only the upper one-sided CUSUM charting statistic defined in Equation 3.3. Assume that the observations follow a standard normal distribution so that μ_0 in Equation 3.3 equals zero. Brook and Evans (1972) proposed a method for evaluating "not only the average run lengths for the scheme, but also moments and percentage points of the run-length distribution and exact probabilities of run length" for the upper one-sided CUSUM chart by treating the CUSUM as a MC with the state space a subset of $\{0, 1, \ldots, H\}$. MC techniques have a great advantage as they are adjustable to many runs-related problems, and they often simplify the solutions to the specific problems that they are applied to. Fu, Spiring, and Xie (2002) present three results that must be satisfied before implementing the finite-state MC approach. Let S_t^+ be a finite-state homogeneous MC on the state space Ω^+ with a transition probability matrix such that (i) $\Omega^+ = \{\zeta_0, \zeta_1, \ldots, \zeta_{r+s-1}\}$, where $0 = \zeta_0 < \zeta_1 < \cdots < \zeta_{r+s-1} = H$ and ζ_{r+s-1} is an absorbing state; (ii) the transition probability matrix is given by $M = [p_{ij}]$ for $i = 0, 1, \ldots, r+s-1$ and $j = 0, 1, \ldots, r+s-1$, where r denotes the number of non-absorbing states and s denotes the number of absorbing states, respectively, and (iii) the starting value should be in the "dummy" state with probability one, that is, $P(S_0^+ = \zeta_0) = 1$, to ensure that the process starts IC. Assume that the MC S_t^+ satisfies conditions (i), (ii), and (iii). Then from Fu, Spiring, and Xie (2002) and Fu and Lou (2003), it follows that the run-length distribution and its various characteristics can be obtained using Equations 3.21–3.24.

At this point, it should be noted that, with the CUSUM charts, the number of non-absorbing states is a fixed number (see the example below). There is no need to discretize the interval between the LCL and the UCL into $\nu = 2s + 1$ subintervals (as explained in Step 1 earlier), since, as stated before, for CUSUM charts the number of non-absorbing states will be a fixed number.

The time that the procedure signals is the first time such that the finite-state MC S_t^+ enters one of the absorbing states where the state space is given by

$$\begin{aligned} &\Omega^+ = \{\zeta_0, \zeta_1, \ldots, \zeta_{r+s-1}\}, S_0^+ = 0 \\ &\text{and} \\ &S_t^+ = \min\{H, \max\{0, S_{t-1}^+ + X_t - K\}\}. \end{aligned} \tag{3.27}$$

The state corresponding to a signal given by the CUSUM chart is called an absorbing state. Clearly, there is only one absorbing state, since the chart signals when S_t^+ falls on or above H, that is, $s = 1$.

When considering the MC approach for the CUSUM chart, Brook and Evans (1972) start by first considering the discrete case. Let X be a positive

integer valued random variable and let the reference value, K, and the decision interval, H, both take on positive integer values. Fractional values can be covered by appropriate rescaling. By defining X, K, and H in this manner, the CUSUM $\sum(X - K)$ can only take on one of the integer values $\{0, 1, \ldots, H\}$, for example, for $H = 4$ and $K = 1$, the state space is calculated in Table 3.9.

It should be noted that only values of $X_t = -5(1)5$ are shown. Any X_t value less than -5 will result in $S_t^+ = 0$ and any X_t value greater than $+5$ will result in $S_t^+ = 4$. Thus, the state space is given Table 3.10.

From Table 3.10, we see that there are three non-absorbing states, that is, $r = 3$, and one absorbing state, that is, $s = 1$. Therefore, the corresponding transition probability matrix will be a 4×4 matrix. Using three IC states,

Table 3.9 Calculation of the state space when $H=4$ and $K=1$.

X_t	$S_{t-1}^+ + X_t - K$	$\max\{0, S_{t-1}^+ + X_t - K\}$	$S_t^+ = \min\{H, \max\{0, S_{t-1}^+ + X_t - K\}\}$
-5	$0 + (-5) - 1 = -6$[3]	$\max\{0, -6\} = 0$	$\min\{4, 0\} = 0$
-4	$0 + (-4) - 1 = -5$	$\max\{0, -5\} = 0$	$\min\{4, 0\} = 0$
-3	$0 + (-3) - 1 = -4$	$\max\{0, -4\} = 0$	$\min\{4, 0\} = 0$
-2	$0 + (-2) - 1 = -3$	$\max\{0, -3\} = 0$	$\min\{4, 0\} = 0$
-1	$0 + (-1) - 1 = -2$	$\max\{0, -2\} = 0$	$\min\{4, 0\} = 0$
0	$0 + 0 - 1 = -1$	$\max\{0, -1\} = 0$	$\min\{4, 0\} = 0$
1	$0 + 1 - 1 = 0$	$\max\{0, 0\} = 0$	$\min\{4, 0\} = 0$
2	$0 + 2 - 1 = 1$	$\max\{0, 1\} = 1$	$\min\{4, 1\} = 1$
3	$1 + 3 - 1 = 3$	$\max\{0, 3\} = 3$	$\min\{4, 3\} = 3$
4	$3 + 4 - 1 = 6$	$\max\{0, 6\} = 6$	$\min\{4, 6\} = 4$
5	$4 + 5 - 1 = 8$	$\max\{0, 8\} = 8$	$\min\{4, 8\} = 4$

Table 3.10 Classifications and descriptions of the states.

State number	Description of the state	Absorbing (A)/ Non-absorbing (NA)
0	$S_t^+ = 0$	NA
1	$S_t^+ = 1$	NA
2	$S_t^+ = 3$	NA
3	$S_t^+ = 4$	A

[3] Note: Since only the state space needs to be described, S_{t-1}^+ can be any value from Ω^+ and we therefore take, without loss of generality, $S_{t-1}^+ = 0$. Any other possible value for S_{t-1}^+ would lead to the same Ω^+.

the transition probability matrix is given by

$$M_{4\times 4} = \begin{pmatrix} Q_{3\times 3} & \mid & \underline{p}_{3\times 1} \\ - & - & - \\ \underline{0}'_{1\times 3} & \mid & 1_{1\times 1} \end{pmatrix}.$$

Recall that only the essential transition probability matrix, Q, is needed in order to obtain the run-length distribution characteristics using Equations 3.22–3.24. Here, the essential transition probability matrix is given by $Q_{3\times 3} = \begin{pmatrix} p_{00} & p_{01} & p_{02} \\ p_{10} & p_{11} & p_{12} \\ p_{20} & p_{21} & p_{22} \end{pmatrix}$. The calculation of the one-step transition probabilities for the essential transition probability matrix is given in Table 3.11.

Here, the essential transition probability matrix is given by $Q_{3\times 3} = \begin{pmatrix} 0.841 & 0 & 0 \\ 0.500 & 0 & 0 \\ 0.023 & 0 & 0 \end{pmatrix}$, $\underline{1}_{3\times 1} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, $\underline{\xi}_{1\times \nu} = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}$, and $I_{3\times 3} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. By substituting these into Equations 3.22 and 3.23, the IC *ARL* and *SDRL* are found to be 6.29 and 5.77, respectively. Thus, when the process is IC, the first false alarm would be observed, on average, at every sixth plotted point. This, of course, is not good since a high IC *ARL* is desired. However, for illustration purposes, the values of the design parameters, K and H, were chosen such that a small essential transition probability matrix was obtained. A small essential transition probability matrix, here of order 3×3, is desirable for illustration purposes, since this leads to only showing nine calculations of one-step transition probabilities in Table 3.11. Continuing with the calculations of the IC run-length characteristics, some representative percentiles (say, the 5th, 25th, 50th, 75th, and 95th) can be calculated using Equation 3.24 resulting that the IC 5th percentile equals 1, the IC 25th percentile/first quartile equals 2, the IC 50th percentile/median equals 5, the IC 75th percentile/third quartile equals 9, and the IC 95th percentile equals 18. For a better understanding of the percentiles, consider the first quartile, which equals 2, and the median, which equals 5. Since the first quartile is 2, we know that a false alarm will not occur within the first two plotted points, with a probability of 75% at most. For the median, since the median is 5, we know that first false alarm will be observed within the first five plotted points at least 50% of the time. In this example, we illustrated the IC case; however, calculations for when the process is OOC can easily be done. This is left as an exercise for the reader.

We illustrate the MC approach for calculating the run-length distribution of an EWMA chart. Recall that, in this case, we first have to discretize the infinite state transition probability matrix.

Example 3.7 Run-length Distribution of an EWMA Chart for the Normal Mean with Individual Data

After discretizing the transition probability matrix, we need to find the one-step transition probabilities; $M = [p_{ij}]$ for $i, j = -s, -s+1, \ldots, s-$

Table 3.11 Calculation of the one-step transition probabilities of the essential transition probability matrix.

p_{00} $= P(S_t^+ = 0 \mid S_{t-1}^+ = 0)$ $= P(\min\{4, \max\{0, 0 +$ $X_t - 1\}\} = 0)$ $= P(\max\{0, 0 +$ $X_t - 1\} = 0)$ $= P(X_t - 1 \leq 0)$ $= P(X_t \leq 1)$ $= 0.841$	p_{01} $= P(S_t^+ = 1 \mid S_{t-1}^+ = 0)$ $= P(\min\{4, \max\{0, 0 +$ $X_t - 1\}\} = 1)$ $= P(\max\{0, 0 +$ $X_t - 1\} = 1)$ $= P(X_t - 1 = 1)$ $= P(X_t = 2)$ $= 0$	p_{02} $= P(S_t^+ = 3 \mid S_{t-1}^+ = 0)$ $= P(\min\{4, \max\{0, 0 +$ $X_t - 1\}\} = 3)$ $= P(\max\{0, 0 +$ $X_t - 1\} = 3)$ $= P(X_t - 1 = 3)$ $= P(X_t = 4)$ $= 0$
p_{10} $= P(S_t^+ = 0 \mid S_{t-1}^+ = 1)$ $= P(\min\{4, \max\{0, 1 +$ $X_t - 1\}\} = 0)$ $= P(\max\{0, 1 +$ $X_t - 1\} = 0)$ $= P(1 + X_t - 1 \leq 0)$ $= P(X_t \leq 0)$ $= 0.500$	p_{11} $= P(S_t^+ = 1 \mid S_{t-1}^+ = 1)$ $= P(\min\{4, \max\{0, 1 +$ $X_t - 1\}\} = 1)$ $= P(\max\{0, 1 +$ $X_t - 1\} = 1)$ $= P(1 + X_t - 1 = 1)$ $= P(X_t = 1)$ $= 0$	p_{12} $= P(S_t^+ = 3 \mid S_{t-1}^+ = 1)$ $= P(\min\{4, \max\{0, 1 +$ $X_t - 1\}\} = 3)$ $= P(\max\{0, 1 +$ $X_t - 1\} = 3)$ $= P(1 + X_t - 1 = 3)$ $= P(X_t = 3)$ $= 0$
p_{20} $= P(S_t^+ = 0 \mid S_{t-1}^+ = 3)$ $= P(\min\{4, \max\{0, 3 +$ $X_t - 1\}\} = 0)$ $= P(\max\{0, 3 +$ $X_t - 1\} = 0)$ $= P(3 + X_t - 1 \leq 0)$ $= P(X_t \leq -2)$ $= 0.023$	p_{21} $= P(S_t^+ = 1 \mid S_{t-1}^+ = 3)$ $= P(\min\{4, \max\{0, 3 +$ $X_t - 1\}\} = 1)$ $= P(\max\{0, 3 +$ $X_t - 1\} = 1)$ $= P(3 + X_t - 1 = 1)$ $= P(X_t = -1)$ $= 0$	p_{22} $= P(S_t^+ = 3 \mid S_{t-1}^+ = 3)$ $= P(\min\{4, \max\{0, 3 +$ $X_t - 1\}\} = 3)$ $= P(\max\{0, 3 +$ $X_t - 1\} = 3)$ $= P(3 + X_t - 1 = 3)$ $= P(X_t = 1)$ $= 0$

$1, s$. In order to calculate these probabilities, we assume that the charting statistic is equal to S_i whenever it is in state i. For j non-absorbing, we obtain

$$p_{ij} = P(S_j - \tau < Z_k \leq S_j + \tau \mid \text{in } S_i) \ \ \forall j \ (j \neq s)$$
$$\text{and} \tag{3.28}$$
$$p_{ij} = P(S_j - \tau < Z_k < S_j + \tau \mid \text{in } S_i) \ \ \text{for } j = s.$$

Note that the subscript k does not indicate that the charting statistic is time dependent. It simply refers to the charting statistic under consideration. For j absorbing, we use the fact that the rows of an M must add to one, and therefore the probabilities of going from a transient state to the absorbing state are found by subtraction. Finding the essential transition probability matrix, Q, involves finding the transition probabilities of going from one transient state to another. In order to calculate the transition probabilities for the parametric EWMA control chart, we use the charting statistic and substitute this into Equation 3.28 to obtain

$$p_{ij} = P(S_j - \tau < \lambda X_k + (1-\lambda) Z_{k-1} \leq S_j + \tau \mid Z_{k-1} = S_i) \ \ \forall j \ (j \neq s)$$
and
$$p_{ij} = P(S_j - \tau < \lambda X_k + (1-\lambda) Z_{k-1} < S_j + \tau \mid Z_{k-1} = S_i) \ \ \text{for } j = s. \tag{3.29}$$

By rewriting and substituting $Z_{k-1} = S_i$ into Equation 3.29, we obtain

$$p_{ij} = P\left(\frac{S_j - \tau - (1-\lambda) S_i}{\lambda} < X_k \leq \frac{S_j + \tau - (1-\lambda) S_i}{\lambda} \right) \ \ \forall j \ (j \neq s)$$
and
$$p_{ij} = P\left(\frac{S_j - \tau - (1-\lambda) S_i}{\lambda} < X_k < \frac{S_j + \tau - (1-\lambda) S_i}{\lambda} \right) \ \ (j = s) \tag{3.30}$$

which can be obtained using

$$p_{ij} = \Phi\left(\frac{S_j + \tau - (1-\lambda) S_i}{\lambda} \right) - \Phi\left(\frac{S_j - \tau - (1-\lambda) S_i}{\lambda} \right) \ \ \forall j \ (j \neq s)$$

where Φ denotes the cumulative standard normal distribution function. At this point, it is important to note that one only needs the underlying process distribution in order to calculate the one-step transition probabilities, which are needed to calculate the run-length distribution of the control chart.

Since the values τ, λ, S_i, and S_j are known constants (λ is chosen by the practitioner and τ, S_i, and S_j are calculated), the required probabilities

can easily be calculated. For more details on the MC approach, the reader is referred to Appendix 3.2.

For a numerical illustration, suppose we consider the parametric EWMA chart for individual data from a normal distribution with design parameters $\lambda = 0.05$ and $L = 2.703$ (this leads to an IC ARL of 370; see Table 3.2), and $v = 2s + 1 = 9$ so that $s = 4$, that is, we divide the interval between the control limits into nine subintervals (see Figure 3.22) resulting in a "small" transition probability matrix with only 10 rows and 10 columns, respectively.

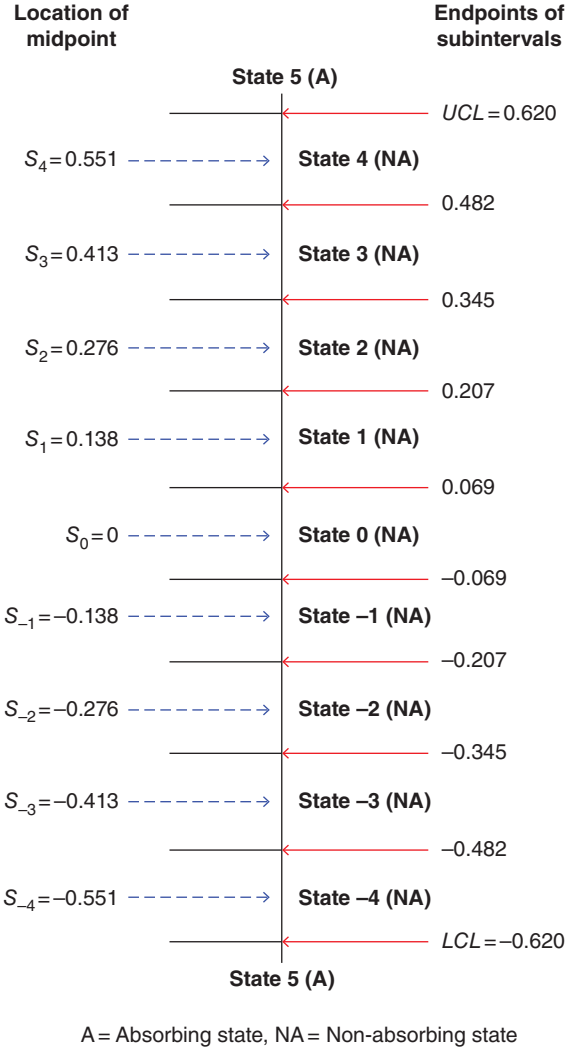


FIGURE 3.22 *Partitioning of the interval between the* LCL *and the* UCL *into* $\nu = 9$ *subintervals with* $\lambda = 0.10$ *and* $L = 2.703$.

Substituting $\mu_0 = 0$ and $\sigma_0 = 1$ for a standard normal distribution and design parameters $\lambda = 0.1$ and $L = 2.703$ into Equation 3.12, the CL and the steady-state control limits are given by

$$UCL = \mu_0 + L\sigma_0\sqrt{\frac{\lambda}{2-\lambda}} = 0 + 2.703(1)\sqrt{\frac{0.1}{2-0.1}} = 0.620$$
$$CL = \mu_0 = 0$$
$$LCL = \mu_0 - L\sigma_0\sqrt{\frac{\lambda}{2-\lambda}} = 0 - 2.703(1)\sqrt{\frac{0.1}{2-0.1}} = -0.620.$$

Dividing the interval between the LCL and the UCL into $\nu = 9$ subintervals of width $2\tau = (UCL - LCL)/\nu = (0.620 - (-0.620))/9 = 0.138$ so that $\tau = 0.069$. The partitioning is illustrated in Figure 3.22.

For nine IC states, the transition probability matrix is given by

$$M_{10\times 10} = \begin{pmatrix} Q_{9\times 9} & | & \underline{p}_{9\times 1} \\ - & - & - \\ \underline{0}'_{1\times 9} & | & 1_{1\times 1} \end{pmatrix}$$

$$= \left(\begin{array}{ccccccccc|c} p_{(-4)(-4)} & p_{(-4)(-3)} & p_{(-4)(-2)} & p_{(-4)(-1)} & p_{(-4)0} & p_{(-4)1} & p_{(-4)2} & p_{(-4)3} & p_{(-4)4} & p_{(-4)5} \\ p_{(-3)(-4)} & p_{(-3)(-3)} & p_{(-3)(-2)} & p_{(-3)(-1)} & p_{(-3)0} & p_{(-3)1} & p_{(-3)2} & p_{(-3)3} & p_{(-3)4} & p_{(-3)5} \\ p_{(-2)(-4)} & p_{(-2)(-3)} & p_{(-2)(-2)} & p_{(-2)(-1)} & p_{(-2)0} & p_{(-2)1} & p_{(-2)2} & p_{(-2)3} & p_{(-2)4} & p_{(-2)5} \\ p_{(-1)(-4)} & p_{(-1)(-3)} & p_{(-1)(-2)} & p_{(-1)(-1)} & p_{(-1)0} & p_{(-1)1} & p_{(-1)2} & p_{(-1)3} & p_{(-1)4} & p_{(-1)5} \\ p_{0(-4)} & p_{0(-3)} & p_{0(-2)} & p_{0(-1)} & p_{00} & p_{01} & p_{02} & p_{03} & p_{04} & p_{05} \\ p_{1(-4)} & p_{1(-3)} & p_{1(-2)} & p_{1(-1)} & p_{10} & p_{11} & p_{12} & p_{13} & p_{14} & p_{15} \\ p_{2(-4)} & p_{2(-3)} & p_{2(-2)} & p_{2(-1)} & p_{20} & p_{21} & p_{22} & p_{23} & p_{24} & p_{25} \\ p_{3(-4)} & p_{3(-3)} & p_{3(-2)} & p_{3(-1)} & p_{30} & p_{31} & p_{32} & p_{33} & p_{34} & p_{35} \\ p_{4(-4)} & p_{4(-3)} & p_{4(-2)} & p_{4(-1)} & p_{40} & p_{41} & p_{42} & p_{43} & p_{44} & p_{45} \\ - & - & - & - & - & - & - & - & - & - \\ p_{5(-4)} & p_{5(-3)} & p_{5(-2)} & p_{5(-1)} & p_{50} & p_{51} & p_{52} & p_{53} & p_{54} & p_{55} \end{array}\right)$$

$$= \left(\begin{array}{ccccccccc|c} 0.447 & 0.380 & 0.063 & 0.002 & 0 & 0 & 0 & 0 & 0 & 0.107 \\ 0.129 & 0.473 & 0.342 & 0.048 & 0.001 & 0 & 0 & 0 & 0 & 0.007 \\ 0.009 & 0.158 & 0.493 & 0.303 & 0.036 & 0.001 & 0 & 0 & 0 & 0 \\ 0 & 0.014 & 0.190 & 0.505 & 0.264 & 0.026 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.019 & 0.226 & 0.509 & 0.226 & 0.019 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.026 & 0.264 & 0.505 & 0.190 & 0.014 & 0 & 0 \\ 0 & 0 & 0 & 0.001 & 0.036 & 0.303 & 0.493 & 0.158 & 0.009 & 0 \\ 0 & 0 & 0 & 0 & 0.001 & 0.048 & 0.342 & 0.473 & 0.129 & 0.007 \\ 0 & 0 & 0 & 0 & 0 & 0.002 & 0.063 & 0.380 & 0.447 & 0.107 \\ - & - & - & - & - & - & - & - & - & - \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right)$$

where the transition probabilities are calculated using Equation 3.30. The calculation of p_{01} is shown below.

$$\begin{aligned} p_{01} &= \Phi\left(\frac{S_1 + \tau - (1-\lambda)S_0}{\lambda}\right) - \Phi\left(\frac{S_1 - \tau - (1-\lambda)S_0}{\lambda}\right) \\ &= \Phi\left(\frac{0.138 + 0.069 - (1-0.1)0}{0.1}\right) - \Phi\left(\frac{0.138 - 0.069 - (1-0.1)0}{0.1}\right) \\ &= \Phi(2.067) - \Phi(0.689) \\ &= 0.226. \end{aligned}$$

Once the one-step transition probabilities are calculated and the essential transition probability matrix Q is found, the IC run-length characteristics are obtained using Equations 3.22–3.24. This yields an IC *ARL* and an IC *SDRL* equal to 270.26 and 263.07, respectively, with the IC 5th, 25th, 50th, 75th, and 95th percentiles being equal to 21, 83, 190, 372, and 795, respectively. Thus, when the process is IC, the first false alarm would be observed, on average, at every 270th plotted point. Since the first quartile is 83, we know that a false alarm will not occur within the first 83 plotted points, with a probability of 75% at most. For the median, since the median is 190, we know that the first false alarm will be observed within the first 190 plotted points at least 50% of the time.

It should be noted at this point that a more accurate answer could be obtained if the number of subintervals between the control limits were to be increased, for example, for $\nu = 1001$ we find $ARL_{\text{IC}} = 371.88$, which is closer to the desired nominal value of 370. However, for this example, a small number of subintervals were taken to illustrate the procedure.

Next we illustrate the MC approach for calculating the run-length distribution of a Shewhart chart with runs-type signaling rules. Recall that, in this case, we first have to discretize the infinite state transition probability matrix.

Example 3.8 Run-length Distribution of a Shewhart Control Chart for Monitoring Process Mean with Supplementary Runs-rules

Continuing with the same data used to illustrate the Shewhart control chart for monitoring the process mean, here we illustrate a Shewhart $\overline{X}$ control chart with runs-type signaling rules. The data and graph are omitted to conserve space; however, the reader is referred to the *1-of-1* Shewhart $\overline{X}$ control chart in Section 3.3.1, Example 3.1 for that data, the charting statistics, etc. Recall that the *2-of-2* and *2-of-3* runs-type signaling rules require at least the last two or three charting statistics to signal, respectively. However, from Figure 3.3 it can be seen that all of the charting statistics are between

the control limits and, consequently, the process is functioning in statistical control with respect to the target mean.

Here, we illustrate the MC approach to calculating the run-length distribution of a two-sided *2-of-2* Shewhart control chart for the mean that signals when two out of the last two consecutive charting statistics plot beyond the control limits. This chart was proposed by Klein (2000). The *2-of-2* chart is one of several types of runs-type signaling rules that have been proposed to enhance the performance of the basic Shewhart chart. We discuss these in more detail in Section 3.5.2.

Note that the *2-of-2* chart only signals when both of the two consecutive charting statistics plot on or above the *UCL* (see Panel (a) of Figure 3.17), or when both plot on or below the *LCL* (see Panel (b) of Figure 3.17). In the first case, an increase or an upward shift in the mean is signaled, while in the latter case, a decrease or downward shift is detected.

The MC approach for calculating the run-length distribution consists of first obtaining the transition probability matrix. To this end, we need to define the number of absorbing and non-absorbing states. It should be emphasized that, with runs-type signaling rules, the number of non-absorbing states is defined by the type of rule used, for example, *2-of-2* or *2-of-3*. There is no need to discretize the interval between the *LCL* and the *UCL* into $\nu = 2s + 1$ subintervals (as explained in Step 1 earlier), since, as stated before, for runs-type signaling rules the number of non-absorbing states depends on the type of rule. To this end, we first define the three non-absorbing states of the MC, namely:

State 1: No points are beyond either of the control limits.

State 2: Only one point is on or above the *UCL*.

State 3: Only one point is on or below the *LCL*.

The MC approach contains one absorbing state, namely:

State 4: Once two consecutive points are on or outside either one of the control limits, the process reaches the absorbing state.

Next, we define the probabilities corresponding to each of these states. Let p denote the probability that the process is in State 1, that is, the charting statistic plots between the *LCL* and the *UCL*. Similarly, let pU and pL denote the probabilities that the stochastic process is in State 2 or State 3, respectively, that is, the charting statistic plots on or above the *UCL* or on or below the *LCL*. The states and the associated probabilities are illustrated in Figure 3.23 for the *2-of-2* runs-type chart.



FIGURE 3.23 *Illustration of the states and the associated probabilities for the* 2-of-2 *runs-type chart.*

The one-step transition probabilities for the *2-of-2* runs-type signaling rule are shown in Klein (2000), and the reader is referred to his paper for details. However, only the essential transition probability matrix, Q, is needed in order to obtain the run-length distribution and the associated characteristics using Equations 3.22–3.24. For our example, the essential transition probability matrix is given by $Q_{3\times 3} = \begin{pmatrix} p & pU & pL \\ p & 0 & pL \\ p & pU & 0 \end{pmatrix}$. Thus, for illustration purposes, supposing that when the process is IC, $p = 0.93$, $pL = pU = 0.035$ so that $Q_{3\times 3} = \begin{pmatrix} 0.93 & 0.035 & 0.035 \\ 0.93 & 0 & 0.035 \\ 0.93 & 0.035 & 0 \end{pmatrix}$, $\underline{1}_{3\times 1} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, $\underline{\xi}_{1\times 3} = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}$, and $I_{3\times 3} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. By substituting these into Equations 3.22 and 3.23, the IC *ARL* and *SDRL* are found to be 422.45 and 420.98, respectively. Thus, when the process is IC, the first false alarm would be observed, on average, at every 422nd time or plotted point. Some representative percentiles (say, the 5th, 25th, 50th, 75th, and 95th) can be calculated using the run-length cdf in Equation 3.24, and we find that the IC 5th percentile equals 23, the IC 25th percentile/first quartile equals 123, the IC 50th percentile/median equals 293, the IC 75th percentile/third quartile equals 585, and the IC 95th percentile equals 1263. For a better understanding of the percentiles, consider the first quartile, which equals 123, and the median, which equals 293. Since the first quartile is 123, we know that a false alarm will not occur within the first 123 plotted points, with a probability of 75% at most. For the median, since the median is 293, we know that the first false alarm will be observed within the first 293 plotted points at least 50% of the time. Similar calculations can be made when the process is OOC. All we need to do is to find the p, pL, and pU, set

up the essential transition probability matrix and apply the formulas given in Equations 3.22–3.24. Note, however, that, in general, the distribution of the charting statistic is needed to calculate the transition probabilities. For example, note that $pU = P(\overline{X} > UCL)$ so that we need the distribution of $\overline{X}$ or the distribution of the population. For the normal means problem, the population is assumed to be normal, and the probability pU can be easily calculated. This is an important point, which will be discussed again in Chapter 4 for the nonparametric charts.

Above, we considered the *2-of-2* runs-type signaling rule. It is important to note that, in general, one can have a *w-of-*$(w+v)$ runs-type signaling rule where, for example, take $w = 2$ and $v = 1$, then we have the *2-of-3* runs-type signaling rule where a signal is given when two out of the most successive three points plot OOC. This approach is similar to the *2-of-2* runs-type signaling rule; however, note that the number of states increases to eight. At this point, the reader is reminded that, with runs-type signaling rules, the number of non-absorbing states is defined by the type of rule used, for example, *2-of-2* or *2-of-3*. There is no need to discretize the interval between the LCL and the UCL into $\nu = 2s + 1$ subintervals (as explained in Step 1 earlier), since, as stated before, for runs-type signaling rules, the number of non-absorbing states depends on the type of rule. Here, we consider the *2-of-3* runs-type signaling rule where the eight states are as follows:

State (OO): Two successive points between both control limits.

State (OU): The first point between both control limits and the second point above the UCL.

State (OL): The first point between both control limits and the second point below the LCL.

State (UL): The first point above the UCL and the second point below the LCL.

State (UO): The first point above the UCL and the second point between the control limits.

State (LO): The first point below the LCL and the second point between the control limits.

State (LU): The first point below the LCL and the second point above the UCL.

The MC approach contains one absorbing state, namely:

State (OOC): Two of three consecutive points either below the LCL or above the UCL.

The one-step transition probabilities for the *2-of-3* runs-type signaling rule are shown in Klein (2000), and we refer the reader to his paper for more detail. However, as in the case of the *2-of-2* rule, only the essential transition probability matrix, Q, is needed in order to obtain the run-length distribution and the associated characteristics using Equations 3.22–3.24. For our example, the essential transition probability matrix is given by

$$Q_{7\times 7} = \begin{pmatrix} p & pU & pL & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & pL & p & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & p & pU \\ 0 & 0 & 0 & 0 & 0 & p & 0 \\ p & 0 & pL & 0 & 0 & 0 & 0 \\ p & pU & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & p & 0 & 0 \end{pmatrix}, \text{ and for illustration purposes,}$$

supposing that when the process is IC, $p = 0.93$, $pL = pU = 0.035$ so that

$$Q_{7\times 7} = \begin{pmatrix} 0.93 & 0.035 & 0.035 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.035 & 0.93 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.93 & 0.035 \\ 0 & 0 & 0 & 0 & 0 & 0.93 & 0 \\ 0.93 & 0 & 0.035 & 0 & 0 & 0 & 0 \\ 0.93 & 0.035 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.93 & 0 & 0 \end{pmatrix}, \quad \underline{1}_{7\times 1} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix},$$

$\underline{\xi}_{1\times 7} = (1\ 0\ 0\ 0\ 0\ 0\ 0\ )$ and $I_{7\times 7} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$. Now, by substituting these into Equations 3.22 and 3.23, the IC *ARL* and *SDRL* are found to be 222.13 and 220.23, respectively. Thus, when the process is IC, the first false alarm would be observed, on average, at every 222nd time or plotted point. Some representative percentiles (say, the 5th, 25th, 50th, 75th, and 95th) can be calculated using the run-length cdf in Equation 2.24, and we find that the IC 5th percentile equals 13, the IC 25th percentile/first quartile equals 65, the IC 50th percentile/median equals 155, the IC 75th percentile/third quartile equals 307, and the IC 95th percentile equals 662. For a better understanding of the percentiles, consider the first quartile, which equals 65, and the median, which equals 155. Since the first quartile is 65, we know that a false alarm will not occur within the first 65 plotted points, with a probability of 75% at most. For the median, since the median is 155, we know that first false alarm will be observed within the first 155 plotted points at least 50% of the time.

Similar calculations can be made when the process is OOC. All we need to do is to find the p, pL, and pU, set up the essential transition probability matrix, and apply the formulas given in Equations 3.22–3.24.

3.6.1.3 The Integral Equation Approach

The integral equation approach utilizes mathematics and combinatorics to find a closed form expression of the run-length distribution. This approach is sometimes challenging in that the expression obtained is typically complex or difficult to evaluate numerically. Often, the exact expression of the run-length distribution can be found after a considerable amount of work, but here a simulation is done instead, since it is much easier.

3.6.1.4 The Computer Simulations (the Monte Carlo) Approach

Monte Carlo simulations can be used to calculate the characteristics of the run-length distribution. The popularity of this method stems from the fact that, no matter how complicated the run-length distribution is, theoretically, computer simulations can almost always be used with relative ease to calculate the run-length distribution and its associated characteristics fairly accurately, provided that the simulation size is big enough. In this body of work, we use 100 000 simulations since it is well known that the error of a run-length characteristic can be bounded by increasing the simulation size sufficiently.

The generic steps of a computer simulation procedure to calculate the run-length distribution for a two-sided control chart, where the charting statistic is calculated from a random sample, is given as follows. This can be implemented in any software package such as R, SAS, or Matlab.

Step 1: Specify the necessary parameters, such as the subgroup size, nominal IC average run-length (or FAR); calculate the control limits.

Step 2: Assume some process distribution, such as the normal distribution and specify parameters. Generate (simulate) subgroups of random observations from the specified distribution.

Step 3: Calculate the charting statistic for each subgroup and compare it to the control limits calculated in Step 1.

Step 4: The number of subgroups needed until the first charting statistic plots on or outside the control limits is recorded as an observation from the run-length distribution.

Step 5: Repeat Steps 1 to 4 a total of 100 000 times.

Step 6: Once we obtain a "data set" with 100 000 observations from the run-length distribution, the run-length distribution characteristics of interest, which are descriptive statistics, such as the average, standard deviation, median, percentiles, etc., can be calculated. For example, one can use proc univariate of SAS to find these values. Minitab or R can be used to do the same and graph the boxplots or histograms of the run-length values for a better understanding of the run-length distribution.

It may be noted that simulating the run-length distribution and the associated characteristics may be particularly time consuming in the IC case since the run-length distribution is heavily right-skewed, which means that there may be some very large run-length values, which can take a very long time to be realized. In some cases, one has to be careful that the run-length characteristic of interest, say, the ARL, is finite, which may not necessarily be the case. A possible shortcut to calculating the run-length distribution may be to use the explicit expressions to calculate the probability of a signal (and its reciprocal) within the simulations, which may be available for some charts under the assumed process distribution.

Note 3.2 Number of Subintervals to Be Used for the EWMA Chart

When applying the MC approach to calculate the run-length distribution in EWMA charts, one has to make a decision regarding the number of subintervals between the control limits. Here, we show an illustration using the parametric EWMA chart. However, a similar pattern unfolds for all EWMA charts. We also contrast the findings from the MC approach with those of the simulation approach.

For the MC approach, when the number of subintervals ν is sufficiently large, it provides an effective method that allows the run-length properties of the EWMA control scheme to be accurately approximated. In practice, values of ν around 100 yield satisfactory approximations (see Yu, 2007, p. 20). However, simulation results (see Table 3.12) suggest that the discrepancies between the simulated values and the values found using the MC approach can be somewhat large for $\nu \leq 100$. For larger values of ν greater than 100, the discrepancies seem small and they are particularly small when a larger value, such as $\nu = 1001$, is employed. In principle, taking larger values of ν should result in more accurate answers while using the MC approach, but in doing so, some run-length characteristics could not be computed within a practical period of time. The (λ, L)-combinations used in Table 3.12 are from Table 3.2, and the numbers in each cell show the attained ARL_{IC} value and the percentage difference from the nominal ARL_{IC} value of 370. On balance, however, the simulation method is recommended for calculating the run-length distribution of the EWMA control chart.

Table 3.12 ARL_{IC} values and the absolute percentage difference between the simulated values[4] and the values obtained using the MC approach for the parametric (Normal Theory) EWMA chart for various values of ν.

			$\lambda = 0.05$, $L = 2.492$		$\lambda = 0.10$, $L = 2.703$	
			Attained		Attained	
			ARL	Diff (%)	*ARL*	Diff (%)
Markov chain approach	ν	51	439.07	(18.62%)	384.92	(4.06%)
		59	386.23	(4.34%)	420.36	(13.64%)
		67	379.34	(2.48%)	337.55	(8.75%)
		75	410.76	(10.97%)	360.21	(2.62%)
		83	410.48	(10.90%)	423.25	(14.42%)
		101	396.12	(7.02%)	357.42	(3.38%)
		151	380.61	(2.83%)	358.61	(3.05%)
		201	391.25	(5.70%)	330.66	(10.61%)
		251	368.50	(0.45%)	354.17	(4.26%)
		301	369.42	(0.20%)	379.63	(2.63%)
		401	365.71	(1.20%)	370.68	(0.21%)
		501	377.47	(1.98%)	377.33	(2.01%)
		1001	369.49	(0.18%)	373.80	(1.05%)
100 000 simulations			370.15		369.91	

From Table 3.12, we can see that, the larger the value of ν, the smaller the percentage difference between the simulated values and the values obtained using the MC approach. When the design parameters are taken to be $\lambda = 0.05$ and $L = 2.492$, for example, we have a large percentage difference of 18.62% when ν is taken to be small ($\nu = 51$) as opposed to a small percentage difference of 0.18% when ν is taken to be large ($\nu = 1001$). The recommendation would be to take $\nu = 1001$, which may be a bit time consuming, however, the results will be more accurate. From a practical standpoint, a value of ν equal to 251 or 301 also seems to work reasonably well.

[4] Note that steady-state control limits were used in both the simulations and the MC approach in order to make the comparisons fair.

3.7 PARAMETER ESTIMATION PROBLEM AND ITS EFFECTS ON THE CONTROL CHART PERFORMANCE

In the previous sections, we studied the situations where the process is assumed to be normally distributed (or the charting statistic is approximately normally distributed) with parameters that are specified or known. In many practical situations, however, even if normality may be acceptable, the process parameters may not be known and would have to be estimated before process monitoring can start. This situation is referred to as the "standard(s) unknown" or the unknown parameter case and is denoted by Case U. It is well known that ignoring the effects of parameter estimation can be unwise and costly as the run-length properties of the chart are greatly impacted, which can lead to, for example, many more false alarms than nominally expected (see, for example, Jensen *et al.*, 2006 and Psarakis *et al.*, 2014), which reduces the effectiveness of control charts and increases costs.

Typically, the unknown parameters are estimated using a fixed size reference sample obtained from a Phase I analysis. We discuss the derivation of the run-length distribution in Case U for the three main types of control charts below. It turns out that the traditional view of the run-length distribution needs to be re-examined in light of the correct treatment of parameter estimators used in the control charts (limits).

By the run-length distribution of a control chart, when parameter estimators are used, one generally refers to what is now known as the unconditional or the marginal run-length distribution, which is obtained by averaging the run-length distribution for a given set of estimators over the distribution of these estimators. The run-length distribution obtained for a given set of estimators is called the conditional run-length distribution. In the earlier days of SPC, parameter estimates were typically calculated and substituted (plugged-in) for the parameters, and control limits (and control charts) were set up for prospective (Phase II) monitoring of a process. The effect of parameter estimation was subsumed basically in a mathematical-statistical step in the derivation of the (unconditional) run-length distribution. One typically used the run-length distribution and the associated characteristics (ARL etc.) to characterize chart performance. In recent years, it has been recognized that the plug-in control charts perform quite differently than their known parameter counterparts, unless a huge number of data were used to calculate the

parameter estimates. The amount of data needed can be substantial and may, in fact, be unaffordable or unavailable in a practical situation. While the approach of using the unconditional run-length distribution and associated properties such as the unconditional ARL may have had merits in providing information about the "average" chart performance, much more practical insight may be provided by the conditional run-length distribution and its attributes, which help explain the performance of a control chart for the user with the particular parameter estimates for their reference samples. In this context, it is of interest to examine the mean or the expectation of the conditional run-length distribution, which is the conditional average run-length, in fact, since the conditional ARL itself is a random variable. However, one also needs to examine other properties, such as the variance of the conditional ARL distribution, to get a better sense of chart performance under parameter estimation. Note that the distinction between the unconditional and the conditional run-length distribution goes away in the parameter known case when they are both equal to the one and the only run-length distribution of the control chart.

The overarching theme in all of this is, of course, is the amount of Phase I data that is required for parameter estimation so that predictable chart performance can be assured. It might also be necessary to take another look at measures of chart performance and how to design the control charts (i.e., obtain the correct charting constants) for the amount of data at hand to guarantee such performance.

We now revisit the control charts that were discussed earlier in this chapter for the known parameter case (Case K) and in the case when the parameters are unknown, that is, Case U. Note that we still assume that the underlying process distribution is normal. When the parameters are unknown, they are usually estimated from a retrospective analysis of historical data, often called a Phase I analysis. Control charts play an important role in Phase I analysis; we discuss some Phase I charts in Chapter 5. Once the Phase I analysis is completed, a set of reference data is obtained, which is assumed to be derived from an IC process. These data are used to estimate parameters and the control limits, which are then used prospectively to monitor incoming new or test data. This part of statistical process control is called a Phase II analysis. Thus, in this section, we first illustrate Phase II control charts where unknown process parameters are estimated from Phase I. We also discuss properties of the Case U control charts and illustrate the so-called conditioning/unconditioning (CUC) method to derive the run-length distribution and its various attributes, such as the average run-length. Note that a lot of work has been done in this area recently, yet more work is in progress and further new results and a clearer understanding are expected.

3.8 PARAMETRIC VARIABLES CONTROL CHARTS IN CASE U

3.8.1 Shewhart Control Charts in Case U

3.8.1.1 Shewhart Control Charts for the Mean in Case U

In our discussion of parametric variables control charts in Case U, we assume that the process follows a normal distribution, but at least one (or both) of the process parameters, the mean and the standard deviation, is (are) unknown. Although other process distributions can be assumed in practice, we discuss the case of the normal distribution to be consistent with the earlier developments in the chapter for the known parameter case (see Section 3.2). Before continuing, we make some finer classifications within Case U. When two parameters are in play, as is the case here, we can have four cases, namely, Case KK (mean known and standard deviation known), which is often referred to as just Case K, Case KU (mean known and standard deviation unknown), Case UK (mean unknown and standard deviation known), and Case UU (both mean and standard deviation unknown). In this section, we consider perhaps the most practical situation, Case UU, when both of these parameters are unknown. So Case U refers to one of the three cases, Case KU, UK or UU. Other situations can be treated in a similar way (see for example, Chakraborti, 2000). In addition, we assume that a reference (Phase I) sample comprising m samples, each of size n, is available after a successful Phase I analysis to estimate these unknown parameters.

Suppose that $\overline{X}_1, \overline{X}_2, \ldots, \overline{X}_m$, and $S_1, S_2, \ldots, S_m$ denote the m Phase I sample means and standard deviations, respectively. Also, let $R_1, R_2, \ldots, R_m$ denote the m Phase I sample ranges. The process mean μ is typically estimated by the average of the m sample means or the grand mean, $\overline{\overline{X}} = (\overline{X}_1 + \overline{X}_2 + \cdots + \overline{\overline{X}}m)/m$. Conversely, several estimators have been proposed to estimate the process standard deviation σ. We use some of these estimators to illustrate various points. Letting $\hat{\sigma}$ denote the chosen Phase I estimator of σ, the control limits for the two-sided Phase II k-sigma Shewhart $\overline{X}$ control chart for the mean in Case UU are given by

$$\begin{aligned} UCL &= \overline{\overline{X}} + k\frac{\widehat{\sigma}}{\sqrt{n}} \\ CL &= \overline{\overline{X}} \\ LCL &= \overline{\overline{X}} - k\frac{\widehat{\sigma}}{\sqrt{n}}. \end{aligned} \tag{3.31}$$

The control limits in Equation 3.13 may be referred to as the plug-in limits as they are of the same form as in Case K, with estimators of the mean

and the standard deviations used in place of the known IC values of the parameters. The choice of the charting constant k is an interesting question. For many years, the value 3 was used in analogy with a 3-sigma chart in Case K. It is now understood that, with plug-in limits, $k = 3$ is not a good choice in that it increases the number of false alarms, unless there is a very large number of Phase I data available, which may be in the hundreds or even in the thousands. In fact, in Case U, the choice of the charting constant depends on m, n, and the given nominal IC average run-length. In Case K, alternatively, the charting constant depends only on n and the nominal IC average run-length. More details about this are provided later.

3.8.1.2 Shewhart Control Charts for the Standard Deviation in Case U

As in Case K, the variation of a process must be monitored to ensure that it remains IC. The Shewhart control limits for the process mean in Case U depend on the process standard deviation. Thus, unless the standard deviation remains IC, the control chart for the mean will not be informative and useful. So, we need to monitor the variance or the standard deviation using a control chart.

As noted earlier, there are several possible estimators $\widehat{\sigma}$ that can be used to monitor the standard deviation σ. The popular choices include functions of the sample range (R), or the sample standard deviation (S), or the sample variance (S^2). Typically, we use a control chart to monitor the process mean together with a control chart to monitor the process variation. If the variation is IC, we go ahead and examine the control chart for the mean.

A Shewhart $\overline{X}$ chart for the mean, for example, is often used together with a Shewhart R chart for the spread. Note that, for illustration purposes, we use the Shewhart R chart for σ, even though recent literature recommends using a different chart, such as the one based on the standard deviation (see, for instance, Mahmoud *et al.*, 2010 and Epprecht, Loureiro, and Chakraborti, 2016). We do this because the Shewhart R chart is simple and continues to be used in the industry.

Let $\overline{R} = (R_1 + R_2 + \cdots + R_m)/m$ denote the average range. The CL and the control limits for the Shewhart R control chart for σ in the unknown parameter case are given by (see Montgomery, 2009, p. 229)

$$\begin{aligned} UCL &= D_4\overline{R} \\ CL &= \overline{R} \\ LCL &= D_3\overline{R} \end{aligned} \tag{3.32}$$

where

$$D_4 = 1 + 3\frac{d_3(n)}{d_2(n)},\ D_3 = 1 - 3\frac{d_3(n)}{d_2(n)}$$

and $d_2(n)$ and $d_3(n)$ are, respectively, the mean and the standard deviation of a range in a random sample of size n from the standard normal distribution. We provide the formulas here for completeness as these are not available in many textbooks.

$$d_2(n) = E(R) = \int_0^\infty r f_R(r) dr$$
$$d_3(n) = \sqrt{var(R)} = \sqrt{E(R^2) - [E(R)]^2}$$

where $E(R^2) = \int_0^\infty w^2 f_R(w) dw$ and

$$f_R(w) = n(n-1) \int_{-\infty}^{\infty} [\Phi(v) - \Phi(v-w)]^{n-2} \phi(v-w)\phi(v) dv, \;\; w > 0,$$

denotes the pdf of the range of the sample of size n from the standard normal distribution with pdf ϕ and cdf Φ (see, for example, Gibbons and Chakraborti, 2010). The integrals involved in these expressions can be calculated numerically or estimated in a simulation study in order to evaluate the constants d_2 and d_3. They both depend on the sample size n and are tabulated in Table C of Appendix A. Note that $\overline{R}/d_2(n)$ is an unbiased estimator of σ so that $d_2(n)$ is called an unbiasing constant.

The control limits given in Equation 3.31 are of the same form as in Case K but the values of the charting constants D_4 and D_3 are different. However, the control limits in Equations 3.32 are the 3-sigma limits and have some serious deficiencies, see, for example, Diko *et al.* (2016), particularly when n is small. We consider some alternatives to the R chart control limits based on probability limits later in this section.

Using a similar idea, but employing the standard deviation of the sample and not the range, the CL and the control limits for the Shewhart S control chart for σ in the unknown parameter case are given by (see Montgomery, 2009, p. 253)

$$\begin{aligned} UCL &= B_4 \overline{S} \\ CL &= \overline{S} \\ LCL &= B_3 \overline{S} \end{aligned} \tag{3.33}$$

where $\overline{S} = \dfrac{S_1 + S_2 + \cdots + S_m}{m}$ is the average standard deviation and the constants are given by

$$B_4 = 1 + \frac{3}{c_4(n)}\sqrt{1 - c_4^2(n)}, \; B_3 = 1 - \frac{3}{c_4(n)}\sqrt{1 - c_4^2(n)} \text{ and}$$

$$c_4(n) = \frac{\Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{n-1}{2}\right) * \sqrt{\frac{n-1}{2}}},$$

where Γ denotes the gamma function with $\Gamma(n) = (n-1)!$ when n is a positive integer. Note that $\overline{S}/c_4(n)$ is also an unbiased estimator of σ. The constants B_3 and B_4 can be easily calculated from the formulas and are tabulated in Table H in Appendix A.

Again, note that these control limits are the 3-sigma limits and, as in the case of the R charts, also have some deficiencies, particularly when n is small. We also consider some alternatives to the S chart control limits.

It is well known that the probability limits are more appropriate for the standard deviation since the distribution of the estimator $\hat{\sigma}$ (used in the charting statistic) is right-skewed. The skewness is more pronounced when the Phase II sample size n is smaller, which is typically the case in SPC. Intuitively, the lower control limit on a chart for σ should be positive (and not zero) so that the control chart is able to detect both increases and decreases in σ. The increase in σ indicates process degradation and is perhaps more important, but a decrease in σ can also be meaningful as it indicates possible improvements in the process. However, the standard 3-sigma limits used in Case K, which are based on the normal approximation to the distribution of $\hat{\sigma}$ can lead to a negative LCL, which is then typically reset to 0. This is, indeed, the case for $n \leq 6$, which includes the most common subgroup used in practice, namely, $n = 5$. Although the 3-sigma limits are perhaps more popular due to their simplicity, note that many authors have advocated their use with charting statistics with a skewed distribution. Montgomery (2009, p. 242) mentions the probability limits and cites Grant and Leavenworth (1980) for the constants needed to set up such control limits. However, note that these limits are correct and appropriate in Case K, the known parameter case. These constants need to be adjusted or adapted for the unknown parameter case, which was recently considered in Diko *et al.* (2016). We present a brief discussion on this important practical problem next.

To this end, note that the Phase II probability limits for the R chart (with estimator $\overline{R}/d_2$) and the S chart (with estimator $\overline{S}/c_4$) in Case U are given by

$$\begin{aligned} UCL &= D_4^*(m,n)\overline{R} \\ CL &= \overline{R} \\ LCL &= D_3^*(m,n)\overline{R} \end{aligned}$$

and

$$\begin{aligned} UCL &= B_4^*(m,n)\overline{S} \\ CL &= \overline{S} \\ LCL &= B_3^*(m,n)\overline{S} \end{aligned}$$

respectively, where the charting statistics are the range R_i and the standard deviation S_i of the ith Phase II sample and the constants (D_3^*, D_4^*) and (B_3^*, B_4^*) are obtained for given values of m, n and a nominal IC average run-length value, such as 370. These constants are provided in Table H in Appendix A. Some details about the derivations of these constants are provided in Appendix 3.3. The value p shown in these tables represents the total tail probability beyond the lower and the upper control limits, under the IC distribution of the charting statistic. This is needed while calculating the probability limits.

Finally, consider $S_p = \sqrt{\sum_1^m S_i^2/m}$, as the "pooled" estimator of σ based on averaging the Phase I sample variances. This is the estimator recommended (see, for example, Mahmoud *et al.*, 2010) in the literature because of its superior statistical properties. The CL and the control limits for the Phase II S chart with S_p used as an estimator of σ in the unknown parameter case can be written as

$$\begin{aligned} UCL &= H_4^*(m,n)S_p \\ CL &= S_p \\ LCL &= H_3^*(m,n)S_p \end{aligned} \tag{3.34}$$

where the charting statistic is S_i and the constants $H_3^*(m,n)$ and $H_4^*(m,n)$ are given in Table H in Appendix A for various values of m and $n = 5$, and 10, and for nominal IC average run-length values of 370 and 500, respectively. Note that one can also use the unbiased version of S_p in the control charts, but it can be seen that the unbiasing constant is approximately equal to one for typical m and n values. Again, some details about the derivations of these constants are provided in Appendix 3.3.

3.8.2 CUSUM Chart for the Mean in Case U

In Case U, the charting constants, k and H, depend on the factors m, n and are obtained for a given nominal IC average run-length; see Table 3.20. The choice of the charting constants for the parametric CUSUM chart is not as straightforward as in Case K. This is shown with an example in Section 3.9.

3.8.3 EWMA Chart for the Mean in Case U

For the EWMA chart in Case U, the charting constants, λ and L, depend on the factors m, n and are obtained for a given nominal IC average run-length; see Table 3.23. As in the case of the CUSUM chart, the choice of the charting constants for the parametric EWMA chart is not as straightforward as in Case K. This is illustrated with an example in Section 3.9.

3.9 TYPES OF PARAMETRIC CONTROL CHARTS IN CASE U: ILLUSTRATIVE EXAMPLES

In this section, we provide some examples of classical parametric, that is, normal theory control charts in Case U.

3.9.1 Charts for the Mean

Example 3.9 A Phase II Shewhart $\overline{X}$ Control Chart for the Mean When Both μ And σ Are Unknown

Column (a) of Table 3.13 presents some simulated data from a normal distribution, which represent measurements taken from $m = 25$ independent random samples, each of size $n = 5$ on a type of wafer. Suppose that these are the reference data from an IC process that were obtained after a careful Phase I analysis. The mean and the standard deviation of each reference sample are shown in Columns (b) and (c) of Table 3.13, respectively.

The estimator of the mean μ is the mean of the Phase I sample means or the grand mean

$$\overline{\overline{X}} = \frac{1.5119 + 1.4951 + \cdots + 1.5264}{25} = 1.5056.$$

As noted above, the estimators $\hat{\sigma}$ used in the control limits are based on either (i) the average of the sample ranges, $\overline{R} = 0.3256$ or (ii) the average of sample standard deviations, $\overline{S} = 0.1316$ or the pooled estimator $S_p = \sqrt{\frac{\sum_{i=1}^{m} S_i^2}{m}} = = 0.1391$. Note that all three estimators of σ are close to each other.

The Phase II control charts for the mean are given below.

$$\begin{aligned} UCL &= \overline{\overline{X}} + k\frac{\hat{\sigma}}{\sqrt{n}} \\ CL &= \overline{\overline{X}} \\ LCL &= \overline{\overline{X}} - k\frac{\hat{\sigma}}{\sqrt{n}}. \end{aligned} \tag{3.35}$$

The charting constants for the Shewhart chart in Case U are tabulated and shown in Table 3.14. For convenience, the same table is given in Table G among the tables in Appendix A. These are based on the works in Chakraborti (2000, 2006) using the unconditional method. From Table G, it can be seen that the charting constant for $m = 25$, $n = 5$ is equal to 2.9725 and 3.0574 for a nominal $ARL_{\text{IC}} = 370$ and 500, respectively. Thus, for

example, the Phase II control limits for the mean using the pooled estimator S_p are given by $UCL = 1.5056 + 2.9725 \times \dfrac{0.1391}{\sqrt{5}} = 1.6905$, $CL = 1.5056$ and $LCL = 1.5056 - 2.9725 \times \dfrac{0.1391}{\sqrt{5}} = 1.3207$, respectively.

It is interesting to examine the entries in Table 3.14. Clearly, the required charting constant is not equal to its counterpart in Case K unless a large number of Phase I samples is used. For example, for $ARL_{\text{IC}} = 370$, only when m is about 100 or more, for $n = 5$, does the charting constant get closer to 3, the value for the 3-sigma limit, which is appropriate for the known parameter case. Also, note that the unconditional ARL_{IC} driven control limits seem to be slightly narrower than those in Case K.

Table 3.13 Measurements used to construct the Shewhart $\overline{X}$ control chart.

Sample	(a)					(b)	(c)	(d)
i	X_1	X_2	X_3	X_4	X_5	$\overline{X}_i$	S_i	R_i
1	1.3235	1.4128	1.6744	1.4573	1.6914	1.5119	0.1635	0.3679
2	1.4314	1.3592	1.6075	1.4666	1.6109	1.4951	0.1111	0.2517
3	1.4284	1.4871	1.4932	1.4324	1.5674	1.4817	0.0565	0.1390
4	1.5028	1.6352	1.3841	1.2831	1.5507	1.4712	0.1389	0.3521
5	1.5604	1.2735	1.5265	1.4363	1.6441	1.4882	0.1412	0.3706
6	1.5955	1.5451	1.3574	1.3281	1.4198	1.4492	0.1168	0.2674
7	1.6274	1.5064	1.8366	1.4177	1.5144	1.5805	0.1614	0.4189
8	1.4190	1.4303	1.6637	1.6067	1.5519	1.5343	0.1077	0.2447
9	1.3884	1.7277	1.5355	1.5176	1.3688	1.5076	0.1439	0.3589
10	1.4039	1.6697	1.5089	1.4627	1.5220	1.5134	0.0988	0.2658
11	1.4158	1.7667	1.4278	1.5928	1.4181	1.5242	0.1548	0.3509
12	1.5821	1.3355	1.5777	1.3908	1.7559	1.5284	0.1682	0.4204
13	1.2856	1.4106	1.4447	1.6398	1.1928	1.3947	0.1699	0.4470
14	1.4951	1.4036	1.5893	1.6458	1.4969	1.5261	0.0937	0.2422
15	1.3589	1.2863	1.5996	1.2497	1.5471	1.4083	0.1568	0.3499
16	1.5747	1.5301	1.5171	1.1839	1.8662	1.5344	0.2423	0.6823
17	1.3680	1.7269	1.3957	1.5014	1.4449	1.4874	0.1432	0.3589
18	1.4163	1.3864	1.3057	1.6210	1.5573	1.4573	0.1289	0.3153
19	1.5796	1.4185	1.6541	1.5116	1.7247	1.5777	0.1195	0.3062
20	1.7106	1.4412	1.2361	1.3820	1.7601	1.5060	0.2230	0.5240
21	1.4371	1.5051	1.3485	1.5670	1.4880	1.4691	0.0819	0.2185
22	1.4738	1.5936	1.6583	1.4973	1.4720	1.5390	0.0832	0.1863
23	1.5917	1.4333	1.5551	1.5295	1.6866	1.5592	0.0922	0.2533
24	1.6399	1.5243	1.5705	1.5563	1.5530	1.5688	0.0431	0.1156
25	1.5797	1.3663	1.6240	1.3732	1.6887	1.5264	0.1482	0.3224

Table 3.14 Charting constants for the Shewhart $\overline{X}$ control chart for the mean in Case UU for $n = 5$, varying m and $ARL_{\text{IC}} = 370$ and 500, respectively, using the unconditional method.

m	$ARL_{\text{IC}} = 370$	$ARL_{\text{IC}} = 500$
5	2.7838	2.8463
10	2.9083	2.9844
20	2.9630	3.0465
25	2.9725	3.0574
30	2.9784	3.0643
35	2.9823	3.0690
40	2.9852	3.0723
45	2.9873	3.0748
50	2.9889	3.0768
75	2.9933	3.0821
100	2.9952	3.0845
200	2.9977	3.0877
300	2.9985	3.0886

Suppose now that at the end of the Phase I analysis, 10 prospective (Phase II) independent random samples are made available. The sample means are shown in Column (b) of Table 3.15.

Figure 3.24 displays the means of these 10 prospective sample means, $\overline{X}_i$, plotted on a Shewhart $\overline{X}$ control chart together with both sets of control

Table 3.15 Phase II data for the Shewhart $\overline{X}$ control chart and charting statistics.

Sample	(a)					(b)	(c)	(d)
i	X_1	X_2	X_3	X_4	X_5	$\overline{X}_i$	R_i	S_i
1	1.4231	1.3457	1.3211	1.4654	1.3845	1.3880	0.5099	0.0581
2	1.4213	1.4620	1.3394	1.6338	1.1239	1.3961	0.2970	0.1863
3	1.4885	1.5635	1.5446	1.5252	1.2665	1.4777	0.2576	0.1213
4	1.4158	1.5190	1.6259	1.3683	1.4117	1.4681	0.2999	0.1041
5	1.5129	1.3825	1.5784	1.3347	1.2785	1.4174	0.5099	0.1249
6	1.4660	1.3371	1.5378	1.6740	1.6637	1.5357	0.3369	0.1413
7	1.7342	1.2176	1.4747	1.6500	1.3623	1.4878	0.5166	0.2097
8	1.4146	1.2826	1.2620	1.3767	1.4036	1.3479	0.1526	0.0707
9	1.2262	1.4307	1.4523	1.5234	1.4574	1.4180	0.2971	0.1127
10	1.4625	1.4568	1.6827	1.4944	1.4810	1.5155	0.2259	0.0947

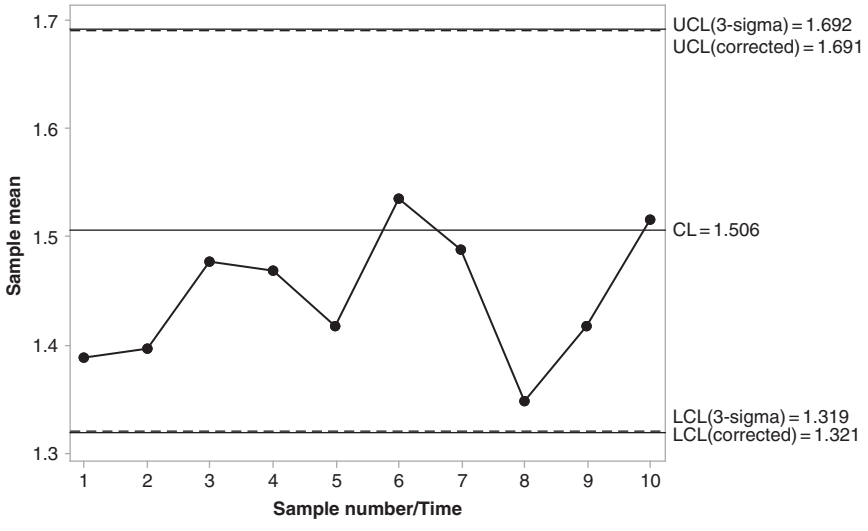


FIGURE 3.24 *A Shewhart $\overline{X}$ control chart for monitoring process mean when both the process mean and standard deviation are unknown.*

limits, the 3-sigma limits, and the adjusted (corrected) limits. This display is typically referred to as a Shewhart $\overline{X}$ control chart or an "X-bar" chart.

Note that the estimator of σ used in the chart was the pooled estimator S_p. As always, once the control chart is drawn, a decision is to be made about the status of the process. In Figure 3.24, none of the sample means plots outside the control limits and the points do not seem to exhibit any non-random pattern. Hence, we conclude that the process is functioning in a state of statistical control with respect to the mean. It is also seen that the two sets of limits are quite close to one another, with the corrected limits being slightly shorter.

3.9.2 Charts for the Standard Deviation

Here we illustrate the R and the S charts which can be used to monitor the spread and the standard deviation, respectively. We start with an example of the Shewhart R chart.

Example 3.10 A Phase II Shewhart *R* Control Chart for the Standard Deviation in the Unknown Parameter Case

The same data that were used to illustrate the Shewhart $\overline{X}$ chart are now used to illustrate the Shewhart R chart in Case U. Recall that Column (a) of Table 3.15 presents the measurements taken from 25 independent Phase I samples on wafers that are each of size ($n = 5$) from a normal distribution. The range of each sample is shown in Column (d) of

Table 3.15. For these data, $\overline{R} = 0.325$. First, for $n = 5$, we find from Table C in Appendix A that $D_3 = 0$ and $D_4 = 2.114$. Thus, with $\overline{R} = 0.325$, the 3-sigma Phase II R chart control limits in Case U for σ are given by $LCL = 0$ and $UCL = 2.114 \times 0.325 = 0.688$ with $CL = 0.325$. However, these control limits do not properly account for parameter estimation and are not expected to be accurate unless one has a large number of Phase I data. Conversely, from Table H in Appendix A, for $m = 25$, $n = 5$ and $ARL_0 = 370$, we find that $D_3^*(25, 5) = 0.16603$ and $D_4^*(25, 5) = 2.32788$. Hence the probability limits-based Phase II Shewhart R control limits for σ, using the estimator $\overline{R}/d_2$, are given by $LCL = D_3^*(m, n)\overline{R} = (0.16603)(0.325) = 0.054$ and $UCL = D_4^*(m, n)\overline{R} = (2.32788)(0.325) = 0.757$, with $CL = \overline{R} = 0.325$. Note that, while the LCL for the 3-sigma R chart is to be reset to 0 as it is negative, the LCL for the probability limits chart is positive and no such adjustment is necessary. Figure 3.25 displays the sample ranges, R_i, of Column (b), which are plotted on a Shewhart R control chart together with the control limits.

In Figure 3.25, all of the points plot between the control limits and there are no anomalous patterns in the data. Hence, we conclude that the process is functioning in statistical control with respect to spread. Note that the adjusted control limits are wider in the unknown parameter case, which is expected since additional variation is introduced through the parameter estimates.

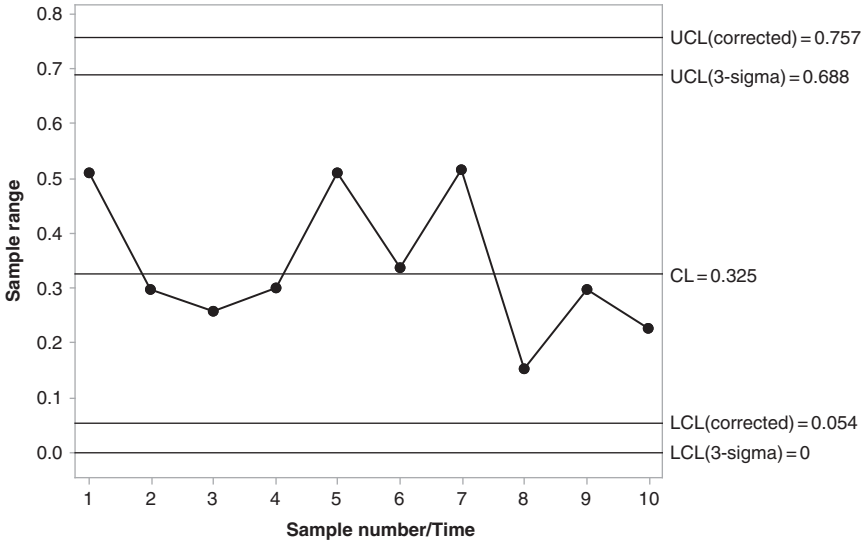


FIGURE 3.25 *Shewhart R chart for monitoring process spread when parameters are unknown.*

Example 3.11 Phase II Shewhart S Control Charts for the Standard Deviation in the Unknown Parameter Case Using the Estimator $\overline{S}$

The same data that were used to illustrate the Shewhart $\overline{X}$ chart and the R chart are now used to illustrate the Shewhart S chart in Case U. Recall that Column (a) of Table 3.15 presents the measurements taken from 25 independent Phase I samples on wafers each of size ($n = 5$) from a normal distribution. The standard deviation of each Phase II sample is shown in Column (c) of Table 3.15. For the standard 3-sigma Shewhart S control chart in Case U, using the estimator $\overline{S}/c_4$, the control limits are obtained (see Table C in Appendix A) as follows. For $n = 5$, we find $B_3 = 0$ and $B_4 = 2.089$. Hence, with $\overline{S} = 0.1316$, the control limits are given by $LCL = 0$ and $UCL = 2.089 \times 0.1316 = 0.2749$ with $CL = 0.1316$. Again, these limits are not advisable for use in practice unless one has a large amount of Phase I data. Instead, from Table H in Appendix A, we find for $m = 25$, $n = 5$, and nominal $ARL_{\text{IC}} = 370$, $B_3^*(25, 5) = 0.1685$ and $B_4^*(25, 5) = 2.2587$. Hence, the probability limits-based Phase II Shewhart S chart, using the estimator $\overline{S}/c_4$, has control limits $LCL = (0.1685)(0.1391) = 0.0234$ and $UCL = (2.2587)(0.1391) = 0.3142$ with $CL = 0.1391$. Notice, again, that the probability limits-based LCL is positive and hence this S chart is able to detect both increases and decreases in the standard deviation.

In Figure 3.26, all of the points plot between the control limits and there are no anomalous patterns in the data. Hence, we conclude that the process

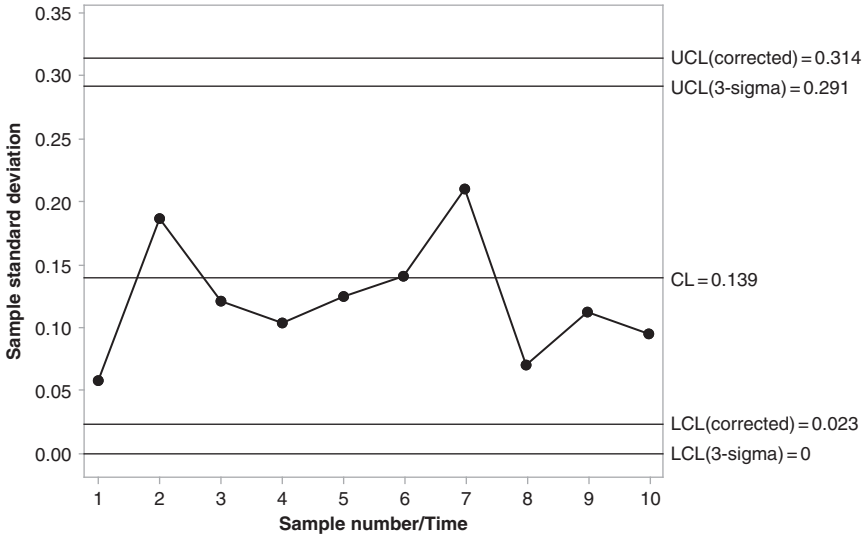


FIGURE 3.26 *Shewhart S chart with $\overline{S}$ as estimator for monitoring process standard deviation when parameters are unknown.*

is functioning in statistical control with respect to variation. Again, note that the control limits are wider in the unknown parameter case, which is expected since additional variation is introduced through the parameter estimates.

3.9.2.1 Using the Estimator S_p

Next, we illustrate the probability limits based Shewhart S chart with S_p as the estimator of σ. Note that, for this data set, the Phase I estimator $S_p = \sqrt{\sum_{i=1}^{m} S_i^2/m} = 0.1391$. From Table H in Appendix A, the needed charting constants are found to be $H_3^*(25, 5) = 0.1581$ and $H_4^*(25, 5) = 2.1240$ for a nominal $ARL_{\text{IC}} = 370$. Thus, the probability limits-based Shewhart S chart is given by $LCL = H_3^*(m, n)S_p = (0.1581)(0.1391) = 0.0220$ and $UCL = H_4^*(m, n)S_p = (2.1240)(0.1391) = 0.2955$ with $CL = 0.1391$. Note that this chart is slightly narrower than the probability limits-based Shewhart S chart using the estimator $\overline{S}/c_4$ shown in Figure 3.26. This is expected since S_p is a more efficient estimator than $\overline{S}/c_4$.

In Figure 3.27, all of the points plot between the control limits and there are no anomalous patterns in the data. Hence, we conclude that the process is functioning in statistical control with respect to variation.

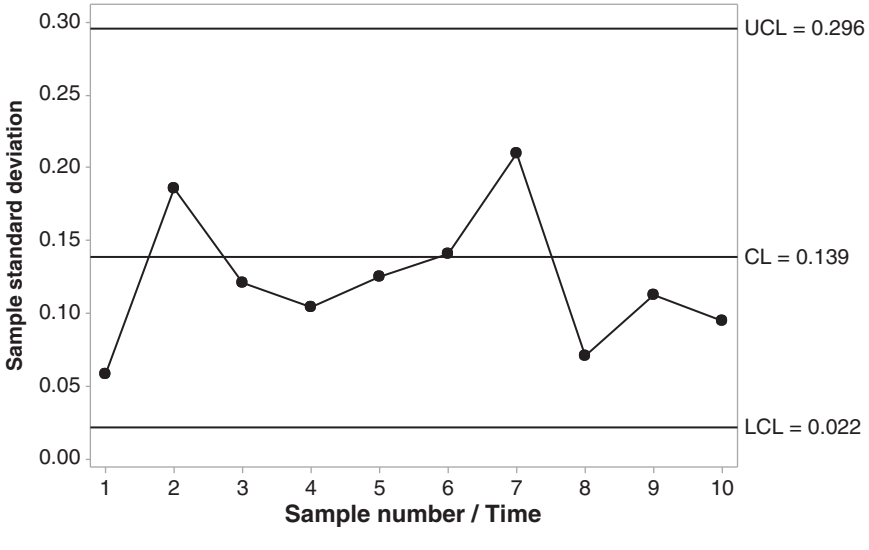


FIGURE 3.27 *Shewhart S chart with S_p as estimator for monitoring process standard deviation when parameters are unknown.*

Example 3.12 A Parametric (Normal Theory) CUSUM Chart for the Mean in the Unknown Parameter Case with Individual Data

Column (a) of Table 3.16 presents 20 observations generated from a process that is normally distributed. Suppose that these are the reference data from an IC process, obtained after a careful Phase I analysis. Suppose then that we are interested in detecting an increase (a shift) in the mean of size 1σ, which is a medium size shift. Since σ is unknown and needs to be estimated, we make use of moving ranges,that is, $MR_i = |x_i - x_{i-1}|$,

Table 3.16 Measurements used to construct the parametric CUSUM control chart for the individual data and the charting statistic.

Period, i	(a) x_i	(b) MR_i
1	9.45	
2	7.99	1.46
3	9.29	1.30
4	11.66	2.37
5	12.16	0.50
6	10.18	1.98
7	8.04	2.14
8	11.46	3.42
9	9.20	2.26
10	10.34	1.14
11	9.03	1.31
12	11.47	2.44
13	10.51	0.96
14	9.40	1.11
15	10.08	0.68
16	9.37	0.71
17	10.62	1.25
18	10.31	0.31
19	8.52	1.79
20	10.84	2.32

since we are working with individual measurements, that is, $n = 1$. The moving ranges are shown in Column (b) of Table 3.16. Then, the estimator of σ is given by $\dfrac{\overline{MR}}{d_2}$ with $d_2 = 1.128$ (see Table C in Appendix A with $n = 2$ since 2 observations are used to calculate the moving range). Thus, we find $\overline{MR} = \dfrac{1.46 + 1.30 + \cdots + 2.32}{19} = 1.550$ so that $\hat{\sigma} = \dfrac{\overline{MR}}{d_2} = \dfrac{1.550}{1.128} = 1.374$. We also need an estimate for the mean, which is given by $\overline{X} = \dfrac{x_1 + x_2 + \cdots + x_{20}}{20} = 9.996$.

Suppose now that at the end of the Phase I analysis that 10 prospective independent measurements are made available.

In his Table 3.1, Hawkins (1993) recommends that, for a shift of about 1σ in the process mean, taking $k = 0.5$ and $h = 4.77$ so that $K = k\hat{\sigma} = 0.5\hat{\sigma}$ and $H = h\hat{\sigma} = 4.77\hat{\sigma}$, gives an $ARL_{\text{IC}} = 370$. In doing so, we find the design parameters $K = k\hat{\sigma} = 0.5 \times 1.374 = 0.687$ and $H = h\hat{\sigma} = 4.77 \times 1.374 = 6.554$, respectively.

We monitor the process mean prospectively in Phase II based on the 10 samples given in Table 3.17. It will be seen from Table 3.18 that the charting constant h for $m = 20$, $n = 1$, $k = 0.5$, and a nominal $ARL_0 = 370$ is equal to 4.1 and not 4.77, as in Case K. At this point, note that the values in Table 3.18 were obtained using simulation, and show the impact of the estimation of parameters on the Phase II control limits. The correct value of H is thus $H = h\hat{\sigma} = 4.1 \times 1.374 = 5.634$. For larger values of m, around 100, the h values seem to converge to their values in Case K.

Table 3.17 Phase II data for the parametric CUSUM control chart.

Period, i	x_i
1	10.90
2	9.33
3	12.29
4	11.50
5	10.60
6	11.08
7	10.38
8	11.62
9	11.31
10	10.52

Table 3.18 The charting constant h for the parametric (Normal Theory) CUSUM control chart for the mean in Case UU for n =1, $k = 0.5$, varying m and $ARL_{\text{IC}} = 370$ and 500.

m	$ARL_{\text{IC}} = 370$	$ARL_{\text{IC}} = 500$
20	4.100	4.300
25	4.390	4.603
50	4.470	4.980
75	4.476	5.132
100	4.476	5.132
125	4.770	5.132
150	4.770	5.132
200	4.770	5.132
250	4.770	5.132
300	4.770	5.132
500	4.770	5.132
1000	4.770	5.120

The CUSUM charting statistics, C_i^+ and C_i^-, are shown in Columns (b) and (d) of Table 3.19, respectively, whereas the corresponding quantities, N^+ and N^-, are given in Columns (c) and (e), respectively. To illustrate the calculations, consider period 1. The charting statistics are calculated as in the parameter known case, except that we use $\overline{X}$ for μ_0. Thus,

$$C_1^+ = \max[0,\ X_1 - \overline{X} - K + C_0^+] = \max[0,\ 10.9 - (9.996 + 0.687) + 0]$$
$$= \max[0, 0.217] = 0.217$$

and

$$C_1^- = \min[0,\ X_1 - \overline{X} + K + C_0^-] = \min[0, 10.9 - 9.996 + 0.687 + 0]$$
$$= \min[0, 1.591] = 0.$$

Figure 3.28 displays the CUSUM statistics, C_i^+ and C_i^-, of Columns (a) and (c), respectively, which are plotted on a parametric CUSUM control chart together with both sets of control limits, one with the 3-sigma limits and the other with the corrected for parameter estimation control limits.

As always, once the control chart is drawn, a decision is to be made about the status of the process. In Figure 3.28, none of the charting statistics plot outside the control limits and the points do not seem to exhibit any non-random pattern. Hence, we conclude that the process is functioning in a state of statistical control with respect to the mean.

Table 3.19 Phase II data for the parametric CUSUM control chart.

Period, i	(a) x_i	(b) C_i^+	(c) N^+	(d) C_i^-	(e) N^-
		0		0	
1	10.90	0.217	1	0	0
2	9.33	0.000	2	0	0
3	12.29	1.607	3	0	0
4	11.50	2.424	4	0	0
5	10.60	2.341	5	0	0
6	11.08	2.738	6	0	0
7	10.38	2.435	7	0	0
8	11.62	3.372	8	0	0
9	11.31	3.999	9	0	0
10	10.52	3.836	10	0	0

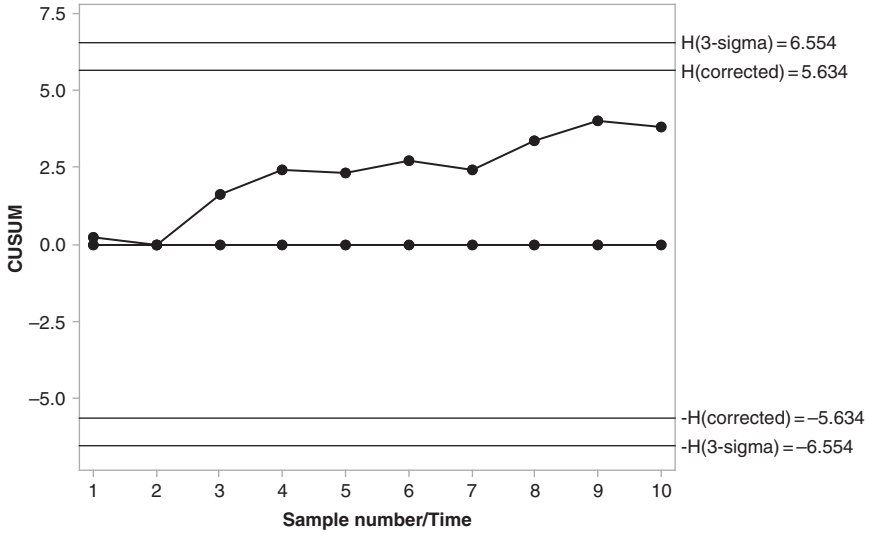


FIGURE 3.28 *A parametric CUSUM control chart for monitoring the process mean when both the process mean and standard deviation are unknown.*

In this example, we've illustrated the parametric CUSUM for individual measurements. However, if rational subgroups of size $n > 1$ are taken, then we would simply replace X_i with $\overline{X}_i$ and σ_0 with $\sigma_{\overline{X}} = \dfrac{\sigma_0}{\sqrt{n}}$ in the previous equations. In this case, the optimal k and h pair may differ from the pair used in this example for individual data. In Table 3.18, the charting constant

Table 3.20 The charting constant h for the parametric CUSUM control chart for the mean in Case U for $n = 5$, $k = 0.5$, for varying m, and $ARL_{\text{IC}} = 370$ and 500, respectively.

m	$ARL_{\text{IC}} = 370$	$ARL_{\text{IC}} = 500$
20	4.290	4.310
25	4.300	4.455
50	4.750	5.050
75	4.900	5.220
100	4.950	5.247
125	4.950	5.250
150	4.950	5.250
200	4.950	5.250
250	4.950	5.250
300	4.900	5.250
500	4.900	5.250
1000	4.900	5.250

h was given for $n = 1$. In Table 3.20, we provide the corresponding values for $n = 5$.

Example 3.13 A Parametric EWMA Chart for the Mean in the Unknown Parameter Case

The same data used to illustrate the parametric CUSUM chart are used to illustrate the parametric EWMA chart in Case U. For illustration purposes, we make use of $\lambda = 0.1$. Montgomery (2009; Table 9.13, p. 425) recommends that, for $\lambda = 0.1$, taking $L = 2.703$ gives an $ARL_{\text{IC}} = 370$.

We monitor the process mean prospectively in Phase II based on the 10 samples given in Table 3.21. It will be seen that, from Table 3.23, the charting constant L for $m = 20$, $n = 1$, $\lambda = 0.1$, and a nominal $ARL_{\text{IC}} = 370$ is equal to 2.650 and not 2.703, as in Case K. It should be noted that the values in Table 3.21 were obtained via simulation and show the impact of the estimation of parameters on the Phase II control limits.

The EWMA charting statistic is given in Column (b) of Table 3.22, whereas the time-varying lower and upper control limits are given in Columns (c) and (d), respectively. We take starting value $Z_0 = \hat{\mu} = 9.996$. To illustrate the calculations, consider observation number 1. The first charting statistic is calculated as follows

$$Z_1 = \lambda X_1 + (1 - \lambda)Z_0 = (0.1)(10.9) + (1 - 0.1)(9.996) = 10.086.$$

Table 3.21 The charting constant L for the parametric EWMA control chart for the mean in Case UU for $n = 1$, $\lambda = 0.05$, 0.10, 0.15, and 0.20, for varying m, and $ARL_{\text{IC}} = 370$ and 500, respectively.

	$ARL_{\text{IC}} = 370$			
m	$\lambda = 0.05$	$\lambda = 0.10$	$\lambda = 0.15$	$\lambda = 0.20$
20	2.600	2.650	2.690	2.710
25	2.629	2.700	2.727	2.737
50	2.659	2.776	2.815	2.840
75	2.650	2.777	2.830	2.855
100	2.640	2.762	2.825	2.863
125	2.632	2.762	2.820	2.858
150	2.618	2.758	2.818	2.854
200	2.595	2.751	2.815	2.852
250	2.575	2.744	2.812	2.850
300	2.570	2.732	2.810	2.849
500	2.535	2.720	2.800	2.846
1000	2.515	2.710	2.787	2.840
	$ARL_{\text{IC}} = 500$			
20	2.700	2.720	2.740	2.760
25	2.718	2.780	2.799	2.820
50	2.777	2.871	2.910	2.920
75	2.777	2.877	2.927	2.947
100	2.765	2.875	2.930	2.962
125	2.760	2.874	2.929	2.961
150	2.752	2.872	9.925	2.960
200	2.730	2.867	2.920	2.959
250	2.715	2.852	2.919	2.958
300	2.710	2.850	2.912	2.958
500	2.665	2.835	2.905	2.955
1000	2.640	2.810	2.895	2.945

Substituting $\hat{\mu} = 9.996$, $\hat{\sigma} = 1.374$, $\lambda = 0.1$, $L = 2.65$, and $i = 1$ into Equation 3.11, the CL and the exact control limits for the data (for $i = 1$) in Table 3.22 are given by

$$UCL = 9.996 + 2.65(1.374)\sqrt{\left(\frac{0.1}{2-0.1}\right)(1-(1-0.1)^{2(1)})} = 10.360$$

$$CL = 9.996$$

$$LCL = 9.996 - 2.65(1.374)\sqrt{\left(\frac{0.1}{2-0.1}\right)(1-(1-0.1)^{2(1)})} = 9.632.$$

Table 3.22 Measurements used to construct the parametric (Normal Theory) EWMA control chart, the charting statistic and the control limits.

Observation number i	(a) X_i	(b) Z_i	(c) LCL_i	(d) UCL_i
		9.996		
1	10.90	10.086	9.632	10.360
2	9.33	10.011	9.506	10.486
3	12.29	10.239	9.424	10.568
4	11.50	10.365	9.366	10.626
5	10.60	10.388	9.322	10.670
6	11.08	10.458	9.288	10.704
7	10.38	10.450	9.262	10.730
8	11.62	10.567	9.242	10.750
9	11.31	10.641	9.226	10.766
10	10.52	10.629	9.213	10.779

The steady-state control limits can be calculated using Equation 3.12 and are given by $LCL = 9.161$ and $UCL = 10.831$, respectively.

Figure 3.29 displays the EWMA statistics, Z_i, given in Column (b) of Table 3.22, plotted on a parametric EWMA control chart together with two

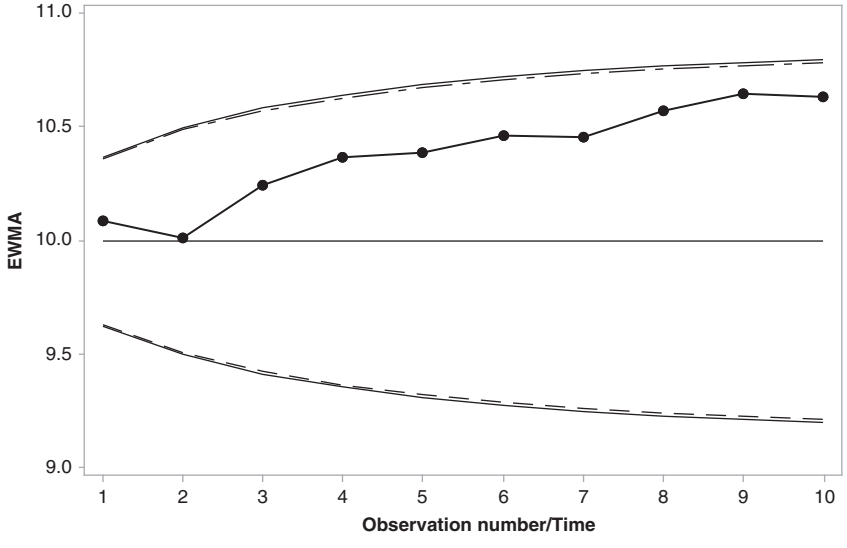


FIGURE 3.29 *A parametric EWMA control chart for monitoring the process mean with 3-sigma and time-varying control limits when both the process mean and standard deviation are unknown.*

Table 3.23 The charting constant L for the parametric (Normal Theory) EWMA control chart for the mean in Case U for $n = 5$, $\lambda = 0.05$, 0.10, 0.15, and 0.20, for varying m, and $ARL_{\text{IC}} = 370$ and 500, respectively.

	$ARL_{\text{IC}} = 370$			
m	$\lambda = 0.05$	$\lambda = 0.10$	$\lambda = 0.15$	$\lambda = 0.20$
20	2.600	2.660	2.675	2.690
25	2.660	2.720	2.721	2.740
50	2.740	2.820	2.845	2.850
75	2.737	2.840	2.879	2.880
100	2.735	2.835	2.877	2.900
125	2.720	2.835	2.875	2.900
150	2.700	2.830	2.873	2.900
200	2.680	2.810	2.870	2.895
250	2.660	2.805	2.862	2.890
300	2.650	2.795	2.855	2.887
500	2.600	2.760	2.845	2.882
1000	2.550	2.730	2.820	2.860
	$ARL_{\text{IC}} = 500$			
20	2.700	2.740	2.750	2.760
25	2.761	2.942	2.810	2.810
50	2.840	2.942	2.960	2.950
75	2.850	2.941	2.985	2.980
100	2.860	2.940	2.985	2.990
125	2.850	2.937	2.985	3.000
150	2.835	2.935	2.985	3.000
200	2.810	2.930	2.977	3.000
250	2.795	2.915	2.960	3.000
300	2.790	2.900	2.955	3.000
500	2.725	2.880	2.932	2.980
1000	2.690	2.850	2.920	2.970

sets of time-varying control limits, one with the 3-sigma limits (solid line) and the other with the corrected for parameter estimation control limits (dotted line).

As always, once the control chart is drawn, a decision is to be made about the status of the process. In Figure 3.29, none of the charting statistics plots outside the control limits and the points do not seem to exhibit any non-random pattern. Hence, we conclude that the process is functioning in a state of statistical control with respect to the mean.

In this example, we've illustrated the parametric EWMA chart for individual measurements. However, if rational subgroups of size $n > 1$ are taken, then simply replace X_i with $\overline{X}_i$ and σ_0 with $\sigma_{\overline{X}} = \dfrac{\sigma_0}{\sqrt{n}}$ in the previous equations. In Table 3.21, the charting constant L was given for $n = 1$. Here we provide a table for $n = 5$.

As discussed in Case K, an important point to keep in mind is that, while monitoring the process mean, the process mean and the standard deviation are both monitored, since the standard deviation appears in the control limits for the mean and thus must be IC when estimated and used in the calculation of the control limits. This is a well-recognized matter in practice, but the impact of this "joint" monitoring is not always explicitly discussed or understood.

3.10 RUN-LENGTH DISTRIBUTION IN THE UNKNOWN PARAMETER CASE (CASE U)

3.10.1 Methods of Calculating the Run-length Distribution and Its Properties: The Conditioning/Unconditioning Method

3.10.1.1 The Shewhart Chart for the Mean in Case U

The conditioning-unconditioning (CUC) method, which was first explicitly coined and used in Chakraborti (2000), is explained here for the parametric Shewhart $\overline{X}$ control chart for the mean, assuming the normal distribution. This development is important since the "standards unknown" case, that is, Case U, is the situation often encountered in practice, and the Shewhart $\overline{X}$ control chart for the mean is one of the most popular charts used in practice. A brief background is given before going into detail. Recall that in the "standards known" case, that is, when process parameters are known (or Case K), the signaling events are mutually independent so that the run-length distribution is geometric with the probability of a success (which, in SPC, is a signal) equal to, say, some θ. This result completely characterizes the performance of the Shewhart control chart in Case K, so that all performance properties of the chart can be obtained from the properties of the $GEO(\theta)$ distribution. Thus, the expected value of the run-length distribution is the reciprocal of θ. For the IC run-length distribution, the ARL_{IC}, simply equals the reciprocal of the FAR, which is the probability of a signal θ when the process is IC. The IC signal probability is denoted by α and therefore, $ARL_{\text{IC}} = 1/\alpha$ so that specifying the FAR specifies the ARL_{IC} and vice

versa. This simple relationship makes understanding the performance of the Shewhart $\overline{X}$ chart easier in Case K. Conversely, when the process parameters are unknown and need to be estimated to set up the control limits before Phase II process monitoring can begin, the signaling events are no longer independent so that the run-length distribution is no longer geometric. As a consequence, for example, the ARL is no longer the reciprocal of the probability of a signal. This is the fundamental conceptual difference between Case K and Case U, which has important implications since some may find it tempting to use the results of Case K to design control charts even when the underlying process parameters are unknown. Practitioners are cautioned against this (see, for example, Quesenberry, 1993) practice and we thus discuss how to handle the situation properly. To this end, we use an important tool called the "conditioning-unconditioning method" proposed in Chakraborti (2000) and used extensively in the literature by many researchers. Note that Chen (1997), among others, has also used similar ideas.

Four cases can arise with the monitoring of the normal mean. First, the simplest case is when both the mean and the standard deviation are known or specified. This is called the standards known case and was referred to as Case K earlier. Henceforth, this is referred to as Case KK, each letter denoting the status/assumption about each parameter, first for the mean and the second for the standard deviation, respectively. Next, perhaps the most common case is when the process mean is specified or known (K) but the process standard deviation is unknown (U). This will be referred to as Case KU. The other two cases are, respectively, Case UK, where the mean is unknown but the standard deviation is known, and finally, the important case, Case UU, where both the process mean and the process standard deviation are unknown. Admittedly, Case UK is more academic and is perhaps less practical, thus we leave the analysis of this case as an exercise for the interested reader.

In each case, the key quantities to be dealt with are the signaling event, the probability of a signaling event, and the run-length distribution. Once the run-length distribution is obtained, chart performance characteristics associated with the run-length distribution, such as the average (expected value), the standard deviation, etc., are obtained easily.

We begin with Case KK for simplicity and to set the stage. In this case, the two-sided Shewhart control limits are given by

$$LCL = \mu_0 - z_{\alpha/2}\frac{\sigma_0}{\sqrt{n}} \text{ and } UCL = \mu_0 + z_{\alpha/2}\frac{\sigma_0}{\sqrt{n}}$$

where μ_0 and σ_0 denote the IC known process mean and standard deviation, respectively. The process is declared OOC or, equivalently, the control chart signals when a charting statistic falls outside of the control limits. Thus, in

Case KK, a signaling event is

$$\overline{X}_i \geq UCL \text{ or } \overline{X}_i \leq LCL,\ i = 1, 2, \ldots$$

or, equivalently, if

$$\frac{\sqrt{n}(\overline{X}_i - \mu_0)}{\sigma_0} \geq z_{\alpha/2} \text{ or } \frac{\sqrt{n}(\overline{X}_i - \mu_0)}{\sigma_0} \leq -z_{\alpha/2},\ i = 1, 2, \ldots$$

Hence, the probability of a signal is

$$\begin{aligned}\theta &= P\left(\frac{\sqrt{n}(\overline{X}_i - \mu_0)}{\sigma_0} \geq z_{\alpha/2}\right) + P\left(\frac{\sqrt{n}(\overline{X}_i - \mu_0)}{\sigma_0} \leq -z_{\alpha/2}\right) \\ &= 1 - \Phi(z_{\alpha/2}) + \Phi(-z_{\alpha/2}).\end{aligned}$$

Thus, in Case KK, the run-length distribution of the Shewhart $\overline{X}$ control chart is geometric with success probability θ, that is, $GEO(\theta)$. This follows from the fact that, for Case KK, the four conditions for the geometric distribution are satisfied, namely, the experiments/trials must be independent Bernoulli trials with a constant probability of success, where a trial/experiment here refers to a comparison of the sample mean to the control limits. Further, at each stage, we observe a Bernoulli trial, which is either a "signal," when $\overline{X} \geq UCL$ or $\overline{X}_i \leq LCL$, which may be classified as a success (S), or a "no signal"; when $LCL < \overline{X}_i < UCL$, which may be classified as a failure (F). Moreover, the probability of success is $P(S) = \theta$ and the trials at different points in time are independent since the samples are drawn independently at each time point. Finally, the run-length refers to the random variable, the number of trials until the first success, among this sequence of independent Bernoulli trials. Thus, the distribution of run-length follows a geometric distribution with success probability $P(S) = \theta$.

Since the run-length distribution is geometric, it is completely characterized by the success probability θ, and hence all properties of the distribution, such as the mean (ARL), the standard deviation ($SDRL$), and the median run-length (MRL), etc., can all be found in terms of θ, for example, the ARL and the $SDRL$ are given by

$$ARL = \frac{1}{\theta} \text{ and } SDRL = \frac{\theta}{(1-\theta)^2}$$

respectively. As noted earlier in Chapter 2, the $100p$th percentile $Q(p)$ can be calculated from

$$Q(p) = inf\{\text{integer } j\colon P(N \leq j) \geq p\} = inf\left\{\text{integer } j \geq \frac{ln(1-p)}{ln(1-\theta)}\right\},\ j \geq 1$$

where *inf* denotes the infimum and $0 < p < 1$ denotes the order of the percentile. Thus, $Q(p)$ is the smallest integer that is at least equal to $\frac{ln(1-p)}{ln(1-\theta)}$. For example, for the median run-length MRL, we have $p = 0.5$, and thus

$$MRL = Q(0.5) = inf\left\{\text{integer } j \geq \frac{\ln(0.5)}{ln(1-\theta)}\right\}$$

$$= inf\left\{\text{integer } j \geq \frac{-\ln(2)}{ln(1-\theta)}\right\}.$$

Next, consider the case when the mean is known or specified and the variance is unknown, that is, Case KU. This situation can arise in problems where the mean is specified by some production or regulatory constraint but the standard deviation of the process is unknown. There is more than one way to start the monitoring process in this situation. Typically, the unknown parameter (here, the variance) is first estimated from a retrospective (Phase I) analysis of data that are already available from the process when it was deemed to be IC. This analysis is called a Phase I analysis and such data are called reference data, which consist of a fixed size sample. Other options could include estimating parameters "on the go" with what are known as self-starting and sequential sampling schemes, but we focus here on the more traditional approach of using a fixed size reference sample. Control charts are used in Phase I to get a process under control, but the construction of control limits is not the same as in Phase II. We discuss control charts for the Phase I analysis later in Chapter 5.

In Phase II, the control limits obtained from Phase I are applied to monitor the status of the process on an ongoing basis. Assume that in Phase I the process follows a normal distribution so that $X_{ij} \sim N(\mu_0, \sigma^2)$ for $i = 1, 2, 3, \ldots, m$ and $j = 1, 2, 3, \ldots, n$. In addition, assume that we estimate the unknown IC process standard deviation σ_0 with the point estimator $S = \sqrt{S^2} = \sqrt{\frac{\sum_{i=1}^{m}(X_i - \mu_0)^2}{m}}$. The two-sided control limits in Case KU are given by substituting (plugging-in) the point estimator in the Case KK control limits

$$\widehat{LCL} = \mu_0 - z_{\alpha/2}\frac{S}{\sqrt{n}} \text{ and } \widehat{UCL} = \mu_0 + z_{\alpha/2}\frac{S}{\sqrt{n}}.$$

In this case, note that the signals and the non-signals refer to events in Phase II, that is, for sample $(m + 1)$ and onward. Assume that the Phase II samples follow a $N(\mu_1, \sigma_0^2)$ distribution. Thus, the signaling event in Phase II can be written as

$$\overline{X}_i \leq \widehat{LCL} \text{ or } \overline{X}_i \geq \widehat{UCL} \text{ for } i = m+1, m+2, m+3, \ldots$$

which can be re-expressed as

$$\frac{\sqrt{n}|\overline{X}_i - \mu_0|}{S} \geq z_{\alpha/2} \text{ for} = m+1, m+2, m+3, \ldots$$

The probability of a signaling event in Case KU is calculated in two steps. First, the probability of a signal is calculated conditionally given the parameter estimator S (or S^2) from Phase I, which is a random variable. Hence, this probability is called the conditional probability of a signal, and plays a key role in the subsequent developments. Note that when substituting the estimate of the standard deviation in the Case KK control limits to construct the Case UU limits, one could use a charting constant from a t distribution with $n-1$ degrees of freedom, particularly since n is usually small. Such an idea has been considered in the literature but we don't pursue these details here.

Let $\mu_1 = \mu_0 + \delta\sigma_1$, where the quantity δ characterizes whether there is a shift in the mean and hence this formulation allows us to handle both the IC ($\delta = 0$) and the OOC ($\delta \neq 0$) cases. Thus, the conditional probability of a signal given S, in Phase II, is

$$P\left(\frac{\sqrt{n}|\overline{X}_i - \mu_0|}{s} \geq z_{\alpha/2}|\mu_1 = \mu_0 + \delta\sigma_1\right) \text{ for} = m+1, m+2, m+3, \ldots$$

This probability can be rewritten as

$$= 1 - P\left(\mu_0 - z_{\alpha/2}\frac{S}{\sqrt{n}} < \overline{X}_i < \mu_0 + z_{\alpha/2}\frac{S}{\sqrt{n}}|\mu_1 = \mu_0 + \delta\sigma_1\right).$$

Now, assuming that, along with the underlying normal distributions, the variances in Phase I and II remain unchanged (i.e., the shift is expected to be only in the mean assuming $\sigma_1 = \sigma_0 = \sigma$), this last probability can rewritten as

$$= 1 - P\left(\frac{\mu_0 - z_{\alpha/2}\frac{S}{\sqrt{n}} - (\mu_0 + \delta\sigma)}{\frac{\sigma}{\sqrt{n}}} < \frac{\overline{X}_i - \mu_1}{\frac{\sigma}{\sqrt{n}}} < \frac{\mu_0 + z_{\alpha/2}\frac{S}{\sqrt{n}} - (\mu_0 + \delta\sigma)}{\frac{\sigma}{\sqrt{n}}}|S\right)$$

$$= 1 - P\left(-z_{\alpha/2}\frac{S}{\sigma} - \delta\sqrt{n} < Z < z_{\alpha/2}\frac{S}{\sigma} - \delta\sqrt{n}|S\right)$$

$$= 1 - P\left(-z_{\alpha/2}\frac{1}{\sqrt{v}}\sqrt{\frac{vS^2}{\sigma^2}} - \delta\sqrt{n} < Z < z_{\alpha/2}\frac{1}{\sqrt{v}}\sqrt{\frac{vS^2}{\sigma^2}} - \delta\sqrt{n}|S\right)$$

$$= \beta(\alpha, \delta, m, n, S), \text{ say.}$$

Finally, writing $Y = \dfrac{vS^2}{\sigma^2}$, which follows a chi-square distribution with $\nu = mn$ degrees of freedom, we can further rewrite the last expression as

$$\beta(\alpha, \delta, m, n, y) = 1 - P\left(-z_{\alpha/2}\sqrt{\frac{Y}{v}} - \delta\sqrt{n} < Z < z_{\alpha/2}\sqrt{\frac{Y}{v}} - \delta\sqrt{n} | Y\right).$$

Hence, since Z follows a $N(0,1)$ distribution, the conditional probability of a signal can be conveniently expressed as

$$\begin{aligned} P_C(Signal) &= \beta(\alpha, \delta, m, n, Y) \\ &= 1 - \Phi\left(z_{\alpha/2}\sqrt{\frac{Y}{v}} - \delta\sqrt{n}\right) - \Phi\left(-z_{\alpha/2}\sqrt{\frac{Y}{v}} - \delta\sqrt{n}\right). \end{aligned}$$

The conditional probability clearly shows that the chart performance will be affected by the value of the random variable Y, which corresponds to the estimator of σ used and obtained from the Phase I data. As noted earlier, the conditional probability of a signal plays an important role in the analysis of the performance of the control chart under estimated parameters. We comment more on this later. However, since this probability is a function of the random variable Y, it is a random variable itself with its own probability distribution. This is the main effect of parameter estimation on the performance of control charts and is crucial to the conceptual understanding. Thus, when the control limits utilize parameter estimate(s), in Phase II, the signaling events are dependent due to the fact that the same Phase I parameter estimate of the standard deviation is used in the control limits and each Phase II sample is compared with the same control limits involving this estimate. Thus, the run-length distribution of the Phase II control chart no longer follows a geometric distribution, since the Bernoulli trials (signal or no signal at each sample or time) involved in this case are no longer independent and hence, typically, one does not use the *FAR* to describe chart performance in this case since the *FAR* depends on the type and the value of the estimates (obtained, for example, from the Phase I analysis). In the IC case $\delta = 0$, and the quantity $\beta(\alpha, 0, m, n, Y)$ denotes the conditional IC probability of a signal, or the conditional *FAR* in Phase II.

Thus, to summarize, in Case KU, that is, when the mean is known or specified but the variance is unknown and hence estimated, for the Shewhart $\overline{X}$ chart,

1. *Conditionally*, the Phase II run-length distribution is geometric with success probability $\beta(\alpha, \delta, m, n, y)$, where α is the nominal *FAR*, $\delta = (\mu_1 - \mu_0)/\sigma$ and y is the realization of a chi-square random variable Y

with $\nu = mn$ degrees of freedom. In this context, note that the corresponding random variable Y is a scaled version of the estimator S^2. Other estimators of σ^2 (such as the one based on the average of the sample ranges) could be considered, but the theory for such estimators, although similar in principle, is somewhat more involved and is thus omitted here.

2. *Conditionally*, the properties of the Phase II Shewhart $\overline{X}$ chart in Case KU can be obtained from the properties of the geometric distribution.

3. So, for example, the conditional run-length distribution is given by its probability mass function (pmf) $P_C(N = j|Y = y) = \beta(y)[1 - \beta(y)]^{j-1}$, $j = 1, 2, \ldots$ where we write, for brevity, $\beta(y) = \beta(\alpha, \delta, m, n, y)$.

Hence, using properties of the geometric distribution, we can completely characterize the conditional run-length distribution in Case KU and calculate various attributes of it, for example, the conditional average run-length (ARL_{C}) of the Phase II chart is given by $[\beta(y)]^{-1}$. Note that, as a function of Y, ARL_{C} is a random variable given by $[\beta(Y)]^{-1}$, with a value $[\beta(y)]^{-1}$, for a value y of Y. Similarly, the variance and the standard deviation of the conditional run-length distribution are both random variables, given by

$$var_C(N|Y) = \frac{1 - \beta(Y)}{[\beta(Y)]^2}$$

and

$$SDRL_C(N|Y) = \frac{\sqrt{(1 - \beta(Y))}}{\beta(Y)}$$

respectively.

The conditional run-length distribution and its associated characteristics, such as the average, the standard deviation, and the percentiles, etc., can help us better understand the effects of parameter estimation on the performance of the Phase II control chart. This is an important point and is common to all Phase II control charts when parameters are estimated, including the Shewhart charts we discuss here. It is because the parameter estimator(s), which is a (are) random variable(s), when simply plugged into the Case KK control limits, significantly alter the performance of the resulting control chart relative to that in Case KK. Moreover, since the estimator is a random variable, two independent reference samples, both from an IC process, would likely produce two different parameter estimates, which in turn will produce two different control charts (limits) and hence would yield different Phase II performance. Thus, there will be variation in the performance of the control charts even though they may use estimates from independent reference samples from the same IC process, and even when the

charts have all been calibrated to achieve the same nominal ARL_{IC}. This performance variation has been called user-to-user variation. Admittedly, this can be unsettling, and various issues come to the surface, particularly from a practitioner's point of view. To this end, several recent articles in the literature have suggested studying the conditional run-length distribution in more detail and hence better understanding the impact of user-to-user variation on Phase II control charts. We cover some of these aspects in later chapters.

While the conditional run-length distribution can help us better understand the effects of parameter estimation, one can further average the conditional distribution over the distributions of the parameter estimator, a process known as "unconditioning," and study the "average" chart performance. For example, the unconditional probability of a signal in Case KU is given by

$$P_{\text{U}}(\text{Signal}) = \int_0^\infty \beta(y) f_{\chi^2_\nu}(y)\, dy$$

where $f_{\chi^2_\nu}(y)$ denotes the pdf of a chi-square distribution with ν degrees of freedom. Similarly, the unconditional or the marginal run-length distribution is given by taking the conditional run-length distribution and averaging over the distributions of the random variable, Y, which are independent for the normal distribution. Hence, we have the unconditional run-length pmf given by

$$P_{\text{U}}(N = j) = \int_0^\infty \beta(y)[1 - \beta(y)]^{j-1} f_{\chi^2_\nu}(y)\, dy \text{ for } j = 1, 2, 3, \ldots$$

which is not the pmf of a geometric distribution.

This type of derivation of the marginal or the unconditional run-length distribution and its associated characteristics is referred to as the conditioning-unconditioning (CUC) method. The derivation proceeds by first conditioning over the parameter estimator and then by taking the expectation of the result with respect to the probability distribution of the estimator.

Thus, in Case KU, using the fact that the conditional run-length distribution is geometric with success probability $\beta(\alpha, \delta, m, n, y)$, and applying the CUC method, the average of the unconditional run-length distribution is given by

$$E_{\text{U}}(N) = \int_0^\infty \frac{1}{[1 - \beta(y)]} f_{\chi^2_\nu}(y)\, dy.$$

This is the *unconditional ARL* of the Shewhart $\overline{X}$ control chart denoted by ARL_{U}. Note that the same result could be obtained from the unconditional run-length pmf $P_{\text{U}}(N = j)$ given earlier.

In addition to the *ARL*, the variance (or the standard deviation) of the unconditional run-length distribution is an important performance measure.

The derivation of the unconditional variance is interesting and, as such, is explained further. The derivation is facilitated by using the result mentioned in Chapter 1, that for any random variable N, the unconditional variance can be obtained using the conditional mean, the conditional variance, and the formula

$$var_U(N) = E(var(N|Y)) + var(E(N|Y))$$

(see also, for example, Hogg, McKean, and Craig, 2005) where the expectation is taken over the distribution of the random variable Y. For the $var(N|Y)$ and the $E(N|Y)$ terms, recall that, given Y, the run-length distribution is geometric with probability $\beta(Y)$ and hence from the properties of the geometric distribution

$$var(N|Y) = \frac{\beta(Y)}{(1-\beta(Y))^2}$$

and

$$E(N|Y) = \frac{1}{1-\beta(Y)}$$

respectively. Thus,

$$\begin{aligned} var(E(N|Y)) &= E((E(N|Y))^2) - (E(E(N|Y)))^2 \\ &= E\left(\left(\frac{1}{1-\beta(Y)}\right)^2\right) - \left(E\left(\frac{1}{1-\beta(Y)}\right)\right)^2 \end{aligned}$$

and by substitution we get

$$var_U(N) = E\left(\frac{\beta(Y)}{(1-\beta(Y)^2}\right) + E\left(\left(\frac{1}{1-\beta(Y)}\right)^2\right) - \left(E\left(\frac{1}{1-\beta(Y)}\right)\right)^2.$$

Rewriting this expression in terms of integrals, we obtain the unconditional variance of the run-length distribution in Case KU

$$var_U(N) = \int_0^\infty \frac{1+\beta(y)}{(1-\beta(y))^2} f_{\chi^2_\nu}(y)\,dy + \left(\int_0^\infty \frac{1}{(1-\beta(y))} f_{\chi^2_\nu}(y)\,dy\right)^2$$

where the reader should recall that $\beta(y) = \beta(\alpha, \delta, m, n, y)$, as defined earlier. Note that, although the run-length distribution is always discrete, the monitored variable, and therefore the corresponding parameter estimator(s), can be continuous (as in the present case) as well as discrete, as in the case of attributes charts (see Montgomery, 2009, Chapter 7). In the latter case, the estimator Y will be discrete and thus the unconditional run-length distribution, its average, and the variance can be obtained in a similar manner using the CUC method, except that now the integral will be replaced by

summation over the mass points of the estimator Y, and the chi-square pdf in the integration will be replaced by the appropriate pmf of Y. The reader is referred to Human and Chakraborti (2006) for examples.

Other moments and characteristics of the run-length distribution can also be obtained similarly using the CUC method, which takes advantage of the facts that (i) the conditional run-length given the parameter estimate is the geometric distribution and (ii) the unconditioning involves averaging (taking expectation) over the distribution of the parameter estimator, which follows some known probability distribution.

Typically, one refers to the unconditional run-length distribution as *the* run-length distribution of a control chart, but the conditional run-length distribution is gaining more prominence as it captures and explains the impact of the inherent variation in the parameter estimators on the performance of the control chart. The reader is referred to the recent literature for more details.

In summary, while the unconditional run-length properties, such as the mean and the standard deviation, are fixed, and describe the average performance of the control chart over many reference samples, the conditional run-length properties vary based on the values of the parameter estimators for a given reference sample and show the impact of parameter estimation based on the value of the estimator that the users have for their particular situation. Both the conditional and the unconditional performance may have a role to play in the practical implementation of control charts with estimated parameters. The Case KK run-length distribution and its associated characteristics have been the benchmarking norm in the design of a control chart for over 50 years, until recently, when the unconditional run-length and its associated characteristics gained prominence in light of the effects of parameter estimation, but, as noted earlier, it is recognized that, although the averaging process may produce some useful overall guidance, the conditional run-length and its associated characteristics may be very useful while implementing and understanding chart performance in practice when estimated parameters are used in control charts. The same summary applies to all types of control charts, univariate or multivariate, variables or attributes, parametric or nonparametric, where unknown parameters are estimated from the data.

Next, consider the case when both parameters are unknown, that is, Case UU. The analysis is similar to that in Case KU, except that now two parameters and their corresponding estimators are involved, which raises the complexity. We will try to point out the similarities with Case KU and highlight the main differences. As we noted earlier, in Phase II, the parameter estimators and hence the control limits are obtained from Phase I and are applied to monitor the status of the process in Phase II on an ongoing basis. Again, as in Case KU, assume that in Phase I the process follows a normal distribution so that $X_{ij} \sim N(\mu_0, \sigma_0^2)$ for $i = 1, 2, 3, \ldots, m$

and $j = 1, 2, 3, \ldots, n$. In addition, we assume that we estimate the unknown IC process mean μ_0 and the unknown IC process standard deviation σ_0 with the overall mean which is the grand mean of the means of the reference samples

$$\overline{\overline{X}} = \frac{1}{m}\sum_{i=1}^{m}\overline{X}_i = \frac{1}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n}X_{ij}$$

and the pooled estimator

$$S_p = \sqrt{\frac{\sum_{i=1}^{m} S_i^2}{m}}.$$

Thus, unlike in Case KU, two estimators, namely $\overline{\overline{X}}$ and S, are involved in the analysis in Case UU. Note also that, unlike in Case KU, the estimator of σ^2, namely S^2, has a scaled chi-square distribution but with $m(n-1)$, degrees of freedom. Thus, one loses m degrees of freedom (recall that in Case KU the degree of freedom was mn) while estimating the mean μ using the m Phase I samples.

Again, the two-sided control limits are given by substituting the point estimators in the control limits for Case KK

$$\widehat{LCL} = \overline{\overline{X}} - z_{\alpha/2}\frac{S_p}{\sqrt{n}} \text{ and } \widehat{UCL} = \overline{\overline{X}} + z_{\alpha/2}\frac{S_p}{\sqrt{n}}\,.$$

As before, assume that the Phase II samples follow an $N(\mu_1, \sigma_1^2)$ distribution. Thus, the signaling events in Phase II can be written as

$$\overline{X}_{\boldsymbol{i}} < \widehat{LCL} \text{ or } \overline{X}_{\boldsymbol{i}} > \widehat{UCL} \text{ for } i = m+1, m+2, m+3, \ldots$$

which can be re-expressed as

$$\frac{\sqrt{n}|\overline{X}_i - \overline{\overline{X}}|}{S_p} \geq z_{\alpha/2} \text{ for} = m+1, m+2, m+3, \ldots$$

Compare this signaling event with that in Case KU. As in Case KU, under the CUC method, the probability of a signaling event in Case UU is calculated in two steps. First, the probability of a signal is calculated given (conditioned on) the two parameter estimators, $\overline{\overline{X}}$ and S_p, which are both random variables. This probability is called the conditional probability of a signal and it plays a key role in the subsequent developments.

To show this, we note that the probability of a signal in Case UU is given by $P\left(\frac{\sqrt{n}|\overline{X}_i - \overline{\overline{X}}|}{S_p} \geq z_{\alpha/2}\right)$ for $= m+1, m+2, m+3, \ldots$ This can

be written as

$$= 1 - P\left(\frac{-z_{\alpha/2}S_p}{\sqrt{n}} + \overline{\overline{X}} < \overline{X}_i < \frac{z_{\alpha/2}S_p}{\sqrt{n}} + \overline{\overline{X}}\right)$$

$$= 1 - P\left(-\frac{z_{\alpha/2}S_p}{\sigma_1} + \frac{\sqrt{n}(\overline{\overline{X}} - \mu_0)}{\sigma_1} - \frac{\sqrt{n}(\mu_1 - \mu_0)}{\sigma_1} < \frac{\sqrt{n}(\overline{X}_i - \mu_1)}{\sigma_1}\right.$$

$$\left. < \frac{z_{\alpha/2}S_p}{\sigma_1} + \frac{\sqrt{n}(\overline{\overline{X}} - \mu_0)}{\sigma_1} - \frac{\sqrt{n}(\mu_1 - \mu_0)}{\sigma_1}\right).$$

Now, writing $Z = \dfrac{\sqrt{mn}(\overline{\overline{X}} - \mu_0)}{\sigma_0}$, $Z_1 = \dfrac{\sqrt{n}(\overline{X}_i - \mu_1)}{\sigma_1}$, and $\delta = \dfrac{(\mu_1 - \mu_0)}{\sigma_0}$, we can rewrite the above probability as

$$= 1 - P\left(\frac{\sigma_0}{\sigma_1}\left(-\frac{z_{\alpha/2}S_p}{\sigma_0} + \frac{Z}{\sqrt{m}} - \delta\sqrt{n}\right) < Z_1\right.$$

$$\left. < \frac{\sigma_0}{\sigma_1}\left(\frac{z_{\alpha/2}S_p}{\sigma_1} + \frac{Z}{\sqrt{m}} - \delta\sqrt{n}\right)\right).$$

Finally, writing $Y = \dfrac{vS_p^2}{\sigma_0^2}$, which follows a chi-square distribution with $\nu = m(n-1)$ degrees of freedom and denoting $\tau = \dfrac{\sigma_0}{\sigma_1}$, we can further rewrite the last expression as

$$= 1 - P\left(\left(-z_{\alpha/2}\sqrt{\frac{Y}{v}} + \frac{Z}{\sqrt{m}} - \delta\sqrt{n}\right)\tau < Z_1\right.$$

$$\left. < \left(z_{\alpha/2}\sqrt{\frac{Y}{v}} + \frac{Z}{\sqrt{m}} - \delta\sqrt{n}\right)\tau\right).$$

When $\sigma_1 = \sigma_0$ the quantity $\tau = 1$. Finally, since Z_1 follows a $N(0,1)$ distribution, the conditional probability of a signal in Phase II for the mean chart in Case UU is given by

$$1 - \Phi\left(\left(z_{\alpha/2}\sqrt{\frac{Y}{v}} + \frac{Z}{\sqrt{m}} - \delta\sqrt{n}\right)\tau\right)$$

$$+ \Phi\left(\left(-z_{\alpha/2}\sqrt{\frac{Y}{v}} + \frac{Z}{\sqrt{m}} - \delta\sqrt{n}\right)\tau\right) = \beta(\alpha, \delta, \tau, m, n, z, y)$$

where Z and Y are independent random variables. The variable Z follows a standard normal distribution and Y follows a chi-square distribution with $v = m(n-1)$ degrees of freedom. Again, other estimators of σ^2 could be considered, and one would have to work with its distribution in the sequel. We leave this up to the reader.

In any case, also in Case UU, the conditional probability of a signal plays an important role in the analysis of the performance of the control chart under estimated parameters. We comment more on this later. In the IC case, we have $\delta = 0$ and under the typical assumption that $\sigma_1 = \sigma_0$, that is, $\tau = 1$, the quantity $\beta(\alpha, 0, 1, m, n, z, y)$ denotes the IC conditional probability of a signal (the *FAR*) in Phase II. However, since this probability is a function of random variables Z and Y, it is a random variable with its own probability distribution. This is why, as we noted earlier, when parameters are estimated in Phase II, typically, one does not use the *FAR* to describe chart performance since the *FAR* depends on the type of estimates obtained from the Phase I analysis and is itself a random variable with a probability distribution.

Moreover, as in Case KU, the Phase II signaling events are dependent in Case UU due to the fact that the same Phase I parameter estimates in the control limits and each Phase II sample is compared with the same control limits involving these estimates. Thus, in Case UU, as in Case KU, the run-length distribution of the Phase II control chart no longer follows a geometric distribution since the Bernoulli trials involved in this case are no longer independent. However, again, as in Case KU, given the Phase I parameter estimators (or conditioning over the corresponding random variables), the signaling events are independent. This observation facilitates a lot of the statistical calculations of the Phase II chart properties in Case UU, as it did in Case KU. Thus, to summarize, in Case UU, for the Shewhart $\overline{X}$ chart,

1. The *conditional* Phase II run-length distribution is geometric with success probability $\beta(\alpha, \delta, \tau, m, n, z, y)$.

2. Hence, conditionally, properties of the Phase II Shewhart $\overline{X}$ chart follow from the properties of the geometric distribution, for example, the conditional run-length distribution is given by its probability mass function (pmf)

$$P_{\mathrm{C}}(N = j|z, y) = \beta(z, y)[1 - \beta(z, y)]^{j-1},\ j = 1, 2, \ldots$$

   where we write, for brevity, $\beta(z, y) = \beta(\alpha, \delta, \tau, m, n, z, y)$. Now, using properties of the geometric distribution, we can easily obtain various attributes of the conditional run-length distribution. The conditional average run-length (ARL_{C}) of the Phase II chart, for

example, is given by $[\beta(z,y)]^{-1}$. Similarly, the variance and the standard deviation are given by $var_C(N|Z,Y) = \dfrac{1-\beta(Z,Y)}{[\beta(Z,Y)]^2}$ and $SDRL_C(N|Z,Y) = \dfrac{\sqrt{(1-\beta(Z,Y))}}{\beta(Z,Y)}$, respectively.

As noted earlier, parameter estimates plugged into the control limits can significantly alter/affect the performance of the control chart and the conditional run-length distribution. Its associated characteristics, such as the average, the standard deviation, etc., can help to describe the effects of parameter estimation on the performance of the Phase II control chart. The estimators are random variables, and two independent reference samples from the same IC process would most likely produce two different sets of parameter estimates, which in turn will produce different control charts (limits), and hence different Phase II chart performance. The bottom line is that there will be variation in the performance of the control charts among users, when they use estimates from independent reference samples from the same IC process, and even though the control charts may have been calibrated to achieve the same nominal ARL_{IC}. This variation may be termed as user-to-user variation, which is not easy to control unless there are lots of data, which may not be practical in all situations. To this end, as noted before, several recent articles have suggested studying the conditional run-length distribution and the impact of user-to-user variation on Phase II control charts. We cover some of these aspects in later chapters.

Averaging over the distributions of the parameter estimates, a process known as "unconditioning," we can study the "average" chart performance, for example, the unconditional probability of a signal in Case UU is given by

$$P_{\text{U}}(\text{Signal}) = \int_{-\infty}^{\infty}\int_{0}^{\infty} \beta(z,y)g(z,y)\,dydz = \int_{-\infty}^{\infty}\int_{0}^{\infty} \beta(z,y)\phi(z)f_{\chi^2_\nu}(y)\,dydz$$

where $f_{\chi^2_\nu}(y)$ denotes the pdf of a chi-square distribution with ν degrees of freedom. Note that for the normal distribution the estimators of the mean and the variance are statistically independent. Therefore, the joint distribution factors into the product of the individual distributions, one a standard normal and one a chi-square. This may not be the situation in all cases, however. Similarly, using the CUC method, the unconditional run-length pmf can be obtained by integrating the conditional run-length pmf over the joint pdf of Z and Y, which is the product of the marginal pdf's of Z and Y, respectively since Z and Y are independent. Hence

$$P_{\text{U}}(N=j) = \int_{-\infty}^{\infty}\int_{0}^{\infty} \beta(z,y)[1-\beta(z,y)]^{j-1}\phi(z)f_{\chi^2_\nu}(y)\,dydz \text{ for } j = 1,2,3,\ldots$$

Similarly, using the fact that the conditional run-length distribution is geometric with a success probability $\beta(z,y)$, and applying the CUC method,

the expected value of the unconditional run-length distribution is given by

$$E_{\mathrm{U}}(N) = \int_{-\infty}^{\infty}\int_{0}^{\infty} \frac{1}{[1-\beta(z,y)]}\phi(z) f_{\chi^2_\nu}(y)\, dydz.$$

This is the unconditional ARL of the Shewhart $\overline{X}$ control chart in Case UU. In general, for higher-order moments, say, the kth non-central moment of the run-length distribution can be obtained from the kth conditional non-central moment, that is, $E(N^k) = E(E(N^k|Z,Y))$ using the fact that $N|Z,Y$ follows a geometric distribution. In particular, the variance (or the standard deviation) of the unconditional run-length distribution is an important performance measure. This can be shown to be equal to $var_U(N) =$

$$\int_{-\infty}^{\infty}\int_{0}^{\infty} \frac{1+\beta(z,y)}{(1-\beta(z,y))^2}\phi(z) f_{\chi^2_\nu}(y)\, dydz - \left(\int_{-\infty}^{\infty}\int_{0}^{\infty} \frac{1}{(1-\beta(z,y))}\phi(z) f_{\chi^2_\nu}(y)\, dydz\right)^2.$$

The derivation of the unconditional variance (and standard deviation) is interesting and is explained further. This follows from the result that for any random variable N, $var_U(N) = E(var(N|Z,Y)) + var(E(N|Z,Y))$ (see, for example, Hogg, McKean, and Craig, 2005) where the expectation is taken over the distributions of the two independent random variables Z and Y. Furthermore, under the assumption of a normal distribution, Z and Y are independent. For the $var(N|Z,Y)$ and the $E(N|Z,Y)$ terms, we note that, given Z and Y, the run-length distribution is geometric with probability $\beta(Z,Y)$ and hence from the properties of the geometric distribution, $var(N|Z,Y) = \dfrac{\beta(Z,Y)}{(1-\beta(Z,Y))^2}$ and $E(N|Z,Y) = \dfrac{1}{1-\beta(Z,Y)}$. Thus

$$\begin{aligned} var(E(N|Z,Y)) &= E((E(N|Z,Y))^2) - (E(E(N|Z,Y)))^2 \\ &= E\left(\left(\frac{1}{1-\beta(Z,Y)}\right)^2\right) - \left(E\left(\frac{1}{1-\beta(Z,Y)}\right)\right)^2 \end{aligned}$$

and by substitution we get

$$var_U(N) = E\left(\frac{\beta(Z,Y)}{(1-\beta(Z,Y))^2}\right) + E\left(\left(\frac{1}{1-\beta(Z,Y)}\right)^2\right) - \left(E\left(\frac{1}{1-\beta(Z,Y)}\right)\right)^2.$$

Rewriting this expression in terms of integrals, we obtain the unconditional variance

$$var_U(N) = \int_{-\infty}^{\infty}\int_{0}^{\infty} \frac{\beta(Z,Y)+1}{(1-\beta(Z,Y))^2}\phi(z)f_{\chi^2_\nu}(y)\,dydz + \int_{-\infty}^{\infty}\int_{0}^{\infty} \left(\frac{1}{1-\beta(Z,Y)}\right)^2 \phi(z)f_{\chi^2_\nu}(y)\,dydz$$

shown above. Hence, the unconditional run-length standard deviation is given by $SDRL_U(N) = \sqrt{var_U(N)}$.

Typically, one refers to the unconditional run-length distribution as *the* run-length distribution of a control chart, which has been used in the traditional evaluation of a control chart performance. However, the conditional run-length distribution has gained more prominence recently as it captures and explains the impact of the inherent variation in the parameter estimators on the performance of the control chart. Other moments and characteristics of the run-length distribution can be obtained similarly using the CUC method, which takes advantage of the facts that (i) the conditional run-length given the parameter estimates is the geometric distribution, and (ii) the unconditioning involves averaging (taking expectation) over the distributions of the two parameter estimators, which are independent, and follows some known probability distributions. In summary, while the unconditional run-length properties, such as the mean and the standard deviation, are fixed and describe the average performance of the control chart over many samples, the conditional run-length properties are not and show the impact of parameter estimation based on the estimators that the users have for their situation. Both conditional and unconditional performance may have a role to play in the practical implementation of control charts with estimated parameters.

For Case UK, the probability of a signal, the conditional run-length distribution, its various moments, the unconditional run-length distribution, and its various moments can be obtained similarly and are left for the reader as exercises. More details can be found in Chakraborti (2000).

Having considered the Shewhart chart for the mean in detail, we are now in a better position to study some of the more advanced or sophisticated charts. These include the CUSUM and the EWMA charts, and Shewhart charts with supplementary runs-rules, which are expected to be more effective in detecting small, persistent changes in the underlying parameter. The calculation of the run-length distribution and associated properties for these charts are more involved in Case U, but one can again apply the CUC method. This is briefly outlined as follows, first in general terms. Recall that, for these charts, expressions for the run-length distribution and some of the associated characteristics in Case K were given in Equations 3.21–3.24

using the MC approach. This is the starting point in Case U. As we noted earlier, a key to using these expressions is the construction of the essential transition probability matrix for each chart. In Case U, we first use these formulas conditionally given the parameter estimates, and then apply the CUC method to find the unconditional run-length distribution and its associated properties. Suppose, for example, that there is a single unknown parameter to be estimated with the corresponding Phase I estimator T. Now we calculate the elements of the essential probability matrix Q, the so-called transition probabilities, by using the conditional distribution of the charting statistic in Phase II, given the estimator T. Thus, each element of the transition probability matrix is calculated conditional on T, and let $Q(T)$ denote the resulting conditional essential probability matrix. Next, we can apply the expressions given in Equations 3.22–3.24 conditionally on T, and get

$$P(N = t|T) = \underline{\xi}Q(T)^{t-1}(I - Q(T))\underline{1} \text{ for } t = 1, 2, 3, \ldots \tag{3.36}$$

$$E(N|T) = \underline{\xi}(\mathrm{I} - Q(T))^{-1}\underline{1} \tag{3.37}$$

$$SD(N|T) = \sqrt{\underline{\xi}(I + Q(T))(I - Q(T))^{-2}\underline{1} - (E(N|T))^2} \tag{3.38}$$

$$P(N \leq j|T) = 1 - \underline{\xi}Q(T)^{j}\underline{1} \text{ for } j = 1, 2, 3, \ldots \tag{3.39}$$

Finally, using the CUC method, the unconditional run-length distribution is obtained by taking the expectation (averaging or integrating) over the distribution of T. So, for example, the unconditional ARL of the chart is given by

$$\int_{-\infty}^{\infty} \{\underline{\xi}[\mathrm{I} - Q(t)]^{-1}\underline{1}\} f(t)dt \tag{3.40}$$

where T is assumed to be continuous with a pdf $f(t)$. Other moments and associated run-length distribution characteristics can be found in a similar manner. Note that a similar analysis is possible if case T is discrete, in which case one would obtain these expressions for the unconditional run-length distribution by summing over the pmf $f(t)$ of T. Interested readers are asked to try out some examples. Some examples can be found in Chakraborti, Eryilmaz and Human (2009).

3.10.1.2 The Shewhart Chart for the Variance in Case U

The run-length distribution of the Shewhart charts for the variance can be obtained similarly, as in the case of the mean, using the CUC method. We

sketch a brief example here for the S chart with S_p as the estimator of σ and derive the IC run-length distribution.

Note that, in this case, the probability limits of the Phase II chart are given by $LCL = H_3^*(m,n)S_p$, $CL = S_p$, and $UCL = H_4^*(m,n)S_p$, where $H_3^*(m,n) = \sqrt{\chi^2_{\alpha(m,n)/2,n-1}/(n-1)}$, $H_4^* = \sqrt{\chi^2_{1-\alpha(m,n)/2,n-1}/(n-1)}$, and $\alpha(m,n)$ is the unconditional probability that the Phase II charting statistic, the standard deviation, S_i plots outside either the lower or the upper control limit when the process is IC. Note that, in the parameter known case, this is the probability of a false alarm or the *FAR* and it only depends on the sample size n, but in the unknown parameter case, the probability $\alpha(m,n)$ depends on both m and n. Once the quantity $\alpha(m,n)$ is determined, the probability limits are easily obtained using the corresponding percentiles of the chi-square distribution with $n-1$ degrees of freedom. Although we may refer to $\alpha(m,n)$ as the *FAR* (probability of a signal when the process is IC) in Case U, for convenience, note that it does not have the same clear interpretation as in Case K since the signaling events are dependent in Case U. Some details about the derivation of $\alpha(m,n)$ are given in Appendix 3.3. The $\alpha(m,n)$ values are calculated and shown in Table H in Appendix A along with the corresponding control limits $H_3^*(m,n)$ and $H_4^*(m,n)$ for $n = 5$ and 10 and some values of m from 5 to 100, with nominal ARL_{IC} values of 370 and 500. Note that the $\alpha(m,n)$ values depend both on m and n, and for a given value of n, they increase for increasing values of m (increasing the number of Phase I samples).

Other moments of the conditional and the unconditional IC run-length distribution can be found in a similar way. Note that when other estimators of σ, such as the one based on the $\overline{R}$ or the $\overline{S}$ are used in the Phase I control limits, or a different chart such as the R chart is used in Phase II, a similar approach can be used to derive the run-length distribution and its various moments by applying the CUC method. We basically need the probability distributions of the charting statistic and of the Phase I estimator of σ. Note that the same approach can be also used to derive the OOC run-length distribution, which is useful in studying the performance of these charts in shift detection and comparisons. We leave the details to the reader.

3.10.1.3 The CUSUM Chart for the Mean in Case U

The reader is referred back to Example 3.6 in Section 3.6.1.2. In this example, the underlying process distribution is assumed to be $N(0,1)$. Let us consider, for example, the one-step transition probability, p_{00}. It was shown that $p_{00} = P(X_t \leq 1) = 0.841$. However, recall that the $P(X_t \leq 1) = 0.841$ was obtained using the cdf of a $N(0,1)$ distribution, which requires not only the knowledge of the normality of the process

distribution but also that the mean and the standard deviation, assumed to be 0 and 1, respectively. Suppose now that we are in Case KU, that is, the process mean is known (say, is equal to zero), but the process variance is unknown. In this case, we have a single unknown parameter σ to be estimated by some estimator T (which is typically obtained from a Phase I analysis) and we calculate the elements of the essential probability matrix Q, that is, the transition probabilities, conditionally, given the estimator T. Now $p_{00} = P(X_t \leq 1)$ is calculated as $\hat{p}_{00} = P(X_t \leq 1|T) = \Phi(1/T)$ where T is some estimator of σ. The same applies to the other one-step transition probabilities. Thus, $Q(T)$, the estimated conditional essential probability matrix, is obtained and substituted into Equations 3.36–3.39. Finally, using the CUC method, the unconditional run-length distribution is obtained by taking expectation over the distribution of T, as explained above and illustrated in Equation 3.40 for the unconditional *ARL*. Case UK follows similarly. However, this time the process mean needs to be estimated and dealt with and, for Case UU, where both the process mean and the process variance need to be estimated, we condition on, say, T_1 and T_2, where T_1 denotes the estimator for the process mean and T_2 denotes the estimator for the process variance. The reader is also referred to Jones, Champ, and Rigdon (2004) for an alternative derivation of the run-length distribution of the CUSUM chart with estimated parameters.

3.10.1.4 The EWMA Chart for the Mean in Case U

The reader is referred back to Example 3.7 in Section 3.6.1.2. In this example, the underlying process distribution is assumed to be $N(0,1)$. Let us consider the one-step transition probability, p_{01}, in that example. It was shown that p_{01} is equal to 0.226 but $0.226 = \Phi(2.067) - \Phi(0.689)$ is obtained using the cdf of a $N(0,1)$ distribution. However, even when we assume normality, one or more of the parameters can be unknown and would need to be estimated. In such a case, we proceed as in the case of the CUSUM chart discussed above, and calculate the elements of the transition probability matrix conditioned on the parameter estimators. Then, the resulting conditional essential probability matrix is obtained and substituted into Equations 3.36–3.39. Finally, using the CUC method, the unconditional run-length distribution is obtained by taking expectation over the distribution of the estimator(s), as explained above and illustrated in Equation 3.40 for the unconditional *ARL*. The reader is referred to Jones, Champ, and Rigdon (2001) for a detailed discussion on the run-length distribution of the EWMA chart with estimated parameters.

In the unknown parameter case, the CUC method has been successfully used not only for various parametric charts but also for nonparametric charts, as will be seen in the next chapter.

3.11 CONTROL CHART ENHANCEMENTS

3.11.1 Run-length Calculation for Runs-type Signaling Rules in Case U

Zhang and Castagliola (2010) considered the Shewhart $\overline{X}$ chart with runs rules when the process parameters are unknown. Recall that, if the IC process mean and standard deviation μ_0 and σ_0 are both unknown, in Case UU, they are estimated from a Phase I reference data set of size m ($\geq$1) subgroups, each of size n ($\geq$1). We can use the commonly used estimators $\widehat{\mu}_0 = \overline{\overline{X}}$ and $\widehat{\sigma}_0 = S_p$ where recall that $\overline{\overline{X}}$ is the mean of the Phase I sample means and S_p is the so-called pooled estimator of the standard deviation. Let $\{Y_{i1}, \ldots, Y_{in}\}$ denote a Phase II sample and let $\overline{Y}_i$ denote the mean of that sample. We now plug in the estimators into the well-known Shewhart k-sigma control limits and get the estimated limits $\widehat{LCL}/\widehat{UCL} = \widehat{\mu}_0 \pm k\widehat{\sigma}_0/\sqrt{n}$ as shown earlier. Since $\widehat{\mu}_0$ and $\widehat{\sigma}_0$ are random variables, $\widehat{LCL}$ and $\widehat{UCL}$ are also random variables. The conditional signaling probabilities can be found as, namely, $p_{\mathrm{L}} = P(\overline{Y}_i < \widehat{LCL} | \widehat{\mu}_0, \widehat{\sigma}_0)$ and $p_{\mathrm{U}} = P(\overline{Y}_i > \widehat{UCL} \,|\, \widehat{\mu}_0, \widehat{\sigma}_0)$ so that $p = P(\widehat{LCL} \leq \overline{Y}_i \leq \widehat{UCL} \,|\, \widehat{\mu}_0, \widehat{\sigma}_0) = 1 - p_{\mathrm{L}} - p_{\mathrm{U}}$. By substituting $\widehat{LCL}/\widehat{UCL} = \widehat{\mu}_0 \pm k\widehat{\sigma}_0$ into the expressions for p_{L} and p_{U} we find $p_{\mathrm{L}} = P(\overline{Y}_i < \widehat{\mu}_0 - k\widehat{\sigma}_0 | \, \widehat{\mu}_0, \widehat{\sigma}_0)$ and $p_{\mathrm{U}} = P(\overline{Y}_i > \widehat{\mu}_0 + k\widehat{\sigma}_0 \mid \widehat{\mu}_0, \widehat{\sigma}_0)$, respectively. We don't discuss p_{IC} in detail from this point forward because it can be found using $p = 1 - p_{\mathrm{L}} - p_{\mathrm{U}}$. By subtracting the Phase II mean $\mu_1 = \mu_0 + \delta\sigma_0$ and dividing each side of the inequalities by $\sigma_0/\sqrt{n}$ we find

$$p_{\mathrm{L}} = P\left((\overline{Y}_i - \mu_0 - \delta\sigma_0)\frac{\sqrt{n}}{\sigma_0} < (\widehat{\mu}_0 - k\widehat{\sigma}_0 \; - \mu_0 - \delta\sigma_0)\frac{\sqrt{n}}{\sigma_0} | \widehat{\mu}_0, \widehat{\sigma}_0 \right)$$

and

$$p_{\mathrm{U}} = P\left((\overline{Y}_i - \mu_0 - \delta\sigma_0)\frac{\sqrt{n}}{\sigma_0} > (\widehat{\mu}_0 + k\widehat{\sigma}_0\mu_0 - \delta\sigma_0)\frac{\sqrt{n}}{\sigma_0} \mid \widehat{\mu}_0, \widehat{\sigma}_0 \right)$$

respectively.

By definition, under the assumption that the Phase II distribution has a mean shift, $\overline{Y}_i \sim N(\mu_1 = \mu_0 + \delta\sigma_0, \sigma_0/\sqrt{n})$, so that we can write the above expressions in terms of the cdf of the standard normal distribution

$$p_{\mathrm{L}} = \Phi\left((\widehat{\mu}_0 - \mu_0)\frac{\sqrt{n}}{\sigma_0} - k\frac{\widehat{\sigma}_0\sqrt{n}}{\sigma_0} - \delta\sqrt{n} \right)$$

and

$$p_{\mathrm{U}} = \Phi\left(-(\widehat{\mu}_0 - \mu_0)\frac{\sqrt{n}}{\sigma_0} - k\frac{\widehat{\sigma}_0\sqrt{n}}{\sigma_0} + \delta\sqrt{n}\right)$$

respectively.

Define the random variables $U = (\widehat{\mu}_0 - \mu_0)\dfrac{\sqrt{n}}{\sigma_0}$ and $V = \dfrac{\widehat{\sigma}_0\sqrt{n}}{\sigma_0}$, respectively. Now we can write the signaling probabilities in a simpler format, namely

$$p_{\mathrm{L}} = \Phi(U - kV - \delta\sqrt{n}) \text{ and } p_{\mathrm{U}} = \Phi(-U - kV + \delta\sqrt{n}),$$

respectively. The pdf of the random variable U is equal to $f_U(u) = \phi(u|0, 1/\sqrt{m})$, where ϕ is the pdf of the standard normal distribution, since

$$\widehat{\mu}_0 \sim N(\mu_0, \sigma_0/\sqrt{mn}) \text{ and} U \sim N(0, 1/\sqrt{m}).$$

For the random variable V, we make use of the fact that

$$\widehat{\sigma}_0^2/\sigma_0^2 \sim GAM(m(n-1)/2, 2/m(n-1))$$

so that

$$V^2 \sim GAM(m(n-1)/2, 2n/m(n-1))$$

and hence the pdf of V is given by

$$f_V(v) = 2v f_\gamma(v^2|m(n-1)/2, 2n/m(n-1))$$

where $f_\gamma(\cdot)$ is the pdf of a gamma distribution with parameters $m(n-1)/2$ and $2n/m(n-1)$, respectively. This line of derivation is based on Zhang and Castagliola (2010) but one can also follow Chakraborti (2000) and rewrite the equations in terms of the more familiar chi-square distribution.

The unconditional properties of the run-length distribution can now be found using the CUC technique discussed in Section 3.7. The idea we wanted to introduce here is that the CUC technique discussed in Section 3.10.1 can also be conveniently used for charts with runs-type signaling rules. Note that as alluded to earlier, since the random variable V can be rescaled so that its distribution may be written in terms of a chi-square distribution, the above expressions can be rewritten in terms of the chi-square distribution with some appropriate degrees of freedom. This may be more useful from a practical point of view since the chi-square pdf is more readily available. We leave the verification of these details to the reader.

3.12 PHASE I CONTROL CHARTS

3.12.1 Phase I $\overline{X}$-chart

As discussed earlier, Phase I analysis is a very important part of the overall SPC regime, and control charts, particularly Shewhart charts, play an important role in monitoring the Phase I subgroup. In this section we first discuss the Phase I $\overline{X}$-chart. Suppose there are m Phase I subgroups each of size n, and X_{ij} denotes the jth observation from the ith subgroup. Let $\overline{X}_i$ and S_i^2 denote the mean and the variance, respectively, of the ith subgroup. The charting statistic for the Phase I $\overline{X}$-chart is the mean, $\overline{X}$, of each subgroup and the control limits are given by

$$LCL/UCL = \overline{\overline{X}} \pm k\hat{\sigma}/\sqrt{n}$$

where $\overline{\overline{X}} = \dfrac{1}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n} X_{ij}$ is the overall (or the grand) mean and $\hat{\sigma} = \dfrac{1}{c_4}\left[\dfrac{1}{m}\sum_{i=1}^{m} S_i^2\right]^{\frac{1}{2}}$ is an unbiased estimator of the process standard deviation, based on the average subgroup variances with the constant c_4 (see Table C in Appendix A). The constant $k > 0$ is the chart design constant which is determined for a given nominal *FAP* value, m and n. Champ and Jones (2004) discussed this problem for a number of standard deviation estimators including the $\hat{\sigma}$ above and show that k can be obtained from the cdf of a multivariate t distribution with a special correlation structure. The interested reader is referred to their paper for more details. They tabulated the constant k for $m = 4(1)10$, $m = 15$, and $n = 3(1)10$ and a nominal *FAP* of 0.01, 0.05, and 0.10. However, recently, the number of Phase I subgroups, m, is recommended to be much larger, say, 50 or even 100 or more in order to properly account for parameter estimation. Champ and Jones (2004) also discussed some approximations for k when m is large; however, recent advances in computer software available for the calculation of the cdf of the multivariate normal and the multivariate t distribution now make it possible to calculate k exactly for large values of m, even for $m = 300$. Recently, Yao *et al.* (2017) has extended their tables to larger values of m and has provided an R package to make the calculations on demand.

Example 3.14 Illustration of the Phase I $\overline{X}$-chart

For illustration purposes, the data set from Montgomery (2005, Table 5.3, p. 223), on the inside diameters of piston rings manufactured by a forging process, is used for many of the examples in this book. We use it here as well. However, in Montgomery (2005)'s data set there are 25 subgroups (m =25) each of size 5 (n = 5), but we only use the first 15 subgroups in

order to make use of the tables of Champ and Jones (2004). Column (a) of Table 3.24 contains 15 retrospective or Phase I subgroups, each of size five, that were collected when the process was thought to be IC. Thus, $N = mn = 15 \times 5 = 75$. Columns (b) and (c) of Table 3.24 show the means and standard deviations for $i = 1, 2, \ldots, m$. The value of the design parameter k (in this case $k = 2.6728$) is found in Table III of Champ and Jones (2004) for $m = 15$ and $n = 5$ and nominal $FAP = 0.10$.

The graphical display of the Phase I $\overline{X}$ chart is shown in Figure 3.30. All of the charting statistics plot between the control limits with no discernible pattern, indicating that the process is IC. The data may now by referred to as the IC or reference data from which any necessary parameters can be estimated and then be used to find appropriate control limits, which are used for prospective process monitoring in Phase II.

Table 3.24 Phase I data for the $\overline{X}$ chart and charting statistics.

	(a)					(b)	(c)
Subgroup, i	X_{i1}	X_{i2}	X_{i3}	X_{i4}	X_{i5}	$\overline{X}_i$	S_i
1	74.030	74.002	74.019	73.992	74.008	74.010	0.015
2	73.995	73.992	74.001	74.011	74.004	74.001	0.008
3	73.998	74.024	74.021	74.005	74.002	74.010	0.012
4	74.002	73.996	73.993	74.015	74.009	74.003	0.009
5	73.992	74.007	74.015	73.989	74.014	74.003	0.012
6	74.009	73.994	73.997	73.985	73.993	73.996	0.009
7	73.995	74.006	73.994	74.000	74.005	74.000	0.006
8	73.985	74.003	73.993	74.015	73.988	73.997	0.012
9	74.008	73.995	74.009	74.005	74.004	74.004	0.006
10	73.998	74.000	73.990	74.007	73.995	73.998	0.006
11	73.994	73.998	73.994	73.995	73.990	73.994	0.003
12	74.004	74.000	74.007	74.000	73.996	74.001	0.004
13	73.983	74.002	73.998	73.997	74.012	73.998	0.010
14	74.006	73.967	73.994	74.000	73.984	73.990	0.015
15	74.012	74.014	73.998	73.999	74.007	74.006	0.007
16	74.000	73.984	74.005	73.998	73.996	73.997	0.008
17	73.994	74.012	73.986	74.005	74.007	74.001	0.011
18	74.006	74.010	74.018	74.003	74.000	74.007	0.007
19	73.984	74.002	74.003	74.005	73.997	73.998	0.008
20	74.000	74.010	74.013	74.020	74.003	74.009	0.008
21	73.982	74.001	74.015	74.005	73.996	74.000	0.012
22	74.004	73.999	73.990	74.006	74.009	74.002	0.007
23	74.010	73.989	73.990	74.009	74.014	74.002	0.012
24	74.015	74.008	73.993	74.000	74.010	74.005	0.009
25	73.982	73.984	73.995	74.017	74.013	73.998	0.016

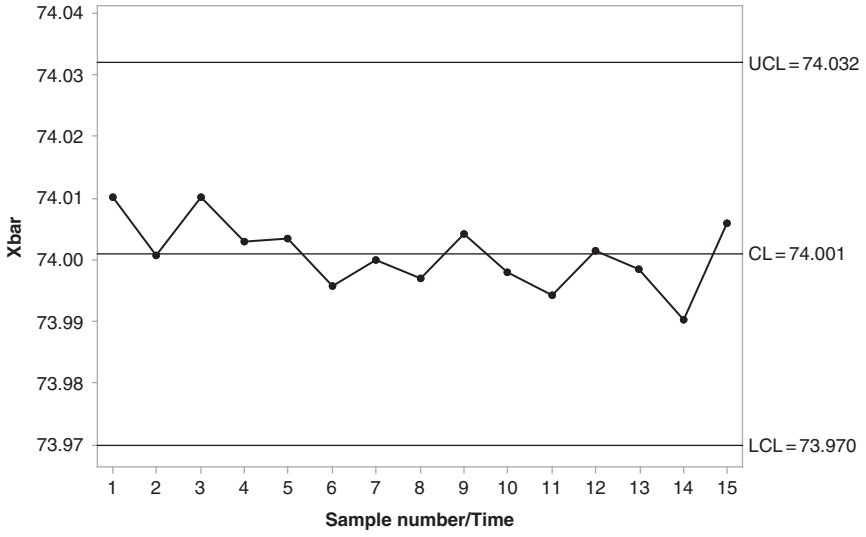


FIGURE 3.30 *A Phase I-* $\overline{X}$ *chart.*

For monitoring Phase I variances, the reader is referred to Human, Chakraborti, and Smit (2010a). Recall that Phase I charts are applied iteratively, investigating and possibly discarding OOC subgroups and recalculating the limits, constructing until all subgroups plot IC.

3.13 SIZE OF PHASE I DATA

The Phase I reference data set is typically made up of m ($\geq$1) subgroups, each of size n ($\geq$1). Thus, there is a total of mn Phase I observations (reference data) available for parameter estimation and for setting up the control limits which are used in Phase II. There are generally two approaches to obtaining a Phase I reference data set. It may be that it is already available (as a retrospective or a historical data set) or that the process needs to be first brought IC and then reference data are gathered. Getting a process IC and thus obtaining the reference data involves a careful Phase I analysis (where the control limits are viewed as trial limits and are used iteratively, until the charting statistics all plot IC) including the use of control charts and other statistical tools. The readers are referred to Jones-Farmer *et al.* (2014) and Chakraborti, Human, and Graham (2009) for more details about Phase I control charting. The bottom line is that the effects of parameter estimation are real, it varies from practitioner to practitioner and often result in more false alarms than what is nominally expected. If a large number of reference data are available, the Phase II limits can perform like their known parameter counterparts. Conversely, the control limits can be calculated for the data at hand. The underlying issues include knowledge of the form/shape of the underlying distribution, the parameter of interest,

the type of estimator, the type of chart, and so on. When estimating process variability, for example, one typically needs much larger sample sizes than when estimating the process mean. Recall that (from the Phase I and Phase II analyses discussion in Section 2.1.12) one of the major goals of a Phase I analysis is to estimate process variability.

To summarize, an accurate and precise estimation of parameters in Phase I is critical for the satisfactory performance of the control chart in the monitoring phase, namely, Phase II, as the run-length properties are greatly impacted by estimation, which can lead to, for example, many more false alarms than are nominally expected based on the pre-set IC average run-length; see, for example, Jensen *et al.* (2006) and Psarakis, Vyniou, and Castagliola (2014). These and other authors have concluded that the number of the Phase I samples, m, must often be quite large (in many hundreds of observations) in order to have reasonable confidence that the performance of the control chart will be close to that of the known performance in Case K or KK. Note that there is no magic number for m since the choice of m depends largely on the type of control chart to be implemented and the parameter estimators employed. Some work in the literature (e.g., Chakraborti, 2006; Saleh *et al.*, 2015; Epprecht, Loureiro, and Chakraborti, 2016) explores the various issues and shows that it takes a much larger Phase I sample size, often a few times larger than what has been typically recommended in textbooks, to obtain consistent chart properties that are close to the known parameter case. The known parameter case is held as a standard since those values don't vary. Thus, one has to be very careful about starting a Phase II charting procedure with a limited amount of historical data. The other approach may be to adjust the control limits for the given amount of data at hand. More research on this topic is in progress.

3.14 ROBUSTNESS OF PARAMETRIC CONTROL CHARTS

A robust statistical procedure is a procedure that performs well not only under ideal conditions (under which it is designed and mathematically derived), but also with some departure from the ideal. An important question about a statistical hypothesis test is its Type-I error robustness since without that the power properties of the test are questionable. In the same spirit, the IC robustness of a control chart is an important property that needs to be assessed and addressed since without it the shift detection performance of the chart is basically meaningless. The issue is serious with parametric charts since they are typically constructed under an assumed distribution (model) and hence their properties become suspect when that assumption doesn't hold or can't be adequately justified in a given situation. Many parametric charts, including the Shewhart, the CUSUM, and the EWMA charts, have been shown to suffer from a lack of IC robustness, to

various degrees, under non-normality. A consequence of this is that the IC average run-length is shortened, so much so, that many more false alarms are observed than are nominally expected. This can reduce the efficacy of any control chart and hurt the entire monitoring regime. Although some authors have suggested that some parametric control charts can be designed (tuned) to be robust against a number of distributions (see Borror, Montgomery, and Runger, 1999, about the traditional parametric EWMA control chart), this, however, can be problematic as some knowledge about the shape of the distribution and the expected shift that are necessary for tuning may not always be available. In fact, Human, Kritzinger, and Chakraborti (2011) showed that the traditional parametric EWMA control chart can lack IC robustness for some non-normal distributions such as the symmetric bi-modal and the contaminated normal distribution. Their observations call into question routine applications of the traditional parametric EWMA control chart in practice. The issue of IC robustness can be a major concern for practitioners and, in such cases, nonparametric control charts can be applied. They are IC robust by definition since their IC run-length distribution remains unchanged for all continuous distributions or continuous and symmetric distributions. This is one major practical advantage of the nonparametric charts. We recommend that, if the distributional assumption is in doubt or cannot be justified for lack of available information or data, a nonparametric or distribution-free control chart should be used. This should be done even if there is a possibility of some loss of efficiency when the true model is correctly specified, since the true model is usually unknown.

Appendix 3.1 SOME DERIVATIONS FOR THE EWMA CONTROL CHART

Result 3.1 The variance of the EWMA charting statistic
The exponentially weighted moving average is defined as

$$Z_i = \lambda X_i + (1-\lambda) Z_{i-1} \text{ with } Z_0 = \mu_0. \tag{A3.1}$$

By expanding Equation A3.1, recursively, we find

$$\begin{aligned} Z_i &= \lambda X_i + (1-\lambda) Z_{i-1} \\ &= \lambda X_i + (1-\lambda)(\lambda X_{i-1} + (1-\lambda) Z_{i-2}) \\ &\vdots \\ &= \lambda \sum_{j=0}^{i-1} (1-\lambda)^j X_{i-j} + (1-\lambda)^i Z_0. \end{aligned}$$

Thus, by continuing to recursively substitute for Z_{i-j}, $j = 2, 3, \ldots, t$, we obtain

$$Z_i = \lambda \sum_{j=0}^{i-1} (1-\lambda)^j X_{i-j} + (1-\lambda)^i Z_0 \tag{A3.2}$$

so that

$$\begin{aligned} var(Z_i) &= var\left(\lambda \sum_{j=0}^{i-1} (1-\lambda)^j X_{i-j} + (1-\lambda)^i Z_0\right) \\ &= \lambda^2 var\left(\sum_{j=0}^{i-1} (1-\lambda)^j X_{i-j}\right) \text{ since } (1-\lambda)^i Z_0 \text{ is constant} \\ &= \lambda^2 (var(X_i + (1-\lambda)X_{i-1} + (1-\lambda)^2 X_{i-2} \\ &\quad + \cdots + (1-\lambda)^{i-2} X_2 + (1-\lambda)^{i-1} X_1)) \\ &= \lambda^2 (\sigma^2 + (1-\lambda)^2\sigma^2 + (1-\lambda)^4\sigma^2 \\ &\quad + \cdots + (1-\lambda)^{2i-4}\sigma^2 + (1-\lambda)^{2i-2}\sigma^2). \end{aligned}$$

Thus

$$var(Z_i) = \lambda^2\sigma^2(1 + (1-\lambda)^2 + (1-\lambda)^4 + \cdots + (1-\lambda)^{2i-4} + (1-\lambda)^{2i-2}). \tag{A3.3}$$

In general, for a finite geometric series, the sum of the first n terms is

$$\sum_{j=0}^{n-1} ar^j = a\frac{(1-r^n)}{1-r} \text{ with } |r| < 1. \tag{A3.4}$$

Applying this result to Equation A3.3 we obtain

$$\begin{aligned} var(Z_i) &= \lambda^2\sigma^2\left(\frac{1-((1-\lambda)^2)^i}{1-(1-\lambda)^2}\right) \\ &= \lambda^2\sigma^2\left(\frac{1-(1-\lambda)^{2i}}{2\lambda-\lambda^2}\right) \\ &= \lambda\sigma^2\left(\frac{1-(1-\lambda)^{2i}}{2-\lambda}\right). \end{aligned}$$

Thus

$$var(Z_i) = \sigma^2\left(\frac{\lambda}{2-\lambda}\right)(1-(1-\lambda)^{2i}). \tag{A3.5}$$

Result 3.2 The EWMA is an unbiased estimator of the process mean

Assume that the process is IC with mean equal to μ_0. Using recursive substitution, we have shown $Z_i = \lambda \sum_{j=0}^{i-1} (1-\lambda)^j X_{i-j} + (1-\lambda)^i Z_0$. Assume either that the starting value Z_0 is zero or that i is so large that the starting value Z_0 has no effect (is zero). Then we have $Z_i = \lambda \sum_{j=0}^{\infty} (1-\lambda)^j X_{i-j}$. Thus

$$E(Z_i) = E\left(\lambda \sum_{j=0}^{\infty} (1-\lambda)^j X_{i-j}\right) = \lambda \sum_{j=0}^{\infty} (1-\lambda)^j E(X_{i-j})$$

by using a property of expected value. Since $E(X_{i-j}) = \mu_0$

$$\begin{aligned} E(Z_i) &= \lambda \sum_{j=0}^{\infty} (1-\lambda)^j \mu_0 \\ &= 1 \times \mu_0 = \mu_0 \end{aligned}$$

using the fact that $\sum_{j=0}^{\infty} (1-\lambda)^j = 1/\lambda$, which follows from the properties of an infinite geometric series. This sum of an infinite geometric series result can be seen from (A3.4), by taking the limit of the sum as $n \to \infty$.

Appendix 3.2 MARKOV CHAINS

Definition 3.1 A **stochastic process** is a collection of random variables $\{Y_t : t \in T\}$. All possible values that can be attained by $\{Y_t,\ t \in T\}$ are called the **states** of the stochastic process and the set of all possible states is called the **state space**, denoted by Ω. The set of all indexing parameters T is called the **parameter space**. If t represents time, the value of $\{Y_t\}$ at time t is referred to as the state in which the process is at time t.

Definition 3.2 The sequence $\{Y_t\}$ is called a **Markov chain** (MC) if, for any sequence

$$\{Y_0 = \omega_0, Y_1 = \omega_1, \ldots, Y_{t-1} = \omega_{t-1}, Y_t = \omega_t\}, \quad t = 1, 2, 3, \ldots,$$

it is true that

$$P(Y_t = \omega_t | Y_{t-1} = \omega_{t-1}, \ldots, Y_0 = \omega_0) = P(Y_t = \omega_t | Y_{t-1} = \omega_{t-1})$$

where $\{\omega_0, \omega_1, \omega_2, \ldots, \omega_t\} \in \Omega$.

Thus, the sequence $\{Y_t\}$ is a MC if the probability that the system enters the state ω_t at time t depends only on the immediately preceding state ω_{t-1}

where the system was at time $t-1$. In the SPC context, the system refers to the process under monitoring.

Definition 3.3 At each step (i.e., point in time) the process $\{Y_t,\, t = 1, 2, 3, \ldots\}$ may remain in the current state or move to another state according to some probability distribution. The state changes are called **transitions** and the probabilities associated with these transitions are called **transition probabilities.** These probabilities are denoted by

$$P(Y_t = \omega_j | Y_{t-1} = \omega_i) = p_{i,j}(t), \text{ where } \omega_j, \omega_i \in \Omega$$

at time t. The **transition probabilities** $p_{i,j}(t), 1 \leq i, j \leq m$, may be represented as an $m \times m$ matrix

$$\mathbf{M}_t = [p_{i,j}(t)] = \begin{bmatrix} p_{1,1}(t) & p_{1,2}(t) & \cdots & p_{1,m}(t) \\ p_{2,1}(t) & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ p_{m,1}(t) & p_{m,2}(t) & \cdots & p_{m,m}(t) \end{bmatrix}_{m \times m}$$

$\mathbf{M}_t,\ \ t = 1, 2, 3, \ldots$, are the **one-step transition probability matrices**. Note that this transition probability matrix depends on time; this implies that $p_{i,j}(t)$ is not necessarily equal to $p_{i,j}(s)$ for $t \neq s$.

Definition 3.4 A MC $\{Y_0, Y_1, Y_2, \ldots\}$ is **homogeneous** if the transition probabilities are constant over time (i.e. independent of time), that is, $P(Y_t = \omega_j | Y_{t-1} = \omega_i) = p_{i,j}$ for any $\omega_i, \omega_j \in \Omega$, and all $t = 1, 2, 3, \ldots$, where the transition probabilities $p_{i,j}$ are independent of the time index t.

Definition 3.5 State $\omega_i \in \Omega$ of a MC is called an **absorbing state** if, once the system enters state ω_i, it can never leave state ω_i, that is, $p_{\omega_i,\omega_i} \equiv 1$ and $p_{\omega_i,\omega_j} \equiv 0$ for any $i \neq j$.

Definition 3.6 A state $\omega_j \in \Omega$ of a MC is called a **transient state** if, given that the process is in state ω_j, there is a positive probability that the process will never return to state ω_j.

Interpreting the absorbing and the transient states on a control chart The movement of a MC between the transient state(s) is interpreted (associated) as (with) the charting statistic moving in a stochastic manner on the control chart, without causing the chart to signal. This indicates that the process is (remains) IC and the state(s) associated with the process being IC is(are) identified with the transient state(s). There is a positive probability that the charting statistic can plot OOC at any point in time with the result that the chart gives a signal. The latter is equivalent to saying that the stochastic process $\{Y_t\}$ has moved to an absorbing state (then the process

cannot move back to any of the transient states). This means that once there is a signal on the chart the cause of the signal is identified and fixed and the process monitoring starts all over again from an IC state.

Proof: **Markov chains**
From mathematical statistics, by differentiating the mgf $M_X(t)$ of any random variable X with respect to t and setting $t = 0$, the moments of the random variable X can be found, that is

$$\frac{d}{dt}M_X(t)|_{t=0} = M'_X(0) = E(X)$$
$$\frac{d^2}{dt^2}M_X(t)|_{t=0} = M''_X(0) = E(X^2).$$

In the SPC setting, one of the key random variables of interest is the run-length N. From Theorem 7.4 in Fu and Lou (2003, page 143), we have

$$M_N(t) = (e^t - 1)\xi(\mathbf{I} - e^t\mathbf{Q})^{-1}\mathbf{1} + 1,$$

so that

$$\begin{aligned}
M'_N(t) &= \frac{d}{dt}M_N(t) \\
&= (1)(e^t - 1)\boldsymbol{\xi}(\mathbf{I} - e^t\mathbf{Q})^{-1}\mathbf{1} + (e^t - 1)\boldsymbol{\xi}(-1)(\mathbf{I} - e^t\mathbf{Q})^{-2}(-e^t\mathbf{Q})\mathbf{1} \\
&= e^t\boldsymbol{\xi}(\mathbf{I} - e^t\mathbf{Q})^{-1}\mathbf{1} + (e^t - 1)\boldsymbol{\xi}(\mathbf{I} - e^t\mathbf{Q})^{-2}(e^t\mathbf{Q})\mathbf{1} \\
&= e^t\boldsymbol{\xi}(\mathbf{I} - e^t\mathbf{Q})^{-1}\mathbf{1} + (e^{2t} - e^t)\boldsymbol{\xi}(\mathbf{I} - e^t\mathbf{Q})^{-2}\mathbf{1} \\
M''_N(t) &= \frac{d^2}{dt^2}M_N(t) \\
&= \frac{d}{dt}M'_N(t) \\
&= e^t\boldsymbol{\xi}(\mathbf{I} - e^t\mathbf{Q})^{-1}\mathbf{1} + e^t\boldsymbol{\xi}(-1)(\mathbf{I} - e^t\mathbf{Q})^{-2}(-e^t\mathbf{Q})\mathbf{1} \\
&\quad + (1)(e^{2t} - e^t)^0(e^{2t}2 - e^t)\boldsymbol{\xi}(\mathbf{I} - e^t\mathbf{Q})^{-2}\mathbf{Q}1 \\
&\quad + (e^{2t} - e^t)\boldsymbol{\xi}(-2)(\mathbf{I} - e^t\mathbf{Q})^{-3}(-e^t\mathbf{Q})\mathbf{Q}1 \\
&= e^t\boldsymbol{\xi}(\mathbf{I} - e^t\mathbf{Q})^{-1}\mathbf{1} + e^{2t}\boldsymbol{\xi}(\mathbf{I} - e^t\mathbf{Q})^{-2}\mathbf{Q}1 \\
&\quad + (2e^{2t} - e^t)\boldsymbol{\xi}(\mathbf{I} - e^t\mathbf{Q})^{-2}\mathbf{Q}1 \\
&\quad + 2(e^{3t} - e^{2t})\boldsymbol{\xi}(\mathbf{I} - e^t\mathbf{Q})^{-3}\mathbf{Q}^2\mathbf{1}
\end{aligned}$$

so that substituting $t = 0$,

$$\begin{aligned} M'_N(0) &= e^0\xi(\mathbf{I} - e^0\mathbf{Q})^{-1}\mathbf{1} + (e^{2(0)} - e^0)\xi(\mathbf{I} - e^0\mathbf{Q})^{-2}\mathbf{Q}\mathbf{1} \\ &= 1\,\xi\,(\mathbf{I} - 1\mathbf{Q})^{-1}\mathbf{1} + (1-1)\xi(\mathbf{I} - 1\mathbf{Q})^{-2}\mathbf{Q}\mathbf{1} \\ &= \xi(\mathbf{I} - \mathbf{Q})^{-1}\mathbf{1} + (0)\xi(\mathbf{I} - \mathbf{Q})^{-2}\mathbf{Q}\mathbf{1} \\ &= \xi(\mathbf{I} - \mathbf{Q})^{-1}\mathbf{1}. \end{aligned}$$

Similarly

$$\begin{aligned} M''_N(0) &= e^0\xi(\mathbf{I} - e^0\mathbf{Q})^{-1}1 + e^{2(0)}\xi(\mathbf{I} - e^0\mathbf{Q})^{-2}\mathbf{Q}1 \\ &\quad + (2e^{2(0)} - e^0)\xi(\mathbf{I} - e^0\mathbf{Q})^{-2}\mathbf{Q}1 + 2(e^{3(0)} - e^{2(0)})\xi(\mathbf{I} - e^0\mathbf{Q})^{-3}\mathbf{Q}^2 1 \\ &= 1\xi(\mathbf{I} - 1\mathbf{Q})^{-1}1 + 1\xi(\mathbf{I} - 1\mathbf{Q})^{-2}\mathbf{Q}1 + (2(1) - 1)\xi(\mathbf{I} - 1\mathbf{Q})^{-2}\mathbf{Q}1 \\ &\quad + 2(1-1)\xi(\mathbf{I} - 1\mathbf{Q})^{-3}\mathbf{Q}^2 1 \\ &= \xi(\mathbf{I} - \mathbf{Q})^{-1}1 + 1\xi(\mathbf{I} - \mathbf{Q})^{-2}\mathbf{Q}1 + \xi(\mathbf{I} - \mathbf{Q})^{-2}\mathbf{Q}1 \\ &= \xi(\mathbf{I} + (\mathbf{I} - \mathbf{Q})^{-1}\mathbf{Q} + (\mathbf{I} - \mathbf{Q})^{-1}\mathbf{Q})(\mathbf{I} - \mathbf{Q})^{-1}1 \\ &= \xi((\mathbf{I} - \mathbf{Q})\mathbf{I} + \mathbf{Q} + \mathbf{Q})(\mathbf{I} - \mathbf{Q})^{-1}(\mathbf{I} - \mathbf{Q})^{-1}1 \\ &= \xi(\mathbf{I} - \mathbf{Q} + \mathbf{Q} + \mathbf{Q})(\mathbf{I} - \mathbf{Q})^{-2}1 \\ &= \xi(\mathbf{I} + \mathbf{Q})(\mathbf{I} - \mathbf{Q})^{-2}1. \end{aligned}$$

Hence, we have the first two moments of the run-length N

$$\begin{aligned} E(N) &= M'_N(0) = \xi(\mathbf{I} - \mathbf{Q})^{-1}\mathbf{1} \\ E(N^2) &= M''_N(0) = \xi(\mathbf{I} + \mathbf{Q})(\mathbf{I} - \mathbf{Q})^{-2}\mathbf{1} \end{aligned}$$

which yield

$$\begin{aligned} var(N) &= E(N^2) - (E(N))^2 \\ &= \xi(\mathbf{I} + \mathbf{Q})(\mathbf{I} - \mathbf{Q})^{-2}\mathbf{1} - (\xi(\mathbf{I} - \mathbf{Q})^{-1}\mathbf{1})^2 \\ \therefore var(N) &= \xi(\mathbf{I} + \mathbf{Q})(\mathbf{I} - \mathbf{Q})^{-2}\mathbf{1} - (\xi(\mathbf{I} - \mathbf{Q})^{-1}\mathbf{1})^2 \\ SD(N) &= \sqrt{var(N)} = \sqrt{\xi(\mathbf{I} + \mathbf{Q})(\mathbf{I} - \mathbf{Q})^{-2}1 - (\xi(\mathbf{I} - \mathbf{Q})^{-1}1)^2}. \end{aligned}$$

■

Appendix 3.3 SOME DERIVATIONS FOR THE SHEWHART DISPERSION CHARTS

The charting constants for the Shewhart charts for the variance can be obtained in two steps. First, setting the unconditional IC average run-length equal to some nominal value we solve for the probability $\alpha(m,n)$. Then, in the second step, charting constants (and hence the control limits) for the Phase II dispersion charts are obtained from the corresponding percentiles of the chi-square distribution (exactly or approximately) with $n-1$ degrees of freedom. These limits are the unconditional probability limits. We sketch some details here using the S chart with S_p as the Phase I estimator of σ, using the CUC method. Note that in the section on the run-length distribution, we showed that the probability limits are given by $H_3^*(m,n) = \sqrt{\chi^2_{\alpha(m,n)/2,n-1}/(n-1)}$ and $H_4^*(m,n) = \sqrt{\chi^2_{1-\alpha(m,n)/2,n-1}/(n-1)}$, where $\alpha(m,n)$ is the probability that the charting statistic S_i plots outside either the lower or the upper control limit when the process is IC. Thus, in order to get the charting constants H_3^* and H_4^*, the $\alpha(m,n)$ values are needed.

To this end, first note that

$$\alpha(m,n) = 1 - P(H_3^*(m,n)S_p < S_i < H_4^*(m,n)\, S_p)|IC)$$
$$= 1 - P\left(\sqrt{\chi^2_{\alpha(m,n)/2,n-1}/(n-1)}\, S_p < S_i < \sqrt{\chi^2_{1-\alpha(m,n)/2,n-1}/(n-1)}\, S_p|IC\right)$$

which can be re-written as

$$\begin{aligned}\alpha(m,n) &= 1 - P\left(\frac{\chi^2_{\alpha(m,n)/2,n-1}}{n-1}\frac{S_p^2}{\sigma^2} < \frac{S_i^2}{\sigma^2} < \frac{\chi^2_{1-\alpha(m,n)/2,n-1}}{n-1}\frac{S_p^2}{\sigma^2}|IC\right)\\ &= 1 - P\left(\chi^2_{\alpha(m,n)/2,n-1}\frac{U}{m} < \frac{(n-1)S_i^2}{\sigma^2} < \chi^2_{1-\alpha(m,n)/2,n-1}\frac{U}{m}|IC\right)\\ &= 1 - P\left(\chi^2_{\alpha(m,n)/2,n-1}\frac{U}{m} < V < \chi^2_{1-\alpha(m,n)/2,n-1}\frac{U}{m}\right)\end{aligned}$$

where $V = \dfrac{(n-1)S_i^2}{\sigma^2}$ and $U = \dfrac{m(n-1)S_p^2}{\sigma^2}$ are independent random variables and each follows a chi-square distribution with $(n-1)$ and $m(n-1)$ degrees of freedom, respectively, in the IC case.

The last probability can be calculated by conditioning on S_p or, equivalently on U. This is called the conditional false alarm rate ($CFAR$), which

is a function of U, m, and n. Hence, the *CFAR* can be written as

$$CFAR(U, m, n, \alpha(m, n)) = 1 - F_{\chi^2_{n-1}}\left(\chi^2_{\alpha(m,n)/2,n-1}\frac{U}{m}\right) + F_{\chi^2_{n-1}}\left(\chi^2_{1-\alpha(m,n)/2,n-1}\frac{U}{m}\right) \quad \text{(A3.6)}$$

so that the conditional run-length distribution is geometric with probability *CFAR* (U,m,n,p). Thus, by applying the CUC method, the unconditional IC run-length distribution (pmf) is given by

$$P(N = k) = \int_0^\infty CFAR(U, m, n, \alpha(m, n))[1 - CFAR(U, m, n, \alpha(m, n))]^{k-1} \times f_{\chi^2_{m(n-1)}}(u)du,\ k = 1, 2, \ldots$$

Note that this unconditional run-length distribution is not geometric.

In a similar way, using the CUC method, the IC conditional average run-length is given by $[CFAR(U, m, n, \alpha(m, n))]^{-1}$ and as such, the IC unconditional IC average run-length is given by

$$ARL_{IC} = \int_0^\infty [CFAR(U, m, n, \alpha(m, n))]^{-1} f_{\chi^2_{m(n-1)}}(u)du. \quad \text{(A3.7)}$$

Again, note that the ARL_{IC} is not the reciprocal of the *CFAR*. Now in the unconditional method setting the ARL_{IC} expression equal to some nominal value, such as 370, we solve the equation $ARL_{IC} = 370$ numerically, for $\alpha(m, n)$ for given values of m and n. Then we find the 100*$\alpha(m, n)$ and the 100*[1 – α(m,n)/2] percentiles of a chi-square distribution with $n - 1$ degrees of freedom. The required charting constants $H_3^*(m, n)$ and $H_4^*(m, n)$ are found by dividing each of these percentiles by $n - 1$ and taking the square root. These values are shown in Table H in Appendix A. We leave the verification of the rest of this table for the other charts and with other Phase I estimators of σ to the reader.

Finally, note that this formulation can accommodate a Phase II sample size, say, n_1, not equal to the Phase I sample size m. The difference in this case will be in the degrees of freedom of the distribution of V, which will now be n_1-1, in the first chi-square distribution, while defining the *CFAR* in Equation A3.6.

CHAPTER 4

Nonparametric (Distribution-free) Univariate Variables Control Charts

Chapter Overview

In Chapter 3, we focused on univariate parametric control charts for variables data. In this chapter we consider nonparametric control charts. As in the statistics literature, these charts are also called distribution-free and we use the terms "distribution-free" and "nonparametric" interchangeably. The vast majority of available nonparametric charts are for the location parameter, which represents the center of a continuous distribution. A location parameter, for example, can be the mean or the median of the process distribution.

4.1 INTRODUCTION

In the framework of statistical process control (SPC), the pattern of chance causes or the process itself is often assumed to follow some *known* parametric distribution. Control charts designed and implemented under an assumed parametric distribution are called parametric charts. For instance, the most common assumption is that of a normal distribution. The statistical properties of these control charts are exact only if the assumption of normality is actually satisfied. Yet, in many applications, the true underlying process distribution is not known or not normal, and consequently the properties

Nonparametric Statistical Process Control, First Edition.
Subhabrata Chakraborti and Marien Alet Graham.
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Companion website: www.wiley.com/go/chakraborti/Nonparametric_Statistical_Process_Control

or the characteristics of these standard (parametric) control charts can be highly affected in such situations. Hence, the development and the application of control charts that do not depend on the normality, or other specific parametric distributional assumptions, are needed. The distribution-free charts can serve this purpose. Still, despite the evidence in favor of using the nonparametric or distribution-free control charts, their implementation and development have been somewhat slow. There are several reasons for this.

First, it can be argued that the central limit theorem will ensure that a control chart will perform as expected. Although this might be true for control charts based on the averages of certain sample statistics, generally, it is not true. Where control charts need to be applied to individual observations, for example, the central limit theorem is not applicable and cannot be invoked. Second, past unavailability of adequate computing facilities and the perception that one necessarily needs to sacrifice "efficiency" when using these (simple) techniques, that are often based on counting and/or ranking, have also played a role. However, the former is no longer a problem considering today's computer age, whereas the latter is not necessarily true. For instance, it is well known in the statistical literature that for some heavy-tailed and/or asymmetric distributions, some distribution-free procedures outperform their parametric counterparts. Remarkably, even when the underlying distribution is normal, the efficiency of some nonparametric tests relative to the corresponding (optimal) normal theory methods can be as high as 0.955 (see, for example, Gibbons and Chakraborti, 2010, p. 218). For some other heavy-tailed and skewed distributions, the efficiency can be 1.0 or even higher. Finally, a lack of familiarity and training among SPC personnel has also contributed to nonparametric charts being unattractive in practice; this is changing.

A formal definition of a nonparametric or a distribution-free control chart is given in terms of its run-length distribution, that is, the distribution of the number of samples or subgroups that need to be collected before the first out-of-control (OOC) signal is given by the control chart.

Definition If the in-control (IC) run-length distribution is the same for every continuous (process) distribution, the control chart is called distribution-free (or nonparametric).

It should be noted that the assumption of symmetry is necessary in the definition above in some cases, for example, for control charts based on the signed-rank statistic; the reader is referred to Section 4.2.1.2.

Obviously, the main advantage of these types of control charts is the flexibility of not having to assume any parametric probability distribution for the underlying process – at least as far as *implementing* and *establishing* these types of control charts are concerned. This is very beneficial in

the field of SPC, particularly in start-up or the short-runs situations, where not many data are available to use any specific parametric procedure. In addition, the distribution-free control charts are likely to share the robustness properties or the robustness characteristics of distribution-free tests and confidence intervals. They are therefore also far more likely to be less impacted by outliers or unusual observations than the standard parametric control charts.

To summarize, some of the *advantages* of distribution-free or nonparametric control charts are: (i) their simplicity; (ii) no need to assume any particular probability distribution for the underlying process, except that the distribution is continuous and in some cases symmetric; (iii) the IC robustness property since the IC run-length distribution is the same for all continuous (symmetric) distributions, which is then also true for the false alarm rate (FAR) and the IC average run-length (ARL_{IC}), for example; (iv) they are more robust and resistant to outliers; (v) they are more efficient in detecting changes in the process distribution, especially when the true process distribution is noticeably non-normal, particularly with heavier tails; and (vi) there is no need to estimate the process variance to set up a distribution-free control chart for a location parameter.

4.2 DISTRIBUTION-FREE VARIABLES CONTROL CHARTS IN CASE K

4.2.1 Shewhart Control Charts

4.2.1.1 Shewhart Control Charts Based on Signs

4.2.1.1.1 A Shewhart Sign Chart Consider the situation in which samples or subgroups of n (≥ 1) observations are taken randomly at regular time intervals from an *unknown* continuous process distribution with a parameter of interest μ. Generally, μ can be any parameter, but it is typically taken to be the median of the process. Suppose that the *known* or *specified* IC value of μ is μ_0. Thus, if it happens that the distribution is symmetric, the target value is equal to the process mean. Let $X_{i1}, X_{i2}, \ldots, X_{in}$, $i = 1,2,\ldots$ be the sample or the subgroup at the ith point in time, and suppose that each observation X_{ij}, $j = 1,2,\ldots,n$ is compared to μ_0 and the number of observations above and below μ_0 is recorded for each sample. Let n_i^+ (n_i^-) denote the number of observations x_{ij} greater (less) than μ_0 in the ith sample. Then, let

$$SN_i = \sum_{j=1}^{n} \varphi(X_{ij} - \mu_0) \tag{4.1}$$

where

$$\varphi(t) = \begin{cases} 1 & \text{if} \quad t > 0 \\ 0 & \text{if} \quad t = 0 \\ -1 & \text{if} \quad t < 0 \end{cases}. \tag{4.2}$$

The statistic SN_i can be used as the charting statistic for the sign chart. It can be seen that $SN_i = n_i^+ - n_i^-$, that is, SN_i is the difference between the number of observations greater than μ_0 and the number of observations less than μ_0 in the ith sample. For the case of individual data, that is, $n = 1$, the SN_i statistic equals $+1$ when the observation is greater than μ_0 and -1 when the observation is less than μ_0. From $SN_i = n_i^+ - n_i^-$, it follows that $SN_i + n = 2n_i^+$ assuming no ties, so that $n_i^+ = \dfrac{SN_i + n}{2}$. Note that we can write n_i^+, the number of observations greater than the specified μ_0, as $n_i^+ = T_i = \sum_{j=1}^{n} \Psi(X_{ij} - \mu_0)$, where $\Psi(A)$ denotes the indicator function $\Psi(A) = \begin{cases} 1 & \text{if} \quad A > 0 \\ 0 & \text{if} \quad A \leq 0 \end{cases}$. From the relationship $n_i^+ = \dfrac{SN_i + n}{2}$, we get $SN_i = 2T_i - n$, since $n_i^+ = T_i$, so that the sign chart can be equivalently expressed in terms of the statistic T_i.

One advantage to using the sign test based on SN_i is the fact that its expected value in the IC case is 0 (this is shown later on). However, in the (nonparametric) literature, the statistic T_i is the more well-known version (since the distribution of T_i is binomial) on which the sign test is based, and is referred to as the **sign test statistic** (see, for example, Gibbons and Chakraborti, 2010, p. 168). For the Shewhart sign chart (denoted Shewhart-SN), the statistic SN_i is used as the charting statistic.

Since the process distribution is assumed to be continuous, $P(X_{ij} = \mu_0) = 0$ and the scenario where $\varphi(X_{ij} - \mu_0) = 0$, that is, "zeros" or zero differences with μ_0, should at least, in theory, never occur. However, where observations are rounded off, the occasional "zero" might be observed; but, as long as the number of "zeros" does not occur too often, we can safely compute SN_i as defined in Equation 4.1.

The control limits and the center line of the two-sided **Shewhart-SN chart** are

$$\begin{aligned} UCL &= a \\ CL &= 0 \\ LCL &= -a \end{aligned} \tag{4.3}$$

where a is a positive integer such that $0 < a \leq n$. Therefore, the Shewhart-SN chart signals that a shift occurred if $|SN_i| \geq a$, that is, if $SN_i \geq a$ or if $SN_i \leq -a$. Alternatively, as long as $-a < SN_i < a$, the process is considered IC at the target median value μ_0. The one-sided and two-sided forms of Shewhart charts are introduced in Section 3.2.1 and here, a lower one-sided Shewhart-SN chart signals that a downward shift

in the median has occurred if $SN_i \leq -a$ (the chart only has a lower control limit, or *LCL*), whereas an upper one-sided Shewhart-SN chart signals that an upward shift has occurred if $SN_i \geq a$ (the chart only has an upper control limit, or *UCL*). Here, we consider two-sided charts since they are more general (as the practitioner may not know the direction of the shift of the median).

Calculation of the Charting Constant a The charting constant a is the smallest integer between $0 < a \leq n$ such that $P_{\mu_0|\text{IC}}(SN_i \geq a) \leq \dfrac{FAR}{2}$. The subscript $\mu_0|$IC indicates that the control charting constant is found in Case K and under the IC case. At this point, it is important to note that, since the distribution of SN_i is discrete, chances are that the equality above will not be attained exactly. Hence, using a conservative approach, a is found so that the attained *FAR* is less than or equal to the specified or nominal *FAR*. For example, setting a nominal *FAR*, denoted FAR_0, equal to 0.0027 we get $\dfrac{FAR_0}{2} = \dfrac{0.0027}{2} = 0.00135$ so that a is the smallest integer such that $P_{\mu_0|\text{IC}}(SN_i \geq a) \leq 0.00135$.

Note 4.1 Using Binomial Probabilities to Calculate the Charting Constant *a* Shewhart-SN Chart

We focus on the "right-tail" entry in the Binomial table (see Table A in Appendix A) since we are looking for the *UCL*, that is, we are focusing on the upper-tail probability $P_{\mu_0|\text{IC}}(T_i \geq t)$ in order to calculate $P_{\mu_0|\text{IC}}(SN_i \geq a)$.

Next, we consider two examples. The first one for $n = 5$ illustrates a case where we can't find an attained *FAR* value that is smaller than 0.0027. This is typically the case for small values of n and the nominal *FAR*. Following this, the second example for $n = 20$ illustrates a case where an attained *FAR* of less than 0.0027 can be obtained. Thus, the size of the sample (subgroup) being monitored is important when it comes to nonparametric control charts and choosing nominal *FAR* values. We note that this is generally the case for charts based on charting statistics with a discrete distribution, which is also the case for attributes charts (not discussed in this book).

For $n = 5$

Across the "right-tail" entry of 5 we have $\dfrac{FAR}{2} = 0.03125$, which is the smallest probability associated with $n = 5$. Although this probability is not less than or equal to 0.00135, it is the smallest probability that we can obtain when working with a sample of size 5. Now, to obtain the *UCL*, that is, a, we use the linear relationship $SN_i = 2T_i - n$ so that $a = 2(5) - 5 = 5$ with $FAR = (2)(0.03125) = 0.0625$. Thus, for $n = 5$ we have $LCL\ /\ UCL = \pm 5$ and $CL = 0$ with a corresponding $FAR = 0.0625$, which is the lowest *FAR* achievable. Since the IC value of the median is known or specified (Case K) $ARL_{\text{IC}} = \dfrac{1}{FAR} = \dfrac{1}{0.0625} = 16$. Again, this is not good, since it is far from the

typical desired value of 370. However, this is the highest ARL_{IC} achievable when working with a sample of size 5.

In fact, it can be seen that, unless the sample size is 10 or more, the sign chart may be somewhat unattractive from the point of view of a small FAR (or a large ARL_{IC}) typically used in SPC. The reader is asked to compute the FAR values for $n = 6$, 7, 8, 9, and 10 to verify this.

For $n = 20$

Across the "right-tail" entry of 17, we have $\frac{FAR}{2} = 0.001288$, which is less than 0.00135. Now, to obtain the UCL, that is, a, we use the linear relationship $SN_i = 2T_i - n$ so that $a = 2(17) - 20 = 14$ with a $FAR = (2)(0.001288) = 0.002576$. Thus, for $n = 20$ we have LCL / $UCL = \pm$ 14 and $CL = 0$ with a corresponding $FAR = 0.002576$ and $ARL_{\text{IC}} = \frac{1}{FAR} = \frac{1}{0.002576} \approx 388$, which is good since it is close to the typical desired value of 370.

4.2.1.1.2 A Shewhart Sign Chart with Runs-type Signaling Rules

While the Shewhart-SN chart is powerful in detecting large shifts in the process location parameter μ, like any Shewhart chart it is expected to lack sensitivity in detecting smaller shifts. Hence, as in the case of the parametric Shewhart chart, we can improve the effectiveness of the Shewhart-SN chart in detecting small shifts by adding runs-type signaling rules. The standard *k-of-k* runs-type signaling rule signals when either k successive points are plotted on or above the UCL or when k successive points are plotted on or below the LCL. An illustration is given in Section 4.3. The reader is also referred to Human, Chakraborti and Smit (2010b) for more examples and details on the standard *2-of-2* signaling rule-based Shewhart-SN charts.

4.2.1.1.3 Methods of Calculating the Run-length Distribution

The Exact Approach For the Shewhart and Shewhart-type sign control charts, it is possible to calculate the run-length distribution and various run-length characteristics exactly. At this point it should be noted that the original Shewhart charts are for the parametric (normal distribution) case and some authors have used the term **Shewhart-type** charts in the nonparametric case to emphasize that these are analogous. For simplicity, however, we use the term **Shewhart** throughout for both parametric and nonparametric charts. As in the case of the parametric Shewhart chart in Case K, the run-length distribution of Shewhart-SN chart is $GEO(\theta)$, that is, geometric with probability $\theta = P(SN_i \geq a \text{ or } SN_i \leq -a)$, which is the probability of a signal. Since $SN_i = 2T_i - n$, the probability θ can be conveniently expressed in terms of the distribution of T_i, which is known to be binomial with parameters n and $p = P(X_{ij} \geq \mu_0)$ in general. Table 4.1 shows the distributional properties of the statistics, T_i and SN_i, respectively.

Table 4.1 Distributional properties of the statistics T_i and SN_i, respectively.

	T_i	SN_i
Expected value	$E(T_i) = np$	$E(SN_i)$ $= E(2T_i - n)$ $= 2E(T_i) - n$ $= 2np - n$ $= n(2p - 1)$
Standard deviation	$SD(T_i) = \sqrt{np(1-p)}$	$SD(SN_i)$ $= 2\sqrt{np(1-p)}$

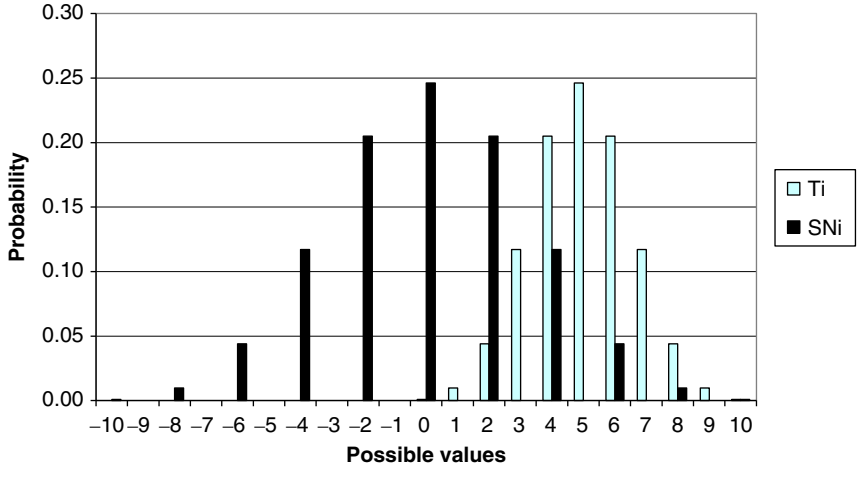


FIGURE 4.1 *The IC probability distribution of T_i and SN_i for* n = *10.*

Figure 4.1 illustrates the IC probability distribution of T_i and SN_i for $n = 10$. Note that in the IC the case with $p = 0.5$ and probability distribution of T_i is symmetric about its expected value $E(T_i) = np = (10)(0.5) = 5$. Conversely, the IC probability distribution of SN_i is symmetric about its expected value $E(SN_i) = n(2p - 1) = 10(2(0.5) - 1) = 0$.

The IC run-length distribution of the two-sided Shewhart-SN chart, for example, follows a geometric distribution, $GEO(\theta)$, with success probability (the *FAR*), θ, where $\theta = (0.5)^n \left[\sum_{\left[\frac{a+n}{2}\right]}^{n} \binom{n}{i} + \sum_{0}^{\left[\frac{n-a}{2}\right]} \binom{n}{i} \right]$, and $[d]$ denotes the largest integer not exceeding d. This is because, in the IC case, the T_i follows a binomial distribution with parameters n and $p = P(X_{ij} \geq \mu_0) = 0.5$.

As in the case of the parametric charts in Case K, the IC *ARL* of the Shewhart-SN chart is the reciprocal of the *FAR*, and the other moments of the IC run-length distribution can be found using properties of the geometric distribution. The OOC run-length distribution of the Shewhart-SN chart is also geometric with success probability θ, which may be calculated in a similar way using the binomial distribution.

Next, consider the Shewhart-SN chart with runs-type signaling rules. Human *et al.* (2010b) considered using the *1-of-1*, the *k-of-k*, and the *k-of-w* runs-rules, and the reader is referred to their paper for more details. Recall that the only assumptions that need to be satisfied for the SN charts are that the observations must be independent and have a continuous distribution. Note that the charting statistic used by Human *et al.* (2010b) is not the SN_i statistic that was introduced earlier (see Equation 4.1), but rather the classical sign statistic T_i (see, for example, Gibbons and Chakraborti, 2010, p. 168). The control limits and *CL* are given by $UCL = n - b$, $LCL = a$, and $CL = n\theta_0$, respectively, where a and b are values between 0 and n, which are typically chosen to obtain some desired ARL_{IC}. Since this is the first runs-rule chart to be considered in this chapter, we start with the simplest case, that is, the *1-of-1* runs-rule, in order to introduce some notation. For the upper one-sided *1-of-1* SN chart, the 1-0 indicator variables Z_h^+ are defined associated with the charting statistic T_h, for $h = 1, 2, 3, \ldots$, each being either a success (denoted by 1), indicating a signal, or a failure (denoted by 0), indicating no signal, with the corresponding probabilities p^+ and $1 - p^+$. This *1-of-1* SN chart signals when $T_i \geq UCL$. Similarly, for the lower one-sided *1-of-1* chart, the indicator variable Z_h^- denotes a success (1) or a failure (0) with probabilities p^- and $1 - p^-$ respectively; this chart signals when $T_i \leq LCL$. For the two-sided *1-of-1* chart, the 1-0 indicator variables Z_h,$h = 1, 2, 3, \ldots$, again denote a success (1) (i.e., a signal) or a failure (0) (i.e., a no signal) with probabilities p and $1 - p$; this chart signals when $T_i \leq LCL$ or $T_i \geq UCL$.

Now, with the basics defined, we consider the upper one-sided Shewhart-SN chart supplemented with the *2-of-2* runs-rule. A one-sided chart is explained first since this is the first runs-rule enhanced chart to be considered in this chapter. Two-sided charts will be discussed in a later section. Let State S_1 represent the area on and above the *UCL* and let State S_0 represent the area below the *UCL*. Let p denote the probability of a signal, that is, the probability of a charting statistic plotting on or above the *UCL*. Hence, the probability of a signal $p = P(T_i \geq UCL) = P(T_i \geq n - b) = I_p(n - b, b + 1)$, where $I_p(n - b, b + 1)$ denotes the incomplete beta integral or the cdf of a beta distribution at p with parameters $n - b$ and $b + 1$. The relation between the upper tail probability of a binomial distribution and the incomplete beta integral is well known; see, for example, Gibbons and Chakraborti (2010, p. 66). Also, let $q = 1 - p$ denote the probability of no signal and ϕ denote the dummy state. The upper one-sided

Table 4.2 Essential TPM for the upper one-sided control chart supplemented with the *2-of-2* runs-type signaling rule.

States	ϕ	S_0	S_1
ϕ	0	p_0	p_1
S_0	0	p_0	p_1
S_1	0	p_0	0

Shewhart-SN chart supplemented with the *2-of-2* runs-rules signals when two consecutive SN charting statistics plot on or above the *UCL*. It is left for the reader to show that the essential transition probability matrix (TPM) in this case is given by the entries in Table 4.2; the interested reader is referred to a paper by Human *et al.* (2010b) for more details and more examples.

Once the essential TPM is obtained, the run-length distribution and various characteristics of it, such as the *ARL*, etc., can be calculated using the MC Theory, as shown in Equations 3.22 to 3.24. Again, the reader is referred to Human *et al.* (2010b) for details. It can be seen, for example, that the $ARL = (1 + p)/(p)^2$ and $FAR = (p_0)^2$, where p_0 is the probability of a signal when the process is IC. Thus, the MC Theory plays a useful role in the derivation of the run-length distribution for the Shewhart charts enhanced with runs-rules, where the charting statistics are dependent.

The Simulation Approach Although for the Shewhart-SN chart the run-length distribution and its various characteristics can be calculated analytically, in general, the simulation may be useful and preferable. The following simulation algorithm can be used to obtain the run-length characteristics for the Shewhart-SN chart. The corresponding SAS® program is given in Appendix B, SAS program number 1, and the output is shown below in Figure 4.2.

Step 1: After specifying the subgroup size ($n \geq 1$) and the size of the shift (δ) to be detected, we generate random subgroups from some continuous distribution, for example, the standard normal, Student's t, Laplace, Logistic, Gamma, etc. Any continuous distribution, whether symmetric or asymmetric, may be used for data generation.

Step 2: Calculate the SN_t statistics for each subgroup.

Step 3: Calculate the control limits, that is, the charting constant a, and compare SN_t to the control limits. The calculation of the charting constant a is explained in detail in Section 4.2.1.1.1.

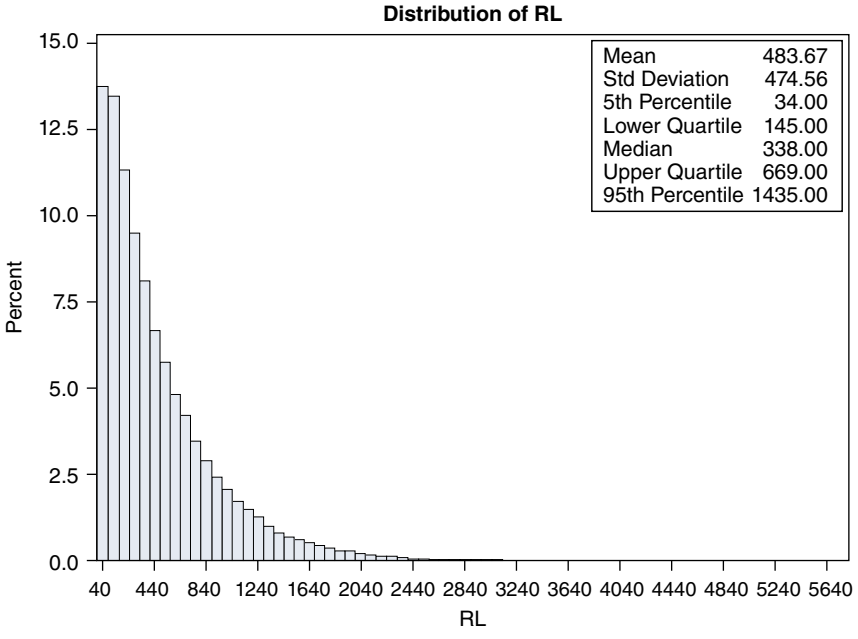


FIGURE 4.2 *SAS output of the RL distribution where the design parameters* (n = *10,* a = *10) of the Shewhart-SN chart were chosen for a nominal* $\text{ARL}_{IC} = 500$.

Step 4: The number of subgroups needed until SN_t plots on or outside the control limits for the first time is recorded as an observation from the run-length distribution.

Step 5: Repeat Steps 1 to 4 a total of 100 000 times.

Step 6: Once we have obtained a "data set" with 100 000 observations from the run-length distribution, proc univariate of SAS can be used to obtain the run-length characteristics. For $\delta = 0$, this would be the IC run-length characteristics, whereas for $\delta \neq 0$ this would be the OOC run-length characteristics.

4.2.1.2 Shewhart Control Charts Based on Signed-ranks

On the one hand, while the sign test is a versatile nonparametric test of location, it is one of the simplest and, since it obviously doesn't use a lot of information from the data, it may not be very powerful. On the other hand, the sign test can be applied with minimal assumptions about the distribution, and even with dichotomized data. The latter may be a situation where the only information available for each unit tested is whether the

measurement was higher (or lower) than the target value of the parameter of interest. This may be a plus in some practical situations.

A popular alternative to the sign test is the signed-rank test. This test uses both the signs and the ranks of the data, and is a popular nonparametric test for the median of a symmetric continuous population. It is well known that the signed-rank test is more efficient than the sign test, but the additional efficiency comes at the price of the additional assumption of symmetry. The same discussion applies to the sign and signed-rank control charts, which are based on the corresponding tests. Furthermore, note that the versatile sign test (control chart) can be used for all percentiles, whereas the signed-rank test (control chart) can only be used for the median.

4.2.1.2.1 A Shewhart Signed-rank Chart We begin with the Shewhart chart based on the signed-rank statistic (denoted Shewhart-SR chart). Let $X_{i1}, X_{i2}, \ldots, X_{in}$, with $i = 1,2,\ldots$ and $n > 1$ be samples or subgroups of independent observations taken at random (and sequentially) from an *unknown* but continuous process distribution symmetric about a *known* IC value μ_0 – typically called the target value. Since the process distribution is assumed to be symmetric, the mean and the median are the same and, in this way, the signed-rank chart setting for the median is comparable to the parametric Shewhart $\overline{X}$ chart setting, which assumes normality (which is, of course, a symmetric distribution). Let R_{ij} denote the rank of the absolute deviation (difference) $|x_{ij} - \mu_0|$ among $(|x_{i1} - \mu_0|, |x_{i2} - \mu_0|, \ldots, |x_{in} - \mu_0|)$ for $i = 1, 2, 3, \ldots$ and $j = 1, 2, 3, \ldots, n$. Then let

$$T_n^+ = \sum_{j=1}^{n} \Psi(x_{ij} - \mu_0) R_{ij} \tag{4.4}$$

where $\Psi(t)$ is the indicator function defined earlier. The statistic T_n^+ is the well-known Wilcoxon signed-rank statistic (see, for example, Gibbons and Chakraborti, 2010, p. 195). Clearly, T_n^+ is the sum of the ranks of the absolute values of the deviations corresponding to the positive differences. Now define

$$SR_i = \sum_{j=1}^{n} \varphi(x_{ij} - \mu_0) R_{ij} \tag{4.5}$$

which is the *difference* between the sum of the ranks associated with the *positive differences* (i.e., those observations for which $X_{ij} - \mu_0 > 0$) and the sum of the ranks associated with the *negative differences* (i.e., those observations for which $X_{ij} - \mu_0 < 0$). The SR_i statistic is linearly related to the more classical Wilcoxon signed-rank statistic T_n^+ through the formula

$$SR_i = 2T_n^+ - \frac{n(n+1)}{2}$$

since $T_n^+ + T_n^- = n(n+1)/2$.

Now we continue to introduce the Shewhart-SR chart. The control limits and the center line of the two-sided **Shewhart-SR chart** are

$$\begin{aligned} UCL &= a \\ CL &= 0 \\ LCL &= -a \end{aligned} \tag{4.6}$$

where a is some positive integer between $0 < a \leq \dfrac{n(n+1)}{2}$. Therefore, the Shewhart-SR chart signals that a shift in the median has occurred if $|SR_i| \geq a$, that is, if $SR_i \geq a$ or if $SR_i \leq -a$. Alternatively, as long as $-a < SR_i < a$, the process is considered IC at the target value μ_0.

The one-sided and two-sided forms of Shewhart charts are introduced in Section 3.2.1 and, here, a lower one-sided Shewhart-SR chart signals that a downward shift occurred if $SR_i \leq -a$ (the chart only has an LCL), whereas an upper one-sided Shewhart-SR chart signals that an upward shift occurred if $SR_i \geq a$ (the chart only has a UCL). Here we consider two-sided charts, since they are more general (as the practitioner may not know the direction of the shift) and since one-sided charts have received little attention in the literature.

The calculation of the charting constant a The charting constant a is the smallest integer between $0 < a \leq \dfrac{n(n+1)}{2}$ such that $P_{\mu_0|\text{IC}}(SR_i \geq a) \leq \dfrac{FAR}{2}$. As in the case of the sign chart, again, since the distribution of the charting statistic SR_i is discrete, chances are that the equality above will not be attained exactly. Thus, using a conservative approach as before, a is found so that the attained FAR is less than or equal to the specified FAR. Setting the ideal FAR equal to 0.0027 with $\dfrac{FAR}{2} = \dfrac{0.0027}{2} = 0.00135$ so that a is the smallest integer such that $P_{\mu_0|\text{IC}}(SR_i \geq a) \leq 0.00135$.

Note 4.2 Using Wilcoxon Signed-rank Probabilities to Calculate the Charting Constant *a* for the Shewhart-SR Chart

We focus on the "right-tail" entry in Table B in Appendix A since we are focusing on the UCL, that is, we are focusing on the upper-tail probability $P_{\mu_0|\text{IC}}(T_n^+ \geq t)$ in order to ultimately obtain $P_{\mu_0|\text{IC}}(SR_i \geq a)$.

As in the case of the sign chart, next, we consider two examples. The first one for $n = 5$ illustrates a case where we can't find an attained FAR value that is smaller than a nominal FAR value of 0.0027. As noted earlier, this typically is the case for small values of n. Following this, the next example for $n = 13$ illustrates a case where an attained FAR value less than 0.0027 can be obtained. This typically is the case for larger values of n, as was seen for the sign chart.

For $n = 5$

Across the "right-tail" entry of 15, we have $\frac{FAR}{2} = 0.031$, which is the smallest probability associated with $n = 5$. Although this probability is not less than or equal to 0.00135, it is the smallest probability that we can obtain when working with a sample of size 5. Now, to obtain the *UCL*, that is, a, we use the linear relationship $SR_i = 2T_n^+ - n(n+1)/2$ so that $a = 2(15) - 15 = 15$ with a $FAR = (2)(0.031) = 0.062$. Thus, for $n = 5$ we have $LCL \;/\; UCL = \pm 15$ and $CL = 0$ with a corresponding $FAR = 0.062$, which is the lowest *FAR* achievable. Since the IC value of the median is known or specified (Case K) $ARL_{\text{IC}} = \frac{1}{FAR} = \frac{1}{0.062} \approx 16$. Again, this is not good, since it is far from the typical desired value of 370. However, this is the highest ARL_{IC} achievable when working with a sample of size 5.

For $n = 13$

Across the "right-tail" entry of 86 we have $\frac{FAR}{2} = 0.001$, which is less than 0.00135. Now, to obtain the *UCL*, that is, a, we use the linear relationship $SR_i = 2T_n^+ - n(n+1)/2$ so that $a = 2(86) - \frac{(13)(14)}{2} = 172 - 91 = 81$ with a $FAR = (2)(0.001) = 0.002$, which is less than 0.0027. Thus, for $n = 13$, we have $LCL \;/\; UCL = \pm 81$ and $CL = 0$ with a corresponding $FAR = 0.002$, which is the lowest *FAR* achievable. Since the IC value of the median is known or specified (Case K), $ARL_{\text{IC}} = \frac{1}{FAR} = \frac{1}{0.002} = 500$, which is good, since it is close to the typical desired value of 500.

4.2.1.2.2 A Shewhart Signed-rank Chart with Runs-type Signaling Rules As noted earlier, the Shewhart charts are known to lack sensitivity in detecting small shifts and so, as in the case of the sign chart, we can improve the effectiveness of the Shewhart-SR chart for detecting smaller shifts by adding runs-type signaling rules. This is studied in Chakraborti and Eryilmaz (2007). An illustration of an improved *2-of-2* Shewhart-SR chart is given in Section 4.3.

4.2.1.2.3 Methods of Calculating the Run-length Distribution *The Exact Approach* As we noted earlier, for Shewhart control charts and some Shewhart-type control charts, it is possible to calculate the characteristics of the run-length distribution exactly. Again, the run-length distribution of the Shewhart signed-rank control chart is geometric with the success probability equal to the probability of a signal. For an example, the reader is referred to Bakir (2004) where the *ARL* of the nonparametric Shewhart signed-rank control chart is computed exactly.

The properties of the statistic T_n^+ are well known (see, for example, Gibbons and Chakraborti, 2010, p. 196) and they are given in Table 4.3.

Table 4.3 Distributional properties of the statistics T_n^+ and SR_i, respectively.

	T_n^+	SR_i
Expected value	$E(T_n^+) = \dfrac{n(n+1)}{4}$	$E(SR_i) = E\left(2T_n^+ - \dfrac{n(n+1)}{2}\right) = 0$
Standard deviation	$SD(T_n^+) = \sqrt{\dfrac{n(n+1)(2n+1)}{24}}$	$SD(SR_i) = \sqrt{\dfrac{n(n+1)(2n+1)}{6}}$

We can find the distribution of SR_i via the linear relationship ($SR_i = 2T_n^+ - n(n+1)/2$) and this is given in the last column of Table 4.3.

The probability distributions of T_n^+ and SR_i are both symmetric[1] in the IC case when the median is equal to μ_0. Hence, when the process is IC we have that:

- The probability distributions, given by the pmf's, are referred to as the IC probability distributions; and
- Since the IC distribution of the charting statistic SR_i is symmetric about 0, the control limits will be equal distances away from 0, assuming that the importance of detecting an upward and downward shift is the same.

Figure 4.3 illustrates the IC probability distributions of T_n^+ and SR_i for $n = 5$. It is seen that the two discrete distributions are symmetric about their means, that is, T_n^+ is symmetric around its mean of $\dfrac{n(n+1)}{4} = \dfrac{(5)(6)}{4} = 7.5$ and SR_i is symmetric around its mean of 0.

The exact run-length distribution of Shewhart-SR charts can be found theoretically but it's more involved, unlike the sign chart, because the distribution of the charting statistic does not have a closed form and thus it's more convenient to use computer simulations. This is described in the next section. In passing, we note that one can also find the exact run-length distribution of nonparametric Shewhart-SR control charts with runs-rules using the Markov chain (MC) approach. This has been done in Chakraborti and Eryilmaz (2007) and the reader is referred to their paper for more details.

[1] T_n^+ and SR_i are symmetrically distributed about $n(n+1)/4$ and zero, respectively, as long as the IC median remains at μ_0.

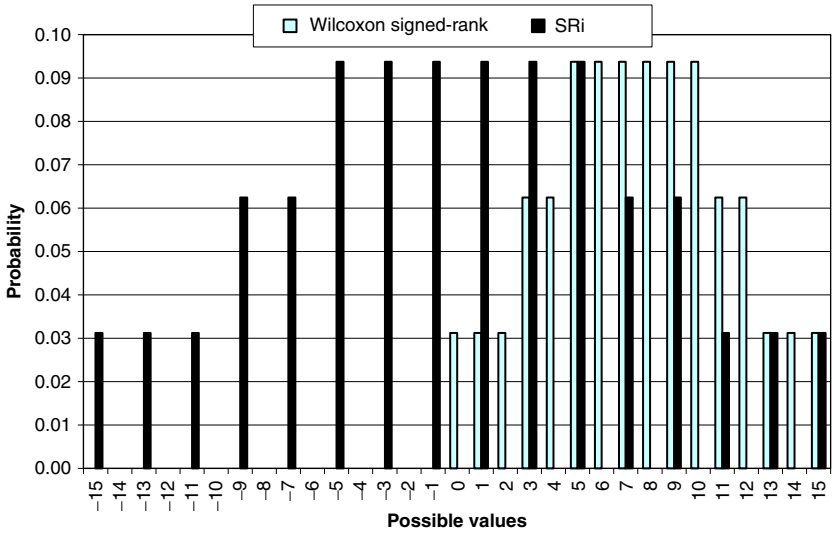


FIGURE 4.3 *The IC probability distribution of* T_n^+ *and* SR_i *for* $n = 5$.

The Simulation Approach The following simulation algorithm can be used to obtain the run-length characteristics for the Shewhart-SR chart. The corresponding SAS program is given in Appendix B, SAS program number 2, and the output is shown below in Figure 4.4.

Step 1: After specifying the subgroup size ($n > 1$) and the size of the shift (δ) to be detected, we generate random subgroups from some symmetric continuous distribution, for example, the standard normal, Student's t, Laplace, Logistic, etc. Any symmetric continuous distribution may be used.

Step 2: Calculate the SR_t statistic for each subgroup.

Step 3: Calculate the control limits, that is, the charting constant a, and compare SR_t to the control limits. The calculation of the charting constant a is explained in detail in Section 4.2.1.2.1.

Step 4: The number of subgroups needed until SR_t plots on or outside the control limits for the first time is recorded as an observation from the run-length distribution.

Step 5: Repeat Steps 1 to 4 a total of 100 000 times.

Step 6: Once we have obtained a "data set" with 100 000 observations from the run-length distribution, we can use, for example, proc univariate of

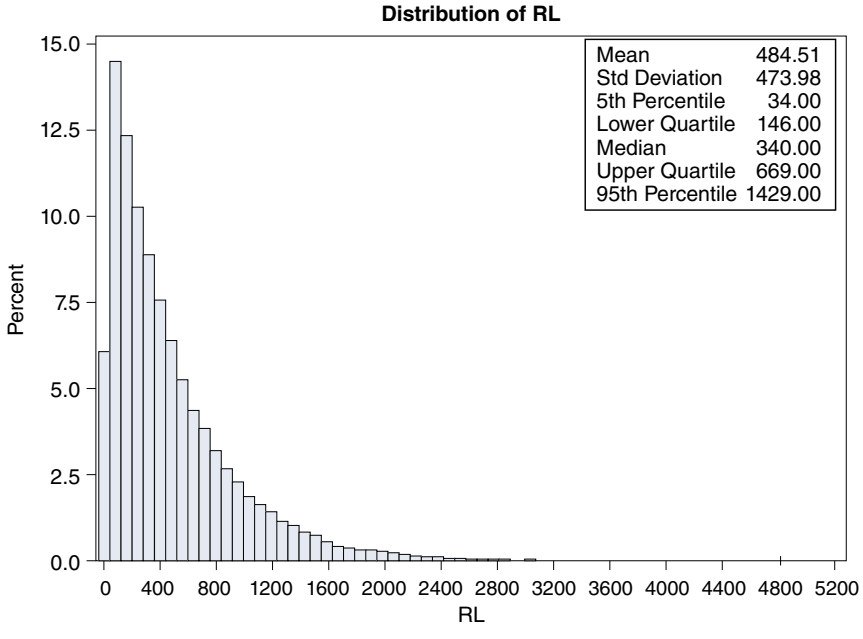


FIGURE 4.4 *SAS output of the RL distribution where the design parameters* (n = *13,* a = *81) of the Shewhart-SR chart were chosen for a nominal* $\text{ARL}_{IC} = 500$.

SAS to obtain the run-length characteristics. For $\delta = 0$ this would be the IC run-length characteristics, whereas for $\delta \neq 0$ this would be the OOC run-length characteristics.

4.2.2 CUSUM Control Charts

While the Shewhart charts are widely known and most often used in practice because of their simplicity and global performance, other classes of charts, such as the CUSUM charts, are useful and sometimes more naturally appropriate in the process control environment in view of the sequential nature of data collection. These charts, typically based on the cumulative sums of a charting statistic that are obtained as data accumulate over time, are known to be more efficient for detecting certain types of shifts in the process.

4.2.2.1 CUSUM Control Charts Based on Signs

The idea behind a nonparametric or distribution-free CUSUM chart is basically the same as that for a parametric chart. Thus, a CUSUM chart based on the sign test statistic (denoted CUSUM-SN) can be obtained by replacing

the statistic $X_i - \mu_0$ in Equations 3.3 and 3.4 with the sign statistic SN_t. In other words, for an **upper one-sided** CUSUM-SN chart, for detecting positive differences from the IC or known target value μ_0, we use

$$SN_t^+ = \max(0, SN_{t-1}^+ + SN_t - k) \text{ for } t = 1, 2, 3 \ldots \tag{4.7}$$

with the starting value $SN_0^+ = 0$ and which signals at the first t for which $SN_t^+ \geq h$.

Similarly, for a **lower one-sided** CUSUM-SN chart, for detecting negative differences from the IC or known target value μ_0, we use

$$SN_t^- = \min(0, SN_{t-1}^- + SN_t + k) \text{ for } t = 1, 2, 3 \ldots \tag{4.8}$$

with the starting value $SN_0^- = 0$ and which signals at the first t for which $SN_t^- \leq -h$.

A corresponding **two-sided** CUSUM-SN chart signals at the first t for which either of the one-sided chart signals, that is, whenever $SN_t^+ \geq h$ or $SN_t^- \leq -h$. Note that, here, the decision interval $h > 0$ and the reference value $k > 0$ are the parameters of the control chart procedure.

Note that the CUSUM-SN chart is constructed following the idea of a parametric CUSUM $\overline{X}$ chart. However, while the CUSUM $\overline{X}$ chart is for the mean and assumes normality, the CUSUM-SN chart is for the median (or for any location parameter) and does not require any assumption about the underlying process, except continuity. When the process distribution is symmetric, the mean and the median are the same and the parametric and nonparametric charts become direct competitors.

4.2.2.2 A CUSUM Sign Control Chart with Runs-type Signaling Rules

Although runs-type signaling rules are generally used to enhance the performance of Shewhart control charts, Riaz, Abbas, and Does (2011) proposed two runs-rules schemes for parametric CUSUM charts. However, to date, enhancing the CUSUM-SN chart with runs-rules schemes has not been considered, and this could be a topic for further research.

4.2.2.3 Methods of Calculating the Run-length Distribution

4.2.2.3.1 The Markov Chain Approach The time that the procedure signals is the first time such that the finite-state MC S_t^+ enters one of the absorbent states where the state space is given by $\Omega^+ = \{\varsigma_0, \varsigma_1, \ldots, \varsigma_{r+s-1}\}$, $S_0^+ = 0$ and

$$S_t^+ = \min\{h, \max\{0, S_{t-1}^+ + SN_t - k\}\}. \tag{4.9}$$

The state that corresponds to a signal by the CUSUM-SN chart is called an absorbent state. Clearly, there is only one absorbent state since the chart signals when S_t^+ falls on or above h.

The distribution of SN_t can easily be obtained from the binomial distribution; recall that $SN_i = 2T_i - n \forall i$, where T_i is binomially distributed with parameters n and $p = P(X_{ij} \geq \mu_0)$. An illustrative example is given in Section 4.3.

An illustration of using the MC approach is given later. Although the exact run-length distribution and various associated characteristics of interest can be found using the MC approach, it is usually more convenient to use computer simulations. This is described next.

4.2.2.3.2 The Simulation Approach The following simulation algorithm can be used to obtain the run-length characteristics for the CUSUM-SN chart. The corresponding SAS program is given in Appendix B, SAS program number 3, and the output is shown in Figure 4.5. Other software such as R® can also be used for the simulation.

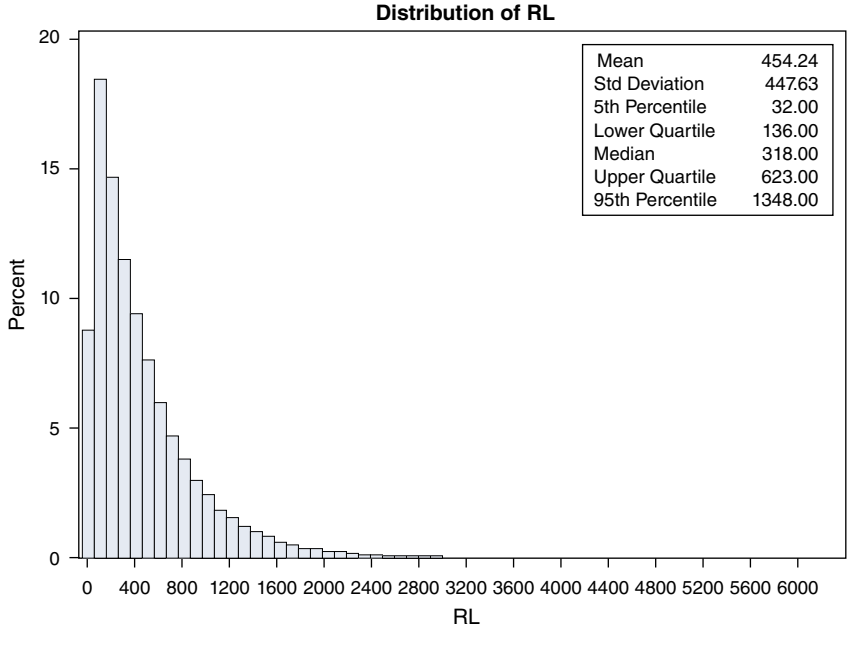


FIGURE 4.5 *SAS output of the RL distribution where the design parameters (*n = *10,* k = *6,* h = *4) of the CUSUM-SN chart were chosen for a nominal* $\text{ARL}_{IC} = 500$.

Step 1: After specifying the subgroup size ($n \geq 1$) and the size of the shift (δ) to be detected, we generate random subgroups from some continuous distribution, for example, the standard normal, Student's t, Laplace, Logistic, Gamma, etc. Any continuous distribution, whether symmetric or asymmetric, may be used.

Step 2: Specify the two design parameters, k and h (the reader is referred to Section 3.2.2 for guidance on choosing the design parameters).

Step 3: Calculate the SN_t statistics and the corresponding SN_t^- and SN_t^+ charting statistics for each subgroup.

Step 4: Compare the charting statistics to the decision interval h.

Step 5: The number of subgroups needed until SN_t^- plots on or below $-h$ the first time or until SN_t^+ plots on or above h the first time is recorded as an observation from the run-length distribution.

Step 6: Repeat Steps 1 to 5 a total of 100 000 times.

Step 7: Once we have obtained a "data set" with 100 000 observations from the run-length distribution, proc univariate of SAS can be used to obtain the run-length characteristics. For $\delta = 0$, this would be the IC run-length characteristics, whereas for $\delta \neq 0$ this would be the OOC run-length characteristics.

4.2.2.4 CUSUM Control Charts Based on Signed-ranks

As mentioned previously, when defining the CUSUM-SN chart, the CUSUM charts are useful and are sometimes more naturally appropriate in the process control environment in view of the sequential nature of data collection. These charts, typically based on the cumulative sums of a charting statistic that are obtained as data accumulate over time, are known to be more efficient for detecting certain types of shifts in the process. Here, we describe the CUSUM signed-rank charts (denoted by CUSUM-SR), which was proposed by Bakir and Reynolds (1979).

4.2.2.4.1 A CUSUM Signed-rank Control Chart An **upper one-sided** CUSUM-SR chart for detecting positive deviations from the IC or known target value μ_0 is defined as

$$SR_t^+ = \max(0, SR_{t-1}^+ + SR_t - k) \text{ for } t = 1, 2, 3 \ldots \tag{4.10}$$

with the starting value $SR_0^+ = 0$, which signals at the first t for which $SR_t^+ \geq h$.

Similarly, for a **lower one-sided** CUSUM-SR chart to detect negative deviations from the IC or known target value μ_0, we define

$$SR_t^- = \min(0, SR_{t-1}^- + SR_t - k) \text{ for } t = 1, 2, 3 \ldots \tag{4.11}$$

with the starting value $SR_0^- = 0$, which signals at the first t for which $SR_t^- \leq -h$.

A corresponding **two-sided** CUSUM-SR chart signals as the first t for which either of the one-sided chart signals, that is, whenever $SR_t^+ \geq h$ or $SR_t^- \leq -h$. Note that, here, the decision interval $h > 0$ and the reference value $k > 0$ are (as in the past) the parameters of the control chart procedure. This chart is used to detect a shift in the mean in any direction, higher or lower.

4.2.2.4.2 A CUSUM Signed-rank Control Chart with Runs-type Signaling Rules Although runs-type signaling rules are generally used to enhance the performance of Shewhart control charts, Riaz, Abbas, and Does (2011) proposed two runs-rules schemes for parametric CUSUM charts. However, to date, enhancing the CUSUM-SR chart with runs-rules schemes has not been considered.

4.2.2.4.3 Methods of Calculating the Run-length Distribution *The Markov Chain Approach* The time that the procedure signals is the first time that the finite-state MC S_t^+ enters the state ς_{r+s-1} where the state space is given by $\Omega^+ = \{\varsigma_0, \varsigma_1, \ldots, \varsigma_{r+s-1}\}$, $S_0^+ = 0$ and

$$S_t^+ = \min\{h, \max\{0, S_{t-1}^+ + SR_t - k\}\}. \tag{4.12}$$

Note that, although Equation 4.12 is not explicitly given in Bakir and Reynolds (1979), it is derived from the same principle as used for Equation 4.9, which was for the CUSUM-SN chart, and that formula is taken from Amin, Reynolds, and Bakir (1995) in their Equation 4.3.

The state corresponding to a signal by the CUSUM chart is called an absorbent state. Clearly, there is only one absorbent state since the chart signals when S_t^+ falls on or above h. The distribution of SR_t can easily be obtained from the distribution of the Wilcoxon signed-rank statistic T_n^+ (recall that $SR_i = 2T_n^+ - \dfrac{n(n+1)}{2} \forall i$). A detailed example is given in Section 4.3.

A detailed illustration of the implementation of Equation 4.12, that is, making use of the MC approach, is given later (see Example 4.10 and, specifically, Table 4.13). Although the exact run-length distribution and its

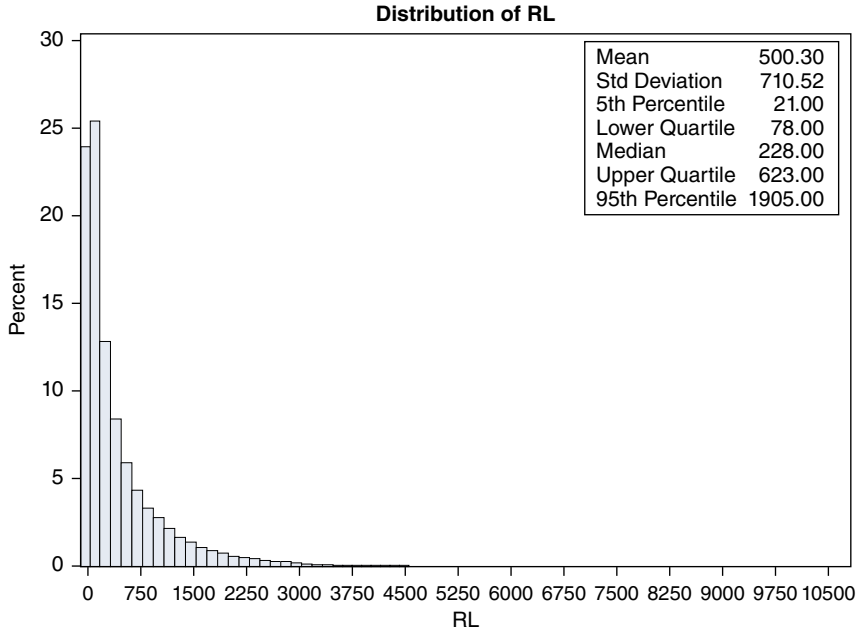


FIGURE 4.6 *SAS output of the RL distribution where the design parameters (*n = *10,* k = *41,* h = *24) of the CUSUM-SR chart were chosen so that nominal* $\text{ARL}_{IC} = 500$.

characteristics of interest can be found using the MC approach, it is usually more convenient to use computer simulation. This is described next.

The Simulation Approach The following simulation algorithm can be used to obtain the run-length characteristics for the CUSUM-SR chart. The corresponding SAS program is given in Appendix B, SAS program number 4, and the output is shown in Figure 4.6. Again, other software such as R can also be used for the simulation.

Step 1: After specifying the subgroup size ($n > 1$) and the size of the shift (δ) to be detected, we generate random subgroups from some symmetric continuous distribution, for example, the standard normal, Student's t, Laplace, Logistic, etc. Any symmetric continuous distribution may be used.

Step 2: Specify the two design parameters, k and h (the reader is referred to Section 3.2.2 for guidance on choosing the design parameters).

Step 3: Calculate the SR_t statistics and the corresponding SR_t^- and SR_t^+ charting statistics for each subgroup.

Step 4: Compare the charting statistics to the decision interval h.

Step 5: The number of subgroups needed until SR_t^- plots on or below $-h$ for the first time or until SR_t^+ plotting on or above h for the first time is recorded as an observation from the run-length distribution.

Step 6: Repeat Steps 1 to 5 a total of 100 000 times.

Step 7: Once we have obtained a "data set" with 100 000 observations from the run-length distribution, proc univariate of SAS can be used to obtain the run-length characteristics. For $\delta = 0$ this would be the IC run-length characteristics, whereas for $\delta \neq 0$ this would be the OOC run-length characteristics.

In addition to the Shewhart and CUSUM charts, as noted for the parametric charts, another popular class of control charts used in practice is the EWMA charts. The EWMA charts also take advantage of the sequentially (time ordered) accumulating nature of the data arising in a typical SPC environment, and are known to be efficient in detecting smaller shifts but are easier to implement than the CUSUM charts. We discuss distribution-free EWMA charts based on the SN and the SR statistics below.

4.2.3 EWMA Control Charts

4.2.3.1 EWMA Control Charts Based on Signs

First, we discuss the EWMA sign control chart (denoted EWMA-SN) by considering individual observations. Again, following the idea of a parametric EWMA chart, an EWMA control charting procedure based on the sign statistic, SN_i, accumulates the sign statistics $SN_1, SN_2, SN_3, \ldots$ Recall that for individual measurements SN_i can only take on the value of -1 or $+1$. The EWMA-SN charting statistic is defined as

$$Z_t = \lambda SN_t + (1-\lambda)Z_{t-1} \tag{4.13}$$

with the starting value $Z_0 = 0$ and the weighting constant $0 < \lambda \leq 1$.

The center line and the **exact** control limits for a two-sided EWMA-SN chart based on individual measurements are given by

$$\begin{aligned} UCL &= L\sqrt{\frac{\lambda}{2-\lambda}(1-(1-\lambda)^{2t})} \\ CL &= 0 \\ LCL &= -L\sqrt{\frac{\lambda}{2-\lambda}(1-(1-\lambda)^{2t})}. \end{aligned} \tag{4.14}$$

The control limits simplify when $t \to \infty$, that is, for large values of t, and are given by

$$\begin{aligned} UCL &= +L\sqrt{\frac{\lambda}{2-\lambda}} \\ LCL &= -L\sqrt{\frac{\lambda}{2-\lambda}}. \end{aligned} \tag{4.15}$$

These control limits are called the **asymptotic** or the **steady-state** control limits, which are applicable when the process has reached a steady state after running for some length of time. The process is considered IC when the charting statistic Z_t, $t = 1,2,3\ldots$, falls between the two control limits, but as soon as a charting statistic falls on or outside the UCL or the LCL the process is declared OOC and typically a search for assignable causes would be started. The derivation of the exact control limits is given below for the more general case where subgrouped data, n, can be set equal to 1 to get the expression above.

The two-sided EWMA can be modified to form a one-sided statistic in much the same way a two-sided CUSUM can be made into a one-sided statistic. An upper one-sided EWMA, for example, is given by $Z_i^+ = \max[0, \lambda SN_t + (1-\lambda)Z_{t-1}]$ for $t = 1,2,3\ldots$ with starting value $z_0^+ = 0$. If the charting statistic Z_t plots on or above the UCL, the process is considered OOC and a search for assignable causes is necessary. Note that the charting statistics for a lower one-sided EWMA chart can be set up in a similar way. This is left as an exercise for the reader.

Next, we consider the case of subgrouped data. The EWMA-SN charting statistic based on subgrouped data is similar to that of the EWMA-SN chart based on individual measurements. However, now SN_t can take on integer values between (and including) $-n$ to $+n$. The center line (CL) and the **exact** control limits for a two-sided EWMA-SN chart based on subgrouped data are given by

$$\begin{aligned} UCL &= L\sqrt{\frac{\lambda n}{2-\lambda}(1-(1-\lambda)^{2t})} \\ CL &= 0 \\ LCL &= -L\sqrt{\frac{\lambda n}{2-\lambda}(1-(1-\lambda)^{2t})}. \end{aligned} \tag{4.16}$$

The control limits simplify when $t \to \infty$, which is for large values of t, and are given by

$$\begin{aligned} UCL &= +L\sqrt{\frac{\lambda n}{2-\lambda}} \\ LCL &= -L\sqrt{\frac{\lambda n}{2-\lambda}}. \end{aligned} \tag{4.17}$$

These control limits are called the **asymptotic** or the **steady-state** control limits, which are applicable when the process has reached a steady state after running for some length of time. Similar to the case with individual measurements, the process is considered IC when the charting statistic Z_t, $t = 1,2,3\ldots$, falls between the two control limits. However, as soon as a charting statistic falls on or outside the UCL or the LCL, the process is declared OOC and typically a search for assignable causes would be started.

4.2.3.2 EWMA Control Charts Based on Signs with Runs-type Signaling Rules

Although runs-type signaling rules are generally used to enhance the performance of Shewhart control charts, Abbas, Riaz, and Does (2011) proposed two runs-rules schemes for the parametric EWMA charts. However, to date, enhancing the EWMA-SN chart with runs-rules schemes has not been considered and this is a topic for further research.

4.2.3.3 Methods of Calculating the Run-length Distribution

4.2.3.3.1 The Markov Chain Approach Recall that we are interested in calculating the one-step transition probabilities; $M = [p_{ij}]$ for $i, j = -s, -s+1, \ldots, s-1, s$. In order to calculate these probabilities, we assume that the charting statistic is equal to S_i whenever it is in state i. For j non-absorbing, we obtain

$$\begin{aligned} &p_{ij} = P(S_j - \tau < Z_k \leq S_j + \tau \,|\, Z_{k-1} = S_i) \;\; \forall j \, (j \neq s) \\ &\text{and} \\ &p_{ij} = P(S_j - \tau < Z_k < S_j + \tau \,|\, Z_{k-1} = S_i) \;\; \text{for} \;\; j = s. \end{aligned} \tag{4.18}$$

Note that the subscript k does not indicate that the charting statistic is time dependent; it simply refers to the charting statistic under consideration. For j absorbing, we use the fact that each row of an M must add to one and therefore the probabilities of going from a transient state to the absorbent state are found by subtraction. Finding the essential transition probability matrix, Q, involves finding the transition probabilities of going from one transient state to another. In order to calculate the transition probabilities for the EWMA-SN chart, we use the charting statistic and

substitute this into Equation 4.18 to obtain

$$p_{ij} = P\left(\left(\left(\frac{(S_j - \tau) - (1-\lambda)S_i}{\lambda}\right) + n\right)/2 < T_k \right.$$
$$\left. \leq \left(\left(\frac{(S_j + \tau) - (1-\lambda)S_i}{\lambda}\right) + n\right)/2\right) \forall j\,(j \neq s)$$

and (4.19)

$$p_{ij} = P\left(\left(\left(\frac{(S_j - \tau) - (1-\lambda)S_i}{\lambda}\right) + n\right)/2 < T_k \right.$$
$$\left. < \left(\left(\frac{(S_j + \tau) - (1-\lambda)S_i}{\lambda}\right) + n\right)/2\right) \quad \text{for } j = s.$$

Since the values n, τ, λ, S_i, and S_j are known constants (λ and n are chosen by the practitioner and τ, S_i, and S_j are calculated), the binomial probabilities in Equation 4.19 can easily be calculated. The proof of Equation 4.19 is given below, and an illustrative example is given in Section 4.3.

Proof: For j non-absorbing, that is, $j = -s, -s+1, \ldots, s-1$, we obtain the one-step transition probabilities $p_{ij} = P(S_j - \tau < Z_k \leq S_j + \tau \,|\, Z_{k-1} = S_i)$. This is the probability that Z_k is within state j, conditioned on Z_{k-1} being equal to the midpoint of state i. We use the definition of the charting statistic and the fact that $SN_k = 2T_k - n$, where T_k is binomially distributed with parameters n and $p = P(X_{ij} > \mu_0)$. This transition probability can be written as

$$\begin{aligned}
p_{ij} &= P(S_j - \tau < \lambda SN_k + (1-\lambda)Z_{k-1} \leq S_j + \tau | Z_{k-1} = S_i) \\
&= P(S_j - \tau < \lambda SN_k + (1-\lambda)S_i \leq S_j + \tau) \\
&= P\left(\frac{(S_j - \tau) - (1-\lambda)S_i}{\lambda} < SN_k \leq \frac{(S_j + \tau) - (1-\lambda)S_i}{\lambda}\right) \\
&= P\left(\frac{(S_j - \tau) - (1-\lambda)S_i}{\lambda} < 2T_k - n \leq \frac{(S_j + \tau) - (1-\lambda)S_i}{\lambda}\right) \\
&= P\left(\left(\frac{(S_j - \tau) - (1-\lambda)S_i}{\lambda} + n\right)/2 < T_k \right. \\
&\qquad \left. \leq \left(\frac{(S_j + \tau) - (1-\lambda)S_i}{\lambda} + n\right)/2\right).
\end{aligned}$$

Once the p_{ij} are calculated, the MC approach can be used to calculate the exact run-length distribution, and an illustration is given in Section 4.3. However, as we have noted before, it is usually more convenient to use computer simulations, and this is described next. ■

4.2.3.3.2 The Simulation Approach The following simulation algorithm can be used to obtain the run-length characteristics for the EWMA-SN chart. The corresponding SAS program is given in Appendix B, SAS program number 5, and the output is shown in Figure 4.7.

Step 1: After specifying the subgroup size ($n \geq 1$) and the size of the shift (δ) to be detected, we generate random subgroups from some continuous distribution, for example, the standard normal, Student's t, Laplace, Logistic, Gamma, etc. Any continuous distribution, whether symmetric or asymmetric, may be used.

Step 2: Specify the two design parameters, λ and L (the reader is referred to Section 3.2.3 for guidance on choosing the design parameters).

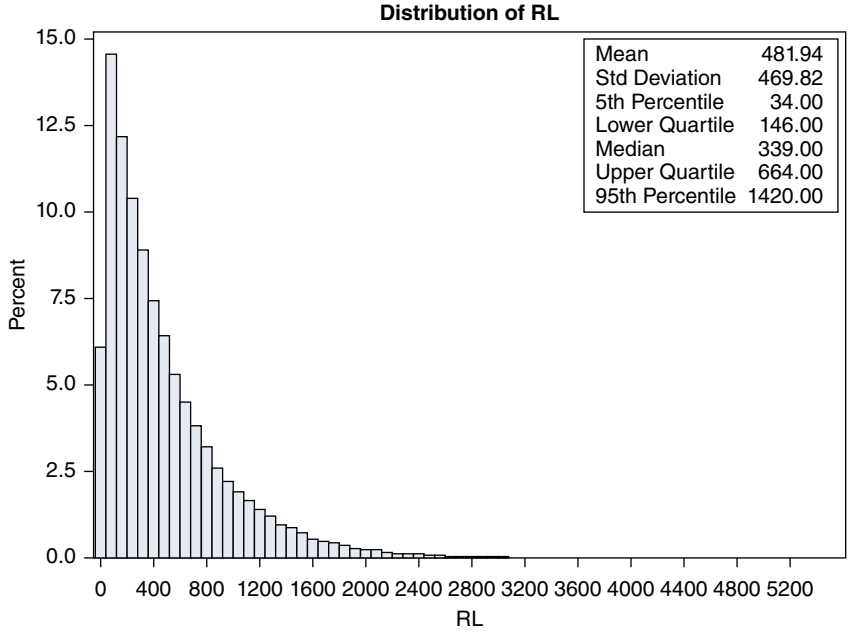


FIGURE 4.7 *SAS output of the RL distribution where the design parameters* (n = *1,* $\lambda = 0.10$, L = *2.667) of the EWMA-SN chart were chosen for a nominal* $\text{ARL}_{IC} = 500$.

Step 3: Calculate the SN_t statistics and the corresponding charting statistics Z_t for each subgroup.

Step 4: Calculate the exact control limits (see Equation 4.16) or the steady-state control limits (see Equation 4.17) and compare Z_t to the control limits.

Step 5: The number of subgroups needed until Z_t plots on or outside the control limits for the first time is recorded as an observation from the run-length distribution.

Step 6: Repeat Steps 1 to 5 a total of 100 000 times.

Step 7: Once we have obtained a "data set" with 100 000 observations from the run-length distribution, proc univariate of SAS can be used to obtain the run-length characteristics. For $\delta = 0$, this would be the IC run-length characteristics, whereas for $\delta \neq 0$ this would be the OOC run-length characteristics.

Note 4.3 The Number of Subintervals Between the Control Limits

When implementing the MC approach to the EWMA-SN chart, a question arises regarding the number of subintervals, that is, the value of ν that one should use. When ν is sufficiently large, the MC approach provides an effective method that allows the run-length properties of the EWMA control scheme to be accurately approximated. In practice, values of ν around 100 yield satisfactory approximations (see Yu, 2007, p. 20). Here, we verify this claim by considering the EWMA-SN chart.

Verification of the MC approach using a Monte Carlo simulation suggests that the discrepancies are within 2% of the simulated values when $\nu = 1001$ (see Table 4.4). Taking larger values of ν would result in more accurate answers, but in doing so, some run-length characteristics could not be computed within a practical period of time.

These results are similar to what was found for the parametric (normal theory) EWMA chart (see Table 3.12), where it was found that the larger the value of ν the smaller the percentage difference between the simulated values and the values obtained using the MC approach.

From Table 4.4, we can see that, as might be expected, the larger the value of ν, the smaller the percentage difference between the simulated values and the values obtained using the MC approach. When the design parameters are taken to be $\lambda = 0.05$ and $L = 2.472$, for example, we have a large percentage difference of 18.62% when ν is taken to be small ($\nu = 51$) as opposed to a small percentage difference of 0.18% when ν is taken to

Table 4.4 ARL_{IC} values and the absolute percentage difference between the simulated values[2] and the values obtained using the MC approach for the EWMA-SN chart for various values of ν.

			$\lambda = 0.05$ $L = 2.472$		$\lambda = 0.10$ $L = 2.588$	
Markov chain approach	ν		Attained		Attained	
			ARL	Diff (%)	ARL	Diff (%)
		51	439.07	(18.62%)	384.92	(4.06%)
		59	386.23	(4.34%)	420.36	(13.64%)
		67	379.34	(2.48%)	337.55	(8.75%)
		75	410.76	(10.97%)	360.21	(2.62%)
		83	410.48	(10.90%)	423.25	(14.42%)
		101	396.12	(7.02%)	357.42	(3.38%)
		151	380.61	(2.83%)	358.61	(3.05%)
		201	391.25	(5.70%)	330.66	(10.61%)
		251	368.50	(0.45%)	354.17	(4.26%)
		301	369.42	(0.20%)	379.63	(2.63%)
		401	365.71	(1.20%)	370.68	(0.21%)
		501	377.47	(1.98%)	377.33	(2.01%)
		1001	369.49	(0.18%)	373.80	(1.05%)
100 000 simulations			370.15		369.91	

be large ($\nu = 1001$). It appears that $\nu \geq 501$ should be fairly satisfactory in practice.

4.2.3.4 EWMA Control Charts Based on Signed-ranks

Like the EWMA-SN chart, we can consider the EWMA chart based on signed-rank denoted EWMA-SR. The EWMA-SR chart was first introduced by Amin and Searcy (1991) and this chart is discussed further in this section.

An EWMA-SR chart is based on the signed-rank statistic, SR_i, which is defined in Equation 4.5 in the general EWMA formula and accumulates the statistics $SR_1, SR_2, SR_3, \ldots$ Thus, the EWMA-SR chart is defined as

$$Z_t = \lambda SR_t + (1 - \lambda)Z_{t-1} \tag{4.20}$$

with the starting value $Z_0 = 0$ and the weighting constant $0 < \lambda \leq 1$.

[2] Note that steady-state control limits were used in both the simulations and the MC approach in order to make the comparisons fair.

The center line and the **exact** control limits are given by

$$\begin{aligned} UCL &= L\sigma_{SR_i}\sqrt{\frac{\lambda}{2-\lambda}(1-(1-\lambda)^{2t})} \\ CL &= 0 \\ LCL &= -L\sigma_{SR_i}\sqrt{\frac{\lambda}{2-\lambda}(1-(1-\lambda)^{2t})} \end{aligned} \tag{4.21}$$

whereas the **asymptotic** or the **steady-state** control limits, that is, when $t \to \infty$, are given by

$$\begin{aligned} UCL &= L\sigma_{SR_i}\sqrt{\frac{\lambda}{2-\lambda}} \\ LCL &= -L\sigma_{SR_i}\sqrt{\frac{\lambda}{2-\lambda}} \end{aligned} \tag{4.22}$$

where $\sigma_{SR_t} = \sqrt{\dfrac{n(n+1)(2n+1)}{6}}$ is the IC standard deviation of the SR_i statistic if there are no ties *within* a sample or subgroup (see Section 4.2.1.2.3 for a discussion on the properties of the Wilcoxon signed-rank statistic, the SR_i statistic, and see Section 4.2.1.1.1 for the discussion on the subject of ties). Note that σ_{SR_i} is obtained using the linear relationship between SR_i and T_n^+ as follows

$$\begin{aligned} \sigma^2_{SR_t} &= VAR(SR_t) \\ &= VAR\left(2T_n^+ - \frac{n(n+1)}{2}\right) = 4\left(\frac{n(n+1)(2n+1)}{24}\right) \\ &= \frac{n(n+1)(2n+1)}{6}. \end{aligned}$$

The process is considered IC, while all the charting statistics Z_t, $t = 1,2,3\ldots$, fall between the two control limits, but as soon as a charting statistic falls on or outside the UCL or the LCL, the process is declared OOC and typically a search for assignable causes would be started.

The reader is referred to Section 3.2.3 for a detailed discussion on the choice of the design parameters λ and L. In short, the recommendation is to choose a small λ, say, equal to 0.05, when small shifts are of interest, choose $\lambda = 0.10$ if moderate shifts are of concern, and choose $\lambda = 0.20$ if larger shifts are of interest. Thus, the choice of $\lambda = 0.05$ indicates that small shifts are of interest. After λ is chosen, the second step involves finding L so that a desired ARL_{IC} is attained.

A one-sided EWMA-SR chart can be constructed from the two-sided EWMA chart in much the same way as for the CUSUM chart, for example, an upper one-sided EWMA charting statistic is given by

$Z_i^+ = \max[0, \lambda SR_t + (1-\lambda)Z_{t-1}]$ for $t = 1,2,3,\ldots$ with starting value $Z_0^+ = 0$. If the charting statistic Z_t plots on or above the UCL, the process is considered OOC and a search for assignable causes is necessary. Note that the UCL for the upper one-sided EWMA chart will probably differ from the UCL of the two-sided EWMA chart. A lower one-sided EWMA charting statistic can be defined similarly. The derivation of the exact control limits is given in Appendix 4.3.

4.2.3.5 An EWMA-SR control chart with runs-type signaling rules

Although runs-type signaling rules are generally used to enhance the performance of Shewhart control charts, Abbas, Riaz, and Does (2011) proposed two runs-rules schemes for parametric EWMA charts. However, to date, enhancing the EWMA-SR chart with runs-rules schemes has not been considered. This is another topic for further research.

4.2.3.6 Methods of Calculating the Run-length Distribution

4.2.3.6.1 The Markov Chain Approach Again, as in the case of the CUSUM chart, we can use the MC approach to calculate the exact run-length distribution of the EWMA chart and, to this end, recall that we are interested in calculating the one-step transition probabilities $M = [p_{ij}]$ for $i, j = -s, -s+1, \ldots, s-1, s$. In order to calculate these probabilities, we assume that the charting statistic is equal to S_i whenever it is in state i. For j non-absorbing, we obtain

$$\begin{aligned} &p_{ij} = P(S_j - \tau < Z_k \leq S_j + \tau \,|\, Z_{k-1} = S_i) \;\; \forall j \, (j \neq s) \\ &\text{and} \\ &p_{ij} = P(S_j - \tau < Z_k < S_j + \tau \,|\, Z_{k-1} = S_i) \;\; \text{for } j = s. \end{aligned} \tag{4.23}$$

Note that the subscript k does not indicate that the charting statistic is time dependent; it simply refers to the charting statistic under consideration. For j absorbing, we use the fact that each row of M must add to one and therefore the probabilities of going from a transient state to the absorbent state are found by subtraction. Finding the essential transition probability matrix, Q, involves finding the transition probabilities of going from one transient state to another. In order to calculate the transition probabilities for the EWMA-SR chart, we use the charting statistic and substitute this

into Equation 4.23 to obtain

$$p_{ij} = P\left(\left(\frac{(S_j-\tau)-(1-\lambda)S_i}{\lambda}+\frac{n(n+1)}{2}\right)/2 < T_n^+ \right.$$
$$\left. \leq \left(\frac{(S_j+\tau)-(1-\lambda)S_i}{\lambda}+\frac{n(n+1)}{2}\right)/2\right) \quad \forall j\,(j \neq s)$$

and (4.24)

$$p_{ij} = P\left(\left(\frac{(S_j-\tau)-(1-\lambda)S_i}{\lambda}+\frac{n(n+1)}{2}\right)/2 < T_n^+ \right.$$
$$\left. < \left(\frac{(S_j+\tau)-(1-\lambda)S_i}{\lambda}+\frac{n(n+1)}{2}\right)/2\right) \quad \text{for } j = s.$$

Since the values n, τ, λ, S_i, and S_j are known constants (λ and n are chosen by the practitioner and τ, S_i, and S_j are calculated), the Wilcoxon signed-rank probabilities in Equation 4.24 can easily be calculated. The proof of Equation 4.24 is given below, and an illustrative example is given in Section 4.3.

Proof: For j non-absorbing, that is, $j = -s, -s+1, \ldots, s-1$, we obtain the one-step transition probabilities $p_{ij} = P(S_j - \tau < Z_k \leq S_j + \tau \,|\, Z_{k-1} = S_i)$. This is the probability that Z_k is within state j, conditioned on Z_{k-1} being equal to the midpoint of state i. By using the definition of the charting statistic and the fact that $SR_k = 2T_n^+ - \dfrac{n(n+1)}{2}$ where T_n^+ is the well-known Wilcoxon signed-rank statistic, this transition probability can be written as

$$\begin{aligned}
p_{ij} &= P(S_j - \tau < \lambda SR_k + (1-\lambda)Z_{k-1} \leq S_j + \tau | Z_{k-1} = S_i) \\
&= P(S_j - \tau < \lambda SR_k + (1-\lambda)S_i \leq S_j + \tau) \\
&= P\left(\frac{(S_j-\tau)-(1-\lambda)S_i}{\lambda} < SR_k \leq \frac{(S_j+\tau)-(1-\lambda)S_i}{\lambda}\right) \\
&= P\left(\frac{(S_j-\tau)-(1-\lambda)S_i}{\lambda} < 2T_n^+ - \frac{n(n+1)}{2} \leq \frac{(S_j+\tau)-(1-\lambda)S_i}{\lambda}\right) \\
&= P\left(\left(\frac{(S_j-\tau)-(1-\lambda)S_i}{\lambda}+\frac{n(n+1)}{2}\right)/2 < T_n^+ \right. \\
&\qquad \left. \leq \left(\frac{(S_j+\tau)-(1-\lambda)S_i}{\lambda}+\frac{n(n+1)}{2}\right)/2\right).
\end{aligned}$$

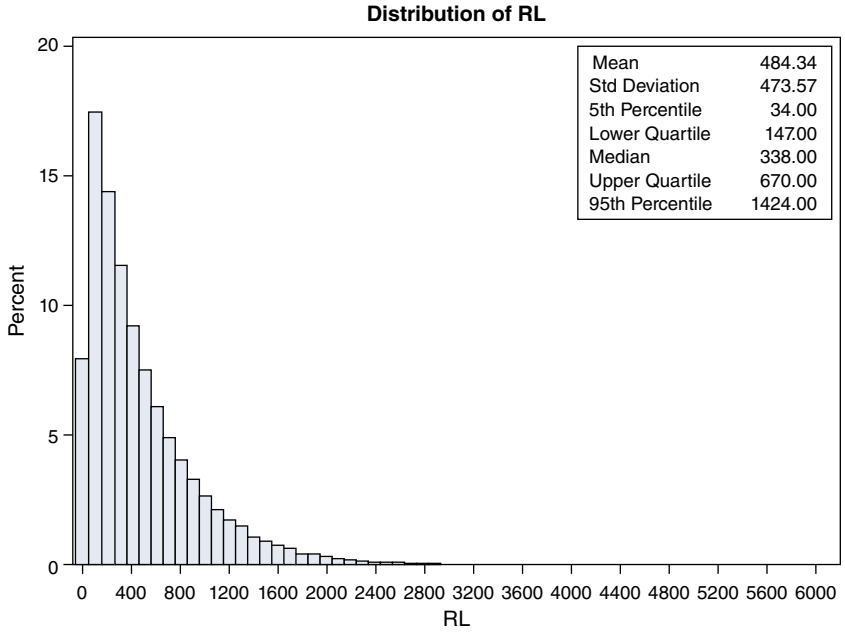


FIGURE 4.8 *SAS output of the RL distribution where the design parameters* ($n = 10$, $\lambda = 0.1$, $L = 2.794$) *of the EWMA-SR chart were chosen for a nominal* $\text{ARL}_{IC} = 500$.

Once the p_{ij} are calculated, the MC approach can be used to calculate the exact run-length distribution, and an illustration is given in Section 4.3. However, as we have noted before, it is usually more convenient to use computer simulations, and this is described next. ■

The Simulation Approach The following simulation algorithm can be used to obtain the run-length characteristics for the EWMA-SR chart. The corresponding SAS program is given in Appendix B, SAS program number 6, and the output is shown below in Figure 4.8.

Step 1: After specifying the subgroup size ($n > 1$) and the size of the shift (δ) to be detected, we generate random subgroups from some symmetric continuous distribution, for example, the standard normal, Student's t, Laplace, Logistic, etc. Any symmetric continuous distribution may be used.

Step 2: Specify the two design parameters, λ and L (the reader is referred to Section 3.2.3 for guidance on choosing the design parameters).

Step 3: Calculate the SR_t statistics and the corresponding Z_t charting statistics for each subgroup.

Step 4: Calculate the exact control limits (see Equation 4.21) or the steady-state control limits (see Equation 4.22) and compare Z_t to the control limits.

Step 5: The number of subgroups needed until Z_t plots on or outside the control limits for the first time is recorded as an observation from the run-length distribution.

Step 6: Repeat Steps 1 to 5 a total of 100 000 times.

Step 7: Once we have obtained a "data set" with 100 000 observations from the run-length distribution, proc univariate of SAS can be used to obtain the run-length characteristics. For $\delta = 0$, this would be the IC run-length characteristics, whereas for $\delta \neq 0$ this would be the OOC run-length characteristics.

Note 4.4 The Number of Subintervals Between the Control Limits

When implementing the MC approach to the EWMA-SR chart, a question arises about the number of subintervals, that is, the value of ν. When the number of subintervals ν is sufficiently large, the MC approach provides an effective method that allows the run-length properties of the EWMA control scheme to be accurately approximated. In Section 4.2.3 it was shown that, for the EWMA-SN chart, taking larger values of ν would result in more accurate answers, but in doing so, some run-length characteristics could not be computed within a practical period of time. The same findings hold for the EWMA-SR chart.

4.3 DISTRIBUTION-FREE CONTROL CHARTS IN CASE K: ILLUSTRATIVE EXAMPLES

4.3.1 Shewhart Control Charts

Example 4.1 Shewhart-SN Chart Based on Individual Measurements

Column (a) of Table 4.5 shows a data set from Montgomery (2009, Table 9.1, p. 401). There are 30 observations, the first 20 of which were drawn at random from a normal distribution with mean $\mu_0 = 10$ and standard deviation $\sigma_0 = 1$. The remaining 10 observations were drawn from a normal

Table 4.5 Data for the Shewhart-SN chart based on individual measurements and the SN statistics.

Observation number i	(a) X_i	(b) SN_i
1	9.45	−1
2	7.99	−1
3	9.29	−1
4	11.66	1
5	12.16	1
6	10.18	1
7	8.04	−1
8	11.46	1
9	9.20	−1
10	10.34	1
11	9.03	−1
12	11.47	1
13	10.51	1
14	9.40	−1
15	10.08	1
16	9.37	−1
17	10.62	1
18	10.31	1
19	8.52	−1
20	10.84	1
21	10.90	1
22	9.33	−1
23	12.29	1
24	11.50	1
25	10.60	1
26	11.08	1
27	10.38	1
28	11.62	1
29	11.31	1
30	10.52	1

distribution with mean $\mu_1 = 11$ and standard deviation $\sigma_0 = 1$. Thus $\mu_1 = \mu_0 + \delta\sigma_0 = 10 + (1)(1)$ so that $\delta = 1$. Column (b) of Table 4.5 contains the values of the charting statistic, SN_i, for $i = 1,2,\ldots,30$. The first observation, for example, equals 9.45, which is less than 10. Thus, $SN_1 = -1$.

For individual measurements (data), from Table A in Appendix A across the "right-tail" entry of 1 we have $\frac{FAR}{2} = 0.5$, which is the smallest probability associated with $n = 1$. Although this probability is not less than or equal to 0.00135, it is the smallest probability that we can find when working with individual measurements. Now, to obtain the UCL, that is, a, we use the linear relationship $SN_i = 2T_i - n$ so that $a = 2(1) - 1 = 1$ with a $FAR = (2)(0.5) = 1$. Thus, for $n = 1$ we have $LCL \ / \ UCL = \pm 1$ and $CL = 0$ with a corresponding $FAR = 1$, which is the lowest FAR achievable. Since the IC value of the median is known or specified (Case K), the $ARL_{\text{IC}} = \frac{1}{FAR} = \frac{1}{1} = 1$. Again, this is not acceptable since it is far from the typically desired value of 370. In fact, we are obtaining the highest possible FAR and the lowest possible ARL_{IC}.

Thus, we have that the control limits and CL are given by

$$UCL = +1$$
$$CL = 0$$
$$LCL = -1$$

with the resulting Shewhart-SN chart shown in Figure 4.9.

Since we obtain the highest possible FAR (of 1) and the lowest possible ARL_{IC} (also of 1), it is not advisable to use a Shewhart-SN chart when

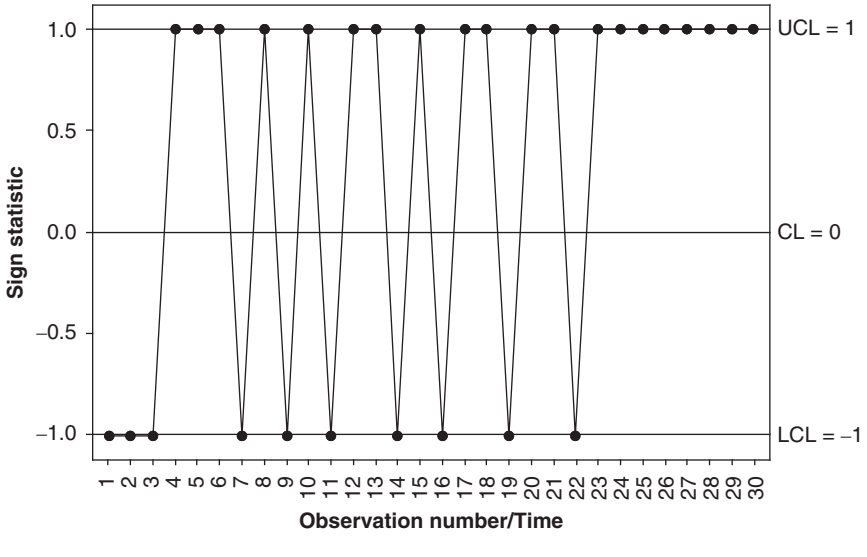


FIGURE 4.9 *A Shewhart-SN chart based on individual measurements.*

working with individual measurements. Later on, it will be shown that the EWMA chart based on the sign statistic (denoted EWMA-SN) and the CUSUM chart based on the sign statistic (denoted CUSUM-SN) perform better for individual measurements.

Example 4.2 A Shewhart-SN Chart Based on Subgrouped Data Although the Shewhart-SN chart is not practically appealing for individual data, as shown above, it can be used for subgrouped data with a moderate size. Column (a) of Table 4.6 displays the observations of 15 independent samples or subgroups from an *unknown* continuous process distribution, each of size $n = 10$, which are to be used to monitor the *known* IC median value $\mu_0 = 0$. Column (b) of Table 4.6 contains the values of the charting statistic, SN_i, for $i = 1,2,\ldots,15$. For example, in sample number $i = 4$, six of the 10 values are larger than zero, four are less than zero, with none equal to zero. Thus, $SN_4 = \sum_{j=1}^{10}(X_{4j} - 0) = 2$.

For a sample size of 10, from Table A in Appendix A across the "right-tail" entry of 10 we have $\frac{FAR}{2} = 0.000977$, which is less than 0.00135. Now, to obtain the UCL, that is, a, we use the linear relationship $SN_i = 2T_i - n$ so that $a = 2(10) - 10 = 10$ with an $FAR = (2)(0.000977) = 0.001994$, which is less than 0.0027. Thus, for $n = 10$, we have $LCL / UCL = \pm$ 10 and $CL = 0$ with a corresponding $FAR = 0.001994$ and $ARL_{\text{IC}} = \frac{1}{FAR} = \frac{1}{0.001994} \approx 502$, which is good since it is close to the typically desired value of 500; typical industry choices include $ARL_{\text{IC}} = 370$ or 500.

Thus, we have that the control limits and center line are given by

$$UCL = +10$$

$$CL = 0$$

$$LCL = -10$$

with the resulting Shewhart-SN chart shown in Figure 4.10.

Since none of the charting statistics falls on or outside the two control limits, the process is considered IC at the target value of the median, $\mu_0 = 0$, and no corrective action or search for assignable causes is necessary. Note that this Shewhart-SN chart is applied without making any assumption about the shape or form of the underlying distribution. For example, we don't need to make or justify that the data follow a normal distribution, as we must for the parametric Shewhart chart for the mean.

It is emphasized that the distribution-free Shewhart-SN chart is one of the simplest and most versatile control charts that can be used to monitor the location of any continuous distribution without knowledge about its variance.

Table 4.6 Data for the Shewhart-SN chart based on subgrouped data and the *SN* statistics.

Sample number	(a)										(b)
i	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9	X_{10}	SN_i
1	−0.30	−1.28	0.24	1.28	1.20	1.73	−2.18	−0.23	1.10	−1.09	0
2	−0.69	−1.69	−1.85	−0.98	−0.77	−2.12	−0.57	−0.40	0.13	−0.37	−8
3	−0.33	−0.37	1.34	−0.09	−0.19	−0.51	1.97	0.87	2.38	−0.65	−2
4	1.66	−1.61	0.54	0.90	1.92	−0.08	−0.52	0.68	−0.38	0.76	2
5	−1.44	−0.85	−1.52	−0.36	−0.03	0.03	−0.32	2.19	−1.74	−0.74	−6
6	−2.58	1.45	−1.28	−0.65	0.76	0.47	0.87	0.60	−1.37	−1.12	0
7	0.69	0.32	−0.94	−0.24	0.13	0.56	0.14	−0.91	1.88	0.49	4
8	0.07	0.83	0.86	−0.64	−0.92	1.11	−1.20	−1.56	0.71	0.64	2
9	2.21	1.44	1.30	0.11	0.00	0.45	−0.03	−1.05	−1.77	0.83	3
10	0.44	0.62	0.21	−1.03	1.24	−0.31	−0.84	−0.82	−0.43	−0.45	−2
11	−0.52	0.85	0.51	−0.61	1.30	−1.76	0.55	−0.12	0.04	−0.65	0
12	−0.55	0.85	0.80	0.46	0.69	1.63	0.30	0.59	1.85	−0.34	6
13	1.04	0.14	1.14	−0.15	−0.78	1.08	−0.58	0.53	0.55	−0.32	2
14	−0.44	−1.37	1.99	−0.57	0.09	−0.23	2.84	1.25	0.88	1.33	2
15	0.19	0.54	−0.25	−1.22	1.27	−0.29	−1.31	0.76	0.78	0.43	2

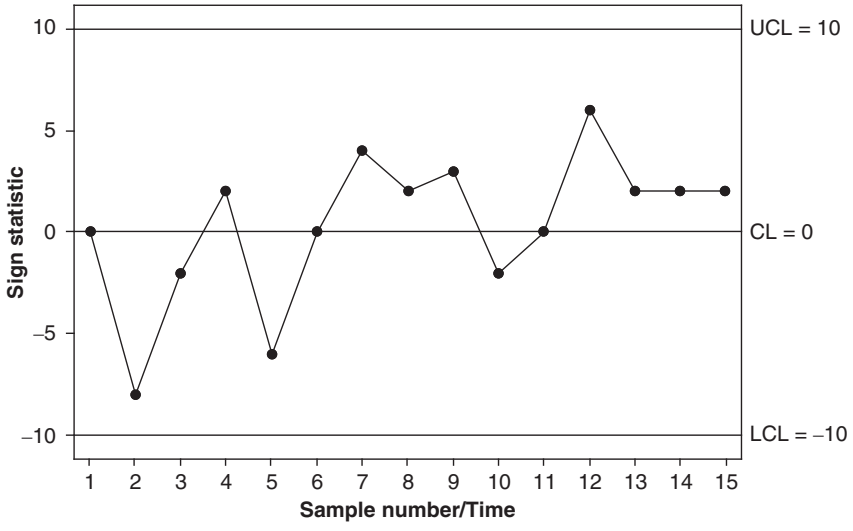


FIGURE 4.10 *A Shewhart-SN chart based on subgrouped data.*

Example 4.3 A Shewhart-SN Chart with Runs-type Signaling Rules

Next, as with the parametric Shewhart $\overline{X}$ chart, the Shewhart-SN chart is enhanced with the improved *2-of-2* runs-type signaling rule and shown here for illustration. The point of this is twofold. First, the simplicity of the distribution-free Shewhart-SN is there and, second, the performance of this simple chart is enhanced. This chart signals when either one point plots on or beyond the LCL/UCL or when two successive points are plotted on or beyond the LWL/UWL. As stated earlier, we only consider the *improved rule* since it has been shown in the literature to outperform the standard rule.

For illustration, an improved *2-of-2* Shewhart-SN chart will be constructed using the same data in Table 4.6. If we now introduce warning limits at, say, ±6, that is, the upper and the lower warning limits are given by $UWL = +6$ and $LWL = -6$, respectively. The resulting improved *2-of-2* Shewhart-SN chart is shown in Figure 4.11.

It should be noted that, typically, the warning limits are found as follows. First, one would set the control limits at some fixed value, say, ±10, and then the warning limits are found via simulation, for example, such that some nominal ARL_{IC} value is attained.

Since there are no two consecutive points plotting on or beyond the LWL/UWL, or no single point plotting on or beyond the LCL/UCL, the process is considered IC at the target median value $\mu_0 = 0$, and no corrective action or search for assignable causes is necessary.

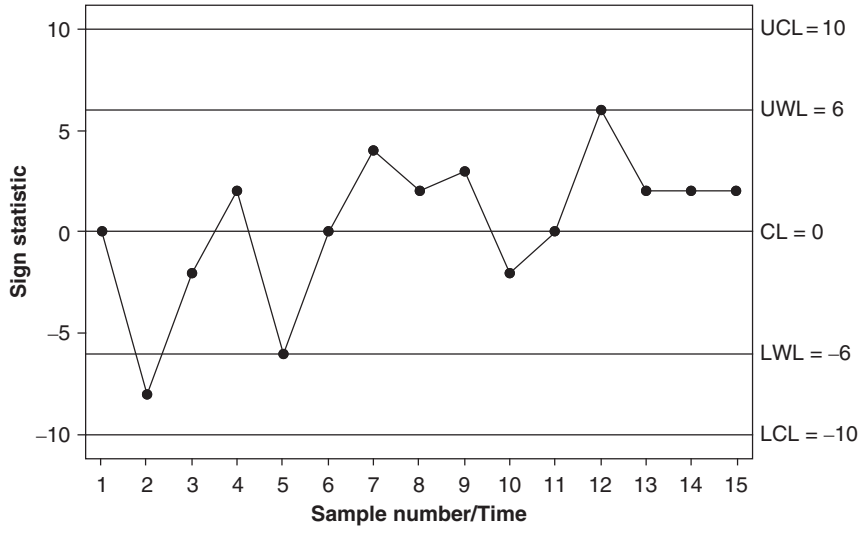


FIGURE 4.11 *A Shewhart-SN chart with runs-type signaling rules.*

Next, consider the distribution-free Shewhart-SR chart to monitor the median of a continuous symmetric distribution, which is based on the signed-rank test statistic.

Example 4.4 A Shewhart Signed-rank Chart

Column (a) of Table 4.7a contains the observations of 20 independent samples, each of size $n = 4$, from an *unknown* continuous distribution, symmetric about the *known* IC median value μ_0, which is taken to be equal to 0 without any loss of generality. Thus, the unknown continuous distribution is assumed to be symmetric about its median 0. The corresponding absolute ranks and the Wilcoxon signed-ranks are shown in column (c) of Table 4.7a and column (b) of Table 4.7b, respectively. The charting statistics for the Shewhart-SR chart are shown in column (c) of Table 4.7b.

If we select a to be the largest value that the charting statistic can take on, that is, $a = \dfrac{n(n+1)}{2} = \dfrac{(4)(5)}{2} = 10$, we have that the control limits and the center line given by

$$UCL = 10$$
$$CL = 0$$
$$LCL = -10.$$

For $n = 4$ and $a = 10$ from Table B in Appendix A across the "right-tail" entry of 10, we have $\dfrac{FAR}{2} = 0.062$, which is the smallest probability

Table 4.7a Data for the Shewhart-SR chart.

Sample number	(a)				(b)				(c)			
i	x_{i1}	x_{i2}	x_{i3}	x_{i4}	$\lvert x_{i1}\rvert$	$\lvert x_{i2}\rvert$	$\lvert x_{i3}\rvert$	$\lvert x_{i4}\rvert$	R_{i1}	R_{i2}	R_{i3}	R_{i4}
1	−0.39	0.03	0.25	0.30	0.39	0.03	0.25	0.30	4	1	2	3
2	−0.61	−1.32	−2.84	−0.07	0.61	1.32	2.84	0.07	2	3	4	1
3	2.04	0.40	0.54	1.12	2.04	0.40	0.54	1.12	4	1	2	3
4	−0.51	−1.63	0.28	−0.96	0.51	1.63	0.28	0.96	2	4	1	3
5	−0.20	−0.01	−0.21	0.82	0.20	0.01	0.21	0.82	2	1	3	4
6	2.03	0.70	−1.21	−0.26	2.03	0.70	1.21	0.26	4	2	3	1
7	0.64	1.48	0.91	1.23	0.64	1.48	0.91	1.23	1	4	2	3
8	0.19	−2.74	−0.85	1.25	0.19	2.74	0.85	1.25	1	4	2	3
9	−0.51	−1.77	−1.69	0.24	0.51	1.77	1.69	0.24	2	4	3	1
10	−0.84	−0.71	−0.82	0.19	0.84	0.71	0.82	0.19	4	2	3	1
11	0.29	0.47	0.04	0.39	0.29	0.47	0.04	0.39	2	4	1	3
12	0.38	2.18	1.24	−0.80	0.38	2.18	1.24	0.80	1	4	3	2
13	−0.86	0.61	−0.70	−1.13	0.86	0.61	0.70	1.13	3	1	2	4
14	−0.31	1.09	−0.43	0.69	0.31	1.09	0.43	0.69	1	4	2	3
15	0.42	1.35	−0.51	−1.80	0.42	1.35	0.51	1.80	1	3	2	4
16	0.39	−0.65	1.57	−0.60	0.39	0.65	1.57	0.60	1	3	4	2
17	1.31	−0.59	−0.21	0.22	1.31	0.59	0.21	0.22	4	3	1	2
18	0.64	0.27	0.26	0.08	0.64	0.27	0.26	0.08	4	3	2	1
19	−0.33	−0.49	0.38	−1.54	0.33	0.49	0.38	1.54	1	3	2	4
20	−0.48	−0.54	1.73	−1.00	0.48	0.54	1.73	1.00	1	2	4	3

Table 4.7b Calculations for the Shewhart-SR chart.

Sample	(a)				(b)				(c)
i	$sign\ (x_{i1}-\mu_0)$	$sign\ (x_{i2}-\mu_0)$	$sign\ (x_{i3}-\mu_0)$	$sign\ (x_{i4}-\mu_0)$	$sign\ (x_{i1}-\mu_0)\ R_{i1}$	$sign\ (x_{i2}-\mu_0)\ R_{i2}$	$sign\ (x_{i3}-\mu_0)\ R_{i3}$	$sign\ (x_{i4}-\mu_0)\ R_{i4}$	SR_i
1	−1	1	1	1	−4	1	2	3	2
2	−1	−1	−1	−1	−2	−3	−4	−1	−10
3	1	1	1	1	4	1	2	3	10
4	−1	−1	1	−1	−2	−4	1	−3	−8
5	−1	−1	−1	1	−2	−1	−3	4	−2
6	1	1	−1	−1	4	2	−3	−1	2
7	1	1	1	1	1	4	2	3	10
8	1	−1	−1	1	1	−4	−2	3	−2
9	−1	−1	−1	1	−2	−4	−3	1	−8
10	−1	−1	−1	1	−4	−2	−3	1	−8
11	1	1	1	1	2	4	1	3	10
12	1	1	1	−1	1	4	3	−2	6
13	−1	1	−1	−1	−3	1	−2	−4	−8
14	−1	1	−1	1	−1	4	−2	3	4
15	1	1	−1	−1	1	3	−2	−4	−2
16	1	−1	1	−1	1	−3	4	−2	0
17	1	−1	−1	1	4	−3	−1	2	2
18	1	1	1	1	4	3	2	1	10
19	−1	−1	1	−1	−1	−3	2	−4	−6
20	−1	−1	1	−1	−1	−2	4	−3	−2

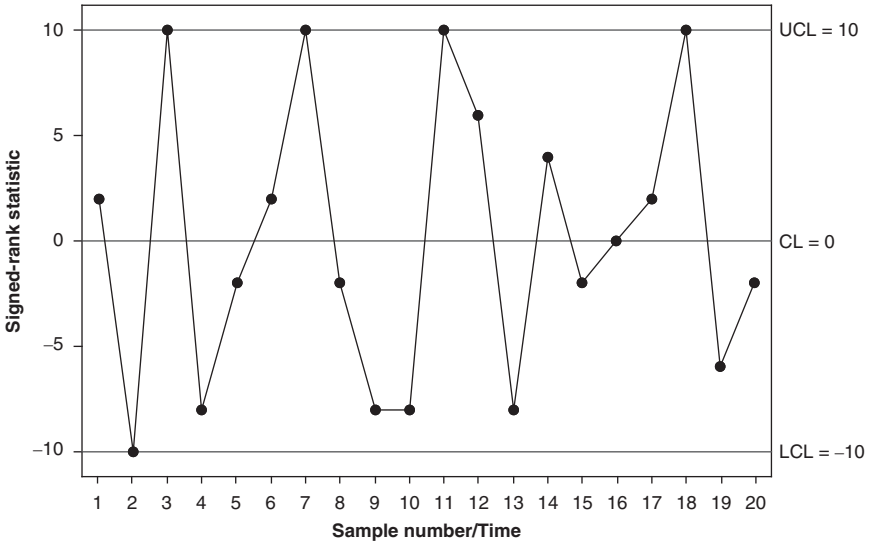


FIGURE 4.12 *A Shewhart-SR chart.*

associated with $n = 4$. Although this probability is not less than or equal to 0.00135, it is the smallest probability that we can obtain when working with a sample of size 4. Now, to obtain the *UCL*, that is, a, we use the linear relationship $SR_i = 2T_n^+ - n(n+1)/2$ so that $a = 2(10) - 10 = 10$ with an $FAR = (2)(0.062) = 0.124$. Thus, for $n = 4$ we have $LCL/UCL = \pm 10$ and $CL = 0$ with a corresponding $FAR = 0.124$, which is the lowest *FAR* achievable. Since the IC value of the median is known or specified (Case K) $ARL_{\text{IC}} = \dfrac{1}{FAR} = \dfrac{1}{0.124} \approx 8$. Again, this low value of the IC *ARL* may not be satisfactory from a practical point of view since it is far from the typically desired value of 370. However, this is the highest ARL_{IC} achievable for this nonparametric chart when working with a sample of size 4. The resulting Shewhart-SR chart is shown in Figure 4.12.

From Figure 4.12, we can see that the process is OOC starting at sample number 2 since the charting statistic plots on the *LCL*. Thus, corrective action and a search for assignable causes are necessary. It should be noted that higher values of the ARL_{IC}, closer to typical values such as 370 or 500, can be achieved for higher values of the subgroup size n. See Tables 2a and 2b of Bakir and Reynolds (1979), for example.

Example 4.5 A Shewhart-SR Chart with Runs-type Signaling Rules

For illustration, an improved *2-of-2* Shewhart-SR chart will be constructed using the same data as in Table 4.7a. If we now introduce warning limits at,

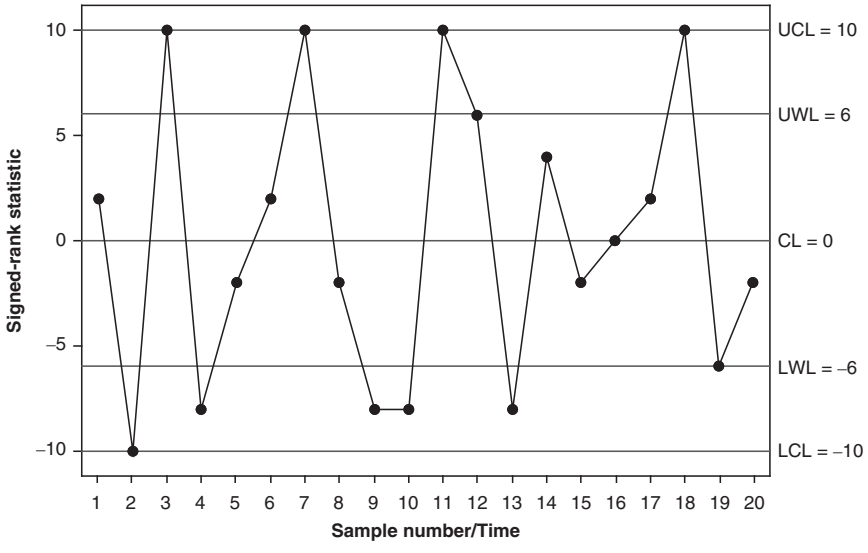


FIGURE 4.13 *A Shewhart-SR chart with runs-type signaling rules.*

say, ±6, that is, the upper warning limit $UWL = +6$ and the lower warning limit $LWL = -6$, respectively, the resulting improved *2-of-2* Shewhart-SR chart is shown in Figure 4.13. It should be noted that the warning limits are found as follows. Typically, one would set the control limits at some fixed value, say ±10, and then the warning limits are found via simulation such that some nominal ARL_{IC} value is attained.

The *2-of-2* Shewhart-SR chart signals when two out of two consecutive charting statistics plot on or above the *LWL/UWL*. Since the 9th and the 10th charting statistics plot between the *LWL* and the *LCL*, the process is declared to be OOC and a search for assignable causes is necessary. Furthermore, it can be seen that the 11th and the 12th charting statistics also give an OOC signal. For more examples on the Shewhart-SR chart with runs-type signaling rules, see Chakraborti and Eryilmaz (2007).

4.3.2 CUSUM Control Charts

Example 4.6 A CUSUM-SN Control Chart Based on Individual Measurements

For the data of Table 4.5, the two one-sided CUSUM charting statistics SN_t^+ and SN_t^- with $SN_0^+ = SN_0^- = 0$ and $k = 0.5$ are calculated and are shown in Table 4.8. The decision interval h was chosen to be 3. The reader is referred to Section 3.3.2 for guidance on choosing the design parameters. To illustrate the calculations, consider observation number 1. The equations

Table 4.8 A CUSUM-SN chart based on individual measurements.

Observation number t	(a) SN_t	(b) SN_t^+	(c) N^+	(d) SN_t^-	(e) N^-
1	−1	0.0	0	−0.5	1
2	−1	0.0	0	−1.0	2
3	−1	0.0	0	−1.5	3
4	1	0.5	1	0.0	0
5	1	1.0	2	0.0	0
6	1	1.5	3	0.0	0
7	−1	0.0	0	−0.5	1
8	1	0.5	1	0.0	0
9	−1	0.0	0	−0.5	1
10	1	0.5	1	0.0	0
11	−1	0.0	0	−0.5	1
12	1	0.5	1	0.0	0
13	1	1.0	2	0.0	0
14	−1	0.0	0	−0.5	1
15	1	0.5	1	0.0	0
16	−1	0.0	0	−0.5	1
17	1	0.5	1	0.0	0
18	1	1.0	2	0.0	0
19	−1	0.0	0	−0.5	1
20	1	0.5	1	0.0	0
21	1	1.0	2	0.0	0
22	−1	0.0	0	−0.5	1
23	1	0.5	1	0.0	0
24	1	1.0	2	0.0	0
25	1	1.5	3	0.0	0
26	1	2.0	4	0.0	0
27	1	2.5	5	0.0	0
28	1	3.0	6	0.0	0
29	1	3.5	7	0.0	0
30	1	4.0	8	0.0	0

for the charting statistics SN_1^+ and SN_1^- are

$$SN_1^+ = \max(0,\ SN_0^+ + SN_1 - k) = \max(0,\ 0 + (-1) - 0.5) = 0$$

and

$$SN_1^- = \min(0,\ SN_0^- + SN_1 + k) = \min(0,\ 0 + (-1) + 0.5) = -0.5$$

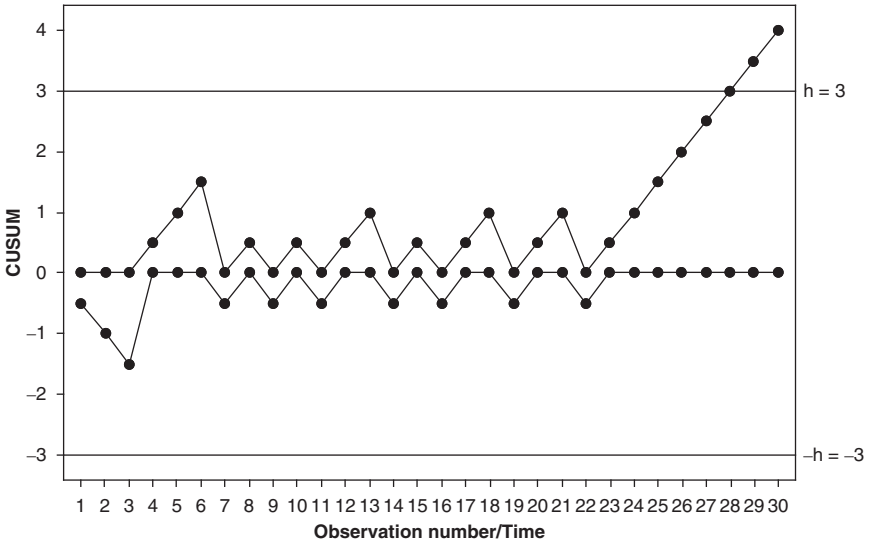


FIGURE 4.14 *A CUSUM-SN chart based on individual measurements.*

The remaining calculations are summarized in columns (b) and (d) of Table 4.8. Along with SN_t^+ and SN_t^-, the quantity N^+, which counts the number of consecutive periods since SN_t^+ became non-zero and N^-, which counts the number of consecutive periods since SN_t^- became non-zero, are calculated and shown. The graphical display of the CUSUM-SN chart is shown in Figure 4.14. The chart signals at observation number 28 with $N^+ = 6$, indicating that the shift most likely occurred at observation number 22 (since $28 - 6 = 22$). A search for assignable causes and corrective action is necessary.

It should be noted that, for the same data set, the parametric (normal theory) CUSUM chart signals at observation number 29 with $N^+ = 7$, indicating that the shift most likely occurred at observation number 22 (since $29 - 7 = 22$). Thus, the parametric counterpart signals one observation later and both charts indicate that the shift in the location most likely occurred at observation number 22.

The example illustrates the value of a nonparametric control chart in general, and the nonparametric CUSUM-SN chart in particular, for the individual data. If one uses the traditional CUSUM chart, the normality assumption must be made, which can present a challenge, particularly for the individual data situation since the central limit theorem cannot be counted on. Moreover, it is known that the CUSUM chart is not very robust in the normality assumption, which may bring into question, for example, the actual *FAR* of the chart. Researchers have shown that the nonparametric CUSUM-SN chart can outperform the normal theory CUSUM-SN chart in situations where the underlying distribution has heavier tails than

the normal distribution. The calculations for the nonparametric chart are not complex. In this example, the nonparametric chart signaled one period earlier than the parametric chart. In short, the nonparametric CUSUM-SN chart is a safer and a capable alternative control chart to be used in practice.

Next, we illustrate the CUSUM-SN chart for subgrouped data.

Example 4.7 A CUSUM-SN Control Chart Based on Subgrouped Data

For the data of Table 4.6 with $n = 10$, the two one-sided CUSUM charting statistics SN_t^+ and SN_t^- with $SN_0^+ = SN_0^- = 0$ and $k = 2$ were calculated and are shown in Table 4.9. Recall that the reference value, k, is selected first and this typically is some small value. Following this, the decision interval, h, is found so that a nominal ARL_{IC} is attained. The reader is referred to Amin, Reynolds, and Bakir (1995), Table 5, for several (h,k) combinations for various nominal ARL_{IC} values for a sample size of $n = 10$. For this example, h was chosen to be 8. To illustrate the calculations, consider sample number 1. The equations for the charting statistics SN_1^+ and SN_1^- are

$$SN_1^+ = \max(0,\ SN_0^+ + SN_1 - k) = \max(0,\ 0 + 0 - 2) = 0$$

Table 4.9 A CUSUM-SN chart based on subgrouped data.

Sample t	(a) SN_t	(b) SN_t^+	(c) N^+	(d) SN_t^-	(e) N^-
1	0	0	0	0	0
2	−8	0	0	−6	1
3	−2	0	0	−6	2
4	2	0	0	−2	3
5	−6	0	0	−6	4
6	0	0	0	−4	5
7	4	2	1	0	0
8	2	2	2	0	0
9	3	3	3	0	0
10	−2	0	0	0	0
11	0	0	0	0	0
12	6	4	1	0	0
13	2	4	2	0	0
14	2	4	3	0	0
15	2	4	4	0	0

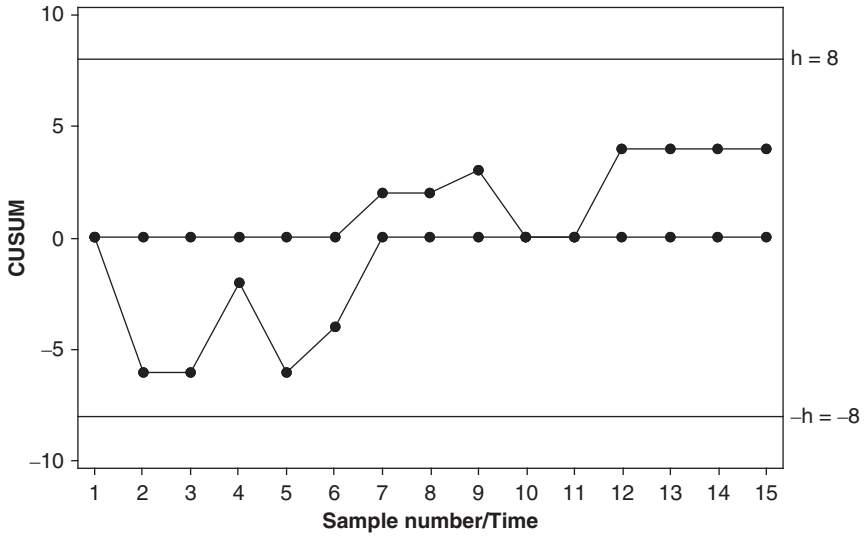


FIGURE 4.15 *A CUSUM-SN chart based on subgrouped data.*

and

$$SN_1^- = \min(0,\ SN_0^- + SN_1 + k) = \min(0, 0 + 0 + 2) = 0.$$

The rest of the calculations are summarized in columns (b) and (d) of Table 4.9. Along with SN_t^+ and SN_t^-, the quantity N^+, which counts the number of consecutive periods since SN_t^+ became non-zero and N^-, which counts the number of consecutive periods since SN_t^- became non-zero, are calculated and shown. The graphical display of the CUSUM-SN chart is shown in Figure 4.15. This chart does not signal or show any obvious patterns and hence neither a search for assignable causes nor any corrective action is necessary. It should be noted that, for the same data set, the parametric CUSUM chart also does not give a signal.

Example 4.8 A CUSUM-SN Control Chart: Markov Chain Approach

Consider an upper one-sided CUSUM-SN chart with a decision interval of 4 ($h = 4$), a reference value of 1 ($k = 1$), and a sample size of 5 ($n = 5$). Regarding the choice of the combination $(h, k) = (4, 1)$, recall that the reference value, k, is selected first and this typically is some small value. Following this, the decision interval, h, is found so that a nominal ARL_{IC} is attained. Note that Amin, Reynolds, and Bakir (1995), in Table 5, only provide (h,k) combinations for a sample size of $n = 10$. Thus, a simulation algorithm was used in order to find an (h,k) combination for $n = 5$ that gives an attained ARL_{IC} that is appropriate for SPC applications,

Table 4.10 Calculation of the state space when $h=4$, $k=1$, and $n=5$.

SN_t	$S_{t-1}^+ + SN_t - k$	$\max\{0, S_{t-1}^+ + SN_t - k\}$	$S_t^+ = \min\{h, \max\{0, S_{t-1}^+ + SN_t - k\}\}$
-5	-6[a]	0	0
-3	-4	0	0
-1	-2	0	0
1	0	0	0
3	2	2	2
5	4	4	4

[a] Since only the state space needs to be described, S_{t-1}^+ can be any value from Ω^+ and we therefore take, without loss of generality, $S_{t-1}^+ = 0$. Any other possible value for S_{t-1}^+ would lead to the same Ω^+.

Table 4.11 Classifications and descriptions of the states.

State number	Description of the state	Absorbent (A) / Non-absorbent (NA)
0	$S_t^+ = 0$	NA
1	$S_t^+ = 2$	NA
2	$S_t^+ = 4$	A

such as the industry standards of 370 or 500. See Section 4.2.2.3.2 for the step-by-step details of the simulation approach for the CUSUM-SN chart. We illustrate the calculations of the IC run-length distribution characteristics of this chart using the MC approach. For n odd, the reference value is taken to be odd because this leads to the sum $\sum(SN_i - k)$ being equal to even values, which reduces the size of the state space for the MC. This will halve the size of the matrix with the transition probabilities. For $h = 4$ the state space is $\Omega^+ = \{\varsigma_0, \varsigma_1, \varsigma_2\} = \{0, 2, 4\}$ with $0 = \varsigma_0 < \varsigma_1 < \varsigma_2 = h$. The state space is calculated using Equation 4.9 and the calculations are shown in Table 4.10.

From Table 4.11, we see that there are two non-absorbent states, that is, $v = 2$, and one absorbent state. Therefore, the corresponding transition probability matrix is a 3×3 matrix. Using two IC states, the IC transition probability matrix is given by

$$M_{3\times 3} = \begin{pmatrix} Q_{2\times 2} & \mid & \underline{p}_{2\times 1} \\ - & - & - \\ \underline{0}'_{1\times 2} & \mid & 1_{1\times 1} \end{pmatrix}$$

$$= \begin{pmatrix} p_{00} & p_{01} & | \, p_{02} \\ p_{10} & p_{11} & | \, p_{12} \\ - & - & - \\ p_{20} & p_{21} & | \, p_{22} \end{pmatrix}$$

$$= \frac{1}{32} \begin{pmatrix} 26 & 5 & | \, 1 \\ 16 & 10 & | \, 6 \\ - & - & - \\ 0 & 0 & | \, 32 \end{pmatrix}$$

where the transition probabilities are calculated using Equation 4.9. The calculation of p_{01}, when the process is IC, is shown below for illustration.

$$\begin{aligned} & p_{01} \\ & = P(S_t = 2 \mid S_{t-1} = 0) \\ & = P(\min\{4, \max\{0, 0 + SN_t - 1\}\} = 2) \\ & = P(\max\{0, SN_t - 1\} = 2) \\ & = P(SN_t - 1 = 2) \\ & = P(SN_t = 3) \\ & = P(2T_t - 5 = 3) \\ & = P(T_t = 4) \\ & = \frac{5}{32}. \end{aligned}$$

The second-to-last equality follows from the fact that the random variable, T_t, having a $BIN(n = 5,\ p = 0.5)$ distribution in the IC case, takes on only non-negative integer values ($T_t \in \{0, 5\}$).

Once the one-step transition probabilities are calculated and the essential transition probability matrix Q is found, the IC run-length characteristics are obtained using Equations 3.22–3.24. This yields $ARL_{\text{IC}} = 16.62$ and $SDRL_{\text{IC}} = 15.51$ with the IC 5th, 25th, 50th, 75th, and 95th percentiles being equal to 2, 6, 12, 23, and 48, respectively.

The ARL_{IC} value indicates that, when the process is IC, the first signal would be observed, on average, at every 17th plotted point on this chart. The first quartile is 6, so we know that a false alarm will not occur within the first six plotted points, with a probability at most 75%. As another example, the $MRL_{\text{IC}} = 12$, which indicates that the first false alarm will be observed within the first 12 plotted points at least 50% of the time.

Note that, although the IC characteristics are illustrated here, the OOC characteristics can easily be found by changing the value of p so that $p \neq 0.5$, calculating the p_{ij} values, forming the essential transition probability matrix Q, and then using the same MC formulas. We leave the details for the reader to verify.

Example 4.9 A CUSUM-SR Control Chart

For the data of Table 4.7a, the two one-sided CUSUM charting statistics SR_t^+ and SR_t^- with $SR_0^+ = SR_0^- = 0$, and $k = 3$ were calculated together with two counters N^+ and N^-, which are shown in Table 4.12. The decision interval h was chosen to be equal to 8.

To illustrate the calculations, consider sample number 1. The expressions for the CUSUM charting statistics SR_1^+ and SR_1^- are

$$\begin{aligned} SR_1^+ &= \max(0, SR_0^+ + SR_1 - k) \\ &= \max(0, 0 + 2 - 3) \\ &= 0 \end{aligned}$$

Table 4.12 A CUSUM-SR chart.

Sample	(a)		(b)	
i	SR_t^+	N^+	SR_t^-	N^-
1	0	0	0	0
2	0	0	−7	1
3	7	1	0	0
4	0	0	−5	1
5	0	0	−4	2
6	0	0	0	0
7	7	1	0	0
8	2	2	0	0
9	0	0	−5	1
10	0	0	−10*	2
11	7	1	0	0
12	10*	2	0	0
13	0	0	−5	1
14	1	1	0	0
15	0	0	0	0
16	0	0	0	0
17	0	0	0	0
18	7	1	0	0
19	0	0	−3	1
20	0	0	−2	2

[a] The two-sided CUSUM-SR chart signals.

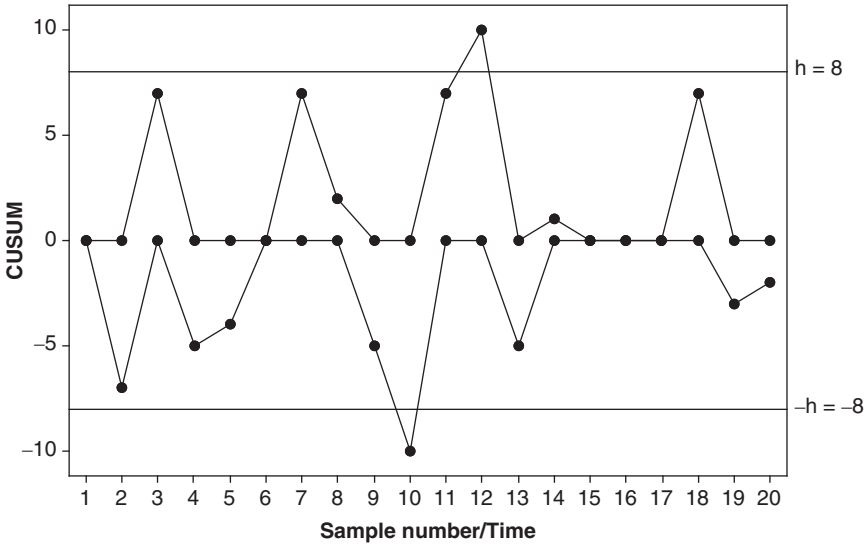


FIGURE 4.16 *A CUSUM-SR chart for subgrouped data.*

and

$$\begin{aligned} SR_1^- &= \min(0, SR_0^- + SR_1 + k) \\ &= \min(0, 0 + 2 + 3) \\ &= 0. \end{aligned}$$

The remaining calculations are summarized in columns (a) and (b) of Table 4.12. The CUSUM-SR chart, displayed in Figure 4.16, signals twice. The first signal is given on sample number 10 when $SR_{10}^- = -10$ plots below the lower decision interval, that is, $h = -8$, with a counter of $N^- = 2$, indicating that the shift most likely occurred at sample number 8. The second signal is given shortly afterwards on sample number 12 when $SR_{12}^+ = 10$ falls above the upper decision interval, that is, $h = 8$, with a counter of $N^+ = 2$, again indicating that the shift most likely occurred at sample number 8.

If follow-up investigations reveal any assignable cause(s) for the first signal, the lower one-sided CUSUM charting statistic would be restarted from zero, while the upper one-sided CUSUM charting statistics would have continued "as usual". The opposite would have been done if any assignable cause(s) were found for the second signal. However, considering the current situation, that is, a signal on the lower side (indicating a possible downward

shift) on sample number 10 followed shortly afterwards with a signal on the upper side (indicating a possible upward shift) on sample number 12, these two signals were considered false alarms and neither of the CUSUM charting statistics was reset to zero. After all, although not impossible, it is unlikely that a downward shift in a process will be followed "so soon" by an upward shift unless there was "over correction" in the positive direction from one of the control chart personnel.

It should be noted that for the same data set the parametric CUSUM chart signals on the lower CUSUM at sample number 2 with a counter of $N^+ = 1$, indicating that the shift most likely occurred at start-up. Thus, the CUSUM-$\overline{X}$ chart outperforms the CUSUM-SR chart, which is consistent with the fact that a goodness-of-fit test for normality is not rejected for these data. Typically, in practice, however, normality can be in doubt or may not be justifiable for lack of information or data, and a nonparametric method may be more desirable.

Next, we illustrate the calculations of some run-length distribution characteristics of an upper one-sided CUSUM-SR chart.

Example 4.10 A CUSUM-SR Control Chart: Markov Chain Approach

Consider an upper one-sided CUSUM-SR chart with a decision interval of 6 ($h = 6$), a reference value of 2 ($k = 2$), and a sample size of 4 ($n = 4$). Again, regarding the choice of the combination $(h, k) = (6, 2)$, recall that the reference value, k, is selected first and this is typically some small value. Following this, the decision interval, h, is found so that a nominal ARL_{IC} is attained. The reader is referred to Bakir and Reynolds (1979) for several (h,k) combinations for various nominal ARL_{IC} values for a sample size of $n =$ 4, 6, and 10. As with the CUSUM-SN chart, we illustrate the calculations of the IC run-length distribution characteristics of the CUSUM-SR chart using the MC approach.

For $h = 6$, we have $\Omega^+ = \{\varsigma_0, \varsigma_1, \varsigma_2, \varsigma_3\} = \{0, 2, 4, 6\}$ with $0 = \varsigma_0 < \varsigma_1 < \varsigma_2 < \varsigma_3 = h$. The state space is calculated using Equation 4.12 and the calculations are shown in Table 4.13.

From Table 4.14, we see that there are two non-absorbent states, that is, $v = 3$, and one absorbent state. Therefore, the corresponding transition probability matrix will be a 4×4 matrix. Using three IC states, the IC transition probability matrix is given by

$$M_{4\times4} = \begin{pmatrix} Q_{3\times3} & \vdots & \underline{p}_{3\times1} \\ - & - & - \\ \underline{0}'_{1\times3} & \vdots & 1_{1\times1} \end{pmatrix} = \begin{pmatrix} p_{00} & p_{02} & p_{04} & \vdots & p_{06} \\ p_{20} & p_{22} & p_{24} & \vdots & p_{26} \\ p_{40} & p_{42} & p_{44} & \vdots & p_{46} \\ - & - & - & & - \\ p_{60} & p_{62} & p_{64} & \vdots & p_{66} \end{pmatrix} = \frac{1}{16} \begin{pmatrix} 11 & 2 & 1 & \vdots & 2 \\ 9 & 2 & 2 & \vdots & 3 \\ 7 & 2 & 2 & \vdots & 5 \\ - & - & - & & - \\ 0 & 0 & 0 & \vdots & 16 \end{pmatrix}$$

Table 4.13 Calculation of the state space when $h=6$, $k=2$, and $n=4$.

SR_t	$S_{t-1}^+ + SR_t - k$	$\max\{0, S_{t-1}^+ + SR_t - k\}$	$S_t^+ = \min\{h, \max\{0, S_{t-1}^+ + SR_t - k\}\}$
-10	-12^*	0	0
-8	-10	0	0
-6	-8	0	0
-4	-6	0	0
-2	-4	0	0
0	-2	0	0
2	0	0	0
4	2	2	2
6	4	4	4
8	6	6	6
10	8	8	6

[a] Since only the state space needs to be described, S_{t-1}^+ can be any value from Ω^+ and we therefore take, without loss of generality, $S_{t-1}^+ = 0$. Any other possible value for S_{t-1}^+ would lead to the same Ω^+.

Table 4.14 Classifications and descriptions of the states.

State number	Description of the state	Absorbent (A) / Non-absorbent (NA)
0	$S_t^+ = 0$	NA
1	$S_t^+ = 2$	NA
2	$S_t^+ = 4$	NA
3	$S_t^+ = 6$	A

where the transition probabilities are calculated using Equation 4.12. To illustrate, the calculation of p_{02} when the process is IC is shown below

$$\begin{aligned} p_{02} &= P(S_t = 2 \mid S_{t-1} = 0) \\ &= P(\min\{6, \max\{0, 0 + SR_t - 2\}\} = 2) \\ &= P(\max\{0, SR_t - 2\} = 2) \\ &= P(SR_t - 2 = 2) \\ &= P(SR_t = 4) \\ &= P(2T_n^+ - 10 = 4) \\ &= P(T_n^+ = 7) \\ &= \frac{2}{16}. \end{aligned}$$

The second-to-last equality follows from the IC probability distribution of the well-known Wilcoxon signed-rank statistic, T_n^+, details of which are given in Table 4.3 in Section 4.2.1.2.3.

Once the one-step transition probabilities are calculated and the essential transition probability matrix Q is found, the IC run-length characteristics are obtained using Equations 3.22–3.24. This yields $ARL_{\text{IC}} = 6.81$ and $SDRL_{\text{IC}} = 6.11$ with the IC 5th, 25th, 50th, 75th, and 95th percentiles being equal to 1, 2, 5, 9, and 19, respectively. These values help us understand the performance of the control chart via its run-length distribution, for example, the ARL_{IC} value indicates that, when the process is IC, the first signal would be observed, on average, at every seventh plotted point. The first quartile is 2, so we know that a false alarm will occur within the first two plotted points, with a probability of at least 75%. As another example, the $MRL_{\text{IC}} = 5$, which indicates that the first false alarm will be observed within the first five plotted points at least 50% of the time.

Again, it should be noted that, although only the results for IC run-length distribution characteristics are shown here, one can follow similar steps to obtain the OOC run-length distribution characteristics for which one needs to calculate the OOC distribution and the transition probabilities p_{ij} for the Wilcoxon signed-rank statistic, T_n^+.

Next, we illustrate an EWMA-SN chart for individual data.

Example 4.11 An EWMA-SN Control Chart Based on Individual Measurements

A nonparametric two-sided CUSUM-SN chart for the data in Table 4.8 was shown in Example 4.6. As an alternative to this, a nonparametric EWMA control chart is now illustrated. For this purpose, we use $\lambda = 0.10$ to calculate the charting statistics and $L = 2.820$ to find the control limits. The reader is referred to Section 3.2.3 for a detailed discussion on the choice of the design parameters λ and L. In short, the recommendation is to choose a small λ, say, equal to 0.05, when small shifts are of interest, choose $\lambda = 0.10$ if moderate shifts are of concern, and choose $\lambda = 0.20$ if larger shifts are of interest. Thus, the choice of $\lambda = 0.10$ indicates that moderate shifts are of interest. After λ is chosen, the second step involves finding L, so that a desired ARL_{IC} is attained. For individual measurements and $\lambda = 0.10$, it has been shown that $L = 2.820$ gives an attained ARL_{IC} close to 500 (see Graham, Human, and Chakraborti, 2009)[3].

[3] Graham, MA, Human, SW, and Chakraborti, S (2009) A nonparametric EWMA control chart based on the sign statistic. Technical report, 09/04, Department of Statistics, University of Pretoria, Pretoria, South Africa, ISBN 978-1-86854-777-7. A hard copy of this technical report is available at the University of Pretoria, Pretoria, South Africa. A soft copy can be request by emailing the author at marien.graham@up.ac.za.

To illustrate the calculations, consider observation number 1 with $SN_1 = -1$ and suppose that the starting value $Z_0 = 0$. Then, the first charting statistic on the EWMA-SN chart in Figure 4.16 is

$$\begin{aligned} Z_1 &= \lambda SN_1 + (1-\lambda)Z_0 \\ &= 0.10(-1) + (1-0.10)0 = -0.1. \end{aligned}$$

The value of the second charting statistic is

$$\begin{aligned} Z_2 &= \lambda SN_2 + (1-\lambda)Z_1 \\ &= 0.10(-1) + (1-0.10)(-0.1) \\ &= -0.19 \end{aligned}$$

with the remaining calculations summarized in column (b) of Table 4.15.

For simplicity, we use the steady-state control limits on the EWMA-SN chart, which is found from Equation 4.15, that is

$$\begin{aligned} LCL &= -L\sqrt{\frac{\lambda}{2-\lambda}} \\ &= -2.820\sqrt{\frac{0.10}{2-0.10}} \\ &= -0.647 \end{aligned}$$

and

$$\begin{aligned} UCL &= L\sqrt{\frac{\lambda}{2-\lambda}} \\ &= 2.820\sqrt{\frac{0.10}{2-0.10}} \\ &= 0.647 \end{aligned}$$

respectively.

The resulting EWMA-SN chart is shown in Figure 4.17. Since none of the charting statistics falls on or outside the *LCL* or the *UCL*, the process is considered IC with no follow-up (remedial) action needed. However, the pattern of the charting statistics does produce some concern that the location may be increasing.

It should be noted that, for the same data set, the CUSUM-SN chart signaled on observation number 28 with $N^+ = 6$, indicating that the shift most likely occurred at observation number 22. The parametric CUSUM

Table 4.15 Calculations for the EWMA-SN chart based on individual measurements.

Sample t	(a) SN_t	(b) $Z_t = \lambda SN_t + (1-\lambda)Z_{t-1}$
1	−1	−0.100
2	−1	−0.190
3	−1	−0.271
4	1	−0.144
5	1	−0.030
6	1	0.073
7	−1	−0.034
8	1	0.069
9	−1	−0.037
10	1	0.066
11	−1	−0.040
12	1	0.064
13	1	0.157
14	−1	0.042
15	1	0.137
16	−1	0.024
17	1	0.121
18	1	0.209
19	−1	0.088
20	1	0.179
21	1	0.261
22	−1	0.135
23	1	0.222
24	1	0.300
25	1	0.370
26	1	0.433
27	1	0.489
28	1	0.540
29	1	0.586
30	1	0.628

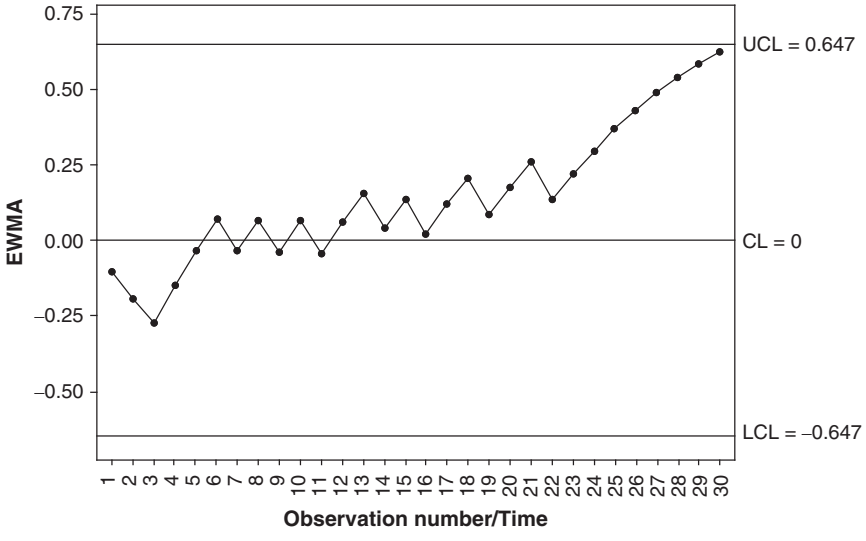


FIGURE 4.17 *An EWMA-SN chart based on individual measurements.*

chart, however, signaled at observation number 29 with $N^+ = 7$, indicating that the shift most likely occurred at observation number 22. Although the EWMA-SN chart doesn't signal, a clear upward trend is visible starting from observation number 22 with the chart almost signaling at the 30th (last) observation.

4.3.3 EWMA Control Charts

Example 4.12 An EWMA-SN Control Chart Based on Subgrouped Data

A two-sided CUSUM-SN chart for the data in Table 4.9 was shown in Example 4.7. As an alternative, an EWMA control chart is now illustrated. For this purpose, we will use $\lambda = 0.05$ to calculate the charting statistics and $L = 2.612$ to find the control limits (see Graham, Chakraborti, and Human, 2011a). The reader is referred to Section 3.2.3 for a detailed discussion on the choice of the design parameters λ and L. For samples of size 10 and $\lambda = 0.05$, it has been shown that $L = 2.612$ gives an attained ARL_{IC} value close to 500.

To illustrate the calculations, consider sample number 1 with $SN_1 = 0$ and suppose that the starting value $Z_0 = 0$. Then, the first charting statistic on the EWMA-SN chart in Figure 4.18 is

$$\begin{aligned} Z_1 &= \lambda SN_1 + (1-\lambda)Z_0 \\ &= 0.05(0) + (1-0.05)0 \\ &= 0. \end{aligned}$$

Table 4.16 Calculations for the EWMA-SN chart based on subgrouped data.

Sample t	(a) SN_t	(b) $Z_t = \lambda SN_t + (1-\lambda)Z_{t-1}$
1	0	0.000
2	−8	−0.400
3	−2	−0.480
4	2	−0.356
5	−6	−0.638
6	0	−0.606
7	4	−0.376
8	2	−0.257
9	3	−0.094
10	−2	−0.190
11	0	−0.180
12	6	0.129
13	2	0.222
14	2	0.311
15	2	0.396

The value of the second charting statistic is

$$\begin{aligned} Z_2 &= \lambda SN_2 + (1-\lambda)Z_1 \\ &= 0.05(-8) + (1-0.05)0 \\ &= -0.4 \end{aligned}$$

with the remaining calculations summarized in column (b) of Table 4.16.

For simplicity, we use the steady-state control limits on the EWMA-SN chart, which are found from Equation 4.17, that is

$$\begin{aligned} LCL &= -L\sqrt{\frac{\lambda n}{2-\lambda}} \\ &= -2.612\sqrt{\frac{(0.05)(10)}{2-0.05}} \\ &= -1.323 \end{aligned} \qquad \begin{aligned} UCL &= L\sqrt{\frac{\lambda n}{2-\lambda}} \\ &= 2.612\sqrt{\frac{(0.05)(10)}{2-0.05}} \\ &= 1.323. \end{aligned}$$

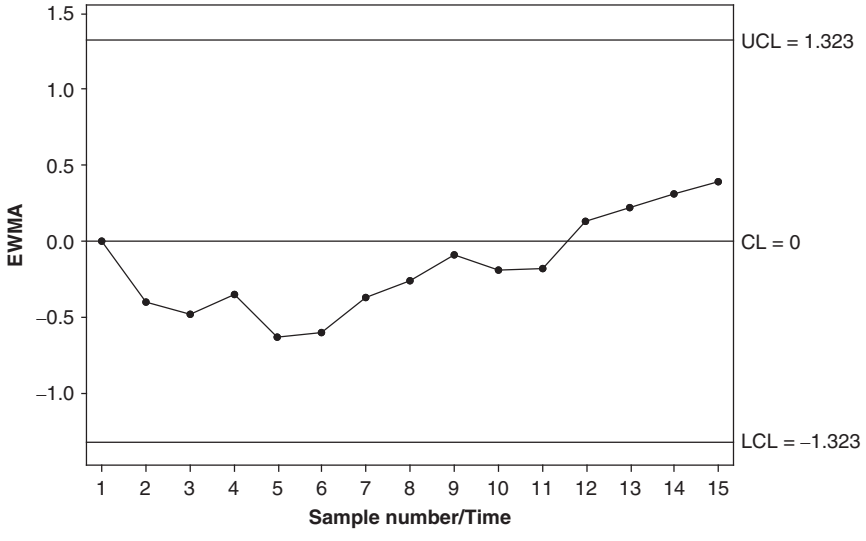


FIGURE 4.18 *An EWMA-SN chart based on subgrouped data.*

The resulting EWMA-SN chart is shown in Figure 4.18. Since none of the charting statistics falls on or outside the LCL and/or the UCL or shows no obvious pattern of concern, the process is considered IC with no remedial action needed.

It should be noted that, for the same data set, the parametric CUSUM chart and the distribution-free CUSUM-SN chart also do not give a signal.

Example 4.13 An EWMA-SN Control Chart: Markov Chain Approach

Here, we consider an EWMA-SN chart for individual data with design parameters $\lambda = 0.05$, $L = 2$, and $v = 2s + 1 = 9$ so that $s = 4$, that is, we divide the interval between the control limits into nine subintervals (see Figure 4.19), resulting in a "small" transition probability matrix with only 10 rows and 10 columns, respectively. Then we illustrate how the run-length distribution can be calculated using the MC method. Note that, for this chart, the steady-state control limits are calculated from Equation 4.15 and are given by $LCL = -0.320$ and $UCL = 0.320$.

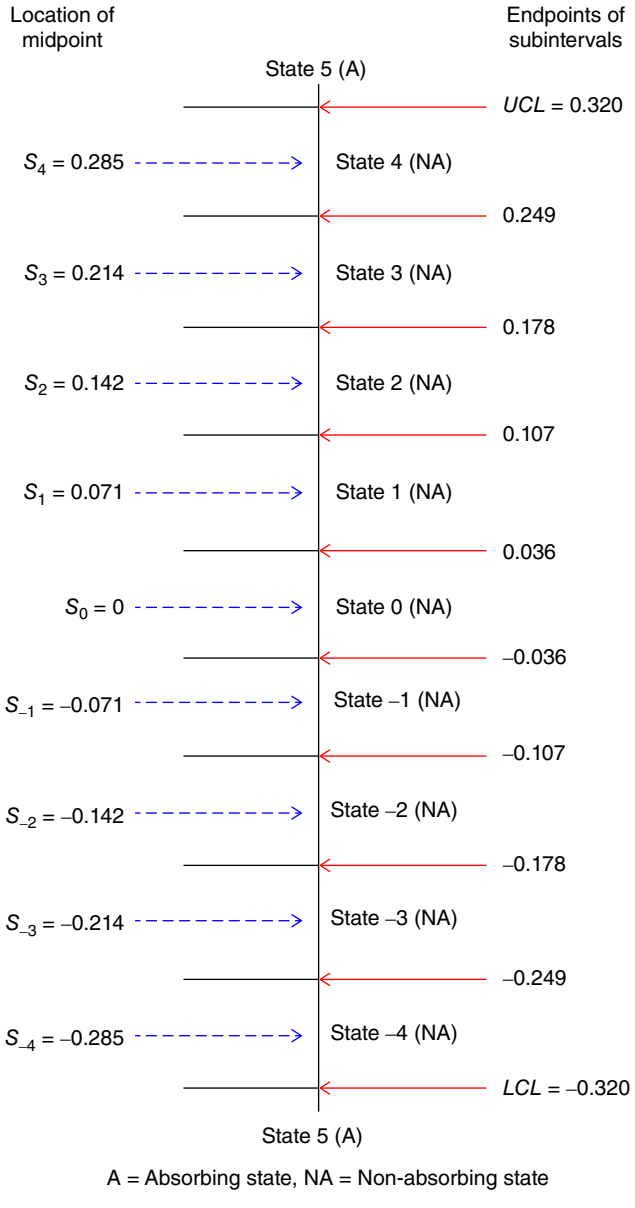


FIGURE 4.19 *Partitioning of the interval between the* LCL *and the* UCL *into* $\nu = 9$ *subintervals for* $\lambda = 0.05$ *and* $L = 2$.

Using these nine states, the transition probability matrix is given by

$$M_{10\times 10} = \begin{pmatrix} Q_{9\times 9} & \vdots & \underline{p}_{9\times 1} \\ \cdots & \cdots & \cdots \\ \underline{0}'_{1\times 9} & \vdots & 1_{1\times 1} \end{pmatrix}$$

$$= \left(\begin{array}{ccccccccc:c} p_{(-4)(-4)} & p_{(-4)(-3)} & p_{(-4)(-2)} & p_{(-4)(-1)} & p_{(-4)0} & p_{(-4)1} & p_{(-4)2} & p_{(-4)3} & p_{(-4)4} & p_{(-4)5} \\ p_{(-3)(-4)} & p_{(-3)(-3)} & p_{(-3)(-2)} & p_{(-3)(-1)} & p_{(-3)0} & p_{(-3)1} & p_{(-3)2} & p_{(-3)3} & p_{(-3)4} & p_{(-3)5} \\ p_{(-2)(-4)} & p_{(-2)(-3)} & p_{(-2)(-2)} & p_{(-2)(-1)} & p_{(-2)0} & p_{(-2)1} & p_{(-2)2} & p_{(-2)3} & p_{(-2)4} & p_{(-2)5} \\ p_{(-1)(-4)} & p_{(-1)(-3)} & p_{(-1)(-2)} & p_{(-1)(-1)} & p_{(-1)0} & p_{(-1)1} & p_{(-1)2} & p_{(-1)3} & p_{(-1)4} & p_{(-1)5} \\ p_{0(-4)} & p_{0(-3)} & p_{0(-2)} & p_{0(-1)} & p_{00} & p_{01} & p_{02} & p_{03} & p_{04} & p_{05} \\ p_{1(-4)} & p_{1(-3)} & p_{1(-2)} & p_{1(-1)} & p_{10} & p_{11} & p_{12} & p_{13} & p_{14} & p_{15} \\ p_{2(-4)} & p_{2(-3)} & p_{2(-2)} & p_{2(-1)} & p_{20} & p_{21} & p_{22} & p_{23} & p_{24} & p_{25} \\ p_{3(-4)} & p_{3(-3)} & p_{3(-2)} & p_{3(-1)} & p_{30} & p_{31} & p_{32} & p_{33} & p_{34} & p_{35} \\ p_{4(-4)} & p_{4(-3)} & p_{4(-2)} & p_{4(-1)} & p_{40} & p_{41} & p_{42} & p_{43} & p_{44} & p_{45} \\ \hdashline p_{5(-4)} & p_{5(-3)} & p_{5(-2)} & p_{5(-1)} & p_{50} & p_{51} & p_{52} & p_{53} & p_{54} & p_{55} \end{array}\right)$$

$$= \frac{1}{2}\left(\begin{array}{ccccccccc:c} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ \hdashline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{array}\right)$$

in the IC case, where the transition probabilities are calculated using Equation 4.19. The calculation of p_{01} is given below for illustration purposes.

$$\begin{aligned} p_{01} & \\ &= P\left(\left(\left(\frac{(S_1-\tau)-(1-\lambda)S_0}{\lambda}\right)+n\right)/2 < T_k\right) \\ &\leq \left(\left(\frac{(S_1+\tau)-(1-\lambda)S_0}{\lambda}\right)+n\right)/2\Bigg) \\ &= P\left(\left(\left(\frac{(0.071-0.036)-(1\text{–}0.05)0}{0.05}\right)+1\right)/2 < T_k\right) \\ &\leq \left(\left(\frac{(0.071+0.036)-(1\text{–}0.05)0}{0.05}\right)+1\right)/2\Bigg) \\ &= P(0.856 < T_k \leq 1.568) = P(T_k = 1) = \frac{1}{2}. \end{aligned}$$

The second-to-last equality follows from the fact that the random variable, T_k, having a $BIN(n=1,\ p=0.5)$ distribution in the IC case, takes on only non-negative integer values ($T_k \in \{0, 1\}$). Once the one-step transition probabilities are calculated and the essential transition probability matrix Q is found, the IC run-length characteristics are obtained using Equations 3.22–3.24. This yields $ARL_{\text{IC}} = 25.00$ and $SDRL_{\text{IC}} = 20.00$ with the IC 5th, 25th, 50th, 75th, and 95th percentiles being equal to 5, 11, 19, 33, and 65, respectively.

The ARL_{IC} value indicates that when the process is IC the first signal would be observed, on average, at every 25th plotted point. The first quartile is 11, so we know that a false alarm will not occur within the first 11 plotted points, with a probability of 75% at most. As another example, the $MRL_{\text{IC}} = 19$, which indicates that the first false alarm will be observed within the first 19 plotted points at least 50% of the time.

Again, it should be noted that, although only the IC characteristics are shown here, the OOC characteristics can easily be found by setting $p \neq 0.5$ in the formulas above.

Example 4.14 An EWMA-SR Control Chart

A two-sided CUSUM-SR chart was shown in Example 4.9 using the data given in Table 4.12. As an alternative, an EWMA-SR control chart is now illustrated. For this purpose, we use $\lambda = 0.15$ to calculate the charting statistics and $L = 2.25$ to find the control limits (see Graham, Chakraborti, and Human, 2011b). To illustrate the calculations, consider sample number 1 with $SR_1 = 2$ and suppose that the starting value $Z_0 = 0$. Then, the first charting statistic on the EWMA-SR chart in Figure 4.19 is

$$\begin{aligned} Z_1 &= \lambda SR_1 + (1-\lambda)Z_0 \\ &= 0.15(2) + (1-0.15)0 \\ &= 0.30. \end{aligned}$$

The value of the second charting statistic is

$$\begin{aligned} Z_2 &= \lambda SR_2 + (1-\lambda)Z_1 \\ &= 0.15(-10) + (1-0.15)0.30 \\ &= -1.245 \end{aligned}$$

with the remaining calculations summarized in column (b) of Table 4.17.

Table 4.17 Calculations for the EWMA-SR chart.

Sample number i	(a) SR_i	(b) $Z_t = \lambda SR_t + (1-\lambda)Z_{t-1}$
1	2	0.300
2	−10	−1.245
3	10	0.442
4	−8	−0.825
5	−2	−1.001
6	2	−0.551
7	10	1.032
8	−2	0.577
9	−8	−0.709
10	−8	−1.803
11	10	−0.033
12	6	0.872
13	−8	−0.459
14	4	0.210
15	−2	−0.121
16	0	−0.103
17	2	0.212
18	10	1.681
19	−6	0.528
20	−2	0.149

For simplicity, we use the steady-state control limits on the EWMA-SR chart, which is found from Equation 4.22, that is

$$\begin{aligned} LCL &= -L\sigma_{SR_i}\sqrt{\frac{\lambda}{2-\lambda}} \\ &= -2.25\sqrt{\frac{4(4+1)(2\times 5+1)}{24}}\sqrt{\frac{0.15}{2-0.15}} \\ &= -1.94 \end{aligned}$$

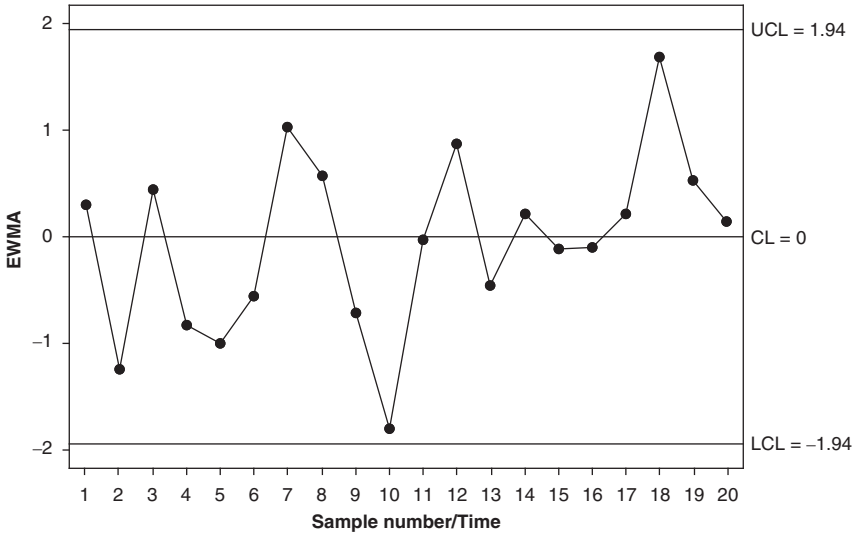


FIGURE 4.20 *An EWMA-SR chart.*

and

$$\begin{aligned} UCL &= L\sigma_{SR_i}\sqrt{\frac{\lambda}{2-\lambda}} \\ &= 2.25\sqrt{\frac{4(4+1)(2\times 5+1)}{24}}\sqrt{\frac{0.15}{2-0.15}} \\ &= 1.94, \end{aligned}$$

respectively.

The resulting EWMA-SR chart is shown in Figure 4.20. Since none of the charting statistics falls on or outside the LCL or the UCL, the process is considered IC with no remedial action needed.

It should be noted that, for the same data, the CUSUM-SR chart signaled on sample number 10 with a counter of $N^{-}= 2$, indicating that the shift most likely occurred at sample number 8. Although the EWMA-SR chart does not signal, from Figure 4.20 it can be seen that the EWMA-SR chart almost signals at sample number 10. A similar pattern is found for the parametric EWMA chart for this data, that is, it almost signals at sample number 10.

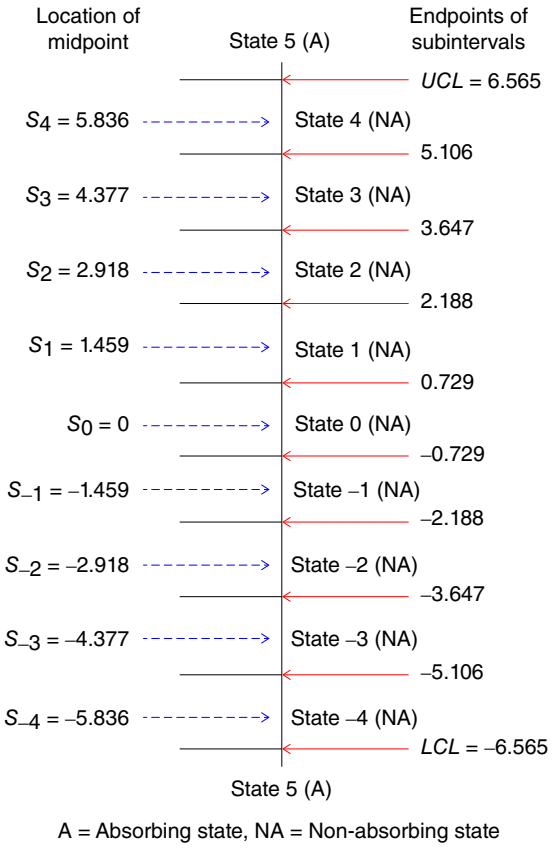


FIGURE 4.21 *Partitioning of the interval between the* LCL *and the* UCL *into* $\nu = 9$ *subintervals for* $\lambda = 0.1$ *and* $L = 3$.

Example 4.15 An EWMA-SR Control Chart: Markov Chain Approach

Finally, we consider an EWMA-SR chart for $n = 6$ with design parameters $\lambda = 0.1$, $L = 3$, and $v = 2s + 1 = 9$ so that $s = 4$, that is, we divide the interval between the control limits into nine subintervals (see Figure 4.21), resulting in a "small" transition probability matrix with only 10 rows and 10 columns, respectively. Then we illustrate how the run-length distribution characteristics can be calculated using the MC method. The steady-state control limits are calculated from Equation 4.22 and are given by $LCL = -6.565$ and $UCL = 6.565$.

Using the nine states, the transition probability matrix is given by

$$M_{10\times 10} = \begin{pmatrix} Q_{9\times 9} & | & \underline{p}_{9\times 1} \\ - & - & - \\ \underline{0}'_{1\times 9} & | & 1_{1\times 1} \end{pmatrix}$$

$$= \left(\begin{array}{ccccccccc|c} p_{(-4)(-4)} & p_{(-4)(-3)} & p_{(-4)(-2)} & p_{(-4)(-1)} & p_{(-4)0} & p_{(-4)1} & p_{(-4)2} & p_{(-4)3} & p_{(-4)4} & p_{(-4)5} \\ p_{(-3)(-4)} & p_{(-3)(-3)} & p_{(-3)(-2)} & p_{(-3)(-1)} & p_{(-3)0} & p_{(-3)1} & p_{(-3)2} & p_{(-3)3} & p_{(-3)4} & p_{(-3)5} \\ p_{(-2)(-4)} & p_{(-2)(-3)} & p_{(-2)(-2)} & p_{(-2)(-1)} & p_{(-2)0} & p_{(-2)1} & p_{(-2)2} & p_{(-2)3} & p_{(-2)4} & p_{(-2)5} \\ p_{(-1)(-4)} & p_{(-1)(-3)} & p_{(-1)(-2)} & p_{(-1)(-1)} & p_{(-1)0} & p_{(-1)1} & p_{(-1)2} & p_{(-1)3} & p_{(-1)4} & p_{(-1)5} \\ p_{0(-4)} & p_{0(-3)} & p_{0(-2)} & p_{0(-1)} & p_{00} & p_{01} & p_{02} & p_{03} & p_{04} & p_{05} \\ p_{1(-4)} & p_{1(-3)} & p_{1(-2)} & p_{1(-1)} & p_{10} & p_{11} & p_{12} & p_{13} & p_{14} & p_{15} \\ p_{2(-4)} & p_{2(-3)} & p_{2(-2)} & p_{2(-1)} & p_{20} & p_{21} & p_{22} & p_{23} & p_{24} & p_{25} \\ p_{3(-4)} & p_{3(-3)} & p_{3(-2)} & p_{3(-1)} & p_{30} & p_{31} & p_{32} & p_{33} & p_{34} & p_{35} \\ p_{4(-4)} & p_{4(-3)} & p_{4(-2)} & p_{4(-1)} & p_{40} & p_{41} & p_{42} & p_{43} & p_{44} & p_{45} \\ - & - & - & - & - & - & - & - & - & - \\ p_{5(-4)} & p_{5(-3)} & p_{5(-2)} & p_{5(-1)} & p_{50} & p_{51} & p_{52} & p_{53} & p_{54} & p_{55} \end{array}\right)$$

$$= \frac{1}{64}\left(\begin{array}{ccccccccc|c} 32 & 24 & 3 & 0 & 0 & 0 & 0 & 0 & 0 & 5 \\ 7 & 30 & 25 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 10 & 32 & 20 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 14 & 32 & 17 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 14 & 36 & 14 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 17 & 32 & 14 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 20 & 32 & 10 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 25 & 30 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 24 & 32 & 5 \\ - & - & - & - & - & - & - & - & - & - \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 64 \end{array}\right)$$

where the transition probabilities are calculated in the IC case using Equation 4.24. The calculation of p_{01} is given below for illustration purposes.

$$\begin{aligned} p_{01} &= P\left(\left(\frac{((S_1-\tau)-(1-\lambda)S_0}{\lambda}+\frac{n(n+1)}{2}\right)/2 < T_n^+ \right. \\ &\qquad \left. \leq \left(\frac{((S_1+\tau)-(1-\lambda)S_0}{\lambda}+\frac{n(n+1)}{2}\right)/2\right) \\ &= P\left(\left(\frac{(1.459-0.729)-(1\text{–}0.1)0}{0.1}+\frac{6(6+1)}{2}\right)/2 < T_n^+ \right. \\ &\qquad \left. \leq \left(\frac{((1.459+0.729)-(1-0.1)0}{0.1}+\frac{6(6+1)}{2}\right)/2\right) \\ &= P(14.145 < T_n^+ \leq 21.440) \end{aligned}$$

$$= P(T_n^+ = 15) + P(T_n^+ = 16) + P(T_n^+ = 17) + P(T_n^+ = 18)$$
$$+P(T_n^+ = 19) + P(T_n^+ = 20) + P(T_n^+ = 21)$$
$$= \frac{17}{64}.$$

The second-to-last equality follows from the fact that the random variable, T_n^+, takes on only non-negative integer values. Also, note that the probabilities are calculated using the IC distribution of T_n^+ given in Table B in Appendix A.

As before, once the one-step transition probabilities are calculated and the essential transition probability matrix Q is found, the IC run-length characteristics are obtained using Equations 3.22–3.24. This yields $ARL_{\text{IC}} = 699.49$ and $SDRL_{\text{IC}} = 690.55$ with the IC 5th, 25th, 50th, 75th, and 95th percentiles being equal to 44, 208, 488, 966, and 2078, respectively.

The ARL_{IC} value indicates that when the process is IC the first signal would be observed, on average, roughly at every 700th plotted point. The first quartile is 44, so we know that a false alarm will not occur within the first 44 plotted points, with a probability of 75% at most. As another example, the $MRL_{\text{IC}} = 488$, which indicates that the first false alarm will be observed within the first 488 plotted points at least 50% of the time.

Again, it should be noted that, although only the IC characteristics are shown here, the OOC characteristics can be found similarly using the OOC distribution of T_n^+.

So far, we have discussed some distribution-free control charts in Case K, the known IC parameter value case. However, in practice, there are situations where the IC parameter values are not known. In this situation, we are in Case U, which is considered next.

4.4 DISTRIBUTION-FREE VARIABLES CONTROL CHARTS IN CASE U

In this section, we describe some popular univariate distribution-free control charts when the underlying location parameter of an unknown continuous distribution is unknown. We start with a Shewhart chart. We also consider CUSUM and EWMA charts later in this chapter. Recall that a location parameter represents the center or a typical value of a distribution. It can, for example, be the mean or the median.

In the case of the parametric charts, we saw that when the IC parameter value(s) is (are) unknown one typically uses a Phase I analysis and the resulting reference data to estimate the parameter(s) and set up the

control limits in Phase II. The same is true for distribution-free charts. The Phase I analysis is an important component of the overall SPC and monitoring regime, and control charts play an important role in this phase. We discuss some of the available distribution-free Phase I control charts later in this chapter. Here, paralleling what was done for parametric charts, we assume that a reference sample is available after a successful Phase I analysis from which a class of distribution-free Phase II control charts, called the precedence charts, can be constructed and prospective monitoring can start.

4.4.1 Shewhart Control Charts

4.4.1.1 Shewhart Control Charts Based on the Precedence Statistic

In Case U, one can construct a control chart adapting a two-sample test. One popular two-sample distribution-free test is the precedence test (see Gibbons and Chakraborti, 2010, p. 253) which may be viewed as a generalization of the one-sample sign test to the two-sample case, and we first discuss an adaptation of the precedence test to construct distribution-free Shewhart control charts for location in Case U. The reader is referred to Balakrishnan and Ng (2006) for the vast literature on precedence/exceedance tests. There are many distribution-free, two-sample tests in the literature so one question that naturally arises is the efficacy of the choice of the precedence test over some other tests such as the more well-known WRS test. To this end, these authors (2006, p. 51) note that "Wilcoxon's rank-sum test performs better than the precedence tests if the underlying distributions are close to symmetry, such as the normal distribution, gamma distribution with large values of shape parameter, and lognormal distribution with small values of shape parameter. However, under some right-skewed distributions such as the exponential distribution, gamma distribution with shape parameter 2.0, and lognormal distribution with shape parameter 0.5, the precedence tests have higher power values than the Wilcoxon's rank-sum test for small values of r. It is evident that the more right-skewed the underlying distribution is, the more powerful the precedence test is." It turns out that both the precedence test and the rank-sum test have been adapted to construct distribution-free control charts.

For the Shewhart chart based on the precedence statistic (denoted as the Shewhart-Prec chart) for monitoring location, order statistics from a fixed size reference sample are used as control limits and a specified order statistic from each incoming Phase II or test sample is used as the charting statistic. Note that this formulation is different from the setting of parametric charts. Also note that, in the distribution-free framework under Case

U, one typically assumes that the underlying distributions, in addition to being continuous, are either in a location (shift) model, or a scale model, or a location-scale model. In this section, we consider the location model so that the Phase I (reference) and the Phase II process distributions are assumed to have a cdf $F(t) = H(t - \theta_1)$ and $G(t) = H(t - \theta_2)$, respectively, where H is some unknown continuous cdf and where θ_1 and θ_2 represent the respective location parameters. For example, if we assume that $H(0) = 0.5$, then the parameters θ_1 and θ_2 represent the median of F and G, respectively. The primary interest then is to detect a shift in the locations, which is to check if $\theta_1 = \theta_2$ (the process is IC), or if $\theta_1 \neq \theta_2$, (that is, if the process has gone OOC). Note that Case U fits the classical two-sample problem and the location model parallels, but generalizes, the common parametric model, that is, both Phase I and Phase II distributions are normal with possibly unknown and unequal means, but with equal variances.

Suppose that a random sample of m observations, $X_1, X_2, \ldots, X_m$, hereafter called a reference sample or the reference data, is available from a Phase I analysis from an IC process, with a cdf $F(x) = H(x - \theta_1)$. Assume for simplicity that there are no ties among the data. The m observations are arranged in an ascending order (i.e., from the smallest to the biggest) and two order statistics $X_{(a:m)}$ and $X_{(b:m)}$, for a given a and b, with $1 \leq a < b \leq m$, are found using some chart performance criteria. These order statistics form the control limits of the Shewhart-Prec chart and are given by

$$L\hat{C}L = X_{(a:m)} \text{ and } U\hat{C}L = X_{(b:m)}. \tag{4.25}$$

In Phase II, test samples are randomly drawn independently from one another and independently from the reference sample, and are assumed to be from an unknown continuous distribution with cdf $G_i(y) = H(y - \theta_i)$ for $i = 2, 3, \ldots$ We are interested in monitoring the process in Phase II, that is, to monitor whether the process operates IC. In other words, we are interested in monitoring whether $\theta_i = \theta_1$ for $i = 2, 3, \ldots$ or, equivalently, whether $\delta = \theta_i - \theta_1 = 0$ for $i = 2, 3, \ldots$ (process is IC) or, whether $\delta = \theta_i - \theta_1 \neq 0$ for $i = 2, 3, \ldots$ (i.e., θ_i has shifted from θ_1 and is equal to $\theta_1 + \delta$, i.e., the process is OOC) at the ith test sample. In this case we use a two-sided control chart. Note that most of what we discuss in this section is about two-sided charts, but it is possible that one is interested in a directional change, say, that $\theta_i > \theta_1$(or $\theta_i < \theta_1$), in either of these cases we would be using a one-sided chart.

For simplicity, assume that the test samples are of size n, the ith sample denoted by $Y_{i1}, Y_{i2}, \ldots, Y_{in}$. The equal sample size assumption will be relaxed later. Again, assume no ties and from each test sample, calculate the charting statistic $Y_{(j:n)}$, the jth order statistic for a given value of j, where $j = 1, 2, \ldots, n$. For example, $Y_{(j:n)}$ can be the median, or the 25th or the

75th percentile (each of which is an order statistic) of the ith test sample. Note that, although we present the Shewhart-Prec chart in the general framework where any test sample order statistic can be used as the charting statistic, guidance is clearly needed about which order statistic to use in practice. From a practical point of view, the median of the test sample is a good choice and is often used. The resulting Shewhart-Prec chart, originally proposed by Janacek and Meikle (1997) and studied by Chakraborti, van der Laan, and Van de Wiel (2004) is called the **median Shewhart-Prec** chart and is denoted as the **Med chart**. This is important for (at least) two reasons: (i) The sample median is less sensitive to outliers or measurement errors than other measures of location, such as the sample mean for instance, and (ii) the sample median is a much more flexible estimator of location and is applicable in a diverse number of situations, unlike the sample mean. Other choices for $Y_{(j:n)}$ are possible and one example is discussed in Section 4.5. It may be noted that Qiu (2014) has given a nice account of some distribution-free control charts in Case U, including the Shewhart-Prec charts; however, we provide a more comprehensive and focused account, including many nuts and bolts type details, that will be useful to the reader.

Getting back to the Shewhart-Prec chart, the $Y_{(j:n)}$ is compared with the control limits given in Equation 4.25. The Shewhart-Prec chart declares the process to be IC if

$$X_{(a:m)} \leq Y_{(j:n)} \leq X_{(b:m)}$$

and OOC if

$$Y_{(j:n)} < X_{(a:m)} \quad \text{or} \quad Y_{(j:n)} > X_{(b:m)}.$$

Before going any further, we discuss what are known as the precedence statistics in the nonparametric literature. Then we show that the statistical properties of the control chart described above can be conveniently studied in terms of the precedence statistics and hence the name Shewhart-Prec is used. More precisely, the run-length distribution of the control chart in Equation 4.26 can be expressed in terms of the distribution of the precedence statistics. This establishes the connection between the control chart, which is operationally convenient to be applied in terms of the order statistics, and a class of well-known statistics that are more convenient to deal with while studying the chart properties. This also allows us to show that the control chart is distribution-free in a straightforward way.

To this end, let W_j, $j = 1, 2, \ldots, n$, denote the number of X observations in the Phase I reference sample that precede, that is, are less than or equal to $Y_{(j:n)}$, the jth order statistic from the ith Phase II test sample. The statistic W_j is called a **precedence statistic**. When the process is IC, that is, when $G_i = F$ or $\theta_i = \theta_1$ in the location model, the exact probability distribution

of the precedence statistic W_j can be obtained by either mathematical statistical techniques or by using combinatorial arguments (see Chakraborti, van der Laan, and Van de Wiel 2004, p. 445 Equation 1). This shows that the IC probability distribution of W_j is given by

$$P_{\text{IC}}(W_j = w) = \frac{\binom{w+j-1}{w}\binom{m+n-j-w}{m-w}}{\binom{m+n}{m}} \tag{4.26}$$

where $j = 1, 2, \ldots, n$, $w = 0, 1, \ldots, m$ and the subscript IC indicates that it is the IC probability distribution of W_j.

We see from Equation 4.26 that the IC probability distribution of the precedence statistic W_j is free of the underlying distributions and depends only on the number of reference samples observations, m, the number of observations in each test sample, n, and the index of the selected test sample order statistic, j. Thus, the precedence statistic W_j is a distribution-free

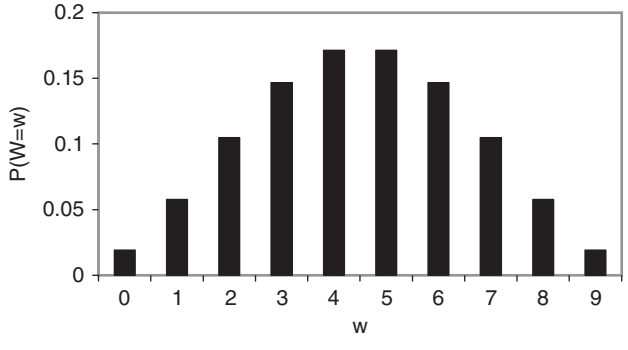


FIGURE 4.22a *The IC distribution of W_j for $m = 9$, $n = 7$, and $j = 4$.*

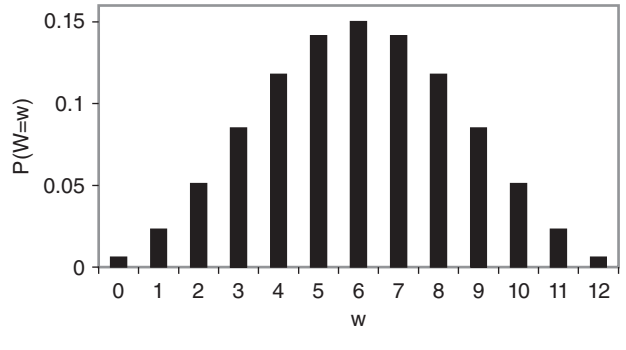


FIGURE 4.22b *The IC distribution of W_j for $m = 12$, $n = 9$, and $j = 5$.*

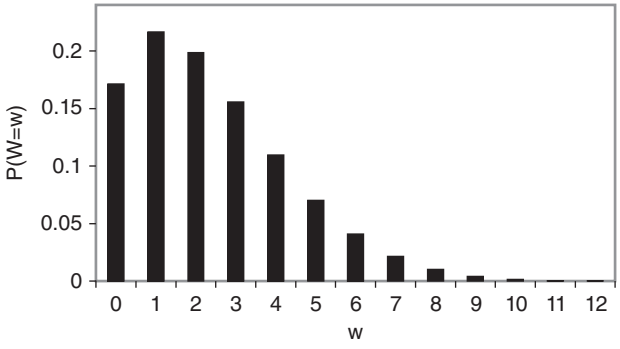


FIGURE 4.22c *The IC distribution of W_j for $m = 12$, $n = 9$, and $j = 2$.*

statistic. To illustrate, Figures 4.22a–4.22c show the IC probability distribution of W_j for selected values of m (the size of the reference sample), n (the size of a future sample or a test sample) and some selected values of j. Figure 4.22b, for example, displays the IC distribution of W_j for $m = 12$, $n = 9$, and $j = 5$, that is, when the jth test sample order statistic is the sample median. Substituting these values in Equation 4.26 yields

$$P_{\text{IC}}(W_5 = w) = \frac{\binom{w+4}{w}\binom{16-w}{12-w}}{\binom{21}{12}}$$

for $w = 0, 1, \ldots, 12$. Figure 4.22a shows the IC distribution of W_j when $m = 9$, $n = 7$, and $j = 4$. Note that, when n is odd and the test sample order statistic is chosen to be the median, that is, $j = (n + 1)/2$, for the Med chart, the IC probability distribution of W_j is symmetric about its mean, which in the case of Figures 4.22a and 4.22b are equal to 4.5 and 6, respectively. The reader can find the formulas for the mean and the standard deviations of the IC distribution of W_j. Figure 4.22c shows the IC distribution of W_j for $m = 12$, $n = 9$, and $j = 2$, that is, when the second order statistic of the test sample is the charting statistic. This distribution is skewed to the right.

As we noted already, W_j is a distribution-free statistic, which means that its distribution when the process is IC is free from the underlying distributions F and G. It will be shown that the run-length distribution of the control chart in Equation 4.26 can be obtained from the distribution of the precedence statistic W_j, and hence the IC run-length distribution of the control chart given in Equation 4.26 is distribution-free, and this distribution-free chart is typically referred to as the Shewhart-Prec chart.

Note that the IC probability distribution of W_j can easily accommodate test samples of unequal size, for example, suppose that each test sample is

of size n_i. Then Equation 4.26 will still hold with n being substituted by n_i with $j = 1, 2, \ldots, n_i$. Thus, a Shewhart-Prec chart is flexible enough to accommodate variable test sample sizes.

To show the connection between the control chart defined in Equation 4.25 in terms of the precedence statistic, note that, under the continuity assumption on F and G, which allows no ties, $Y_{(j:n)} \leq X_{(b:m)}$ if and only if at most $b-1$ X observations precede $Y_{(j:n)}$, that is, if $W_j \leq b-1$. Similarly, $X_{(a:m)} \leq Y_{(j:n)}$ if and only if at least a X observations precede $Y_{(j:n)}$, that is, if $W_j \geq a$. Thus, the non-signaling event $\{X_{(a:m)} \leq Y_{(j:n)} \leq X_{(b:m)}\}$ is equivalent to the event $\{a \leq W_j \leq b-1\}$ and hence the control chart in Equation 4.25 can be expressed in terms of the precedence statistic W_j. The process should be declared IC if $\{a \leq W_j \leq b-1\}$ and OOC if $\{W_j \leq a-1\}$ or $\{W_j \geq b\}$. As a result, the control chart in Equation 4.25 is called a **Shewhart-Prec** chart. Because of the equivalence of the non-signaling events, the probability that the Shewhart-Prec chart does not signal

$$p = P(X_{(a:m)} \leq Y_{(j:n)} \leq X_{(b:m)}) = P(\text{No Signal})$$

can be expressed, in terms of distribution of W_j, as

$$P(X_{(a:m)} \leq Y_{(j:n)} \leq X_{(b:m)}) = P(a \leq W_j \leq b-1) = p(m, n, j; F, G),$$

say. It is shown later on that the IC run-length distribution of the Shewhart-Prec chart can be expressed in terms of the IC distribution of the precedence statistic and hence the Shewhart-Prec chart is distribution-free since W_j has been shown to be distribution-free. This is also shown later on when the run-length distribution is studied.

4.4.1.1.1 Methods of Calculating the Run-length Distribution of the Shewhart-Prec Control Chart To evaluate the performance of the precedence control chart, we need to derive and study its run-length distribution. In Case U, since we use a reference sample to estimate the control limits, the control limits are random variables that introduce additional variability, which needs to be accounted for in deriving the run-length distribution. Thus, as in Case U for the normal theory $\overline{X}$ chart, we use the conditioning/unconditioning (CUC) method discussed in Chapter 3 to derive the run-length distribution. There are two main steps to this end. First, we condition on the two reference sample order statistics $X_{(a:m)}$ and $X_{(b:m)}$, the two control limits, to find the conditional run-length distribution. Then, we find the unconditional run-length distribution by taking the expectation (averaging) over the distribution of the two order statistics. However, a big difference between the normal theory case and the nonparametric case is the fact that, in the normal case, the estimators

of the mean and the standard deviation that define the control limits are independent, but the two order statistics that define the control limits for the Shewhart-Prec chart are not. So, the expectation (averaging or unconditioning) step, while finding the unconditional run-length distribution in the nonparametric case involves integrating over the joint distribution of the two order statistics (the control limits), whereas, in the normal case, it involves the product of the individual probability density functions (pdf's) of the two estimators which are independent.

We first start with the conditional run-length distribution.

4.4.1.1.2 The Conditional Run-length Distribution of the Two-sided Shewhart-Prec Control Chart The two reference sample order statistics $X_{(a:m)}$ and $X_{(b:m)}$ define the control limits of the Shewhart-Prec chart. Conditionally, on these order statistics, that is, when these are fixed quantities, the signaling events in Phase II are mutually independent since the charting statistics $Y_{(j:n)}$ come from independent Phase II samples. Thus, conditionally on the reference sample order statistics, the run-length distribution of the Shewhart-Prec control chart is a geometric distribution with parameter $1 - p_{\mathrm{C}}$, where p_{C} is the conditional probability of no signal, $p_{\mathrm{C}} = P_{\mathrm{C}}(x_{(a:m)} \leq Y_{(j:n)} \leq x_{(b:m)} | X_{(a:m)} = x_{(a:m)}.\ X_{(b:m)} = x_{(b:m)})$ is given by

$$\begin{aligned} p_{\mathrm{C}} &= P_{\mathrm{C}}(F^{-1}(u_{(a:m)}) \leq G^{-1}(U_{(j:n)}) \leq F^{-1}(u_{(b:m)})) \\ &= P_{\mathrm{C}}(GF^{-1}(u_{(a:m)}) \leq U_{(j:n)} \leq GF^{-1}(u_{(b:m)})\ ) \end{aligned} \tag{4.27}$$

where $u_{(e:l)}$ denotes the observed value of the eth order statistic $U_{(e:l)}$ in a sample of size l from the Uniform(0,1) distribution. Note the use of the probability integral transformation (PIT) (see Section 1.4) in this derivation, according to which, for example, $G(Y_{(j:n)})$ is distributed as $U_{(j:n)}$, the jth order statistic in a sample of size n from the Uniform(0,1) distribution, which can be shown to follow a beta distribution with parameters j and $n - j + 1$ (see Section 1.4) that has the pdf $f(u) = \dfrac{1}{\mathrm{B}(j, n-j+1)} u^{j-1}(1-u)^{n-j}$.

Thus

$$p_{\mathrm{c}} = \int_{GF^{-1}(u_{(a:m)})}^{GF^{-1}(u_{(b:m)})} f(u)du = F_2(u_{(a:m)}, u_{(b:m)}, j, n, F, G)$$

for example.

Hence, the conditional distribution of the run-length random variable N, that is, $(N \mid X_{(a:m)},\ X_{(b:m)})$ is $\mathrm{GEO}(1 - p_{\mathrm{C}})$ and therefore the pmf of the conditional run-length distribution is given by

$$P_{\mathrm{C}}(N = k \mid X_{(a:m)}, X_{(b:m)}) = p_{\mathrm{C}}^{k-1}(1 - p_{\mathrm{C}}) = p_{\mathrm{C}}^{k-1} - p_{\mathrm{C}}^{k} \tag{4.28}$$

for $k = 1, 2, 3, \ldots$ Note that the complete notation for the condition is $X_{(a:m)} = x_{(a:m)}, X_{(b:m)} = x_{(b:m)}$; however, for brevity, we write $X_{(a:m)}, X_{(b:m)}$. All properties of the conditional run-length distribution then follow from the properties of the geometric distribution, for example, the cdf of the conditional run-length distribution is

$$P_{\mathrm{C}}(N \leq k \mid X_{(a:m)}, X_{(b:m)}) = 1 - p_{\mathrm{C}}^{k} \tag{4.29}$$

for $k = 1, 2, 3, \ldots$ The conditional average run-length is

$$ARL_{\mathrm{C}} = E(N \mid X_{(a:m)}, X_{(b:m)}) = \frac{1}{1 - p_{\mathrm{C}}}. \tag{4.30}$$

Similarly, the conditional variance of the run-length is

$$VRL_{\mathrm{C}} = var(N \mid X_{(a:m)}, X_{(b:m)}) = \frac{p_{\mathrm{C}}}{(1 - p_{\mathrm{C}})^2}. \tag{4.31}$$

So, it follows that the conditional standard deviation of the run-length is

$$SDRL_{\mathrm{C}} = SD(N \mid X_{(a:m)}, X_{(b:m)}) = \sqrt{\frac{p_{\mathrm{C}}}{(1 - p_{\mathrm{C}})^2}}. \tag{4.32}$$

Note that for Equations 4.29–4.32, the subscript C stands for "conditional" as in the case of the parametric charts in Chapter 3.

When the process is IC, p_{C} reduces to

$$p_{\mathrm{C,IC}} = P_{\mathrm{C,IC}}(u_{(a:m)} \leq U_{(j:n)} \leq u_{(b:m)} \mid U_{(a:m)}, U_{(b:m)}) = \int_{u_{(a:m)}}^{u_{(b:m)}} f(u)du. \tag{4.33}$$

Again, note that the complete notation for the condition is $U_{(a:m)} = u_{(a:m)}$, $U_{(b:m)} = u_{(b:m)}$; however, for brevity, we write $U_{(a:m)}, U_{(b:m)}$.

In Equation 4.33, the subscript C,IC stands for "conditional in-control". Thus, $p_{\mathrm{C,IC}}$ denotes the conditional IC probability of no signal. So, the cdf, the average, the variance, and the standard deviation of the IC conditional run-length distribution can be obtained from Equations 4.29–4.32 by replacing p_{C} with $p_{\mathrm{C,IC}}$.

Furthermore, expanding the $(1 - u)^{n-j}$ term in the beta pdf in the expression for p_{C} as a binomial sum and then integrating it, it can be shown that we can express the conditional probability of no signal as

$$\begin{aligned} p_{\mathrm{C}} &= F_2(u_{(a:m)}, u_{(b:m)}, j, n, F, G) \\ &= \frac{1}{\mathrm{B}(j, n - j + 1)} \sum_{h=0}^{n-j} \frac{(-1)^h}{j + h} \binom{n - j}{h} \\ &\quad \times \left\{ (GF^{-1}(u_{(b:m)}))^{j+h} - (GF^{-1}(u_{(a:m)}))^{j+h} \right\}. \end{aligned} \tag{4.34}$$

The last expression may be useful for computational purposes. The conditional run-length distribution and its various attributes, such as its average, standard deviation, percentiles, etc., are random variables and these can provide useful insight into the effect of parameter estimation on the control chart, in particular the Shewhart-Prec chart. The conditional run-length distribution itself, for example, can be analyzed to study the impact of the reference sample size on the performance of the control chart. A number of researchers in the recent SPC literature have suggested placing more emphasis on the conditional run-length distribution when it comes to designing a Phase II chart, but we don't pursue these details here. Next, we derive the unconditional run-length distribution of the Shewhart-Prec chart. The unconditional run-length distributions and its attributes have been more traditionally used to study chart performance in the literature.

4.4.1.1.3 The Unconditional Run-length Distribution of the Two-sided Shewhart-Prec Control Chart The unconditional run-length distribution pmf, that is, $P_{\mathrm{U}}(N = k)$, and its various characteristics can easily be found using the CUC method. Note that the subscript U refers to "unconditional."

For example, the unconditional pmf is given by

$$P_{\mathrm{U}}(N = k) = E_{X_{(a:m)}, X_{(b:m)}}(P_{\mathrm{C}}(N = k \mid X_{(a:m)}, X_{(b:m)})) \tag{4.35}$$

which, using Equation 4.28, simplifies to

$$P_{\mathrm{U}}(N = k) = E_{X_{(a:m)}, X_{(b:m)}}(p_{\mathrm{C}}^{k-1}) - E_{X_{(a:m)}, X_{(b:m)}}(p_{\mathrm{C}}^{k}) = D^*(k-1) - D^*(k) \tag{4.36}$$

for $k = 1, 2, 3, \ldots$, where

$$\begin{aligned} D_{\mathrm{U}}(k) &= E_{X_{(a:m)}, X_{(b:m)}}(p_{\mathrm{C}}^{k}) \\ &= \int_0^1 \int_0^{u_{(b:m)}} \left(\int_{GF^{-1}(u_{(a:m)})}^{GF^{-1}(u_{(b:m)})} f(u)du \right)^k f(u_{(a:m)}, u_{(b:m)}) du_{(a:m)} du_{(b:m)} \\ &= \int_0^1 \int_0^{u_{(b:m)}} (F_2(u_{(a:m)}, u_{(b:m)}, j, n, F, G)^k \\ &\quad \times f(u_{(a:m)}, u_{(b:m)}) du_{(a:m)} du_{(b:m)} \end{aligned} \tag{4.37}$$

with $D_{\mathrm{U}}(0) = 1$ and

$$f(u_{(a:m)}, u_{(b:m)}) = C(u_{(a:m)})^{a-1}(u_{(b:m)} - u_{(a:m)})^{b-a-1}(1 - u_{(b:m)})^{m-b}$$

where $C = \dfrac{m!}{(a-1)!(b-a-1)!(m-b)!}$ and $f(u_{(a:m)}, u_{(b:m)})$ is the joint pdf of $U_{(a:m)}$ and $U_{(b:m)}$ in a sample of size m from the Uniform(0,1) distribution.

The IC unconditional run-length distribution of the Shewhart-Prec control chart is obtained by setting $G = F$, and we denote this by $P_{\mathrm{U,\,IC}}$. Thus, the unconditional IC run-length pmf is given by

$$P_{\mathrm{U,\,IC}}(N = k) = D_{\mathrm{U}}(k-1) - D_{\mathrm{U}}(k) \tag{4.38}$$

for $k = 1, 2, 3, \ldots$, where

$$\begin{aligned} D_{\mathrm{U}}(k) = \int_0^1 \int_0^{u_{(b:m)}} & (F_2(u_{(a:m)}, u_{(b:m)}, j, n, F, F)^k \\ & \times f(u_{(a:m)}, u_{(b:m)}) du_{(a:m)} du_{(b:m)} \end{aligned} \tag{4.39}$$

with $D_{\mathrm{U}}(0) = 1$. This formula can be used, recursively, to find the IC unconditional run-length pmf.

The cdf of the unconditional run-length distribution, in general, can be obtained from Equations 4.35–4.39. Hence, for example, the unconditional IC run-length cdf can be obtained and is given by

$$P_{\mathrm{U,\,IC}}(N \leq k) = 1 - D_{\mathrm{U}}(k) \tag{4.40}$$

for $k = 1, 2, \ldots$

4.4.1.1.4 The Unconditional Run-length Performance Characteristics of the Two-sided Shewhart-Prec Control Chart Various chart performance characteristics, such as the unconditional average run-length, the unconditional median run-length, and the unconditional standard deviation of the run-length, etc., can be found from the run-length distribution pmf or cdf. Alternatively, for the unconditional average and the standard deviation of the run-length, we can directly use the CUC method, taking advantage of the fact that the conditional run-length distribution is geometric. This approach may be advantageous in other situations as far as calculating the first two moments of the unconditional run-length distribution. To illustrate this, first note that, as shown earlier in Equation 4.30, the conditional average run-length is equal to

$$ARL_{\mathrm{C}} = \frac{1}{1 - p_{\mathrm{C}}} \tag{4.41}$$

and hence the unconditional average run-length of the two-sided Shewhart-Prec chart, ARL_{U}, can be obtained by taking the expectation of

ARL_{C} over the joint distribution of $U_{(a:m)}$ and $U_{(b:m)}$, two order statistics in a sample of size m from the Uniform(0,1) distribution. Hence,

$$\begin{aligned} ARL_{\mathrm{U}} &= \int_0^1 \int_0^{u_{(b:m)}} \left(\frac{1}{1-p_{\mathrm{C}}}\right) f(u_{(a:m)}, u_{(b:m)}) du_{(a:m)} du_{(b:m)} \\ &= \int_0^1 \int_0^{u_{(b:m)}} \left(\frac{1}{1-F_2(u_{(a:m)}, u_{(b:m)}, j, n, F, G)}\right) \\ &\quad \times f(u_{(a:m)}, u_{(b:m)}) du_{(a:m)} du_{(b:m)}. \end{aligned} \tag{4.42}$$

The unconditional IC average run-length is obtained by setting $G = F$. Thus, Equation 4.42 reduces to

$$\begin{aligned} ARL_{\mathrm{U,IC}} = \int_0^1 \int_0^{u_{(b:m)}} &\left(\frac{1}{1-F_2(u_{(a:m)}, u_{(b:m)}, j, n, F, F)}\right) \\ &\times f(u_{(a:m)}, u_{(b:m)}) du_{(a:m)} du_{(b:m)}. \end{aligned} \tag{4.43}$$

Similarly, for the variance, we can also use the CUC method. To this end, as noted in Chapter 3, the variance of any random variable T can be obtained from the formula

$$var(T) = E(var(T|W)) + var(E(T|W)).$$

Applying this formula with the run-length variable N as T and the conditioning variable $W = (X_{(a:m)}, X_{(b:m)})$, the conditional variance and the expectation on the right-hand side of this formula are found from Equations 4.31 and 4.30, respectively. Hence, the unconditional run-length variance is equal to

$$VRL_{\mathrm{U}} = E(VRL_C) + var(ARL_{\mathrm{C}}) = E\left(\frac{p_{\mathrm{C}}}{(1-p_{\mathrm{C}})^2}\right) + var\left(\frac{1}{1-p_{\mathrm{C}}}\right)$$

where the expectation is again taken over joint pdf of $U_{(a:m)}$ and $U_{(b:m)}$. Thus, the CUC method can be a very useful technique in calculating the run-length distribution and its first two moments.

Furthermore, as we noted earlier, the median run-length (MRL) is an additional important chart performance characteristic that should play a more prominent role in SPC since the run-length distribution is right skewed. The MRL of the two-sided Shewhart-Prec chart in general can be conveniently calculated from the run-length cdf, for example, the unconditional IC MRL can be found by solving for the lowest integer MRL_{IC} so that $P_{U,IC}(N \leq MRL_{\mathrm{IC}}) \geq 0.5$, which, using Equation 4.40

leads to solving $D_U(MRL_{\text{IC}}) \leq 0.5$ where $D_U(\cdot)$ is defined in Equation 4.39. It can then be shown, and this is left as an exercise for the reader, that the MRL_{IC} is given by the solution of the equation

$$\int_0^1 \int_0^{u_{(b:m)}} F_2(u_{(a:m)}, u_{(b:m)}, m, n, F, F)^{MRL_{\text{IC}}} \times f(u_{(a:m)}, u_{(b:m)}) du_{(a:m)} du_{(b:m)} = 0.5 \tag{4.44}$$

where $F_2(u_{(a:m)}, u_{(b:m)}, m, n, F, F)$ is given by Equation 4.34.

Some non-linear methods would need to be used to solve Equation 4.44 for the MRL_{IC} (or the MRL in general); we leave the details to the reader. Other IC run-length distribution percentiles can be obtained similarly. It may be noted that, finding the OOC percentiles of the unconditional run-length distribution requires specifying the conversion function GF^{-1}, which is needed in order to calculate the run-length pmf or the cdf. We touch on this point in the performance comparison section later.

While we have exact formulas to calculate the run-length pmf or the cdf of the Shewhart-Prec control chart that require numerical integration for evaluation, as is often the case in practice with many control charts, simulations can provide a quicker alternative. We outline the steps in the next section.

4.4.1.1.5 The Simulation Approach The following simulation algorithm can be used to obtain the run-length characteristics for the Shewhart-Prec chart. The corresponding SAS program is given in Appendix B, SAS program number 7, and the output is shown in Figure 4.23.

Step 1: After specifying the Phase I reference sample size ($m \geq 1$), the Phase II test sample size ($n \geq 1$) and the size of the shift (δ) to be detected, we specify two distributions for generating the Phase I and Phase II samples, respectively. For the IC case, the two distributions are identical (we say that the shift equals zero ($\delta = 0$)), so any continuous distribution can be used to generate the observations. For the OOC case, recall that the distribution for the Phase II sample is assumed to be of the same form as that for the Phase I sample, but with a shift in the mean in units of the population standard deviation. The Phase I sample, for example, may be drawn from a normal distribution with mean 0 and standard deviation 1 and then the Phase II sample may be drawn from a normal distribution with mean $\mu = \delta \neq 0$ and standard deviation 1 (we say that the shift δ in the mean μ is not equal to zero ($\delta \neq 0$)). Any continuous common distribution, whether symmetric or asymmetric, may be used to generate the observations.

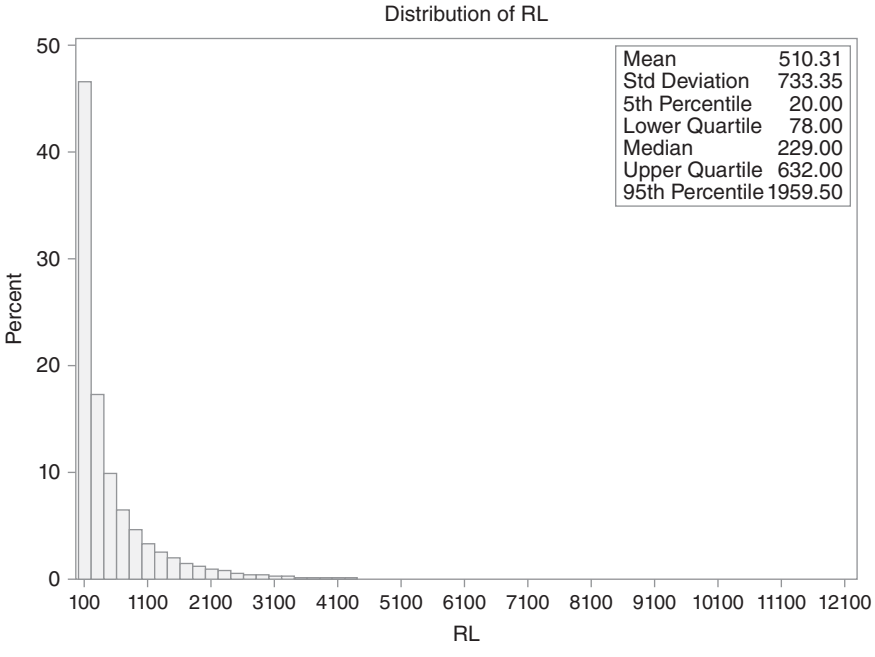


FIGURE 4.23 *SAS output of the RL distribution where the design parameters (*m = *1000,* n = *5,* j = *3,* a = *48,* b = *953) of the Shewhart-Prec chart were for a nominal* $\mathrm{ARL}_{IC} = 500$.

Step 2: Select the size of a test sample, n, and the value of j for the desired order statistic in the test sample. This is the charting statistic, for example, the sample median can be chosen; this would give the Med chart.

Step 3: Specify a value for the desired nominal ARL_{IC}, such as 370 or 500.

Step 4: The charting constants, a and b, are found by one of two ways. The first method involves making use of the tables provided in Chakraborti, van der Laan, and Van de Wiel (2004). However, they only provide the charting constants for $m = 50$, 100, 500, and 1000, respectively, and for $n = 5$, 11, and 25, respectively, for desired nominal ARL_{IC} values equal to 370, 500, and 1000. Should the reader have a different (m, n) combination or wish to make use of a different desired nominal ARL_{IC} value, then the second method, which involves a "search algorithm," should be used. For this method, the ARL_{IC} is calculated for different (a, b) combinations and the combination closest to the desired nominal ARL_{IC} is chosen. More details on the "search algorithm" are given in the next section, where the implementation of the chart is discussed in detail.

Step 5: The estimated Shewhart control limits are given by the corresponding order statistics in the reference sample $\hat{LCL} = X_{(a:m)}$ and $\hat{UCL} = X_{(b:m)}$, respectively.

Step 6: The number of subgroups needed until W_j plots on or outside the control limits for the first time is recorded as an observation from the run-length distribution.

Step 7*:* Repeat Steps 1 to 6 a total of 100 000 times.

Step 8: Once we have obtained a "data set" with 100 000 observations from the run-length distribution, proc univariate of SAS can be used to obtain the run-length characteristics. For $\delta = 0$, this would be the IC run-length characteristics, whereas, for $\delta \neq 0$, this would be the OOC run-length characteristics.

4.4.1.1.6 Implementation of the Two-sided Shewhart-Prec Chart

In order to implement the two-sided Shewhart-Prec, we need values for the indices a and b so that the control limits, that is, the two reference sample order statistics $X_{(a:m)}$ and $X_{(b:m)}$, can be found from the reference data. As in the case of parametric charts, these control limits are typically determined for a given nominal ARL_{IC} value. Thus, while designing a two-sided precedence control chart with a specified nominal ARL_{IC} value, say, 370, we find two positive integers a and b, $1 \leq a < b \leq m$, for a given m, n, and j, such that $ARL_{\text{IC}} = 370$.

Recall that the unconditional IC average run-length is given in Equation 4.43. Solving $ARL_{\text{IC}} = 370$ for a and b, in general, can be done using a "search algorithm" (in which a and b are both varied so that $1 \leq a < b \leq m$) that stops when the attained IC average run-length is close to 370. To this end, it is helpful to note that, in general, the ARL is *increasing* in b and *decreasing* in a. For the Med chart, the required calculations are relatively simpler since here we can use the symmetry of the IC distribution of the precedence statistic, set $b = m - a + 1$ and solve $ARL_{\text{IC}} = 370$ for a. This is discussed in the next section.

4.4.1.1.7 The Two-sided Med Chart

In order to implement the two-sided Med chart, as noted before, we set $b = m - a + 1$ and solve $ARL_{\text{IC}} =$ 370, 500, and 1000, respectively, for a. The values shown in Table 4.18 are taken from Chakraborti, van der Laan, and Van de Wiel (2004).

Note that, since the IC run-length distribution of the precedence chart is discrete, it's most likely that the nominal ARL_{IC} is not exactly achieved. We use a conservative approach so that the corresponding attained ARL_{IC} is "close" to the nominal value; for example, for $m = 100$, $n = 5$, and $j = 3$ with a nominal $ARL_{\text{IC}} = 370$, from Table 4.18 the control limits of the Med

Table 4.18 The control chart indices (a, b) the attained ARL_{IC} and the attained FAR for the two-sided Med chart for a given IC average run-length, m, n, and j.

Desired			m			
ARL_{IC}	n	j	100	200	500	1000
			(a,b)	(a,b)	(a,b)	(a,b)
			Attained ARL_{IC}	Attained ARL_{IC}	Attained ARL_{IC}	Attained ARL_{IC}
			Attained FAR	Attained FAR	Attained FAR	Attained FAR
370	5	3	(6,95)	(11,190)	(27,474)	(53,948)
			359.6	385.5	365.5	374.4
			0.00554	0.00375	0.00332	0.00287
	11	6	(14,87)	(27,174)	(67,434)	(132,869)
			367.8	372.4	355.3	377.3
			0.00627	0.00412	0.00332	0.00288
	25	13	(24,77)	(46,155)	(114,387)	(227,774)
			316.5	375.7	367.3	377.0
			0.01108	0.00526	0.00354	0.00301
			(a,b)	(a,b)	(a,b)	(a,b)
			Attained ARL_{IC}	Attained ARL_{IC}	Attained ARL_{IC}	Attained ARL_{IC}
			Attained FAR	Attained FAR	Attained FAR	Attained FAR
500	5	3	(6,95)	(10,191)	(24,477)	(48,953)
			359.6	520.1	520.3	501.9
			0.00554	0.00291	0.00227	0.00216
	11	6	(13,88)	(26,175)	(63,438)	(125,876)
			574.5	460.5	497.2	506.9
			0.00449	0.00342	0.00242	0.00225
	25	13	(23,78)	(45,156)	(111,390)	(219,782)
			510.8	472.5	480.1	492.2
			0.00804	0.00436	0.00276	0.00233
			(a,b)	(a,b)	(a,b)	(a,b)
			Attained ARL_{IC}	Attained ARL_{IC}	Attained ARL_{IC}	Attained ARL_{IC}
			Attained FAR	Attained FAR	Attained FAR	Attained FAR

Table 4.18 (*continued*)

Desired			m			
ARL_{IC}	n	j	100	200	500	1000
1000	5	3	(5,96)	(8,193)	(19,482)	(38,963)
			678.4	1067.5	1055.8	1005.9
			0.00352	0.00161	0.00109	0.00111
	11	6	(11,90)	(23,178)	(56,445)	(110,891)
			940.6	927.8	953.2	1021.7
			0.00212	0.00188	0.00131	0.00109
	25	13	(22,79)	(42,159)	(103,398)	(206,795)
			854.6	986.2	1033.6	1004.4
			0.00573	0.00240	0.00136	0.00117

chart are given by the 6th and the 95th order statistics of the reference sample. The attained ARL_{IC} for this chart is 359.6 and the corresponding attained *FAR* is 0.00554. The 359.6 value compares well with the traditional 370, but note that the attained *FAR* is not 0.0027 but almost twice that. This is one effect of parameter estimation that is inherent in Case U.

The statistical properties of the Med chart, including the run-length distribution and its associated characteristics, can be obtained from the general discussions given earlier about the Shewhart-Prec charts. The run-length characteristics for the Med chart can be found by substituting n with $2s + 1$, j with $s + 1$, and b with $m - a + 1$ into Equations 4.26–4.44 and this is left as an exercise for the reader.

4.4.1.1.8 The Performance of the Two-sided Shewhart-Prec Control Chart The performance of any control chart is of paramount interest to the practitioner as well as the researcher. Thus, as we've noted before, the run-length distribution is of interest when studying the performance of the precedence control chart. For instance, if the process location shifts from its IC value, we might be interested in the probability that the chart will detect the shift (or signal a change) on the first test sample, or on the second test sample, or more generally, on the kth subsequent test sample. Similarly, for example, we are interested in finding out the OOC average run-length (ARL_{OOC}), which is, on an average, how long (how many Phase II samples) it takes before a location shift is detected by the chart. To answer these questions, the OOC run-length distribution and its associated characteristics are needed. Here, we focus mainly on the unconditional run-length distribution as it is the norm in the current literature, but, as noted earlier, a conditional analysis is also possible, and is perhaps more insightful.

As we have seen already, the unconditional run-length distribution for the Shewhart-Prec chart is not geometric. In Equation 4.28, we provide the formula for the run-length pmf. This can be evaluated to answer questions like those listed above. Note, however, that in order to calculate the run-length distribution in cases other than the IC situation, for example, under a shift, we need to specify the $GF^{-1}(u_{(b:m)})$ function. An example is now provided.

Two (very) popular characteristics of the run-length distribution are the average run-length or the mean of the run-length distribution and the probability of a signal or, equivalently, the probability of no signal. Thus, we concentrate on these characteristics as measures of the performance of the precedence control chart. It should be noted that a practical consideration for the implementation of the control chart is the fact that, for both the two-sided and the one-sided control charts, certain conditions on the chart constants a and b, as well as on m, n, and j need to be satisfied in order for the ARL_{IC} to be **finite**. These conditions are given in Chakraborti, van der Laan, and Van de Wiel (2004) as:

1. For the two-sided control chart:

$$ARL_{\text{IC}} < \infty \Leftrightarrow (a-j)(n-j+1) + j(m-b+1) > 0;$$

and

2. For the one-sided control chart:

$$ARL_{\text{IC}} < \infty \Leftrightarrow (m-b) - (n-j) > 0.$$

For example, if we intend to construct a two-sided Med chart with test samples of size $n = 5$ using $m = 50$ reference observations and with a specified FAR of 0.0027, Table 4.18 yields values of $a = 1$ and $b = 50$ with the exact FAR equal to 0.00076. However, for these values we find that $(a-j)(n-j+1) + j(m-b+1) = -3$, which is less than zero. and therefore, we find from Table 1 of Chakraborti, van der Laan, and Van de Wiel (2004) that $ARL_{\text{IC}} = \infty$.

While studying the performance of any control chart, one should first study its robustness properties under possible violations of the assumptions under which the chart is designed. To this end, the IC robustness of a control chart is most important, as without this, the shift detection properties of the chart, no matter how attractive, would be questionable. One can draw an analogy of the size and the power of a test of hypothesis. If a test does not maintain its size, that is, it's not Type-I error robust to the underlying assumptions, its power properties are of little value. The distribution-free precedence control charts are, by definition, IC robust in that

their IC properties remain the same for all continuous distributions. We now take a closer look at the Med chart from this point of view.

4.4.1.1.9 The In-control Robustness of the Two-sided Med Chart As pointed out earlier, one of the most appealing properties of distribution-free control charts is the stability of the IC run-length distribution. By contrast, recall that the IC average run-length of the parametric control charts, such as the standard parametric Shewhart $\overline{X}$ control chart, the CUSUM, and the EWMA control charts, can be (sometimes greatly) affected by changes in the underlying process distribution. In fact, IC performance characteristics of standard parametric charts can be significantly affected when there are changes from the parametric distribution (such as normal) under which the chart was designed.

Table 4.19, for example, displays the IC ARL values of the two-sided Med control chart. For comparison purposes, we also include the corresponding results for the two-sided (parametric) CUSUM and the EWMA control charts – all based on the normal distribution with mean μ and variance σ^2. These results are from Chakraborti, van der Laan, and Van de Wiel (2004). The CUSUM chart constants, h and k, as well as the EWMA control chart constants, λ and L, were chosen such that they are either good at detecting a small shift, that is, $\frac{1}{2}\sigma$ in this case, or a large process shift, that is, 2σ. The reader is referred to Sections 3.2.2 and 3.2.3 for a detailed discussion on the choice of the design parameters (h, k) and (λ, L) for the CUSUM and EWMA charts, respectively.

In the robustness study, Chakraborti, van der Laan, and Van de Wiel (2004) included distributions of various shapes: normal, non-normal, but normal like with heavier and lighter tails and skewed distributions. For the non-normal but normal like distributions, the Student's t-distribution and the Laplace distribution (also known as the double exponential distribution) were used. These two distributions represent the effect of **heavy** tails (heavier than the normal) on the ARL_{IC}, whereas the uniform distribution was used to study the effect of **light** tails (lighter than the normal). In addition, to study the effect that **skewness** can have on the ARL_{IC}, two different gamma distributions were included. Finally, to study the effects of a **change in the process variance**, the average run-lengths for the $N(0,1)$ distribution were compared to those for the $N(0,1.1)$ distribution.

The results from Table 4.19 are instructive. We observe that the ARL_{IC} of the CUSUM chart is particularly sensitive to departures from the assumption of normality and for a change in the variance of the process, especially when the CUSUM chart is designed specifically to detect a large shift in the location of the process. Conversely, the EWMA chart is relatively robust against departures from normality, especially when the weighting constant λ is small. However, the EWMA chart is not particularly

Table 4.19 The ARL_{IC} values for several distributions.

	Med	CUSUM		EWMA	
	$m = 1000$	*Large shift*	*Small shift*	*Large shift*	*Small shift*
	$n = 5$, j $= 3$	$h = 4.61$	$h = 0.86$	$L = 2.81$	$L = 3.05$
	$P_0 = 0.0078$	$k = 0.56$	$k = 2.24$	$\lambda = 0.1$	$\lambda = 0.4$
	$a = 48$,	$n = 5$	$n = 5$	$n = 5$	$n = 5$
Distribution	$b = 953$				
$N(0,1)$	501.89	500.00	500.00	500.00	500.00
Laplace(0,1)	501.89	390.64	196.18	418.77	289.82
Uniform(0,1)	501.89	549.16	1820.13	494.73	703.72
$N(0,1.1)$	501.89	206.38	196.69	214.49	199.79
Gamma(1,1)	501.89	310.74	121.53	404.27	207.62
Gamma(4,1)	501.89	462.43	345.68	472.33	370.20
$t(4)$	501.89	266.86	131.90	380.98	190.39
$t(40)$	501.89	456.76	413.44	470.87	380.63

robust for a change in the variance of the process distribution. However, in all the cases, the ARL_{IC} of the Shewhart-Prec chart remains constant at 501.89 for a given nominal ARL_{IC} value of 500. More recently, several studies have looked at the question of IC robustness of many parametric control charts and concluded that the distributional assumption can be crucial in the implementation of these charts, in that, for example, many more false alarms are observed under these charts when the distributional assumption is violated. This is not consistent with the advertised *FAR* and can perhaps ruin the effectiveness of much of the SPC scheme. As a result, there is a renewed interest in distribution-free control charts. We next examine the OOC performance comparisons.

4.4.1.1.10 The Out-of-control Average Run-length of the Two-sided Med Chart For comparison of the OOC average run-length of the two-sided precedence control chart with that of the standard parametric Shewhart $\overline{X}$ chart, the CUSUM chart and the EWMA chart, Tables 4.20a–4.20c present the average run-lengths for the $N(0,1)$, the *Gamma*(1,1) as well as the $t(4)$ distributions. These results are again taken from Chakraborti, van der Laan, and Van de Wiel (2004). From Tables 4.20a and 4.20b, we see that the CUSUM and the EWMA charts are better than the precedence control chart in detecting (especially) small shifts in a normal distribution and a *Gamma*(1,1) distribution. However, for the normal distribution the differences between the precedence control chart and the parametric CUSUM and EWMA charts are not very large

for shifts larger than 1.5σ, say. Nonetheless, this shouldn't be surprising considering that the Shewhart-Prec chart is a Shewhart control chart.

From Table 4.20c, we see that for the $t(4)$ distribution, that is, for a symmetric distribution with heavier tails than the standard normal distribution, small shifts are better detected by the CUSUM chart (with a small value for the chart constant k), as well as the EWMA chart (with a small value for the weighting constant λ), but larger shifts are once again detected sooner by the precedence control chart. Note however, that a reference sample of 1000 observations is used for these comparisons. It is generally true that nonparametric methods require more data but of course they do not require the assumption of a parametric distribution for the model. The choice comes down to particular applications, available resources and priorities.

Although much of our discussion has focused on two-sided control charts following the traditional practice, we note that one-sided control charts are sometimes needed and they are useful in situations where only an upward shift or only a downward shift in a particular process parameter is of interest. We might, for example, be monitoring the breaking strength of a cable used for supporting a heavy weight, or we might be monitoring the (remaining or future) lifetime of a light bulb. Many of the two-sided control charts, both parametric and nonparametric, can be modified to fit this situation. However, the precedence charts seem to be particularly suitable as they are based on ordered values. We touch on this briefly in the next section.

Table 4.20a The average run-length values for the $N(0,1)$ distribution.

	Med	CUSUM		EWMA	
	$m = 1000$	Large	Small	Large	Small
	$n = 5, j = 3$	$h = 4.61$	$h = 0.86$	$L = 2.81$	$L = 3.05$
Shift	$FAR = 0.0078$	$k = 0.56$	$k = 2.24$	$\lambda = 0.1$	$\lambda = 0.4$
σ	$a = 48, b = 953$	$n = 5$	$n = 5$	$n = 5$	$n = 5$
0.00	501.89	500.00	500.00	500.00	500.00
0.25	240.93	33.35	162.72	25.75	56.05
0.50	71.70	8.78	36.09	8.86	11.00
0.75	24.22	4.86	10.60	5.33	4.76
1.00	9.79	3.40	4.21	3.87	2.97
1.50	2.70	2.22	1.56	2.57	1.77
2.00	1.37	1.77	1.09	2.04	1.25
2.50	1.07	1.34	1.01	1.81	1.04
3.00	1.01	1.06	1.00	1.40	1.00
4.00	1.00	1.00	1.00	1.01	1.00

Table 4.20b The average run-length values for the *Gamma*(1,1) distribution.

	Med	CUSUM		EWMA	
	$m = 1000$	Large	Small	Large	Small
	$n = 5, j = 3$	$h = 5.18$	$h = 1.75$	$L = 2.81$	$L = 3.05$
Shift	$FAR = 0.0078$	$k = 0.56$	$k = 2.24$	$\lambda = 0.1$	$\lambda = 0.4$
σ	$a = 48, b = 953$	$n = 5$	$n = 5$	$n = 5$	$n = 5$
0.00	501.89	496.43	498.22	509.76	505.68
0.25	439.01	41.40	182.85	25.18	80.83
0.50	256.45	10.28	96.94	7.66	18.73
0.75	125.31	5.44	24.50	3.91	9.98
1.00	61.87	3.75	9.05	2.47	3.70
1.50	15.81	2.43	2.32	1.38	1.76
2.00	4.83	1.90	1.36	1.03	1.20
2.50	1.63	1.61	1.02	1.00	1.00
3.00	1.02	1.15	1.00	1.00	1.00
4.00	1.00	1.00	1.00	1.00	1.00

Table 4.20c The average run-length values for the $t(4)$ distribution.

	Med	CUSUM		EWMA	
	$m = 1000$	Large	Small	Large	Small
	$n = 5, j = 3$	$h = 5.38$	$h = 1.81$	$L = 3.04$	$L = 3.65$
Shift	$FAR = 0.0078$	$k = 0.56$	$k = 2.24$	$\lambda = 0.1$	$\lambda = 0.4$
σ	$a = 48, b = 953$	$n = 5$	$n = 5$	$n = 5$	$n = 5$
0.00	501.89	495.23	499.47	507.91	504.44
0.25	300.09	41.88	340.58	28.28	128.87
0.50	32.70	10.24	136.21	7.86	22.45
0.75	11.16	5.54	35.50	3.98	7.11
1.00	2.26	3.86	9.84	2.47	3.77
1.50	1.16	2.49	2.31	1.38	1.20
2.00	1.02	1.98	1.34	1.06	1.03
2.50	1.00	1.63	1.06	1.01	1.00
3.00	1.00	1.20	1.01	1.00	1.00
4.00	1.00	1.01	1.00	1.00	1.00

Note 4.5 One-sided Shewhart-Prec Control Charts
In the first scenario, if, for example, the breaking strength of a cable decreases with a weight put on it, the cable might just snap as it might not be able to carry or hold the weight and, subsequently, this may become a hazard. However, in contrast, if the breaking strength of the cable increases, that is, shifts upward, it will be a benefit since then the cable will be able to support a heavier weight than originally required. Similarly, if the average lifetime of a certain kind of light bulb is less than the advertised average lifetime, more customers might decide not to buy the particular brand and, consequently, this might lead to a loss in revenue for the manufacturer. Therefore, it would be wise to guard against a downward shift in the average lifetime of a light bulb. In both of these scenarios, we are interested only in detecting a downward shift in a process parameter of interest, in this case the location parameter, and therefore, a lower one-sided control chart is desired. Alternatively, in a situation where only an upward shift might be of any real importance, an upper one-sided chart will be needed. The reader is referred to Section 3.2.1 for a discussion on when a lower one-sided, an upper one-sided, or a two-sided chart is appropriate. The problem of when a one- or two-sided control chart is needed is similar to the problem of choosing between a one- or two-sided confidence interval.

The adjustments required to go from a two-sided precedence control chart to a one-sided chart are relatively simple. For example, if detecting only an upward shift is of interest, that is, whether the location parameter has increased (or shifted to the right), we only use an $U\hat{C}L = X_{(b:m)}$ as an estimated UCL with no LCL. However, if only detecting a downward shift is of interest, that is, whether the location parameter has decreased (or shifted to the left), we only use $L\hat{C}L = X_{(a:m)}$ as an estimated LCL with no UCL. We invite the reader to complete the discussion about one-sided charts, including the charting constants and deriving chart properties by following the provided details for the two-sided charts. Some further details can be found in Chakraborti, van der Laan, and Van de Wiel (2004).

4.4.1.2 Shewhart Control Charts Based on the Mann–Whitney Test Statistic

While the precedence tests are well-known distribution-free tests and useful in some situations, the Mann–Whitney (denoted MW; see Mann and Whitney (1947)) test is another, perhaps more popular, two-sample distribution-free test used in practice as an alternative to the classical two-sample t-test. It is known that the MW test (see Gibbons and Chakraborti, 2010, p. 261) is more efficient than the precedence test for light-tailed distributions. There is a large amount of literature on the MW test. We describe the MW statistic first and then discuss a Shewhart chart based on the MW statistic.

4.4.1.2.1 The Mann–Whitney Test Statistic A Phase II distribution-free Shewhart chart for location based on the MW statistics was proposed by Chakraborti and Van de Wiel (2008). For the setting, as in the case of the precedence test, consider two continuous distributions (the X and the Y distributions) with cdfs F and G, respectively, where F represents the reference distribution cdf and G the cdf of the distribution from which the test sample is drawn. A random sample of size m, that is, $X_1, X_2, \ldots, X_m$, is drawn from F and an independent random sample of size n, that is, $Y_1, Y_2, \ldots, Y_n$, is drawn from G. The MW test is based on the relative position of the X's and the Y's in the combined sample. To this end, define the mn indicator random variables as

$$D_{ij} = \begin{cases} 1 & \text{if } X_i > Y_j \\ 0 & \text{if } X_i < Y_j \end{cases} \text{ for } i = 1, 2, \ldots, m \text{ and } j = 1, 2, \ldots, n.$$

The MW test is based on the U statistic, which is defined as the number of times a Y observation precedes an X observation in the combined ordered arrangement of the X's and the Y's. This means that each X observation is compared with each of the Y observations in a pairwise manner and whenever the X is greater than the Y, we score that comparison as 1; otherwise, we score it as a 0. The MW statistic counts the total number of 1's in these mn comparisons and can be written as $M_{YX} = \sum_{i=1}^{m} \sum_{j=1}^{n} D_{ij}$. Thus M_{XY} represents the number of pairs (X, Y) where the X observation is bigger than the Y. We start with an example of how to calculate the MW statistic.

Example 4.16 Calculation of the MW Statistic

Suppose $m = 3$ and $n = 3$. After ranking the set of $m + n$ observations jointly from the smallest to the biggest, suppose, we obtain the following configuration: $x_1 y_1 y_2 x_2 x_3 y_3$. Based on this, we calculate the U-test of MW as follows. First, note that

$$\begin{array}{lll} D_{11} = 0 \text{ since } X_1 < Y_1 & D_{21} = 1 \text{ since } X_2 > Y_1 & D_{31} = 1 \text{ since } X_3 > Y_1 \\ D_{12} = 0 \text{ since } X_1 < Y_2 & D_{22} = 1 \text{ since } X_2 > Y_2 & D_{32} = 1 \text{ since } X_3 > Y_2 \\ D_{13} = 0 \text{ since } X_1 < Y_3 & D_{23} = 0 \text{ since } X_2 < Y_3 & D_{33} = 0 \text{ since } X_3 < Y_3 \end{array}$$

Thus $M_{YX} = \sum_{i=1}^{m} \sum_{j=1}^{n} D_{ij} = 4$.

4.4.1.2.2 The Shewhart Chart Based on the Mann–Whitney Statistic For the Shewhart chart based on the MW statistic (hereafter denoted Shewhart-MW chart) of Chakraborti and Van de Wiel (2008), let $X_1, X_2, \ldots, X_m$ represent the Phase I reference sample from an IC process and let $Y_1, Y_2, \ldots, Y_n$ denote a Phase II (test) sample of size n. The charting statistic is the total number of (X, Y) pairs where the Y

observation is greater than the X observation (or the X precedes the Y), $M_{XY} = \sum_{i=1}^{m} \sum_{j=1}^{n} I(Y_j > X_i)$, where $I(\cdot)$ denotes the usual indicator function. The fact that we are counting Y's greater than the X's (and not the other way around, as in the earlier description) is done merely for convenience. Note that M_{XY} takes on values between 0 and mn, and larger (smaller) values of M_{XY} indicate a positive (negative) shift in the location. The smallest value of 0 is attained when all the Y's are less than the X's and the largest value of mn is attained when all the Y's are greater than the X's. If the Y's are stochastically larger (smaller) than the X's, the M_{XY} will be significantly large (small) and thus the Shewhart-MW chart signals an OOC situation when $M_{XY} > b_{mn}$ or $M_{XY} < a_{mn}$, where b_{mn} and a_{mn} are the UCL and the LCL, respectively. In the location model, the Y's are stochastically larger (smaller) than the X's according to the corresponding location parameter $\theta_i < \theta_1$ (or $\theta_i > \theta_1$). In other words, when the Y-median has decreased (or increased) from that of the X-median, the process is OOC.

While the MW test is a well-known distribution-free test, a more familiar two-sample distribution-free test for location is the Wilcoxon rank-sum (WRS) test proposed by Wilcoxon (1945), published earlier than the paper by Mann and Whitney (1947). The WRS statistic (W_{XY}) is one member of a class of what are called "linear rank statistics" (see Gibbons and Chakraborti, 2010, page 275). The WRS statistic is the sum of the ranks of the Y's in the combined sample of the X's and the Y's. It can be shown that $W_{XY} = M_{XY} + m(m+1)/2$ (see, for example, Gibbons and Chakraborti, 2010, page 293) so that the two tests are equivalent.

Thus, the two-sided Shewhart-MW chart of Chakraborti and Van de Wiel (2008) can also be described in terms of the WRS statistic W_{XY}. To this end, note that, if there are no ties, the mean and variance of W_{XY} in the IC case can be found (see Gibbons and Chakraborti, 2010, p. 291, for example) as

$$E(W_{XY}) = \frac{m(m+n+1)}{2} \quad \text{and} \quad VAR(W_{XY}) = \frac{mn(m+n+1)}{12}. \quad (4.45)$$

Hence, the Shewhart-Wilcoxon chart gives an OOC signal when

$$W_{XY} > b_{mn} + m(m+1)/2 \quad \text{or} \quad W_{XY} < a_{mn} + \frac{m(m+1)}{2}.$$

We illustrate the calculation of the MW and the WRS statistics for a small data set.

Example 4.17 Calculation of the MW and the WRS Statistics

Using the same data as in the previous example, suppose $m = 3$ and $n = 3$ and there are no ties. After ranking the set of $m + n = 6$ observations jointly

from smallest to largest, recall that we obtain the following ordered combined sample: $x_1y_1y_2x_2x_3y_3$. For these data, it is easy to see that $M_{XY} = \sum_{i=1}^{m}\sum_{j=1}^{n} I(Y_j > X_i) = 5$. Note that we found $M_{YX} = 4$ earlier and $M_{YX} + M_{XY} = mn = 9$. Hence, $W_{XY} = \sum_{i=1}^{m}\sum_{j=1}^{n} I(Y_j > X_i) + \dfrac{m(m+1)}{2} = 5 + 6 = 11$.

Next, we study the run-length distribution of the Shewhart-MW chart.

4.4.1.2.3 Methods of Calculating the Run-length Distribution Following the methodology of Chakraborti and Van de Wiel (2008), let $p(\underline{x})$ denote the probability of a signal with any test (Phase II) sample, given the reference sample $\underline{X} = (X_1, X_2, \ldots, X_m) = (x_1, x_2, \ldots, x_m) = \underline{x}$. Therefore, $p(\underline{x})$ is a function of the cdf G, written as $p_G(\underline{x})$. The conditional run-length distribution is geometric, given by

$$P_C(N = k|\underline{X} = \underline{x}) = (1 - p_G(\underline{x}))^{k-1}(p_G(\underline{x})), \text{ for} = 1, 2, \ldots \tag{4.46}$$

Consequently, the conditional ARL is given by

$$ARL_{\mathrm{C}} = E(N|\underline{X} = \underline{x}) = \frac{1}{p_G(\underline{x})}. \tag{4.47}$$

It follows that the unconditional ARL is given by

$$ARL_{\mathrm{U}} = \int_{-\infty}^{\infty} \ldots \int_{-\infty}^{\infty} \frac{1}{p_G(\underline{x})}\, dF(x_1) \ldots dF(x_m). \tag{4.48}$$

4.4.1.2.4 The Unconditional In-control ARL of the Shewhart-MW Chart The unconditional IC ARL is found by substituting $G = F$ into Equation 4.48. Thus

$$ARL_{\mathrm{U,IC}} = \int_{-\infty}^{\infty} \ldots \int_{-\infty}^{\infty} \frac{1}{p_F(\underline{x})}\, dF(x_1) \ldots dF(x_m). \tag{4.49}$$

Where $p(\underline{x})$ is now a function of F since $G = F$. Using the PIT, Equation 4.49 can be written as

$$ARL_{\mathrm{U,IC}} = \int_{0}^{1} \ldots \int_{0}^{1} \frac{1}{p_U(\underline{u})}\, du_1 \ldots du_m \tag{4.50}$$

where $u_i = F(x_i)$, $i = 1, 2, \ldots, m$, $\underline{u} = (u_1, u_2, \ldots, u_m)$ and the subscript U on p_U denotes the fact that the probability of a signal $p(\underline{x})$ is calculated under the common distribution $F = G$, based on the u_i's, which are realizations from the $U(0,1)$ distribution.

4.4.1.2.5 The Out-of-control ARL In the OOC case, the unconditional OOC ARL is given by

$$ARL_{\mathrm{U,OOC}} = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \frac{1}{p_G(\underline{x})}\, dF(x_2) \ldots dF(x_m) \tag{4.51}$$

where $\delta = \theta_i - \theta_1$ represents the difference (shift) between the location parameters of G and F. Thus, we can find both $ARL_{\mathrm{U,IC}}$ and $ARL_{\mathrm{U,OOC}}$ using the CUC method.

We need to calculate the $ARL_{\mathrm{U,IC}}$ and the $ARL_{\mathrm{U,OOC}}$ respectively, in order to implement and evaluate the performance of the Shewhart-MW chart. To this end, note that we do not have an exact compact formula for the ARL. Unlike for the Shewhart-Prec chart, the ARL expressions shown above are m-dimensional integrals. Accordingly, it is difficult and time-consuming to calculate the unconditional ARL analytically using these formulas. As a practical alternative, and, as mentioned before on several occasions, we again recommend using simulations to estimate and evaluate the ARL or any other run-length distribution characteristic. The general steps for simulating the run-length characteristics for any control chart are described below. Note that these steps apply for any chart, parametric or nonparametric, the only difference is that the specific charting statistic at hand and its calculation will change.

The Simulation Approach The following simulation algorithm can be used to obtain the run-length characteristics for the Shewhart-MW chart. The corresponding SAS program is given in Appendix B, SAS program number 8, and the output is shown below in Figure 4.24.

Step 1: Specify the Phase I reference sample size m (≥ 1), the Phase II test sample size n (≥ 1), the desired nominal ARL_{IC}, and the size of the shift (δ) to be detected. Next, specify the two distributions for generating the Phase I and Phase II samples, respectively. For the IC case, the two distributions are identical (we say that the shift equals zero ($\delta = 0$)). For the OOC case, the distribution for the Phase II sample is of the same form as that for the Phase I sample, but with a shift in the location in units of the population standard deviation. The Phase I sample, for example, may be drawn from a normal distribution with mean 0 and standard deviation 1, whereas in the OOC case, the Phase II samples may be drawn from a normal distribution with some mean δ ($\neq 0$) and standard deviation 1 (we say that the mean shift is δ ($\neq 0$)). Any continuous distribution, whether symmetric or asymmetric, may be used for the two underlying distributions.

Step 2: Calculate the control limits, that is, the charting constant b_{mn} as a_{mn} can be obtained from $a_{mn} = mn - b_{mn}$. The input factors are m, n,

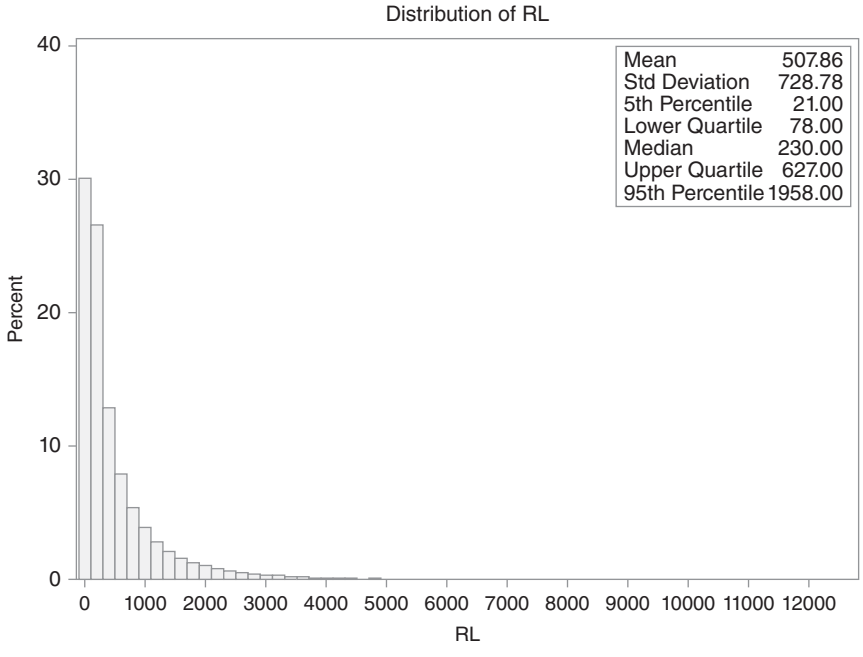


FIGURE 4.24 *SAS output where the design parameters* ($m = 500$, $n = 5$, $b_{mn} = 2172$) *of the Shewhart-MW chart were chosen for a nominal* $\text{ARL}_{IC} = 500$.

and the nominal ARL_{IC}. Typically, a search algorithm (using a software package such as SAS or R) is used to obtain the charting constant b_{mn}.

Step 3: Calculate the MW (M_{XY}) charting statistic for the Phase II sample and the reference sample generated in Step 1.

Step 4: If, on one hand, the charting statistic M_{XY} plots on or outside the control limits, an observation of 1 is recorded as an observation from the run-length distribution. If, on the other hand, the charting statistic M_{XY} plots inside the control limits, Steps 1 and 2 are repeated with the same reference sample but with a newly generated Phase II sample each time, until the charting statistic M_{XY} plots on or outside the control limits for the first time so that a signal is observed. At that point, the number of Phase II samples that were needed to produce that first signal is recorded as an observation from the run-length distribution.

Step 5: Repeat Steps 1 to 4 a total of 100 000 times (for instance). This creates 100 000 observations from the run-length distribution of the MW (or any other Phase II) control chart.

Step 6: The observations obtained in Step 5 can be analyzed to calculate estimates of the run-length distribution characteristics such as the average, the median, the standard deviation, etc.

A software package such as SAS or R can be used to program the steps of the simulation. Note that, when we use $\delta = 0$, we get the IC run-length characteristics, whereas for $\delta \neq 0$ we get the OOC run-length characteristics.

4.4.2 CUSUM Control Charts

4.4.2.1 CUSUM Control Charts Based on the Exceedance Statistic

While the Shewhart-Prec chart is a Shewhart chart and is therefore well known to be effective in detecting larger and more abrupt shifts, charts that are more sensitive to smaller and more sustained types of shifts may be desirable in some situations. As discussed in Chapter 3, there are two classes of such charts considered in the parametric control charting literature, namely, the CUSUM and the EWMA charts. It is thus natural to consider nonparametric analogues of these charts in Case U, and we consider two such charts here. Although it is possible to consider a nonparametric CUSUM chart based on the precedence statistic, for convenience we consider using some related statistics, called the exceedance statistics following the literature. The exceedance statistic is the number of observations from one of the two samples (Phase I sample and the test sample) that exceed the rth order statistic of the other sample. Thus, the precedence and exceedance statistics are linearly related. While tests based on precedence statistics are called precedence tests, the tests based on exceedance statistics are called exceedance tests. The latter have been found to be useful in a number of applications, including quality control and reliability studies with lifetime data, where the exceedance probability can be associated with the so-called "warranty time" of a product (see, for example, Chakraborti and van der Laan, 2000).

As before, assume that a Phase I reference sample $X_1, X_2, \ldots, X_m$ $(m > 1)$ is available from an IC process with a cdf $F(x)$. Let $Y_{j1}, Y_{j2}, \ldots, Y_{jn}$, $(n \geq 1)$ $j = 1, 2, \ldots,$ denote the jth test (Phase II) sample of size n from a cdf $G(y)$. Both F and G are unknown continuous distribution functions and the process is IC when $F = G$. To detect a change in the location, we assume that the cdfs $F(t) = H(t - \theta_1)$ and $G(t) = H(t - \theta_2)$, respectively, where H is some unknown continuous cdf and where θ_1 and θ_2 represent the respective location parameters. Let $U_{j,r}$ denote the number of Y observations in the jth Phase II sample that exceed $X_{(r)}$, the rth ordered observation in the Phase I reference sample. The statistic $U_{j,r}$ is called an exceedance statistic and the probability

$p_r = P(Y > X_{(r)} | X_{(r)})$ is called an exceedance probability. Note that this exceedance probability is a conditional probability of a Y observation exceeding $X_{(r)}$, for a given value of $X_{(r)}$.

A CUSUM chart based on the exceedance statistic (denoted CUSUM-EX) is considered next. This discussion is derived from the work of Mukherjee, Graham, and Chakraborti (2013), which can be consulted for more details.

An **upper one-sided** CUSUM-EX chart is defined in terms of the charting statistic

$$C_j^+ = \max[0, C_{j-1}^+ + (U_{j,r} - nd) - k] \text{ for } j = 1, 2, 3 \ldots \tag{4.52}$$

with the starting value $C_0^+ = 0$. This signals at the first j for which $C_j^+ \geq h$.

Similarly, for a **lower one-sided** CUSUM-EX chart, the charting statistic is given by

$$C_j^- = \min[0, C_{j-1}^- + (U_{j,r} - nd) + k] \text{ for } \mathrm{j} = 1, 2, 3 \ldots \tag{4.53}$$

with the starting value $C_0^- = 0$. This signals at the first j for which $C_j^- \leq -h$.

The key idea behind the proposed charts is that given $X_{(r)} = x_{(r)}$, each $U_{j,r}$ follows a binomial distribution with parameters (n, p_r). Hence, conditionally on $X_{(r)}$, and following the idea of existing CUSUM charts for a binomial probability, a binomial-type CUSUM chart can be constructed based on the $U_{j,r}$'s to monitor the process location (via the exceedance probabilities). Now, from the properties of the binomial distribution, $E(U_{j,r}|X_{(r)}) = np_r$ for any $j = 1,2,3,\ldots$ Since the conditional probability p_r is unknown and depends on the underlying distributions, it has been suggested in the literature that p_r be replaced by its unconditional IC value given by $P(Y > X_{(r)}) = \dfrac{(m - r + 1)}{(m + 1)} = d$.

The corresponding **two-sided** CUSUM-EX chart signals at the first j for which either of the one-sided charts signal, that is, whenever $C_j^+ \geq h$ or $C_j^- \leq -h$. Note that, here, the decision interval $h > 0$ and the reference value $k > 0$ are the parameters of the control chart procedure.

If $X_{(r)}$ is taken to be the median of the reference sample, we find that $d = 0.5$. By substituting $d = 0.5$ into Equations 4.52 and 4.53, the **upper one-sided** CUSUM-EX chart, based on the reference sample median, is given by the charting statistic

$$C_j^+ = \max[0, C_{j-1}^+ + (U_{j,r} - n/2) - k] \text{ for } j = 1, 2, 3 \ldots \tag{4.54}$$

and the **lower one-sided** CUSUM-EX chart, based on the reference sample median, is given by the charting statistic

$$C_j^- = \min[0, C_{j-1}^- + (U_{j,r} - n/2) + k] \text{ for } j = 1, 2, 3 \ldots \tag{4.55}$$

Again, the corresponding **two-sided symmetric** CUSUM-EX of control chart signals as the first j for which either of the one-sided charts signal, that is, whenever $C_j^+ \geq h$ or $C_j^- \leq -h$. The CUSUM-EX chart that is based on the reference sample median will be referred to as the **median CUSUM-EX chart** from this point forward.

Note 4.6 Implementation of the Chart: Choice of Design Parameters

The implementation of the CUSUM-EX chart requires specifying the following quantities: (i) m – the size of the IC Phase I reference sample, (ii) n – the size of each Phase II test sample (the subgroup size), (iii) the desired ARL_{IC}, (iv) k – the reference value, essentially a rational number, and (v) h – the decision interval/UCL depending on m, n, k, and the desired ARL_{IC}. It is up to the experimenter to specify the parameters m, n, and the desired ARL_{IC} in a given situation. The choice of the IC Phase I reference sample size can be profound and is discussed in detail in Appendix B of Mukherjee, Graham, and Chakraborti (2013). The design parameters (k, h) are chosen so that the chart has a specified nominal ARL_{IC} and is capable of detecting a shift, specially a small shift, as soon as possible. The first step is to choose k. In Section 3.2.2, we considered the parametric CUSUM chart with individual data ($n = 1$) with no reference sample. We examined the impact of k (the reader is referred to Figure 2.5 and the discussion on it) and concluded that, when there is little or no a priori information regarding the size of the shift, a smaller value of k is the safest choice (to protect against any unnecessary delays in detection). Mukherjee, Graham, and Chakraborti (2013) draw similar conclusions for the CUSUM-EX chart. The next step is to choose h in conjunction with the chosen k so that a desired nominal ARL_{IC} is attained. An example is shown below.

4.4.2.1.1 Methods of Calculating the Run-length Distribution

Analytical expressions for the run-length distribution of the CUSUM-EX chart have been obtained by Mukherjee, Graham, and Chakraborti (2013). However, it is more convenient to use simulations, as outlined below.

The Simulation Approach The following simulation algorithm can be used to obtain the run-length characteristics for the CUSUM-EX chart. The corresponding SAS program is given in Appendix B, SAS program number 9, and the output is shown below in Figure 4.25.

Step 1: Specify the Phase I reference sample size m (≥ 1), the Phase II test sample size n (≥ 1), the desired nominal ARL_{IC}, and the size of the shift (δ) to be detected. Next, specify the two distributions for generating the Phase I and Phase II samples, respectively. For the IC case, the two distributions are identical (we say that the shift equals zero ($\delta = 0$)). For

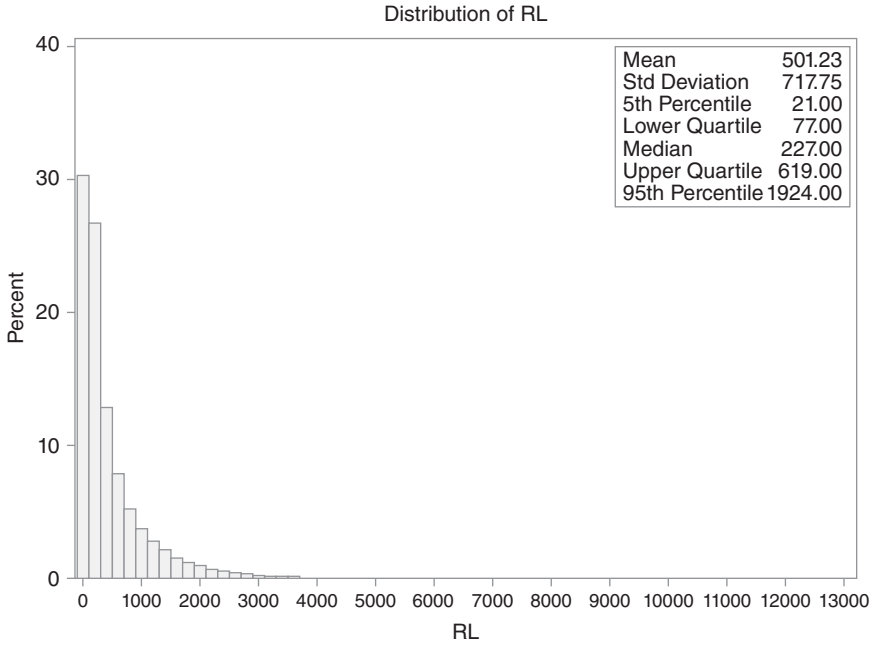


FIGURE 4.25 *SAS output where the design parameters* ($m = 125$, $n = 5$, $j = 3$, $h = 9.775$, $k = n(d^* - d)$) *of the CUSUM-EX chart were chosen for a nominal* $\text{ARL}_{IC} = 500$.

the OOC case, the distribution for the Phase II sample is of the same form as that for the Phase I sample, but with a shift in the location in units of the population standard deviation. The Phase I sample, for example, may be drawn from a normal distribution with mean 0 and standard deviation 1, whereas in the OOC case, the Phase II sample may be drawn from a normal distribution with some mean $\delta(\neq 0)$ and standard deviation 1 (we say that the mean shift is δ $(\neq 0)$). Any continuous distribution, whether symmetric or asymmetric, may be used.

Step 2: Specify the two design parameters, k and h (the reader is referred to Section 3.2.2 for guidance on choosing the design parameters).

Step 3: Calculate the $U_{j,r}$ statistic and the corresponding C_j^- and C_j^+ charting statistics for each subgroup.

Step 4: Compare the charting statistic to the decision interval h.

Step 5: The number of subgroups needed until C_j^- plots on or below $-h$ or until C_j^+ plots on or above h for the first time is recorded as an observation from the run-length distribution.

Step 6: Repeat Steps 1 to 5 a total of 100 000 times.

Step 7: The observations obtained in Step 5 can be analyzed to calculate estimates of the run-length distribution characteristics such as the average, the median, the standard deviation, etc.

A software package such as SAS or R can be used to program the simulation steps. Note that when $\delta = 0$ we get the IC run-length characteristics, whereas for $\delta \neq 0$ we get the OOC run-length characteristics.

4.4.2.2 CUSUM Control Charts Based on the Wilcoxon Rank-sum Statistic

As in the case of the precedence/exceedance statistic, it is natural to consider CUSUM and EWMA charts based on the MW or, equivalently, the WRS statistics. Li, Tang, and Ng (2010) did just that, following the work of Chakraborti and Van de Wiel (2008), and because of the equivalence of the WRS statistics and MW statistics indicated earlier, we call these charts the Wilcoxon–Mann–Whitney (WMW) charts hereafter. These charts are expected to be more sensitive than the Shewhart charts for small, sustained shifts in the location. First, consider the CUSUM charts.

Distribution-free CUSUM charts based on the WMW statistic are called CUSUM-rank charts from this point forward. An upper **one-sided** CUSUM-rank chart is defined as

$$C_j^+ = \max\left[0, C_{j-1}^+ + \left(W_j - \frac{n(m+n+1)}{2}\right) - k\right] \quad \text{for } j = 1,2,3\ldots \tag{4.56}$$

with the starting value $C_0^+ = 0$. This chart signals at the first j for which $C_j^+ \geq h$.

Conversely, for a lower **one-sided** CUSUM-rank chart, we define

$$C_j^- = \min\left[0, C_{j-1}^- + \left(W_j - \frac{n(m+n+1)}{2}\right) + k\right] \quad \text{for } j = 1,2,3\ldots \tag{4.57}$$

with the starting value $C_0^- = 0$. This signals at the first j for which $C_j^- \leq -h$.

The corresponding **two-sided** CUSUM-rank chart signals at the first j for which either of the one-sided charts signal, that is, whenever $C_j^+ \geq h$ or $C_j^- \leq -h$. Note that, here, the decision interval $h > 0$ and the reference value $k > 0$ are (as in the past) the parameters of the control chart procedure.

4.4.2.2.1 Methods of Calculating the Run-length Distribution

Expressions for the run-length distribution of the CUSUM-rank chart have been obtained by Li, Tang, and Ng (2010) using the CUC technique. We

provide some derivations in Appendix 4.2. However, it is more convenient to use the simulations, as outlined below.

The Simulation Approach The following simulation algorithm can be used to obtain the run-length characteristics for the CUSUM-rank chart. The corresponding SAS program is given in Appendix B, SAS program number 10, and the output is shown below in Figure 4.26.

Step 1: Specify the Phase I reference sample size m (≥ 1), the Phase II test sample size n (≥ 1), and the size of the shift (δ) to be detected. Next, specify the two distributions for generating the Phase I and Phase II samples, respectively. For the IC case, the two distributions are identical (we say that the shift equals zero ($\delta = 0$)). For the OOC case, the distribution for the Phase II sample is of the same form as that for the Phase I sample, but with a shift in the location in units of the population standard deviation. The Phase I sample, for example, may be drawn from a normal distribution with mean 0 and standard deviation 1, whereas in the OOC case, the Phase II sample may be drawn from a normal distribution with some mean $\delta(\neq 0)$ and standard deviation 1 (we say that the mean shift is δ ($\neq 0$)). Any continuous distribution, whether symmetric or asymmetric, may be used.

Step 2: Specify the two design parameters, k and h (the reader is referred to Section 3.2.2 for guidance on choosing the design parameters).

Step 3: Calculate the W_j statistics and the corresponding C_j^- and C_j^+ charting statistics for each subgroup.

Step 4: Compare the charting statistics to the decision interval h.

Step 5: The number of subgroups needed until C_j^- plots on or below $-h$ or until C_j^+ plots on or above h for the first time is recorded as an observation from the run-length distribution.

Step 6: Repeat Steps 1 to 5 a total of 100 000 times. This creates 100 000 observations from the run-length distribution.

Step 7: The observations obtained in Step 5 can be analyzed to calculate estimates of the run-length distribution characteristics such as the average, the median, the standard deviation, etc.

Again, a software package such as SAS or R can be used to program the steps of the simulations. Note that when $\delta = 0$ we get the IC run-length characteristics, whereas for $\delta \neq 0$ we get the OOC run-length characteristics.

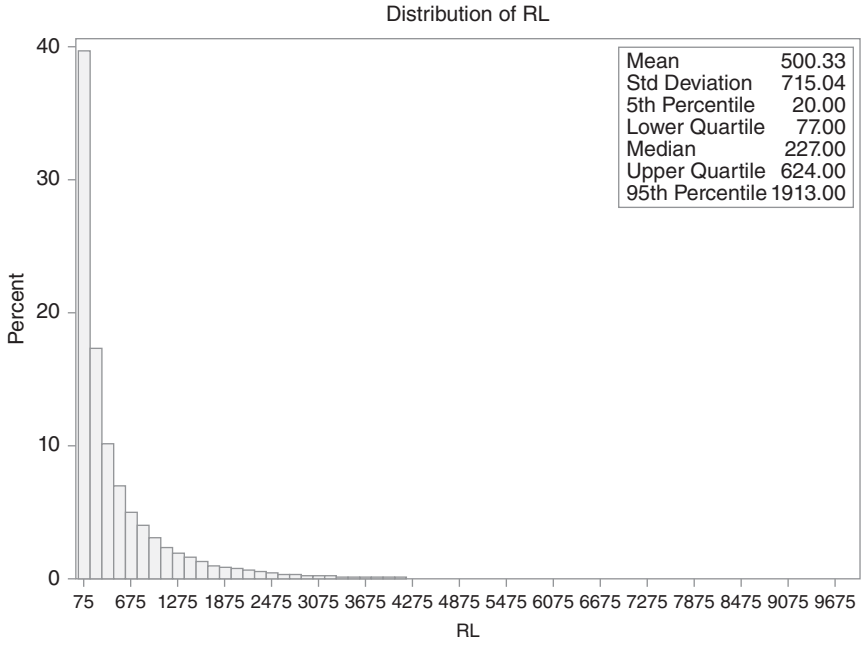


FIGURE 4.26 *SAS output where the design parameters* ($m=100$, $n=5$, $h=353$, $k=0$) *of the CUSUM-rank chart were chosen for a nominal* $\text{ARL}_{IC}=500$.

Next, consider some distribution-free EWMA charts in Case U. This discussion is from Graham, Mukherjee, and Chakraborti (2012).

4.4.3 EWMA Control Charts

4.4.3.1 EWMA Control Charts Based on the Exceedance Statistic

An EWMA control chart procedure based on the exceedance statistic, $U_{j,r}$, accumulates the statistics $U_{1,r}, U_{2,r}, U_{3,r}, \ldots$ with the EWMA charting statistics defined as

$$Z_j = \lambda\, U_{j,r} + (1-\lambda) Z_{j-1} \text{ for } j = 1,2,3, \ldots \tag{4.58}$$

with the weighting constant $0 < \lambda \leq 1$.

The *CL* and the **exact** control limits are given by

$$UCL = n(1-a)(1-(1-\lambda)^j) + L\sqrt{\left(\frac{na(1-a)}{m+2}\right)\left(n(1-(1-\lambda)^j)^2 + \frac{\lambda(m+1)}{2-\lambda}(1-(1-\lambda)^{2j})\right)}$$

$$
\begin{aligned}
CL &= n(1-a)(1-(1-\lambda)^j) \\
LCL &= n(1-a)(1-(1-\lambda)^j) \\
&\quad -L\sqrt{\left(\frac{na(1-a)}{m+2}\right)\left(n(1-(1-\lambda)^j)^2+\frac{\lambda(m+1)}{2-\lambda}(1-(1-\lambda)^{2j})\right)}
\end{aligned} \tag{4.59}
$$

where $a = r/(m+1)$ and the starting value $Z_0 = E(U_{j-k,r}|X_{(r)}) = 0$ when exact control limits are used. See Appendix 4.3.3, Result 4.10, for the detailed derivations of the exact time-varying control limits for the EWMA-EX chart. The **asymptotic** or the **steady-state** control limits, that is, when $j \to \infty$, are given by

$$
\begin{aligned}
UCL &= n(1-a)+L\sqrt{\left(\frac{na(1-a)}{m+2}\right)\left(n+\frac{\lambda(m+1)}{2-\lambda}\right)} \\
CL &= n(1-a) \\
LCL &= n(1-a)-L\sqrt{\left(\frac{na(1-a)}{m+2}\right)\left(n+\frac{\lambda(m+1)}{2-\lambda}\right)}
\end{aligned} \tag{4.60}
$$

where the starting value $Z_0 = E(U_{j-k,r}|X_{(r)}) = np_r$ when steady-state control limits are used. See Appendix 4.3.3, Result 4.11, for the detailed derivations of the steady-state control limits for the EWMA-EX chart. The process is considered IC while all the charting statistics Z_j, $j = 1,2,3,\ldots$ fall between the two control limits, but as soon as a charting statistic falls on or outside the upper or the LCL, the process is declared OOC and typically a search for assignable causes would be started.. The derivation of the exact control limits is given in Appendix 4.3.3, Result 4.10.

4.4.3.1.1 Methods of Calculating the Run-length Distribution

Expressions for the run-length distribution of the EWMA-EX chart have been obtained by Graham, Mukherjee, and Chakraborti (2012), which the interested reader can consult. However, again, it is more convenient to use the simulations, as outlined below.

The Simulation Approach The following simulation algorithm can be used to obtain the run-length characteristics for the EWMA-EX chart. The corresponding SAS program is given in Appendix B, SAS program number 11, and the output is shown below in Figure 4.27.

Step 1: Specify the Phase I reference sample size m (≥ 1), the Phase II test sample size n (≥ 1), the desired nominal ARL_{IC} and the size of the shift (δ) to be detected. Next, specify the two distributions for generating

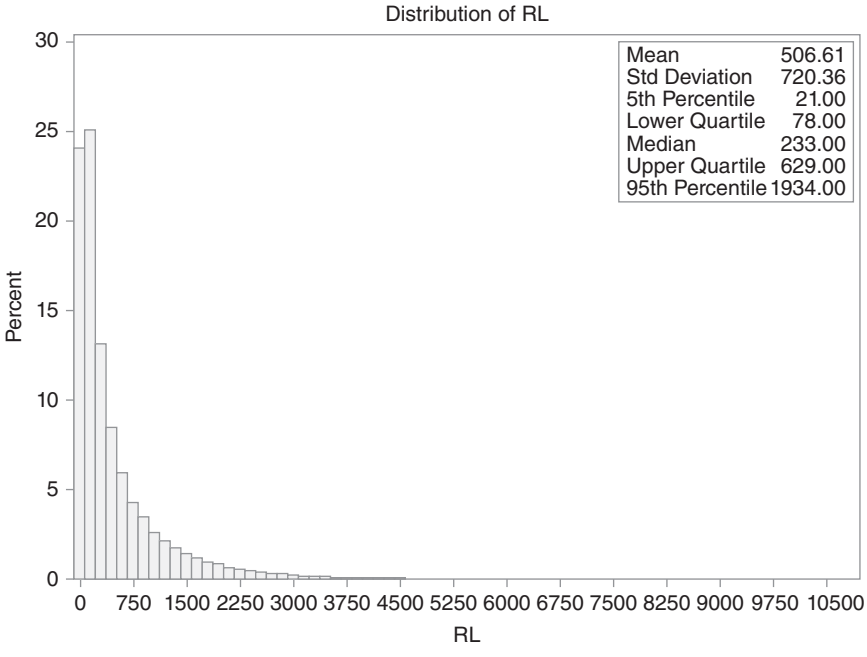


FIGURE 4.27 *SAS output where the design parameters* ($m = 100$, $n = 5$, $\lambda = 0.10$, $L = 2.220$) *of the EWMA-EX chart were chosen for a nominal* $\text{ARL}_{IC} = 500$.

the Phase I and Phase II samples, respectively. For the IC case, the two distributions are identical (we say that the shift equals zero ($\delta = 0$)). For the OOC case, the distribution for the Phase II sample is of the same form as that for the Phase I sample, but with a shift in the location in units of the population standard deviation. The Phase I sample, for example, may be drawn from a normal distribution with mean 0 and standard deviation 1, whereas in the OOC case the Phase II sample may be drawn from a normal distribution with some mean $\delta(\neq 0)$ and standard deviation 1 (we say that the mean shift is δ ($\neq 0$)). Any continuous distribution, whether symmetric or asymmetric, may be used.

Step 2: Specify the two design parameters, λ and L (the reader is referred to Section 3.2.3 for guidance on choosing the design parameters).

Step 3: Calculate the $U_{j,r}$ statistic and the corresponding Z_j charting statistic for each subgroup.

Step 4: Compare the charting statistics to the control limits.

Step 5: The number of subgroups needed until Z_j plots on or outside the control limits for the first time is recorded as an observation from the run-length distribution.

Step 6: Repeat Steps 1 to 5 a total of 100 000 times. This creates 100 000 observations from the run-length distribution.

Step 7: The observations obtained in Step 5 can be analyzed to calculate estimates of the run-length distribution characteristics such as the average, the median, the standard deviation, etc.

As noted before, a software package such as SAS or R can be used to program the simulation study. Note that when $\delta = 0$ we get the IC run-length characteristics, whereas for $\delta \neq 0$ we get the OOC run-length characteristics.

Similar to the CUSUM-rank chart, one can consider an EWMA-rank chart based on the WRS statistic. This is described next.

4.4.3.2 EWMA Control Charts Based on the Wilcoxon Rank-sum Statistic

Li *et al.* (2010) considered an EWMA control chart based on the WRS statistic W_j; we call this the EWMA-rank chart. This chart accumulates the statistics $W_1, W_2, W_3, \ldots$ in accordance with the EWMA formula. The resulting EWMA-rank charting statistic is defined as

$$Z_j = \lambda\, W_j + (1-\lambda) Z_{j-1} \text{ for } j = 1,2,3, \ldots \tag{4.61}$$

where the weighting constant $0 < \lambda \leq 1$.

The center line and the **exact** control limits are given by

$$\begin{aligned} UCL &= \mu_W + L\sqrt{\sigma_W^2 \left(\frac{\lambda}{2-\lambda}(1-(1-\lambda)^{2j})\right)} \\ CL &= \mu_W \\ LCL &= \mu_W - L\sqrt{\sigma_W^2 \left(\frac{\lambda}{2-\lambda}(1-(1-\lambda)^{2j})\right)} \end{aligned} \tag{4.62}$$

where $\mu_W = \dfrac{n(m+n+1)}{2}$ and $\sigma_W^2 = \dfrac{mn(m+n+1)}{12}$ denote the IC mean and variance of W, respectively, and the starting value $Z_0 = \mu_W$ when exact control limits are used. The **asymptotic** or the **steady-state** control limits, that is, when $j \to \infty$, are given by

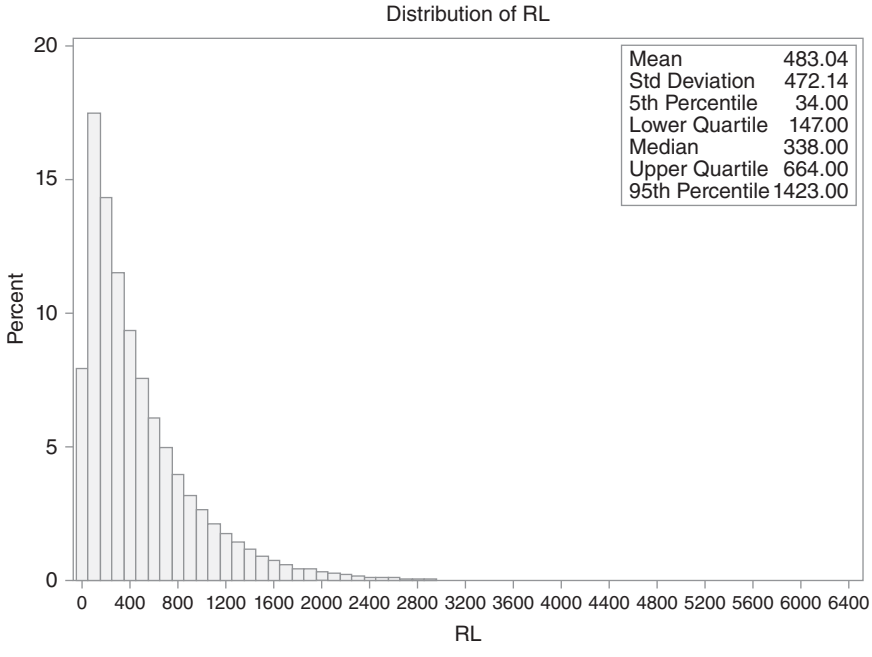


FIGURE 4.28 *SAS output where the design parameters* ($m = 100$, $n = 5$, $\lambda = 0.20$, $LCL = 195.5$, $UCL = 334.5$) *of the EWMA-rank chart were chosen for a nominal* $\text{ARL}_{IC} = 370$.

$$\begin{aligned} UCL &= \mu_W + L\sqrt{\sigma_W^2\left(\frac{\lambda}{2-\lambda}\right)} \\ CL &= \mu_W \\ LCL &= \mu_W - L\sqrt{\sigma_W^2\left(\frac{\lambda}{2-\lambda}\right)} \end{aligned} \tag{4.63}$$

where the starting value $Z_0 = \mu_W$ when the steady-state control limits are used. The process is considered IC while all the charting statistics Z_j, $j = 1,2,3,\ldots$ fall between the two control limits, but as soon as a charting statistic falls on or outside the *UCL* or the *LCL*, the process is declared OOC and typically a search for assignable causes would be started.

4.4.3.2.1 Methods of Calculating the Run-length Distribution The steps are similar to those of obtaining the conditional and the unconditional run-length distribution for the CUSUM-rank chart as described in Li, Tang, and Ng (2010). The reader is referred to Appendix 4.2 for details. However, using simulations is much easier and more straightforward. An

algorithm similar to the ones for the Shewhart and the CUSUM charts is shown below.

The Simulation Approach The following simulation algorithm can be used to obtain the run-length characteristics for the EWMA-rank chart. The corresponding SAS program is given in Appendix B, SAS program number 12, and the output is shown below in Figure 4.28.

Step 1: Specify the Phase I reference sample size m (≥ 1), the Phase II test sample size n (≥ 1), the desired nominal ARL_{IC}, and the size of the shift (δ) to be detected. Next, specify the two distributions for generating the Phase I and Phase II samples, respectively. For the IC case, the two distributions are identical (we say that the shift equals zero ($\delta = 0$)). For the OOC case, the distribution for the Phase II sample is of the same form as that for the Phase I sample, but with a shift in the location in units of the population standard deviation. The Phase I sample, for example, may be drawn from a normal distribution with mean 0 and standard deviation 1, whereas in the OOC case the Phase II sample may be drawn from a normal distribution with some mean $\delta(\neq 0)$ and standard deviation 1 (we say that the mean shift is δ ($\neq 0$)). Any continuous distribution, whether symmetric or asymmetric, may be used.

Step 2: Specify the design parameter, λ, (the reader is referred to Section 3.2.3 for guidance on choosing this design parameter) and the LCL and UCL such that a nominal ARL_{IC} is obtained. It should be noted at this point that, typically, for an EWMA chart, we have two design parameters, λ and L, that need to be specified. However, Li, Tang, and Ng (2010) appear to have directly selected the control limits (LCL and UCL) in their paper as opposed to selecting the width of the control limits (L).

Step 3: Calculate the W_j statistic and the corresponding Z_j charting statistic for each subgroup.

Step 4: Compare the charting statistics to the control limits.

Step 5: The number of subgroups needed until Z_j plots on or outside the control limits for the first time is recorded as an observation from the run-length distribution.

Step 6: Repeat Steps 1 to 5 a total of 100 000 times. This creates 100 000 observations from the run-length distribution.

Step 7: The observations obtained in Step 5 can be analyzed to calculate estimates of the run-length distribution characteristics such as the average, the median, the standard deviation, etc.

A software package such as SAS or R can be used to program the simulation study. Note that when $\delta = 0$ we get the IC run-length characteristics, whereas for $\delta \neq 0$ we get the OOC run-length characteristics.

Next, we provide some illustrative examples of the distribution-free charts in Case U.

4.5 DISTRIBUTION-FREE CONTROL CHARTS IN CASE U: ILLUSTRATIVE EXAMPLES

4.5.1 Shewhart Control Charts

Example 4.18 Shewhart Med Chart

Table 4.21 displays $m = 100$ reference observations (arranged in ascending order) from an unknown continuous process distribution. Suppose that we wish to monitor the location of the process distribution using the sample medians of future test samples, each of size $n = 5$. In addition, suppose that the nominal ARL_{IC} value is 370. Hence, using Table 4.18, we find that $a = 6$ and $b = 100 - 6 + 1 = 95$ so that the control limits of the Med chart are given by the 6th and the 95th reference sample order statistics, $\hat{LCL} = X_{(6:100)}$ and $\hat{UCL} = X_{(95:100)}$, which equal 4.39 and 9.83, respectively, and yield an attained ARL equal to 359.6. Note that the attained ARL_{IC} is not equal to the nominal value of 370 but is close enough. This raises an important and interesting question for many control charts used in practice whose charting statistics have a discrete probability distribution. This is indeed the case for the precedence charts in particular, and nonparametric control charts in general. In fact, this is also the case for control charts with attributes data. In these cases, one often can't achieve the specified nominal ARL_{IC} exactly and one has to live with some value close enough, or use the idea of randomized test (see Gibbons and Chakraborti, 2010, p. 25), which is not recommended in practice. We adopt the conservative approach, where the constant a is found so that the attained ARL_{IC} is no larger than the nominal ARL_{IC}.

Next, 25 test samples each of size $n = 5$ were obtained and process monitoring continued based on the sample median $Y_{(3:5)}$. These values are displayed in column (a) and column (b), respectively, of Table 4.22 with the two-sided Med control chart shown in Figure 4.29.

From Figure 4.29 it can be seen that there is no lack of statistical control and, accordingly, no corrective action is necessary; so process monitoring continues.

Table 4.21 Phase I data and the order statistics for the Med control chart.

i	$X_{(i:100)}$	i	$X_{(i:100)}$	i	$X_{(i:100)}$	i	$X_{(i:100)}$	i	$X_{(i:100)}$
1	3.13	21	5.59	41	6.45	61	7.32	81	8.30
2	3.72	22	5.62	42	6.46	62	7.37	82	8.31
3	3.82	23	5.63	43	6.51	63	7.48	83	8.35
4	4.23	24	5.73	44	6.52	64	7.67	84	8.64
5	4.34	25	5.74	45	6.53	65	7.68	85	8.67
6	4.39	26	5.77	46	6.55	66	7.70	86	8.80
7	4.46	27	5.84	47	6.64	67	7.73	87	8.86
8	4.58	28	5.90	48	6.65	68	7.81	88	8.91
9	4.66	29	5.96	49	6.72	69	7.84	89	8.96
10	4.72	30	6.02	50	6.87	70	7.89	90	9.01
11	4.83	31	6.02	51	6.87	71	7.93	91	9.17
12	4.94	32	6.05	52	6.95	72	7.96	92	9.17
13	5.08	33	6.15	53	6.96	73	8.01	93	9.49
14	5.08	34	6.21	54	7.00	74	8.04	94	9.60
15	5.20	35	6.23	55	7.04	75	8.07	95	9.83
16	5.33	36	6.32	56	7.11	76	8.14	96	9.88
17	5.37	37	6.36	57	7.17	77	8.14	97	9.96
18	5.42	38	6.39	58	7.20	78	8.24	98	10.29
19	5.46	39	6.43	59	7.20	79	8.24	99	10.31
20	5.53	40	6.44	60	7.21	80	8.29	100	10.56

Example 4.19 Shewhart-MW Chart

Table 4.23 shows 25 retrospective or Phase I samples, each of size five, that were collected when the process was thought to be IC, that is, $m = 125$. These data are considered the Phase I reference data for which a goodness-of-fit test for normality is not rejected. From Chakraborti and Van de Wiel (2008), we use $b_{mn} = 540$ and $a_{mn} = mn - b_{mn} = (125)(5) - 540 = 85$. We discuss how to find the control limits later in this section.

Column (a) of Table 4.24 contains 15 prospective (Phase II) samples each of size $n = 5$ on the inside diameters of piston rings manufactured by a forging process. The corresponding MW charting statistics are shown in Column (b).

The graphical display of the Shewhart-MW control chart is shown in Figure 4.30 and signals at sample number 12, indicating that a search for assignable causes is necessary. In fact, samples 12, 13, and 14 all seem to be OOC.

Next, consider the distribution-free CUSUM control charts.

Table 4.22 Phase II data and the corresponding charting statistics for the Med control chart.

	(a)					(b)
i	$Y_{i:1}$	$Y_{i:2}$	$Y_{i:3}$	$Y_{i:4}$	$Y_{i:5}$	$Y_{(3:5)}$
1	7.97	8.08	8.62	6.82	6.56	7.97
2	8.80	7.89	2.82	8.56	7.13	7.89
3	6.70	7.26	6.51	8.00	5.61	6.70
4	3.82	5.85	7.32	7.47	6.92	6.92
5	6.77	5.13	7.13	5.96	8.25	6.77
6	7.37	8.18	4.53	7.14	6.87	7.14
7	6.01	6.78	9.14	7.85	4.10	6.78
8	6.44	5.26	6.72	7.38	7.25	6.72
9	6.87	5.19	6.78	5.69	7.37	6.78
10	6.54	7.46	8.60	7.25	8.14	7.46
11	8.66	7.92	6.54	6.18	7.97	7.92
12	4.43	4.20	6.34	6.24	6.13	6.13
13	7.35	7.54	10.28	7.69	6.30	7.54
14	4.54	9.60	5.66	6.55	6.85	6.55
15	8.43	7.32	5.65	9.72	8.77	8.43
16	8.51	8.60	7.09	7.66	6.83	7.66
17	9.11	5.06	6.09	8.54	7.61	7.61
18	7.52	9.15	7.18	5.97	8.44	7.52
19	6.55	6.05	5.41	4.68	8.00	6.05
20	7.07	8.29	6.59	4.55	3.75	6.59
21	5.81	4.89	6.57	6.53	6.21	6.21
22	6.86	8.65	6.67	7.99	7.60	7.60
23	5.47	7.29	5.92	9.25	9.32	7.29
24	8.20	7.62	5.97	3.72	5.83	5.97
25	6.26	7.98	6.38	6.09	7.45	6.38

4.5.2 CUSUM Control Charts

Example 4.20 Two-sided Median CUSUM-EX Chart

Column (a) of Table 4.25 contains 15 prospective (Phase II) samples each of size $n = 5$ on the inside diameters of piston rings manufactured by a forging process. For the data of Table 4.25, the two one-sided CUSUM charting statistics C_j^+ and C_j^- with $C_0^+ = C_0^- = 0$ are calculated together with two counters N^+ and N^- and are shown in columns (c) and (d) of Table 4.25. We use $k = \delta STDEV(U_{j,r})$ with $= \gamma\sigma/\sqrt{n}$. Assuming that a small shift is to be detected, we take γ to be small, say, $\gamma = 0.25$. Recall from Note 4.6 in Section

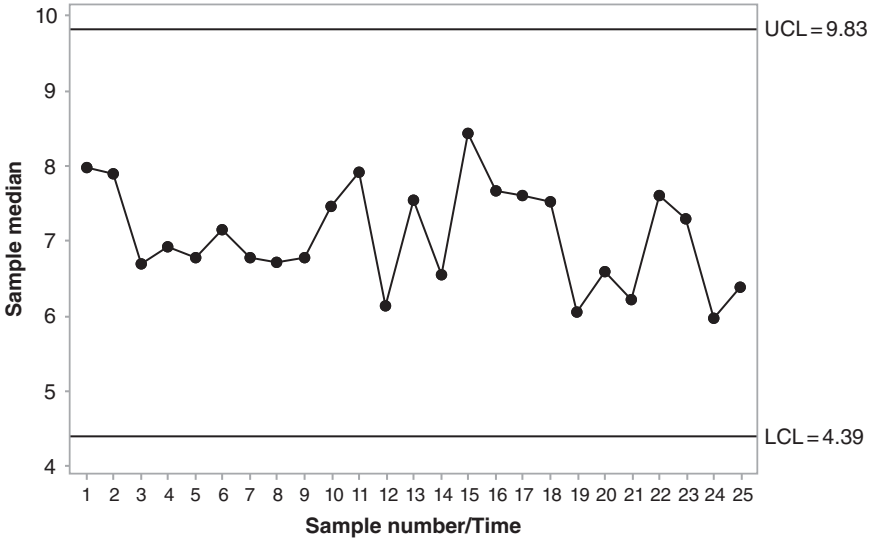


FIGURE 4.29 *A Med control chart.*

4.4.2 that the safest choice for k is taking a smaller value of k or even $k = 0$; here, the value of k is to be 0. The next step is to choose h in conjunction with the chosen k so that a desired nominal ARL_{IC} is attained. Here, the decision interval was taken to be $h = 7.5$. Again, the reader is referred to Note 4.6 in Section 4.4.2 and to the paper by Mukherjee, Graham, and Chakraborti (2013) for a detailed discussion on the choice of the design parameters.

To illustrate the calculations, consider sample number 1. The expressions for the CUSUM charting statistics C_1^+ and C_1^- are

$$\begin{aligned} C_1^+ &= \max\left[0, C_0^+ + \left(U_{1,r} - \frac{n}{2}\right) - k\right] \\ &= \max[0, 0 + (3 - 5/2) - 0] \\ &= 0.5 \end{aligned}$$

and

$$\begin{aligned} C_1^- &= \min\left[0, C_0^- + \left(U_{1,r} - \frac{n}{2}\right) + k\right] \\ &= \min[0, 0.5 + (3 - 5/2) + 0] \\ &= 0. \end{aligned}$$

The graphical display of the median CUSUM-EX control chart is shown in Figure 4.31. From Figure 4.31, we can see that the process is OOC starting at sample number 13, since the corresponding charting statistic plots above

Table 4.23 Phase I data for the Shewhart-MW chart.

j	X_{j1}	X_{j2}	X_{j3}	X_{j4}	X_{j5}
1	74.030	74.002	74.019	73.992	74.008
2	73.995	73.992	74.001	74.011	74.004
3	73.998	74.024	74.021	74.005	74.002
4	74.002	73.996	73.993	74.015	74.009
5	73.992	74.007	74.015	73.989	74.014
6	74.009	73.994	73.997	73.985	73.993
7	73.995	74.006	73.994	74.000	74.005
8	73.985	74.003	73.993	74.015	73.988
9	74.008	73.995	74.009	74.005	74.004
10	73.998	74.000	73.990	74.007	73.995
11	73.994	73.998	73.994	73.995	73.990
12	74.004	74.000	74.007	74.000	73.996
13	73.983	74.002	73.998	73.997	74.012
14	74.006	73.967	73.994	74.000	73.984
15	74.012	74.014	73.998	73.999	74.007
16	74.000	73.984	74.005	73.998	73.996
17	73.994	74.012	73.986	74.005	74.007
18	74.006	74.010	74.018	74.003	74.000
19	73.984	74.002	74.003	74.005	73.997
20	74.000	74.010	74.013	74.020	74.003
21	73.982	74.001	74.015	74.005	73.996
22	74.004	73.999	73.990	74.006	74.009
23	74.010	73.989	73.990	74.009	74.014
24	74.015	74.008	73.993	74.000	74.010
25	73.982	73.984	73.995	74.017	74.013

the *UCL*. Thus, some corrective action and a search for assignable causes are necessary. Also, since the counter $N^+ = 8$, at sample number 13 and $N^+ = 0$ at sample number 5, the shift most likely occurred at sample number 5.

Example 4.21 Two-sided CUSUM-rank Chart

Column (a) of Table 4.26 contains 15 prospective (Phase II) samples each of size $n = 5$ on the inside diameters of piston rings manufactured by a forging process. For the data of Table 4.26, the two one-sided CUSUM charting statistics C_j^+ and C_j^- with $C_0^+ = C_0^- = 0$ were calculated together with two counters N^+ and N^- and are shown in Table 4.24. We use the reference value $k = \tau_l SD(W_j)$ with $\tau_l = c_l\delta$, where c_l is a positive constant and δ is the amount of the shift to be detected in terms of the process standard deviation units. To illustrate, the desired shift to be detected was taken to

Table 4.24 Phase II data for the Shewhart-MW chart and the corresponding charting statistics.

Sample	(a)					(b)
j	X_{j1}	X_{j2}	X_{j3}	X_{j4}	X_{j5}	M_{YX}
1	74.012	74.015	74.030	73.986	74.000	429
2	73.995	74.010	73.990	74.015	74.001	333
3	73.987	73.999	73.985	74.000	73.990	143
4	74.008	74.010	74.003	73.991	74.006	371
5	74.003	74.000	74.001	73.986	73.997	242
6	73.994	74.003	74.015	74.020	74.004	411
7	74.008	74.002	74.018	73.995	74.005	393
8	74.001	74.004	73.990	73.996	73.998	241
9	74.015	74.000	74.016	74.025	74.000	471
10	74.030	74.005	74.000	74.016	74.012	486
11	74.001	73.990	73.995	74.010	74.024	341
12	74.015	74.020	74.024	74.005	74.019	561
13	74.035	74.010	74.012	74.015	74.026	576
14	74.017	74.013	74.036	74.025	74.026	602
15	74.010	74.005	74.029	74.000	74.020	485

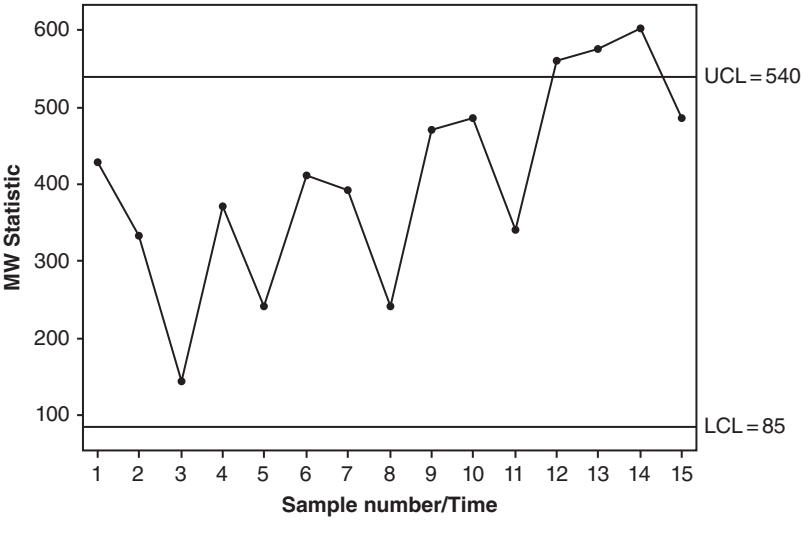


FIGURE 4.30 *A Shewhart-MW chart.*

Table 4.25 Data for the median CUSUM-EX chart and the corresponding charting statistics.

Sample	(a)					(b)	(c)		(d)	
j	X_{j1}	X_{j2}	X_{j3}	X_{j4}	X_{j5}	$U_{j,r}$	C_j^-	N^-	C_j^+	N^+
1	74.012	74.015	74.030	73.986	74.000	3	0.0	0	0.5	1
2	73.995	74.010	73.990	74.015	74.001	2	−0.5	1	0.0	0
3	73.987	73.999	73.985	74.000	73.990	0	−3.0	2	0.0	0
4	74.008	74.010	74.003	73.991	74.006	4	−1.5	3	1.5	1
5	74.003	74.000	74.001	73.986	73.997	1	−3.0	4	0.0	0
6	73.994	74.003	74.015	74.020	74.004	4	−1.5	5	1.5	1
7	74.008	74.002	74.018	73.995	74.005	4	0.0	0	3.0	2
8	74.001	74.004	73.990	73.996	73.998	1	−1.5	1	1.5	3
9	74.015	74.000	74.016	74.025	74.000	3	−1.0	2	2.0	4
10	74.030	74.005	74.000	74.016	74.012	4	0.0	0	3.5	5
11	74.001	73.990	73.995	74.010	74.024	2	−0.5	1	3.0	6
12	74.015	74.020	74.024	74.005	74.019	5	0.0	0	5.5	7
13	74.035	74.010	74.012	74.015	74.026	5	0.0	0	8.0	8
14	74.017	74.013	74.036	74.025	74.026	5	0.0	0	10.5	9
15	74.010	74.005	74.029	74.000	74.020	4	0.0	0	12.0	10

be small, $\delta = 0.25$ and $c_l = 1.00$ so that

$$k = \tau_l SD(W_j) = (1)(0.25)\sqrt{\frac{(125)(5)(125+5+1)}{12}} = 20.650.$$

The decision interval was taken to be $h = 946$; an explanation follows.

A quick recap on the choice of the design parameters (k, h) follows. The (k, h) are chosen so that the chart has a specified nominal $ARL_{\text{IC}} = t$ (or a specified nominal $MRL_{\text{IC}} = s$) and is capable of detecting a shift, especially a small shift, as soon as possible. Note that Mukherjee, Graham, and Chakraborti (2013), among others, recommend using the MRL over ARL as a chart performance characteristic. The first step is to choose k. In Section 3.2.2, we considered the parametric CUSUM chart with individual data ($n = 1$) with no reference sample. We examined the impact of k (the reader is referred to Figure 3.2 and the discussion on it) and concluded that when there is little or no a priori information regarding the size of the shift a smaller value of k is the safest choice (to protect against any unnecessary delays in detection). Mukherjee, Graham, and Chakraborti (2013) found similar conclusions for nonparametric CUSUM charts. The next step is to choose h in conjunction with the chosen k so that a desired nominal ARL_{IC}

Table 4.26 Data for the CUSUM-rank chart and the corresponding charting statistics.

Sample	(a)					(b)	(c)		(d)	
j	X_{j1}	X_{j2}	X_{j3}	X_{j4}	X_{j5}	W_j	C_j^-	N^-	C_j^+	N^+
1	74.012	74.015	74.030	73.986	74.000	429	0	0	80.850	1
2	73.995	74.010	73.990	74.015	74.001	333	0	0	65.700	2
3	73.987	73.999	73.985	74.000	73.990	143	−164.350	1	0	0
4	74.008	74.010	74.003	73.991	74.006	371	−100.700	2	22.350	1
5	74.003	74.000	74.001	73.986	73.997	242	−166.049	3	0	0
6	73.994	74.003	74.015	74.020	74.004	411	−62.399	4	62.350	1
7	74.008	74.002	74.018	73.995	74.005	393	0	0	107.200	2
8	74.001	74.004	73.990	73.996	73.998	241	−66.350	1	0	0
9	74.015	74.000	74.016	74.025	74.000	471	0	0	122.850	1
10	74.030	74.005	74.000	74.016	74.012	486	0	0	260.700	2
11	74.001	73.990	73.995	74.010	74.024	341	0	0	253.049	3
12	74.015	74.020	74.024	74.005	74.019	561	0	0	465.899	4
13	74.035	74.010	74.012	74.015	74.026	576	0	0	693.249	5
14	74.017	74.013	74.036	74.025	74.026	602	0	0	946.599	6
15	74.010	74.005	74.029	74.000	74.020	485	0	0	1082.948	7

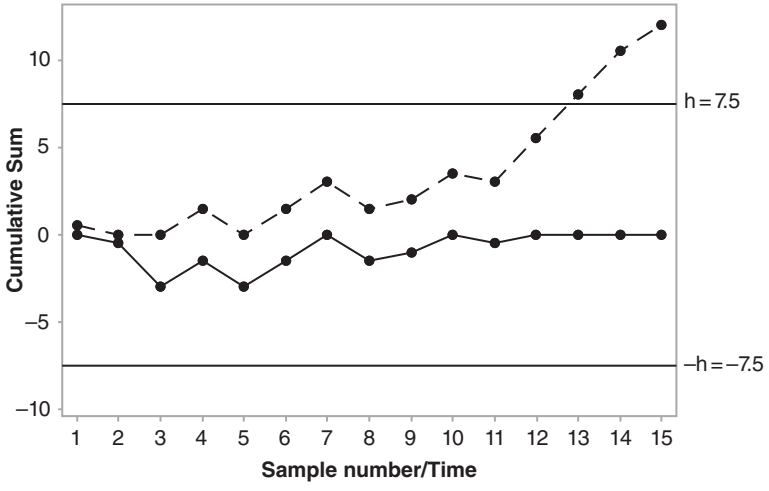


FIGURE 4.31 *A two-sided median CUSUM-EX chart.*

or MRL_{IC} is attained. In Mukherjee, Graham, and Chakraborti (2013), the h and k were selected for a nominal $MRL_{\text{IC}} \approx 350$. To illustrate the calculations for the CUSUM-rank chart, consider sample number 1. The expressions for the CUSUM charting statistics C_1^+ and C_1^- are

$$\begin{aligned} C_1^+ &= \max[0, C_{j-1}^- - \left(W_j - \frac{n(m+n+1)}{2}\right) \\ &= \max\left[0, 0 + \left(429 - \frac{5(125+5+1)}{2}\right) - 20.650\right] \\ &= 80.850 \end{aligned}$$

and

$$\begin{aligned} C_1^- &= \min\left[0, C_{j-1}^- + \left(W_j - \frac{n(m+n+1)}{2}\right) + k\right] \\ &= \min\left[0, 0 + \left(429 - \frac{5(125+5+1)}{2}\right) + 20.650\right] \\ &= 0. \end{aligned}$$

These values and the corresponding counters are shown in columns (c) and (d) of Table 4.26. The graphical display of the CUSUM-rank chart is shown in Figure 4.32 and it shows that the process goes OOC at the second last sample, that is, it signals at sample number 14 with a counter of $N^+ = 6$, meaning that the shift most likely occurred at sample number 8.

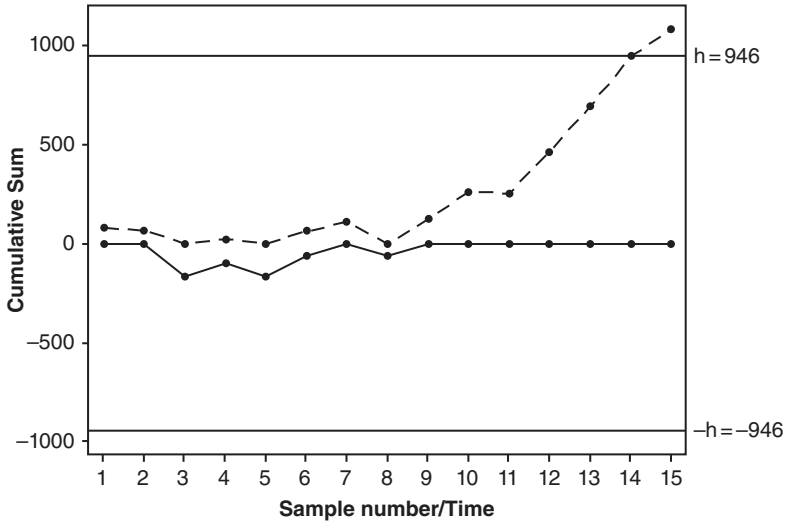


FIGURE 4.32 *A two-sided CUSUM-rank chart.*

4.5.3 EWMA Control Charts

Example 4.22 Two-sided Median EWMA-EX Chart

A two-sided median CUSUM-EX chart was discussed in Example 4.20 using the data given in Table 4.25. As an alternative, an EWMA control chart is now applied to the same data. To illustrate, we use $\lambda = 0.1$ to calculate the charting statistics and $L = 2.31$ to find the control limits. To illustrate the calculations, consider sample number 1 with $U_{1,r} = 3$ and suppose that the starting value $Z_0 = np_r = (5)(0.5) = 2.5$. Then, the first charting statistic on the EWMA-EX chart is

$$\begin{aligned} Z_1 &= \lambda\, U_{1,r} + (1-\lambda)Z_0 \\ &= 0.1(3) + (1-0.1)2.5 \\ &= 2.550. \end{aligned}$$

The value of the second charting statistic is

$$\begin{aligned} Z_2 &= \lambda\, U_{2,r} + (1-\lambda)Z_1 \\ &= 0.1(3) + (1-0.1)2.550 \\ &= 2.595 \end{aligned}$$

with the remaining calculations summarized in column (b) of Table 4.27.

Table 4.27 Calculations for the median EWMA-EX chart.

Sample j	(a) $U_{j,r}$	(b) $Z_j = \lambda U_{j,r} + (1-\lambda)Z_{j-1}$
1	3	2.550
2	3	2.595
3	0	2.336
4	4	2.502
5	2	2.452
6	4	2.607
7	4	2.746
8	2	2.671
9	3	2.704
10	4	2.834
11	3	2.850
12	5	3.065
13	5	3.259
14	5	3.433
15	4	3.490
16	3	2.550
17	3	2.595
18	0	2.336
19	4	2.502
20	2	2.452

For simplicity, we use the steady-state control limits on the EWMA-EX chart, which is found from Equation 4.60, that is

$$\begin{aligned} UCL &= n(1-a) + L\sqrt{\left(\frac{na(1-a)}{m+2}\right)\left(n + \frac{\lambda(m+1)}{2-\lambda}\right)} \\ &= (5)(1-0.5) + 2.31\sqrt{\left(\frac{(5)(0.5)(1-0.5)}{125+2}\right)\left(5 + \frac{(0.1)(125+1)}{2-0.1}\right)} \\ &= 3.282 \end{aligned}$$

and

$$\begin{aligned} LCL &= n(1-a) - L\sqrt{\left(\frac{na(1-a)}{m+2}\right)\left(n + \frac{\lambda(m+1)}{2-\lambda}\right)} \\ &= (5)(1-0.5) - 2.31\sqrt{\left(\frac{(5)(0.5)(1-0.5)}{125+2}\right)\left(5 + \frac{(0.1)(125+1)}{2-0.1}\right)} \\ &= 1.718 \end{aligned}$$

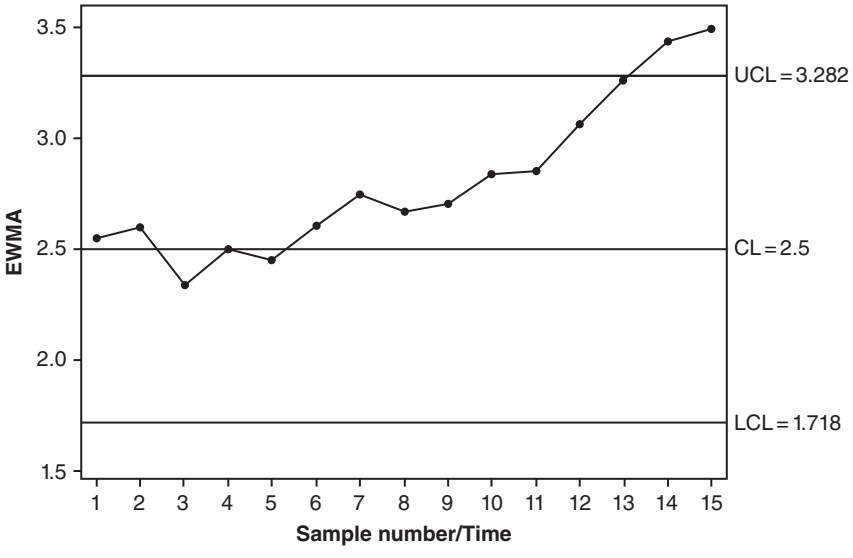


FIGURE 4.33 *A two-sided median EWMA-EX chart.*

with $a = \dfrac{r}{m+1} = \dfrac{63}{125+1} = 0.5$, and the CL is given by $CL = n(1-a) = (5)(1-0.5) = 2.5$.

The resulting EWMA-EX chart is shown in Figure 4.33, and we can see that the process is OOC, starting at sample number 14 since the charting statistic plots on the *UCL*. Thus, corrective action and a search for assignable causes are necessary.

Example 4.23 Two-sided EWMA-rank Chart

A two-sided CUSUM-rank chart was created in Example 4.21 using the data given in Table 4.26. Here we consider, as an alternative, an EWMA control chart. For this purpose, we use $\lambda = 0.1$ to calculate the charting statistics and $L = 2.814$ to find the control limits. To illustrate the calculations, consider sample number 1 with $W_1 = 429$ and suppose that the starting value $Z_0 = \mu_W = \dfrac{n(m+n+1)}{2} = \dfrac{5(125+5+1)}{2} = 327.5$.

Then, the first charting statistic on the EWMA-rank chart is

$$\begin{aligned} Z_1 &= \lambda\, W_1 + (1-\lambda) Z_0 \\ &= 0.1(429) + (1-0.1)327.5 \\ &= 337.65. \end{aligned}$$

Table 4.28 Calculations for the EWMA-rank chart.

Sample j	(a) W_j	(b) $Z_j = \lambda W_j + (1-\lambda)Z_{j-1}$
1	429	337.65
2	333	337.19
3	143	317.72
4	371	322.99
5	242	314.85
6	411	324.41
7	393	331.27
8	241	322.19
9	471	337.07
10	486	351.97
11	341	350.82
12	561	371.84
13	576	392.20
14	602	413.13
15	485	420.27

The value of the second charting statistic is

$$\begin{aligned} Z_2 &= \lambda W_2 + (1-\lambda)Z_1 \\ &= 0.1(333) + (1-0.1)337.65 \\ &= 337.19 \end{aligned}$$

with the remaining calculations summarized in column (b) of Table 4.28.

For simplicity, we use the steady-state control limits on the EWMA-rank chart, which is found from Equation 4.63, that is

$$\begin{aligned} UCL &= \mu_W + L\sqrt{\sigma_W^2\left(\frac{\lambda}{2-\lambda}\right)} \\ &= \frac{n(m+n+1)}{2} + L\sqrt{\left(\frac{mn(m+n+1)}{12}\right)\left(\frac{\lambda}{2-\lambda}\right)} \\ &= \frac{5(125+5+1)}{2} + 2.814\sqrt{\left(\frac{(125)(5)(125+5+1)}{12}\right)\left(\frac{0.1}{2-0.1}\right)} \\ &= 380.825 \end{aligned}$$

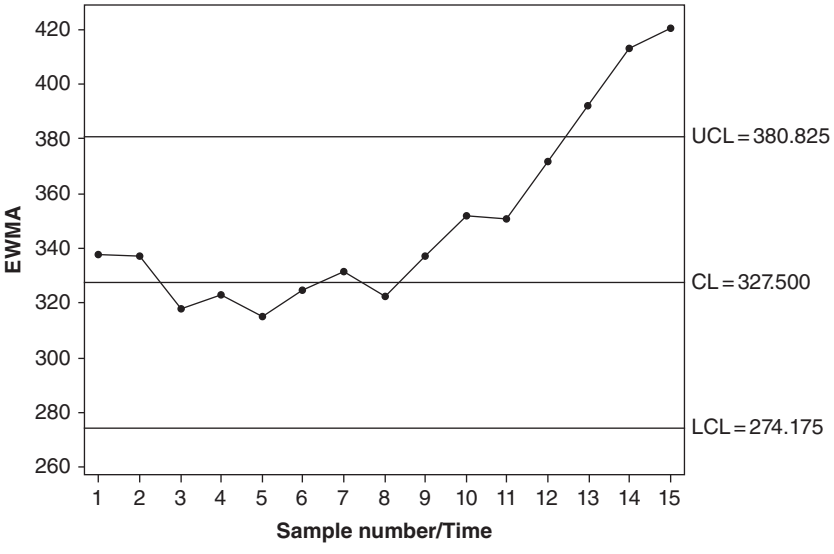


FIGURE 4.34 *A two-sided EWMA-rank chart.*

and

$$\begin{aligned} LCL &= \mu_W - L\sqrt{\sigma_W^2\left(\frac{\lambda}{2-\lambda}\right)} \\ &= \frac{n(m+n+1)}{2} - L\sqrt{\left(\frac{mn(m+n+1)}{12}\right)\left(\frac{\lambda}{2-\lambda}\right)} \\ &= \frac{5(125+5+1)}{2} - 2.814\sqrt{\left(\frac{(125)(5)(125+5+1)}{12}\right)\left(\frac{0.1}{2-0.1}\right)} \\ &= 274.175 \end{aligned}$$

and the CL is given by $CL = \mu_W = \dfrac{n(m+n+1)}{2} = \dfrac{5(125+5+1)}{2} = 327.5$.

The resulting EWMA-rank chart is shown in Figure 4.34 and we can see that the process is OOC with an increase in the location parameter, starting at sample number 13, since the charting statistic plots above the *UCL*. Thus, a search for assignable causes and some follow-up corrective actions may be necessary.

4.6 EFFECTS OF PARAMETER ESTIMATION

The effects of parameter estimation on the performance of the Phase II control chart can be profound. The reader is referred to Jensen *et al.* (2006) for a literature overview and some broad recommendations in this area. Here, we simply note that the effect of the size and the quality of the reference sample is important not only for the nonparametric control charts but for all Phase II charts, including the normal theory charts, when the parameters are estimated from Phase I reference data. The impact and effects of parameter estimation continue to be an active area of current research in SPC and the reader is referred to these works. We conclude by stating that, even though it is well known that the use of estimated parameters significantly affects the statistical performance of Phase II charts, the research regarding the ways and means to actually obtain an IC reference sample, that is, the Phase I sample, has received less emphasis and should be considered in future research.

4.7 SIZE OF PHASE I DATA

The reader is referred to Jensen *et al.* (2006) for a literature overview and some recommendations on this area. A lot more recent research has been done on this subject. The consensus is that much more reference data are needed in Phase I than the typically recommended 20 to 30 samples of size 5 in order to achieve comparable performance in Case K, the known parameter case. It is important to note that the size of the Phase I reference data depends on many factors, such as whether it's an attributes chart or a variables chart, the process distribution, the type of the control chart (Shewhart, CUSUM, or EWMA), and also the dimensionality of the data. In some cases where the required amount of reference data may not be available, the so-called self-starting control charts may be considered, which make use of successive observations to simultaneously update parameter estimates and the charting statistic. Some recent publications on self-starting charts include Maboudou-Tchao and Hawkins (2011), Zhang, Xie, and Jin (2012), and Keefe, Woodall, and Jones-Farmer (2015). The other option may be to adjust the charting constants in some way. Both IC and OOC performance need to be satisfactory. We don't go into these topics here, the reader can consult some of the recent papers on effects of parameter estimation on control charts.

4.8 CONTROL CHART ENHANCEMENTS

4.8.1 Sensitivity and Runs-type Signaling Rules

Runs-type signaling rules are generally used to enhance the performance of Shewhart control charts. Very little has been done in terms of adding signaling rules to CUSUM and EWMA control charts. For the latter, the reader is referred to Riaz, Abbas, and Does (2011), who proposed two runs-rules schemes for parametric CUSUM charts, and Abbas, Riaz, and Does (2011), who propose two runs-rules schemes for the parametric EWMA charts. However, these papers are for parametric control charts and, to the best of our knowledge, no one has considered adding signaling rules to the CUSUM-EX, CUSUM-rank, EWMA-EX, and EWMA-rank charts to date. Chakraborti, Eryilmaz, and Human (2009) considered a Shewhart-Prec chart with runs-type signaling rules and conclude that they outperform the typical *1-of-1* Shewhart-Prec chart. Malela-Majika, Chakraborti, and Graham (2016) propose a Shewhart-MW chart with runs-type signaling rules and also found that it outperforms the *1-of-1* Shewhart-MW chart, which is to be expected. The reader is referred to these papers for further details. Here, we only give one example to illustrate the Shewhart charts with signaling rules in Case U. We do this by first obtaining the unconditional ARL for the nonparametric Shewhart-Prec chart, with runs-type signaling rules, as proposed by Chakraborti, Eryilmaz, and Human (2009).

Again, it is assumed that a reference sample of size m is available from an IC process with an unknown continuous distribution function F. Let $X_{1:m} < X_{2:m} < \ldots < X_{m:m}$ denote order statistics from the reference sample with $LCL = X_{a:m}$ and $UCL = X_{b:m}$ where $1 \leq a < b \leq m$. In Phase II, test samples of size n are drawn sequentially and independently from each other, as well as independently from the reference sample. Recall that, for the Shewhart-Prec chart, the charting statistic for the hth test sample, $Y^h_{j:n}$, can be any order statistic, but the median is chosen for simplicity and the resulting chart is called the Med chart. The following indicator variables are defined for the hth test sample: $Z_h = 1$ if $Y^h_{j:n} \notin (LCL,\, UCL)$ and $Z_h = 0$ if $Y^h_{j:n} \in (LCL,\, UCL)$ with $h = 1, 2, 3 \ldots$ Since the control limits are order statistics from the reference sample, the signaling indicators $Z_1, Z_2, \ldots$ are dependent binary variables. It is also assumed that the test sample originates from a continuous distribution with cdf G.

As an example for the *1-of-1* runs-type signaling rule (which is the original Med chart) the conditional probability of no signal given $X_{a:m} = x_1$ and $X_{b:m} = x_2$ is $P(Z_h = 0|X_{a:m} = x_1,\, X_{b:m} = x_2) = P(Z_h = 0|X_{a:m} = x_1,\, X_{b:m} = x_2) = P(x_1 < Y_{j:n} < x_2|X_{a:m} = x_1,\, X_{b:m} = x_2) = G_j(x_2) - G_j(x_1)$, where G_j denotes the cdf of the jth order

statistic in a sample of size n from a distribution with cdf G. Since $G_j(x) = I_{G(x)}(i, n-j+1)$, where $I_x(y,z)$ denotes the incomplete beta integral (which is the cdf of a beta distribution), the probability of a signal can be rewritten as $I_{G(x_2)}(i, n-j+1) - I_{G(x_1)}(i, n-j+1)$.

Hence, using the CUC method, the unconditional probability of no signal can be found by averaging the conditional probability over the joint distribution of $X_{a:m}$ and $X_{b:m}$. Although it is not discussed here in detail, the interested reader is referred to Chakraborti (2000) for a detailed discussion on the CUC method.

The performance of the control charts is evaluated by considering their run-length distribution, which can be viewed as the waiting time until the first signal. The waiting times are defined as follows

$$T_1 = \min(t : Z_t = 1).$$
$$T_2 = \min(t : Z_{t-1} = 1, Z_t = 1).$$
$$T_2' = \min(T_2^{(1)}, T_2^{(2)})$$

where $T_2^{(1)} = \min(t : Z'_{t-1} = 1, Z'_t = 1)$, $T_2^{(2)} = \min(t : Z'_{t-1} = 1, Z'_t = 2)$, and the Z'_h indicator variables are $Z'_h = 0$ if $Y_{j:n} \in (X_{a:m},\ X_{b:m})$, $Z'_h = 1$ if $Y_{j:n} \geq X_{a:m}$, $Z'_h = 2$ if $Y_{j:n} \leq X_{a:m}$.

Relating this back to runs-type signaling rules, T_1 is the run-length of a *1-of-1* runs-rules chart, whereas T_2 represents that of the *2-of-2* runs-rules chart. Using the MC method, the conditional distribution of T_2^*, which is the waiting time for the first signal (or the run-length) for the *2-of-2* KL control chart, can be obtained by applying Theorem 5.2 of Fu and Lou (2003, p. 68)

$$P(T_2^* = x | X_{a:m}, X_{b:m}) = \underline{\xi} Q^{x-1}(I - Q)\underline{1},\ x \geq 2.$$

where $\underline{\xi} = \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix}$ and the essential TPM (Q) is provided in Table 4.29 (the interested reader is also referred to Chakraborti, Human, and Graham, 2009, p. 1058).

Table 4.29 Essential TPM (Q) for the two-sided control chart supplemented with the *2-of-2* runs-type signaling rule.

States	S_ϕ	S_0	S_U	S_L
S_ϕ	0	$1 - p_L - p_U$	p_U	p_L
S_0	0	$1 - p_L - p_U$	p_U	p_L
S_U	0	$1 - p_L - p_U$	0	p_L
S_L	0	$1 - p_L - p_U$	p_U	0

Note that the transition states in Table 4.29 are denoted by S_ϕ, S_0, S_U, and S_L, where George and Bowman's (1995) definition of the equation on p. 83 defines S_n as the total number of successes in a sequence of n binary trials. The transition states referred to in Table 4.29 are defined as the empty state denoted by S_ϕ, the state denoted by S_0, and defined as the state entered if the charting statistic plots between the UCL and the LCL, the state denoted by S_U, and defined as the state entered if the charting statistic plots on or above the UCL. Lastly, the state denoted by S_L is defined as the state entered if the charting statistic plots on or below the LCL.

Applying the CUC method, the unconditional run-length (waiting time) distribution of T_2' is obtained from that of T_2^* (by averaging over the joint distribution of $X_{a:m}$ and $X_{b:m}$)

$$P(T_2' = x) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} P(T_2^* = x|X_{a:m} = x_1, X_{b:m} = x_2)h_{a,b}(x_1, x_2)d_{x1}d_{x2},\ x > 2$$

where $P(T_2^* = x|X_{a:m}, X_{b:m}) = \underline{\xi}Q^{x-1}(I - Q)\underline{1},\ \mathrm{x} \geq 2$ and $h_{a,b}(x_1, x_2)$ is the joint pdf of $X_{a:m}$ and $X_{b:m}$.

Chakraborti, Human, and Graham (2009) derived expressions for the exact *ARLs* for both the *DR* and *KL* charts. The idea is, again, based on the CUC. Given $X_{a:m}$ and $X_{b:m}$, the random variable T_2 has the so-called geometric distribution of order 2; see Balakrishnan and Koutras (2002). Appealing to the properties of this distribution, it can be shown that the conditional average run-length (the expected value) $E(T_2|X_{a:m} = x_1, X_{b:m} = x_2)$ is equal to $\dfrac{1+p}{p^2}$ where

$$\begin{aligned} p &= 1 - (G_j(x_2) - G_j(x_1)) \\ &= 1 - (I_{G(x_2)}(j, n-j+1) - I_{G(x_1)}(j, n-j+1)). \end{aligned}$$

Using the CUC method, the unconditional *ARL* for the *DR* chart is obtained by taking the expectation of the conditional expectation, and is given by

$$\begin{aligned} ARL_{DR} &= E_{X_{a:m},X_{b:m}} E(T_2|X_{a:m}, X_{b:m}) \\ &= \int_0^1 \int_0^{x_2} \left(\frac{2 - (G_j(F^{-1}(x_2)) - G_j(F^{-1}(x_1))}{(1 - (G_j(F^{-1}(x_2)) - G_j(F^{-1}(x_1))))^2} \right) \\ &\quad \times f_{a,b}(x_1, x_2)dx_1dx_2. \end{aligned}$$

To obtain the exact *ARL* of the *2-of-2 KL* chart, the derivation of the *ARL* in Klein (2000) can be used along with the CUC method. Thus, given $X_{a:m}$ and $X_{b:m}$, the conditional expected value of T_2' is given by

$E(T_2'|X_{a:m}, X_{b:m}) = \dfrac{1}{p - \dfrac{p_U}{1+p_U} - \dfrac{p_L}{1+p_L}}$, where p_U and p_L were defined earlier. Using this result, the unconditional ARL is then given by

$$ARL_{KL} = \int_0^1 \int_0^{x_2} \left[\frac{(1+(1-G_j(F^{-1}(x_2))))(1+G_j(F^{-1}(x_1)))}{(G_j(F^{-1}(x_1)))^2(1+(1-G_j(F^{-1}(x_2))) \\ +(1-G_j(F^{-1}(x_2)))^2(1+G_j(F^{-1}(x_1)))} \right] \\ \times f_{a,b}(x_1, x_2)dx_1dx_2.$$

Appendix 4.1 SHEWHART CONTROL CHARTS

Appendix 4.1.1 The Shewhart-Prec Control Chart

Result 4.1 The unconditional exceedance probability $P(Y > X_{(r)})$ equals $\dfrac{m-r+1}{m+1}$ when the process is IC

Proof: Note that $P(Y > X_{(r)}) = E(P(Y > X_{(r)}|X_{(r)})) = E(p_r)$. When the process is IC, $F = G$ and then $p_r = 1 - F(X_{(r)}) = 1 - V_r$ (say). Since $X_{(r)}$ is the rth order statistic in a random sample of size m from F, using the PIT, it can be shown that V_r follows the distribution of the rth order statistic in a random sample of size m from the $U(0,1)$ distribution as long as F is continuous. This latter distribution is known to be a beta distribution with parameters r and $m - r + 1$, respectively. Thus, when the process is IC, $p_r = 1 - V_r$ follows a beta distribution with parameters $m - r + 1$ and r. Thus

$$E(p_r) = P(Y > X_{(r)}) = 1 - \frac{r}{m+1} = \frac{m-r+1}{m+1} = d\,(\text{say})$$

when the process is IC using the expectation formula for a beta distribution. ■

Result 4.2 The run-length distribution of the Shewhart-Prec chart and its properties under some specific alternatives

Proof: It can be seen that the run-length distribution and its associated properties depend, in general, on the cdf F of the reference distribution and the cdf G of the Phase II distribution through the composite function $\Psi = GF^{-1}$. Thus, in order to evaluate the run-length distribution and/or

any of the performance measures, we need to specify the function Ψ. For instance, when the process is IC, that is, when $G = F$, then $\Psi(t) = t$, and the IC run-length distribution can be obtained and is given in Equation 4.26. However, in the case where $G \neq F$, which means that the process is OOC in some way, the following results are useful to find the Ψ function:

1. For **location** alternatives, when the process follows the same parametric distribution in Phase I and Phase II with merely a shift in the location parameter, let $F(x) = H(x - \theta_1)$ and $G(x) = H(x - \theta_2)$, where H is a continuous cdf, $x \in \Re$ and $\theta_1, \theta_2 \in \Re$, so that $\Psi(t) = H(\theta_1 - \theta_2 + H^{-1}(t))$. Note that in the IC case, $\Psi(t) = t$.

2. For **scale** alternatives, which is when the process follows the same parametric distribution in Phase II as in Phase I, but in this case, with only a possibility of a change in the process variation or spread, let $F(x) = H\left(\frac{x}{\gamma_1}\right)$ and $G(x) = H\left(\frac{x}{\gamma_2}\right)$, where H is a continuous cdf, $x \in \Re$ and $\gamma_1, \gamma_2 \in \Re^+$, so that $\Psi(t) = H\left(\frac{\gamma_1}{\gamma_2} H^{-1}(t)\right)$. Note that in the IC case, $\Psi(t) = t$.

3. For **location-scale** alternatives, let $F(x) = H\left(\frac{x - \theta_1}{\gamma_1}\right)$ and $G(x) = H\left(\frac{x - \theta_2}{\gamma_2}\right)$, where H is a continuous cdf, $x \in \Re$, $\theta_1, \theta_2 \in \Re$ and $\gamma_1, \gamma_2 \in \Re^+$, so that $\Psi(t) = H\left(\frac{\theta_1 - \theta_2}{\gamma_2} + \frac{\gamma_1}{\gamma_2} H^{-1}(t)\right)$. Note that in the IC case, again, $\Psi(t) = t$.

Thus, it is clear that the run-length distribution can only be calculated and/or evaluated when the change from the IC to the OOC process distribution is known or, in general, if we know the relationship between G and F through the composite function Ψ, as well as the amount of shift in the location parameters in terms of the scale parameter of the G distribution and the amount of shift in the scale parameters.

■

Appendix 4.2 CUSUM CONTROL CHARTS

Appendix 4.2.1 The CUSUM-EX Control Chart

Result 4.3 Given (conditionally on) $X_{(r)}$, the rth reference sample order statistic, the $U_{j,r}$'s are independently and binomially distributed with parameters (n, p_r) for any $j = 1, 2, 3, \ldots$

Proof: Since $U_{j,r}$ is the number of Y observations in the jth Phase II sample that exceeds $X_{(r)}$, given $X_{(r)}$, the random variable $U_{j,r}$ follows a binomial distribution with parameters (n, p_r), where $p_r = P[Y > X_{(r)} | X_{(r)}]$. Independence of $U_{j,r}$ follows since the test samples are independent.

■

Result 4.4 The (unconditional) IC distribution of $U_{j,r}$ is distribution-free and is given by the pmf

$$P(U_{j,r} = u) = \frac{\binom{u+m-r}{u}\binom{n-u+r-1}{n-u}}{\binom{m+n}{n}} \text{ with } u = 0, 1, 2, \ldots, n.$$

Proof: Using Result 4.3 and averaging over the distribution of $X_{(r)}$, we find the unconditional distribution of $U_{j,r}$. Thus

$$P(U_{j,r} = u) = E[P(U_{j,r} = u|X_{(r)}] = \binom{n}{u} \int p_r^u (1 - p_r)^{n-u} dF(X_{(r)})$$

$$= \binom{n}{u} \int [1 - G(t)]^u [G(t)]^{n-u} dF_{X_{(r)}}(t).$$

When the process is IC we have $F = G$ and the above integral can be shown to simplify, via a beta function, to the given result. Hence, the statistics $U_{j,r}$ are unconditionally distribution-free when the process is IC. The same result can be obtained by combinatorial arguments, the details of which are skipped here. Note that the pmf of $U_{j,r}$ shown above is known as the negative hypergeometric distribution.

Next, we extend Result 4.4 and show that the (unconditional) joint distribution of $U_{j,r}$ for $j = 1, 2, \ldots, v$ is distribution-free when the process is IC. This establishes that the proposed chart is distribution-free.

■

Result 4.5 The unconditional IC distribution of $U_{j,r}$ for $j = 1, 2, \ldots, v$, where $v > 1$ is a positive integer, is distribution-free.

Proof: Noting that by Result 4.3, given $X_{(r)}$, the $U_{j,r}, j = 1, 2, \ldots, v$ are independent binomial (n, p_r) variables, the joint distribution of $(U_{1,r}, U_{2,r}, \ldots, U_{v,r})$, when the process is IC, is given by

$$P[U_{1,r} = u_1, U_{2,r} = u_2, \ldots, U_{v,r} = u_v]$$
$$= E[P[U_{1,r} = u_1 U_{2,r} = u_2, \ldots, U_{v,r} = u_v] | X_{(r)}]$$

$$= \binom{n}{u_1}\binom{n}{u_2}\cdots\binom{n}{u_v}\int (1-p_r)^{n\nu}\left[\frac{p_r}{1-p_r}\right]^{\sum_{j=1}^{v} u_j} dF(X_{(r)})$$

$$= \binom{n}{u_1}\binom{n}{u_2}\cdots\binom{n}{u_v}\frac{B\left(\sum_{j=1}^{v} u_j + m - r + 1,\ n\nu + r - \sum_{j=1}^{v} u_j\right)}{B(r,\ m-r+1)}.$$

The last result follows again by using the fact that $F = G$ when the process is IC, and simplifying the integral via a beta function. Hence, the unconditional IC joint distribution of exceedance statistics from any number of independent Phase II samples is distribution-free. This proves that the proposed chart is distribution-free.

■

Appendix 4.2.2 The CUSUM-rank Control Chart

Result 4.6 The conditional run-length distribution for the CUSUM-rank chart

Let us consider the upper one-sided CUSUM-rank chart proposed by Li, Tang, and Ng (2010). From Equation 4.56, we have that the charting statistic for this chart is given by

$$C_j^+ = \max\left[0, C_{j-1}^+ + \left(W_j - \frac{n(m+n+1)}{2}\right) - k\right]$$

for $j = 1,2,3,\ldots$ Suppose that the chart starts from the value u, that is, $C_0^+ = u$, with $0 \le u \le h$. For the run-length (N) equal to one, we have

$$\begin{aligned}
&P_{\mathrm{C}}(N = 1 \,|\, u, X)\\
&\quad = P_{\mathrm{C}}(C_1^+ \ge h \,|\, u, X)\\
&\quad = P_{\mathrm{C}}\left(u + W_1 - \frac{n(m+n+1)}{2} - k \ge h \,|\, u, X\right)\\
&\quad = P_{\mathrm{C}}\left(W_1 \ge \frac{n(m+n+1)}{2} + k + h - u \,|\, u, X\right)\\
&\quad = P_{\mathrm{C}}\left(W \ge \frac{n(m+n+1)}{2} + k + h - u \,|\, u, X\right).
\end{aligned}$$

For $t > 1$, we have that

$$\begin{aligned}
&P_{\mathrm{C}}(N = t \,|\, u, X) \\
&\quad = P_{\mathrm{C}}(N = t,\, C_1^+ = 0 \,|\, u, X) + P_{\mathrm{C}}(N = t,\, 0 < C_1^+ < h \,|\, u, X) \\
&\quad = P_{\mathrm{C}}(N = t | C_1^+ = 0,\, u, X) \cdot P_{\mathrm{C}}(C_1^+ = 0 \,|\, u, X) \\
&\qquad + P_{\mathrm{C}}(N = t \,|\, 0 < C_1^+ < h,\, u, X) \cdot P_{\mathrm{C}}(0 < C_1^+ < h \,|\, u, X)
\end{aligned}$$

by properties of marginal, joint, and conditional probabilities.

Now, when $C_1^+ = 0$, we have that $P_{\mathrm{C}}(N = t, | C_1^+ = 0,\, u, X) = P_{\mathrm{C}}(N = t - 1, | \, u = 0, X)$.

Knowing that

$$C_1^+ = \max\left[0, u + \left(W_1 - \frac{n(m+n+1)}{2}\right) - k\right]$$

and when $C_1^+ = 0$, we can infer that

$$\begin{aligned}
&P_{\mathrm{C}}(C_1^+ = 0 \,|\, u, X) \\
&\quad = P_{\mathrm{C}}\left(u + \left(W_1 - \frac{n(m+n+1)}{2}\right) - k \le 0 \,|\, u, X\right) \\
&\quad = P_{\mathrm{C}}\left(W_1 \le k + \frac{n(m+n+1)}{2} - u \,|\, u, X\right).
\end{aligned}$$

Therefore

$$\begin{aligned}
&P_{\mathrm{C}}(N = t, | C_1^+ = 0,\, u, X) \cdot P_{\mathrm{C}}(C_1^+ = 0 \,|\, u, X) \\
&\quad + P_{\mathrm{C}}(N = t \,|\, 0 < C_1^+ < h,\, u, X) \cdot P_{\mathrm{C}}(0 < C_1^+ < h \,|\, u, X) \\
&= P_{\mathrm{C}}(N = t - 1 \,|\, u = 0, X) \cdot P_{\mathrm{C}}\left(W_1 \le k + \frac{n(m+n+1)}{2} - u \,|\, u, X\right) \\
&\quad + \sum_{v \varepsilon V} (P_{\mathrm{C}}(N = t \,|(C_1^+ = v), (C_0^+ = u), X) \cdot P_{\mathrm{C}}(C_1^+ = v \,|(C_0^+ = u), X)) \\
&= P_{\mathrm{C}}(N = t - 1 \,|u = 0, X) \cdot P_{\mathrm{C}}\left(W \le k + \frac{n(m+n+1)}{2} - u \,|\, u, X\right) \\
&\quad + \sum_{v \varepsilon V} \left(P_{\mathrm{C}}(N = t-1 | v, X) \cdot P_{\mathrm{C}}\left(W = v + \frac{n(m+n+1)}{2} + k - u \,|\, u, X\right)\right)
\end{aligned}$$

where the set V contains all the values that C_1^+ can take strictly between 0 and h, excuding 0 and h. The conditional run-length distribution can

then be calculated recursively. The conditional run-length distribution for the lower one-sided CUSUM-rank chart can be obtained similarly and is shown below. Again, suppose that u signifies that the chart starts from value u, that is, $C_0^- = u$, with $0 \le u \le h$. We have that $C_1^- = \max[0, C_0^- - \left(W_1 - \frac{n(m+n+1)}{2}\right) - k]$ so that

$$\begin{aligned} P_{\mathrm{C}}(N = 1 \,|\, u, X) &= P_{\mathrm{C}}(C_1^- \ge h \,|\, u, X) \\ &= P_{\mathrm{C}}\left(u - \left(W_1 - \frac{n(m+n+1)}{2}\right) - k \ge h | \, u, X\right) \\ &= P_{\mathrm{C}}\left(W_1 \ge k - u - \frac{n(m+n+1)}{2} + h | \, u, X\right) \\ &= P_{\mathrm{C}}\left(W \ge k - u - \frac{n(m+n+1)}{2} + h | \, u, X\right). \end{aligned}$$

Although not all the steps are shown for $t > 1$ for the lower one-sided CUSUM-rank chart, the derivations are similar to those of the upper one-sided CUSUM-rank chart and only the final expression is given below. For $t > 1$ we have that

$$\begin{aligned} &P_{\mathrm{C}}(N = t | \, u, X) \\ &= P_{\mathrm{C}}(N = t - 1 \,|\, u = 0, X) \cdot P_{\mathrm{C}}\left(W \ge u + \frac{n(m+n+1)}{2} - k \,|\, u, X\right) \\ &\quad + \sum_{v \varepsilon V} \left(P_{\mathrm{C}}(N = t-1 | v, X) \cdot P_{\mathrm{C}}\left(W = u + \frac{n(m+n+1)}{2} - k - v \,|\, u, X\right)\right) \end{aligned}$$

where the set V contains all the values that C_1^- can take strictly between 0 and h, excuding 0 and h.

Result 4.7 The unconditional run-length distribution for the CUSUM-rank chart

In the paper by Li, Tang, and Ng (2010), the unconditional run-length distribution can be found from

$$P_{\mathrm{U}}(N = t) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} P_{\mathrm{C}}(N = t | \, u, X) dF(x_1) \cdots dF(x_n).$$

Since this is an n-dimentional integral and $P_{\mathrm{C}}(N = t | \, u, X)$ is generally not in a closed form, a direct numerical solution to the n-dimentional integral above is difficult to obtain. Alternatively, one can approximate the

n-dimentional integral by making use of

$$P_{\mathrm{U}}(N=t) \approx \frac{1}{Sim}\sum_{i=1}^{Sim} P_{\mathrm{C}}(N=t\,|\,u, X_j)$$

where Sim is the number of Monte Carlo simulations and X_j is the jth randomly generated reference sample.

Appendix 4.3 EWMA CONTROL CHARTS

Appendix 4.3.1 The EWMA-SN Control Chart

Result 4.8 Derivation of the CL and Control Limits of the EWMA-SN Control Chart

By the definition of the charting statistic (see Equation 4.13), and using recursive substitution, we find that

$$\begin{aligned} E(Z_i|\mathrm{IC}) &= E(\lambda SN_i + (1-\lambda)Z_{i-1}|\mathrm{IC}) \\ &= E\left(\lambda\sum_{j=0}^{i-1}(1-\lambda)^j SN_{i-j} + (1-\lambda)^i Z_0|\mathrm{IC}\right). \end{aligned}$$

Now, using the fact that $E(SN_{i-j}|\mathrm{IC}) = 0$ and $Z_0 = 0$, it follows that

$$E(Z_i|\mathrm{IC}) = \lambda\sum_{j=0}^{i-1}(1-\lambda)^j E(SN_{i-j}|\mathrm{IC}) + (1-\lambda)^i Z_0 = 0.$$

Hence the result. In order to calculate the variance of the EWMA charting statistic, similar steps are followed, that is, we once again use the definition of the charting statistic (see Equation 4.13), and recursive substitution. However, we also use the result for the sum of a finite geometric series, properties of variance of independent random variables, and the fact that $var(SN_{i-j}|\mathrm{IC}) = n$ when $p = 0.5$. Consequently

$$\begin{aligned} var(Z_i|\mathrm{IC}) &= var\left(\lambda\sum_{j=0}^{i-1}(1-\lambda)^j SN_{i-j} + (1-\lambda)^i Z_0|\mathrm{IC}\right) \\ &= \lambda^2\sum_{j=0}^{i-1}(1-\lambda)^{2j} var(SN_{i-j}|\mathrm{IC}) \end{aligned}$$

$$= n\lambda^2 \sum_{j=0}^{i-1} (1-\lambda)^{2j}$$
$$= n\lambda^2 \left(\frac{1-(1-\lambda)^{2i}}{1-(1-\lambda)^2} \right)$$
$$= n\lambda \left(\frac{1-(1-\lambda)^{2i}}{2-\lambda} \right).$$

Therefore, $SD(Z_i|\text{IC}) = \sigma_{Z_i|\text{IC}} = \sqrt{\frac{\lambda n}{2-\lambda}(1-(1-\lambda)^{2i})}$.

Finally, since $LCL/UCL = E(Z_i|\text{IC}) \pm L\sigma_{Z_i|\text{IC}}$ and $CL = E(Z_i|\text{IC})$, we obtain the control limits and CL given in Equation 4.14.

Appendix 4.3.2 The EWMA-SR Control Chart

Result 4.9 Derivation of the CL and Control Limits of the EWMA-SR Control Chart

By the definition of the charting statistic (see Equation 4.20), and using recursive substitution, we obtain the following result

$$E(Z_i|\text{IC}) = E(\lambda SR_i + (1-\lambda)Z_{i-1}|\text{IC})$$
$$= E\left(\lambda \sum_{j=0}^{i-1} (1-\lambda)^j SR_{i-j} + (1-\lambda)^i Z_0 |\text{IC} \right).$$

Using the fact that $E(SR_{i-j}|\text{IC}) = 0$ and $Z_0 = 0$, we have that

$$E(Z_i|\text{IC}) = \lambda \sum_{j=0}^{i-1} (1-\lambda)^j E(SR_{i-j}|\text{IC}) + (1-\lambda)^i Z_0 = 0.$$

In order to obtain the variance, similar steps are followed, that is, we once again use the definition of the charting statistic (see Equation 4.20) and recursive substitution. However, we also use the result for the sum of a finite geometric series and the fact that $var(SR_i|\text{IC}) = \frac{n(n+1)(2n+1)}{6}$.

Consequently, we have that

$$\begin{aligned} var(Z_i|\text{IC}) &= var\left(\lambda\sum_{j=0}^{i-1}(1-\lambda)^j SR_{i-j} + (1-\lambda)^i Z_0|\text{IC}\right) \\ &= \lambda^2\sum_{j=0}^{i-1}(1-\lambda)^{2j} var(SR_{i-j}|\text{IC}) \\ &= \left(\frac{n(n+1)(2n+1)}{6}\right)\lambda^2\sum_{j=0}^{i-1}(1-\lambda)^{2j} \\ &= \left(\frac{n(n+1)(2n+1)}{6}\right)\lambda^2\left(\frac{1-(1-\lambda)^{2i}}{1-(1-\lambda)^2}\right). \end{aligned}$$

Therefore

$$SD(Z_i|\text{IC}) = \sigma_{Z_{i|\text{IC}}} = \sqrt{\left(\frac{n(n+1)(2n+1)}{6}\right)\left(\frac{\lambda}{2-\lambda}\right)(1-(1-\lambda)^{2i})}.$$

Finally, since $LCL/UCL = E(Z_i|\text{IC}) \pm L\sigma_{Z_i|\text{IC}}$ and $CL = E(Z_i|\text{IC})$, we obtain the control limits and CL given in Equation 4.21.

Appendix 4.3.3 The EWMA-EX Control Chart

Result 4.10 The exact time-varying control limits for the EWMA-EX chart

For exact time-varying control limits, we have

$$E(Z_j|\text{ IC}) = n(1-a)(1-(1-\lambda)^j)$$

and

$$\begin{aligned} &SD(Z_j|\text{ IC}) \\ &= \sqrt{\left(\frac{na(1-a)}{m+2}\right)\left(n(1-(1-\lambda)^j)^2 + \frac{\lambda(m+1)}{2-\lambda}(1-(1-\lambda)^{2j})\right)} \end{aligned}$$

with $LCL/UCL = E(Z_j|\text{ IC}) \pm L \times SD(Z_j|\text{ IC})$ and $CL = E(Z_j|\text{ IC})$.

Proof:
The conditional IC expected value of the charting statistic $Z_j = \lambda U_{j,r} + (1-\lambda)Z_{j-1}$ By using properties of expected values, recursive substitution,

and a finite geometric series together with the fact that $E(U_{j,r}|X_{(r)}) = np_r \; \forall j \, (j = 1, 2, \dots)$ and $Z_0 = 0$, we obtain the following

$$\begin{aligned} &E(Z_j|X_{(r)}) \\ &= E(\lambda U_{j,r} + (1-\lambda)Z_{j-1}|X_{(r)}) = E\left(\lambda \sum_{k=0}^{j-1} (1-\lambda)^k U_{j-k,r} + (1-\lambda)^j Z_0 | X_{(r)}\right) \\ &= np_r(1-(1-\lambda)^j). \end{aligned}$$

The conditional IC variance and standard deviation of the charting statistic $Z_j = \lambda U_{j,r} + (1-\lambda)Z_{j-1}$ By using properties of variance, recursive substitution, and a finite geometric series, along with the fact that $VAR(U_{j,r}|X_{(r)}) = np_r(1-p_r) \forall j \, (j = 1, 2, \dots)$, we obtain

$$\begin{aligned} var(Z_j|X_{(r)}) &= var\left(\lambda \sum_{k=0}^{j-1} (1-\lambda)^k U_{j-k,r} + (1-\lambda)^j Z_0 | X_{(r)}\right) \\ &= np_r(1-p_r)\lambda^2 \left(\frac{1-(1-\lambda)^{2j}}{1-(1-\lambda)^2}\right). \end{aligned}$$

The last expression simplifies to $np_r(1-p_r)\lambda\left(\dfrac{1-(1-\lambda)^{2j}}{2-\lambda}\right)$ so that

$$SD(Z_j|X_{(r)}) = \sqrt{\frac{\lambda}{2-\lambda}(1-(1-\lambda)^{2j})np_r(1-p_r)}.$$

In order to obtain the unconditional properties of the charting statistic, we first have to provide some properties of p_r.

Properties of p_r In the IC case, $F(X_{(r)}) = 1 - p_r$ follows a Beta distribution with parameters r and $m + 1 - r$.

Hence

$$E(p_r) = \frac{m+1-r}{m+1}$$

and $E(p_r^2) = \dfrac{m^2+3m+2-2rm-3r+r^2}{(m+1)(m+2)}$ so that $var(p_r) = \dfrac{r(m+1-r)}{(m+1)^2(m+2)}$.

The unconditional IC expected value of the charting statistic $Z_j = \lambda U_{j,r} + (1-\lambda)Z_{j-1}$

$$\begin{aligned} E(Z_j) &= E_{X_{(r)}} E(Z_j \mid X_{(r)}) \\ &= E_{X_{(r)}}(np_r(1-(1-\lambda)^j)) = n(1-(1-\lambda)^j)E_{X_{(r)}}(p_r) \\ &= n(1-(1-\lambda)^j)\left(\frac{m+1-r}{m+1}\right) = n(1-(1-\lambda)^j)(1-a) \end{aligned}$$

where $a = r/(m+1)$.

The unconditional IC variance and standard deviation of the charting statistic $Z_j = \lambda U_{j,r} + (1-\lambda)Z_{j-1}$

$$
\begin{aligned}
var(Z_j) &= var[E(Z_j|X_{(r)})] + E[var(Z_j|X_{(r)})] \\
&= var[np_r(1-(1-\lambda)^j)] + E\left[\frac{\lambda}{2-\lambda}(1-(1-\lambda)^{2j})np_r(1-p_r)\right] \\
&= n^2(1-(1-\lambda)^j)^2 var(p_r) + \frac{\lambda}{2-\lambda}(1-(1-\lambda)^{2j})n(E(p_r)-E(p_r^2)) \\
&= n^2\frac{r(m-r+1)}{(m+1)^2(m+2)}(1-(1-\lambda)^j)^2 \\
&\quad + \frac{\lambda}{2-\lambda}(1-(1-\lambda)^{2j})n\frac{r(m-r+1)}{(m+1)(m+2)} \\
&= n^2\frac{r(m-r+1)}{(m+1)(m+1)(m+2)}(1-(1-\lambda)^j)^2 \\
&\quad + \frac{\lambda}{2-\lambda}(1-(1-\lambda)^{2j})n\frac{r(m-r+1)}{(m+1)(m+2)} \\
&= n\frac{a(1-a)}{(m+2)}\left(n(1-(1-\lambda)^j)^2 + \frac{\lambda(m+1)}{2-\lambda}(1-(1-\lambda)^{2j})\right)
\end{aligned}
$$

where $a = r/(m+1)$.

Therefore

$$
\begin{aligned}
&SD(Z_j|\text{ IC}) \\
&\quad = \sqrt{\left(\frac{na(1-a)}{m+2}\right)\left(n(1-(1-\lambda)^j)^2 + \frac{\lambda(m+1)}{2-\lambda}(1-(1-\lambda)^{2j})\right)}.
\end{aligned}
$$

■

Result 4.11 The steady-state control limits for the EWMA-EX chart

For steady-state control limits, we have

$$E(Z_j|\text{ IC}) = n(1-a)$$

and

$$SD(Z_j|\text{ IC}) = \sqrt{\left(\frac{na(1-a)}{m+2}\right)\left(n + \frac{\lambda(m+1)}{2-\lambda}\right)}$$

with $LCL/UCL = E(Z_j|\text{ IC}) \pm L \times SD(Z_j|\text{ IC})$ and $CL = E(Z_j|\text{ IC})$.

Proof:

The conditional IC expected value of the charting statistic $Z_j = \lambda U_{j,r} + (1-\lambda)Z_{j-1}$ By using properties of expected values, recursive substitution, and a finite geometric series, together with the fact that $E(U_{j,r}|X_{(r)}) = np_r \; \forall j \; (j = 1, 2, \dots)$ and $Z_0 = np_r$, we obtain the following

$$
\begin{aligned}
E(Z_j|X_{(r)}) &= E(\lambda U_{j,r} + (1-\lambda)Z_{j-1}|X_{(r)}) \\
&= E\left(\lambda \sum_{k=0}^{j-1} (1-\lambda)^k U_{j-k,r} + (1-\lambda)^j Z_0 | X_{(r)}\right) \\
&= np_r(1-(1-\lambda)^j) + np_r(1-\lambda)^j = np_r
\end{aligned}
$$

The conditional IC variance and standard deviation of the charting statistic $(Z_j = (\lambda U_{j,r} + (1-\lambda)Z_{j-1})$ By using properties of variance, recursive substitution, and a finite geometric series, along with the fact that $var(U_{j,r}|X_{(r)}) = np_r(1-p_r)\forall j\,(j = 1, 2, \dots)$, we obtain the following

$$
\begin{aligned}
&var(Z_j|X_{(r)}) \\
&= var\left(\lambda \sum_{k=0}^{j-1} (1-\lambda)^k U_{j-k,r} + (1-\lambda)^j Z_0 | X_{(r)}\right) \\
&= np_r(1-p_r)\lambda^2 \left(\frac{1-(1-\lambda)^{2j}}{1-(1-\lambda)^2}\right).
\end{aligned}
$$

The last expression simplifies to $np_r(1-p_r)\lambda\left(\dfrac{1-(1-\lambda)^{2j}}{2-\lambda}\right)$ so that

$$
SD(Z_j|X_{(r)}) = \sqrt{\frac{\lambda}{2-\lambda}(1-(1-\lambda)^{2j})np_r(1-p_r)}.
$$

However, the term $(1-(1-\lambda)^{2j})$ approaches unity as $j \to \infty$, so that the term can be dropped.

$$
SD(Z_j|X_{(r)}) = \sqrt{\frac{\lambda}{2-\lambda}np_r(1-p_r)}.
$$

In order to obtain the unconditional properties of the charting statistic, we first have to provide some properties of p_r.

Properties of p_r In the IC case, $F(X_{(r)}) = 1-p_r$ follows a Beta distribution with parameters r and $m+1-r$.

Hence

$$
E(p_r) = \frac{m+1-r}{m+1} \quad \text{and} \quad E(p_r^2) = \frac{m^2+3m+2-2rm-3r+r^2}{(m+1)(m+2)}
$$

$$
\text{so that} \quad var(p_r) = \frac{r(m+1-r)}{(m+1)^2(m+2)}.
$$

The unconditional IC expected value of the charting statistic $Z_j = \lambda U_{j,r} + (1-\lambda)Z_{j-1}$

$$\begin{aligned} E(Z_j) &= E_{X_{(r)}}(Z_j \mid X_{(r)}) = E_{X_{(r)}}(np_r) = nE_{X_{(r)}}(p_r) \\ &= n\left(\frac{m+1-r}{m+1}\right) = n(1-a) \end{aligned}$$

where $a = r/(m+1)$.

The unconditional IC variance and standard deviation of the charting statistic $Z_j = \lambda U_{j,r} + (1-\lambda)Z_{j-1}$

$$\begin{aligned} var(Z_j) &= var[E(Z_j|X_{(r)})] + E[var(Z_j|X_{(r)})] \\ &= var[np_r] + E\left[\frac{\lambda}{2-\lambda}np_r(1-p_r)\right] \\ &= n^2 var(p_r) + \frac{\lambda}{2-\lambda}n(E(p_r) - E(p_r^2)) \\ &= n^2\frac{r(m-r+1)}{(m+1)^2(m+2)} + \frac{\lambda}{2-\lambda}n\frac{r(m-r+1)}{(m+1)(m+2)} \\ &= n^2\frac{r(m-r+1)}{(m+1)(m+1)(m+2)} + \frac{\lambda}{2-\lambda}n\frac{r(m-r+1)}{(m+1)(m+2)} \\ &= n\frac{a(1-a)}{(m+2)}\left(n + \frac{\lambda(m+1)}{2-\lambda}\right) \end{aligned}$$

where $a = r/(m+1)$.

Therefore

$$SD(Z_j|\text{ IC}) = \sqrt{\left(\frac{na(1-a)}{m+2}\right)\left(n + \frac{\lambda(m+1)}{2-\lambda}\right)}.$$ ■

Appendix 4.3.4 The EWMA-rank Control Chart

Result 4.12 The conditional run-length distribution for the EWMA-rank chart

The charting statistic for the EWMA-rank chart, proposed by Li, Tang, and Ng (2010), is given by $Z_j = \lambda\, W_j + (1-\lambda)Z_{j-1}$ for $j = 1, 2, 3, \ldots$ (see Equation 4.61). Suppose that u represents the initial value of the charting statistic, that is, $Z_0 = u$.

For the run-length (N) to be equal to one, we have that

$$P_{\text{C}}(N = 1 \mid u, X)$$

$$
\begin{aligned}
&= P_{\mathrm{C}}\left(\left(Z_1 - \frac{n(m+n+1)}{2} \le LCL\right)\right. \\
&\qquad \left. \cup \left(Z_1 - \frac{n(m+n+1)}{2} \ge UCL\right) \mid u, X\right) \\
&= P_{\mathrm{C}}\left(\left(\lambda\, W_1 + (1-\lambda)u - \frac{n(m+n+1)}{2} \le LCL\right)\right. \\
&\qquad \left. \cup \left(\lambda\, W_1 + (1-\lambda)u - \frac{n(m+n+1)}{2} \ge UCL\right) \mid u, X\right) \\
&= P_{\mathrm{C}}\left(W_1 \le \frac{n(m+n+1) + 2LCL - 2(1-\lambda)u}{2\lambda} \mid u, X\right) \\
&\quad + P_{\mathrm{C}}\left(W_1 \ge \frac{n(m+n+1) + 2UCL - 2(1-\lambda)u}{2\lambda} \mid u, X\right) \\
&= P_{\mathrm{C}}\left(W \le \frac{n(m+n+1) + 2LCL - 2(1-\lambda)u}{2\lambda} \mid u, X\right) \\
&\quad + P_{\mathrm{C}}\left(W \ge \frac{n(m+n+1) + 2UCL - 2(1-\lambda)u}{2\lambda} \mid u, X\right).
\end{aligned}
$$

Making use of the recursive approach, we have that

$$
\begin{aligned}
&P_{\mathrm{C}}(N = t \mid u, X) \quad \text{for} \quad t = 1, 2, 3 \ldots \\
&= P_{\mathrm{C}}\left(N = t,\ \frac{n(m+n+1)}{2} + LCL < Z_1 < \frac{n(m+n+1)}{2} + UCL \mid u, X\right) \\
&= P_{\mathrm{C}}\left(N = t \mid \left(\frac{n(m+n+1)}{2} + LCL < Z_1 < \frac{n(m+n+1)}{2} + UCL\right), u, X\right) \\
&\quad \cdot P_{\mathrm{C}}\left(\frac{n(m+n+1)}{2} + LCL < Z_1 < \frac{n(m+n+1)}{2} + UCL \mid u, X\right) \\
&= \sum_{v \epsilon V} (P_{\mathrm{C}}(N = t \mid Z_1 = v, Z_0 = u, X)\ P_{\mathrm{C}}(Z_1 = v) \mid Z_0 = u, X)
\end{aligned}
$$

where the set V contains all possible values that Z_1 can take on between the LCL and the UCL. Since it is known that Z_1 takes on a value between the LCL and the UCL, we can make use of the fact that the process is IC at time 1 and we get

$$
\begin{aligned}
&= \sum_{v \epsilon V} (P_{\mathrm{C}}(N = t - 1 \mid Z_0 = u, X)\ P_{\mathrm{C}}(Z_1 = v) \mid Z_0 = u, X) \\
&= \sum_{v \epsilon V} \left(P_{\mathrm{C}}(N = t - 1 \mid v, X)\ P_{\mathrm{C}}\left(W = \frac{v - (1-\lambda)u}{\lambda} \mid u, X\right)\right).
\end{aligned}
$$

CHAPTER 5

Miscellaneous Univariate Distribution-free (Nonparametric) Variables Control Charts

Chapter Overview

In this chapter, we consider some miscellaneous univariate distribution-free variables control charts. Examples include Phase I control charts, adaptive control charts, bootstrap control charts, and change-point models.

5.1 INTRODUCTION

So far, we have discussed most of the available univariate nonparametric control charts that researchers can use in practice. Here, we cover some miscellaneous charts, which range from Phase I control charts to adaptive control charts and some others. Nonparametric Phase I charts are of particular interest since typically not much knowledge about the process, including the distribution of the output, is initially available. The adaptations considered are along the lines of their parametric counterparts recently considered in the literature. This includes variable sample size (VSS) or variable sampling interval (VSI) or possibly a combination of both (VSSI), which have shown to be faster at detecting shifts in the process than the traditional

Nonparametric Statistical Process Control, First Edition.
Subhabrata Chakraborti and Marien Alet Graham.
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fixed sampling size (FSS) and fixed sampling interval (FSI) control charts. More research on these interesting topics is currently underway.

5.2 OTHER UNIVARIATE DISTRIBUTION-FREE (NONPARAMETRIC) VARIABLES CONTROL CHARTS

5.2.1 Phase I Control Charts

5.2.1.1 Introduction

As we noted in Chapter 2, there are usually two phases in the overall statistical process control (SPC) regime, Phase I and II. In Phase I, a retrospective analysis is performed on existing or historical process data, mainly to establish control. The Phase I analysis includes planning, administration, data collection, data management, and exploratory data analysis, among other steps. Control charts are also used in this phase to establish process control by eliminating points that plot out-of-control (OOC) and can be traced to assignable causes. Thus, control charts (limits) in Phase I may be initially viewed as trial limits, which are possibly updated (recalculated) until control is established, which means that all charting statistics plot within the control limits, target quality is achieved and no non-random pattern is detected. At that stage, the data at hand make up an in-control (IC) or a reference sample from which parameter estimates and control limits are calculated. As noted earlier, Phase I is now recognized as a key step in the overall SPC regime and nonparametric charts are thought to be particularly suitable in this phase since the underlying distribution is unknown and not much information about it is available initially. In Phase II, the control limits developed in Phase I are used for the prospective process monitoring of incoming (new/test) samples of data. Recall that, if the process parameters are known or specified, we refer to the situation as Case K. Alternatively, if the process parameters are unknown and need to be estimated, which is usually done after a successful Phase I analysis that yields a reference sample, we refer to that situation as Case U.

While similar types of control charts are used in both Phase I and II, the most notable difference are the criteria (the measures or the metrics) that are used to implement and evaluate the process in the two phases, consistent with the objectives of the particular process monitoring. In Phase I, we use the false alarm probability (FAP) as the chart design/performance measure, whereas in Phase II, we use characteristics of the run-length distribution such as the average run-length (ARL), the standard deviation

of the run-length (*SDRL*), the median run-length (*MRL*), or some other percentiles of the run-length distribution to assess chart performance. On the one hand, the *FAP* is defined as the probability of observing at least one false alarm out of many comparisons (of all Phase I samples) done simultaneously. The false alarm rate (*FAR*), on the other hand, is defined as the probability of observing a single false alarm in a single comparison of the sample at hand to the control limits. The two performance metrics, the *FAP* and the *FAR*, are related as the *FAP* can be expressed in terms of the *FAR* and vice versa, as will be seen later.

Without any loss of generality, let m denote the "final" number of samples at the end of a Phase I analysis when the process is deemed to be IC. Suppose that each of these samples has n observations and so the reference data set has a total of $N = mn$ observations. Note that in Chapter 4, for distribution-free charts in Case U, we assumed that a reference sample of size m was available. This was done mainly to be consistent with the notation in the literature. For a two-sided control chart, an OOC situation is indicated when a charting statistic C_i falls on or outside of either of the two control limits. Recall that this important event is often called a signal or a signaling event. In some cases, it is convenient to consider the complementary event, which is when a subgroup does not signal, called the non-signaling event. Thus, for the ith subgroup, $E_i = \{L\hat{C}L < C_i < U\hat{C}L\}$, $i = 1,2,\ldots,m$, denotes the non-signaling event. Note that $L\hat{C}L$ and $U\hat{C}L$ denote the estimated Phase I control limits and C_i denotes the charting statistic. As we discussed earlier, the *FAP* is usually the recommended chart design criterion adopted in Phase I (see, for example, Chakraborti, Human, and Graham, 2009). This probability can be expressed as follows

$$\begin{aligned} FAP &= P(\text{At least one false alarm from the } m \text{ subgroups}) \\ &= 1 - P(\text{No signal among the } m \text{ subgroups } |\text{IC}) \\ &= 1 - P(E_1, E_2, \ldots, E_m \mid \text{IC}) \\ &= 1 - P\left(\bigcap_{i=1}^{m} \{L\hat{C}L < C_i < U\hat{C}L\} | \text{IC}\right) \\ &= 1 - \int_{L\hat{C}L}^{U\hat{C}L} \int_{L\hat{C}L}^{U\hat{C}L} \cdots \int_{L\hat{C}L}^{U\hat{C}L} f_{C_1,C_2,\ldots,C_m;\text{IC}}(c_1, c_2, ..., c_m) dc_1 ... dc_m, \end{aligned} \tag{5.1}$$

where the charting statistics $C_1, C_2, \ldots, C_m$ are continuous and $f_{C_1,C_2,\ldots,C_m;\text{IC}}(c_1, c_2, \ldots, c_m)$ denotes their joint pdf when the process is IC.

If the charting statistics are discrete, then

$$FAP = 1 - \sum_{c_1=L\hat{C}L+1}^{U\hat{C}L-1} \sum_{c_2=L\hat{C}L+1}^{U\hat{C}L-1} \cdots \sum_{c_m=L\hat{C}L+1}^{U\hat{C}L-1} f_{C_1,C_2,\ldots,C_m}(c_1, c_2, \ldots, c_m) dc_1 \ldots dc_m$$

where $f_{C_1,C_2,\ldots,C_m}(c_1, c_2, \ldots, c_m)$ denotes the joint pmf of the charting statistics C_1, C_2, $\ldots$, C_{m} when the process is IC. Note that if the charting statistics are independent, the joint pdf or the pmf simplifies in to the product of the marginal pdfs or the pmfs, which results in a simplified expression for the *FAP*. But in Phase I, the charting statistics are not independent, as we explain next and this dependence needs to be accounted for in setting up the control charts.

The *FAP* involves the probability of the m non-signaling (or, equivalently, signaling) events simultaneously, and depends on the fact that the control limits are estimated in Case U. These non-signaling (and also the signaling) events are dependent since the charting statistics are all compared simultaneously with the same pair of control limits. Hence, the calculation of the *FAP* requires knowledge of the joint distribution of the charting statistics when the process is IC. The derivation of this distribution and the determination of the control limits (and the associated charting constants, which differ depending on the type of Phase I control chart under consideration) pose some important practical challenges in an SPC context where the number of comparisons (m) can be as low as 25 or as high as 300 or more.

Conversely, the *FAR*, which is the probability of a single charting statistic plotting on or outside the control limits when the process is IC, can be expressed as

$$\begin{aligned} FAR &= 1 - P(L\hat{C}L < C_i < U\hat{C}L|\mathrm{IC}) \\ &= \begin{cases} 1 - \displaystyle\int_{L\hat{C}L}^{U\hat{C}L} g_{C_i}(c_i)dc_i \\ 1 - \displaystyle\sum_{L\hat{C}L+1}^{U\hat{C}L-1} g_{C_i}(c_i) \end{cases} \end{aligned} \tag{5.2}$$

respectively, depending on whether C_i is continuous or discrete with a pdf or a pmf $g_{C_i}(c_i)$ when the process is IC. It can be seen that the *FAR* involves a single non-signaling event and thus the calculation of this probability requires only the marginal distribution of the ith charting statistic C_i when the process is IC. It should be noted that Equation 5.2 is the general formula to calculate a *FAR* regardless of the phase that the process is in. However, the *FAR* has interpretability problems in both phases, since the signaling events are dependent.

Clearly, the success of the overall monitoring regime depends on a careful and well-thought-out Phase I analysis. This is now well recognized in the literature and several authors have recently recommended paying particular attention to the Phase I analysis, including the control charts. Among the many types of available control charts, Shewhart charts are mostly recommended in Phase I because they are easy to construct and they are effective in detecting both large and sustained shifts in the process parameters and outliers, measurement errors, data recording, and/or transmission errors

and the like. Nonparametric charts also seem to be natural in a Phase I setting since the underlying distribution is unknown at the outset, and we want to apply charts that remain valid under broad conditions. Yet, only a few Phase I Shewhart nonparametric charts have been proposed in the literature for the Phase I monitoring of location or scale. We describe some of these control charts below.

A good overview of Phase I SPC and control charts is given in Chakraborti, Human, and Graham (2009), but until recently, there has not been much interest in this important area of SPC. Capizzi and Masarotto (2013) and Jones-Farmer *et al.* (2014) have reviewed the Phase I SPC in a broader context and re-emphasized the points made in Chakraborti, Human, and Graham (2009). However, much more work remains to be done in this important area.

First, we define some concepts and discuss the various available types of Phase I control limits.

Type 1 FAR-based Control Limits

Hillier (1969) and Yang and Hillier (1970) proposed a methodology in which the *FAR* is controlled at a desired level, such as 0.0027, in order to determine the control limits. However, this method is more suitable in Case K where the signaling events are independent. Although they did consider the effects of parameter estimation, they did not explicitly consider (i) the fact that the subgroups are all compared to the same control limits, so the signaling events are statistically dependent and (ii) that many subgroups are all compared to the same control limits at the same time (so there is the issue of simultaneous comparisons that needs to be accounted for). As noted by several authors in the literature, these reasons limit the applicability of their methods in a Phase I setting since there will be too many false alarms with a chart constructed for a fixed *FAR*.

Type 2 FAP-based Control Limits

The *FAP*-based charts are generally recommended in Phase I where *FAP* denotes the false alarm probability, which has been defined earlier as the probability of at least one false alarm. The *FAP*-based control limits are calculated for a specified nominal false alarm probability (FAP_0, say). Note that the *FAP* metric takes into account the fact that the signaling events are dependent and correctly uses the relevant joint (multivariate) distribution of the charting statistics to determine the limits, and thus accounts for the issue of simultaneity. Hence, The *FAP*-based charts are to be used in Phase I, where we basically check for homogeneity of the groups. Typical nominal FAP_0 values used in practice include 0.10 and 0.20, which are higher than a typically used nominal FAR_0 value, such as 0.0027. This is to recognize the fact that multiple groups are compared, simultaneously, to a set of control limits in a Phase I setting.

Type 3 Bonferroni-adjusted (BA) Control Limits

While the *FAR* (see Type 1) is an important chart performance criterion in some applications, particularly in Case K, as noted before, the typical chart performance criterion in Phase I applications is the *FAP* (see Type 2). Since the calculation of *FAP* involves dealing with the joint distribution of the charting statistics, which may or may not be tractable in a given application, a simpler approximation to the *FAP* in terms of the *FAR* can be useful. This can be obtained via the Boole (or the Bonferroni) inequality (see, for example, Casella and Berger, 2002, p. 13) and the corresponding Phase I control limits may be referred to as the Bonferroni-adjusted (*BA*) limits. Note that since

$$FAP = P(\text{at least 1 false alarm}) = P\left(\bigcup_{i=1}^{m} E_i^c | IC\right) \leq \sum_{i=1}^{m} P(E_i^c | IC) = mFAR$$

an upper bound to the *FAP* is *mFAR*. So, if each of the m individual control charts uses an *FAR* of FAP_0/m, the FAP of the Phase I control chart will be approximately equal to a nominal FAP_0. Thus, for example, symmetrically placed two-sided probability based control limits can be derived from the $(FAP_0/2m)100^{th}$ and $(1-(FAP_0/2m))100^{th}$ percentiles of the marginal IC distribution of a charting statistic. Hence this approach may be more convenient in practice since the marginal IC distribution of a single charting statistic is often more readily available.

The three types of control limits are illustrated in Figure 5.1 in Case U. For this illustration, it was assumed that $m = 15$ random samples, each of size $n = 5$, are available for a Phase I analysis. The control limits for each method are given in Table 5.1 and illustrated in Figure 5.1. It is clear that

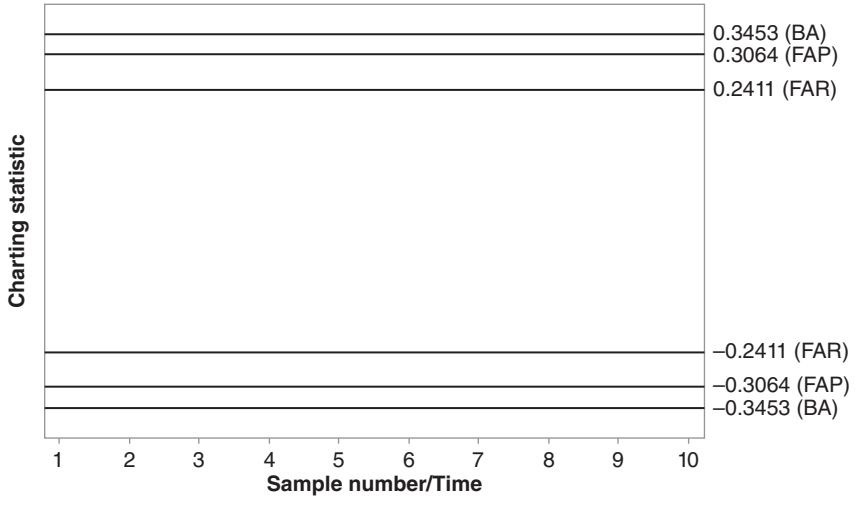


FIGURE 5.1 *The* FAR*-based, the* FAP*-based, and the* BA *Phase I control limits.*

Table 5.1 The *FAR*-based, the *FAP*-based, and the *BA* control limits.

FAR-based control limits	*FAP*-based control limits	*BA* control limits
±0.2411	±0.3064	±0.3453

the *FAR*-based control limits are the narrowest and the *BA* limits are the widest, while the exact limits based on the joint distribution are somewhere in between.

From Figure 5.1, it is seen that there can be many more false alarms if the control limits are based on the *FAR* criterion, that is, Type 1. The *BA* control limits are slightly wider than the *FAP*-based control limits, and it is thus likely that one would observe fewer false alarms if one used the *BA* control limits. Although the *BA* limits are easier to find and fewer false alarms may be appealing from a practical point of view, if the control limits are too wide, the change detection properties of the chart can be affected negatively and unwanted causes of variation might go undetected. Consequently, we will focus on the method that controls the *FAP*.

5.2.1.2 Phase I Control Charts for Location

Shewhart charts are particularly suitable in Phase I applications since they are more sensitive to larger changes. There are two nonparametric Phase I Shewhart control charts for location in the literature. First, we describe the distribution-free mean-rank chart considered by Jones-Farmer, Jordan, and Champ (2009).

5.2.1.2.1 The Mean-rank Chart Let $(X_{i1}, X_{i2}, \ldots, X_{in})$ denote an independent random sample (subgroup) of size $n > 1$, taken from the ith population with a continuous cdf F_i, $i = 1, 2, \ldots, m$. Thus, suppose that there are m independent random samples each of size n and these $N = m \times n$ data values are used in a Phase I analysis. The mean-rank chart is an adaptation of the well-known nonparametric test, called the Kruskal–Wallis test (see Gibbons and Chakraborti, 2010, p. 353).

To construct the Phase I mean-rank chart, first the observations from all the m subgroups are combined into a single array of $N = mn$ observations, arranged from the lowest to the highest and ranked (assume no ties), keeping track of the subgroup where each observation came from. Then, one calculates the total-rank (sum of the ranks) for the observations for each subgroup and hence the mean-ranks (average ranks), by dividing the total-ranks by the subgroup size. The charting statistics for

the nonparametric Phase I Shewhart mean-rank control chart are the standardized subgroup mean-ranks, given by

$$Z_i = \frac{\overline{R}_i - E(\overline{R}_i|IC)}{\sqrt{var(\overline{R}_i|IC)}} = \frac{\overline{R}_i - \left(\frac{N+1}{2}\right)}{\sqrt{(N-n)(N+1)/12n}} \tag{5.3}$$

for $i = 1, 2, \ldots, m$, where $\overline{R}_i = R_i/n$ denotes the mean-rank of the ith subgroup, and R_i denotes the total-rank for the ith subgroup in the combined sample of N observations. The expectation and the variance formulas are shown later in this section.

The values of the lower (LCL) and upper control limits (UCL) are chosen such that a specified nominal $FAP = \alpha$ is attained where α is typically taken to be small, say, $\alpha = 0.01$, 0.05, or 0.10. The process is considered IC while all the charting statistics Z_i, $i = 1, 2, \ldots, m$, fall between the two control limits, but as soon as a charting statistic falls on or outside the UCL or the LCL, the process is declared OOC. The control limits obtained early in this phase are viewed as trial limits and are often revised (as some charting statistics plot outside the control limits) and refined to ensure that the process is IC.

Let R_{ij}, where $i = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, n$, denote the rank of the jth observation from the ith sample in the pooled sample of size N. It can be shown (see Gibbons and Chakraborti, 2010, p. 354) that the expected value and the variance of the ranks, when the process is IC, are given by

$$E(R_{ij}|IC) = \frac{N+1}{2} \tag{5.4}$$

and

$$var(R_{ij}|IC) = \frac{(N-1)(N+1)}{12} \tag{5.5}$$

respectively.

The sum of ranks or the total-rank for the ith subgroup is $R_i = \sum_j R_{ij}$ and the mean-rank is the average rank for the ith subgroup is given by $\overline{R}_i = \left(\sum_j R_{ij}\right)/n$. It follows that, for an IC process, the mean and the variance of $\overline{R}_i$ are given by

$$E(\overline{R}_i|IC) = \frac{N+1}{2} \tag{5.6}$$

and

$$var(\overline{R}_i|IC) = \frac{(N-n)(N+1)}{12n} \tag{5.7}$$

respectively. The mean-rank $\overline{R}_i$ values are standardized by subtracting the IC mean and dividing by the IC standard deviation. This yields the charting statistic Z_i shown above.

Control limits for the mean-rank chart are discussed in Jones-Farmer, Jordan, and Champ (2009). Various approaches are used; one of which is that according to the central limit theorem, the random variable $Z_i = (\overline{R}_i - E(\overline{R}_i|IC))/\sqrt{var(\overline{R}_i|IC)}$ approximately follows a standard normal distribution. However, the m charting statistics $Z_1, Z_2, \ldots, Z_m$ used in the Phase I mean-rank chart are not independent and have a joint probability distribution, which complicates the calculation of the control limits. Although the exact limits can be found by enumeration of the ranks and the mean-ranks, this must be a formidable task in practice, even for moderate values of m and n. Jones-Farmer, Jordan, and Champ (2009) provide the control limits for the mean-rank chart by simulating the empirical distribution of the standardized mean-ranks for $m = 10$, 20, 30, 40, and 50, and for $n = 3$, 4, 5, 10, 15, and 20 using 100 000 iterations in each case. We calculated the limits by simulation for $m = 5$, 6, 7, 8, 9, 10, 25, 30, and 45, and for $n = 5$ and 10 using 100 000 simulations in each case; our results are presented in Table 5.2 and can be viewed as a useful supplement to what is given in Jones-Farmer, Jordan, and Champ (2009).

Jones-Farmer, Jordan, and Champ (2009) also considered using the multivariate normal distribution approximation for $(Z_1, Z_2, \ldots, Z_m)$, as well as a univariate normal approximation since the correlation coefficient between Z_i and Z_j in the approximate joint distribution can be shown to approach 0 quickly as m increases. They showed that the univariate standard normal approximation based control limits are similar to those based on the multivariate normal approximation, for example, for $m = 30$, and a nominal FAP of 0.05, the multivariate normal approximation gives control limits ± 3.128, while the univariate standard normal approximation yields ± 3.137. These approximate limits don't depend on the sample size n. However, note that for $m = 30$ the exact limits shown in Table 5.2, which were obtained via simulating the empirical distribution, are 2.895 and 3.031 for $n = 5$ and 10, respectively, so that both the multivariate and univariate approximations lead to wider control limits.

Example 5.1 Illustration of the Phase I Mean-rank Chart

Column (a) of Table 5.3 contains 25 retrospective or Phase I samples, each of size five, that were collected when the process was thought to be IC. This data set is from Montgomery (2005, Table 5.3, page 223) on the inside diameters of piston rings manufactured by a forging process. In this data set, there are 25 samples ($m = 25$) each of size 5 ($n = 5$). Thus, $N = mn = 25 \times 5 = 125$. Columns (b) and (c) of Table 5.3 show the mean-rank $\overline{R}_i$ values for $i = 1, 2, \ldots, m$ and the standardized mean-rank values, which are the charting statistics. Now, for $n = 5$ and a nominal $FAP = 0.10$ that we get from Table 5.2, the control limits are $\pm$ 2.700.

Table 5.2 Control limits for the mean-rank Phase I control chart.

		Nominal $FAP = 0.05$		Nominal $FAP = 0.1$		Nominal $FAP = 0.15$		Nominal $FAP = 0.2$	
m	n	Control limits ($\pm$)	Simulated FAP	Control limits ($\pm$)	Simulated FAP	Control limits ($\pm$)	Simulated FAP	Control limits ($\pm$)	Simulated FAP
5	5	2.452	0.0481	2.298	0.0866	2.100	0.1480	2.000	0.1882
	10	2.500	0.0494	2.290	0.0931	2.111	0.1468	2.019	0.1879
6	5	2.530	0.0483	2.310	0.0883	2.150	0.1469	2.040	0.1942
	10	2.550	0.0482	2.381	0.0947	2.201	0.1489	2.100	0.1871
7	5	2.551	0.0494	2.400	0.0969	2.223	0.1418	2.120	0.1999
	10	2.645	0.0470	2.450	0.0831	2.224	0.1499	2.125	0.1992
8	5	2.670	0.0416	2.460	0.0832	2.235	0.1500	2.151	0.1915
	10	2.671	0.0478	2.461	0.0945	2.280	0.1486	2.200	0.1859
9	5	2.666	0.0408	2.462	0.0898	2.290	0.1492	2.231	0.1867
	10	2.700	0.0479	2.470	0.0987	2.350	0.1413	2.270	0.1717
10	5	2.720	0.0411	2.474	0.0924	2.352	0.1410	2.271	0.1939
	10	2.750	0.0454	2.520	0.0992	2.399	0.1420	2.320	0.1825
25	5	2.585	0.0499	2.700	0.0983	2.600	0.1460	2.530	0.1851
	10	2.800	0.0443	2.780	0.0987	2.790	0.0953	2.600	0.1811
30	5	2.895	0.0495	2.748	0.0993	2.650	0.1450	2.550	0.1961
	10	3.031	0.0493	2.834	0.0980	2.730	0.1419	2.650	0.1872
45	5	3.149	0.0433	2.950	0.0990	2.735	0.1495	2.750	0.1944
	10	3.150	0.0433	2.960	0.0978	2.736	0.1499	2.751	0.1945

Table 5.3 Phase I data for the mean-rank chart and charting statistics.

Sample	(a)					(b)	(c)
i	X_{i1}	X_{i2}	X_{i3}	X_{i4}	X_{i5}	$\overline{R}_i$	Z_i
1	74.030	74.002	74.019	73.992	74.008	85.2	1.398
2	73.995	73.992	74.001	74.011	74.004	59.3	−0.233
3	73.998	74.024	74.021	74.005	74.002	88.1	1.581
4	74.002	73.996	73.993	74.015	74.009	68.6	0.353
5	73.992	74.007	74.015	73.989	74.014	70.3	0.460
6	74.009	73.994	73.997	73.985	73.993	40.5	−1.417
7	73.995	74.006	73.994	74.000	74.005	57.1	−0.372
8	73.985	74.003	73.993	74.015	73.988	46.3	−1.052
9	74.008	73.995	74.009	74.005	74.004	76.2	0.832
10	73.998	74.000	73.990	74.007	73.995	49.1	−0.876
11	73.994	73.998	73.994	73.995	73.990	31.1	−2.010
12	74.004	74.000	74.007	74.000	73.996	63.6	0.038
13	73.983	74.002	73.998	73.997	74.012	53.5	−0.598
14	74.006	73.967	73.994	74.000	73.984	35.7	−1.720
15	74.012	74.014	73.998	73.999	74.007	81.6	1.172
16	74.000	73.984	74.005	73.998	73.996	46.1	−1.065
17	73.994	74.012	73.986	74.005	74.007	63.3	0.019
18	74.006	74.010	74.018	74.003	74.000	87.1	1.518
19	73.984	74.002	74.003	74.005	73.997	53.2	−0.617
20	74.000	74.010	74.013	74.020	74.003	92.3	1.846
21	73.982	74.001	74.015	74.005	73.996	60.1	−0.183
22	74.004	73.999	73.990	74.006	74.009	65.2	0.139
23	74.010	73.989	73.990	74.009	74.014	68.5	0.346
24	74.015	74.008	73.993	74.000	74.010	78.6	0.983
25	73.982	73.984	73.995	74.017	74.013	54.4	−0.542

To illustrate the calculations, consider sample number 1. The IC expectation and the variance of the ranks are $E(R_{ij}|IC) = \dfrac{N+1}{2} = \dfrac{125+1}{2} = 63$ and $var(\overline{R}_i|IC) = \dfrac{(N-n)(N+1)}{12n} = \dfrac{(125-5)(125+1)}{(12)(5)} = \dfrac{15120}{60} = 252$, respectively, so that the charting statistic $Z_1 = \dfrac{85.2-63}{\sqrt{252}} = 1.398$. The rest of the $Z_i's$ are calculated similarly.

The graphical display of the Phase I mean-rank chart is shown in Figure 5.2. All of the charting statistics plot between the control limits without any obvious pattern, indicating that the process is IC. The data

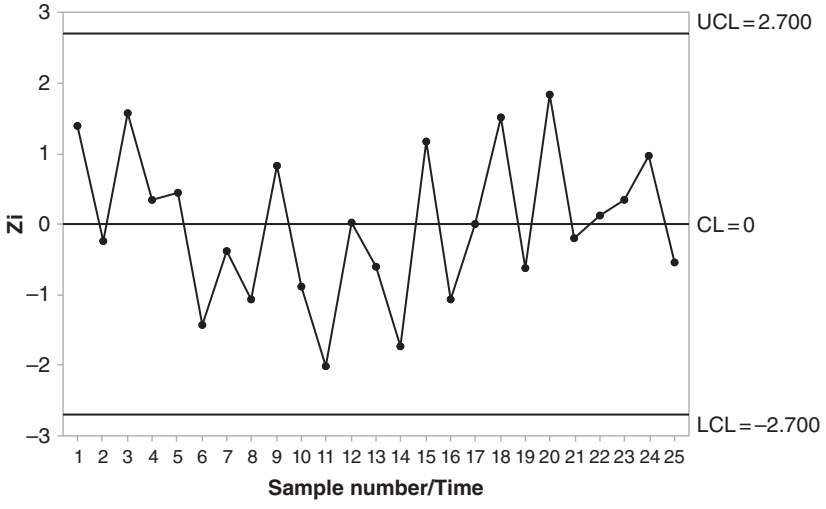


FIGURE 5.2 *A Phase I mean-rank chart.*

may now by referred to as the IC or reference data from which any necessary parameters can be estimated and then be used to find appropriate control limits, which are used for prospective process monitoring in Phase II.

Note that, even though ties are ruled out under the continuity assumptions on the distributions, in practice, it is quite likely that the data will have some tied observations. Ties are generally broken using the average of the tied ranks. If there are not many ties, one can continue to use the standardized average ranks as described here; otherwise, the formula for the IC variance of the average rank needs to be adjusted. See, for example, Gibbons and Chakraborti (2010, p. 356).

Next, we describe the Phase I median chart for location considered by Graham, Human, and Chakraborti (2010).

5.2.1.2.2 The Median Chart The median chart for the Phase I location applies in the same setting as in the case of the mean-rank chart described earlier. Thus, suppose, as before, that there are m independent random samples each of size n and these $N = mn$ reference data values are found using a Phase I analysis.

The median chart is an adaptation of Mood's median test (see Gibbons and Chakraborti, 2010, p. 355). Suppose that M denotes the median of the combined sample of N observations. The process is said to be IC when the m population medians corresponding to the $N = m$ subgroups are all equal. To construct the Phase I median chart, first the observations from the m subgroups are combined into one sample of $N = mn$ observations and arranged from the lowest to the highest. The pooled median is given as

$$M = \begin{cases} X_{((N+1)/2)} & \text{if} \quad N \text{ is odd} \\ (X_{(N/2)} + X_{((N+2)/2)})/2 & \text{if} \quad N \text{ is even} \end{cases}$$

where $X_{(1)} < X_{(2)} < \ldots < X_{(N)}$ denotes the order statistics of the combined sample. The charting statistic for the Phase I median chart is a subgroup precedence statistic, that is, the number of observations in each subgroup that precedes the pooled median, and is given by $U_i = \sum_{j=1}^{n} I(X_{ij} < M)$ for $i = 1, 2, \ldots, m$, where $I(A) = 1$ or 0 for A true or false. The values of the LCL and UCL are chosen such that a specified nominal $FAP = \alpha$ is attained where $\alpha = 0.05$ or 0.10, for example. It may be noted that the U_i resembles the sign statistic introduced in Chapter 4, where the sample median M takes the spot of the specified median. Thus U_i can be viewed as a generalized sign statistic.

An extract of the table of Graham, Human, and Chakraborti (2010) is given in Table 5.4; the reader is referred to their paper for more details. In Example 5.2, the combination of $m = 25$ and $n = 5$ is used, however, the control limits and the corresponding attained FAP values are not given in Graham, Human, and Chakraborti (2010). It should be noted that, because of the discreteness of the IC distribution of the charting statistic, the subgroup size n needs to be larger than the number of subgroups m in order to achieve a small FAP value such as 0.05 or even for an FAP value of 0.1. This can be a potential weakness of the Phase I median chart. However, note that the situation is similar to control charting with attributes data, where typically the subgroup size n is taken to be much larger than the number of subgroups m; see, for example, the attribute control charts in Chapter 7 of Montgomery (2009) and the nonparametric charts presented in Chapter 4, which are all based on charting statistics that have discrete distributions. This is why the combination $m = 25$ and $n = 5$ is not tabulated in Graham, Human, and Chakraborti (2010). However, Example 5.2 is given for illustration purposes, although the (m, n) combination of (25, 5) is not ideal for the median chart.

Example 5.2 Illustration of the Phase I Median Chart

The median chart is illustrated using the same data set from Montgomery (2005, Table 5.3, p. 223) on the inside diameters of piston rings

Table 5.4 Control limits for the median Phase I control chart (adapted from Table 1 of Graham, Human, and Chakraborti (2010)).

		Nominal $FAP = 0.10$		Nominal $FAP = 0.05$	
n	m	Control limits (a, b)	Simulated FAP	Control limits (a, b)	Simulated FAP
15	5	(3, 12)	0.0563	(2, 13)	0.0144
	10	(3, 12)	0.0563	(2, 13)	0.0472
20	5	(5, 15)	0.0681	(4, 16)	0.0243
	10	(5, 15)	0.0681	(3, 17)	0.0152

manufactured by a forging process and used earlier to illustrate the mean-rank chart. Recall that in this data set, there are 25 samples ($m = 25$) each of size 5 ($n = 5$), that is, $N = mn = 25 \times 5 = 125$. The value of the pooled median equals 74.001, that is, $M = 74.001$. Columns (b) and (c) of Table 5.5 shows the $I(X_{ij} < M)$ values for $i = 1, 2, \ldots, m$ and the charting statistics, respectively. We use $UCL = 5$ and $LCL = n - UCL = 5 - 5 = 0$ for illustration purposes; recall that the subgroup size n needs to be larger than the number of subgroups m; however, the same data set is used as in Example 5.1, so we simply take $LCL/UCL = \pm 5$ for illustration.

To illustrate the calculations, consider sample number 1. We have $I(X_{11} < M) = 0$, since $X_{11}(= 74.030) > M(= 74.001)$, $I(X_{12} < M) = 0$

Table 5.5 Phase I data for the median chart and the charting statistics.

Sample	(a)					(b)					(c)
i	X_{i1}	X_{i2}	X_{i3}	X_{i4}	X_{i5}	$I(X_{ij} < M)$					U_i
1	74.030	74.002	74.019	73.992	74.008	0	0	0	1	0	1
2	73.995	73.992	74.001	74.011	74.004	1	1	0	0	0	2
3	73.998	74.024	74.021	74.005	74.002	1	0	0	0	0	1
4	74.002	73.996	73.993	74.015	74.009	0	1	1	0	0	2
5	73.992	74.007	74.015	73.989	74.014	1	0	0	1	0	2
6	74.009	73.994	73.997	73.985	73.993	0	1	1	1	1	4
7	73.995	74.006	73.994	74.000	74.005	1	0	1	1	0	3
8	73.985	74.003	73.993	74.015	73.988	1	0	1	0	1	3
9	74.008	73.995	74.009	74.005	74.004	0	1	0	0	0	1
10	73.998	74.000	73.990	74.007	73.995	1	1	1	0	1	4
11	73.994	73.998	73.994	73.995	73.990	1	1	1	1	1	5
12	74.004	74.000	74.007	74.000	73.996	0	1	0	1	1	3
13	73.983	74.002	73.998	73.997	74.012	1	0	1	1	0	3
14	74.006	73.967	73.994	74.000	73.984	0	1	1	1	1	4
15	74.012	74.014	73.998	73.999	74.007	0	0	1	1	0	2
16	74.000	73.984	74.005	73.998	73.996	1	1	0	1	1	4
17	73.994	74.012	73.986	74.005	74.007	1	0	1	0	0	2
18	74.006	74.010	74.018	74.003	74.000	0	0	0	0	1	1
19	73.984	74.002	74.003	74.005	73.997	1	0	0	0	1	2
20	74.000	74.010	74.013	74.020	74.003	1	0	0	0	0	1
21	73.982	74.001	74.015	74.005	73.996	1	0	0	0	1	2
22	74.004	73.999	73.990	74.006	74.009	0	1	1	0	0	2
23	74.010	73.989	73.990	74.009	74.014	0	1	1	0	0	2
24	74.015	74.008	73.993	74.000	74.010	0	0	1	1	0	2
25	73.982	73.984	73.995	74.017	74.013	1	1	1	0	0	3

since $X_{12}(= 74.002) > M(= 74.001)$ and similarly $I(X_{13} < M) = 0$, $I(X_{14} < M) = 1$ and $I(X_{15} < M) = 0$.

Hence

$$\begin{aligned} U_1 &= \sum_{j=1}^{n} I(X_{1j} < M) \\ &= I(X_{11} < M) + I(X_{12} < M) + I(X_{13} < M) + I(X_{13} < M) \\ &\quad + I(X_{14} < M) + I(X_{15} < M) = 0 + 0 + 0 + 1 + 0 = 1. \end{aligned}$$

This is displayed in column (c) of the first row of Table 5.5; the rest of the rows show the corresponding calculations for the other observations. Note that when an observation is equal to or tied with the value of M, the indicator function yields a value of 0 by definition. There are only two such cases for these data. If however, there are an excessive number of ties, the charting statistic and the chart may have to be revised.

The Phase I median chart is displayed in Figure 5.3 and it shows that the charting statistic for sample 11 plots on the *UCL*, indicating that the process is OOC. Recall that, in Phase I, the control limits are viewed as trial limits and are often revised and refined to ensure that the process is IC. Accordingly, sample number 11 would be investigated and perhaps discarded and the remaining 24 samples will be examined, with a possibly different set of control limits. This process will be continued until there are no OOC points and/or non-random patterns. This final set of data will be referred to as the IC or reference data from which any necessary parameters will

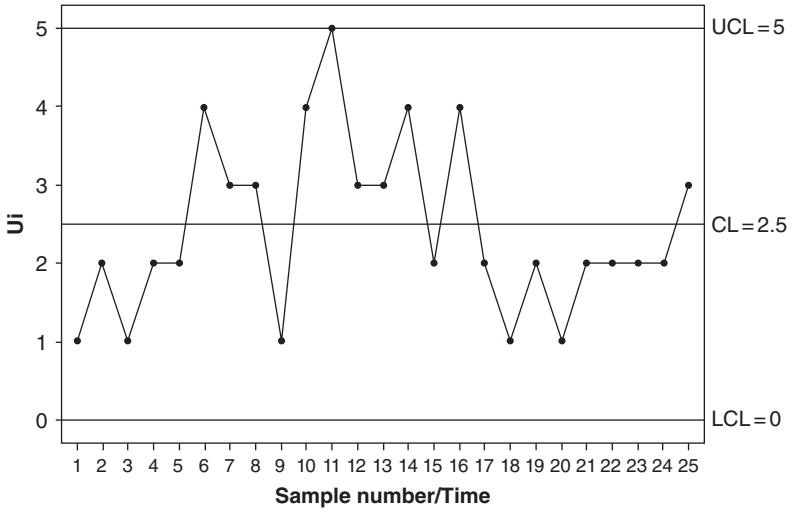


FIGURE 5.3 *A Phase I median chart.*

be estimated and used to calculate the control limits, which are used for prospective process monitoring in Phase II. Finally, note that, contrary to the Phase I median chart, the Phase I mean-rank chart did not find any out-of-control samples for this data set.

5.2.1.2.3 The Scale-rank Chart Most nonparametric charts have been considered for location; however, as in the parametric case, the chart for monitoring the scale is quite useful. To this end, we describe the "scale-rank" Phase I chart proposed by Jones-Farmer and Champ (2010). This chart is based on the work of Fligner and Kileen (1976) on nonparametric tests for the scale parameters in the multi-sample setting. As in the case of the location charts, let $(X_{i1}, X_{i2}, \ldots, X_{in})$ denote the ith reference sample (subgroup), $i = 1, 2, \ldots, m$ of size $n > 1$, and that there are m such preliminary independent random samples that constitute the Phase I data. Recall that, for the Phase I mean-rank chart, we combined the data from the m subgroups into a single array of $N = mn$ observations and arranged them from the lowest to the highest and ranked the data, keeping track of the subgroup where each observation came from. Here, instead of combining and ranking the original observations $X_{ij}, j = 1, 2, \ldots, n; i = 1, 2, \ldots, m$ as in the case of the mean-rank chart, we combine and rank the absolute differences $V_{ij} = |X_{ij} - M|, j = 1, 2, \ldots, n; i = 1, 2, \ldots, m,$ keeping track of the sample i, where M is an estimate of the IC process location. The choice of M will be discussed later. Let S_{ij} denote the rank of the V_{ij} in the combined sample of N observations. Jones-Farmer and Champ (2010) considered four transformations of these ranks and recommended using the transformed rank $T_{2,ij} = S_{ij}^2$ based on performance comparisons for the scale-rank chart. We present the scale-rank chart for this particular transformed rank.

The steps for calculating the scale-rank control chart are as follows. Once the transformed ranks $T_{2,ij}$ are found, calculate the average and the variance of these ranks, $W_i = \sum_{j=1}^{n} T_{2,ij}/n$ and S_{2i}^2 respectively, for each subgroup. These averages of the transformed ranks are the charting statistics for the scale-rank chart and are plotted on a Shewhart Phase I $\overline{X}$ control chart (Champ and Jones, 2004). The control limits are based on the univariate t distribution adjusted for an overall nominal FAP of α.

Let $\hat{\mu} = \sum_{i=1}^{m} \sum_{j=1}^{n} T_{2,ij}/N$ denote the grand average of the transformed ranks and let $\hat{\sigma} = \sqrt{\sum_{i=1}^{n} S_{2i}^2/m}/c_4(v)$ denote an estimate of the standard deviation, where $c_4(\nu) = \sqrt{\frac{2}{\nu}} \frac{\Gamma(\nu+1)/2}{\Gamma(\nu/2)}$ is the unbiasing constant with $v = m(n-1)$. The Phase I nonparametric scale-rank chart control limits are given by

$$\hat{\mu} \pm t_{1-\tau/2,v} \cdot c_4(v) \cdot \left(\sqrt{\frac{m-1}{m}}\right) \cdot \frac{\hat{\sigma}}{\sqrt{n}}$$

where $\tau = 1 - (1 - \alpha)^{1/m}$ denotes the *FAP*, assuming independence of the charting statistics, for a nominal *FAR* α. Hence, these approximate control limits can be found using tables of the univariate t distribution and no special tables are necessary. Note that when m is large, which is often the case in a Phase I analysis, the factors $\dfrac{m-1}{m}$ and $c_4(v)$ both tend to one and hence the formulas for the control limits can be simplified further.

Jones-Farmer and Champ (2010) note that, in order for the chart to be distribution-free, the pooled estimator M needs to be a symmetric statistic based on the combined sample. One such estimator is the median of the combined sample, as used in the median chart. The authors also explain that the use of a pooled estimator for location (which is necessary for the scale-rank chart to be distribution-free) in calculating the transformed ranks requires sort of a paradigm shift in the charting philosophy, namely, that here the location parameter has to be brought under control first, whereas in typical control charting, it is the reverse, as the scale (or the standard deviation) is first to be brought under control before charts for the location (or the mean) are constructed. Thus, for the scale-rank chart, a Phase I location chart such as the mean-rank chart is applied first, perhaps on a trial-and-error basis, until the process is deemed to be IC. At that point, using the data at hand, called the reference data, the estimate M is found and the scale-rank chart calculations are started; this is illustrated below. The authors examined the performance of the scale-rank chart compared with the Normal Theory charts in a simulation study. The scale rank chart performed favorably and maintained the IC *FAP* values at the nominal values for a variety of underlying distributions. The reader is referred to their paper for more details.

Example 5.3 Illustration of the Phase I Scale-rank Chart

We again use the Montgomery data on the inside diameters of piston rings manufactured by a forging process to illustrate the Phase I scale-rank chart. Note that here, $N = m \times n = 25 \times 5 = 125$. First, we need to apply a Phase I chart for location to establish that the process is IC and then estimate the median M from the resulting reference data. To this end, we use the mean-rank chart discussed earlier in this chapter. In Example 5.1, using this chart, the process was found to be IC, and thus the median $M =$ 74.001 was calculated from the 125 observations. Next, following the recipe in Jones-Farmer and Champ (2010), we calculate the scale-rank charting statistic (W_i) for each sample and show them in Column (b) of Table 5.6. In Column (c), we show the standard deviation values (S_{2i}). Note that the average rank is used in these calculations when there are ties. For a *FAP* of 0.10, the *LCL* and the *UCL* are found to be −727.39 and 11 262.97. The reader is asked to verify these calculations for themselves.

Table 5.6 Phase I data for chart for the scale-rank chart and the charting statistics.

Sample	(a)					(b)	(c)
i	X_{i1}	X_{i2}	X_{i3}	X_{i4}	X_{i5}	W_i	S_{2i}
1	74.030	74.002	74.019	73.992	74.008	8013.05	6503.86
2	73.995	73.992	74.001	74.011	74.004	3670.5	3456.58
3	73.998	74.024	74.021	74.005	74.002	6418.05	7853.59
4	74.002	73.996	73.993	74.015	74.009	4947.5	4158.39
5	73.992	74.007	74.015	73.989	74.014	8072.45	3151.95
6	74.009	73.994	73.997	73.985	73.993	5956.55	3864.78
7	73.995	74.006	73.994	74.000	74.005	2222.5	1696.92
8	73.985	74.003	73.993	74.015	73.988	7997.55	4880.37
9	74.008	73.995	74.009	74.005	74.004	2897.25	1888.58
10	73.998	74.000	73.990	74.007	73.995	3145.7	3316.40
11	73.994	73.998	73.994	73.995	73.990	4312.2	2835.46
12	74.004	74.000	74.007	74.000	73.996	1306.5	1364.39
13	73.983	74.002	73.998	73.997	74.012	4837.9	5910.30
14	74.006	73.967	73.994	74.000	73.984	7083.15	6889.59
15	74.012	74.014	73.998	73.999	74.007	4455.1	4304.73
16	74.000	73.984	74.005	73.998	73.996	3531.75	5415.67
17	73.994	74.012	73.986	74.005	74.007	5714.65	4116.39
18	74.006	74.010	74.018	74.003	74.000	4379.9	5313.94
19	73.984	74.002	74.003	74.005	73.997	3290.9	5530.63
20	74.000	74.010	74.013	74.020	74.003	6169.7	6010.87
21	73.982	74.001	74.015	74.005	73.996	5839.15	6502.36
22	74.004	73.999	73.990	74.006	74.009	3416.8	3462.01
23	74.010	73.989	73.990	74.009	74.014	8017.3	1897.27
24	74.015	74.008	73.993	74.000	74.010	5606.65	3990.15
25	73.982	73.984	73.995	74.017	74.013	10391.95	4485.08

The graphical display of the Phase I scale-rank is shown in Figure 5.4, and it can be seen that the process scale is IC. It may be noted that scale-rank chart is a Shewhart-type chart and its LCL can be negative, as is the case with these data. This is, of course, not a reasonable option for a scale parameter and the LCL of the chart should be reset to 0. More work on scale charts is therefore necessary.

Next, we consider two other Phase II nonparametric charts for location. The first is called the Min chart, which is a Shewhart chart that can be used both for individuals and subgroup data. In the latter case, the Min chart is a member of the class of precedence charts based on the subgroup minimum and not the median (as in the case of the precedence charts), as described

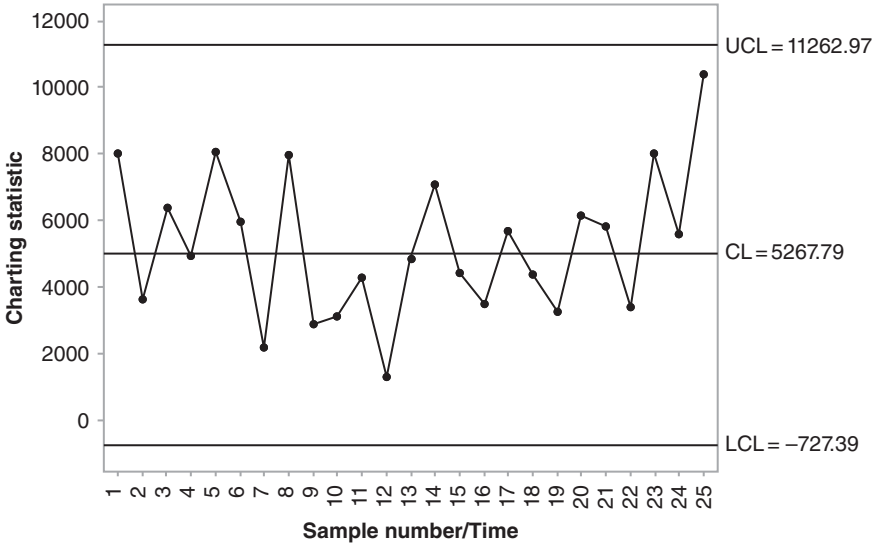


FIGURE 5.4 *A Phase I scale-rank chart.*

in Chapter 4. The second is a CUSUM chart that is based on accumulating the charting statistics of the Min chart.

5.2.2 Special Cases of Precedence Charts

5.2.2.1 The Min Chart

Albers and Kallenberg (2008) proposed a Phase II nonparametric Shewhart control chart to monitor the location of a random variable with an unknown cdf F. Their Min chart can be used for both individual ($n = 1$) and subgroup data ($n > 1$) and, in the latter case for the upper one-sided case, is based on comparing the minimum of the subgroup from the test sample to an UCL, which is given by an order statistic of the reference sample $\widehat{UCL} = X_{(b:m)}$ where $1 < b \leq m$. The subgroup based Min chart is a special case of the class of Phase II precedence charts discussed in Chapter 4. Hence, the run-length distribution and its various properties can be obtained from the results on the precedence charts discussed in detail in Section 4.4.1.1 by setting $j = 1$ into those equations.

Example 5.4 Calculation of the Charting Statistics for the Min Chart

Column (a) of Table 5.7 contains 25 retrospective or Phase I samples, each of size $n = 5$, that were collected when the process was thought to be IC, that is, $m = 125$. Column (a) of Table 4.1 also contains 15 prospective

Table 5.7 Data for the CUSUM-EX chart.

	Sample	(a)					(b)
	j	X_{j1}	X_{j2}	X_{j3}	X_{j4}	X_{j5}	Min
Phase I	1	74.030	74.002	74.019	73.992	74.008	73.992
	2	73.995	73.992	74.001	74.011	74.004	73.992
	3	73.998	74.024	74.021	74.005	74.002	73.998
	4	74.002	73.996	73.993	74.015	74.009	73.993
	5	73.992	74.007	74.015	73.989	74.014	73.989
	6	74.009	73.994	73.997	73.985	73.993	73.985
	7	73.995	74.006	73.994	74.000	74.005	73.994
	8	73.985	74.003	73.993	74.015	73.988	73.985
	9	74.008	73.995	74.009	74.005	74.004	73.995
	10	73.998	74.000	73.990	74.007	73.995	73.990
	11	73.994	73.998	73.994	73.995	73.990	73.990
	12	74.004	74.000	74.007	74.000	73.996	73.996
	13	73.983	74.002	73.998	73.997	74.012	73.983
	14	74.006	73.967	73.994	74.000	73.984	73.967
	15	74.012	74.014	73.998	73.999	74.007	73.998
	16	74.000	73.984	74.005	73.998	73.996	73.984
	17	73.994	74.012	73.986	74.005	74.007	73.986
	18	74.006	74.010	74.018	74.003	74.000	74.000
	19	73.984	74.002	74.003	74.005	73.997	73.984
	20	74.000	74.010	74.013	74.020	74.003	74.000
	21	73.982	74.001	74.015	74.005	73.996	73.982
	22	74.004	73.999	73.990	74.006	74.009	73.990
	23	74.010	73.989	73.990	74.009	74.014	73.989
	24	74.015	74.008	73.993	74.000	74.010	73.993
	25	73.982	73.984	73.995	74.017	74.013	73.982
Phase II	1	74.012	74.015	74.030	73.986	74.000	73.986
	2	73.995	74.010	73.990	74.015	74.001	73.990
	3	73.987	73.999	73.985	74.000	73.990	73.985
	4	74.008	74.010	74.003	73.991	74.006	73.991
	5	74.003	74.000	74.001	73.986	73.997	73.986
	6	73.994	74.003	74.015	74.020	74.004	73.994
	7	74.008	74.002	74.018	73.995	74.005	73.995
	8	74.001	74.004	73.990	73.996	73.998	73.990
	9	74.015	74.000	74.016	74.025	74.000	74.000
	10	74.030	74.005	74.000	74.016	74.012	74.000
	11	74.001	73.990	73.995	74.010	74.024	73.990

(*continued overleaf*)

Table 5.7 (*continued*)

Sample	(a)					(b)
j	X_{j1}	X_{j2}	X_{j3}	X_{j4}	X_{j5}	Min
12	74.015	74.020	74.024	74.005	74.019	74.005
13	74.035	74.010	74.012	74.015	74.026	74.010
14	74.017	74.013	74.036	74.025	74.026	74.013
15	74.010	74.005	74.029	74.000	74.020	74.000

(Phase II) samples, each of size $n = 5$, on the inside diameters of piston rings manufactured by a forging process.

To illustrate the calculation, consider sample number 1 of the prospective phase. The first charting statistic is computed as follows

$$Y_{1:n} = \min(74.012, 74.015, 74.030, 73.986, 74.000) = 73.986.$$

The rest of the charting statistics are given in Column (b) of Table 5.7.

We use $b = 87$ so that the estimated UCL for the Min chart is the 87th ordered value of the reference sample. Thus, $\widehat{UCL} = X_{87:125} = 74.006$. A plot of the minimum values for the chart is shown in Figure 5.5. It will be seen that the Min chart signals on the 13th sample in the prospective phase. Thus, corrective action and a search for assignable causes are necessary.

Since the Min and the traditional Med charts are both nonparametric Shewhart charts in the class of precedence charts, a performance

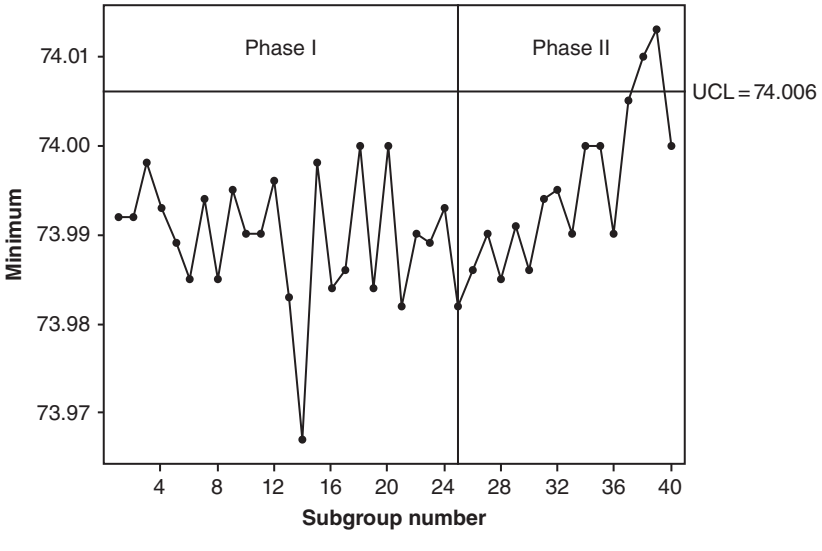


FIGURE 5.5 *The Min control chart.*

comparison between the two is of interest. A recent paper by Malela-Majika, Chakraborti, and Graham (2016) has looked into this, along with further enhancements. It turns out that both charts have their advantages. Other enhancements of the basic precedence charts have been considered by adding runs-type rules. The reader is referred to their paper for more details.

5.2.2.2 The CUMIN Chart

In practice, it is not always possible to obtain a large sample size n, which is especially needed especially for nonparametric procedures. In this case, Albers and Kallenberg (2009) proposed an idea where the Phase II observations may be grouped into smaller groups of size, say $m^* = 3$, 4, or 5 in order to detect smaller shifts, and for larger shifts, one may take $m^* = 1$. Note that Albers and Kallenberg (2009) use the notation m in order to denote the sample size of the Phase II observations. However, since m is typically used to denote the Phase I reference sample size, we replace Albers and Kallenberg's m with m^*. After the groups have been formed, one might consider taking various statistics as charting statistics. For example, if one takes the group mean, it would lead to the traditional Shewhart $\overline{X}$-chart, or for the case of $m^* = 1$, the Shewhart X-chart for individual observations. However, this approach leads to a parametric design, and approximations based on the central limit theorem may be invalidated when m^* isn't large. Consequently, the need to consider a different charting statistic arises. In light of this, Albers and Kallenberg (2009) proposed and investigated plotting the minimum as the charting statistic as an alternative. This is the basis of their Min chart.

Albers and Kallenberg (2009) then went on to consider improving the Min chart for smaller shifts by using the idea of cumulating the charting statistics. This chart is called a CUMIN chart; being a CUSUM chart, it has the benefit of having favorable detection properties. The CUMIN chart works as follows: recall that for the Min chart, each time a complete subgroup of size n is collected, its minimum (the charting statistic) is calculated and compared to a *UCL*. However, as soon as an observation occurs within a subgroup that falls below the *UCL*, it doesn't seem reasonable to complete that group and we could stop immediately. The next observation will then be treated as the first of a new attempt. Clearly, an alarm is given the first time n consecutive observations all exceed the *UCL*, that is, a signaling event occurs for the first k such that $S_k \geq n$ where $S_i = I(X_i > UCL)(1 + S_{i-1})$, $i = 1, 2, \ldots,$ with $S_0 = 0$ where $I(\cdot)$ is the indicator function. Although this representation of the charting statistic has a format similar to a CUSUM chart, it should be noted that the CUMIN chart is not a typical CUSUM control chart.

Example 5.5 Calculation of the Charting Statistics for a CUMIN Chart

Column (a) of Table 5.8 contains the prospective observations of the data in Table 5.8 and column (b) contains the corresponding S_i statistics. We see that $S_{31} \geq 5$, that is, the 31st individual measurement (out of

Table 5.8 The CUMIN chart.

Observation number i	(a) Y_i	(b) S_i
1	74.012	1
2	74.015	2
3	74.030	3
4	73.986	0
5	74.000	0
6	73.995	0
7	74.010	1
8	73.990	0
9	74.015	1
10	74.001	2
11	73.987	0
12	73.999	0
13	73.985	0
14	74.000	0
15	73.990	0
16	74.008	1
17	74.010	2
18	74.003	3
19	73.991	0
20	74.006	1
21	74.003	2
22	74.000	0
23	74.001	1
24	73.986	0
25	73.997	0
26	73.994	0
27	74.003	1
28	74.015	2
29	74.020	3
30	74.004	4
31	74.008	5

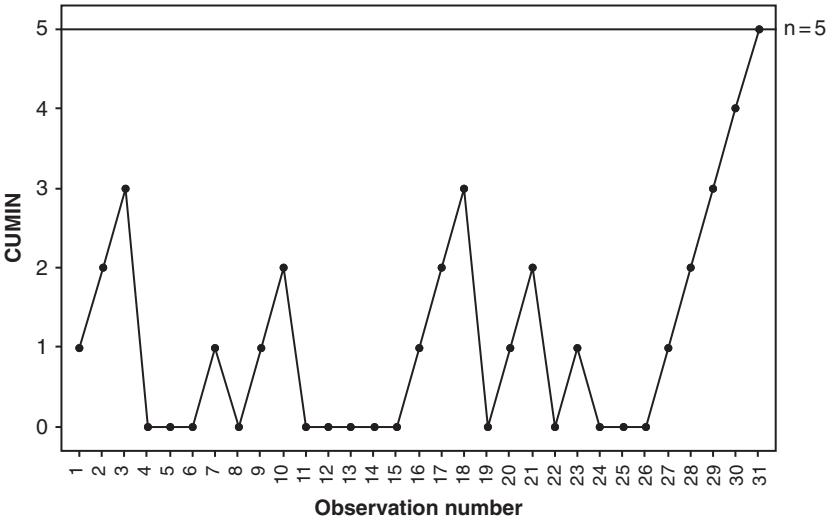


FIGURE 5.6 *A CUMIN control chart.*

$15 \times 5 = 75$ measurements) from the prospective phase gives a signal. Therefore, for the Min chart, we had to sample $13 \times 5 = 65$ individual measurements before a signaling event occurred, whereas, with the CUMIN chart, we only had to sample 31 individual measurements.

To illustrate the calculation, the first charting statistic is computed as follows

$$S_1 = I(X_1 > UCL)(1 + S_0)$$

since $X_1(=74.012) > UCL(=74.006)$ we have that $I(X_1 > UCL) = 1$, so that

$$S_1 = (1)(1 + 0) = 1.$$

The resulting CUMIN chart is shown in Figure 5.6 and it can be seen that the 31st individual measurement from the prospective phase gives a signal. Thus, corrective action and a search for assignable causes are necessary.

Next, we consider another class of control charts which, strictly speaking, do not fall within the traditionally defined class of nonparametric control charts, but they are practically nonparametric since one does not need to make a parametric model assumption about the underlying distribution. These are the control charts based on bootstrapping and are discussed in the next section.

5.2.3 Control Charts Based on Bootstrapping

Since Efron (1979) introduced bootstrap (resampling) methods in statistics for making statistical inference, it has developed and evolved in various ways

to provide powerful practical tools for various problems, such as interval estimation and hypothesis testing. The bootstrap is a computer-intensive technique that uses resampling from a given set of data to make statistical inference, eliminating the need to impose a convenient statistical (parametric) model that may not have a strong scientific basis (Leger, Politis, and Romano, 1992). Note that this is the same idea behind developing and using nonparametric methods. There is now a huge body of literature on bootstrap methods in statistics.

Bootstrap methods have been developed and applied to SPC problems and to construct control charts. Shewhart control charts based on bootstrap methods are discussed by various authors; see, for example, Seppala, *et al.* (1995), Liu and Tang (1996), Willemain and Runger (1996), Wood, Kaye, and Capon (1999), and Ambartsoumian and Jeske (2015). Chatterjee and Qiu (2009) proposed a CUSUM control chart based on bootstrapping. It appears that more work on control charts using bootstrapping may be useful. We explain the basic ideas here.

Seppala *et al.* (1995) summarized the general resampling (bootstrap) algorithm as used by Efron (1979) as follows

1. Initiate an iteration counter $i = 1$ and set B, a large number (1000–2000).

2. Draw (generate) a random sample $X_1^*, X_2^*, \ldots, X_n^*$ of size n from an initial sample $X_1, X_2, \ldots, X_n$ (available from the process) with replacement, and compute the bootstrap value for the statistic of interest, $T_i^* = T(X_1^*, X_2^*, \ldots, X_n^*)$. This could, for example, be taken to be the mean of the sample if the population parameter of interest T was the mean.

3. If i equals B, stop. Otherwise, increment i to $i + 1$, and repeat 2 above.

Thus, using Monte Carlo simulations, B bootstrap values of the statistics $T_1^*, T_2^*, \ldots, T_B^*$ are obtained, which can then be used to do statistical inference; for example, one can find a confidence interval for T.

Seppala *et al.* (1995) used a modified version of this algorithm to calculate the bootstrap control limits for the Shewhart-$\overline{X}$ and the Shewhart-S^2 control chart, respectively. They calculated the residuals for each subgroup, $X_{i1}, X_{i2}, \ldots, X_{in}$, by subtracting the mean of the ith subgroup (denoted as $\overline{X}_i$) from every X_{ij} in the ith subgroup, that is, $\hat{e}_{ij} = X_{ij} - \overline{X}_i$ for all i =1,2,3,...,$k \leq m$. After creating the $N = kn$ estimated residuals, $\hat{e}_{ij}'s$, the grand mean is calculated, $\overline{X} = \frac{1}{N}\sum_{i=1}^{k}\sum_{j=1}^{n} X_{ij}$. They then proceeded to create a resample of size N using the following steps

Step 1: Initiate an iteration counter l =1 and set B, a large number (1000–2000).

Step 2: Draw a random sample $e^*_{i1}, e^*_{i2}, \ldots, e^*_{in}$ of size n from the $N = kn$ estimated residuals $\hat{e}_{11}, \hat{e}_{12}, \ldots, \hat{e}_{1n}, \ldots, \hat{e}_{k1}, \hat{e}_{k2}, \ldots, \hat{e}_{kn}$ with replacement. Set $X^*_{ij} = \bar{\bar{X}} + ae^*_{ij}$ with $a = \left(\frac{N}{n} - 1\right)^{\frac{1}{2}}$ for $j = 1, 2, \ldots, n$.

Step 3: Run Step 2 for $i = 1, 2, \ldots, k$.

Step 4: Compute the bootstrap value for the statistics of interest $T^*_l = T(X^*_{11}, X^*_{12}, \ldots, X^*_{1n}, X^*_{21}, X^*_{22}, \ldots, X^*_{2n}, \ldots, X^*_{k1}, X^*_{k2}, \ldots, X^*_{kn})$.

Step 5: If l equals B, stop. Otherwise, increment l to $l + 1$ and repeat Steps 2–4.

Wood, Kaye, and Capon (1999) used a simpler version of resampling that is very similar to Efron (1979). They drew a sample $X^*_{11}, X^*_{12}, \ldots, X^*_{1n}, \ldots, X^*_{k1}, X^*_{k2}, \ldots, X^*_{kn}$ of size $N = kn$ from the initial sample $X_{11}, \ldots, X_{1n}, \ldots, X_{k1}, \ldots, X_{kn}$ with replacement instead of resampling residuals, as in Seppala *et al.* (1995).

The advantages of using bootstrap are:

1. It is a nonparametric method that does not require a large sample size.
2. It is theoretically sound.
3. It is intuitively appealing.
4. It is easy to implement since it replaces a multitude of different algorithms, statistical tables.
5. Greater flexibility since it is easy to see the effects of using different sample sizes and it is straightforward to produce charts for statistics not usually employed, for example, the median or the interquartile range.
6. Bootstrap control limits may be preferable than the conventional control limits when the process data are not normally distributed or skewed.

A few noticeable disadvantages regarding resampling methods used in SPC include:

1. The bootstrap method can be computationally quite intensive as some percentiles or statistics may require large numbers of replications to get more accurate estimates.

2. Bootstrap-based control limits for the Shewhart-$\overline{X}$ and the Shewhart-S^2 charts are considerably wider (or more conservative) than those for the standard normal theory charts, which can potentially lead to longer OOC run-lengths.

3. Bootstrap-based control limits are usually not symmetric about the center line (CL), which can be a potential problem when it comes to estimating values like the constant k on the Shewhart control charts.

4. Bootstrap-based control limits are not repeatable and thus may lead to different answers for the same data set used by different practitioners, which may be unsettling.

5.2.3.1 Methodology

Let us consider the bootstrap method used by Wood, Kaye, and Capon (1999). Note that, in general, the bootstrap procedure requires some assumptions about the process and the data, namely, that (i) the observations are independent and identically distributed (i.i.d.) from a continuous distribution, (ii) the underlying process distribution is symmetric, (iii) the process is "stable" or IC, and (iv) the initial sample is representative of the population. Thus, we assume that we have a reference sample of observations and suppose that these observations can be organized in a matrix

$$X_{obs} = \begin{pmatrix} x_{11} & x_{12} & x_{13} & \cdots & x_{1n} \\ x_{21} & x_{22} & x_{23} & \cdots & x_{2n} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ x_{m1} & x_{m2} & x_{m3} & \cdots & x_{mn} \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x_m} \end{pmatrix}$$

where x_i denotes the ith row of the observed reference sample matrix for $i = 1,2,3,\ldots,m$, which represents the ith sample of size n within the reference sample. Following this, we resample the entries of the observed reference sample matrix X_{obs} (with replacement) and create a new $m \times n$ matrix, which will be denoted by X_b^*, defined as

$$X_b^* = \begin{pmatrix} x_{11}^* & x_{12}^* & x_{13}^* & \cdots & x_{1n}^* \\ x_{21}^* & x_{22}^* & x_{23}^* & \cdots & x_{2n}^* \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ x_{m1}^* & x_{m2}^* & x_{m3}^* & \cdots & x_{mn}^* \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1^* \\ \mathbf{x}_2^* \\ \vdots \\ \mathbf{x_m^*} \end{pmatrix}$$

where $x^*_{ij} = x_{kl}$ for any $k = 1, 2, 3, \ldots, m$ and $l = 1, 2, 3, \ldots, n$. Suppose, for the construction of the control chart in question, that estimates of the process mean and process standard deviation are needed. The estimators for μ and σ are obtained from X^*_b, which is obtained by resampling from X_{obs} as follows

$$\hat{\mu}^*_b = \overline{\overline{x}}b = \frac{1}{m}\sum_{i=1}^{m}\overline{x}^*_i = \frac{1}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n}x^*_{ij} \tag{5.8}$$

$$\hat{\sigma}^*_b = \frac{S^*_{p,b}}{c_{4,m}} \tag{5.9}$$

where

$$S^*_{p,b} = \sqrt{\frac{\sum_{i=1}^{m}\sum_{j=1}^{n}(x^*_{ij} - \overline{x}^*_i)^2}{m(n-1)}} \tag{5.10}$$

and

$$c_{4,m} = \frac{\sqrt{2}\Gamma\left(\frac{m(n-1)+1}{2}\right)}{\sqrt{m(n-1)}\Gamma\left(\frac{m(n-1)}{2}\right)}. \tag{5.11}$$

The constant $c_{4,m}$ in Equation 5.11 is chosen such that $E\left(\frac{S^*_{p,b}}{c_{4,m}}\middle|\,\text{IC}\right) = \sigma$ and $var\left(\frac{S^*_{p,b}}{c_{4,m}}\right) \leq var(S^*_{p,b})$. These are the same unbiasing constants used with the parametric charts. Note that $\overline{x}^*_i$ is the sample mean for the ith row of the bootstrap sample matrix, $\hat{\mu}^*_b$ and $\hat{\sigma}^*_b$ are calculated from the B different bootstrap samples, where $B \geq 1000$, as suggested by Wood, Kaye, and Capon (1999). Using the set of bootstrap resample estimates for μ, that is, $\hat{\mu}^*_1, \hat{\mu}^*_2, \ldots, \hat{\mu}^*_B$ and the estimates for σ, that is, $\hat{\sigma}^*_1, \hat{\sigma}^*_2, \ldots, \hat{\sigma}^*_B$, we can create approximate estimated confidence intervals for μ and σ using the percentiles of these bootstrap resamples. The final estimates that will be used for μ and σ are then given by

$$\hat{\mu}^* = \frac{1}{B}\sum_{b=1}^{B}\hat{\mu}^*_b \tag{5.12}$$

and

$$\hat{\sigma}^* = \frac{1}{B}\sum_{b=1}^{B}\hat{\sigma}^*_b \tag{5.13}$$

respectively. It can also be argued that more sensible estimates for μ and σ are obtained when using the medians of the bootstrap samples for each respective parameter estimate if the samples imply that the bootstrap distributions are skewed, that is, $\hat{\mu}^* = \text{median}\,\{\hat{\mu}^*_1, \hat{\mu}^*_2, \ldots, \hat{\mu}^*_B\}$ and $\hat{\sigma}^* = \text{median}\,\{\hat{\sigma}^*_1, \hat{\sigma}^*_2, \ldots, \hat{\sigma}^*_B\}$.

Alternatively, when constructing control limits, one can standardize the charting statistic, say, Z_t, and use the fact that the LCL and UCL are the $\left(\frac{FAR}{2}\right)$ 100th and $\left(1-\frac{FAR}{2}\right)$ 100th percentiles of Z_t, respectively. First, we estimate μ and σ using Equations 5.12 and 5.13 and, following this, we resample from the reference sample matrix X_{obs}, like in the previous section. The only difference now is that we calculate $z^*_{b,t}$ (the bth bootstrap charting statistic for subgroup/sample number t) for each of the m subgroups in the bootstrap reference sample matrix. We repeat this procedure several times, say, $B \geq 1000$, until we have a vector of the last observed bootstrap subgroup (m) charting statistics $\mathrm{z}^*_{\mathrm{m}} = (z^*_{1,m}, z^*_{2,m}, \ldots, z^*_{B,m})'$. We then estimate the $\left(\frac{FAR}{2}\right)$ 100th and the $\left(1-\frac{FAR}{2}\right)$ 100th percentiles of Z_m using the bootstrap sample of $z^*_{b,m}$'s, that is, we set

$$\hat{LCL}^* = z^*_{m,\frac{FAR}{2}} \tag{5.14}$$

and

$$\hat{UCL}^* = z^*_{m,1-\frac{FAR}{2}} \tag{5.15}$$

where $z^*_{m,\frac{FAR}{2}}$ and $z^*_{m,1-\frac{FAR}{2}}$ are the $\left(\frac{FAR}{2}\right)$ 100th and the $\left(1-\frac{FAR}{2}\right)$ 100th percentiles of $\mathrm{z}^*_m = (z^*_{1,m}, z^*_{2,m}, \ldots, z^*_{B,m})$.

5.2.4 Change-point Models

Another useful perspective in SPC, in both the parametric and the nonparametric settings, is given by the change-point model formulation. In this approach, we observe a process that is indexed by time, and investigate whether a change or a shift in the process has taken place at a certain point (change-point) in time. The change can be in the process distribution itself or in the parameters of the distribution (such as the mean (location) and/or the variance (scale)). We assume a single change-point here for simplicity, but of course it is possible to have multiple change-points. The simple change-point formulation typically assumes that the observations up to and including the change-point are i.i.d. (with the same mean and variance), while the observations after the change-point are also i.i.d. but with the same distribution prior to the change, and with a possibly different mean and/or variance. When monitoring the location of a process, for example, the change-point model can be formulated as

$$X_i \sim \begin{cases} F(x) & \text{for} \quad i = 1, 2, \ldots, \tau \\ F(x-\theta) & \text{for} \quad i = \tau+1, \ldots, n \end{cases}$$

where X_i denotes an individual observation, $-\infty < \theta < \infty$ represents a possible (sustained) shift in the location, and $1 \leq \tau < n$ is the (unknown) change-point when the location change (from 0 to $\theta \neq 0$) takes place. Thus, the detection of a change-point is synonymous with declaring that a process has gone from IC to OOC and this connects a change-point (model) formulation, typically studied in the more mathematical statistical literature, to the SPC context. The goal then is to locate (detect) the change-point (time) τ as well as to measure (estimate) the magnitude of the change (θ).

There is a large amount of literature on parametric change-point problems and thus we don't consider them here. Hawkins and Deng (2010) developed an interesting nonparametric change-point model based on the Mann–Whitney statistic $U_{k,n} = \sum_{i=1}^{k} \sum_{j=k+1}^{n} sign(X_i - X_j)$. The assumption is that the sample size, n, is fixed and the standardized statistic is used, that is,

$$T_{k,n} = \frac{U_{k,n} - E(U_{k,n}|IC)}{STDEV(U_{k,n}|IC)} \text{ for } 1 \leq k \leq n-1. \tag{5.16}$$

When the process is IC, we have

$$E(U_{k,n}) = 0 \text{ and } STDEV(U_{k,n}) = \sqrt{k(n-k)(n+1)/3},$$

so that

$$T_{k,n} = \frac{U_{k,n}}{\sqrt{k(n-k)(n+1)/3}} \text{ for } 1 \leq k \leq n-1, \tag{5.17}$$

when the process is IC. It is also known that $T_{k,n}$ is approximately $N(0,1)$ when k and $n - k$ both tend to infinity. The test statistic, called the maximized split statistic, used for testing the hypotheses H_0: $\theta = 0$ against H_a: $\theta \neq 0$ (if H_0 is rejected we would declare the presence of a change-point, that is, the process is OOC at time k) is obtained by maximizing the absolute value of $T_{k,n}$ over k, given by $T_{\max,n} = \max_{1 \leq k \leq n-1} |T_{k,n}|$.

Note that in a typical Phase analysis I, we have a fixed size reference sample, so it is appropriate to calculate $T_{max,n}$ for the whole data set $X_1, X_2, \ldots, X_n$. However, in a change-point setting, the observations become available sequentially over time. As a new observation is accrued, the Mann–Whitney test is applied between that observation and the earlier IC data, and the statistic $T_{max,n}$ is re-calculated. Stated differently, as a new observation, X_n, is accumulated and the past data $X_1, X_2, \ldots, X_{n-1}$ remain IC (no change-point is detected), the change-point statistic $T_{max,n}$ is re-computed for the whole sequence $X_1, X_2, \ldots, X_n$ and the monitoring moves thus ahead with every single new observation. The change-point model signals that a change has occurred (i.e., $H_0 : \theta = 0$ is rejected) if $T_{max,n}$ exceeds $UCL = h_{n,\alpha}$. The critical value $h_{n,\alpha}$ is chosen so that the conditional probability of a false alarm, given that there was no false alarm

at the previous stage of testing, is fixed at some nominal value, α. This critical value can be found using a simulation algorithm.

Zhou *et al.* (2009) proposed a Phase II nonparametric EWMA control chart for a sustained step shift in the mean, based on a change-point model and the two-sample Mann–Whitney statistic. They assumed a fixed number of m IC Phase I observations $\{X_1, \ldots, X_m\}$ and n Phase II observations $\{X_{m+1}, \ldots, X_{m+n}\}$. Then, the Mann–Whitney statistic at any time point t, $m \leq t < m+n$ is defined as $MW_{t,m+n} = \sum_{i=1}^{t} \sum_{j=t+1}^{m+n} I(X_j < X_i)$. Zhou *et al.* (2009) used the charting statistic

$$T_{m,n} = \max_{m \leq t < m+n} |SMW_{t,m+n}|$$

which is the maximum of the absolute values of a number of standardized Mann–Whitney statistics, $SMW_{t,m+n} = \dfrac{(MW_{t,m+n} - t(N-t)/2)}{\sqrt{t(N-t)(N+1)/12}}$, $N = m+n$. However, the authors note that, since the possible values that $T_{m,n}$ can attain are limited (especially when n is small), it is difficult to obtain very accurate control limits. Alternatively, the authors considered a Phase II nonparametric EWMA chart of the standardized Mann–Whitney statistics given by

$$Y_j(m,n) = \lambda SMW_{j,m+n} + (1-\lambda) Y_{j-1}(m,n)$$

with $Y_{m-m_0-1}(m,n) = 0$ and $j = m - m_0, \ldots, N-1$. The charting statistic at time n is defined as $Y_{\max}(m,n) = \max_{m-m_0 \leq j < m+n} |Y_j(m,n)|$, where m_0 is a suitably chosen integer, $n = 1, 2, 3, \ldots$, and $Y_{\max}(m,n)$ is compared against a sequence of *UCL*s. If $Y_{\max}(m,n) > h_{m,n}$, an OOC signal is indicated. Note that the sequence of *UCL*s is chosen to obtain a specified ARL_{IC}, but these limits change (sequentially over time) as we continue to monitor the process because it is a function of $n = 1, 2, 3, \ldots$ Again, the $UCL = h_{m,n}$ is found using a simulation algorithm.

Example 5.6 A Change-point Model

The data for the example with a shift after the 20th sample are given in Column (a) of Table 5.9. This data set is from Table 2 of Zhou *et al.* (2009) and the assumption is that the underlying IC process distribution is a chi-square distribution with four degrees of freedom. It is assumed that $m = 15$ and that an increase of 0.75 standard deviations in the mean occurs after the fifth Phase II observation. Suppose that $\lambda = 0.2$ and $m_0 = 4$. Column (b) of Table 5.9 shows the values of the charting statistics $Y_{max}(m,n)$ for $n = 1, 2, \ldots, 13$ and Column (c) of Table 5.9 shows the corresponding sequence of *UCL*s.

The calculation of the first Phase II charting statistic is given for illustration purposes, that is, the calculation of $Y_{max}(m = 15, n = 1)$. For $m = 15$,

Table 5.9 Data for the change-point model and the charting statistics.

n	(a) x_n	(b) $Y_{max}(m, n)$	(c) $h_{10,n-15}(0.005)$
1	2.061	—	—
2	1.113	—	—
3	4.298	—	—
4	6.972	—	—
5	1.675	—	—
6	3.614	—	—
7	3.446	—	—
8	8.057	—	—
9	4.702	—	—
10	2.827	—	—
11	1.637	—	—
12	4.925	—	—
13	7.506	—	—
14	1.170	—	—
15	2.308	—	—
16	3.606	0.249	1.305
17	7.425	0.462	1.430
18	0.277	0.535	1.539
19	5.455	0.153	1.641
20	3.597	0.100	1.719
21	6.068	0.550	0.789
22	4.618	0.647	1.852
23	8.384	1.291	1.906
24	6.015	1.457	1.945
25	5.792	1.607	1.992
26	6.267	1.763	2.016
27	6.134	1.898	2.047
28	9.937	2.195	2.078

we have $m - m_0 = 11$. Now, we compare the values of the observations 12, 13, 14, 15, and 16 with each of the values of the observations 1 through 11 and calculate the corresponding two-sample Mann–Whitney statistic; this is found to be 25. We then standardize the Mann–Whitney statistic using its IC expected value and standard deviation of 27.5 and 8.827, respectively, and get $SMW = -0.283$. Then, using $\lambda = 0.2$ and a starting value of $Y_{j=10}(m = 15, n = 1) = 0$, we find that $Y_{j=11}(m = 15, n = 1) = -0.057$; this value is shown in Column (e) of Table 5.10. Note that we can think of observations

Table 5.10 Calculation of the first Phase II change-point charting statistic.

	j	(a) MW	(b) E(MW)	(c) SD (MW)	(d) SMW	(e) $Y_j(m,n)$
Observation 12–16 vs. 1–11	11	25	27.5	8.827	−0.283	−0.057
Observation 13–16 vs. 1–12	12	26	24.0	8.246	0.243	0.003
Observation 14–16 vs. 1–13	13	28	19.5	7.433	1.144	0.231
Observation 15–16 vs. 1–14	14	16	14.0	6.298	0.318	0.249
Observation 16 vs. 1–15	15	7	7.5	4.610	−0.108	0.232

1 through 11 as the "first" sample, and observations 12 through 16 as the "second" sample.

Next, the above procedure is repeated for observations 13 through 16 against observations 1 through 12 to calculate $Y_{j=12}(m = 15, n = 1) = 0.003$, and then for observations 14 through 16 against observations 1 through 13 to calculate $Y_{j=13}(m = 15, n = 1) = 0.231$, etc. (see Table 5.10). This procedure is repeated five times in total for the first charting statistic and each time we need to calculate the statistic $Y_j(m, n)$ – see Column (e) of Table 5.10. Finally, we calculate the charting statistic $Y_{max}(m, n)$ which is 0.249; this is the maximum absolute value of all the $Y_j(m, n)$ values where $j = 11, 12, \ldots, 15$ and is the charting statistic corresponding to the first $(n = 1)$ Phase II observation. The rest of the Phase II charting statistics for $n = 2, 3, \ldots, 13$ are calculated similarly and are shown in Column (b) of Table 5.10.

5.2.5 Some Adaptive Charts

5.2.5.1 Introduction

Adaptive control charts allow the components of the monitoring scheme to vary in order to obtain improved performance over non-adaptive control charts. This could be components such as the sample size, the warning limits, and/or the control limits. In doing so, the control chart might be more complicated; however, adaptive charts yield considerable improvement over non-adaptive charts.

5.2.5.2 Variable Sampling Interval (VSI) and Variable Sample Size (VSS) Charts

We can partition the area between the UCL and LCL into regions or zones. Theoretically, one can have more zones (than depicted in Figure 5.7), but it may complicate administrating the control process with only marginally improved efficiency relative to only having two zones.

Most control charts make use of an FSS and an FSI. However, one can consider using a VSS, that is, the sample size isn't fixed, and a VSI, that is, the interval between successive samples isn't fixed.

5.2.5.2.1 Variable Sample Size (VSS) A VSS control chart allows the sampling sizes to vary according to what is being observed in the current sample. Characteristics of the run-length distribution, such as the ARL, lose effectiveness in comparing different charts when the sample size is varied. Therefore, we use the average number of observations to signal ($ANOS$) as performance measure. One major advantage of using a VSS scheme is that it detects shifts quicker than an FSS scheme. It works as follows: let C_i denote the charting statistic for $i = 1, 2, 3, \ldots$ and let n_1 and n_2 denote the different sample sizes with $n_1 < n_2$. If $C_i \in$ Zone 1 a larger sample size (n_2) is used to increase power. If $C_i \in$ Zone 2, a smaller sample size (n_1) is used since there is no evidence that the process is demanding adjustment.

5.2.5.2.2 Variable Sampling Interval (VSI) A VSI is when the interval between successive samples is not fixed. The VSI control chart allows the sampling intervals to vary according to what is being observed in the current sample. Characteristics of the run-length distribution lose effectiveness in comparing different charts when the sampling interval is varied. Therefore, we use the average time to signal (ATS) as a performance measure. Using a VSI scheme intuitively makes sense from a cost or economic

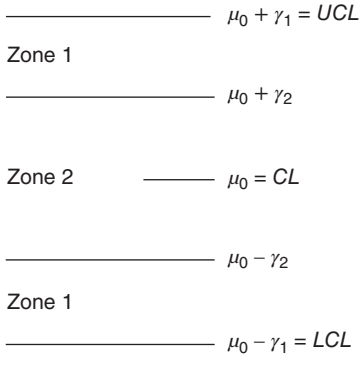


FIGURE 5.7 *The area between the control limits partitioned into zones.*

perspective in that it aims to reduce the time to signal and the number of observations to signal when the process is, in fact, OOC. It also has the advantage of detecting shifts quicker than an FSI scheme. It works as follows: let d_1 and d_2 denote two interval sizes with $d_1 < d_2$. If $C_i \in$ Zone 1 $\rightarrow$ wait d_1 time units until sample $i + 1$. Thus, when the charting statistic plots close to the control limits, a shorter sampling interval is used. If $C_i \in$ Zone 2 $\rightarrow$ wait d_2 time units until sample $i + 1$. Thus, when the charting statistic plots close to the CL, a longer sampling interval is used. It is reasonable to take more time, because there is no evidence that the process is OOC.

5.2.5.3 Other Adaptive Schemes

The first modification to the VSI scheme, to lessen the disadvantage of unpredictable sampling, was introduced by Stoumbos and Reynolds (1996) and is called the VSIFT (variable sampling interval with fixed times) scheme. It requires that samples always be taken at specified fixed times with additional samples allowed between these fixed times when necessary. The variable sampling scheme at fixed times has recently been combined with the sequential probability ratio test, and the sequential probability ratio test at fixed times (SPRTFT) chart was introduced, which detects shifts in the process mean. It was revealed that this control scheme has major administrative advantages and is also extremely efficient.

Typically, control charts are based on samples with a fixed size, taken at fixed-length sampling intervals, for example, taking a sample of 10 observations every 30 minutes. After each sample is taken, a control statistic is calculated that quantifies essential process characteristics and then it is plotted against time. Control limits are assembled in such a way that the control statistic is very improbable to fall outside the control limits when the process is IC. When a control statistic falls on or outside either of the control limits, it is taken as a signal that a process change has taken place and that some proper action is mandatory.

Because of the variation in the process data, a control chart regularly fails to instantly detect a process change and could even mistakenly signal such a change when the process is, in fact, IC. The performance of a control chart usually refers to the delay in detecting process changes. Two different types of performance are typically considered, namely, the zero-time and the steady-state. The detection delay of a process change that is present during the beginning of the control chart at time zero is referred to as zero-time performance, while the steady-state performance refers to the detection delay of a process change that occurs after the chart has been operating long enough for its control statistic to reach a stationary distribution by the time of the change. The steady-state performance of a chart is normally more important than the zero-time performance and these two also tend to differ a lot. The reason for this is that, although there are cases where the process is initially

OOC, in the majority of practical situations, the process would initially be IC and then change at a random time in the future.

The advantages of the FSI scheme include its simplicity of implementation, although it has a dreadfully slow response to process changes. A control chart can be considered an adaptive control chart when at least one of its parameters is allowed to change. Thus, the sample size n, the sampling interval (say, h), and the control limits coefficient (say, k) are allowed to change all, depending on what is observed from the specific process.

There have been numerous developments in VSI control charts where the sampling interval h varies as a function of prior sample results. The VSI chart uses control and warning limits to partition the chart into three separate regions consisting of the safe region, the warning region, and the OOC region. When a sample point falls in the warning region, it is sensible to suspect that the process may have shifted to an OOC state and the time interval until the next sample point should be significantly shorter in order to enhance the control. Conversely, if the sample point falls in the safe region, the next sampling interval can be long, since there is no clear evidence that the process requires adjustment. When a sample point falls in the OOC region, the process might be OOC because of the occurrence of an assignable cause. The sampling interval of the adaptive rule thus requires the introduction of two limit coefficients, namely, a warning limit coefficient, w, (see Equation 5.18) and a control limit coefficient, k (see Equation 5.19).

Typically, for a Shewhart chart, the warning limit coefficient (w) and the control limit coefficient (k) are used to set up the control limits

$$LWL/UWL = E(X) \pm w\sigma_X \tag{5.18}$$

$$LCL/UCL = E(X) \pm k\sigma_X \tag{5.19}$$

Tagaras (1998) presented a comprehensive survey about the development in the design of adaptive charts up until the year 1997. Here, we consider work done before and after 1997. The comprehensive survey on nonparametric adaptive control charting is brought up to date. A brief summary of the findings of each article is given and we also take a look at what the best chart or charting statistic would be in different situations. The main focus will be on showing how performance can be improved by allowing the sample size and the sampling interval to vary.

5.2.5.4 Properties and Performance Measures of Adaptive Charts

The statistical effectiveness of a control chart is generally determined by the speed with which it detects the occurrence of an assignable cause, as well as the amount of false alarms that occur. The most popular statistical measure

of performance is the *ARL*, which is the number of samples before a control chart signals an OOC situation or a false alarm occurs. It is all right to use the *ARL* when comparing control charts as long as the sampling interval h remains equal and constant for all charts. When working with charts with an adaptive sampling interval, the previous condition is not valid since time to signal is not a constant multiple of *ARL*. Some useful measures of performance to overcome this limitation are:

- **Average time to signal (*ATS*)**: The expected time that it would take from the beginning of the process up until the time when the chart signals.
- **Average number of observations to signal (*ANOS*) or average number of samples to signal (*ANSS*)**: The expected number of observations (for *ANOS*) or samples (for *ANSS*) taken from the beginning of the process until the time when the chart signals. Although the definitions of *ARL* and *ANSS* do not differ, *ANSS* is preferred in adaptive h situations since *ARL* tends to be associated with time, as well as the number of samples in conventional charts where h is constant.

If we keep the control limits constant, then the values of h_{i+1} and n_{i+1} depend only on the latest sample statistic, meaning for a Shewhart chart with adaptive h and n it is clear that

$$ANSS = \frac{1}{1-\beta} \tag{5.20}$$

where β is the probability that the sample statistic will fall between the control limits. Moreover, when the process is in a constant state, the adaptive sampling intervals, h_1,h_2,.., will be i.i.d., and, according to Wald's identity

$$ATS = E(h_i)(ANSS) \tag{5.21}$$

assuming that h_1, the interval used before the first sample is drawn from the same distribution as the other h_i's. In practice, the first sampling interval h_1 is usually given. Equation 5.21 holds when the process starts at a certain state and remains in that state for the remainder of the time. In reality, however, the process usually starts IC and then shifts to an OOC state at a random time. Tagaras (1998) adjusted the *ATS* and introduced a new measure of performance:

- **Adjusted average time to signal (*AATS*)**: The expected value of the time it takes when the assignable cause occurs until the chart signals.

$$AATS = E(Y) + (h_i)(ANSS - 1) \tag{5.22}$$

where Y is the time from the process shift until the next sample. *AATS* can also be referred to as the steady state *ATS*, according to Runger and Pignatiello (1991).

5.2.5.5 Adaptive Nonparametric Control Charts

First, we briefly discuss adaptive parametric charts in order to introduce some basic concepts to the reader before moving to a discussion on adaptive nonparametric charts.

5.2.5.5.1 Adaptive Parametric Control Charts to Monitor Location So far, most of the work on adaptive control chart designs focuses on the $\overline{X}$ chart. The VSI $\overline{X}$ control chart was first introduced by Reynolds *et al.* (1988). When implementing the VSI scheme, the sampling interval depends on the value of the statistic of the control chart. When indications of shifts in the process mean occur, samples are taken more frequently. According to Ken (1997), "The VSI scheme seems reasonable when it can be applied practically." Reynolds, Amin, and Arnold (1990) applied the VSI scheme to CUSUM charts, today better known as VSI CUSUM charts, and confirm that these new VSI CUSUM charts are more efficient than FSI CUSUM charts. Shamma, Amin, and Shamma (1991) and Saccucci, Amin, and Lucas (1992) applied the VSI scheme to EWMA charts. Reynolds (1996) also added the VSI feature to the $\overline{X}$ charts with runs rules. The research of Reynolds (1996) on VSI control charts indicates that a VSI control chart can identify the majority of process shifts significantly faster than the corresponding FSI control chart, but only when both charts are set up to have equal average sampling rates as well as the same *FAR*. This research specifically focused on VSI control charts where the samples are constantly taken at specified evenly spaced fixed time points, but further samples are acceptable between these fixed times once specified by the data from the process. Reynolds (1996) used Markov process methods to analyze the performance of VSIFT charts. The VSIFT feature can be considered for the $\overline{X}$ chart, the EWMA chart, as well as the CUSUM chart. VSS is another approach to varying the sample rate as a function of the process data, especially for situations in which more than one observation is taken at each sampling time. To determine the sample size used at each sampling time for VSS charts, one has to look at the data obtained in past samples. An advantage that the VSS chart has is that it can be set up to sample only at specified fixed times. It was shown that VSIFT charts were able to detect most process shifts considerably quicker than FSI charts. It also confirmed that VSIFT charts were just as successful in detecting shifts as standard VSI charts that are not forced to sample at specified fixed times. In comparison to VSS charts, the VSIFT charts were shown to be better at detecting large shifts but

not as good at detecting small shifts. To sum up the basic idea of these modifications, always take samples at some fixed equally spaced time point but at the same time allow for sampling between the particular time points whenever there is some indication of change in the process.

The *ATS* can also be referred to as the performance measure or the expectation of the length of time required for signaling in the control charts. A change point is a shift of the process mean at a certain time point. The *ATS* is thus also the expected difference between the change-point and the time to signal in the control chart. Ken (1997) proposes an alternative approach for one-sided CUSUM charts with VSI where a Markov chain with some dummy states is used. By using this approach, we can determine a few dynamic performance measures of VSI CUSUM charts, such as the *ATS* against a change-point, as well as the distribution of time to signal (*DTS*).

5.2.5.5.2 Adaptive Nonparametric Control Chart to Monitor Location

VSI Sign Control Chart Amin and Widmaier (1999) proposed a VSI scheme for the Shewhart control chart based on the sign test statistic. The reader is referred to Section 4.2.1.1 for more details on the sign statistic. It should be noted that the statistic defined in Equation 4.1 was used as the charting statistic in Section 4.2.1.1, namely, $SN_i = \sum_{j=1}^{n} \varphi(X_{ij} - \mu_0)$, for the Shewhart FSI sign control chart. However, Amin and Widmaier (1999) use the more well-known version of the sign statistic for the Shewhart VSI sign control chart, that is, T_i, which is defined as

$$T_i = \left\{ \sum_{j=1}^{n} sign(X_{ij} - \mu_0) + n \right\} /2, i = 1, 2, \ldots \tag{5.23}$$

where the *sign* function is defined by

$$sign(x) = \begin{cases} 1 & \text{if} \quad x > 0 \\ 0 & \text{if} \quad x = 0 \\ -1 & \text{if} \quad x < 0. \end{cases}$$

A Shewhart VSI sign chart is designed in a similar way as a Shewhart VSI $\overline{X}$-chart. For the sign chart, the statistic T_i is the number of (+) signs in a sample and, unlike the continuous $\overline{X}$-statistic, T_i, can only take on discrete values. If there were a large number of (+) signs it would indicate a possible upward shift of the median. A small number of (+) signs would indicate a possible downward shift in the median, in which case it would be strongly suggested to take the following sample quickly. It is advised to wait longer until the next sample when the number of (+) signs are not too small and not too large.

Note that T_i (see Equation 5.23) is the more well-known version, on which the sign test is based, and is referred to as the **sign test statistic** (see, for example, Gibbons and Chakraborti, 2010, p. 168) and it follows a Binomial distribution with parameters n and $p = P(X_{ij} > \mu_0)$, with $p = 0.5$ when the process is IC.

The control limits for the Shewhart VSI sign chart are given by $UCL = c$ and $LCL = n - c$. The value of c, which is always an integer value, can be chosen such that a sign chart will have a specified IC ARL. The Shewhart VSI sign chart signals that a shift occurred if (i) $T_i > c$ in the one-sided case for detecting increases in p, (ii) $T_i < n - c$ in the one-sided case for detecting decreases in p, and (iii) $T_i > c$ or $T_i < n - c$ in the two-sided case.[1] In the two-sided case for detecting an upper or a lower shift, c is chosen such that

$$1 - \alpha_0 = \sum_{j=n-c}^{c} \binom{n}{j} (0.5)^n$$

and the IC ARL is given as

$$ARL_0 = \frac{2^n}{1 - \sum_{j=n-c}^{c} \binom{n}{j}} = \frac{2^n}{\sum_{j=0}^{n-c-1} \binom{n}{j} + \sum_{j=c+1}^{n} \binom{n}{j}} = \frac{2^{n-1}}{\sum_{j=c+1}^{n} \binom{n}{j}}.$$

Amin and Widmaier (1999) mentioned that the greatest achievable IC ARL value for a Shewhart sign chart with $p = 0.5$ at a given value of n occurs at $c = n - 1$, resulting in an IC ARL of 2^{n-1} for the two-sided case.

For VSI charts, the area between the control limits is divided into (typically) two regions. If the charting statistic falls close to the control limits, then a shorter sampling interval is used as this could help detect a shift earlier. Instead, if the charting statistic falls close to the CL, then a longer sampling interval should be used.

Amin and Widmaier (1999) used the $AATS$ as a measure of performance since this can be used when the shift occurs at any point in the sample and it does not assume that the shift was present at the start of the process, which is what the ATS assumes.

The $AATS$ is given by

$$AATS = \frac{\sum_{j=1}^{2} d_j^2 p_{0j}}{2 \sum_{j=1}^{2} d_j p_{0j}} + \frac{1}{\alpha_1} \sum_{j=1}^{2} d_j p_{1j}. \tag{5.24}$$

In a two-sided Shewhart VSI sign chart, the values $(n - c, \ldots, c)$ correspond to the IC case. Figure 5.8 shows the partitioning of the range

[1] Typically a control chart signals when the charting statistics plots on or above the UCL or on or below the LCL. Amin and Widmaier (1999), however, specify the signal to occur only when the charting statistics plot above the UCL or below the LCL.



FIGURE 5.8 *Regions of the VSI sign control chart.*

of discrete values $(n-c, \ldots, c)$ into two regions, I_1 and I_2, where $I_2 = \{n-k, \ldots, k)$ and $I_1 = \{n-c, \ldots, n-k-1) \cup \{k+l, \ldots, c)$. The signal region corresponds to the values $(0, \ldots, n-c-1) \cup \{c+l, \ldots, n)$; this is visually illustrated in Figure 5.8. The probability of using sampling interval d_2 can be written as

$$P(T_i \in I_2) = P(n-k \le T_i \le k) = \sum_{j=n-k}^{k} \binom{n}{j} p^j (1-p)^{n-j} \tag{5.25}$$

and the probability of using sampling interval d_1 as

$$P(T_i \in I_1) = P(n-c \le T_i < n-k) + P(k < T_i \le c)$$

$$= \sum_{j=n-c}^{n-k-1} \binom{n}{j} p^j (1-p)^{n-j} + \sum_{j=k+1}^{c} \binom{n}{j} p^j (1-p)^{n-j}. \tag{5.26}$$

When the process is IC, the previous expressions can be simplified to

$$p_{02} = P(T_i \in I_2 | \mu = \mu_0) = \sum_{j=n-k}^{k} \binom{n}{j} 0.5^n \tag{5.27}$$

and

$$p_{01} = P(T_i \in I_1 | \mu = \mu_0) = \sum_{j=n-c}^{n-k-1} \binom{n}{j} 0.5^n$$

$$+ \sum_{j=k+1}^{c} \binom{n}{j} 0.5^n = 2 \sum_{j=k+1}^{c} \binom{n}{j} 0.5^n. \tag{5.28}$$

The probability of a signal, in general, is given by

$$\alpha = \sum_{j=0}^{n-c-1} \binom{n}{j} p^j (1-p)^{n-j} + \sum_{j=c+1}^{n} \binom{n}{j} p^j (1-p)^{n-j}. \tag{5.29}$$

This also simplifies in the IC case to

$$\alpha_0 = \sum_{j=0}^{n-c-1} \binom{n}{j} 0.5^n + \sum_{j=c+1}^{n} \binom{n}{j} 0.5^n. \tag{5.30}$$

Example 5.7 A Shewhart VSI Sign Chart

To illustrate the FSI and VSI sign charts, we make use of the same data in Table 4.6, which was used to illustrate the FSI Shewhart-SN chart for subgrouped data (the reader is referred to Example 4.2 in Chapter 4). Recall that, in that example, the charting statistic for the FSI Shewhart-SN chart was SN_i (see Equation 4.1), whereas, here, we make use of the statistic T_i, which is the more well-known version (since the distribution of T_i is binomial) on which the sign test is based (see, for example, Gibbons and Chakraborti, 2010, p. 168). Since we make use of a different charting statistic, we give the values in Table 5.11.

This data set has 15 samples, each consisting of 10 observations (see Table 4.6), making the total number of observations equal to 150. The samples were taken at intervals of 30 minutes. For the Shewhart FSI sign chart, we used every second sample in order to have the samples spaced at time intervals of 1 hour consistently. For the Shewhart VSI sign chart, since the longer time interval should be close to $1.5d$, we used every third sample, making it a time interval of 90 minutes between each sample. As soon as a charting statistic plots outside the warning limits, the next sample is used, decreasing the time interval between the two samples to 30 minutes.

If we select $c = 9$, we have that the control limits are given by

$$UCL = c = 9$$
$$LCL = n - c = 10 - 9 = 1$$

with the resulting Shewhart FSI sign chart shown in Figure 5.9.

Table 5.11 Charting statistics for the Shewhart FSI and VSI sign charts.

Sample (i)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
T_i	5	1	4	6	2	5	7	6	7	4	5	8	6	6	6

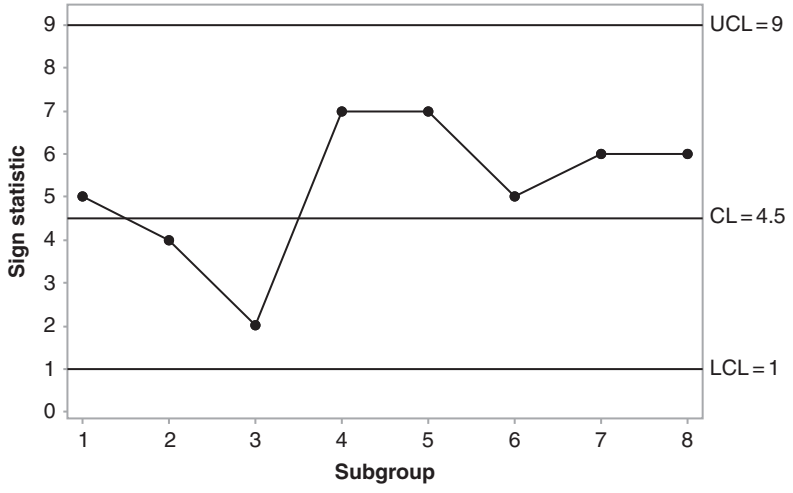


FIGURE 5.9 *A Shewhart FSI sign chart.*

Note that the Shewhart FSI chart does not signal. For the Shewhart VSI sign chart, we add warning limits $LWL = 2$ and $UWL = 8$. Since none of the charting statistics plots on the warning limits of $LWL = 2$ and $UWL = 8$, respectively, the time interval does not have to be decreased from 90 minutes to 30 minutes and we continue using every third sample. The resulting Shewhart VSI sign chart is shown in Figure 5.10.

The important conclusion is that an adaptive control chart using VSS and VSI schemes instead of the traditional FSI and FSS schemes, will most

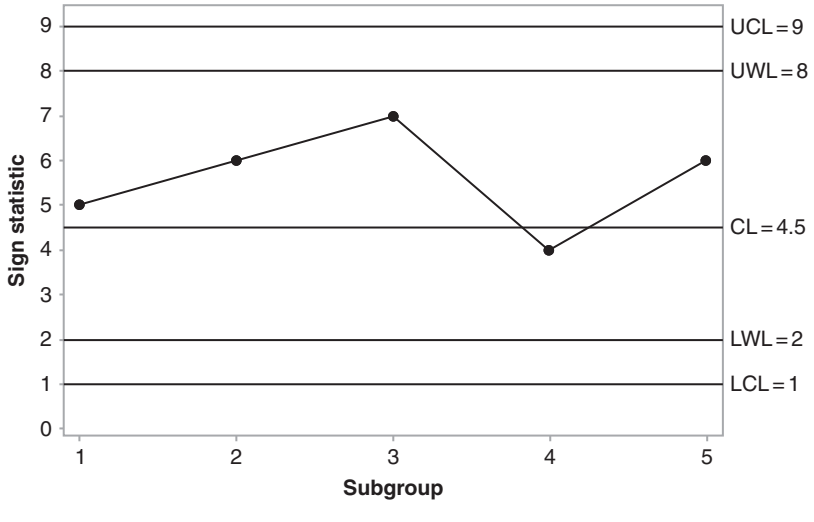


FIGURE 5.10 *A Shewhart VSI sign chart.*

likely result in considerable improvements in the performance of the chart. Thus adaptive control charts can be more effective than the traditional static ones in detecting process shifts, which, in practice, may lead to some economic benefits. In addition, whether we are using parametric or nonparametric control charts, the adaptive schemes are favorable over the traditional fixed schemes. On the other hand, one disadvantage of these adaptive charts is that they might be more complicated to administer. Nevertheless, the potential advantages that VSS and VSI schemes offer are too great to be overlooked.

Although much has been accomplished in the last few years and a lot of work is currently underway, much more remains to be done. From a practical standpoint, the adaptive charting procedures must be made more accessible to the practitioner and, to this end, the ease of implementation is vital. Computer programs, add-ons to popular software packages such as Minitab® and SAS®, and/or websites would greatly help in this effort. In terms of research, more work needs to be done on adaptive charts for scale and on charts for simultaneous monitoring of location and scale parameters. In addition, more work needs to be done on adaptive nonparametric control charts.

Appendix A

Tables

Table A: Binomial distribution – probabilities for the in-control case

Each table entry (i) labelled P is the tail probability, i.e., left tail ($P(X \leq x)$) and right tail ($P(X \geq x)$) (ii) with probability of success $p = 0.5$ on each trial.

n	Left tail	P	Right tail	n	Left tail	P	Right tail	n	Left tail	P	Right tail
1	0	0.5000	1	12	0	0.0002	12	17	0	0.0000	17
2	0	0.2500	2		1	0.0032	11		1	0.0001	16
	1	0.7500	1		2	0.0193	10		2	0.0012	15
3	0	0.1250	3		3	0.0730	9		3	0.0064	14
	1	0.5000	2		4	0.1938	8		4	0.0245	13
4	0	0.0625	4		5	0.3872	7		5	0.0717	12
	1	0.3125	3		6	0.6128	6		6	0.1662	11
	2	0.6875	2	13	0	0.0001	13		7	0.3145	10
5	0	0.0312	5		1	0.0017	12		8	0.5000	9
	1	0.1875	4		2	0.0112	11	18	0	0.0000	18
	2	0.5000	3		3	0.0461	10		1	0.0001	17
6	0	0.0156	6		4	0.1334	9		2	0.0007	16
	1	0.1094	5		5	0.2905	8		3	0.0038	15
	2	0.3438	4		6	0.5000	7		4	0.0154	14
	3	0.6562	3	14	0	0.0000	14		5	0.0481	13

(*continued overleaf*)

Nonparametric Statistical Process Control, First Edition.
Subhabrata Chakraborti and Marien Alet Graham.
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Companion website: www.wiley.com/go/chakraborti/Nonparametric_Statistical_Process_Control

n	Left tail	P	Right tail	n	Left tail	P	Right tail	n	Left tail	P	Right tail
7	0	0.0078	7		1	0.0009	13		6	0.1189	12
	1	0.0625	6		2	0.0065	12		7	0.2403	11
	2	0.2266	5		3	0.0287	11		8	0.4073	10
	3	0.5000	4		4	0.0898	10		9	0.5927	9
8	0	0.0039	8		5	0.2120	9	19	0	0.0000	19
	1	0.0352	7		6	0.3953	8		1	0.0000	18
	2	0.1445	6		7	0.6047	7		2	0.0004	17
	3	0.3633	5	15	0	0.0000	15		3	0.0022	16
	4	0.6367	4		1	0.0005	14		4	0.0096	15
9	0	0.0020	9		2	0.0037	13		5	0.0318	14
	1	0.1950	8		3	0.0176	12		6	0.0835	13
	2	0.8980	7		4	0.0592	11		7	0.1796	12
	3	0.2539	6		5	0.1509	10		8	0.3238	11
	4	0.5000	5		6	0.3036	9		9	0.5000	10
10	0	0.0010	10		7	0.5000	8	20	0	0.0000	20
	1	0.0107	9	16	0	0.0000	16		1	0.0000	19
	2	0.0547	8		1	0.0003	15		2	0.0002	18
	3	0.1719	7		2	0.0021	14		3	0.0013	17
	4	0.3770	6		3	0.0106	13		4	0.0059	16
	5	0.6230	5		4	0.0384	12		5	0.0207	15
11	0	0.0005	11		5	0.1051	11		6	0.0577	14
	1	0.0059	10		6	0.2272	10		7	0.1316	13
	2	0.0327	9		7	0.4018	9		8	0.2517	12
	3	0.1133	8		8	0.5982	8		9	0.4119	11
	4	0.2744	7						10	0.5881	10
	5	0.5000	6								

Note: These values are obtained using Excel's BINOM.DIST function.

Table B: Probabilities for the Wilcoxon signed-rank statistic

Each table entry labelled P is the tail probability from each extreme to the value of T_n, the Wilcoxon signed-rank statistic for sample size n, where T_n is interpreted as either T_n^+ or T_n^-.

n	Left tail	P	Right tail	n	Left tail	P	Right tail	n	Left tail	P	Right tail
2	0	0.250	3	7	0	0.008	28	9	0	0.002	45
	1	0.500	2		1	0.016	27		1	0.004	44
3	0	0.125	6		2	0.023	26		2	0.006	43
	1	0.250	5		3	0.039	25		3	0.010	42
	2	0.375	4		4	0.055	24		4	0.014	41
	3	0.625	3		5	0.078	23		5	0.020	40
4	0	0.062	10		6	0.109	22		6	0.027	39
	1	0.125	9		7	0.148	21		7	0.037	38
	2	0.188	8		8	0.188	20		8	0.049	37
	3	0.312	7		9	0.234	19		9	0.064	36
	4	0.438	6		10	0.289	18		10	0.082	35
	5	0.562	5		11	0.344	17		11	0.102	34
5	0	0.031	15		12	0.406	16		12	0.125	33
	1	0.062	14		13	0.469	15		13	0.150	32
	2	0.094	13		14	0.531	14		14	0.180	31
	3	0.156	12	8	0	0.004	36		15	0.213	30
	4	0.219	11		1	0.008	35		16	0.248	29
	5	0.312	10		2	0.012	34		17	0.285	28
	6	0.406	9		3	0.020	33		18	0.326	27
	7	0.500	8		4	0.027	32		19	0.367	26
6	0	0.016	21		5	0.039	31		20	0.410	25
	1	0.031	20		6	0.055	30		21	0.455	24
	2	0.047	19		7	0.074	29		22	0.500	23
	3	0.078	18		8	0.098	28	10	0	0.001	55
	4	0.109	17		9	0.125	27		1	0.002	54
	5	0.156	16		10	0.156	26		2	0.003	53
	6	0.219	15		11	0.191	25		3	0.005	52
	7	0.281	14		12	0.230	24		4	0.007	51
	8	0.344	13		13	0.273	23		5	0.010	50
	9	0.422	12		14	0.320	22		6	0.014	49
	10	0.500	11		15	0.371	21		7	0.019	48
					16	0.422	20		8	0.024	47
					17	0.473	19		9	0.032	46
					18	0.527	18		10	0.042	45

(*continued overleaf*)

n	Left tail	P	Right tail	n	Left tail	P	Right tail	n	Left tail	P	Right tail
10	11	0.053	44	11	28	0.350	38	13	0	0.000	91
	12	0.065	43		29	0.382	37		1	0.000	90
	13	0.080	42		30	0.416	36		2	0.000	89
	14	0.097	41		31	0.449	35		3	0.001	88
	15	0.116	40		32	0.483	34		4	0.001	87
	16	0.138	39		33	0.517	33		5	0.001	86
	17	0.161	38	12	0	0.000	78		6	0.002	85
	18	0.188	37		1	0.000	77		7	0.002	84
	19	0.216	36		2	0.001	76		8	0.003	83
	20	0.246	35		3	0.001	75		9	0.004	82
	21	0.278	34		4	0.002	74		10	0.005	81
	22	0.312	33		5	0.002	73		11	0.007	80
	23	0.348	32		6	0.003	72		12	0.009	79
	24	0.385	31		7	0.005	71		13	0.011	78
	25	0.423	30		8	0.006	70		14	0.013	77
	26	0.461	29		9	0.008	69		15	0.016	76
	27	0.500	28		10	0.010	68		16	0.020	75
11	0	0.000	66		11	0.013	67		17	0.024	74
	1	0.001	65		12	0.017	66		18	0.029	73
	2	0.001	64		13	0.021	65		19	0.034	72
	3	0.002	63		14	0.026	64		20	0.040	71
	4	0.003	62		15	0.032	63		21	0.047	70
	5	0.005	61		16	0.039	62		22	0.055	69
	6	0.007	60		17	0.046	61		23	0.064	68
	7	0.009	59		18	0.055	60		24	0.073	67
	8	0.013	58		19	0.065	59		25	0.084	66
	9	0.016	57		20	0.076	58		26	0.095	65
	10	0.021	56		21	0.088	57		27	0.108	64
	11	0.027	55		22	0.102	56		28	0.122	63
	12	0.034	54		23	0.117	55		29	0.137	62
	13	0.042	53		24	0.133	54		30	0.153	61
	14	0.051	52		25	0.151	53		31	0.170	60
	15	0.062	51		26	0.170	52		32	0.188	59
	16	0.074	50		27	0.190	51		33	0.207	58
	17	0.087	49		28	0.212	50		34	0.227	57
	18	0.103	48		29	0.235	49		35	0.249	56
	19	0.120	47		30	0.259	48		36	0.271	55
	20	0.139	46		31	0.285	47		37	0.294	54

n	Left tail	P	Right tail	n	Left tail	P	Right tail	n	Left tail	P	Right tail
	21	0.160	45		32	0.311	46		38	0.318	53
	22	0.183	44		33	0.339	45		39	0.342	52
	23	0.207	43		34	0.367	44		40	0.368	51
	24	0.232	42		35	0.396	43		41	0.393	50
	25	0.260	41		36	0.425	42		42	0.420	49
	26	0.289	40		37	0.455	41		43	0.446	48
	27	0.319	39		38	0.485	40		44	0.473	47
					39	0.515	39		45	0.500	46

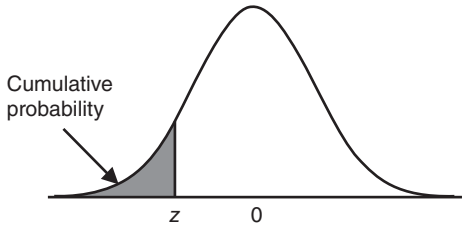
Note: These values are taken from Gibbons and Chakraborti (2010, Table H, p. 567).

Table C: Unbiasing charting constants for the construction of normal-theory variables control charts

n	2	3	4	5	6	7	8	9	10	For larger n
A_2	1.880	1.023	0.729	0.577	0.483	0.419	0.373	0.337	0.308	
A_3	2.659	1.954	1.628	1.427	1.287	1.182	1.099	1.032	0.975	$\frac{3}{c_4\sqrt{n}}$
c_4	0.7979	0.8862	0.9213	0.9400	0.9515	0.9594	0.9650	0.9693	0.9727	$\frac{4(n-1)}{4n-3}$
B_3	0	0	0	0	0.030	0.118	0.185	0.239	0.284	$1-\frac{3}{c_4\sqrt{2(n-1)}}$
B_4	3.267	2.568	2.266	2.089	1.970	1.882	1.815	1.761	1.716	$1+\frac{3}{c_4\sqrt{2(n-1)}}$
B_5	0	0	0	0	0.029	0.113	0.179	0.232	0.276	$c_4-\frac{3}{\sqrt{2(n-1)}}$
B_6	2.606	2.276	2.088	1.964	1.874	1.806	1.751	1.707	1.669	$c_4+\frac{3}{\sqrt{2(n-1)}}$
d_2	1.128	1.693	2.059	2.326	2.534	2.704	2.847	2.970	3.078	
d_3	0.853	0.888	0.880	0.864	0.848	0.833	0.820	0.808	0.797	
D_1	0	0	0	0	0	0.204	0.388	0.547	0.687	
D_2	3.686	4.358	4.698	4.918	5.078	5.204	5.306	5.393	5.469	
D_3	0	0	0	0	0	0.076	0.136	0.184	0.223	
D_4	3.267	2.574	2.282	2.114	2.004	1.924	1.864	1.816	1.777	

Note: These values are taken from Montgomery (2009, Appendix VI, p. 702).

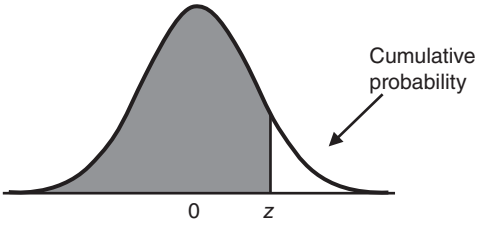
Table D1: Cumulative probabilities for the standard normal distribution



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
−1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
−1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
−1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
−1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
−1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
−1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
−1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
−1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
−1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
−0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
−0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
−0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
−0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
−0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
−0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
−0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
−0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
−0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
−0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Note: These values are obtained using Excel's NORM.S.DIST function.

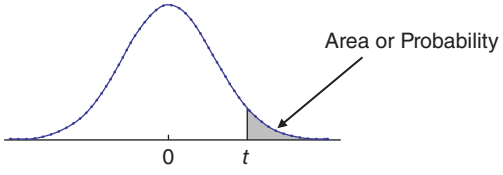
Table D2: Cumulative probabilities for the standard normal distribution continued



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990

Note: These values are obtained using Excel's NORM.S.DIST function.

Table E: Upper tail probabilities for the t distribution



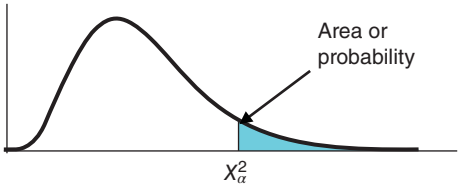
Degrees of freedom	Area in Upper Tail					
	0.20	0.10	0.05	0.025	0.01	0.005
1	1.376	3.078	6.314	12.706	31.821	63.657
2	1.061	1.886	2.920	4.303	6.965	9.925
3	0.978	1.638	2.353	3.182	4.541	5.841
4	0.941	1.533	2.132	2.776	3.747	4.604
5	0.920	1.476	2.015	2.571	3.365	4.032
6	0.906	1.440	1.943	2.447	3.143	3.707
7	0.896	1.415	1.895	2.365	2.998	3.499
8	0.889	1.397	1.860	2.306	2.896	3.355
9	0.883	1.383	1.833	2.262	2.821	3.250
10	0.879	1.372	1.812	2.228	2.764	3.169
11	0.876	1.363	1.796	2.201	2.718	3.106
12	0.873	1.356	1.782	2.179	2.681	3.055
13	0.870	1.350	1.771	2.160	2.650	3.012
14	0.868	1.345	1.761	2.145	2.624	2.977
15	0.866	1.341	1.753	2.131	2.602	2.947
16	0.865	1.337	1.746	2.120	2.583	2.921
17	0.863	1.333	1.740	2.110	2.567	2.898
18	0.862	1.330	1.734	2.101	2.552	2.878
19	0.861	1.328	1.729	2.093	2.539	2.861
20	0.860	1.325	1.725	2.086	2.528	2.845
21	0.859	1.323	1.721	2.080	2.518	2.831

(*continued overleaf*)

Degrees of freedom	Area in Upper Tail					
	0.20	0.10	0.05	0.025	0.01	0.005
22	0.858	1.321	1.717	2.074	2.508	2.819
23	0.858	1.319	1.714	2.069	2.500	2.807
24	0.857	1.318	1.711	2.064	2.492	2.797
25	0.856	1.316	1.708	2.060	2.485	2.787
26	0.856	1.315	1.706	2.056	2.479	2.779
27	0.855	1.314	1.703	2.052	2.473	2.771
28	0.855	1.313	1.701	2.048	2.467	2.763
29	0.854	1.311	1.699	2.045	2.462	2.756
30	0.854	1.310	1.697	2.042	2.457	2.750
31	0.853	1.309	1.696	2.040	2.453	2.744
32	0.853	1.309	1.694	2.037	2.449	2.738
33	0.853	1.308	1.692	2.035	2.445	2.733
34	0.852	1.307	1.691	2.032	2.441	2.728
35	0.852	1.306	1.690	2.030	2.438	2.724
36	0.852	1.306	1.688	2.028	2.434	2.719
37	0.851	1.305	1.687	2.026	2.431	2.715
38	0.851	1.304	1.686	2.024	2.429	2.712
39	0.851	1.304	1.685	2.023	2.426	2.708
40	0.851	1.303	1.684	2.021	2.423	2.704
41	0.850	1.303	1.683	2.020	2.421	2.701
42	0.850	1.302	1.682	2.018	2.418	2.698
43	0.850	1.302	1.681	2.017	2.416	2.695
44	0.850	1.301	1.680	2.015	2.414	2.692
45	0.850	1.301	1.679	2.014	2.412	2.690
46	0.850	1.300	1.679	2.013	2.410	2.687
47	0.849	1.300	1.678	2.012	2.408	2.685
48	0.849	1.299	1.677	2.011	2.407	2.682
49	0.849	1.299	1.677	2.010	2.405	2.680
50	0.849	1.299	1.676	2.009	2.403	2.678
∞	0.842	1.282	1.645	1.960	2.326	2.576

Note: These values are obtained using Excel's T.INV function.

Table F: Upper tail probabilities for the Chi-square distribution



Degrees of freedom	Area in the upper tail									
	0.995	0.99	0.975	0.95	0.9	0.1	0.05	0.025	0.01	0.005
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993

Degrees of freedom	Area in the upper tail									
	0.995	0.99	0.975	0.95	0.9	0.1	0.05	0.025	0.01	0.005
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
35	17.192	18.509	20.569	22.465	24.797	46.059	49.802	53.203	57.342	60.275
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
45	24.311	25.901	28.366	30.612	33.350	57.505	61.656	65.410	69.957	73.166
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
55	31.735	33.570	36.398	38.958	42.060	68.796	73.311	77.380	82.292	85.749
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
65	39.383	41.444	44.603	47.450	50.883	79.973	84.821	89.177	94.422	98.105
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

Note: These values are obtained using Excel's CHISQ.INV function.

Table G: Charting constants for the Phase II Shewhart $\overline{X}$ control chart in Case UU for $n = 5$, varying m, and $ARL_{\text{IC}} = 370$ and 500

m	$ARL_{\text{IC}} = 370$	$ARL_{\text{IC}} = 500$
5	2.7838	2.8463
10	2.9083	2.9844
20	2.9630	3.0465
25	2.9725	3.0574
30	2.9784	3.0643
35	2.9823	3.0690
40	2.9852	3.0723
45	2.9873	3.0748
50	2.9889	3.0768
75	2.9933	3.0821
100	2.9952	3.0845
200	2.9977	3.0877
300	2.9985	3.0886

Note: These values are obtained using a simulation algorithm programmed in SAS.

Table H: Charting constants for Phase II Shewhart R and S control charts in Case UU with three Phase I estimators of standard deviation for nominal ARL_{IC} values of 370 and 500 with varying m and $n = 5, 10$

n	m	R chart with estimator $\overline{R}/d_2$			S chart with estimator S_p			S chart with estimator $\overline{S}/c_4$		
		$\alpha(m,n)$	$D_3^*(m,n)$	$D_4^*(m,n)$	$\alpha(m,n)$	$H_3^*(m,n)$	$H_4^*(m,n)$	$\alpha(m,n)$	$B_3^*(m,n)$	$B_4^*(m,n)$
					$ARL_{\text{IC}} = 370$					
5	**5**	0.00195	0.15703	2.36113	0.00191	0.14887	2.15474	0.00196	0.15947	2.28860
	10	0.00220	0.16176	2.34350	0.00217	0.15375	2.13835	0.00220	0.16420	2.27276
	20	0.00239	0.16518	2.33095	0.00237	0.15724	2.12683	0.00239	0.16766	2.26140
	25	0.00243	0.16603	2.32788	0.00242	0.15811	2.12402	0.00244	0.16851	2.25862
	30	0.00247	0.16665	2.32566	0.00246	0.15873	2.12198	0.00247	0.16913	2.25662
	50	0.00255	0.16800	2.32077	0.00254	0.16009	2.11759	0.00255	0.17049	2.25222
	75	0.00259	0.16876	2.31806	0.00259	0.16083	2.11521	0.00260	0.17126	2.24975
	100	0.00262	0.16917	2.31661	0.00261	0.16121	2.11401	0.00262	0.17167	2.24844
10	**5**	0.00182	0.34818	1.95122	0.00181	0.35341	1.76806	0.00183	0.36388	1.81677
	10	0.00210	0.35450	1.93597	0.00209	0.35979	1.75620	0.00211	0.37022	1.80499
	20	0.00232	0.35905	1.92511	0.00232	0.36439	1.74776	0.00233	0.37479	1.79658
	25	0.00238	0.36017	1.92246	0.00238	0.36551	1.74570	0.00238	0.37592	1.79453
	30	0.00242	0.36097	1.92056	0.00242	0.36633	1.74421	0.00243	0.37673	1.79304
	50	0.00252	0.36275	1.91636	0.00252	0.36811	1.74096	0.00252	0.37852	1.78978
	75	0.00257	0.36374	1.91402	0.00257	0.36910	1.73916	0.00257	0.37953	1.78795
	100	0.00260	0.36427	1.91279	0.00260	0.36964	1.73818	0.00260	0.38007	1.78697
					$ARL_{\text{IC}} = 500$					
5	**5**	0.00144	0.14525	2.40701	0.00140	0.13768	2.19393	0.00144	0.14751	2.33028
	10	0.00162	0.14964	2.38958	0.00159	0.14223	2.17774	0.00162	0.15193	2.31459
	20	0.00176	0.15287	2.37704	0.00174	0.14552	2.16624	0.00176	0.15516	2.30332
	25	0.00179	0.15367	2.37394	0.00178	0.14634	2.16342	0.00180	0.15597	2.30055
	30	0.00182	0.15425	2.37173	0.00181	0.14692	2.16142	0.00182	0.15655	2.29856
	50	0.00188	0.15553	2.36684	0.00187	0.14822	2.15697	0.00188	0.15784	2.29412
	75	0.00192	0.15626	2.36408	0.00191	0.14892	2.15456	0.00192	0.15857	2.29165
	100	0.00194	0.15665	2.36258	0.00193	0.14928	2.15336	0.00194	0.15897	2.29028
10	**5**	0.00133	0.33496	1.98385	0.00133	0.34015	1.79315	0.00134	0.35022	1.84257
	10	0.00154	0.34109	1.96861	0.00154	0.34633	1.78138	0.00155	0.35637	1.83087
	20	0.00171	0.34551	1.95773	0.00171	0.35081	1.77293	0.00171	0.36083	1.82247
	25	0.00175	0.34661	1.95504	0.00175	0.35191	1.77087	0.00176	0.36194	1.82039
	30	0.00178	0.34740	1.95312	0.00178	0.35271	1.76937	0.00179	0.36274	1.81890
	50	0.00186	0.34915	1.94887	0.00186	0.35446	1.76611	0.00186	0.36450	1.81561
	75	0.00190	0.35014	1.94648	0.00190	0.35545	1.76425	0.00190	0.36549	1.81375
	100	0.00192	0.35066	1.94522	0.00192	0.35600	1.76324	0.00192	0.36603	1.81276

Note: These values are obtained using a simulation algorithm programmed in R.

Appendix B

Programmes

The SAS programs used in the text are provided in this section, which are also posted at the companion website for easy downloading. It should be noted that although the SAS programs below are all for the normal distribution, the **call randgen** function can be adjusted to generate observations from any desired distribution.

The following table lists the figures containing SAS output.

Figure number	Figure title	SAS program number
4.2	SAS output of the RL distribution where the design parameters ($n = 10$, $a = 10$) of the Shewhart-SN chart were chosen for a nominal $ARL_{\text{IC}} = 500$	1
4.4	SAS output of the RL distribution where the design parameters ($n = 13$, $a = 81$) of the Shewhart-SN chart were chosen for a nominal $ARL_{\text{IC}} = 500$	2
4.5	SAS output of the RL distribution where the design parameters ($n = 10$, $k = 6$, $h = 4$) of the CUSUM-SN chart were chosen for a nominal $ARL_{\text{IC}} = 500$	3
4.6	SAS output of the RL distribution where the design parameters ($n = 10$, $k = 41$, $h = 24$) of the CUSUM-SR chart were chosen so that nominal $ARL_{\text{IC}} = 500$	4

Nonparametric Statistical Process Control, First Edition.
Subhabrata Chakraborti and Marien Alet Graham.
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Companion website: www.wiley.com/go/chakraborti/Nonparametric_Statistical_Process_Control

Figure number	Figure title	SAS program number
4.7	SAS output of the RL distribution where the design parameters ($n = 1$, $\lambda = 0.10$, $L = 2.667$) of the EWMA-SN chart were chosen for a nominal $ARL_{\text{IC}} = 500$	5
4.8	SAS output of the RL distribution where the design parameters ($n = 10$, $\lambda = 0.1$, $L = 2.794$) of the EWMA-SR chart were chosen for a nominal $ARL_{\text{IC}} = 500$	6
4.23	SAS output of the RL distribution where the design parameters ($m = 1000$, $n = 5$, $j = 3$, $a = 48$, $b = 953$) of the Shewhart-Prec chart were for a nominal $ARL_{\text{IC}} = 500$	7
4.24	SAS output where the design parameters ($m = 500$, $n = 5$, $b_{mn} = 2172$) of the Shewhart-MW chart were chosen for a nominal $ARL_{\text{IC}} = 500$	8
4.25	SAS output where the design parameters ($m = 125$, $n = 5$, $j = 3$, $h = 9.775$, $k = n$(d*-d)) of the CUSUM-EX chart were chosen for a nominal $ARL_{\text{IC}} = 500$	9
4.26	SAS output where the design parameters ($m = 100$, $n = 5$, $h = 353$, $k = 0$) of the CUSUM-Rank chart were chosen for a nominal $ARL_{\text{IC}} = 500$	10
4.27	SAS output where the design parameters ($m = 100$, $n = 5$, $\lambda = 0.10$, $L = 2.220$) of the EWMA-EX chart were chosen for a nominal $ARL_{\text{IC}} = 500$	11
4.28	SAS output where the design parameters ($m = 100$, $n = 5$, $\lambda = 0.10$, $LCL = 219.5$, $UCL = 310.5$) of the EWMA-Rank chart were chosen for a nominal $ARL_{\text{IC}} = 500$	12

SAS program number	Figure title	Figure number
1	SAS output of the RL distribution where the design parameters ($n = 10$, $a = 10$) of the Shewhart-SN chart were chosen for a nominal $ARL_{\text{IC}} = 500$	4.2

SAS program

For the Shewhart-SN chart, the SAS program for the EWMA-SN chart (SAS program number 5 shown below) can be used with $\lambda = 1$.

SAS program number	Figure title	Figure number
2	SAS output of the RL distribution where the design parameters ($n = 13$, $a = 81$) of the Shewhart-SN chart were chosen for a nominal $ARL_{\text{IC}} = 500$	4.4

SAS program

For the Shewhart-SR chart, the SAS program for the EWMA-SR chart (SAS program number 6 shown below) can be used with $\lambda = 1$

SAS program number	Figure title	Figure number
3	SAS output of the RL distribution where the design parameters ($n = 10$, $k = 6$, $h = 4$) of the CUSUM-SN chart were chosen for a nominal $ARL_{\text{IC}} = 500$	4.5

SAS program

```
proc iml;

* Number of simulations;
sim = 10000;

* Design parameters of CUSUM control chart;
n = 10;
H = 4;
k = 6;

* Size of the shift;
shift = 0;

*One-sided or two-sided chart;
*If one-sided, then type = 1;
*If two-sided, then type = 2;
type = 2;

*Constructing the control chart;
runl = j(sim,1,.);

stdev = 1;
xi = j(n,1,.);
ti_vec = j(n,1,.);
```

```
do o = 1 to sim;

      * Initializing values;
      indicator = 0;
      count = 0;
      Ci_1_plus = 0;
      Ci_1_minus = 0;

      do i = 1 to 1000000000000000 until (indicator = 1);
      count = count + 1;

      * Generating observations from the Normal distribution;

      call randgen(xi,'NORMAL',shift,1);

      do u = 1 to nrow(xi);
            if xi[u,]>=0 then ti_vec[u,]=1;
            else ti_vec[u,]=0;
      end;

      ti = ti_vec[+,];
      sni=2*ti - n;

      * Charting statistics;
      Ci_plus = max(0, (Ci_1_plus + sni - k));
      Ci_minus = max(0, (Ci_1_minus - sni - k));

*Comparing the charting statistics to the control limits;
if type = 2 then do;
            if ((Ci_plus >= H) | (Ci_minus >= H)) then indicator = 1;
end;
if type = 1 then do;
            if Ci_plus >= H then indicator = 1;
end;

Ci_1_plus = Ci_plus;
Ci_1_minus = Ci_minus;
end;
runl[o,1] = count;
end;

create CUSUM from runl[colname={RL}];
append from runl;
proc univariate data = CUSUM noprint;
histogram;
inset mean std p5 q1 median q3 p95 / format=10.2 pos = ne;
run;
```

SAS program number	Figure title	Figure number
4	SAS output of the RL distribution where the design parameters ($n = 10$, $k = 41$, $h = 24$) of the CUSUM-SR chart were chosen so that nominal $ARL_{\text{IC}} = 500$	4.6

SAS program

The SAS program for the CUSUM-SR is similar to that of the CUSUM-SN chart above, except that the one line of coding, namely

```
sni = 2*ti - n;
```

should be altered to compute the SR_i statistic instead of the SN_i which can easily be done making use of the RANKTIE function of SAS. This SAS program is not given here, for brevity. Instead, the SAS program for the Markov chain approach is given below so that the interested reader can see the programming complexities of the programming the Markov chain approach in SAS as opposed to making use of simulation.

```
proc iml;
* The reference value;
k = 41;

* The decision interval;
h = 24;

* The sample size;
n = 10;

* The z-value will be used in the calculation of the
pmf, P(N = z), and the cdf, P(N< = z);
z = 1000000;
```

```
* The following values will be used to calculate the 5th,
25th, 50th, 75th and 95th percentiles, respectively;
p5p = 0.05;
p25p = 0.25;
p50p = 0.5;
p75p = 0.75;
p95p = 0.95;

* Calculating the state space;
SRn = do((-n*(n+1)/2),(n*(n+1)/2),2)';
S = j(nrow(SRn),1,1);

do i = 1 to nrow(SRn);
S[i,] = min(h,(max(0,SRn[i,]-k)));
end;

do i = 1 to nrow(S);
do j = 1 to nrow(S);
if i = j then S[i,] = S[i,];
else if S[i,] = S[j,] then S[j,] = 999;
end;
end;

S = S[loc(S<999)];

upper_limit = (n*(n+1))/2 - k;
if h > upper_limit then print "Not possible";

S_S = j(nrow(S),1,0);

do i = 1 to nrow(S);
S_S[i,] = -S[i,];
end;
S = S||S_S;

SS = j(nrow(S)+nrow(S),1,0);

do i = 1 to nrow(SS) by 2;
      SS[i,] = S[(i+1)/2,1];
SS[i+1,] = S[(i+1)/2,2];
end;

SS = SS[2:nrow(SS),];
```

```
* Defining the vector eta used in the formula for the
ARL given by E(N) = eta*inv(I-Q)*one;
eta = j(1,nrow(SS)-2,1);

do i = 1 to nrow(SS)-1;
if i = 1 then eta[1,i] = 1;
else eta[1,i] = 0;
end;

* Calculating the transition probability matrix;
P = j( nrow(SS) , nrow(SS), 0);

* Wilcoxon signed-rank probabilities for a sample size of 4;
if n = 4 then do;
T4 = j((n*(n+1)/2)+1,1,1);
T4[1:3,] = 1;
T4[4:8,] = 2;
T4[9:11,] = 1;
T = T4/(2**n);
end;

* Wilcoxon signed-rank probabilities for a sample size of 5;
if n = 5 then do;
T5 = j((n*(n+1)/2)+1,1,1);
T5[1:3,] = 1;
T5[4:5,] = 2;
T5[6:11,] = 3;
T5[12:13,] = 2;
T5[14:16,] = 1;
T = T5/(2**n);
end;

* Wilcoxon signed-rank probabilities for a sample size of 6;
if n = 6 then do;
T6 = j((n*(n+1)/2)+1,1,1);
T6[1:3,] = 1;
T6[4:5,] = 2;
T6[6,] = 3;
T6[7:9,] = 4;
T6[10:13,] = 5;
T6[14:16,] = 4;
T6[17,] = 3;
T6[18:19,] = 2;
T6[20:22,] = 1;
T = T6/(2**n);
end;
```

```
* Wilcoxon signed-rank probabilities for a sample size of 7;
if n = 7 then do;
T7 = j((n*(n+1)/2)+1,1,1);
T7[1:3,] = 1;
T7[4:5,] = 2;
T7[6,] = 3;
T7[7,] = 4;
T7[8:9,] = 5;
T7[10,] = 6;
T7[11:12,] = 7;
T7[13:17,] = 8;
T7[18:19,] = 7;
T7[20,] = 6;
T7[21:22,] = 5;
T7[23,] = 4;
T7[24,] = 3;
T7[25:26,] = 2;
T7[27:29,] = 1;
T = T7/(2**n);
end;

* Wilcoxon signed-rank probabilities for a sample size of 8;
if n = 8 then do;
T8 = j((n*(n+1)/2)+1,1,1);
T8[1:3,] = 1;
T8[4:5,] = 2;
T8[6,] = 3;
T8[7,] = 4;
T8[8,] = 5;
T8[9,] = 6;
T8[10,] = 7;
T8[11,] = 8;
T8[12,] = 9;
T8[13,] = 10;
T8[14,] = 11;
T8[15,] = 12;
T8[16:18,] = 13;
T8[19,] = 14;
T8[20:22,] = 13;
T8[23,] = 12;
T8[24,] = 11;
T8[25,] = 10;
T8[26,] = 9;
T8[27,] = 8;
T8[28,] = 7;
T8[29,] = 6;
T8[30,] = 5;
T8[31,] = 4;
T8[32,] = 3;
T8[33:34,] = 2;
T8[35:37,] = 1;
T = T8/(2**n);
end;
```

```
* Wilcoxon signed-rank probabilities for a sample size of 9;
if n = 9 then do;
T9 = j((n*(n+1)/2)+1,1,1);
T9[1:3,] = 1;
T9[4:5,] = 2;
T9[6,] = 3;
T9[7,] = 4;
T9[8,] = 5;
T9[9,] = 6;
T9[10,] = 8;
T9[11,] = 9;
T9[12,] = 10;
T9[13,] = 12;
T9[14,] = 13;
T9[15,] = 15;
T9[16,] = 17;
T9[17,] = 18;
T9[18,] = 19;
T9[19:20,] = 21;
T9[21,] = 22;
T9[22:25,] = 23;
T9[26,] = 22;
T9[27:28,] = 21;
T9[29,] = 19;
T9[30,] = 18;
T9[31,] = 17;
T9[32,] = 15;
T9[33,] = 13;
T9[34,] = 12;
T9[35,] = 10;
T9[36,] = 9;
T9[37,] = 8;
T9[38,] = 6;
T9[39,] = 5;
T9[40,] = 4;
T9[41,] = 3;
T9[42:43,] = 2;
T9[44:46,] = 1;
T = T9/(2**n);
end;

* Wilcoxon signed-rank probabilities for a sample size of 10;
if n = 10 then do;
T10 = j((n*(n+1)/2)+1,1,1);
T10[1:3,] = 1;
T10[4:5,] = 2;
T10[6,] = 3;
T10[7,] = 4;
```

```
T10[8,] = 5;
T10[9,] = 6;
T10[10,] = 8;
T10[11,] = 10;
T10[12,] = 11;
T10[13,] = 13;
T10[14,] = 15;
T10[15,] = 17;
T10[16,] = 20;
T10[17,] = 22;
T10[18,] = 24;
T10[19,] = 27;
T10[20,] = 29;
T10[21,] = 31;
T10[22,] = 33;
T10[23,] = 35;
T10[24,] = 36;
T10[25,] = 38;
T10[26:27,] = 39;
T10[28:29,] = 40;
T10[30:31,] = 39;
T10[32,] = 38;
T10[33,] = 36;
T10[34,] = 35;
T10[35,] = 33;
T10[36,] = 31;
T10[37,] = 29;
T10[38,] = 27;
T10[39,] = 24;
T10[40,] = 22;
T10[41,] = 20;
T10[42,] = 17;
T10[43,] = 15;
T10[44,] = 13;
T10[45,] = 11;
T10[46,] = 10;
T10[47,] = 8;
T10[48,] = 6;
T10[49,] = 5;
T10[50,] = 4;
T10[51,] = 3;
T10[52:53,] = 2;
T10[54:56,] = 1;
T = T10/(2**n);
end;

nn = n*(n+1)/2;
```

```
* Calculating the first column of the transition matrix;
do i = 1 to nrow(SS)-2;
if SS[i,]<0 then uppersmall_t = (k-0+nn)/2;
else if SS[i,] > = 0 then uppersmall_t = (k-SS[i,]+nn)/2;
if SS[i,]> = 0 then lowersmall_t = (-k-0+nn)/2;
else if SS[i,] < 0 then lowersmall_t = (-k-SS[i,]+nn)/2;
if lowersmall_t < = uppersmall_t then P[i,1] =
sum(T[(lowersmall_t+1):(uppersmall_t+1),]);
else if lowersmall_t > uppersmall_t then P[i,1] = 0;
end;

* Calculating the middle columns of the transition matrix;
do j = 2 to nrow(SS)-2;
do i = 1 to nrow(SS)-2;
if SS[j,]>0 then if SS[i,]<0 then small_t=ceil((SS[j,]+k-0+nn)/2);
else if SS[i,]> = 0 then small_t = ceil((SS[j,]+k-SS[i,]+nn)/2);
if SS[j,]<0 then if SS[i,]>0 then small_t=ceil((SS[j,]-k-0+nn)/2);
else if SS[i,]< = 0 then small_t = ceil((SS[j,]-k-SS[i,]+nn) / 2);
P[i,j] = T[small_t+1,];
end;
end;

* Calculating the second last column of the transition matrix;
do i = 1 to nrow(SS)-2;
if SS[i,]0 then small_t = ceil((SS[ncol(P)-1,]+k-0+nn) / 2);
else if SS[i,]> = 0 then small_t =
ceil((SS[ncol(P)-1,]+k-SS[i,]+nn) / 2);
P[i,ncol(P)-1] = sum(T[small_t+1:nrow(T),]);
end;

* Calculating the last column of the transition matrix;
do i = 1 to nrow(SS)-2;
if SS[i,]>0 then small_t = ceil((SS[ncol(P),]-k-0+nn) / 2);
else if SS[i,]< = 0 then small_t =
ceil((SS[ncol(P),]-k-SS[i,]+nn)/ 2);
P[i,ncol(P)] = sum(T[1:(small_t+1),]);
end;

P[nrow(P)-1,ncol(P)-1] = 1;
P[nrow(P),ncol(P)] = 1;

* Defining the vector one used in the formula for the ARL given by
E(N) = eta*inv(I-Q)*one;
one = j(nrow(P)-2,1,1);
```

```
* Defining the matrix Q used in the formula for the ARL given by
E(N) = eta*inv(I-Q)*one;
Q = P[1:nrow(P)-2,1:nrow(P)-2];

* Defining the identity matrix I used in the formula for the
ARL given by E(N) = eta*inv(I-Q)*one;
identity = I(nrow(P)-2);

* Calculating the 5th, 25th, 50th, 75th and 95th percentiles;
pmf = j(z,1,1);
cdf = j(z,1,1);
cdf_5th_p = j(z,1,1);
cdf_25th_p = j(z,1,1);
cdf_50th_p = j(z,1,1);
cdf_75th_p = j(z,1,1);
cdf_95th_p = j(z,1,1);

do i = 1 to z;
pmf[i,1] = eta * (Q**(i-1)) * (identity - Q) * one;
cdf[i,1] = sum(pmf[1:i,1]);
end;

index = j(z,1,1);

do i = 2 to z;
index[i,] = index[i-1,]+1;
end;

* Calculating the 5th percentile;
do i = 1 to z;
cdf_5th_p[i,] = cdf[i,];
if cdf_5th_p[i,]> = p5p then cdf_5th_p[i,] = 999;
end;

cdf_5th_p = cdf_5th_p[loc(cdf_5th_p<999)];

if cdf_5th_p[1,] = 999 then percentile_p5p = 1;
else percentile_p5p = nrow(cdf_5th_p)+1;

* Calculating the 25th percentile;
do i = 1 to z;
cdf_25th_p[i,] = cdf[i,];
if cdf_25th_p[i,]> = p25p then cdf_25th_p[i,] = 999;
end;

cdf_25th_p = cdf_25th_p[loc(cdf_25th_p<999)];

if cdf_25th_p[1,] = 999 then percentile_p25p = 1;
else percentile_p25p = nrow(cdf_25th_p)+1;
```

```
* Calculating the 50th percentile;
do i = 1 to z;
cdf_50th_p[i,] = cdf[i,];
if cdf_50th_p[i,]> = p50p then cdf_50th_p[i,] = 999;
end;

cdf_50th_p = cdf_50th_p[loc(cdf_50th_p<999)];

if cdf_50th_p[1,] = 999 then percentile_p50p = 1;
else percentile_p50p = nrow(cdf_50th_p)+1;

* Calculating the 75th percentile;
do i = 1 to z;
cdf_75th_p[i,] = cdf[i,];
if cdf_75th_p[i,]> = p75p then cdf_75th_p[i,] = 999;
end;

cdf_75th_p = cdf_75th_p[loc(cdf_75th_p999)];

if cdf_75th_p[1,] = 999 then percentile_p<75p = 1;
else percentile_p75p = nrow(cdf_75th_p)+1;

* Calculating the 95th percentile;
do i = 1 to z;
cdf_95th_p[i,] = cdf[i,];
if cdf_95th_p[i,]> = p95p then cdf_95th_p[i,] = 999;
end;

cdf_95th_p = cdf_95th_p[loc(cdf_95th_p<999)];

if cdf_95th_p[1,] = 999 then percentile_p95p = 1;
else percentile_p95p = nrow(cdf_95th_p)+1;

* Calculating the average run length (ARL);
ARL = eta * inv(identity-Q) * one;

* Calculating the second moment;
N2 = eta * (identity + Q) * (inv((identity-Q)**2)) * one;

* Calculating the standard deviation;
SDRL = sqrt (N2 - ((ARL)**2) );

* Printing the output;
print_cdf = index||cdf;
print_pmf = index||pmf;
```

```
print      k [label = 'Reference value'],
h [label = 'Decision interval'],
n [label = 'Sample size'],
SS [label = 'State Space'],
P [label = 'Transition probability matrix' format = .3],
ARL [label = 'ARL' format = .2],
SDRL [label = 'SDRL' format = .2],
percentile_p5p [label = '5th'],
percentile_p25p [label = '25th'],
percentile_p50p [label = '50th'],
percentile_p75p [label = '75th'],
percentile_p95p [label = '95th'];
run;
```

SAS program number	Figure title	Figure number
5	SAS output of the RL distribution where the design parameters ($n = 1$, $\lambda = 0.10$, $L = 2.667$) of the EWMA-SN chart were chosen for a nominal $ARL_{\text{IC}} = 500$	4.7

SAS program

```
proc iml;
* Number of simulations;
sim = 10000;
* Design parameters of EWMA control chart;
lambda = 0.1;
L = 2.667;
* Sample size;
n = 1;
* Size of the shift;
delta = 0;

*Constructing the NPEWMA-SN chart;
stdev = 1;
mean = delta * stdev/sqrt(n);
runlength = j(sim,1,.);
xi = j(n,1,.);
ti_vec = j(n,1,.);

* Steady state control limits for the EWMA chart;
UCL_steady = 0 + L*sqrt(n)*sqrt((lambda/(2-lambda)));
LCL_steady = 0 - L*sqrt(n)*sqrt((lambda/(2-lambda)));
```

```
do j = 1 to sim;
* Initializing values;
indicator = 0;
count = 0;
zi_1 = 0;
do i = 1 to 1000000000000000 until (indicator = 1);
count = count + 1;

* Generating observations from the Normal distribution;
call randgen(xi,'normal', mean,1);
do o = 1 to nrow(xi);
if xi[o,]>=0 then ti_vec[o,]=1;
else ti_vec[o,]=0;
end;
ti = ti_vec[+,];
sni = 2*ti - n;

* Charting statistic;
zi = lambda*sni + (1 - lambda) * zi_1;

* Comparing the charting statisitc to the steady state control limits;
if (zi>UCL_steady) | (zi<LCL_steady) then do; indicator = 1; end;
zi_1 = zi;

end;
runlength[j,1] = count;
end;
create EWMA from runlength[colname={RL}];
append from runlength;
proc univariate data = EWMA noprint;
histogram;
inset mean std p5 q1 median q3 p95 / format=10.1;
run;
```

SAS program number	Figure title	Figure number
6	SAS output of the RL distribution where the design parameters ($n = 10$, $\lambda = 0.1$, $L = 2.794$) of the EWMA-SR chart were chosen for a nominal $ARL_{\text{IC}} = 500$	4.8

SAS program

```
proc iml;
* Number of simulations;
sim = 100000;
runl = j(sim,1,.);
```

```
* Design parameters of EWMA control chart;
lambda = 0.1;
L = 2.794;
* Sample size;
n = 10;
* Shift;
gamma = 0;
stdev = 1;
mean = gamma * stdev / sqrt(n);
* Steady state control limits;
UCL = L * sqrt((lambda/(2-lambda)))*sqrt ((n*(n+1)*(2*n+1))/6 );
LCL = -UCL;

do k = 1 to sim;
* Initializing values;
indicator = 0;
count = 0;
zi_1 = 0;
do i = 1 to 10000000000 until (indicator = 1);
count = count + 1;
* Generating observations from the Normal distribution;
tv = 0; * Median of the normal distribution;
xi = j(n,1,.);
sign = j(n,1,.);
abs_diff = j(n,1,.);
sri = j(n,1,.);
call randgen (xi, 'normal', mean, 1);
do j = 1 to n;
if xi[j,1] > tv then sign[j,1] = 1;
if xi[j,1] < tv then sign[j,1] = -1;
if xi[j,1] = tv then sign[j,1] = 0;
end;
do j = 1 to n;
abs_diff[j,1] = abs(xi[j,1]-tv);
end;
rank_abs_diff = rank(abs_diff);
do j = 1 to n;
sri[j,1] = sign[j,1] # rank_abs_diff[j,1];
end;
signed_rank = sum(sri);
*Charting statistic;
zi = lambda*signed_rank + (1-lambda)*zi_1;
* Comparing the charting statistic to the control limits;
if (zi>UCL) | (zi<LCL) then do; indicator = 1; end;
zi_1 = zi;
end;
```

```
runl[k,1] = count;
end;
create EWMA from runl[colname = {RL}];
append from runl;
proc univariate data = EWMA noprint;
histogram;
inset mean std p5 q1 median q3 p95 / format = 10.1 pos = ne;
run;
```

SAS program number	Figure title	Figure number
7	SAS output of the RL distribution where the design parameters ($m = 1000$, $n = 5$, $j = 3$, $a = 48$, $b = 953$) of the Shewhart-Prec chart were for a nominal $ARL_{IC} = 500$	4.23

SAS program

```
proc iml;
*Design parameters of the control chart;
m=1000;
n=5;
a = 48;
b = 953;
*Number of simulations;
sim = 10000;
* Size of shift;
shift = 0.0;

*Constructing the control chart;
rlvec = j(sim,1,.);
j=(n+1)/2;

do k = 1 to sim;
refx = j(m,1,.);
call randgen(refx,'NORMAL',0,1);
call sort (refx,{1});
ucl = refx[b,1];
lcl = refx[a,1];
count = 0;
signal = 0;
count2 = 0;
vec={};

do i = 1 to 1000000000 until (signal=1);
testy = j(n,1,.);
```

```
call randgen (testy,'NORMAL',shift,1);
call sort(testy,{1});
plotstat = testy[j,1];
count2 = count2+1;

if ((plotstat >= ucl)|(plotstat <= lcl)) then signal = 1;
else signal = 0;

rlvec[k,1] = count2;
end;
end;
results = rlvec;
create runlength_precedence from results[colname = {d000}];
append from results;
quit;

data runlength_precedence;
set work.runlength_precedence;
proc univariate data = runlength_precedence noprint;

var d000;
histogram;
inset mean std p5 q1 median q3 p95/format = 10.2;
run;
```

SAS program number	Figure title	Figure number
8	SAS output where the design parameters ($m = 500$, $n = 5$, $b_{mn} = 2172$) of the Shewhart-MW chart were chosen for a nominal $ARL_{\text{IC}} = 500$	4.24

SAS program

```
proc iml;
m = 500;
n = 5;
ucl = 2172;
lcl = m*n-ucl;

sim = 100000;
rlvec = j(sim,13,.);
do d = 1 to 4 by 0.25;
   shift = 4*d-3;
   do k = 1 to sim;
      refx = j(m,1,.);
  xt = j(m,1,.);
```

```
  call randgen(refx,'normal',0,1);
  mn = m*n;
  count2 = 0;
  c = j(mn,1,.);
  testy = j(n,1,.);
      signal = 0;
  yt = j(n,1,.);
      do t = 1 to 1000000000 until (signal = 1);
  count1 = 0;
  call randgen (testy,'NORMAL',d-1,1);
            do i = 1 to m;
          do j = 1 to n;
                      count1 = count1+1;
                        if refx[i,]<testy[j,] then c[count1,] = 1;
                        else c[count1,] = 0;
                end;
       end;
            mxy = sum(c);
       if mxy > = ucl then signal = 1;
            else if mxy < = lcl then signal = 1;

       else signal = 0;
       count2 = count2+1;
            rlvec[k,shift] = count2;
  end;
   end;
end;
create runlength_prec_normal from rlvec[colname = {d000 d025
d050 d075 d100 d125 d150 d175 d200 d225 d250 d275 d300 }];
append from rlvec;
quit;
data runlength_prec_normal;
set work.runlength_prec_normal;

proc univariate data = runlength_prec_normal;
var d000 d025 d050 d075 d100 d125 d150 d175 d200 d225 d250 d275 d300;
histogram;
inset mean std p5 q1 median q3 p95 / format = 10.2 pos = ne;
```

SAS program number	Figure title	Figure number
9	SAS output where the design parameters ($m = 125$, $n = 5$, $j = 3$, $h = 9.775$, $k =$ n(d*-d)) of the CUSUM-EX chart were chosen for a nominal $ARL_{\text{IC}} = 500$	4.25

SAS program

```
proc iml;

*Size of the IC Phase I sample;
m = 125;

* Answer = 0 if m is even and 1 if m is odd;
answer = 1;

*Number of simulations;
sim = 100000;
runlength = j(sim,1,.);

* Winsorization;
wind = 10000000000000000000000000;
wz = j(sim,1,0);

*Size of the Phase II sample;
n = 5;

*One-sided or two-sided chart;
*If one-sided, then type = 1;
*If two-sided, then type = 2;
type = 2;

*Design parameters of CUSUM chart;
d = 0.5;

* For small m;
dstar = 0.5*sqrt( (n*(m+n+1)) / (4*m+8) );
k = n*(dstar - d);
*UCL;
H = 9.775;

*Shift;
delta = 0;

stdev = 1;
mean  = delta * (stdev/sqrt(n));
```

```
*Obtaining the median;
keep = j(2,1,.);
if answer = 1 then r = (m+1)/2;
if answer = 0 then r_1 = (m/2);
if answer = 0 then r_2 = ((m+2)/2);

*Generating an IC Phase I sample;
xi = j(m,1,.);

do j = 1 to sim;

* Generating observations from the Normal distribution;
call randgen(xi,'NORMAL',0,1);

* Finding the median;
xi_rank = xi || rank(xi);
do l = 1 to m;
    *For median odd;
    if answer = 1 then do;
        if xi_rank[l,2] = r then median = xi_rank[l,1];
    end;
    *For median even;
    if answer = 0 then do;
        if xi_rank[l,2] = r_1 then keep[1,] = xi_rank[l,1];
        if xi_rank[l,2] = r_2 then keep[2,] = xi_rank[l,1];
    end;
    if answer = 0 then median = sum(keep)/2;
end;

*Calculating the CUSUM;
count = 0;
indicator = 0;
Ci_1_plus = 0;
Ci_1_minus = 0;

do i = 1 to 100000000000000000 until (indicator = 1);
count = count + 1;

*Generating a Phase II sample;
yi = j(n,1,.);
*Generating observations from the Normal distribution;
call randgen(yi,'NORMAL',mean,1);

*Calculating the precedence statistic;
precedence = j(n,1,0);
        do i = 1 to n;
            if yi[i,]>median then precedence[i,] = 1;
        end;
U = sum(precedence);
```

```
*Charting statistic;
Ci_plus = max (0, (Ci_1_plus + (U - n*d) - k));
Ci_minus = max (0, (Ci_1_minus - (U - n*d) - k));

*Comparing the charting statistics to the control limits;
if type = 2 then do;
      if ((Ci_plus > = H) | (Ci_minus > = H)) then indicator = 1;
end;
if type = 1 then do;
      if Ci_plus > = H then indicator = 1;
end;

Ci_1_plus = Ci_plus;
Ci_1_minus = Ci_minus;

if count > wind then count = wind;

end;

runlength[j,1] = count;
if runlength[j,1] = wind then wz[j,1] = 1;
end;

wind_prop = sum(wz)/sim;
print wind_prop;

create CUSUM from runlength [colname = {RL}];
append from runlength;
proc univariate data = CUSUM noprint;
histogram;
inset mean std p5 q1 median q3 p95 / format = 10.1 pos = ne;
```

SAS program number	Figure title	Figure number
10	SAS output where the design parameters ($m = 100$, $n = 5$, $h = 353$, $k = 0$) of the CUSUM-rank chart were chosen for a nominal $ARL_{\text{IC}} = 500$	4.26

SAS program

```
proc iml;

*Size of the IC Phase I sample;
m = 100;

*Number of simulations;
sim = 10000;
```

```
* Winsorization;
wind = 10000000000000000;

*Size of the Phase II sample;
n = 5;

*Shift;
delta = 0;

*Design parameters of CUSUM chart;
exp_w = (n*(n+m+1))/2;
stdev_w = sqrt ( n*m*(n+m+1)/12);
k = 0.5*stdev_w;

*UCL;
H = 353;

*One-sided or two-sided chart;
*If one-sided, then type = 1;
*If two-sided, then type = 2;
type = 2;

runl = j(sim,1,.);
wz = j(sim,1,0);
stdev = 1;
mean  = delta * (stdev/sqrt(n));

do j = 1 to sim;
* Initializing values;
    count = 0;
    indicator = 0;
    Ci_1_plus = 0;
    Ci_1_minus = 0;
    xi = j(m,1,.);
 zeros = j(m,1,0);
* Generating an IC Phase I sample;
* Generating observations from the Normal distribution;
  call randgen(xi,'NORMAL',0,1);
   xii = zeros||xi;

        do i = 1 to 100000000000000000 until (indicator = 1);
        count = count + 1;
        yi = j(n,1,.);
        * Generating a Phase II sample;
```

```
        * Generating observations from the Normal distribution;
        call randgen(yi,'NORMAL',mean,1);

        *Obtaining the rank-sum statistics;
        ones = j(n,1,1);
        yii = ones||yi;
        comb = xii // yii;
        rank = ranktie(comb[,2]);
        comb_rank = comb || rank;
        call sort (comb_rank, {2 3}, {3});
        W_vec = comb_rank[,1]#comb_rank[,3];
        W = sum(W_vec);

        *Charting statistic;
        Ci_plus = max (0, Ci_1_plus + W - exp_w - k);
        Ci_minus = max (0, Ci_1_minus + exp_w - k - W);

        *Comparing the charting statistics to the control limits;
      if type = 2 then do;
            if ((Ci_plus > = H) | (Ci_minus > = H)) then
                        indicator = 1;
      end;
      if type = 1 then do;
            if Ci_plus > = H then indicator = 1;
      end;

        Ci_1_plus = Ci_plus;
        Ci_1_minus = Ci_minus;

      if count > wind then count = wind;
        end;

runl[j,1] = count;
if runl[j,1] = wind then wz[j,1] = 1;
end;

wind_prop = sum(wz)/sim;
print wind_prop;

create CUSUM from runl [colname = {RL}];
append from runl;
proc univariate data = CUSUM;
histogram;
inset mean std p5 q1 median q3 p95 / format = 10.2 pos = ne;
```

SAS program number	Figure title	Figure number
11	SAS output where the design parameters ($m = 100$, $n = 5$, $\lambda = 0.10$, $L = 2.220$) of the EWMA-EX chart were chosen for a nominal $ARL_{\text{IC}} = 500$	4.27

SAS program

```
proc iml;

*Size of the IC Phase I sample;
m = 100;

*Odd (answer = 1) or Even (answer = 0);
answer = 0;

*Number of simulations;
sim = 100000;
runlength = j(sim,1,.);

*Size of the Phase II sample;
n = 5;

*design parameters of EWMA control chart;
lambda = 0.1;
L = 2.220;

*Shift;
delta = 0;
stdev = 1;
mean  = delta * (stdev/sqrt(n));

*Obtaining the median;
keep = j(2,1,.);
if answer = 1 then r = (m+1)/2;
if answer = 0 then r = m/2;
if answer = 0 then r_1 = (m/2);
if answer = 0 then r_2 = ((m+2)/2);

*Obtaining the value of pr;
pr = (m - r + 1) / (m + 1);

*steady state control limits;
UCL = n*((m - r + 1)/(m + 1 ))+L* sqrt(((n*n)*(r*(m-r+1))/
(((m+1)*(m+1))*(m+2)))+(lambda/(2-lambda))*n*(r*(m-r+1))/
((m+1)*(m+2)));
LCL = n*((m - r + 1)/(m + 1 ))-L* sqrt(((n*n)*(r*(m-r+1))/
(((m+1)*(m+1))*(m+2)))+(lambda/(2-lambda))*n*(r*(m-r+1))/
((m+1)*(m+2)));
```

```
*Generating an IC Phase I sample;
xi = j(m,1,.);

do j = 1 to sim;

* Generating observations from the Normal distribution;
call randgen(xi,'NORMAL',0,1);

* Finding the median;
xi_rank = xi || rank(xi);
do l = 1 to m;
    *For median odd;
    if answer = 1 then do;
        if xi_rank[l,2] = r then median = xi_rank[l,1];
    end;
    *For median even;
    if answer = 0 then do;
        if xi_rank[l,2] = r_1 then keep[1,] = xi_rank[l,1];
        if xi_rank[l,2] = r_2 then keep[2,] = xi_rank[l,1];
    end;
    if answer = 0 then median = sum(keep)/2;
end;

*Calculating the EWMA;
count = 0;
indicator = 0;
zi_1 = n*((m - r + 1)/(m + 1 )); *Starting value;

do i = 1 to 100000000000000000 until (indicator = 1);
count = count + 1;

*Generating a Phase II sample;
yi = j(n,1,.);
*Generating observations from the Normal distribution;
call randgen(yi,'NORMAL',mean,1);

*Calculating the precedence statistic;
precedence = j(n,1,0);
        do i = 1 to n;
            if yi[i,]>median then precedence[i,] = 1;
        end;
U = sum(precedence);

*Charting statistic;
zi = lambda*U + (1-lambda)*zi_1;

if (zi < UCL) & (zi > LCL) then do; indicator = 0; end;
if (zi > UCL) | (zi < LCL) then do; indicator = 1; end;
zi_1 = zi;
end;
```

```
runlength[j,1] = count;
end;

create EWMA from runlength [colname = {RL}];
append from runlength;
proc univariate data = EWMA noprint;
histogram;
inset mean std p5 q1 median q3 p95 / format = 10.2 pos = ne;
run;
```

SAS program number	Figure title	Figure number
12	SAS output where the design parameters ($m = 100$, $n = 5$, $\lambda = 0.10$, $LCL = 219.5$, $UCL = 310.5$) of the EWMA-rank chart were chosen for a nominal $ARL_{\text{IC}} = 500$	4.28

SAS program

```
proc iml;

*Size of the IC Phase I sample;
m = 100;

*Number of simulations;
sim = 100000;
runl = j(sim,1,.);

*Size of the Phase II sample;
n = 5;

*Expected value and stdev of W;
exp_w = (n*(n+m+1))/2;
stdev_w = sqrt ( n*m*(n+m+1)/12);

*Design parameters of EWMA chart;
lambda = 0.1;

*Control limits;
Add_subtract = 45.5;
LCL = exp_w - add_subtract;
UCL = exp_w + add_subtract;

*Shift;
delta = 0;
stdev = 1;
mean  = delta * (stdev/sqrt(n));
```

```
do j = 1 to sim;
* Initializing values;
    count = 0;
    indicator = 0;
    zi_1 = exp_W;
    xi = j(m,1,.);
 zeros = j(m,1,0);
* Generating an IC Phase I sample;
* Generating observations from the Normal distribution;
  call randgen(xi,'NORMAL',0,1);
   xii = zeros||xi;

        do i = 1 to 100000000000000000 until (indicator = 1);
        count = count + 1;
        yi = j(n,1,.);
        * Generating a Phase II sample;
        * Generating observations from the Normal distribution;
        call randgen(yi,'NORMAL',mean,1);

        *Obtaining the rank-sum statistics;
        ones = j(n,1,1);
        yii = ones||yi;
        comb = xii // yii;
        rank = ranktie(comb[,2]);
        comb_rank = comb || rank;
        call sort (comb_rank, {2 3}, {3});
        W_vec = comb_rank[,1]#comb_rank[,3];
        W = sum(W_vec);

        *Charting statistic;
       zi = lambda*W + (1-lambda)*zi_1;

        *Comparing the charting statistics to the control
limits (For the two-sided chart);
        if ((zi > = UCL) | (zi < = LCL)) then indicator = 1;

        zi_1 = zi;

        end;

runl[j,1] = count;

end;

create EWMA from runl [colname = {RL}];
append from runl;
proc univariate data = EWMA noprint;
histogram;
inset mean std p5 q1 median q3 p95 / format = 10.2 pos = ne;
run;
```

Some examples of SAS programs for Phase I control charts

SAS program for the Phase I median control chart

```
proc iml;

* Number of simulations;
z = 100000;
* Number of samples;
m = 5;
* Number of observations;
n = 16;
* LCL;
lcl = 4;
*UCL;
ucl = n-lcl;
* Constant added for the OOC case;
delta = 0;

* Generating observations from the IC N(mu, var) distribution;
* IC mean of Normal distribution;
mu = 0;
* IC variance of Normal distribution;
var = 1;
obs = j(m*z,n,.);
call randgen(obs, "normal", mu, sqrt(var));

do i = 1 to z;
obs[(i*m-(m-1)),] = obs[(i*m-(m-1)),]+delta;
end;

do i = 1 to z;
vector = 0;
apart = j(m,n,.);
apart[1:m,] = obs[(i*m-(m-1)):(i*m),];
do o = 1 to n;
vector = vector//apart[,o];
end;
vector = vector [2:nrow(vector),];
me = median(vector);
do k = 1 to nrow(apart);
do l = 1 to ncol(apart);
if apart[k,l]<me then apart[k,l] = 1; else apart[k,l] = 0;
end;
end;
sum = apart[,+];
signal_ucl = j(nrow(sum),1,.);
signal_lcl = j(nrow(sum),1,.);
do p = 1 to nrow(signal_ucl);
if sum[p,]>=ucl then signal_ucl[p,]=1; else signal_ucl[p,]=0;
if sum[p,]<=lcl then signal_lcl[p,]=1; else signal_lcl[p,]=0;
end;
```

```
signal = signal_lcl||signal_ucl;
at_least = sum(signal);
if at_least>0 then FAP_sum = 1; else FAP_sum = 0;
keep = keep//FAP_sum;
end;

total = sum(keep);
FAP = total/z;

print      "The Median chart for normally distributed observations",
lcl ucl,
m [label = 'Number of samples'],
n [label = 'Number of observations in each sample'],
z [label = 'Number of simulations'],
mu [label = 'mu0'],
var [label = 'sigmasq0'],
delta,
FAP [format = .4];
```

SAS program for the Phase I mean-rank control chart

```
proc iml;
keep = 0;

*Number of simulations;
sim = 100000;

*Number of samples;
m = 45;

*Number of observations in each sample;
n = 10;

*Control limits;
UCL = 2.736;
LCL = -UCL;

*The size of the pooled sample;
upper_N = m*n;

*Expected value of the rank;
exp = (upper_N+1)/2;

*Variance of the rank;
var = ( (upper_N - n) * (upper_N + 1) ) / (12*n);

*Standard deviation of the rank;
stdev = sqrt(var);

do k = 1 to sim;
obs = j(m,n,.);
r_bar_i = j(m,1,.);
zi = j(m,1,.);
```

```
*Generating observations from the N(0,1) distribution;
call randgen(obs, "normal", 0, 1);

*Ranking the observations of the pooled sample of size N = mn;
rank_obs = rank(obs);

*Calculating the average rank of each sample;
do l = 1 to m;
r_bar_i[l,] = sum (rank_obs[l,1:n])/n;
end;

*Calculating the standardized values;
do o = 1 to m;
zi[o,] = (r_bar_i[o,] - exp) / stdev;
end;

*Is the chart signalling?;
signal_ucl = j(m,1,.);
signal_lcl = j(m,1,.);
do w = 1 to m;
if zi[w,]> = ucl then signal_ucl[w,] = 1; else signal_ucl[w,] = 0;
if zi[w,]< = lcl then signal_lcl[w,] = 1; else signal_lcl[w,] = 0;
end;
signal = signal_lcl + signal_ucl;

at_least = sum(signal);
if at_least>0 then FAP_one_sim = 1; else FAP_one_sim = 0;
keep = keep // FAP_one_sim;
end;

FAP = (sum(keep))/sim;

print FAP;
```

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Nonparametric Statistical Process Control, First Edition.
Subhabrata Chakraborti and Marien Alet Graham.

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Companion website: www.wiley.com/go/chakraborti/Nonparametric_Statistical_Process_Control

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